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Preface

In 1978 the first edition of "Basic Electric Circuit Analysis" was published. It was well received by students and instructors, and after a decade of widespread adoption and four editions, a somewhat expanded version entitled "Electric Circuit Analysis" became available in 1989. The book you hold is the third edition of "Electric Circuit Analysis." It is somewhat unusual for an engineering text project to survive and prosper over a duration approaching two decades, and for the loyalty of those who use and have come to rely on it we are grateful. We hope you will find its traditional virtues securely woven into the present edition, accompanied perhaps with a few enhancements in the bargain.

The continued acceptance of this book is, we believe, the consequence of two essential ingredients: a sound initial design, and an ongoing commitment to upgrade and adapt, to regularly reappraise both content and approach in light of actual experience. To do this effectively requires awareness of the needs of instructors and students, not in the abstract but "on the ground." The four authors listed on the spine of this book have among us over one century of circuits teaching experience, and we remain in communication with the larger community of instructors and students beyond our campuses through extensive regular surveys conducted by Prentice Hall. Changes both large and small can be traced to comments made by many insightful colleagues and students, and, yes, even to complaints about specific examples or sections of narrative which could be written more clearly, and, we hope, as a consequence have been. To these individuals, too numerous to list but not to remember, we owe a great debt.

THE INTENDED PURPOSE

This book is intended for a two semester or three quarter introductory course in linear lumped circuit analysis for the sophomore post-secondary year. The first portion of the book may also be used for a one semester or two quarter terminal course for non-majors. The development is calculus-based, as are the physical laws upon which the methods of circuit analysis stand. The exposure to differential and integral calculus routinely acquired by the engineering or physical science student in the freshman year is adequate background. Other mathematical topics are raised in a limited and self-contained manner, and are not assumed or required as prerequisite background. This includes the algebraic manipulation of complex numbers, vector-matrix formulation and solution of

with instructors and students in the future. This is how we learn and move the project closer to the center of your needs and concerns. We have established an electronic mail address which will be read regularly by the authors, and to which we urge and invite you to post reactions, questions, and yes, even gripes. All will be carefully considered as the next edition, or next new book, comes round. The Internet email address is: *peter@eng.buffalo.edu*.

This work has been greatly enriched by the efforts of those who have contributed to earlier editions, whose voice can clearly be heard here, as well as those active in preparing the current edition. Of those in the latter category, special thanks to Profs. Dennis Tyner, Indira Chatterjee, Keith A. Ross and Bruce Wollenberg for their careful reading and many useful suggestions, to Sondra Chavez, Alan Apt and Eric Svendsen for their invaluable editorial help. The last-named author shouldered the heaviest load in the current revision cycle, and he would like to save the final acknowledgment for his family, who somehow understood.

David E. Johnson
John L. Hilburn
Johnny R. Johnson
Peter D. Scott

1 Introduction



Alessandro Volta
1745–1827

This endless circulation of the electric fluid may appear paradoxical, but it is no less true and real, and you may feel it with your hands.

Alessandro Volta

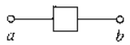
Electric circuit theory had its real beginning on March 20, 1800, when the Italian physicist Alessandro Volta announced his invention of the electric battery. This magnificent device allowed Volta to produce *current electricity*, a steady, continuous flow of electricity, as opposed to *static electricity*, produced in bursts by previous electrical machines such as the Leyden jar and Volta's own *electrophorus*.

Volta was born in the Italian city of Como, then a part of the Austrian Empire, and at age 18 he was performing electrical experiments and corresponding with well-known European electrical investigators. In 1782 he became professor of physics at the University of Padua, where he became involved in controversy with another well-known electrical pioneer, Luigi Galvani, professor of anatomy at Bologna. Galvani's experiments with frogs had led him to believe that current electricity was *animal electricity* caused by the organisms themselves. Volta, on the other hand, maintained that current electricity was *metallic electricity*, the source of which was the dissimilar metal probes attached to the frogs' legs. Both men were right. There is an animal electricity, and Galvani became famous as a founder of nerve physiology. Volta's great invention, however, revolutionized the use of electricity and gave the world one of its greatest benefits, the electric current. Volta was showered with honors during his lifetime. Napoleon made him a senator and later a count in the French Empire. After Napoleon's defeat, the Austrians allowed Volta to return to his Italian estate as a citizen in good stead. Volta was rewarded 54 years after his death when the unit of electromotive force was officially named the *volt*.

1.1 *Definitions and Units*
1.2 *Charge and Current*
1.3 *Voltage, Energy, and Power*
1.4 *Passive and Active Elements*
1.5 *Circuit Analysis and Design*
Summary
Problems

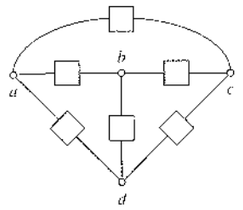
Electric circuit analysis is the portal through which students of electric phenomena begin their journey. It is the first course taken in their majors by most electrical engineering and electrical technology students. It is the primary exposure to electrical engineering, sometimes the only exposure, for students in many related disciplines, such as computer, mechanical, and biomedical engineering. Virtually all electrical engineering specialty areas, including electronics, power systems, communications, and digital design, rely heavily on circuit analysis. The only study within the electrical disciplines that is arguably more fundamental than circuits is electromagnetic (EM) field theory, which forms the scientific foundation upon which circuit analysis stands. However, even the primacy of EM field theory over circuits is incomplete, since many EM field theory problems turn out to be best solved by resort to circuit analysis techniques. It is no exaggeration to say that the ideas and methods of this first circuits course are, for many students, the most important to be encountered in their entire undergraduate curriculum. To begin our study of electric circuits, we need to consider what an electric circuit is and what we mean by its analysis and its design. Since this is a quantitative study, we must identify the quantities associated with it, in what units these quantities are measured, and the basic definitions and conventions to be used throughout. These are the topics we consider in this chapter.

1.1 DEFINITIONS AND UNITS



General two-terminal electrical element.

An electric *circuit*, or electric *network*, is a collection of electrical elements interconnected in some way. Later we shall define the electrical elements in a formal manner, but for the present we shall be content to represent a general *two-terminal element* as shown in Fig. 1.1. The terminals a and b are accessible for connection with other elements. Familiar examples of two-terminal elements are resistors, inductors, capacitors, and sources, each of which will be studied in later sections.



Electric circuit containing six elements.

More general circuit elements may have more than two terminals. Transistors and operational amplifiers are common examples. Also, a number of simple elements may be combined by interconnecting their terminals to form a single package having any number of accessible terminals. We consider some multiterminal elements later, but our main concern will be simple two-terminal elements. An example of an electric circuit with six such elements is shown in Fig. 1.2.

To understand the behavior of a circuit element, we need to consider certain quantities associated with it, such as *current* and *voltage*. These quantities and others, when they arise, must be carefully defined. This can be done only if we have a standard system of units so that when a quantity is described by measuring it, we can all agree on what the measurement means. Fortunately, there is such a standard system of units that is used today by virtually all the professional engineering societies and the authors of most modern engineering textbooks. This system, which we use throughout the book, is the *International System of Units* (abbreviated SI), adopted in 1960 by the General Conference on Weights and Measures.

There are six basic units in the SI, and all other units are derived from them. Four of the basic units—the meter, kilogram, second, and coulomb—are important in the study of circuits, and we shall examine them in some detail. The other two are the kelvin and the candela, which are not essential to our study and thus will not be considered further.

The SI units are very precisely defined in terms of permanent and reproducible quantities. However, the complete definitions are both lengthy and esoteric.* Therefore, we shall simply name the basic units and then relate them to the very familiar *British System of Units*, which includes inches, feet, pounds, and so on. Note, however, that the SI incorporates the *metric system*, favored in all but a few countries in the world, rather than the more cumbersome British System. Prefixes for powers of 10 in the SI system are shown, along with their standard abbreviations, in Table 1.1.

Prefixes in the SI

Multiple	Prefix	Symbol
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f

The basic unit of length in the SI is the *meter*, abbreviated m, which is related to the British unit of inches by the ratio 0.0254001 m/in. (to six significant digits). The basic

*Complete definitions of the basic units may be found, for example, in "IEEE Recommended Practice for Units in Published Scientific and Technical Work," *IEEE Standard 268-1973*, IEEE, New York, 1973; also see updates 268-1978 and 268-1982 from the same source.

unit of mass is the *kilogram* (kg), related to the British pound mass by 0.453593 kg/lb, and the basic unit of time in both systems is the *second* (s).

The fourth basic unit in the SI is the *coulomb* (C), which is the basic unit used to measure electric charge. We defer formal definition of this unit to the next section, where we consider charge and *current* in more detail. The name *coulomb* was chosen to honor the French scientist, inventor, and army engineer Charles Augustin de Coulomb (1736–1806), who was an early pioneer in the fields of friction, electricity, and magnetism. We should note that all SI units named for famous people have abbreviations that are capitalized; otherwise, lowercase abbreviations are used.

The quantities we shall encounter most commonly in our study of circuits are current, voltage, and power. The SI units for these quantities are derived from the four basic units (meter, kilogram, second, and coulomb) introduced above. Units of current, voltage, and power are described next, along with related units of force and energy. Together, these five derived units, along with the four basic units, are all we shall need to describe the fundamental quantitative behavior of electric circuits.*

The fundamental unit of force is the *newton* (N), which is the force required to accelerate a 1-kg mass by 1 meter per second per second (1 m/s^2). Thus 1 N = 1 kg-m/s^2. The newton is named for the great English scientist, astronomer, and mathematician Sir Isaac Newton (1642–1727). Newton's prolific accomplishments include the law of universal gravitation, the wave nature of light, and many other fundamental contributions.

The fundamental unit of work or energy is the *joule* (J), named for the British physicist James P. Joule (1818–1889), who shared in the discovery of the law of conservation of energy and helped establish the idea that heat is a form of energy. A joule is the work done by a constant 1-N force applied through a 1-m distance. Thus 1 J = 1 N-m.

The fundamental unit of current is the *ampere*, named in honor of André Marie Ampere (1775–1836), who discovered the relationship between electric current and magnetic induction. One ampere or amp is the current that flows when 1 coulomb of charge passes each second (1 A = 1 C/s). This topic will be taken up in more detail in the next section.

The fundamental unit of electrical potential is the *volt*, named after Alessandro Volta, whose thumbnail biography is given on page 1. If a charge of 1 coulomb may be moved between two points in space with the expenditure of 1 joule of work, 1 volt is said to be the potential difference existing between these points (1 V = 1 J/C).

The last derived unit we shall need is the *watt* (W), which is the fundamental unit of power, the rate at which work is done or energy expended. The watt is defined as 1 joule per second (1 J/s) and is named in honor of James Watt, the Scottish engineer whose engine design first made steam power practical and triggered the Industrial Revolution.

Example 1.1

Let us convert the speed limit of 55 miles per hour into kilometers per hour.

$$1 \text{ inch} = 0.0254 \text{ meter}$$

so

*Other measurable quantities in electric circuits, such as those associated with magnetic and electrooptic phenomena, will not be needed here. These quantities and their units will therefore not be formally considered.

$$1 \text{ mile} = (0.0254 \text{ m/in.})\ (12 \text{ in./ft})\ (5280 \text{ ft/mi})\ (1/1000 \text{ km/m})$$

or

$$1 \text{ mi} = 1.609 \text{ km}$$

Thus

$$55 \text{ mi/h} = (55 \text{ mi/h})\ (1.609 \text{ km/mi}) = 88.5 \text{ km/h}$$

We note that the units cancel at each step, which may be used to indicate which factors are needed in the conversion process.

PROBLEMS

1.1.1. Find the number of picoseconds in (a) 20 μs, (b) 2 ms, and (c) 100 ns.

Answer: (a) 2×10^7; (b) 2×10^9; (c) 1×10^5

1.1.2. If a microprocessor can perform 4 million instructions per second, how many nanoseconds are required to execute an operation requiring three instructions?

Answer: 750 ns

1.1.3. A 10,000-lb mass spacecraft in deep space is being accelerated at 1.5 g (1 g = 32 ft/s^2). How many newtons of force is the engine applying?

Answer: 6.67×10^4 N

1.1.4. The fastest recorded speed run by a human being for 10 m or more is 26.32 mph, set jointly by two runners in 1987. If this speed were sustained over 100 m, how long would it take to cover that distance? (The 100-m dash record at that time was 9.83 s.)

Answer: 8.50 s

1.2 CHARGE AND CURRENT

We are familiar with the gravitational force of attraction between bodies, which is responsible for holding us on the earth and which causes an apple dislodged from a tree to fall to the ground rather than soar into the sky. There also exist much stronger forces, far out of proportion to the masses of the bodies, which are both attractive and repulsive. These forces cannot be gravitational in nature.

One of the most important of these forces is electrical and is caused by the presence of *electrical charges*.* We explain the existence of electrical forces of both attraction and repulsion by postulating that there are two kinds of charges, positive and negative, and that unlike charges attract and like charges repel.

According to the standard theory, which has been verified by careful and repeated testing, matter is made up of atoms composed of a number of particles. The most

*Others include magnetic force and the so-called weak and strong fundamental forces acting between subatomic particles.

important of these particles are protons (positive charges) and neutrons (neutral, no net charge) found in the nucleus of the atom and electrons (negative charges) moving in orbit about the nucleus. Normally, the atom is electrically neutral; the negative charge of the electrons balances the positive charge of the protons. Atoms may become positively charged by losing electrons and become negatively charged by gaining electrons from other atoms.

As an example, we may transfer negative charge to a balloon by rubbing it against our hair. The balloon will then stick to a wall or ceiling, evidence of the attractive force between the balloon's excess electrons and the wall, which becomes positively charged when its surface electrons drift toward its interior due to repulsion by the net negative charge of the balloon.

We now define the *coulomb* (C), introduced in Section 1.1, by stating that an electron has a negative charge of 1.6021×10^{-19} coulomb. Putting it another way, a coulomb is the aggregate charge of about 6.24×10^{18} electrons. These are, of course, mind-boggling numbers, but their sizes enable us to use more manageable numbers, such as 2 C, in the work to come.

The symbol for charge will be taken as Q or q, the capital letter usually denoting constant charges, such as $Q = 4$ C, and the lowercase letter indicating a time-varying charge. In the latter case we may emphasize the time dependency by writing $q(t)$. This practice will be carried over to other electrical quantities as well.

The primary purpose of an electric circuit is to move charges at desired rates along specified paths. The motion of charges constitutes an *electric current*, denoted by the letters i or I, taken from the French word *intensité*. By definition, current is the time rate of change of charge, or

$$i = \frac{dq}{dt}$$

As introduced in Section 1.1, the basic unit of current is the ampere. One ampere of current is said to flow past a given cross section if the net rate of charge movement crossing it is 1 coulomb per second (1 C/s).

In circuit theory, current is generally specified as the movement of positive charges. This convention was proposed by the great American scientist, inventor, and diplomat Benjamin Franklin (1706–1790), who guessed that electricity traveled from positive to negative. While this is indeed true in some physical media, we now know that in the important case of metallic conductors so common in real-world circuits, the opposite is the case. In metals and most other conductors, free electrons with negative charge are the current carriers, rather than Franklin's postulated positive charge flow. Fortunately from a circuit theory point of view, there is no difference between positive charge motion in one direction and equal but opposite negative charges moving in the other. Thus we shall cling to the traditional convention of positive charge motion by defining *conventional current* and using this to measure current regardless of the true identity of the charge carrier. Conventional current is the *equivalent flow of positive charges* in a given

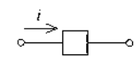
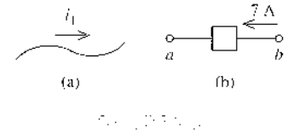


FIGURE 1.5 Current flowing in a general element.

conductor. Unless otherwise specified, all currents are to be understood as conventional currents.

Current flow along a lead or through an element will be specified by two indicators: an arrow, which establishes a *current reference direction*, and a value (variable or fixed), which quantifies the current flow in the reference direction. Figure 1.3(a) depicts a current i_1 flowing left to right through a lead, and Fig. 1.3(b) a current of 7 A flowing from right to left through an element. In the latter case, 7 A of conventional (positive) charge flows right to left, which may physically consist of an equal rate of negative charges flowing left to right. Returning to Fig. 1.3(a), note that if the variable i_1 is positive conventional current flows left to right, whereas if it is negative, current is said to flow from right to left. The arrow does not indicate the *actual* direction of current flow but rather, *the direction of conventional current flow if the algebraic sign of the current value is positive*. The arrow points opposite to the direction of conventional current flow if the sign of the current value is negative.



As another example, suppose that the current in the wire of Fig. 1.4(a) is 3 A. That is, 3 C/s passes some specific point in the wire in the left-to-right direction. In Fig. 1.4(b), −3 A passes right to left. These two cases result in exactly the same charge transfer and are thus equivalent ways to denote precisely the same current flow. *We may always reverse both the algebraic sign and the reference direction of a current without changing either its value or direction.* This corresponds to reversing the sign of a number twice, thus leaving it unchanged. Note that without knowing its reference direction, a current's numerical value alone is not enough to specify the current. Both value and reference direction are needed.



FIGURE 1.4 Two representations of the same current.

Figure 1.5 represents a general circuit element with a current i flowing from the left toward the right terminal. The total charge entering the element between time t_0 and t is found by integrating (1.1). The result is

$$q_T = q(t) - q(t_0) = \int_{t_0}^{t} i \, d\tau \tag{1.2}$$

We should note at this point that we are considering the network elements to be *electrically neutral*. That is, no net positive or negative charge can accumulate in the element. Any positive charges entering must be accompanied by equal positive charges

leaving (or, equivalently, an equal negative charge entering). This property is a consequence of Kirchhoff's current law, which is discussed in Section 1.3. Thus the current shown entering the left terminal in Fig. 1.5 must leave the right terminal. Note that the arrow could as well have been drawn pointing out the right side of the element as into the left with no change in current reference direction.

Example 1.2

As an example, suppose that the current entering a terminal of an element is $i = 4$ A. The total charge entering the terminal between $t = 0$ and $t = 3$ s is given by

$$q = \int_0^3 4dt = 12 \text{ C}$$

Several types of current are frequently encountered, some of which are shown in Fig. 1.6. A constant current, as shown in Fig. 1.6(a), will be termed a *direct current*, or dc. An *alternating current*, or ac, is by definition a sinusoidal current, such as that of Fig. 1.6(b). Sinusoids are discussed in some detail in Chapter 8. Figure 1.6(c) and (d) illustrate, respectively, an *exponential current* and a *sawtooth current*.

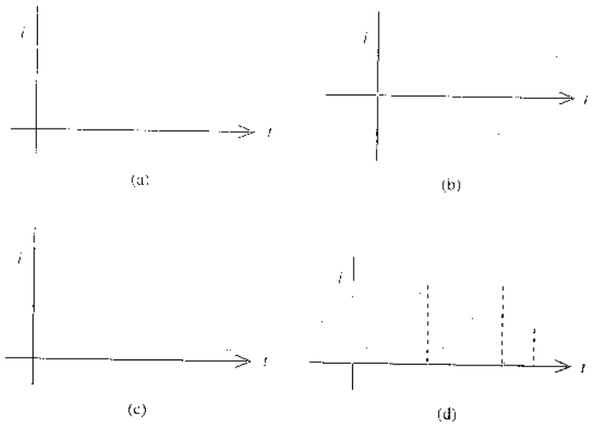


FIGURE 1.6 (a) Dc; (b) ac; (c) exponential; (d) sawtooth current.

There are many commercial uses for dc, such as in flashlights and power supplies for electronic circuits, and ac is the common household current found all over the world. Exponential currents appear quite often (whether we want them or not!) when a switch is actuated to close a path in an energized circuit. Sawtooth waves are useful in devices such as cathode ray tubes (CRTs) used for displaying electrical waveforms visually.

1.2.1. How many electrons are represented by a charge of 0.32042 pC?
Answer: 2 million

1.2.2. The total charge entering a terminal of an element is given by $q = 4t^3 - 5t$ mC. Find the current i at $t = 0$ and at $t = 2$ s.
Answer: -5 mA; 43 mA

1.2.3. The current entering a terminal is given by $i = 1 + \pi \sin 2\pi t$ A. Find the total charge entering the terminal between $t = 0$ and $t = 1.5$ s.
Answer: 2.5 C

1.3 VOLTAGE, ENERGY, AND POWER

Charges in a conductor, exemplified by free electrons, may move in a random manner. However, if we want some concerted motion on their part, such as is the case with an electric current, we must apply an external or *electromotive force (emf)*. Thus work is done on the charges. In Section 1.2 we defined voltage across an element as the work done in moving a unit charge (+1 C) through the element from one terminal to the other.

Voltage across an element will be designated by two indicators: a plus–minus sign pair, which establishes a *voltage reference direction*, and a *value* (variable or fixed), which quantifies the voltage across the element in the specified reference direction. In Fig. 1.7(a) we see a voltage (or potential difference) of value v_6 volts across the element, measured as potential at the left end of the element minus potential at the right end. Thus, if $v_6 > 0$ V, the left end is at the higher potential, and if $v_6 < 0$ V, the right end is at the higher potential. Note that specifying a reference direction does not by itself indicate which end of the element is at the higher potential. We need not try to guess the direction of potential drop when assigning a voltage reference direction for a voltage variable. In Fig. 1.7(b) the right end of the element is +23 V higher than the left, so there is a +23-V drop right to left across the element or, equally well, a +23-V rise left to right across the element.

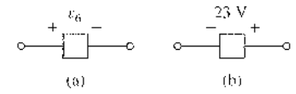


FIGURE 1.7 Specifying voltage value and reference direction.

As examples, Fig. 1.8(a) and (b) are two versions of exactly the same voltage. In (a), terminal A is +5 V above terminal B, and in (b), terminal B is −5 V above A (or +5 V below A).

We may also use a *double-subscript* notation, v_{ab}, to denote the potential of point a with respect to point b. In this case we have, in general, $v_{ab} = -v_{ba}$. Thus in Fig. 1.8(a), $v_{AB} = 5$ V and $v_{BA} = -5$ V.

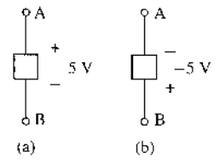
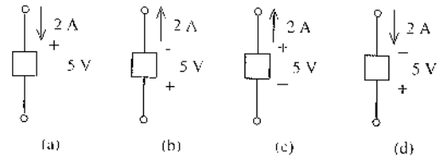


FIGURE 1.8 Two equivalent voltage representations.

In transferring charge through an element, work is being done, as we have said. Or, putting it another way, energy is being supplied. To know whether energy is being supplied *to* the element or *by* the element to the rest of the circuit, we must know both the polarity of the voltage across the element and the direction of the current through the element. If a positive current enters the positive terminal, an external force must be driving the current and is thus supplying or delivering energy to the element. The element is *absorbing* energy in this case. If, on the other hand, a positive current leaves the positive terminal (enters the negative terminal), the element is delivering energy to the external circuit.

As examples, in Fig. 1.9(a) the element is absorbing energy. A positive current enters the positive terminal. This is also the case in Fig. 1.9(b). In Fig. 1.9(c) and (d), a positive current enters the negative terminal, and therefore the element is delivering energy in both cases.



Various voltage–current relationships.

Let us consider now the *rate* at which energy is being delivered to or by a circuit element. If the voltage across the element is v and a small charge Δq is moved through the element from the positive to the negative terminal, the energy absorbed by the element, say Δw, is given by

$$\Delta w = v\,\Delta q$$

If the time involved is Δt, then the rate at which the work is being done, or that the energy w is being expended, is given, as Δt gets smaller and smaller, by

$$\lim_{\Delta t \to 0} \frac{\Delta w}{\Delta t} = \lim_{\Delta t \to 0} v \frac{\Delta q}{\Delta t}$$

or

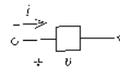
$$\frac{dw}{dt} = v\frac{dq}{dt} = vi \tag{1.3}$$

Since by definition the rate at which energy is expended is power, denoted by p, we have

$$p = \frac{dw}{dt} = vi \tag{1.4}$$

We might observe that (1.4) is dimensionally correct, since the units of vi are (J/C)(C/s) or J/s, which is watts (W), defined earlier.

The quantities v and i are generally functions of time, which we may also denote by $v(t)$ and $i(t)$. Therefore, p given by (1.4) is a time-varying quantity. It is sometimes



Element whose terminal variables satisfy the passive sign convention.

called the *instantaneous power* because its value is the power at the instant of time at which v and i are measured.

Note carefully the relationship between the current and voltage reference directions of the element of Fig. 1.10. *If the current reference direction arrow points into the positive end of the voltage reference direction, the current and voltage so defined are said to satisfy the passive sign convention.* In this case, $vi > 0$ means the element is *absorbing* power; $vi < 0$ means it is *delivering* power at that instant of time. More will be said about this in Section 1.4.

Example 1.3

As examples, in Fig. 1.9(a) and (b) the passive sign convention is satisfied (arrow points into the positive end or equivalently out of the negative end). The element is absorbing power: $p = (5)(2) = 10$ W. In Fig. 1.9(c) and (d) it is *delivering* 10 W to the external circuit, since $vi = +10$ W, but these variables violate the passive sign convention.

Before ending our discussion of power and energy, let us solve (1.4) for the energy w delivered to an element between time t_0 and t. We have, upon integrating both sides between t_0 and t,

$$w(t) - w(t_0) = \int_{t_0}^{t} vi\,d\tau \tag{1.5}$$

Example 1.4

For example, if, in Fig. 1.10, $i = 2t$ A and $v = 6$ V, the energy delivered to the element between $t = 0$ and $t = 2$ s is given by

$$w(2) - w(0) = \int_0^2 (6)(2\tau)\,d\tau = 24 \text{ J}$$

Since the left member of (1.5) represents the energy delivered to the element between t_0 and t, we may interpret $w(t)$ as the energy delivered to the element between the beginning of time and t and $w(t_0)$ as the energy between the beginning of time and t_0. In the beginning of time, which let us say is $t = -\infty$, the energy delivered by that time to the element was zero; that is,

$$w(-\infty) = 0$$

If $t_0 = -\infty$ in (1.5), we shall have the energy delivered to the element from the beginning up to t, given by

$$w(t) = \int_{-\infty}^{t} vi\,dt \tag{1.6}$$

This is consistent with (1.5), since

$$w(t) = \int_{-\infty}^{t} vi\,dt = \int_{-\infty}^{t_0} vi\,dt + \int_{t_0}^{t} vi\,dt$$

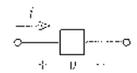
By (1.6), this may be written

$$w(t) = w(t_0) + \int_{t_0}^{t} vi\, dt$$

which is (1.5).

1.3.1. Find v if $i = 6$ mA and the element is (a) absorbing power of $p = 18$ mW and (b) delivering to the external circuit a power $p = 12$ mW.

Answer: (a) 3 V; (b) −2 V



1.3.2. If $i = 3$ A and $v = 6$ V in Exercise 1.3.1, find (a) the power absorbed by the element and (b) the energy delivered to the element between 2 and 4 s.

Answer: (a) 18 W; (b) 36 J

1.3.3. A two-terminal element absorbs w millijoules of energy as shown. If the current entering the positive terminal is $i = 100\cos 1000\pi t$ mA, find the element voltage at $t = 1$ ms and $t = 4$ ms.

Answer: −50 V; 5 V

1.4 PASSIVE AND ACTIVE ELEMENTS

We may classify circuit elements into two broad categories, passive elements and active elements, by considering the energy delivered to or by them. A circuit element is said to be *passive* if it cannot deliver more energy than has previously been supplied to it by the rest of the circuit. That is, referring to (1.6), at each t the net energy absorbed by a passive element up to t must be nonnegative:

$$w(t) = \int_{-\infty}^{t} p(\tau)\, d\tau = \int_{\infty}^{t} vi\, d\tau \geq 0 \tag{1.7}$$

The reference directions of v and i in this equation are assumed to satisfy the passive sign convention introduced in Section 1.3 (the arrow for current reference points into the positive end of the voltage reference direction). Only if the passive sign convention is satisfied will this integral be guaranteed nonnegative for passive elements (which explains the choice of the name "passive sign convention"). As we shall explore later, examples of passive elements are resistors, capacitors, and inductors.

An *active* element is any element that is not passive. That is, (1.7) does not hold at each time t. Examples of active elements are generators, batteries, and electronic devices that require power supplies.

We are not ready at this stage to begin a formal discussion of the various passive elements. This will be done in later chapters. In this section we give a brief discussion of two very important active elements, the independent voltage source and the independent current source.

An independent voltage source is a two-terminal element, such as a battery or a generator, that maintains a specified voltage between its terminals regardless of the rest of the circuit it is inserted into. The voltage is completely independent of the current through the element.

The symbol for a voltage source is shown in Fig. 1.11(a). The value v_s is the *source function* and the plus–minus signs within the source give the *source function reference direction*. The source function v_s, which may be constant or a function of time $v_s(t)$, is assumed known. In addition, we will always define a terminal voltage variable across every element in a circuit, including voltage sources. v is the terminal voltage in Fig. 1.11(a). By definition of an independent voltage source, its terminal voltage simply equals the specified value:

$$v = v_s$$

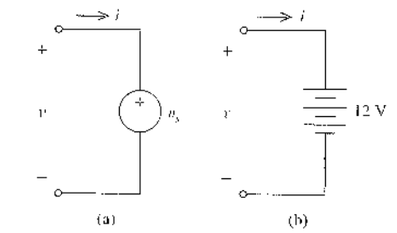


FIGURE 1.11 Independent voltage source.

In other words, *the voltage across an independent voltage source always equals its specified source function, regardless of the current through it.* Note that, if the terminal voltage reference direction and the source function reference direction are opposite one another, a minus sign must be inserted in the element law, $v = -v_s$, since the terminal voltage is defined in this case as the negative of the source function.

Equation (1.8a) is an example of an *element law*, an equation involving the element's terminal voltage and/or current, which describes the behavior of an element. Each element has a distinct element law, as we shall see for current sources, resistors, and each other type of element as it is introduced.

Another symbol that is often used for a constant voltage source, such as a battery with 12 V across its terminals, is shown in Fig. 1.11(b). In the case of constant sources, we shall use Fig. 1.11(a) and (b) interchangeably.

An *independent current source* is a two-terminal element through which a specified current flows. The value of this current is given by its source function (i_s in Fig. 1.12)

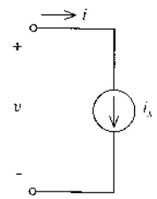
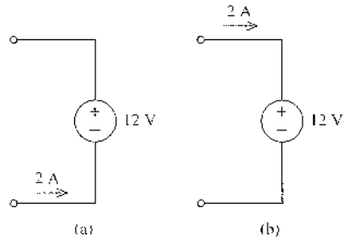


FIGURE 1.12
Independent current source.

and its source function reference direction by the arrow inside the source. As with the independent voltage source, the source function i_s [or $i_s(t)$ in the time-varying case] is assumed known and is related to this element's terminal current variable (i in Fig. 1.12) by the element law for an independent current source:

$$i = i_s$$

If the source function i_s and terminal current i reference directions are opposed rather than in agreement as shown in Fig. 1.13, a minus sign must be introduced into the terminal law, $i = -i_s$.



(a) Source delivering power; (b) source absorbing power.

Example 1.5

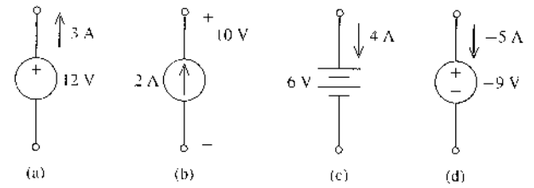
To show that independent sources are, in general, active elements, we need only demonstrate that there exist circumstances under which they may be made to supply rather than absorb net energy. For a current source with source function $i_s(t)$ A and element law $i(t) = i_s(t)$, suppose that we arrange for the voltage across the source to be $v(t) = -i_s(t)$ V, with reference directions as shown in Fig. 1.12. Then, by (1.5), $p(t) = v(t)i(t) = -i_s^2(t)$, and the net energy $w(t)$ absorbed by the source must be nonpositive for each t. If $i_s(t)$ is not identically equal to zero for all t, $w(t)$ must be not just nonpositive but, in fact, negative for some t. This violates the requirement (1.7) that passive elements never absorb negative net energy, in other words never supply net energy, under any circumstances. A similar argument can be used to show that a voltage source is also an active element. Note that this result assumes that the source function $i_s(t)$ is not identically equal to zero for all time t, in which case $w(t)$ would also be zero for all t. Sources with zero source functions are passive elements; all other sources are capable of producing net energy and are thus active elements.

Independent sources are usually meant to deliver power to the external circuit, although in a particular circuit they may be either net suppliers or absorbers of power. In Fig. 1.13(a), $v = +12$ V is the terminal voltage across the source and $i = +2$ A its terminal current. Since the reference directions of these terminal variables do not satisfy the passive sign convention (the arrow points into the minus end), $p = vi = 24$ W > 0 implies that the source is *supplying* 24 W of power to the rest of the circuit. In Fig. 1.13(b), the reference directions do satisfy the passive sign convention, so $p = vi = 24$ W > 0 signifies that the battery is *absorbing* 24 W, as would be the case when it is being charged by a more powerful source somewhere in the circuit.

The sources that we have discussed here, as well as the circuit elements to be considered later, are *ideal elements*. That is, they are mathematical models that approximate the actual or physical elements only under certain conditions. For example, an ideal automobile battery supplies a constant 12 V, no matter what external circuit is connected to it. Since its current is completely arbitrary, it could theoretically deliver an infinite amount of power. This, of course, is not possible in the case of a physical device. A real 12-V automobile battery supplies roughly constant voltage only as long as the current it is required to deliver is low. When the current exceeds a few hundred amperes, the voltage drops appreciably from 12 V.

1.4.1. Find the power being supplied by the sources shown.

Answer: (a) 36 W; (b) 20 W; (c) −24 W; (d) −45 W



1.4.2. The terminal voltage of a voltage source is $v = 6 \sin 2t$ V. If the charge leaving the positive terminal is $q = -2 \cos 2t$ mC, find (a) the power supplied by the source at any time and (b) the energy supplied by the source between 0 and t seconds.

Answer: (a) $24 \sin^2 2t$ mW; (b) $12t - 3 \sin 4t$ mJ

1.5 CIRCUIT ANALYSIS AND DESIGN

Let us now turn to the words *circuit analysis*, which are contained in the title of the book, and the related idea of circuit design, and consider their meaning. Generally, if an electric circuit is subjected to an *input* or excitation in the form of a voltage or a current provided by an independent source, then an *output* or response is elicited. The input and

output are voltages or currents associated with particular elements in the circuit. There may be, of course, more than one input and more than one output.

There are two main branches of circuit theory, and they are closely linked to the fundamental concepts of input, circuit, and output. The first branch is *circuit analysis*, which, given an input and a circuit, is the process of determining the output. The other branch is *circuit design*, which, given a set of inputs and their desired outputs, is the process of creating a circuit that realizes this desired input–output relationship (often subject to additional side constraints such as size, cost, etc).

The balance of emphasis in this book tilts strongly toward analysis. While circuit design is the subject of several chapter sections entitled "Design of ..." and is the sole focus of the final chapter, the balance of this study is devoted to analysis. This may seem a somewhat puzzling choice. After all, design, the creation of something new and useful, is for most of us more satisfying and motivating than analyzing a given circuit by solving for its currents and voltages. The emphasis on analysis perhaps merits some comment.

First consider the process of circuit design, which may be viewed as a cyclic process. Given the desired inputs and outputs, an initial guess of a circuit is made. This circuit is then analyzed to determine its actual outputs for the desired inputs, and the result is compared to the desired input–output relationship. A judgment is made as to whether the design goals have been met satisfactorily. If not, modifications in the circuit are made and this iterative-improvement design cycle is then repeated.

Conceiving a first-cut or initial-guess circuit and selecting the specific modifications to be made during each redesign cycle are seldom matters on which two circuit designers would agree exactly. These choices depend heavily on individual skill, experience, and intuition. Matters largely of judgment and creativity, they constitute the "art" component of circuit design. They cannot be codified into explicit, mechanistic rules any more than the process by which a sculptor works can. *Circuit design is a creative human activity*, like the sculptor's process in designing a statue.

Circuit analysis constitutes the complementary "science" component embedded within the circuit design cycle. Without analysis, the input–output behavior of even the initial-guess circuit could not be determined; thus not a single improvement in the first-cut circuit can be made with confidence. We cannot expect to improve steadily what we don't clearly understand. Thus, while circuit analysis can be performed without reference to circuit design, the opposite is not true.

Since circuit analysis forms the necessary foundation for circuit design, it must be studied before meaningful design is possible. Just as a jazz musician must master the musical scales before creating really interesting and original music, the elements of circuit analysis are our scales and must be mastered if we are to accept the challenge of designing useful and interesting circuits.

Finally, we note that circuit design may be approached on two levels. *Schematic design* is the solution of a design problem by specifying a circuit diagram, while *physical design* is the further conversion of the schematic to actual hardware. In this study we content ourselves with schematic design. Unlike physical design, schematic design does not require a parts inventory, measurement instruments, or access to an electronics laboratory. The elements of schematic design are more abstract than physical elements such as resistors, capacitors, and the like: equations to model the idealized behavior of the elements contained in the circuit diagram, and the mathematical methods to exercise the circuit analytically to predict its idealized input–output behavior. In addition, computer-aided circuit design software tools such as SPICE or PSpice will prove vital to all but the simplest schematic design exercises, speeding the many calculations that must be undertaken at each repetition of the design cycle. Without these simulation tools we could not tackle designs of circuits containing many loops and nodes, that is, we would limit ourselves to a narrow selection of "plain vanilla" design exercises. With these tools in hand, the scope of accessible schematic designs will prove satisfyingly broad and practical. Our goal in this study is to present the systematic methods of analysis upon which design stands, and to offer a first exposure to the challenges and satisfactions of circuit design.

SUMMARY

We begin our study of electrical circuits by defining those electrical quantities that will be of interest, primarily current, voltage, and power, and the selection of a consistent system of units for their measurement. The SI system chosen incorporates the metric system (meters, kilograms, seconds) and uses the coulomb, a measure of charge, as a fundamental unit. SI units of current, voltage, and power are amperes, volts, and watts, respectively.

- Current has both magnitude and direction. Its direction is determined by its current reference direction arrow and the sign of its algebraic value. A positive value indicates equivalent positive current flow in the direction of the arrow, and negative current indicates the opposite direction of flow.
- Voltage, or potential difference, also has magnitude and direction. Its direction is determined by its voltage reference direction plus–minus signs and its algebraic sign. A positive voltage indicated voltage drop in the plus-to-minus direction (higher electric potential at the plus sign), and negative value indicates the opposite direction of voltage drop.
- Passive elements cannot supply more energy on balance than they receive from other elements. Active elements, such as independent sources, can.
- The instantaneous power absorbed by a two-terminal subnetwork equals its current–voltage product if the reference directions satisfy the passive sign convention (current arrow points into voltage plus sign or out of minus sign). Negative power absorbed is equivalent to positive power supplied.
- Circuit analysis is the process of determining output currents and voltages for a specified circuit. Circuit design is the discovery of a circuit with some required input–output behavior.

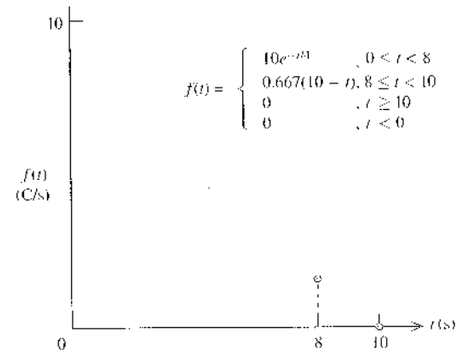
PROBLEMS

1.1. Einstein's formula for conversion of mass to energy is $E = mc^2$, where E is energy, m is mass, and $c = 3.02 \times 10^8$ m/s is the speed of light. How many joules of energy are there in 5 micrograms of matter?

1.2. Taking the speed of light to be $c = 3.02 \times 10^8$ m/s and a year to contain exactly $365\frac{1}{4}$ days, how many kilometers are there in a light-year?

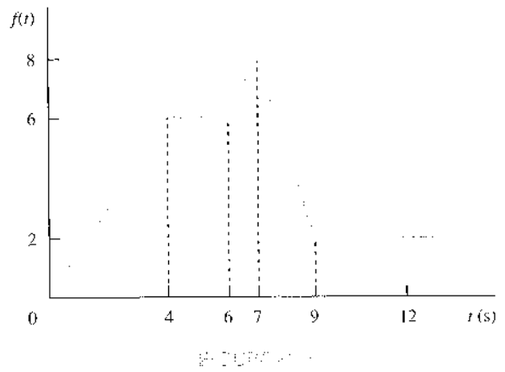
1.5. The fastest recorded speed for any ground vehicle was 6121 mph (an unmanned rocket sled in 1982). For many years, the official air speed record for manned flight (a SR-71 "Blackbird" aircraft in 1964) was 980.4 m/s. Compare these speeds in kilometers per hour.

1.4. $f(t)$ is the charge in coulombs passing a certain cross section along a wire per unit time. Find (a) the total charge passing the cross section, (b) the time at which exactly 1 C has passed, (c) the current at $t = 5$ s, and (d) the current at $t = 8$ s.



1.5. How long would it take a force of 10 N to accelerate a 1000-lb mass object to a speed of 55 mph?

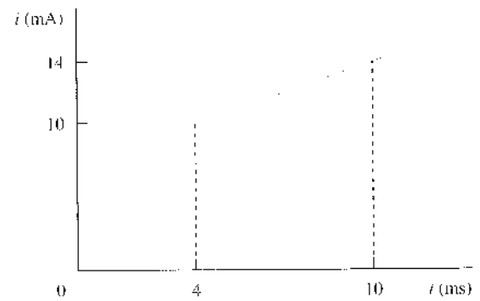
1.6. If the function $f(t)$ is the current in amperes entering the positive terminal of an element at time t (seconds), find (a) the total charge that has entered between 4 and 9 s, (b) the charge entering at 8 s, and (c) the current at 1, 5, and 8 s. Take the charge to be 0 at $t = 0$.



1.7. If $f(t)$ in Problem 1.6 is the charge entering the element in coulombs and t is in seconds, find the charge entering the element between 4 and 9 s, the current at 6.5 s, and the current at 8 s.

1.8. If 10 J of work is required to move $\frac{1}{4}$ C of negative charge from point a to point b, find v_{ab} and v_{ba}.

1.9. If the voltage across an element is 8 V and the current i entering the positive terminal is as shown, find the power delivered to the element at $t = 7$ ms and the total charge and total energy delivered to the element between 0 and 10 ms.



1.10. If the function graphed in Problem 1.9 is the voltage v (volts) across an element versus the time (ms), and the current entering the positive terminal is 3 mA, find the power delivered to the element at 2 ms and at 6 ms.

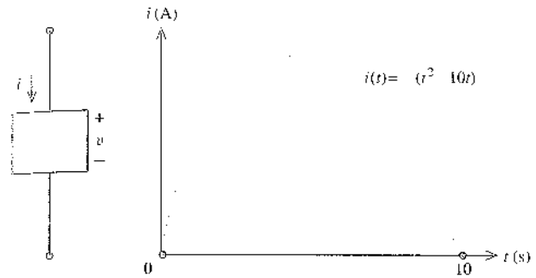
1.11. Find the power delivered to an element at $t = 2$ ms if the charge entering the positive terminal is

$$q = 10\cos 125\pi t \text{ mC}$$

and the voltage is

$$v = 6\sin 125\pi t \text{ V}$$

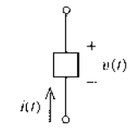
1.12. If $v(t) = 4t$ V and $i(t)$ is as shown, find the power $p(t)$ dissipated by the element and total energy supplied to the element between times $t = 0$ and $t = 10$.



1.13. Repeat Problem 1.12 for $v(t) = 4(t - 5)$ V.

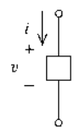
1.14. The power delivered to an element is $p = 10t\cos^2 t$ W and the voltage is $v(t) = 2\cos t$ V. Find the current entering the positive terminal and the charge between times 0 and 2π s.

1.15. An element has $v(t) = 4e^{-t}$ V for $t \geq 0$ s. If we wish this element to supply net energy of $2t$ J for $t \geq 0$, what must $i(t)$ be?



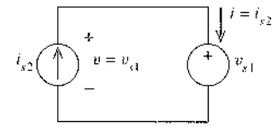
1.16. Show that by definition of a passive element, any element satisfying $v = mi$, where $m \geq 0$ and the element variables i and v satisfy the passive sign convention, is passive.

1.17. The element shown satisfies $v = 2i + 1$. Is this element active or passive? Find the net energy between $t = -\infty$ and $t = 1$ s if $v = 0$ for $t < 0$ and $v = \frac{1}{2}$ V for $t \geq 0$.



1.18. An element has constant current and voltage, $v = +100$ V and $i = +10$ A, with i and v satisfying the passive sign convention. How much energy is required to move each coulomb through the element? How long does it take to move 1 C through? Repeat if $i = -10$ A.

1.19. If $v_{s1} = 100$ V and $i_{s2} = -10$ A, which source is supplying energy and which is dissipating energy? Holding i_{s2} fixed at -10 A, what would v_{s1} be so that 1000 J/h was being supplied by the current source to the voltage source?

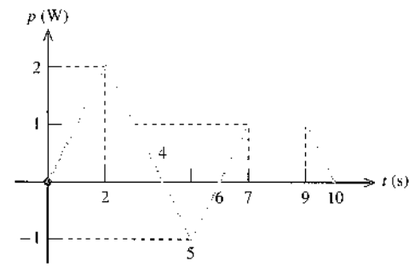


1.20. If a current $i = 0.4$ A is entering the positive terminal of a battery with terminal voltage $v = 12$ V, the battery is in the process of being charged. (It is absorbing rather than delivering power.) Find (a) the energy supplied to the battery and (b) the charge delivered to the battery in 2 h (hours). Note the consistency of the units 1 V = 1 J/C.

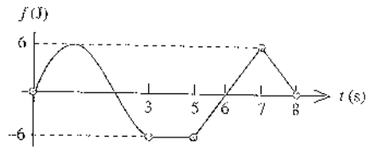
1.21. Suppose that the voltage in Problem 1.20 varies linearly from 6 to 18 V as t varies from 0 to 10 min. If $i = 2$ A during this time, find (a) the total energy supplied and (b) the total charge delivered to the battery.

1.22. If a current source and a voltage source are connected back to back as in Problem 1.19, are there values v_{s1} and i_{s2} for which both sources are supplying equal nonzero power to one another? If so, what values? If not, show why not.

1.23. The power absorbed by a circuit element is shown. At what time is the net energy a maximum? What is the maximum value of the net energy?



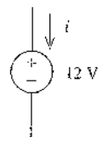
1.24. The net energy $f(t)$ in joules absorbed by an element is shown in the figure. [$f(t)$ equals net energy from $t = -\infty$ to time t.] $f(t) = 6\sin(\pi/2)t$ for $0 \leq t \leq 3$ and is piecewise linear thereafter as shown. What is the peak power supplied to the element? What is the peak power dissipated by the element? Is this a passive or active element?



1.25. Suppose that an element obeys $v = Ki^p$, where i and v satisfy the passive sign convention, $K > 0$, and p is a positive integer. Find those values of p for which the element is passive.

1.26. An element has a power rating of 1 kW. If the element satisfies the equation $v = 2i + di/dt$, what is the maximum voltage $v(t)$ we may have across the element if $i(t) = 4\cos 2t$ A?

1.27. A circuit breaker is set to trip if the current through the voltage source reaches ±20 A (that is, if $|i| > 20$). If the net energy delivered by the source from $t = -\infty$ to time t is 0 for $t < 1$ s and $10(\cos \pi t + 1)$ J for $t \geq 1$ s, will the circuit breaker ever trip? If so, when will it first trip?

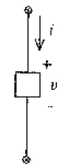


1.28. A certain element with i and v that satisfy the passive sign convention obeys the equation $i(t) = \int_{t_0}^{t} v(\tau)\, d\tau + i(t_0)$. We wish to supply this element 5 J of energy using a constant voltage between $t = 0$ and $t = 1$ s. What should be the value of this voltage? What is the peak power we must be able to supply?

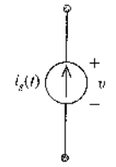
1.29. When fully charged, a car battery's stored charge depends on the temperature. If $q_0(T) = \frac{3}{4}T + 15$, where $q_0(T)$ is the stored charge (units are kilocoulombs) and T the Fahrenheit temperature, how many times can we attempt to start the car in 100°F weather? In 32°F weather? Assume that each attempt lasts 10 s and draws 90 A of current from the battery.

The electrical potential at a distance r away from a certain electrical charge is $20/r$ V (r in meters). How much work is required to move a +1-C charge from 10 m away to 1 m away? If the movement were made at uniform velocity over 9 s, what would be the maximum power required?

The element shown has the relationship $v(t) = |i(t)|$ between its current and voltage variables. Sketch the power $p(t)$ if $i(t) = 10\cos 2\pi t$. Is this element active or passive?



The power delivered by an independent current source is $p(t) = 15e^{-2t}\sin t$ W for $t \geq 0$. If the voltage across the source is $v(t) = 30e^{-t}$ V for $t \geq 0$, find the source function $i_s(t)$ and determine the total energy delivered by the source for $0 \leq t < \infty$.



Determine if the rest of the circuit to which the source described in Problem 1.32 is connected is passive or active. Justify your answer.



André Marie Ampère
1775–1836

I shall call the first "electric tension" [voltage] and the second "electric current."

André Marie Ampère

2 Resistive Circuits

On September 11, 1820, the exciting announcement was read to the French Academy of Sciences of the discovery by the Danish physicist Hans Christian Oersted that an electric current produces a magnetic effect. One member of the academy, André Marie Ampère, a French mathematics professor, was highly impressed and within a week had repeated Oersted's experiment, given a mathematical explanation of it, and—in addition—discovered that electric currents in parallel wires exert a magnetic force on each other.

Ampère was born in Lyons, France, and at an early age had read all the great works in his father's library. At age 12 he was introduced to the Lyons library and because many of its best mathematical works were in Latin, he mastered that language in a few weeks. Despite two crushing personal tragedies—at age 18 he witnessed his father's execution on the guillotine by the French Revolutionaries, and later his young, beloved wife died suddenly after only four years of marriage—Ampère was a brilliant and prolific scientist. He formulated many of the laws of electricity and magnetism and was the father of electrodynamics. The unit of electric current, the *ampere*, was chosen in his honor in 1881.

2.1 Kirchhoff's Laws
2.2 Ohm's Law
2.3 Equivalent Subcircuits
2.4 Series Equivalents and Voltage Division
2.5 Parallel Equivalents and Current Division
2.6 Thevenin and Norton Equivalents
2.7 Practical Sources and Resistors
2.8 Ammeters, Voltmeters, and Ohmmeters
2.9 Design of Resistive Circuits
Summary
Problems

The simplest and most commonly used circuit element is the resistor. All electrical conductors exhibit properties that are characteristic of resistors. When currents flow in metals, for instance, electrons that make up the current collide with the orderly lattice of atoms of the metal. This impedes or resists the motion of the electrons. The larger the number of collisions, the greater the resistance of the conductor. In other materials the charge carriers and their surrounding milieu may differ from free electrons flowing through crystal lattices, but the principle that movement of charge is resisted remains the same. We shall consider a resistor to be any device that exhibits solely a resistance. Materials that are commonly used in fabricating resistors include metallic alloys and carbon compounds and, in the case of resistors that are deposited onto integrated-circuit chip substrates, semiconductors.

In this chapter we first introduce two general laws governing the flow of current and the pattern of voltages in any circuit, called Kirchhoff's current law and Kirchhoff's voltage law. We next introduce the basic relationship linking current and voltage in resistors, Ohm's law. Armed with these tools, we begin our study of systematic circuit analysis. The fundamental concept of equivalent circuits is introduced. Simplifications available through the use of equivalent circuits are then discovered in the context of series–parallel and Thevenin–Norton equivalents.

2.1 KIRCHHOFF'S LAWS

For purposes of circuit analysis, an electric circuit is fully characterized by specifying just two of its characteristics: the elements it contains and how those elements are interconnected. Nothing more need be known to determine the resulting currents and voltages.

In this section we consider two simple laws first stated in 1847 by the German physicist Gustav Kirchhoff (1824–1887), which together define how the interconnection

of elements constrains their possible currents and voltages. These laws are valid for circuits containing elements of all types: resistors, inductors, capacitors, sources, and others. They are called Kirchhoff's current law and Kirchhoff's voltage law. These two interconnection laws, together with element laws describing the behavior of each individual element in the circuit, are all the equations we will need for systematic circuit analysis. The element law for a resistor, Ohm's law, will be taken up in Section 2.2, and systematic circuit analysis methods are developed in Chapter 4.

Our development of Kirchhoff's laws will not include rigorous proofs. These are best supplied in the context of the study of electromagnetic fields, the theory upon which Kirchhoff's laws are based. For the present circuit theory purposes, we content ourselves with defining the laws and justifying their plausability.

A *circuit* consists of two or more circuit elements connected by means of perfect conductors. Perfect conductors are idealized leads or wires that allow current to flow without impediment (no charge accumulations or voltage drops along the leads, no power or energy dissipation). For circuits so defined, the energy can be considered to reside, or be lumped, entirely within each circuit element, and thus the circuit is called a *lumped-parameter circuit*.

A point of connection of two or more circuit elements, together with all the connecting wire* in unbroken contact with this point, is called a *node*. An example of a circuit with three nodes is shown in Fig. 2.1(a). Nodes are indicated in diagrams in two ways: by a dashed line enclosing the node, as with nodes 1 and 3, or by marking a typical point within the node, as with node 2 of this figure. Whichever is used, it is important to remember that a node is all the wire in direct contact with a given point, and thus any two points that can be traversed by moving exclusively along connecting leads are both part of the same node. The points marked a and b in Fig. 2.1(a) are part of the same node, designated node 1. We are now ready to discuss the all-important laws of Kirchhoff.

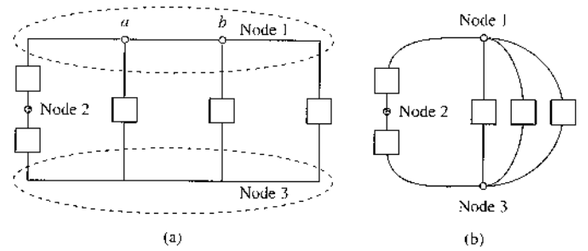


FIGURE 2.1 (a) Three-node circuit; (b) same circuit redrawn.

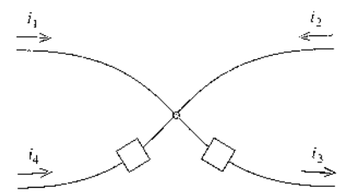
Kirchhoff's current law (KCL) states that

> The algebraic sum of the currents entering any node is zero.

*The leads that connect elements will frequently be called "wires" even though they may take on quite different physical form. For instance, integrated-circuit chips leads are narrow, shallow conducting channels called *traces* deposited onto the chip substrate.

For example, the currents entering the node of Fig. 2.2 are i_1, i_2, $-i_3$, and i_4 (since i_3 leaving is $-i_3$ entering). Therefore, KCL for this case is

$$i_1 + i_2 + (-i_3) + i_4 = 0$$



Current flowing into a node.

For the sake of argument let us suppose that the sum were not zero. In such a case we would have

$$i_1 + i_2 - i_3 + i_4 = \Psi \neq 0$$

where Ψ has units of coulombs per second and must be the rate at which charges are accumulating in the node. However, a node consists of perfect conductors and cannot accumulate charges. In addition, a basic principle of physics states that charges can be neither created nor destroyed (conservation of charge). Therefore, our assumption is not valid, and Ψ must be zero, demonstrating the plausibility of KCL. For every charge carrier entering a node, one is immediately forced out, leaving the net current entering the node at each instant of time exactly equal to zero.

There are several ways that KCL may be stated, all logically equivalent but each phrased slightly differently. Multiplying both sides of the KCL equation by -1, we obtain

$$(-i_1) + (-i_2) + i_3 + (-i_4) = 0$$

Examining this equation, each term on the left is a current *exiting* the node. Generalizing, we arrive at a second form of KCL.

> The algebraic sum of the currents leaving any node is zero.

Next, this same equation may be rearranged in the form

$$i_1 + i_2 + i_4 = i_3$$

where currents whose reference directions point into the node are gathered on one side and those exiting the node on the other. All minus signs disappear, and we have the third form of KCL.

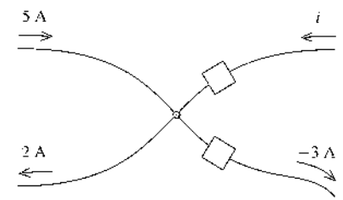
> The sum of the currents entering any node equals the sum of the currents leaving the node.

Depending on the reference directions assigned to currents impinging on a node, one of these three forms may be more convenient to use when KCL is required. In general, KCL may be expressed mathematically as

$$\sum_{n=1}^{N} i_n = 0$$

where i_n is the nth current entering (or leaving) the node and N is the number of node currents.

Example 2.1



Example of KCL.

As an example of KCL, let us find the current i in Fig. 2.3. Summing the currents entering the node, we have

$$5 + i - (-3) - 2 = 0$$

or

$$i = -6 \text{ A}$$

We note that -6 A entering the node is equivalent to 6 A leaving the node. Therefore, it is not necessary to guess the correct current direction when assigning reference directions to current variables. We still arrive at the correct answer in the end.

We may find the current i slightly more conveniently by noting that since i flows into the node, it must equal the sum of all the other currents if they are defined to be flowing out (by reversing their reference directions if necessary), yielding

$$i = -3 + 2 + (-5) = -6 \text{ A}$$

which agrees with the previous answer.

We now move to Kirchhoff's voltage law, which states that

> The algebraic sum of voltage drops around any closed path is zero.

As an illustration, application of this principle to the closed path *abcda* of Fig. 2.4 gives

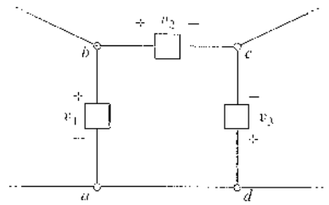
$$-v_1 + v_2 - v_3 = 0$$

where the algebraic sign for each voltage has been taken as positive when going from + to − (higher to lower potential) and negative when going from − to + (lower to higher potential) in traversing the element.

As with KCL, temporarily suppose the contrary, that the sum is not zero. Then

$$-v_1 + v_2 - v_3 = \Phi \neq 0$$

where Φ has units of volts and must equal the potential difference between the initial point and endpoint of the traversal. But this is a closed path, the two points are one



Voltages around a closed path.

and the same, and a node can have no potential difference with itself. The supposition is invalid, and the plausibility of KVL is demonstrated. KVL is in fact an instance of an even more general physical principle that potential differences along a closed path in any conservative energy field must sum to zero. The electric fields produced in lumped-parameter circuits are such fields, and so (for instance) is the field of gravity. If we hike along a path with uphill and downhill sections, the sum of all the altitude changes around any closed path must be zero, for we will have returned to the altitude (gravitational potential) at which we started.

As with KCL, there are two additional equivalent statements of KVL worth noting. Multiplying each term in the *sum of drops* form of KVL stated above by -1 yields the *sum of rises* form.

> The algebraic sum of voltage rises along any closed path equals zero.

The final form of KVL arises when voltages encountered plus-to-minus in the direction of motion around the closed path are summed on one side of the equation and minus-to-plus on the other, as in the loop of Fig. 2.4, yielding

$$v_1 + v_3 = v_2$$

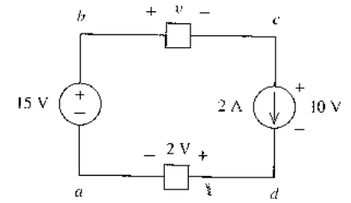
> The sum of voltage drops equals the sum of voltage rises along any closed path.

In general, KVL may be expressed mathematically as

$$\sum_{n=1}^{N} v_n = 0 \qquad (2.5)$$

where v_n is the nth voltage drop (or rise) in a loop containing N voltages. The signs of the terms are as noted in the illustrations, positive for plus-to-minus traversal in the sum of drops form or for minus-to-plus traversal using the sum of rises form, and negative if the traversal is in the opposite sense.

Example 2.2



Circuit to illustrate KVL.

We will use KVL to find v in Fig. 2.5. Traversing the loop clockwise beginning at point a, using the sum of drops form of KVL, yields

$$-15 + v + 10 + 2 = 0$$

or $v = 3$ V. Had we chosen to move counterclockwise from b, we would have

$$+15 - 2 - 10 - v = 0$$

yielding the same result. The sum of rises form yields exactly these same two equations, the first starting from the point b and moving counterclockwise, the second from point a moving clockwise. Finally, equating drops and rises clockwise, we have

$$v + 10 + 2 = 15$$

Clearly, none of these equations is more informative than the others. Their choice is left to convenience or some arbitrary convention. All other things being equal, we will adopt the convention of using the sum of drops form and clockwise traversal.

There is a common situation, however, where a special form of the sum of drops will be useful. Suppose that there is a single unknown voltage in a loop, such as v in Fig. 2.5, which we wish to solve for. Traversing the loop in the plus-to-minus direction of the unknown voltage (clockwise in Fig. 2.5) and summing the drops to zero and then moving all terms but the unknown to the other side of the equation, we have

$$v = +15 - 2 - 10$$

from which the value of v may be immediately computed. Examining this equation, the left-hand side is the voltage drop from the point b to the point c, while the right-hand side is the negative sum of voltage drops from c to back to b, in other words, the sum of voltage drops from b to c by this alternative (counterclockwise) route $badc$. Indeed, the voltage drop between two nodes must be the same regardless of the path used to compute it, in the present case

$$v = v_{bc} = v_{ba} + v_{ad} + v_{dc}$$

or

$$v = 15 - 2 - 10 = 3 \text{ V}$$

Thus, by using the principle that *the voltage drop between two nodes equals the sum of the voltage drops between the same nodes along any path*, a loop containing a single unknown voltage can be conveniently described and the unknown solved.

By use of similar reasoning, it is easily demonstrated that *a desired current into a node equals the sum of all other currents at that node, counting outward flows as positive*. This may be used to solve for a single unknown current at a node and in fact was employed to solve the final example in the earlier discussion of KCL in this section.

Example 2.3

We will use KCL and KVL to find all unknown currents and voltages in Fig. 2.6. Summing voltage drops around the right loop clockwise gives

$$v_1 + 5 + 3 + 2 = 0$$

or

$$v_1 = -10 \text{ V}$$

Repeating for the left loop, we have

$$11 - v_2 - 9 - (-10) = 0$$

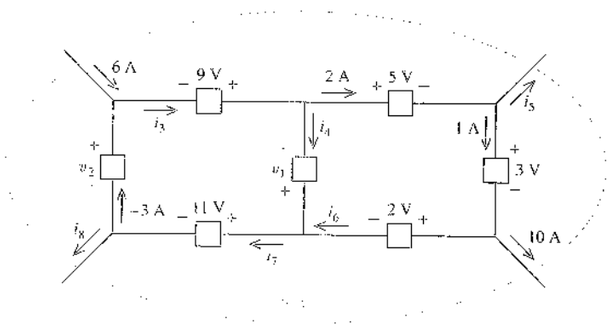
or

$$v_2 = 12 \text{ V}$$

In the left loop equation we have used the fact that $v_1 = -10$ V. We could have solved for v_2 without knowing v_1 by writing the outer loop equation:

$$v_2 = -9 + 5 + 3 + 2 + 11 = 12 \text{ V}$$

Here we equated the voltage drop across an element to the sum of voltage drops between the same two points along another path.



Circuit for Example 2.3.

Next we determine the unknown currents. At the upper-left node, KCL in the sum in equals-sum-out form yields

$$i_3 = 6 + (-3) = 3 \text{ A}$$

At the node immediately to the right,

$$i_4 = 3 + (-2) = 1 \text{ A}$$

where i_4 flows out, and 3 A and −2 A both flow in (+2 A out equals −2 A in). At the upper-right node,

$$i_5 = 2 + (-1) = 1 \text{ A}$$

while at the lower right,

$$i_6 = 1 + (-10) = -9 \text{ A}$$

Finally,

$$i_7 = i_4 + i_6 = -8 \text{ A}$$

$$i_8 = -(-3) + i7 = -5 \text{ A}$$

Note that Kirchhoff's laws are quite indifferent to the *identities* of the elements populating a circuit; they are sensitive only to the circuit *topology*, how the elements are interconnected.

Before concluding our discussion of Kirchhoff's laws, note that if the sum of currents entering any closed region in a circuit, such as that bounded by the surface S in Fig. 2.6, were not zero, charge would be accumulating somewhere in that region. This would imply a violation of KCL at one or more nodes in the region. This illustrates the generalized form of KCL.

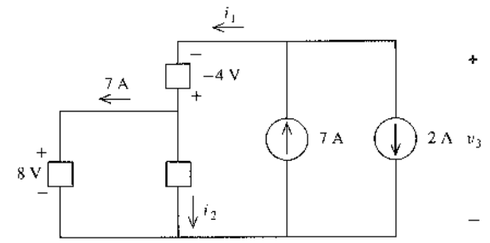
> The algebraic sum of currents entering any closed surface is zero.*

Applying generalized KCL to the circuit of Fig. 2.6, we may sum the currents into the surface S to zero, or

$$6 + (-i_5) + (-10) + (-i_8) = 0$$

In Example 2.3, the values of i_5 and i_8 were computed using the nodal forms of KCL to be $i_5 = 1$ A and $i_8 = -5$ A, which is consistent with the foregoing result.

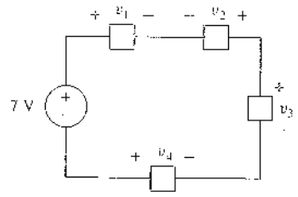
2.1.1. Find i_1 and i_2 for the circuit shown.
ANS: $i_1 = 5$ A; $i_2 = -2$ A



2.1.2. In the circuit for Exercise 2.1.1, find v_3.
ANS: 12 V

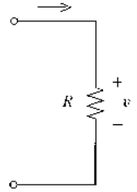
*The surface cannot pass through an element, which is considered to be concentrated at a point in lumped-parameter networks.

2.1.3. Sketch a loop satisfying the equation $v_1 - v_2 + v_3 - v_4 - 7 = 0$. Assume that this equation is KVL in the sum-of-drops form with clockwise traversal.



2.2 OHM'S LAW

Georg Simon Ohm (1787–1854), a German physicist, is credited with formulating the current–voltage relationship for a resistor based on experiments performed in 1826. In 1827 he published the results in a paper titled "The Galvanic Chain, Mathematically Treated." As a result of his work, the unit of resistance is called the *ohm*. It is ironic, however, that Henry Cavendish (1731–1810), a British chemist, discovered the same result 46 years earlier. Had he not failed to publish his findings, the unit of resistance might well be known as the *caven*.



Circuit symbol for the resistor.

Ohm's law states that the voltage across a resistor is directly proportional to the current flowing through the resistor. The constant of proportionality is the resistance value of the resistor in ohms. The circuit symbol for the resistor is shown in Fig. 2.7. For element current i and voltage v defined to satisfy the passive sign convention introduced in Section 1.3 (current reference direction arrow points into the plus end of the voltage reference direction), Ohm's law is

$$v = Ri \tag{2.3}$$

where $R \geq 0$ is the resistance in ohms.

The symbol used to represent the ohm is the capital Greek letter *omega* (Ω). By (2.3), we have $R = v/i$ so, dimensionally,

$$1\ \Omega = 1\ \text{V/A}$$

Example 2.4

If $R = 3\ \Omega$ and $v = 6$ V in Fig. 2.7, the current is

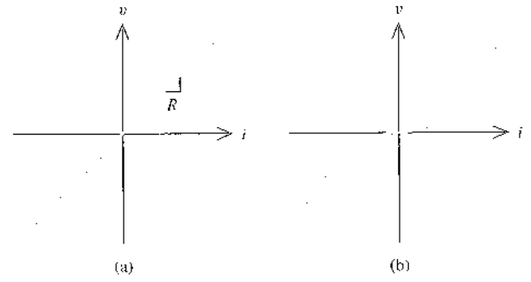
$$i = \frac{v}{R} = \frac{6\ \text{V}}{3\ \Omega} = 2\ \text{A}$$

If R is changed to 1 kΩ (using the power of 10 prefixes of Table 1.1), then with the same voltage the current becomes

$$i = \frac{6\ \text{V}}{1\ \text{k}\Omega} = 6\ \text{mA}$$

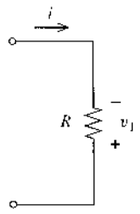
The process is streamlined by noting that $\text{V/k}\Omega = \text{mA}$, $\text{V/M}\Omega = \mu\text{A}$, and so on.

Since R is constant, (2.3) describes a linear current–voltage (i–v) graph, as shown in Fig. 2.8(a). For this reason, a resistor obeying Ohm's law is called a *linear resistor*. Resistors with different i–v laws are called *nonlinear resistors*. In nonlinear resistors, the resistance varies with the current flowing through it. An example is an ordinary incandescent lamp, whose typical i–v graph is shown in Fig. 2.8(b). Since sets of linear equations are far easier to solve than sets containing even a single nonlinear equation, the incorporation of a nonlinear resistor into a network complicates its analysis dramatically.



(a) i–v graph for an example linear resistor; (b) i–v graph for an example nonlinear resistor.

In reality, all practical resistors exhibit nonlinear behavior because the resistance of any resistor is affected by conditions such as temperature, which are in turn linked to past and present current flow through the element (current flow causes heating). Many materials, however, closely approximate an ideal linear resistor over a limited range of currents and environmental conditions. In this book our interest will be exclusively on linear elements, including linear resistors (which we will hereafter simply refer to as "resistors"). Implied by the choice to concentrate on linear elements is the result that our equations and techniques will work only over a limited operating range. A 1-kΩ resistor pulled out of a lab drawer will reliably draw 1 mA of current when placed across a 1-V source, but do not expect this same 1-kΩ resistor to conduct 1 kA of current if you should find a 1-MV power supply with which to excite it. In all likelihood you will measure zero current, because after a brief transient you will be left with just a wisp of smoke and a bad smell in the room (or worse).



Resistor with terminal variables i and v_1 not satisfying the passive sign convention.

It was stated that Ohm's law takes on the familiar form $v = Ri$ if the terminal variables satisfy the passive sign convention. If we choose to state this law using variables that violate the convention, the form taken on by Ohm's law changes accordingly. In Fig. 2.9, suppose that we use the variables i and v_1, rather than i and v as used in Fig. 2.7, as the element current and voltage variables, where $v_1 = -v$. Then, substituting v_1 for v in $v = Ri$, Ohm's law becomes $v_1 = -Ri$. Both $v = Ri$ and this last equation are valid expressions of Ohm's law. To avoid the minus sign, we will almost always choose to define resistor current and voltage variables that satisfy the passive sign convention.

In Section 1.4, passive elements were defined to be those incapable of supplying positive net energy. For resistors the instantaneous power is

$$p = vi = Ri^2 = \frac{v^2}{R} \qquad (2.4)$$

where v and i have been assumed to satisfy the passive sign convention. Note that since p is always positive for a resistor ($R \geq 0$, and the squares of i and v are also nonnegative) the net energy dissipated by this element, the integral of its instantaneous power p, must be nonnegative as well. Thus a resistor does indeed satisfy the definition of a passive element. Resistors dissipate, but under no circumstances supply, electrical power or energy.

Another important quantity very useful in circuit analysis is known as *conductance*, defined by

$$G = \frac{1}{R} \qquad (2.5)$$

The SI unit for conductance is the *siemens*, symbolized by S and named for the brothers Werner and William Siemens, two noted German engineers of the late nineteenth century. Thus 1 S = 1 A/V. (Another unit of conductance, widely used in the United States, is the *mho*, which is *ohm* spelled backward. The symbol for the mho is an inverted omega.) Combining (2.3) to (2.5), we see that Ohm's law and the formulas for instantaneous power may be expressed in terms of conductance as

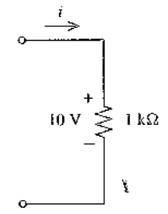
$$i = Gv \qquad (2.6)$$

and

$$p = \frac{i^2}{G} = Gv^2 \qquad (2.7)$$

The important concepts of short circuit and open circuit may be defined in terms of resistance. A *short circuit* is a resistor of zero ohms resistance, in other words a perfect conductor capable of carrying any amount of current without sustaining a voltage drop across it. An *open circuit* is a resistor of zero siemens conductance, in other words a perfect insulator capable of supporting any voltage without permitting any current flow through it. In light of (2.5), an open circuit is equivalent to an infinite resistance, or a break in the wire. Two points may be short-circuited by joining them with a wire, which we always assume to be a perfect conductor. They may be open-circuited by removing all current pathways between them.

Example 2.5

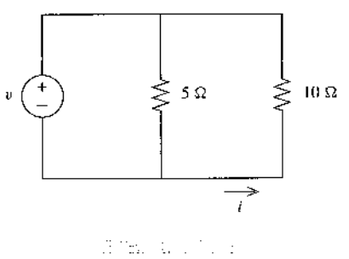


Circuit for Example 2.5.

We wish to determine the current i and the power p absorbed by the 1-kΩ resistor in Fig. 2.10 and the energy it dissipates each half-hour.

Since v and i together satisfy the passive sign convention, we may use Ohm's law in the $v = Ri$ form. Then $i = 10$ V/1000 Ω = 10 mA and $p = vi = (10 \text{ V})(0.01 \text{ A})$, or 100 mW. Since this power is constant over the half-hour, the total energy absorbed during the period t_0 to $t_1 = t_0 + (30 \text{ min})(60 \text{ s/min})$ s is, by (1.7),

$$w(t_1) - w(t_0) = \int_{t_0}^{t_1} p \, dt$$
$$= (0.1 \text{ W})(1800 \text{ s}) = 180 \text{ J}.$$



2.2.1. The terminal voltage of a 10-kΩ resistor is 5 V. Find (a) the conductance, (b) the terminal current, and (c) the power dissipated.

(a) 0.1 mS; (b) 0.50 mA; (c) 2.5 mW

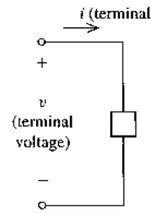
2.2.2. A long cable carries 3 A of current while dissipating 72 W. What are the resistance of the cable and the voltage drop across its length?

$R = 8$ Ω; $v = 24$ V

2.2.3. Find (a) the current i and (b) the power delivered to the resistors if $v = -100$ V.

(a) +10 A; (b) 1 kW to the 10-Ω resistor, 2 kW to the 5-Ω resistor

2.3 EQUIVALENT SUBCIRCUITS



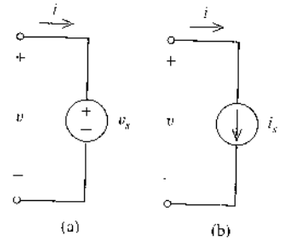
Two-terminal subcircuit. The shaded region may contain any interconnection of elements.

A generally useful strategy in analyzing electric circuits is to *simplify wherever possible*. We shall see how to remove a part of a circuit, replacing it with a simpler subcircuit containing fewer elements, without altering any current or voltage outside that region. The simpler circuit can then be analyzed, and the results will apply equally to the original, more complex, circuit. Such a beneficial trade is possible only when the original and replacement subcircuits are equivalent to one another in a specific sense to be defined.

A *subcircuit* is any part of a circuit. A subcircuit containing any number of interconnected elements, but with only two accessible terminals, is called a *two-terminal subcircuit*. Examples include two-terminal elements, as illustrated in Fig. 1.10, and any arbitrary circuit in which a pair of leads is made accessible for measurement or interconnection with other elements. The voltage across and current into these terminals are called the *terminal voltage* and *terminal current* of the two-terminal subcircuit, as in Fig. 2.11.

The behavior of a two-terminal subcircuit, what it "does" to any circuit containing it, is completely described by its *terminal law*. The terminal law is a function of the form $v = f(i)$ or $i = g(v)$, where v and i are the terminal variables.

Two-terminal subcircuits containing just a single element were introduced in Section 1.1. For these simplest of subcircuits, called *two-terminal elements*, the terminal law is also referred to as the *element law*. The element law for a resistor is Ohm's law, $v = f(i) = Ri$ [or $i = g(v) = Gv$]. The element law for the independent voltage source of Fig. 2.12(a) is, by KVL, $v = v_s$. This element law is of the form $v = f(i)$, where the function $f(i)$ is the constant* equal to v_s. The independence of the terminal voltage v from the current i flowing through the source is what characterizes this element as an ideal voltage source. The element law for the independent current source shown in Fig. 2.12(b) is $i = i_s$ by KCL. An independent current source maintains its terminal current i at the specified source function value i_s regardless of its terminal voltage.



Independent source elements. v_s, i_s are the source functions and v, i the terminal variables.

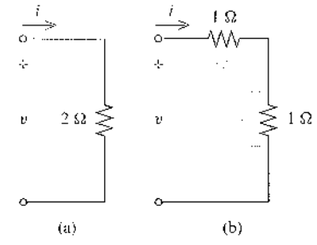
The terminal laws for ideal voltage and current sources are specified in terms of only one terminal variable. All other elements, such as the resistor, as well as subcircuits with several elements, have terminal laws that are equations linking their two terminal variables. The terminal law is what is needed to understand the action of elements and subcircuits.

We come to the central point of this discussion. *Two two-terminal subcircuits are said to be equivalent if they have the same terminal law.* A subcircuit with terminal law $v = f_1(i)$ and a second with terminal law $v = f_2(i)$ are equivalent if $f_1(i) = f_2(i)$ for each i, or if $g_1(v) = g_2(v)$ for every v.

Example 2.6

Let's find the terminal laws for the two subcircuits of Fig. 2.13. That for Fig. 2.13(a) is clearly $v = 2i$. In (b), by Ohm's law applied to each resistor, $v_{21} = i$, $v_{22} = i$, and by KVL, $v = v_{21} + v_{22}$ or $v = i + i = 2i$. Since the terminal laws are identical, the subcircuits are indeed equivalent.

*By constant we mean not dependent on i. The source functions v_s and i_s might in fact vary with time, $v_s = v_s(t)$ and $i_s = i_s(t)$.



Equivalent subcircuits.

The significance of the notion of equivalence is that *equivalent subcircuits may be freely exchanged without altering any external current or voltage.* Suppose that a two-terminal subcircuit A is removed from a circuit and an equivalent subcircuit B inserted in its place as in Fig. 2.14. For any equation that could have been written for circuit (a), the identical equation will be true of circuit (b). For instance, around the outer loop we will have in either case

$$v_2 + v - v_3 - v_1 = 0$$

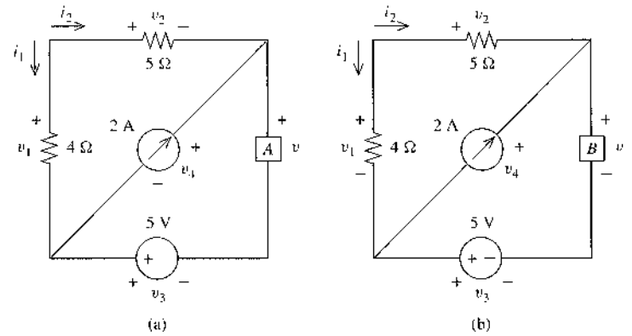
and rewriting this equation in terms of currents by applying the terminal laws yields

$$5i_2 + f_1(i) - 5 - 4i_1 = 0$$

for (a) and

$$5i_2 + f_2(i) - 5 - 4i_1 = 0$$

for (b).



Exchanging equivalent subcircuits.

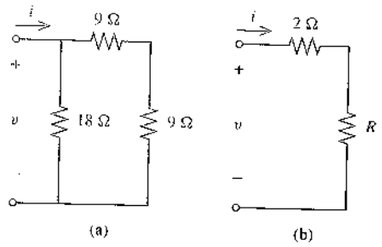
Clearly, these equations are identical if and only if the circuits are equivalent. Since corresponding equations for the two equivalents are all identical, so must be their solutions. We conclude that equivalent circuits result in equal currents and voltages in any circuit to which they are connected.

The notion of equivalent circuits will prove very helpful in simplifying circuit problems. We will proceed by replacing more complex subcircuits by simpler ones. Classes of equivalents to be explored in this chapter include series–parallel and Thevenin–Norton equivalents. The test for equivalence will always be the identity of the terminal laws of the two subcircuits.

While the currents and voltages *external* to equivalent subcircuits will not be changed when one is exchanged for the other in any circuit, the *internal* behavior of the equivalents may be quite different. For instance, in Example 2.6 there is no element in Fig. 2.14(b) with the same terminal voltage as the element in Fig. 2.14(a). All that is guaranteed by their equivalence is that they can be freely interchanged in any circuit without affecting any currents or voltages measurable outside their own boundaries. Based on their effects (resulting currents and voltages) on the rest of the circuit, there is no way to tell which of two equivalents is in the circuit.

Finally, while there is a useful distinction between the terms *subcircuit* and *circuit*, that of part versus whole, it is often clear enough which we are talking about from the context. Wherever this is the case, we will freely use the term *circuit* to refer to either a complete circuit or a subcircuit; thus we will use terms such as *equivalent circuit* and *two-terminal circuit*. Only where it is thought necessary to emphasize the incompleteness of a part of a circuit will the term *subcircuit* be used.

2.3.1. Find the terminal laws for the two circuits sketched.
(a) $v = 9i$; (b) $v = (2 + R)i$



2.3.2. Find a value for R for which the two circuits of Exercise 2.3.1 are equivalent.

$R = 7\ \Omega$

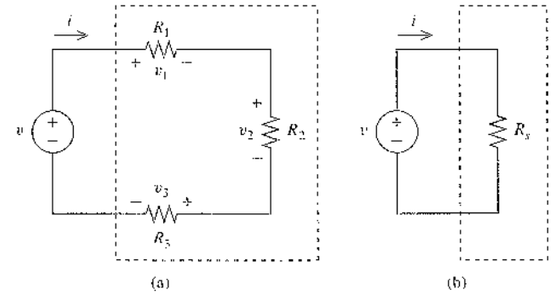
2.3.3. Find an equivalent to the two-terminal circuit shown in Exercise 2.3.1(a) containing just a single element.

9-Ω resistor

2.4 SERIES EQUIVALENTS AND VOLTAGE DIVISION

Now that the laws of Kirchhoff and Ohm and the notion of equivalent circuits have been introduced, we are prepared to analyze resistive circuits. We begin with a *simple circuit*, which we define as one that can be completely described by a single equation. One type, which we consider in this section, is a circuit consisting of a single closed path, or loop, of elements. By KCL, each element has a common current, say i. Then Ohm's law and KVL applied around the loop yield a single equation in i that describes the circuit completely.

Two adjacent elements are said to be connected in *series* if they share a common node that has no other currents entering it. Nonadjacent elements are in series if they are each in series with the same element. So chains of series elements may be formed of any length. For instance, in Fig. 2.15(a) the resistors R_1 and R_2 are in series since they share exclusive access to a common node; similarly, R_2 and R_3 are in series. Since R_1 and R_3 are both in series with the same element R_2, they are in series with one another as well. Note that, by KCL, elements connected in series must all carry the same current.



(a) Single-loop circuit; (b) equivalent circuit.

The single-loop networks of this section consist entirely of elements connected in series. Consider first the circuit of Fig. 2.15(a). The first step in the analysis of this or any circuit is the assignment of current and voltage variables, both names and reference directions, to each element. Since all elements in the loop are in series, they carry a common current, which we will note as i and assign a clockwise reference direction. We could have assigned its reference direction counterclockwise had we preferred; in the end we would draw consistent conclusions. It is never necessary somehow to guess the "right" reference direction in order to arrive at a correct answer. The opposite choice of reference direction for i will just result in the opposite sign when we compute its value. Since, for any number I, i amperes clockwise is the same as $-I$ amperes counterclockwise, the two results must be identical.

We next make the voltage variable assignments v_1, v_2, and v_3 for the resistors R_1, R_2, and R_3, respectively. Note that the voltage reference directions have been chosen to satisfy the passive sign convention jointly with the current reference direction already chosen for their common current i. This will permit us to use Ohm's law for the resistors

without a minus sign. Applying KVL gives

$$v = v_1 + v_2 + v_3$$

Applying Ohm's law to each term on the right side of this equation and gathering the common factor i yields

$$v = (R_1 + R_2 + R_3)i$$

Then, knowing the source function v, we may solve for the loop current i:

$$i = \frac{v}{R_1 + R_2 + R_3}$$

Equation (2.8) or (2.9) also serves as the terminal law for the two-terminal subcircuit shaded in blue in Fig. 2.15(a), since it links its terminal current i and terminal voltage v (by KVL, v is both the source function of the voltage source and the terminal voltage into the shaded region). Examining Fig. 2.15(b), the terminal law of its shaded region is

$$v = R_s i$$

Comparing this result with (2.8), we see that the terminal laws are identical; hence the two shaded circuits of Fig. 2.15 are equivalent if R_s is the sum of the three resistances:

$$R_s = R_1 + R_2 + R_3$$

Generalizing in the obvious way, for N resistors connected in series,

$$R_s = \sum_{i=1}^{N} r_i$$

A chain of series resistors is equivalent to a single resistor whose resistance is the sum of the series resistances. Less formally, series resistance adds. This equivalent resistor can be freely substituted in any circuit without changing any current or voltage external to the equivalent circuits themselves.

Continuing our analysis of Fig. 2.15(a), we substitute the current from (2.9) into Ohm's law $v_1 = R_1 i$ for the first resistor, yielding

$$v_1 = \frac{R_1}{R_s} v$$

Similarly,

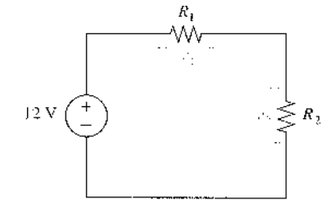
$$v_2 = \frac{R_2}{R_s} v$$

$$v_3 = \frac{R_3}{R_s} v$$

Note that these voltages are in proportion to their resistances. This is the *principle of voltage division: the voltage across series resistors divides up in direct proportion to their resistances.* For this reason, the circuit of Fig. 2.15(a) is called a *voltage divider.* Larger resistances have larger voltage drops in a voltage divider.

Example 2.7

We seek values for R_1 and R_2 in Fig. 2.16 so that the voltage across R_2 is 8 V. With $v_2 = 8$, by KVL $v_1 = 4$. Since R_1 and R_2 form a voltage divider, their voltages are in proportion to their resistance. Then a voltage ratio of $v_2/v_1 = 2$ requires a resistance ratio $R_2/R_1 = 2$. Any R_1 together with $R_2 = 2R_1$, such as $R_1 = 1\ \Omega$ and $R_2 = 2\ \Omega$, will do.



Voltage-divider example.

Example 2.8

As another example, suppose that $v = 120 \sin t$ V in Fig. 2.17. Let us determine the loop current, resistive voltages, and the element power deliveries or dissipations. Replacing the three series resistors by a single 60-Ω equivalent, the loop current is, by Ohm's law,

$$i = \frac{120 \sin t}{60} = 2 \sin t \text{ A}$$

and the voltages are

$$v_1 = R_1 i = 60 \sin t \text{ V}$$

$$v_2 = R_2 i = 40 \sin t \text{ V}$$

$$v_3 = R_3 i = 20 \sin t \text{ V}$$

Although we did not explicitly use the voltage-divider principle, note that these three voltages are in proportion to their resistances.

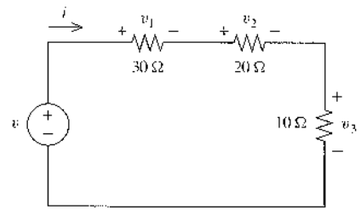
We next compute the powers. Since v_1, v_2, and v_3 each satisfies the passive sign convention relative to i, the power dissipated by these elements is just their vi products, or

$$p_1 = v_1 i = 120 \sin^2 t \text{ W}$$

$$p_2 = v_2 i = 80 \sin^2 t \text{ W}$$

$$p_3 = v_3 i = 40 \sin^2 t \text{ W}$$

Note that the power dissipations are also in proportion to the resistances. This follows from the voltage-divider principle and the fact that series elements carry a common current. Finally, v and i together violate the passive sign convention for the source (the current reference direction points out of, not into, the plus end of the voltage reference direction). Thus a positive vi product means instantaneous power *supplied*, not dissipated, by this element. Then



Single-loop circuit.

the power delivered by the source is

$$vi = 240\sin^2 t \text{ W}$$

Thus the power delivered in this circuit equals, at each instant of time, the total power dissipated. This is an illustration of *Tellegen's theorem*, or conservation of electrical power, which we take up in Chapter 10.

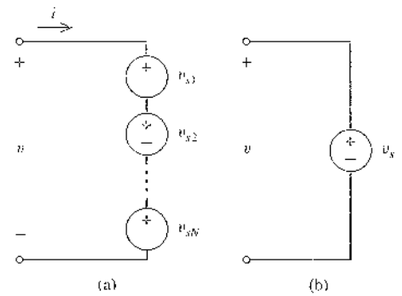
We next consider the series connection of N voltage sources as in Fig. 2.18(a). By KVL, the total voltage v is the algebraic sum of the source functions:

$$v = v_{s1} + v_{s2} + \cdots + v_{sN}$$

This is also the terminal law for the two-terminal circuit. Comparing the terminal law $v = v_s$ for Fig. 2.18(b), we conclude that the two are equivalent if

$$v_s = v_{s1} + v_{s2} + \cdots + v_{sN}$$

A chain of series voltage sources is equivalent to a single voltage source whose source function is the algebraic sum of the series source functions. More concisely, series voltage sources add.



(a) Series voltage sources; (b) equivalent circuit.

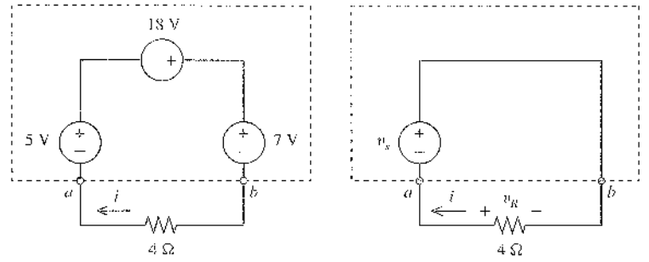
Example 2.9

Find i in Fig. 2.19(a).

Since the three voltage sources are in series, we may replace them by a single equivalent source. With reference directions as indicated in Fig. 2.19,

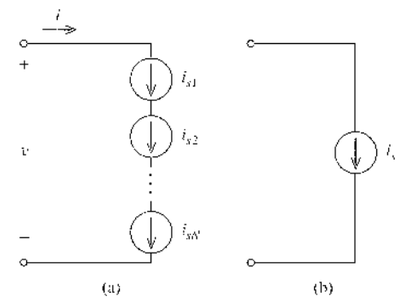
$$v_s = v_{ba} = -7 + 18 + 5 = 16 \text{ V}$$

Then $v_R = 16 = 4i$, so $i = 4$ A. Note that we could have assigned the opposite reference direction for the source function v_s, in which case we would have had $v_s = -16$ V and $v_R + v_s = 0$, so $v_R = +16$ V and once again $i = 4$ A. It is not necessary to guess the best reference direction for an equivalent source function; either will do.



(a) Single-loop circuit; (b) equivalent circuit.

Finally, consider the series interconnection of the remaining type of element found in resistive circuits, the current source. From Fig. 2.20(a) we have $i_{s1} = i_{s2} = \cdots = i_{sN}$, and at the top node i equals this common value $i = i_{s1}$. From Fig. 2.20(b) we see that $i = i_s$, and the circuits are equivalent if the single source function i_s equals the common value of the series current source functions.



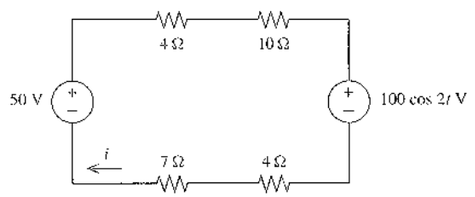
(a) Series current sources; (b) equivalent circuit.

But what if the series current source functions we are given are not all equal? Apparently, charge will be accumulating at the node connecting two unequal sources and KCL will be violated. Yet we know this cannot happen, since we have agreed that KCL will not be violated by any lumped-parameter circuit. This is a case of *inconsistent mathematical assumptions*. Either the given element laws are incorrect or KCL is violated. Once a mathematical inconsistency is discovered in a problem statement, there is nothing more to be done than to note this fact and stop. Inconsistent equations cannot be solved. In the real world, if we were to series-connect two current sources with unequal source functions, they would not behave like ideal current sources whose currents were independent of their voltages. The "give" would be on the part of the element law, bringing the assumptions back to consistency.

We conclude that *a chain of current sources in series must have equal source functions, in which case the chain is equivalent to any one of them. If the given source functions are not all equal, there is a mathematical inconsistency and the circuit cannot be analyzed.*

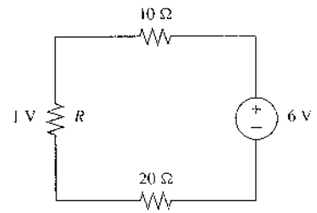
2.4.1. Reduce the circuit to two equivalent elements and find i and the the instantaneous power dissipated by the 50-V source.

$2 - 4\cos 2t$ A; $200\cos 2t - 100$ W



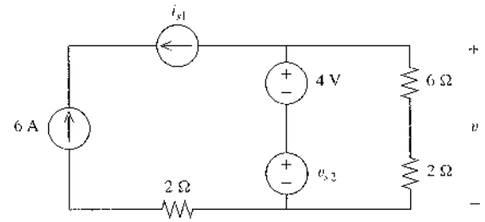
2.4.2. Find R in the circuit shown.

6 Ω



2.4.3. Find values for i_{s1} and v_{s2} so that $v = 4$ V.

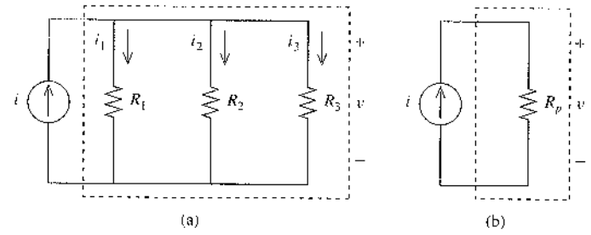
-6 A; 0 V



2.5 PARALLEL EQUIVALENTS AND CURRENT DIVISION

Another important simple circuit is the single-node-pair resistive circuit. Just as single-loop circuits lead naturally to series interconnections and voltage dividers, single-node-pair circuits will be shown in this section to lead to the equally useful ideas of parallel interconnection and current division.

Two elements are connected in *parallel* if together they form a loop containing no other elements. By KVL, elements in parallel all have the same voltage across them. The single-node-pair circuit of Fig. 2.21 contains three resistors and a current source all in parallel, since any pair of the four elements forms a loop containing no other elements.



(a) Single-node-pair circuit; (b) equivalent circuit.

Designate the common voltage of these elements v and define resistive currents to satisfy the passive sign convention relative to v. Applying KCL at the upper node, we obtain

$$i = i_1 + i_2 + i_3$$

Using Ohm's law in the form $i = Gv$ for each term on the right and gathering the common voltage factor yields

$$i = (G_1 + G_2 + G_3)v$$

where each conductance G_i is the inverse of the corresponding resistance R_i. Then, knowing the current source function i, we may solve for v:

$$v = \frac{i}{G_1 + G_2 + G_3}$$

Equation (2.11) or (2.12) also serves as the terminal law for the two-terminal circuit shaded in blue in Fig. 2.21(a), since it links terminal current i and voltage v. The terminal law for the shaded circuit in Fig. 2.21(b) in terms of the conductance $G_p = 1/R_p$ is

$$i = G_p v$$

Comparing this with (2.11), the two shaded circuits of Fig. 2.21 are equivalent if G_p is the sum of the three conductances:

$$G_p = G_1 + G_2 + G_3$$

Generalizing in the obvious way for N resistors connected in parallel, *a set of parallel resistors is equivalent to a single resistor whose conductance is the sum of the parallel conductances*. Informally, parallel conductance adds. By denoting the conductance of this equivalent resistor as

$$G_p = \sum_{i=1}^{N} G_i$$

where $R_p = 1/G_p$ and remembering that $R_i = 1/G_i$ for each i, we have (2.13) in terms of the resistances:

$$\frac{1}{R_p} = \sum_{i=1}^{N} \frac{1}{R_i} \tag{2.14}$$

A set of parallel resistors is equivalent to a single resistor whose resistance is the inverse of the sum of the inverses of the parallel resistances.

This *inverse–inverse* law has an even simpler form when $N = 2$. In this case (2.14) becomes

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

or

$$R_p = \frac{R_1 R_2}{R_1 + R_2} \tag{2.15}$$

The equivalent resistance of two resistors in parallel is the product of their resistances divided by the sum. Unfortunately, this *product-by-sum rule* does not directly apply to sets of more than two resistors in parallel. In this case, (2.13) or (2.14) should be used.

Note by (2.13) that adding parallel resistors can only increase the equivalent conductance, in other words decrease the equivalent resistance. Putting resistors in parallel reduces the overall resistance below that of any of them individually. For N equal R-Ω resistors in parallel, by (2.14), $R_p = R/N$; in other words, putting N equal resistors in parallel reduces the resistance by a factor of N.

Continuing our analysis of Fig. 2.21(a), we substitute the voltage from (2.12) into Ohm's law in the form $i_1 = G_1 v$ for the first resistor, yielding

$$i_1 = \frac{G_1}{G_p} i \tag{2.16a}$$

Similarly,

$$i_2 = \frac{G_2}{G_p} i \tag{2.16b}$$

$$i_3 = \frac{G_3}{G_p} i \tag{2.16c}$$

Note that these currents are all in proportion to their conductances. This is the *principle of current division: the current through parallel resistors divides up in direct proportion to their conductances.* For this reason, the circuit of Fig. 2.18(a) is called a *current divider*. Smaller resistances (larger conductances) have larger current flows in a current divider.

For two resistors in parallel, $G_p = G_1 + G_2$ and (2.16a) and (2.16b) can be rewritten in terms of the resistances $R_1 = 1/G_1$, $R_2 = 1/G_2$, and $R_p = 1/G_p$ as

$$i_1 = \frac{R_2}{R_1 + R_2} i$$

$$i_2 = \frac{R_1}{R_1 + R_2} i$$

For two resistors in parallel, current divides inversely with their resistances. Note that the same caution holds with this as with the product-by-sum rule: it works only in the case of exactly two resistors in parallel.

Example 2.10

Let us determine how the 18 mA delivered by the source divides among the four resistors in Fig. 2.22. Given their resistances, we note their conductances to be $G_1 = 4$ mS, $G_2 = G_3 = 2$ mS, and $G_4 = 1$ mS. Then the currents will be, by the current-divider principle, in the proportions $4 : 2 : 2 : 1$, with R_1 having the largest current and R_4 the smallest. The equivalent parallel conductance is the sum

$$G_p = 4 + 2 + 2 + 1 = 9 \text{ mS}$$

and by (2.16), for the case of four parallel resistors,

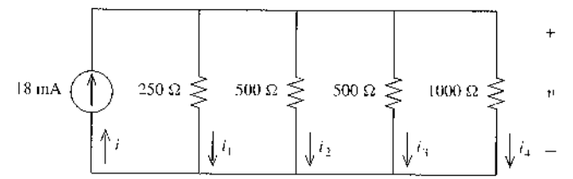
$$i_1 = \frac{G_1}{G_p} i = \frac{4}{9}(18 \text{ mA}) = 8 \text{ mA}$$

$$i_2 = \frac{G_2}{G_p} i = \frac{2}{9}(18 \text{ mA}) = 4 \text{ mA}$$

$$i_3 = \frac{G_3}{G_p} i = \frac{2}{9}(18 \text{ mA}) = 4 \text{ mA}$$

$$i_4 = \frac{G_4}{G_p} i = \frac{1}{9}(18 \text{ mA}) = 2 \text{ mA}$$

Note that the currents are in the predicted proportions $(4 : 2 : 2 : 1)$.



Current-divider example.

Continuing the example, let us see how the power divides. The voltage v is found from the equivalent

$$v = \frac{i}{G_p} = \frac{18 \text{ mA}}{9 \text{ mS}} = 2 \text{ V}$$

or by dividing any of the resistive currents computed above by the conductance of that resistor. Then noting that v and that each resistive current in Fig. 2.22 satisfies the passive sign convention, the dissipated power in R_1 is

$$p_1 = vi_1 = (2 \text{ V})(8 \text{ mA}) = 16 \text{ mW}$$

Similarly, p_2 and p_3 are found to be 8 mW and p_4 is 4 mW. Turning to the source, since v and i together violate the passive sign convention for the source, the power supplied by this element (rather than dissipated by) is given by the vi product,

$$vi = (2 \text{ V})(18 \text{ mA}) = 36 \text{ mW}$$

Thus the 36 mW produced by the source is divided among the resistors comprising the current divider in the same ratio as the currents themselves, and the power supplied equals the power dissipated over the entire circuit.

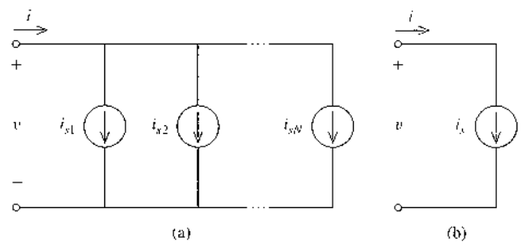
We next consider current sources connected in parallel. Examining Fig. 2.23(a),

$$i = i_{s1} + i_{s2} + \cdots + i_{sN}$$

while, for Fig. 2.23(b), $i = i_s$. The terminal laws are identical if

$$i_s = i_{s1} + i_{s2} + \cdots + i_{sN}$$

A set of parallel current sources is equivalent to a single current source whose source function is the sum of the parallel source functions. Concisely, parallel current sources add.

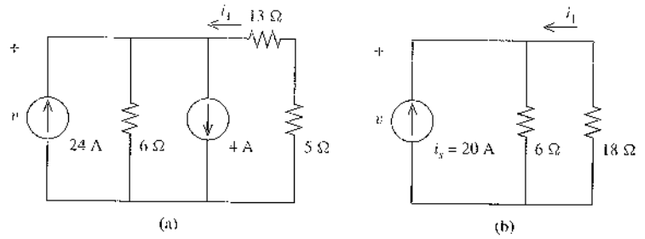


(a) Parallel current sources; (b) equivalent circuit.

Example 2.11

Find v and i_1 in Fig. 2.24.

Since the two current sources are in parallel, they may be replaced by an equivalent current source as shown. Note that with



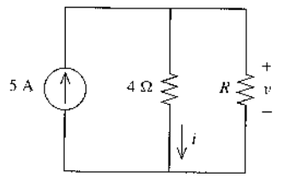
(a) Circuit; (b) its equivalent.

the chosen reference direction for the equivalent source, $i_s = +24 - 4 = +20$ A. The two series resistors are equivalent to a single 18-Ω resistor, which is then in parallel with the 6-Ω resistor, and by the product-by-sum rule for two parallel resistors,

$$R_{\text{eq}} = \frac{(18)(6)}{18 + 6} = \frac{9}{2} \Omega$$

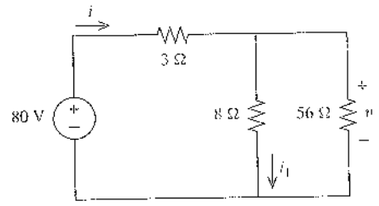
Then $v = R_{\text{eq}} i_s = (9/2)(20) = 90$ V and $i_1 = -v/18 = -5$ A. Note the minus sign in Ohm's law here, since i_1 flows out of the positive end of v and this pair of variables violates the passive sign convention.

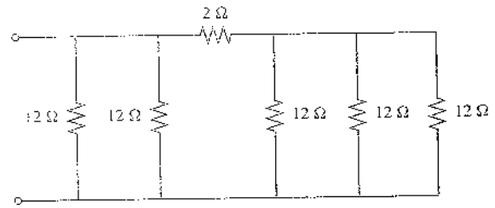
Finally, voltage sources in parallel are vulnerable to the mathematical inconsistency problem that we noted in Section 2.4 with current sources in series. If two or more voltage sources in parallel are all equal, they can be replaced by an equivalent single source of the common value, and if they are not all equal, there is an inconsistency between KVL and the element laws for these sources. *A set of voltage sources in parallel must have equal source functions, in which case any one of them is equivalent to the entire set. If the given source functions are not all equal, there is a mathematical inconsistency and the circuit cannot be analyzed.*



2.5.1. Find R and v if $i = 1$ A.
Answer: 1 Ω; 4 V

2.5.2. Determine the equivalent resistance of the circuit shown.
Answer: 3 Ω





2.5.3. We wish to construct a $\frac{6}{5}$-Ω equivalent resistance by putting some number N_1 of 100-Ω resistors and some number N_2 of 3-Ω resistors in parallel. How many of each are needed?

$N_1 = 50;\ N_2 = 1$

2.5.4. Find the equivalent resistance seen by the source and use this to find i, i_1, and v.

10 Ω; 8 A; 7 A; 56 V

2.6 THEVENIN AND NORTON EQUIVALENTS

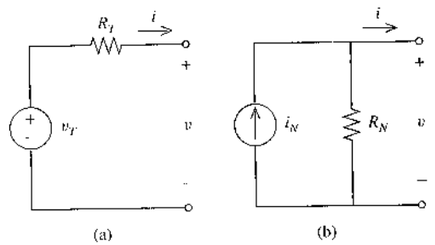
The series and parallel equivalents described thus far are limited to elements of the same type: resistors in series with other resistors, current source in parallel with current sources, and so on. In this section we develop a pair of equivalents, called Thevenin and Norton forms, containing both resistors and sources. They will prove to be of great utility in simplifying many circuit analysis problems.

Consider the pair of two-terminal circuits shown in Fig. 2.25. Applying KVL to (a) gives

$$v = -R_T i + v_T$$

while KCL at the upper node of (b) yields

$$i_N = i + \frac{v}{R_N}$$



(a) Thevenin and (b) Norton subcircuits.

Solving the last for v yields

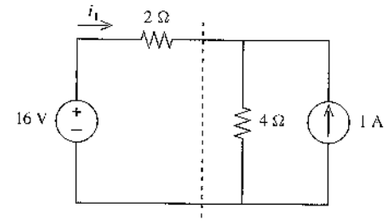
$$v = -R_N i + R_N i_N$$

Comparing the terminal law for the circuit (a) given in (2.18) with that for (b) in (2.19b), they are identical, and thus the two circuits are equivalent if $R_T = R_N$ and $v_T = R_N i_N$. The circuit of Fig. 2.25(a) is called the *Thevenin form*, a series combination of *Thevenin equivalent voltage source* v_s and *Thevenin equivalent resistance* R_T, and that of Fig. 2.25(b) the *Norton form*, a parallel combination of *Norton equivalent current source* i_N and *Norton equivalent resistance* R_N.

> The Thevenin form with voltage source v_T and series resistance R_T is equivalent to the Norton form with current source i_N and parallel resistance R_N, if
>
> (a) $R_T = R_N$ and
>
> (b) $v_T = R_N i_N$

From our previous discussion of equivalents in Section 2.3, we know that two equivalent subcircuits can be freely exchanged in any circuit without altering any current or voltage external to the subcircuits. If it is to our advantage, we may replace a Thevenin form by a Norton form (or vice versa) before computing a desired current or voltage. The following example shows how it may be to our advantage to do so.

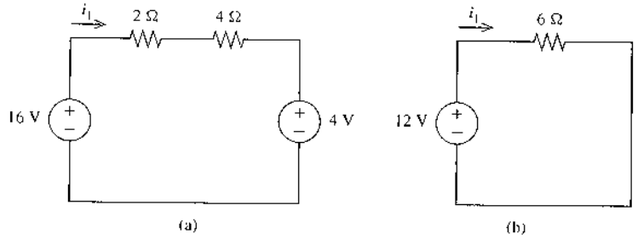
Example 2.12



Circuit for Example 2.12.

Find i_1 in Fig. 2.26.

First, replace the Norton form to the right of the dashed line by its Thevenin equivalent. Since $R_N = 4\ \Omega$ and $i_N = 1$ A, to be equivalent we require that $R_T = 4\ \Omega$ and $v_T = (4)(1) = 4$ V. Performing this replacement, we have the circuit of Fig. 2.27(a). This simple single-loop network may be further simplified by combining series resistors and series sources, yielding the circuit of Fig. 2.27(b). From this circuit, $i_1 = 12/6 = 2$ A. Note that these additional series simplifications were not possible before the Thevenin–Norton transformation.

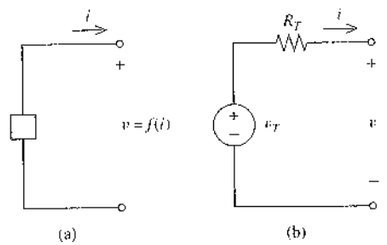


(a) Circuit of Fig. 2.26 after Thevenin–Norton transformation; (b) combining series elements.

Next consider an arbitrary two-terminal resistive circuit with terminal law $v = f(i)$ as shown in Fig. 2.28(a). Comparing the terminal law for the Thevenin form, (2.18), the two circuits of Fig. 2.28(a) and (b) are equivalent if

$$f(i) = -R_T i + v_T$$

Given $f(i)$, we can find slope $(-R_T)$ and intercept (v_T) values to satisfy this requirement as long as the terminal law $v = f(i)$ also describes a straight line. This will always be the case with resistive circuits of interest here, which contain only linear resistors and independent sources. This follows from the linearity of Kirchhoff's laws and the element laws and the fact that combinations of linear equations remain linear.



(a) Two-terminal circuit; (b) Thevenin form.

Equation (2.21) is satisfied by selecting R_T and v_T so that the slope of $f(i)$ equals $-R_T$ and the intercept is v_T. The resulting Thevenin form, Fig. 2.26(b), is then equivalent to the given circuit, Fig. 2.26(a). The Norton equivalent may be found by first finding the Thevenin equivalent and then performing the Thevenin–Norton transformation described by (2.20). Or the Norton equivalent may be found directly by matching the slope of $f(i)$ to $-R_N$ and the intercept to $R_N i_N$.

Example 2.13

The voltage v_0 in Fig. 2.29(a) is required. We first simplify the circuit by replacing the circuit to the right of the dashed line by its Norton equivalent. To do so, we will compute its terminal law. By KCL,

$$i_1 = 10 + i - 6 = i + 4$$

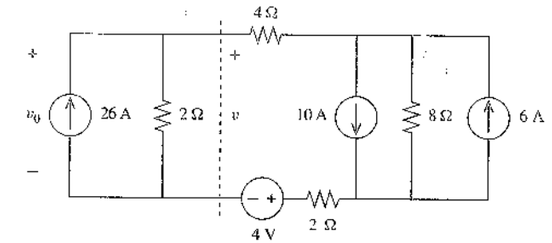
and by KVL

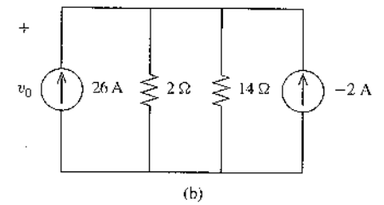
$$v = -4i - 8i_1 - 2i + 4$$

Combining, we have the terminal law

$$v = -14i - 28$$

The equivalent Thevenin form then has $R_T = 14\ \Omega$ and $v_T = -28$ V, and the equivalent Norton form has $R_N = R_T = 14\ \Omega$ and $i_N = v_T/R_N = -2$ A. Choosing the Norton form, the circuit after replacement is shown in Fig. 2.29(b). This is a simple circuit of the





Circuit for Example 2.13.

single-node-pair type, with KCL

$$26 - 2 = \frac{v_0}{2} + \frac{v_0}{14}$$

or $v_0 = 42$ V.

As Example 2.13 illustrates, a circuit with many elements will have Thevenin and Norton equivalents with only two. Replacing the original by its two-element equivalent has the desired effect of simplifying the circuit.

Given the terminal law for a circuit, it is straightforward to find its Thevenin and Norton equivalents. In the previous examples, we derived the terminal law from the circuit diagram by applying Kirchhoff's laws and the element laws. It would be more convenient to determine the equivalents directly from the circuit diagram. Setting $i = 0$ in (2.21) yields

$$f(0) = v_T \tag{2.22}$$

With terminal law $v = f(i)$, $f(0)$ must be the terminal voltage of the given circuit when its terminal current i is zero. Since no current can flow if the terminals of the circuit are open circuited, $f(0)$ is called the *open-circuit voltage* $v = v_{oc}$ of the circuit, and we see by (2.22) that the Thevenin equivalent voltage v_T is equal to the open-circuit voltage of the circuit. If instead of open circuiting the terminals, we short circuited them, a *short-circuit current* $i = i_{sc}$ would flow. Since $v = f(i)$, and across a short circuit there is no voltage $v = 0$, (2.21) in the short-circuit case becomes

$$0 = -R_T i_{sc} + v_T \tag{2.23}$$

which may be solved for R_T. Combining these two results, we can state:

> Given a subcircuit with open-circuit voltage v_{oc} and short-circuit current i_{sc}, its Thevenin equivalent may be found from
>
> (a) $v_T = v_{oc}$ and
>
> (b) $R_T = v_{oc}/i_{sc}$

Use of Eq. (2.20) permits translation of these results to the Norton form.

> Given a subcircuit with open-circuit voltage v_{oc} and short-circuit current i_{sc}, its Norton equivalent may be found from
>
> (a) $i_N = i_{sc}$ and
>
> (b) $R_N = v_{oc}/i_{sc}$

Example 2.14

Find Thevenin and Norton equivalents for the circuit of Fig. 2.30.

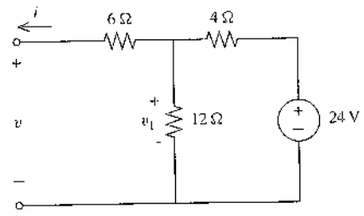


Figure 2.30 Circuit for Example 2.14.

To find v_{oc}, we assume that the terminals are open circuited; in other words, $i = 0$. Then, with no current in the 6-Ω resistor, by voltage division

$$v_1 = \left(\frac{12}{16}\right) 24 = 18 \text{ V}$$

and by KVL around the left loop $v = v_{oc} = 0 + 18 = 18$ V. So the Thevenin equivalent voltage source is $v_T = v_{oc} = 18$ V. To find i_{sc}, we assume that the terminals are short circuited, $v = 0$. Then the two leftmost resistors are in parallel (they form a loop containing no other voltage drops), and by the product-by-sum rule,

$$R_{\text{eq}} = \frac{(12)(6)}{12 + 6} = 4 \ \Omega$$

In this case v_1 is, by voltage division of 24 V between the two 4-Ω resistors,

$$v_1 = \frac{4}{8} 24 = 12 \text{ V}$$

Since v_1 is also the voltage across the 6-Ω resistor under these short-circuit conditions, by Ohm's law $v_1 = 6i$. Then, solving for $i = i_{sc}$ gives

$$i_{sc} = \frac{v_1}{6} = \frac{12}{6} = 2 \text{ A}$$

Hence $R_T = v_{oc}/i_{sc} = 18/2 = 9$ Ω. The Thevenin form and corresponding Norton form ($R_N = R_T = 9$ Ω, $v_T = v_{oc} = 18$ V, $i_N = i_{sc} = \frac{1}{2}$ A) are shown in Fig. 2.31.

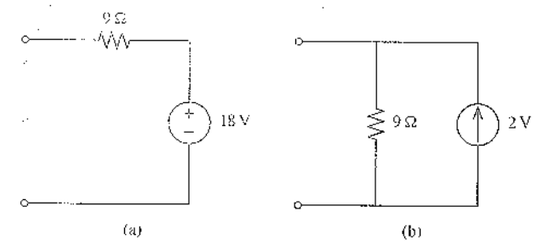


Figure 2.31 (a) Thevenin equivalent; (b) Norton equivalent.

There is another way to find the common resistance $R_T = R_N$ in the Thevenin and Norton forms, which is sometimes simpler than computing the ratio of open-circuit voltage v_{oc} to short-circuit current i_{sc}. Returning to (2.20) and remembering that $f(i)$ for linear resistive circuits describes a straight-line relationship, we note that if all independent sources internal to the subcircuit were set to zero (or "killed," to use a more vivid terminology), this would set the intercept of this straight line to zero; that is, the terminal law $v = -R_T i + v_T$ becomes simply $v = -R_T i$. This follows from the fact that in the absence of internal sources, the open-circuit voltage v_{oc} must be zero, since there is no source of power remaining to create any currents or voltages anywhere in the subcircuit. Then, with all internal sources killed and defining $i' = -i$, we have $v = -R_T/(-i')$, or

$$R_T = \left.\frac{v}{i'}\right|_{\text{sources killed}}$$

Equation (2.25) suggests that R_T (or R_N) *can be computed as the resistance looking into the terminals of the subnetwork when all internal sources have been killed.* The sign change $i' = -i$ was necessary to create a pair of terminal variables (i', v) that together satisfy the passive sign convention so that Ohm's law will hold without a minus sign.

Note that killing a source means setting its source function to zero. Since a 0-V voltage source is simply a short circuit, *independent voltage sources are killed by replacing them by short circuits.* Similarly, since a 0-A current source is an open circuit, *kill independent current sources by replacing them with open circuits.*

Example 2.15

Consider the circuit of Fig. 2.30. After killing internal sources, we have the circuit shown in Fig. 2.32. The resistance looking into its terminals is found by combining the parallel 12- and 4-Ω resistors

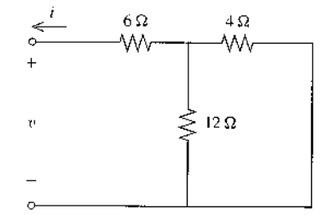
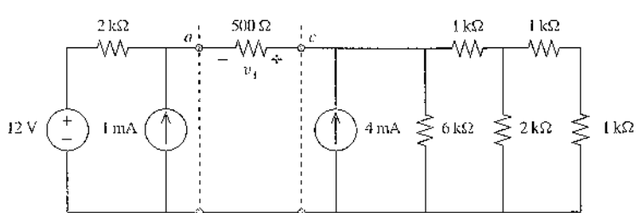


Figure 2.32 Circuit for Example 2.15.

into a single equivalent of 3 Ω and adding this to the 6-Ω resistance that this equivalent is in series with, yielding $R_T = R_N = 9\ \Omega$. This agrees with our calculation in Example 2.14.

Finding Thevenin–Norton equivalents for a given circuit is no more or less difficult than finding any two of its open-circuit voltage, short-circuit current, and look-in resistance with internal sources killed. If the circuit contains many loops and nodes, it may not yet be clear how to find these quantities. Systematic methods for finding any current or voltage in any circuit are developed in Chapter 4. For now we content ourselves with being aware of the power of Thevenin and Norton equivalents to simplify circuit analysis and being able to compute and use these equivalents in simple circuits.

2.6.1. Find the Thevenin equivalent of the circuit to the right of line *cd*.
$v_T = 6$ V; $R_T = 1.5$ kΩ



2.6.2. Repeat Exercise 2.6.1 for the circuit to the left of line *ab*.
$v_T = 14$ V; $R_T = 2$ kΩ

2.6.3. Use the results of Exercises 2.6.1 and 2.6.2 to find v_1.
−1 V

2.7 PRACTICAL SOURCES AND RESISTORS

In Chapter 1 an independent voltage source was defined as an ideal element whose terminal voltage is equal to its source function, $v = v_s$ in Fig. 1.11, regardless of the current flowing through it. Similarly, an ideal current source always carries current exactly equal to its source function, $i = i_s$ in Fig. 1.12, no matter what terminal voltage this requires. One clear sign that these ideal sources are not physically realizable is that such devices must be capable of producing infinite power. This can be seen by connecting an R_L-Ω load resistor across the terminals of an ideal voltage source and considering what happens as R_L goes to zero. Since the voltage across the resistor remains fixed at v_s regardless of R_L, the power v_s^2/R_L goes to infinity as R_L goes to

zero. An ideal current source with R_L Ω across its terminals is similarly required to produce infinite power $i^2 R_L$ as R_L goes to infinity, since its terminal current i is likewise fixed.

Practical sources, such as the familiar dry cells found in flashlights and calculators or dc power supplies in electronics equipment, are not capable of such infinitely powerful behavior. How then are we to model practical, physically realizable independent voltage and current sources? In Section 2.6 it was shown that there exist Thevenin and Norton equivalent circuits for all linear two-terminal subcircuits. We will use this universal modeling capability to represent practical sources by their Thevenin or Norton equivalent circuits. The presence of a resistor in both circuits limits their current, voltage, and power to finite values and makes them useful models for physically realizable sources. Practical sources connected to load resistors are shown in Fig. 2.33.

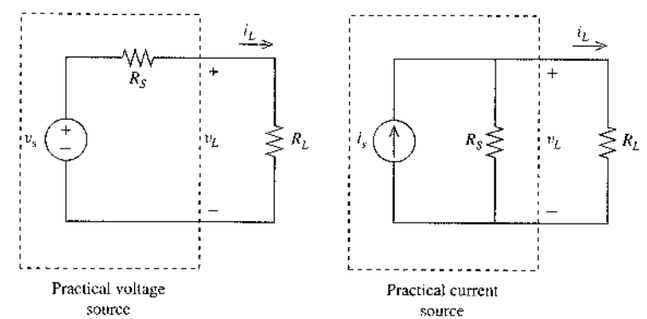


FIGURE 2.33 Practical sources connected to a load resistor: (a) voltage source; (b) current source.

The effect of the source resistance R_S in the practical voltage source model of Fig. 2.33(a) may be seen by plotting load voltage v_L and current i_L versus load resistance R_L as shown in Fig. 2.34. For large values of R_L the source is lightly loaded (little current is being drawn), and the source produces values closely approximating the ideal voltage source $R_S = 0$, as shown. As R_L converges toward zero, v_L decreases from its ideal value, as does i_L. The power dissipated by the load is

$$P = \frac{v_L^2}{R_L} = \left(\frac{R_L}{R_L + R_S} v_s\right)^2 \frac{1}{R_L} = \frac{R_L v_S^2}{(R_L + R_S)^2} \tag{2.26}$$

In the case of an ideal voltage source we have no source resistance ($R_S = 0$) and, examining (2.26) with $R_S = 0$, the power supplied to the load by the ideal source goes to infinity as the load resistance R_L goes to zero. For a practical source we have $R_S > 0$, and by (2.26) the power goes to zero, not infinity, as R_L goes to zero. The source resistance R_S serves to limit the maximum or short-circuit current the source may supply, thereby forcing the power to zero as R_L becomes small.

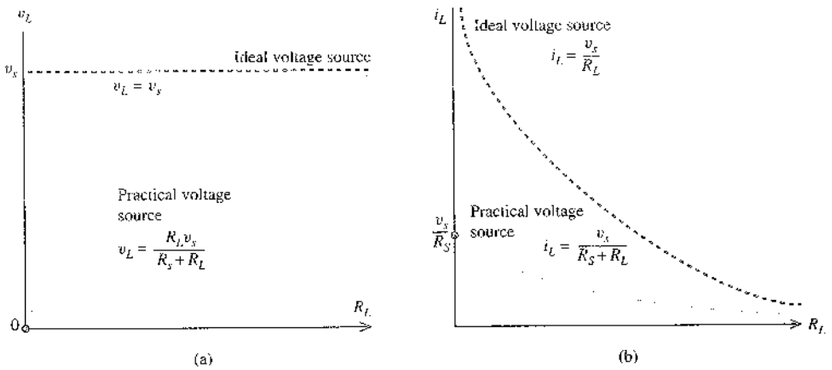


FIGURE 2.34 Comparison of practical and ideal voltage sources: (a) v_L versus R_L; (b) i_L versus R_L.

Similarly, the load current and voltage comparisons for ideal and practical current sources are shown in Fig. 2.35. The practical values are reduced for current sources with R_L large. The power to the load is, noting the current divider in Fig. 2.33(b),

$$P = i_L^2 R_L = \left(\frac{R_S}{R_L + R_S} i_S\right)^2 R_L \tag{2.27}$$

For an ideal current source with R_S infinite, P goes to infinity with R_L. For a practical current source with finite source resistance $R_S > 0$, as R_L goes to infinity in (2.27), P goes to zero. The source resistance R_S serves to limit the open-circuit voltage the practical current source may produce, and thus its power to the load. The comparisons for ideal and practical current sources are shown in Fig. 2.35.

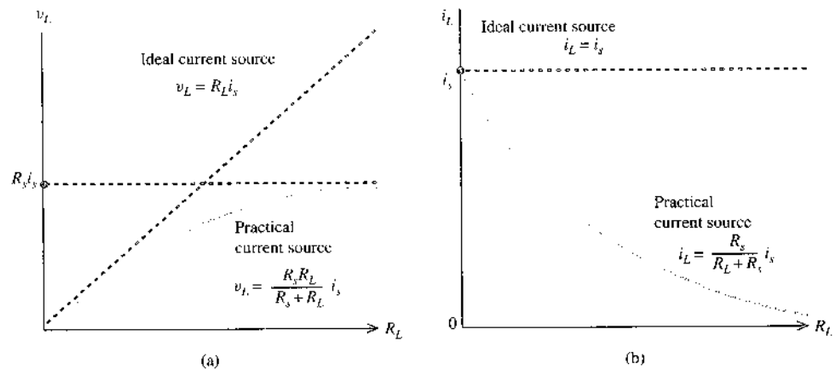


FIGURE 2.35 Comparison of practical and ideal current sources: (a) v_L versus R_L; (b) i_l versus R_L.

Thus by the addition of a single resistor to the ideal source we have a model that accurately reflects the behavior of practical sources. The Thevenin and Norton sources v_S and i_S in Fig. 2.33 may be constants for the dc case, or more general sources for signal generators or other non-dc sources.

The Thevenin or Norton equivalent for a given source may found by measuring any two of the open-circuit voltage, short-circuit current, and look-in resistance with internal sources killed, as discussed in Section 2.6. The last measurement is often difficult to arrange when it is a source that is being modeled. Also, short-circuiting a source to measure the short-circuit current may result in excessive load current, possibly damaging the source. Fortunately, it is not necessary to make these specific measurements. It is possible to determine the Thevenin or Norton parameters from *any* two i–v pairs, as we see in the following example.

Example 2.16

The open-circuit voltage across the terminals of a fresh C cell is measured to be 1.55 V. When a 29-Ω resistor is placed across it, its terminal voltage drops to 1.45 V. Find the Thevenin and Norton equivalents of the C cell, and the maximum current that can be drawn from this battery.

The Thevenin equivalent voltage $v_T = v_{oc}$, its open-circuit voltage. This is 1.55 V as shown in Fig. 2.36(a). Noting that R_T and 29 Ω form a voltage divider, we find that

$$\left(\frac{29}{29 + R_T}\right)(1.55) = 1.45$$

or

$$R_T = 2\ \Omega$$

The Thevenin equivalent of the C cell is shown in Fig. 2.36(b). Maximum current will flow if the device is short-circuited, $v_{ab} = 0$. In this case

$$i_{\max} = i_{sc} = \frac{1.55}{2} = 775\ \text{mA}$$

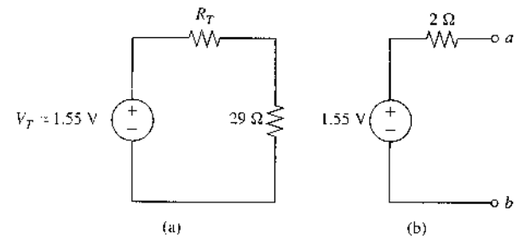


FIGURE 2.36 (a) Circuit for the example; (b) practical source.

Note that while ideal sources can supply power to a circuit without themselves dissipating any fraction, this is not the case for practical sources. For instance, in Example 2.16, if the C cell is supplying 100 mA, there will be $P = 0.1^2(2)$ or 20 mW

power dissipated by the internal source resistance of the cell. Sources feel warm to the touch when they are operating because they always drop some of the power they produce internally across their source resistance. This power is not available to supply to other elements.

Example 2.17

A practical current source with a 1-A short-circuit current and $R_s = 100\ \Omega$ look-in resistance is to be used to deliver $\frac{1}{12}$ A to a 100-Ω load R_L. Determine an external resistance R_{SH} that can be placed in "shunt," or across the source terminals, to achieve this result. Then combine R_{SH} with the original current source in a Norton source model, and determine the maximum power dissipation internal to this new practical source under all possible load conditions.

We first use current division to determine R_{SH}. Let $G_{SH} = 1/R_{SH}$ be the conductance. By current division, to achieve the prescribed load current

$$i_L = \left(\frac{0.01}{0.01 + 0.01 + G_{SH}}\right)(1) = \frac{1}{12}$$

and solving for G_{SH}, we obtain

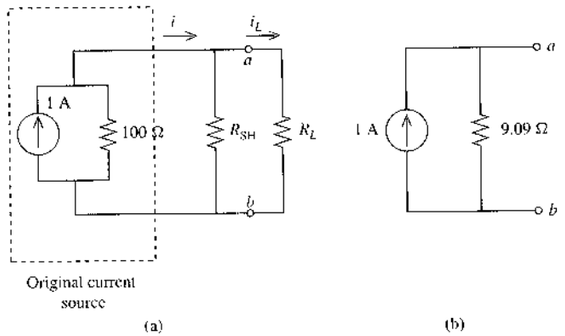
$$G_{SH} = (12)(0.01) - 0.02 = 0.1\ \text{S}$$

So $R_{SH} = 10\ \Omega$. Then the Norton equivalent of the left of ab in Fig. 2.37(a) has Norton equivalent resistance

$$R_N = \frac{(10)(100)}{10 + 100} = 9.09\ \Omega$$

and short-circuit current 1 A. The new source model, including the shunt resistor, is shown in Fig. 2.37(b). Maximum power dissipation will occur when the load is an open circuit, for then the entire 1 A will flow through the source resistance 9.09 Ω:

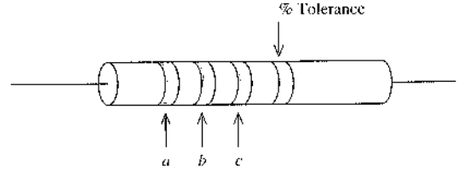
$$P = (1^2)(9.09) = 9.09\ \text{W}$$



(a) Circuit for Example 2.17; (b) Norton source model.

We next turn to practical resistors. Resistors are manufactured from a variety of materials and are available in many sizes and values. Their characteristics include a nominal resistance value, an accuracy with which the actual resistance approximates the nominal value (known as *tolerance*), a power dissipation, and a stability as a function of temperature, humidity, and other environmental factors. They may be produced as discrete elements or deposited onto substrates as part of an integrated circuit containing other elements such as transistors and capacitors.

The most common type of discrete resistor found in electrical circuits is the carbon composition or carbon film resistor. The composition type is made of hot-pressed carbon granules. The carbon film device consists of carbon powder that is deposited on an insulating substrate. A typical resistor of this type is shown in Fig. 2.38. Multicolored bands, shown as a, b, c, and % tolerance, are painted on the resistor body to indicate the nominal value of the resistance. The color code for the bands is given in Table 2.1.



Carbon resistor.

Color Code for Carbon Resistors

Color	Value	Color	Value
Silver[a]	−2	Yellow	4
Gold[a]	−1	Green	5
Black	0	Blue	6
Brown	1	Violet	7
Red	2	Gray	8
Orange	3	White	9

% Tolerance Band	
Gold	± 5%
Silver	± 10%

[a]These colors apply to band c only.

Bands a, b, and c give the nominal resistance of the resistor, and the tolerance band gives the percentage by which the resistance may deviate from its nominal value. Referring to Fig. 2.38, the resistance is

$$R = (10a + b)10^c \pm \%\ \text{tolerance}$$

by which we mean that the % tolerance of the nominal resistance is to be added or subtracted to give the range in which the resistance lies.

Example 2.18

As an example, suppose that we have a resistor with band colors of yellow, violet, red, and silver. The resistor will have a value given by

$$R = (4 \times 10 + 7) \times 10^2 \pm 10\%$$
$$= 4700 \pm 470\ \Omega$$

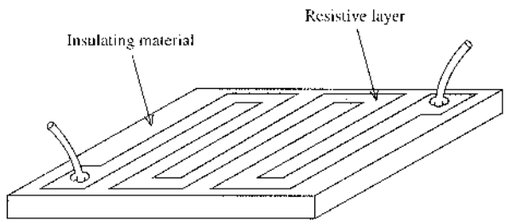
Therefore, the resistance value lies between 4230 and 5170 Ω.

Values of carbon resistors range from 2.7 to 2.2×10^7 Ω, with wattages from $\frac{1}{8}$ to 2 W. For resistance values less than 10 Ω, we see from (2.28) that the third band must be gold or silver. Carbon resistors are inexpensive but have the disadvantage of a relatively high variation of resistance with temperature (typically increasing 0.1% per degree Celsius).

Another resistor type that is commonly used in applications requiring a high power dissipation is the wire-wound resistor. This device consists of a metallic wire, usually a nickel–chromium alloy, wound on a ceramic core. Low-temperature-coefficient wire permits the fabrication of resistors that are very precise and stable, having accuracy and stability on the order of ±1 to ±0.001%.

The metal film resistor is another useful discrete resistor type. These resistors are made by vacuum deposition of a thin metal layer on a low-thermal-expansion substrate. The resistance is then adjusted by etching or grinding a pattern through the film. Accuracy and stability for these resistors approach that of wire-wound types, and high resistance values are much easier to attain.

Integrated circuits are rapidly superceding discrete element circuits for most low-power applications. An integrated circuit is a single monolithic chip of *semiconductor* (material with conducting properties between those of a conductor and an insulator) in which active and passive elements are fabricated by diffusion and deposition processes. Integrated circuits containing hundreds of thousands of transistors, resistors, and other elements are available on chips less than 1 in. square. An integrated-circuit resistor has the typical structure shown in Fig. 2.39. The resistive layer is very thin, typically a fraction of a micrometer (10^{-6} m), and its resistance is set by its length, width, and doping. It consists of the same doped semiconductor material used to form transistors and capacitors on the chip.



Integrated-circuit resistor.

2.7.1. A practical source supplies $i_L = 2$ A dc to a 10-Ω load and $i_L = 1$ A dc to a 50-Ω load. Determine its Thevenin and Norton parameters V_T, R_T, I_N, and R_N.

80 V; 30 Ω; $\frac{8}{3}$ A; 30 Ω

2.7.2. A 500-mA fuse is located in the wire connecting a power supply to its load. If the power supply has an open-circuit voltage of +100 V dc and a source resistance of 30 Ω, what is the range of acceptable load resistances that will not cause the fuse to blow?

$R_L > 170\ \Omega$

2.7.3. An automobile battery in a poor state of charge has an open-circuit voltage of +11.5 V dc and when loaded by the starter motor, which we model as a 0.8-Ω resistor, supplies 10 V across the starter motor terminals. Determine the power delivered to the load and the power being dissipated internally in the source resistance under these conditions.

125 W; 18.75 W

2.7.4. Find the resistance range of carbon resistors having color bands of (a) brown, black, red, silver; (b) red, violet, yellow, silver; and (c) blue, gray, gold, gold.

(a) 900 to 1100 Ω; (b) 243 to 297 kΩ; (c) 6.46 to 7.14 Ω

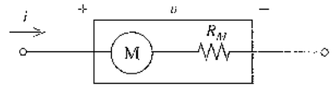
2.8 AMMETERS, VOLTMETERS, AND OHMMETERS

In the real world of physical design and use of electric circuits, those quantities of interest, such as current, voltage, and resistance, must be measured by instruments. Ideally, these measurement devices do not disturb the circuits they are assessing. An *ideal ammeter* measures the current flowing through its terminals and has zero voltage across its terminals. In contrast, an *ideal voltmeter* measures the voltage across its terminals and has a terminal current of zero. An *ideal ohmmeter* measures the resistance connected between its terminals and delivers zero power to the resistance.

Practical measuring instruments only approximate these ideal devices. Practical ammeters, for instance, will not have zero terminal voltages. Similarly, voltmeters will not have zero terminal currents, and ohmmeters will deliver nonzero power to the load they are measuring.

The most common type of ammeter consists of a mechanical movement known as a d'Arsonval meter. This device is constructed by suspending an electrical coil between the poles of a permanent magnet. A dc current passing through the coil causes a rotation of the coil, as a result of magnetic forces, that is proportional to the current. A pointer is attached to the coil so that the rotation, or meter deflection, can be observed visually. d'Arsonval meters are characterized by their *full-scale current*, which is the current that will cause the meter to read its greatest value. Meter movements having full-scale currents from 10 μA to 10 mA are common.

An equivalent circuit for the d'Arsonval meter consists of an ideal ammeter in series with a resistance R_M, as shown in Fig. 2.40. In this circuit, R_M represents the



Equivalent circuit for a d'Arsonval meter.

resistance of the electrical coil. Clearly, a voltage appears across the ammeter terminals as a result of the current i flowing through R_M. R_M is usually a few ohms, and the terminal voltage for a full-scale current is typically from 20 to 200 mV.

The d'Arsonval meter is an ammeter suitable for measuring dc currents not greater than the full-scale current I_{FS}. Suppose that we wish, however, to measure a current that exceeds I_{FS}. We must not allow a current greater than I_{FS} to flow through the device without risking damage to the delicate device. A circuit to accomplish this is shown in Fig. 2.41(a), where R_p is a parallel resistance which reduces the current flowing through the meter coil.

From current division we see that

$$I_{FS} = \frac{R_p}{R_M + R_p} i_{FS}$$

where i_{FS} is the current that produces I_{FS} in the d'Arsonval meter. (Clearly, i_{FS} is the maximum current the ammeter can measure.) Solving for R_p, we have

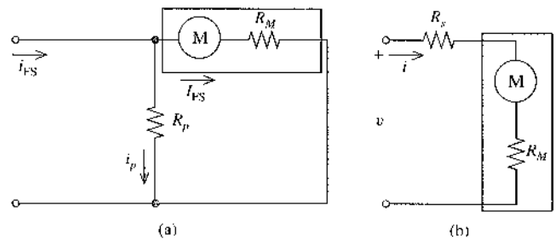
$$R_p = \frac{R_M I_{FS}}{i_{FS} - I_{FS}}$$

A dc voltmeter can be constructed using the basic d'Arsonval meter by placing a large resistance R_s in series with the device, as shown in Fig. 2.41(b). It is obvious that the full-scale voltage, $v = v_{FS}$, occurs when the meter current is I_{FS}. Therefore, from KVL,

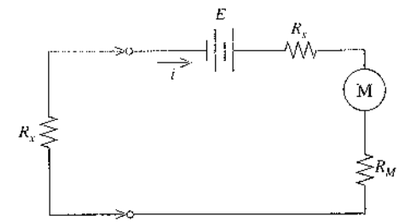
$$-v_{FS} + R_s I_{FS} + R_M I_{FS} = 0$$

from which

$$R_s = \frac{v_{FS}}{I_{FS}} - R_M$$



(a) Ammeter circuit; (b) voltmeter circuit.



Ohmmeter circuit.

The *current sensitivity* of a voltmeter, expressed in ohms per volt, is the value obtained by dividing the resistance of the voltmeter by its full-scale voltage. Therefore,

$$\Omega/\text{V rating} = \frac{R_s + R_M}{v_{FS}} \approx \frac{R_s}{v_{FS}}$$

(*Note:* "≈" means approximately equal to.)

A simple ohmmeter circuit employing a d'Arsonval meter for measuring an unknown resistance R_x is shown in Fig. 2.42. In this circuit, the battery E causes a current i to flow when R_x is connected into the circuit. Applying KVL, we have

$$-E + (R_s + R_M + R_x)i = 0$$

from which

$$R_x = \frac{E}{i} - (R_s + R_M)$$

We select E and R_s such that for $R_x = 0$, $i = I_{FS}$. Therefore,

$$I_{FS} = \frac{E}{R_s + R_M}$$

Combining the last two equations, we find that

$$R_x = \left(\frac{I_{FS}}{i} - 1\right)(R_s + R_M)$$

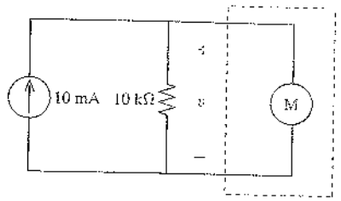
A very popular general-purpose meter which combines the three circuits described previously is the *VOM* (voltmeter–ohmmeter–milliammeter). In the VOM, provisions are made for changing R_p and R_s so that a wide dynamic range of operation is provided. Scales may then be laid out on the face of the instrument to correspond with the resistance values R_x given in the preceding equation for each available R_s.

Electronic volt-ohm-ammeters employ active circuits, such as precision op amp stages, instead of the passive d'Arsonval movements. They can be made to measure with precision unavailable using the d'Arsonval meter, and have the added advantage of coupling naturally with digital readouts.

EXERCISES

2.8.1. A d'Arsonval meter has $I_{FS} = 1$ mA and $R_m = 50\ \Omega$. Determine R_p in Fig. 2.41 so that i_{FS} is (a) 1.0 mA, (b) 10 mA, and (c) 100 mA.

Answers: (a) Infinite; (b) 5.556 Ω; (c) 0.505 Ω



2.8.2. In Fig. 2.26, determine R_s and the Ω/V rating for a voltmeter to have a full scale voltage of 100 V using a d'Arsonval meter with (a) $R_M = 100\ \Omega$ and $I_{FS} = 50\ \mu$A and (b) $R_M = 50\ \Omega$ and $I_{FS} = 1$ mA.

(a) 2 MΩ, 20 kΩ/V; (b) 100 kΩ, 1 kΩ/V

2.8.3. What voltage would each meter design of Exercise 2.8.2 measure in the circuit shown? Why are the two measurements different?

99.5 V, 90.9 V

2.9 DESIGN OF RESISTIVE CIRCUITS

Limited to resistors and independent sources, resistive circuits cannot be expected to have the versatility of those containing other types of circuit elements, such as dependent sources, inductors, and capacitors. These elements will be introduced in the next several chapters, and the wider range of design opportunities they afford will be explored in stages throughout the balance of our study. Resistive circuits are not without important practical applications, however. In this section we present three design examples using resistive circuits.

Series–parallel transformations may be used to reduce a set of interconnected resistors to a single resistor in order to simplify their analysis. They may also be used to expand a single resistor to many for purposes of design. This typically occurs when the single resistor is not of an available resistance, or when its power rating is insufficient, as in the first example.

Example 2.10

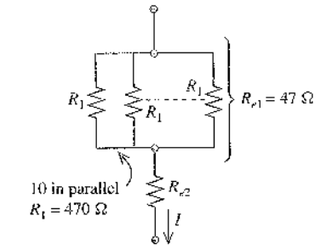
For our first design example, we shall use any number of available resistors, which are of the specific component values 470 Ω and 6.8 Ω, to synthesize a desired 50-Ω load resistor (±1%). Each of the available resistors is rated at $\frac{1}{4}$ W maximum power dissipation, and our 50-Ω load resistor must dissipate at least 2 W.

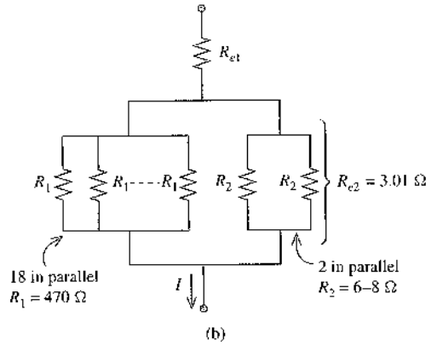
As with most design problems, there is no unique circuit that will satisfy the design specifications, and we begin by seeking a circuit layout that appears capable of doing the job, being prepared to modify as necessary. If we put N identical R-Ω resistors in parallel, adding their conductances yields $N(1/R)$, so the equivalent resistance is R/N. If we put $N = 10$ of the available 470-Ω resistors in parallel, this yields an equivalent of 47 Ω, as shown in Fig. 2.43(a). Then if R_{e2} is put in series, we would need

$$R_{e2} = 50 - 47 = 3\ \Omega$$

How can we synthesize a 3-Ω equivalent resistance from the available ones? We cannot have two or more in series, since even the smallest available resistance exceeds 3 Ω and series resistance adds. They must be in parallel once again. The required conductance is

$$G_{e2} = \frac{1}{R_{e2}} = 333.3\ \text{mS}$$





(a) First step: ten 470-W resistors in parallel; (b) completing the design of a 50-W equivalent load.

and the two available conductances are

$$G_1 = \frac{1}{R_1} = \frac{1}{470} = 2.13\ \text{mS}$$

$$G_2 = \frac{1}{R_2} = \frac{1}{6.8} = 147\ \text{mS}$$

Since parallel conductances add, we seek integers n_1 and n_2 such that to a good approximation, $n_1G_1 + n_2G_2$ yields the required conductance G_{e2}. Setting $n_2 = 2$ and $n_1 = 18$ gives

$$G_{e2} = (2)(147) + (18)(2.13) = 332.3\ \text{mS}$$

This yields

$$R_{e2} = \frac{1}{G_{e2}} = \frac{1}{0.3323} = 3.01\ \Omega$$

so that the series combination

$$R_{e1} + R_{e2} = 47 + 3.01 = 50.01\ \Omega$$

is well within the ±1% requirement. This configuration is shown in Fig. 2.43(b).

We need to determine if this resistor array satisfies the power requirement as well. In order that the circuit dissipate 2 W of power, we must have

$$I^2(50.01) = 2$$

or $I = 0.20$ A. The total current I passing through the circuit divides equally among the 10 resistors in R_{e1}, so for $I = 0.20$ A, the power dissipated in each is

$$P = \left(\frac{0.2}{10}\right)^2 (470) = 0.188 \text{ W}$$

well below the $\frac{1}{4}$ W permitted. The power dissipated in R_{e2} divides proportionally with the conductance of the parallel elements, so each of the two 147-mS conductances dissipate more power than the other conductances. By current division, the power dissipated in each 147-mS (6.8 Ω) element is

$$P = \left[\left(\frac{147}{332.3}\right)(0.2)\right]^2 (6.8) = 0.053 \text{ W}$$

again below the $\frac{1}{4}$ W maximum. Thus we have a solution.

Note that this solution is hardly unique. We could have begun by placing seven 6.8-Ω resistors in series rather than ten 470-Ω resistors in parallel. This may seem to save resistors, but minimizing the number of resistors used was not an explicit design specification. Among distinct solutions to a design problem it is common sense to reduce the component count where possible, and this may seem to favor the series alternative. Note, however, that had we done so, the entire current $I = 0.2$ A needed to meet the 2-W power requirement would flow through each 6.8-Ω resistor, yielding

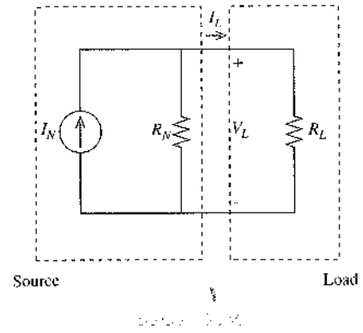
$$P = (0.2^2)(6.8) = 0.272 \text{ W}$$

which would exceed the $\frac{1}{4}$-W limit per resistor. We could work around this problem by substituting a pair of parallel 6.8-Ω resistors in series with another such pair for each of the seven proposed 6.8-Ω resistors, since each four-resistor array would have the same equivalent resistance of 6.8 Ω, but each resistor would dissipate only one-fourth as much power as before. But in this case the component count advantage would be lost.

Design Example 2.2

Design a practical electrical source (by specifying its Thevenin equivalent components) which delivers at least 16 V dc to an 8-Ω load but is current limited to 3 A.

We will determine the Norton source, then transform to get the desired Thevenin values. Connecting the source and load as shown in Fig. 2.44, we wish I_L to be less than or equal to 3 A regardless of



R_L. Setting $R_L = 0$ clearly achieves the greatest value for I_L, which is I_N itself. Thus the Norton equivalent current source $I_N = 3$ A. When $R_L = 8$ Ω, the load voltage should be 16 V,

$$V_L = 16 = \frac{I_N}{G_N + G_L}$$

where G_N and G_L are the conductances. Setting $G_L = \frac{1}{8}$ S and solving for G_N yields

$$G_N = \frac{3}{16} - \frac{1}{8} = \frac{1}{16} \text{ S}$$

or $R_N = 16$ Ω. The Norton source elements are $I_N = 3$ A and $R_N = 16$ Ω, so the Thevenin source has $V_T = (3)(16) = 48$ V and $R_T = 16$ Ω. This source will deliver 16 V to an 8-Ω load and never produce more than 3 A, as required.

One very practical class of resistive circuits are those that supply relatively fixed currents or voltages to variable loads. Our final design example is a circuit of this type.

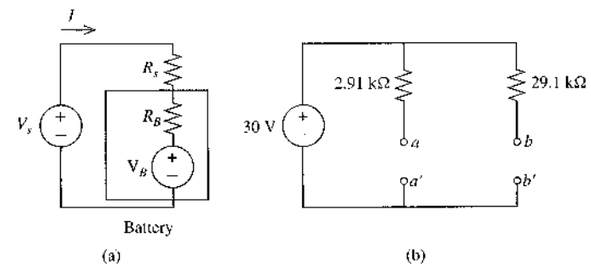
Design Example 2.3

Our goal is to design a battery charger circuit that will deliver controlled levels of dc current to charge batteries of different sizes. We wish to deliver 10 mA of current to larger batteries, which will be connected between terminals a and a', and 1 mA to smaller batteries being charged at terminals b–b'. At any terminal voltage with batteries from zero to full charge 1.5 V, and any battery internal resistance from zero up to 30 Ω, the charging currents must remain within ±5% of the target values.

We begin the design by noting that the circuit shown in Fig. 2.45(a) will deliver

$$I = \frac{V_S - V_B}{R_S + R_B} \qquad (2.29)$$

where V_s and R_s are the source voltage and resistance of the charger circuit, V_B the battery voltage, and R_B its internal resistance. Since I must be relatively constant as V_B and R_B vary, we will set $V_S \gg V_B$



(a) Source and battery; (b) final circuit.

and $R_S \gg R_B$ so, to a good approximation, $I \simeq V_S/R_S$. Now the numerator in (2.29) will vary by 5% as V_B varies from 0 to 1.5 V if $V_S = (20)(1.5) = 30$ V. Then setting $V_s/R_S = 1$ mA yields

$$0.001 = \frac{30}{R_S} \quad \text{or} \quad R_S = 30 \text{ k}\Omega \tag{2.30}$$

By (2.29), the maximum charging current occurs with $V_B = R_B = 0$, and by (2.29), $I_{\max} = 1$ mA. $I_{\min}$ occurs with $V_B = 1.5$ V, $R_B = 30\ \Omega$, or

$$I_{\min} = \frac{30 - 1.5}{30{,}000 + 30} = 0.946 \text{ mA}$$

The worst-case error occurring at $I_{\min}$ is -5%, which is slightly beyond our tolerance. This can easily be remedied by "tuning" this first-cut try.

Since the range of errors is $+0$ to -5%, let's increase the maximum current by, say, 3%. Then retaining $V_S = 30$ V for convenience gives

$$I_{\max} = 0.00103 = \frac{30}{R_S}$$

or $R_S = 29.1$ kΩ. The minimum current is then increased to

$$I_{\min} = \frac{30 - 1.5}{29{,}100 + 30} = 0.978 \text{ mA}$$

The errors now range between $+3$ and -2.2%, well within the specified $\pm 5\%$ tolerance.

Using the same 30-V source to produce the other desired charging current of 10 mA, note from Fig. 2.45(a) that if we scale R_S by $\frac{1}{10}$, I will roughly scale by 10 as desired, since R_S still dominates R_B and V_S dominates V_B. The maximum and minimum currents are

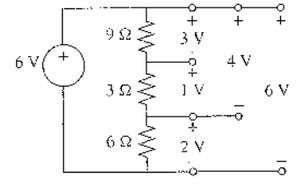
$$I_{\max} = \frac{30}{2910} = 10.3 \text{ mA}, \qquad I_{\min} = \frac{30 - 1.5}{2910 + 30} = 9.7 \text{ mA}$$

Once again the range of errors ($\pm 3\%$) is well within the $\pm 5\%$ tolerance. The final design is shown in Fig. 2.45(b).

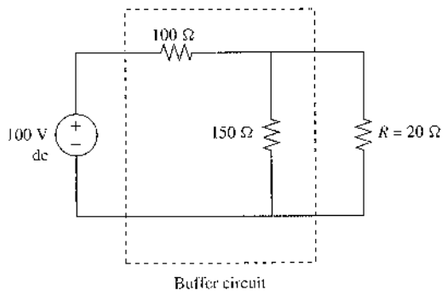
The process of design begins with a complete statement of design specifications and constraints. A first-cut circuit layout is conceived and first-cut element values are selected. Circuit analysis methods are then used to determine the performance of the proposed circuit. In the examples above, the circuit analysis methods invoked included series–parallel and Thevenin–Norton transformations, also current–voltage division. If the resulting element values do not satisfy the specifications and constraints, changes are made and the cycle repeated. Even if a solution has been found, it may often be improved further, unless the optimal solution has been discovered. If desired, another cycle may then be executed to improve performance, as in Design Example 2.3.

In very simple designs it may be possible to solve for solution values of the circuit elements exactly rather than through repeated cut-and-try cycles. This is the case with the examples above. But as the designs get more complex, it is usually preferable to improve iteratively by cut-and-try rather than to attempt to solve a complex system of coupled equations which are often nonlinear in the circuit parameters being sought. This will be the case for most of the design examples in subsequent chapters.

2.9.1. Design a circuit that supplies the following dc voltages from a single voltage source: 1 V, 2 V, 3 V, 4 V, 6 V, all $\pm 5\%$ into resistive loads of 100 Ω or more. Use as few resistors and as low a voltage source as possible. The voltages supplied need not have a common ground terminal.



2.9.2. Design a buffer circuit to go between a 100-V dc source and a variable resistive load with nominal value $R\ \Omega$, which will protect the load from both overvoltage and overcurrent conditions. The load current should never exceed 1 A or the load voltage exceed 60 V, even as the load resistance varies unpredictably from its nominal value during operation. At its nominal value of $R\ \Omega$ the load should dissipate at least 10 W. Specify both the buffer circuit and an acceptable nominal value for the load.



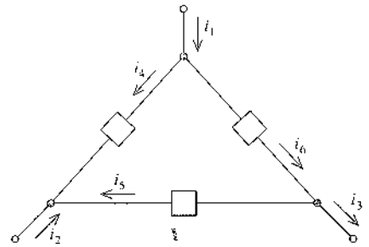
SUMMARY

The behavior of an electrical circuit is a consequence of two factors: the individual behaviors of each element populating the circuit, and the manner in which they are interconnected. There are two interconnection laws, the Kirchhoff current and voltage laws. Each type of element has its characteristic element (or i–v) law, an equation involving its terminal variables. In this chapter we introduce two elements, resistors and independent sources, which when interconnected form resistive circuits. Others are introduced in later chapters.

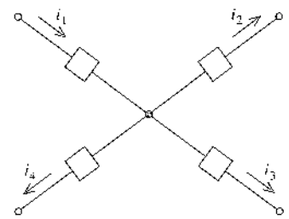
- KCL asserts that the net current entering any node or region is zero.
- KVL asserts that the net voltage drop around any closed loop is zero.
- The i–v law for resistors, Ohm's law, is $v = Ri$, with R the resistance in ohms, and is valid as long as i and v satisfy the passive sign convention.
- Resistors are passive elements; that is, they cannot supply power.
- Two subcircuits are equivalent if they have the same i–v law. Equivalent subcircuits may be freely interchanged without changing any external variable. Examples include series–parallel and Thevenin–Norton equivalents.
- Series resistance adds, parallel conductance adds.
- Parallel resistance obeys the reciprocal–reciprocal law: the equivalent resistance is the reciprocal of the sum of the reciprocals of the parallel resistances.
- Series voltage divides in proportion to resistance, parallel current divides in proportion to conductance (or inversely proportional to resistance).
- A Thevenin form is a two-terminal series interconnection of voltage source and resistor, a Norton form is a parallel interconnection of current source and resistor.
- Any two-terminal subnetwork has a Thevenin equivalent and a Norton equivalent form. These may be found by determining the open-circuit voltage, short-circuit current, and resistance looking into the terminal pair.
- Thevenin and Norton transformations are useful in replacing a selected region of a circuit to be analyzed by a simpler one which will not result in any changed currents or voltages outside its own boundaries.
- Practical voltage and current sources may be modeled accurately by their Thevenin and Norton forms.
- Ammeters, voltmeters, and ohmmeters based upon the d'Arsonval movement are in wide use. The force between a fixed and variable magnetic field moves a pointer.

While KCL, KVL, i–v laws, and equivalent subcircuits are introduced here in the context of resistive circuits studied in the time domain, these fundamental ideas will apply equally well as we broaden our scope in later chapters to include other types of elements and other domains of analysis.

2.1. Write KCL at each node.



2.2. Write KCL for this node:
(a) In sum-entering-equals zero form.
(b) In sum-leaving-equals zero form.
(c) In sum-entering-equals-sum-leaving form.

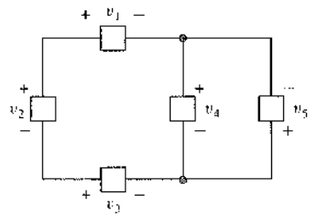


2.3. KCL equations for a certain three-node circuit are:

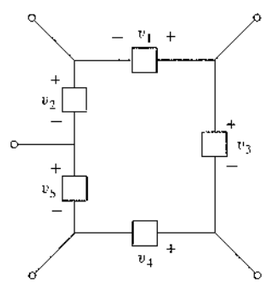
Node 1: $i_1 + i_2 - i_3 = 0$
Node 2: $-i_1 - i_2 - i_4 = 0$
Node 3: $i_3 + i_4 = 0$

Sketch the circuit, indicating current reference directions.

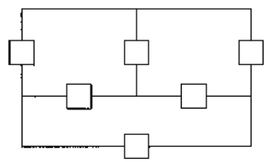
2.4. Write KVL around each of the three loops.



2.5. Write KVL for this loop:
(a) In sum-of-drops-equals-zero form.
(b) In sum-of-rises-equals-zero form.
(c) In sum-of-drops-equals-sum-of-rises form.

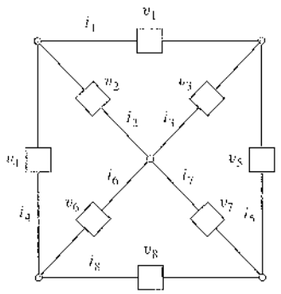


2.6. Assign names and reference directions for all voltages, then write KVL for each of the seven loops in this circuit.



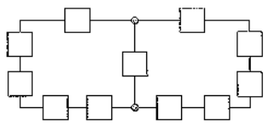
2.7. Supply reference directions so that these KCL equations are all valid and the passive sign convention is satisfied everywhere.

$$i_1 + i_4 - i_2 = 0$$
$$i_7 + i_8 - i_5 = 0$$
$$i_1 + i_3 - i_5 = 0$$
$$i_6 - i_4 - i_8 = 0$$
$$i_2 + i_3 - i_6 - i_7 = 0$$



2.8. In a given circuit, suppose that we write KCL at each node in the sum-entering-equals-zero form.
(a) In how many equations will a given element current appear? With what signs? Justify.
(b) If all these equations were added together, what would we get?
(c) Use part (b) to show that the last node equation is just the negative sum of all the previous ones.

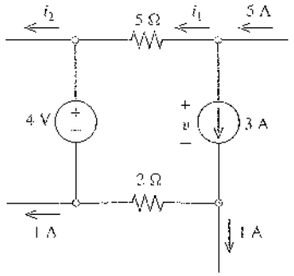
2.9. Assign names and reference directions; then write KCL at each node and KVL around each loop.



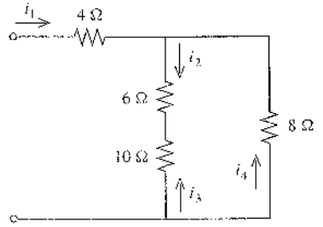
2.10. A toaster is essentially a resistor that becomes hot when it carries a current. If a toaster is dissipating 960 W at a voltage of 120 V, find its current and its resistance.

2.11. Find the energy used by a toaster with a resistance of 12 Ω, which is operated at 120 V for 10 s.

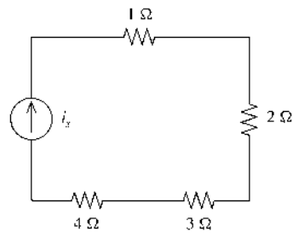
2.12. Find i_1, i_2, and v.



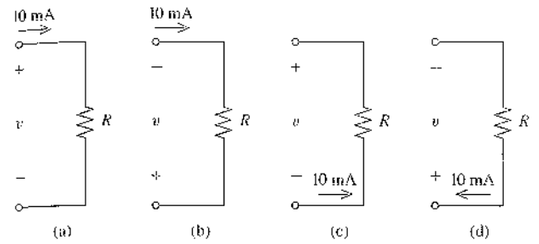
2.13. Assign voltage reference directions for each resistor so that the passive sign convention is satisfied. Determine each voltage if $i_2 = 2$ A and $i_4 = -4$ A.



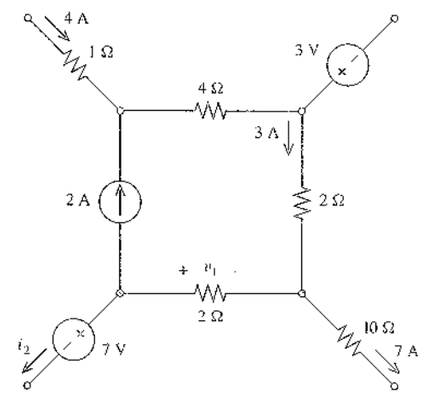
2.14. Assign voltage reference directions for all five elements so that the voltage drop across each element is positive in the clockwise direction; then assign current reference directions that satisfy the passive sign convention.



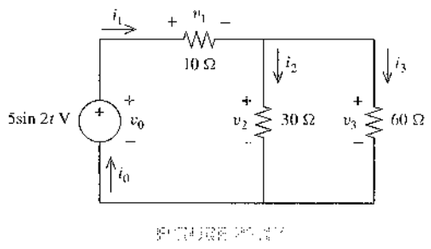
2.15. Find v for each subcircuit. In each case, $R = 1$ kΩ.



2.16. Find v_1, i_2, and the power produced (or dissipated) by the 7-V voltage source.



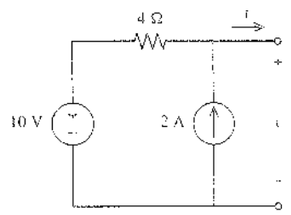
2.17. Find all currents and voltages.



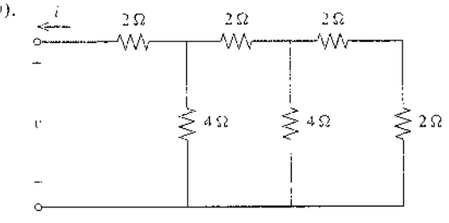
2.18. Determine the power supplied by, or dissipated by, each of the four elements in the circuit of Problem 2.17.

2.19. By what factor would we increase the source function in the circuit of Problem 2.17 if we wished to double the power it supplies to the rest of the circuit? Justify.

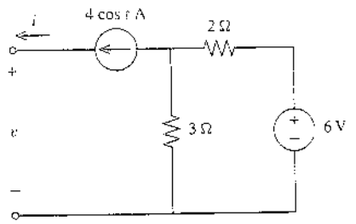
2.20. Find the terminal law $v = f(i)$. Rewrite as $i = g(v)$.



2.21. Find the terminal law $v = f(i)$. Rewrite as $i = g(v)$.



2.22. Find the terminal law for this circuit. Note that only one of the two forms $v = f(i)$ and $i = g(v)$ exists in this case.



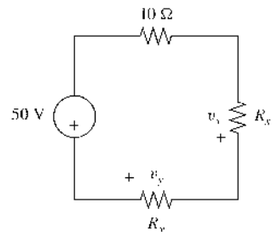
2.23. A circuit has 4 nodes. The resistor connecting nodes a and c is $R_{ac} = 360\Omega$. Similarly $R_{ad} = 180\Omega$, $R_{bc} = 540$V, $R_{bd} = 540\Omega$. Open means no other connections. (a) Find the equivalent resistance looking in terminals $a-b$ if terminals $c-d$ are open, and if terminals $c-d$ are shorted together. (b) Find the equivalent resistance looking in terminals $c-d$ if terminals $a-b$ are open, and if terminals $a-b$ are shorted together.

2.24. Design a voltage divider that delivers +3, +6, +12, and +24 V, all with a common negative terminal. All resistors must be integer multiples of 1 kΩ, and the voltage source should be as low voltage as possible.

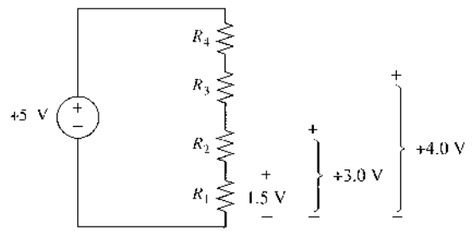
2.25. A 50-V source and two resistors, R_1 and R_2, are connected in series. If $R_2 = 4R_1$, find the voltages across the two resistors.

2.26. A 12-V source in series with a resistive load R carries a current of 60 mA. If a resistor R_1 is added in series with the source and load, find R_1 so that the voltage across it is 8 V.

2.27. If $v_x = 10$ V and $v_y = 20$ V, find R_x and R_y.

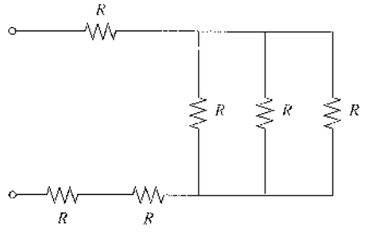


2.28. If all resistors in a voltage divider are scaled by the same factor α, how does this affect the voltages across them? The currents through them? The power? Use this result to design a voltage divider using a +5-V source with outputs of +4.0, +3.0, and +1.5 V (all with respect to a common negative terminal) and a power dissipation of 1 mW. Start by taking R_1 to be 1 Ω; compute the other R's and scale.

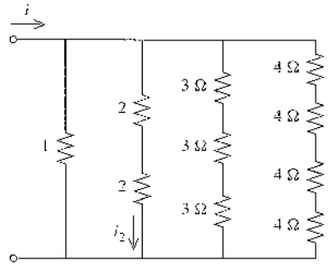


2.29. A voltage divider is to be constructed using a +24-V battery and 1-kΩ resistors. Design a voltage divider with a +18-V output using as few resistors as possible. Design another one with a +18.1-V output.

2.30. What must R be to set the equivalent resistance of this circuit to 300 Ω?

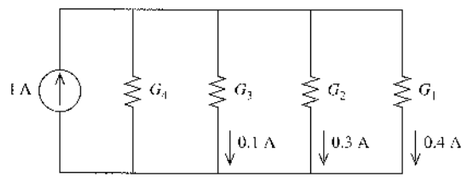


2.31. Find i and the total power dissipated by this circuit if $i_2 = 1$ A.

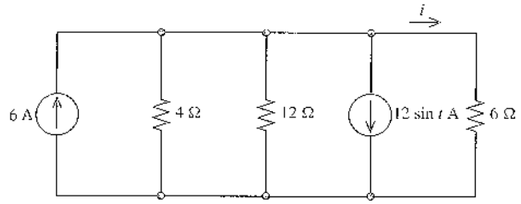


2.32. Determine the equivalent resistance of the circuit of Problem 2.31.

2.33. If the conductances of all resistors in a current divider are scaled by the same factor α, how does this affect the currents through them? The voltages across them? The power? Use this result to design a current divider using a +1-A source with outputs of 0.4, 0.3, and 0.1 A and a power dissipation of 1 mW. Start by taking G_1 to be 1 S; compute the other G's and scale.

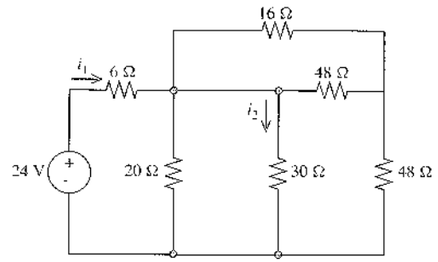


2.34. Find i and the power delivered to the 4-Ω resistor.



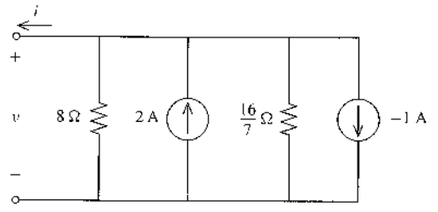
2.35. A current divider consists of the parallel connection of a 20-, a 40-, a 60-, and a 120-kΩ resistor. Find the equivalent resistance of the divider, and if the total current entering the divider is 120 mA, find the current in the 20-kΩ resistor.

2.36. Find i_1 and i_2.

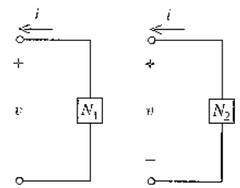


2.37. A current divider is to be constructed using a +2-mA constant current source and 1-MΩ resistors. Design a current divider with a +1.5-mA output. Design another with a +1.51-mA output.

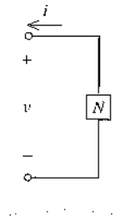
2.38. Find the Thevenin equivalent circuit.



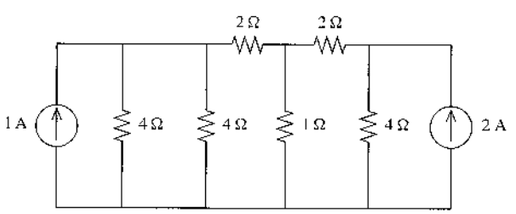
2.39. Suppose that the terminal laws for N_1 and N_2 are $v = -2i + 1$ and $v = -4i - 7$. Find the Thevenin equivalent if N_1 and N_2 are placed in series.



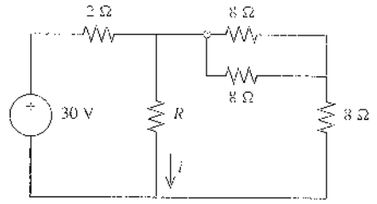
2.40. Ten identical copies of a subcircuit N with terminal law $i = g(v) = -\frac{1}{3}v + 2$ are put in series. Find the terminal law of the new subcircuit and its Norton equivalent.



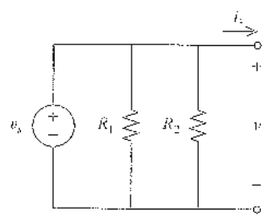
2.41. Use a series of Thevenin–Norton and series–parallel transformations to reduce to a simple two-node circuit the circuit shown. Which variables from the original circuit remain in the simple circuit?



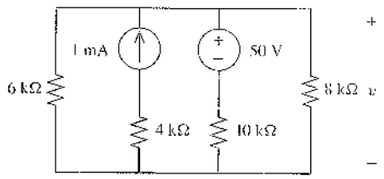
2.42. Find the Thevenin equivalent of everything except the R-Ω resistor. Then find R so that $i = 1$ A.



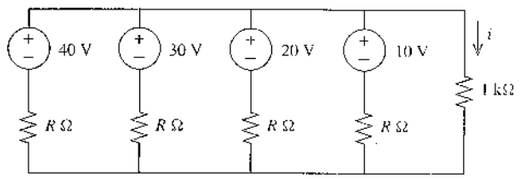
2.43. Find the Thevenin equivalent of this subcircuit. Use this to show that resistors in parallel with voltage sources have no effect on the rest of the circuit. Find a similar relation involving resistors and current sources.



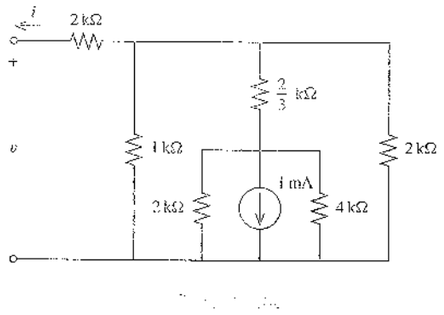
2.44. Find v. Use a Thevenin–Norton transformation to help.



2.45. Find a value for R such that $i = 1$ mA.

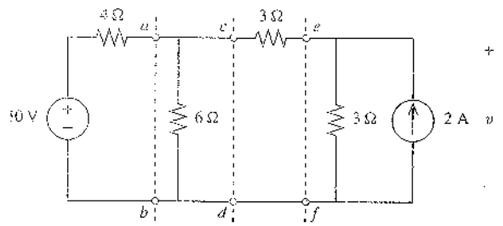


2.46. Find the Thevenin equivalent of this circuit by first finding and using the Thevenin equivalent of the current source and its parallel resistors.

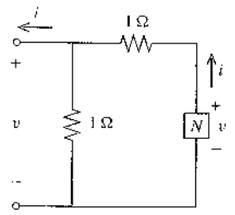


2.47. Find the Norton equivalent of the subcircuit of Problem 2.46.

2.48. Find the Norton equivalent to the left of ab, then the Thevenin equivalent to the left of cd, then the Norton equivalent to the left of ef. Having simplified to a two-node circuit, finally find v.



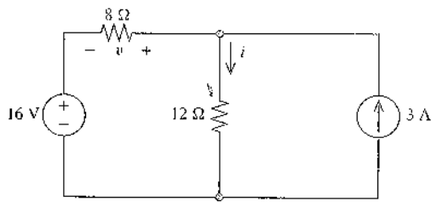
2.49. N satisfies the terminal law $v = -7i + 4$. Find the Thevenin and Norton equivalents for the given circuit.



2.50. Under what circumstances will the Thevenin equivalent of a given subcircuit exist but not the Norton equivalent? Give an example.

2.51. Under what circumstances will the Norton equivalent of a given subcircuit exist but not the Thevenin equivalent? Give an example.

2.52. Find i and v.



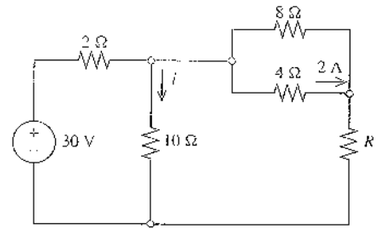
2.53. Using any number of 10 Ω resistors, design a subcircuit whose equivalent resistance is 144.7 Ω.

2.54. Use any number of 2.2-kΩ $\frac{1}{4}$-W and 56-Ω $\frac{1}{8}$-W resistors, together with 1.5-V batteries, to produce a practical source supplying 25 mA of short-circuit current and 15 mA of current to a 50-Ω load (each ±5%).

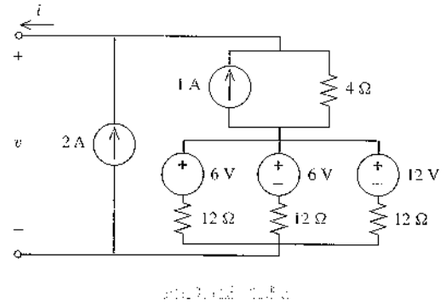
2.55. Design a dc power supply circuit with four pairs of load terminals marked 1–1′ through 4–4′. When a load resistor is connected between 1 and 1′, it should get 1 A dc; between 2 and 2′, 500 mA dc; 3 and 3′, 250 mA dc; and 4 and 4′, 50 mA dc. The load resistor can be any $R_L \leq 50$ Ω. Use one independent current source of 2 A dc and as few internal resistors as possible. Tolerance on the load currents is ±10%.

2.56. An alternator with an open-circuit voltage of 22 V dc and a short-circuit current of 10 A dc is to be used to charge automobile batteries safely with open-circuit voltage in the range 0 to 12 V and Thevenin equivalent impedance in the range 1 to 10 Ω. Design a resistive circuit limiting the power supplied to any battery in these ranges to 10 W while permitting the maximum number of batteries to be charged at the same time.

2.57. Find i and R.



2.58. Find Thevenin and Norton equivalents.

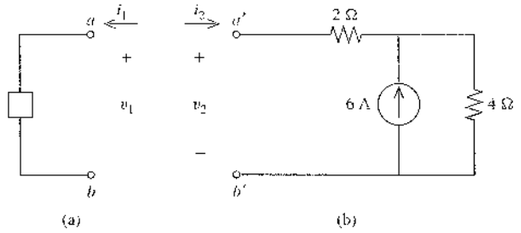


2.59. A certain circuit has seven elements and obeys the KCL equations

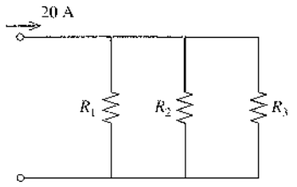
$$i_1 = i_2$$
$$i_2 = i_3 - i_4$$
$$i_7 = i_1 + i_5$$
$$i_3 + i_5 + i_6 = 0$$

(a) Sketch the circuit, labeling currents $i_1, i_2, \ldots, i_7$ and voltages $v_1, \ldots, v_7$. Define the voltages for each element to satisfy the passive sign convention.
(b) Write three distinct KVL equations for this circuit.

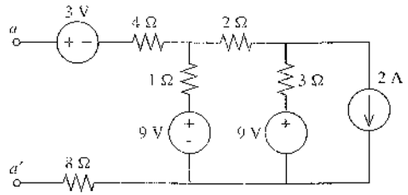
2.60. (a) Find the i–v law $v_2 = f_2(i_2)$ for the circuit shown in (b). (b) If $v_1 = 3i_1 - \sin t$, nodes a–a' are connected together, and nodes b–b' are connected together, find i_1 and v_2.



A total current of 20 A divides among three resistors as shown. If the largest is $R_1 = 1$ kΩ, find R_2 and R_3 so that 12 A flows through R_3 and $R_2 = 2R_3$.



Sketch the Thevenin and Norton equivalents to the circuit shown.





Georg Simon Ohm
1787–1854

I herewith present to the public a theory of galvanic electricity [Ohm's law].

Georg Simon Ohm

3

Dependent Sources and Op Amps

The most basic and most widely used of all the laws of electricity, Ohm's law, was published in 1827 by the German physicist Georg Simon Ohm in his great work, *The Galvanic Chain, Mathematically Treated*. Without Ohm's law we could not analyze the simplest galvanic chain (electric circuit), but at the time of its publication, Ohm's work was denounced by critics as "a web of naked fancies," the "sole effort" of which was "to detract from the dignity of nature."

Ohm was born in Erlangen, Bavaria, the oldest of seven children in a middle-class-to-poor family. He was an early dropout at the University of Erlangen but returned in 1811 and earned his doctorate and the first of several modest, low-paying mathematics teaching positions. To improve his lot, he threw himself into his electrical research at every opportunity allowed by his heavy teaching duties, and his efforts culminated in his famous law. Despite the misplaced criticisms of his work, during his lifetime Ohm received the fame that was due him. The Royal Society of London awarded him the Copley Medal in 1841, and the University of Munich gave him its Professor of Physics chair in 1849. He was also honored after his death when the *ohm* was chosen as the unit of electrical resistance.

3.1 Definitions
3.2 Circuits with Dependent Sources
3.3 Operational Amplifiers
3.4 Role of Negative Feedback
3.5 Op Amp Building Block Circuits
3.6 Interconnecting Op Amp Building Blocks
3.7 Practical Op Amps
3.8 Design of Simple Op Amp Circuits
Summary
Problems

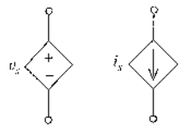
The voltage and current sources of Chapters 1 and 2 are independent sources, as defined in Section 1.4. We may also have *dependent sources*, which are very important in circuit theory, particularly for understanding electronic circuits (circuits containing elements such as transistors or vacuum tubes that need to be coupled to power supplies). In this chapter we shall define dependent sources and consider an additional circuit element, the operational amplifier or *op amp*. Dependent sources are central to the design of electronic amplifiers and to a wide variety of other circuits of great practical interest. Op amps behave like dependent sources, and op amps are a convenient way to fill the requirement for dependent sources in these circuits, and the design of simple op amp circuits.

We shall first analyze a few simple circuits containing resistors and sources, both independent and dependent. As we shall see, the analysis is very similar to that performed in Chapter 2. We shall then turn our attention to the op amp and its behavior as a linear circuit element. Ideal and nonideal op amp models will be presented and the uses of these models in analyzing op amp circuits explored. A set of op amp building block circuits will be introduced, easily analyzed circuits from which complex circuit behaviors can be built by combination. The chapter concludes with a look at practical op amps in the real world of physical packages, finite limitations, and environmental constraints, and the design of simple op amp circuits.

3.1 DEFINITIONS

A *dependent or controlled voltage source* is a voltage source whose terminal voltage depends on, or is controlled by, a voltage or a current defined at some other location in the circuit. Controlled voltage sources are categorized by the type of controlling variable. A *voltage-controlled voltage source (VCVS)* is controlled by a voltage and a *current-controlled voltage source (CCVS)* by a current. The symbol for a dependent voltage source with source function v_s is shown in Fig. 3.1(a).

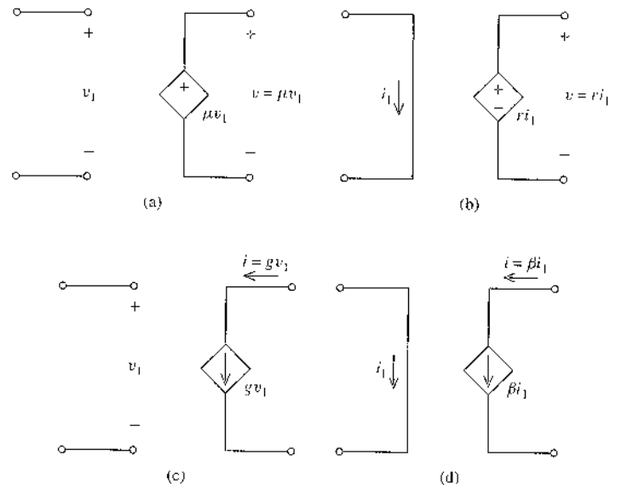
A *dependent or controlled current source* is a current source whose current depends on, or is controlled by, a voltage or a current defined at some other location in the circuit.



(a) Dependent voltage source; (b) dependent current source.

A *voltage-controlled current source (VCCS)* is controlled by a voltage and a *current-controlled current source (CCCS)* by a current. The symbol for a dependent current source with value (source function) i_s is shown in Fig. 3.1(b).

Figure 3.2 illustrates the four types of *linear* controlled sources and shows the voltage or current on which they are dependent. The quantities μ and β are dimensionless constants, commonly referred to as the *voltage gain* and *current gain*, respectively. The constants r and g have dimensions of ohms and siemens, units of resistance and conductance, and are thus called the *transresistance* and *transconductance*, respectively. The prefix *trans* is to remind us that the current and voltage are not measured at the same location. These four parameters each scale the ratio of the controlled variable produced per unit controlling variable.

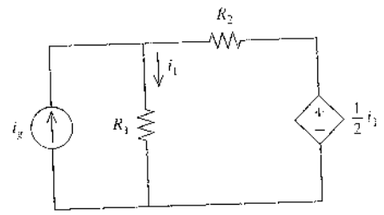


(a) VCVS; (b) CCVS; (c) VCCS; (d) CCCS.

Nonlinear dependent sources are those in which the controlled variable is not simply proportional to the controlling variable as in the *linear dependent sources* above. In the present book we are interested only in studying linear circuits, and thus nonlinear controlled sources will not be considered further.

As an example of a circuit containing a dependent source, in Fig. 3.3 we have an independent source, a dependent source, and two resistors. The dependent source is a CCVS with controlling current i_1 and transresistance $r = 0.5\ \Omega$. Note that if i_1 were redefined to be the current flowing downward through the rightmost arm of this circuit rather than the middle arm, the controlled source would then be an element whose voltage was 0.5 times its own current, and thus it would be identical to an ordinary 0.5-Ω resistor. This illustrates that only if the controlling variable is located *elsewhere* in the circuit does the element truly have to be represented as a controlled source.

Dependent sources are essential for producing *amplifiers*, circuits that produce outputs more powerful than their inputs. They are also integral to active filters (to be studied in Chapter 14) and to electronic circuits of all kinds. Among their many important uses,



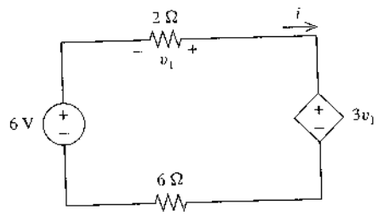
Circuit containing a dependent source.

dependent sources are used to prevent loading, isolate one part of a circuit from another, and provide exotic behaviors such as negative resistance. As we saw in Chapter 2, the resistor is a passive element with positive resistance. However, by means of dependent sources we may create circuits exhibiting negative resistance, as we shall see (Exercise 3.2.3).

3.2 CIRCUITS WITH DEPENDENT SOURCES

Circuits containing dependent sources are analyzed in the same manner as those without dependent sources. All the tools introduced in Chapter 2 may be applied: the Kirchhoff current and voltage laws, Ohm's law for resistors, series–parallel equivalents, Thevenin–Norton equivalents, and voltage and current division. In Chapter 4 we develop systematic strategies for using these tools, following a step-by-step plan that will permit us to solve for any variable in any circuit. For now, we are content to use the tools of circuit analysis on an ad hoc basis and limit our treatment to relatively small circuits.

Example 3.1



Dependent source example.

Find the current i in the circuit of Fig. 3.4. The dependent source is a VCVS with voltage gain of +3 controlled by the voltage v_1.

Applying Kirchhoff's voltage law around the circuit, we have

$$-v_1 + 3v_1 + 6i = 6 \tag{3.1}$$

and by Ohm's law

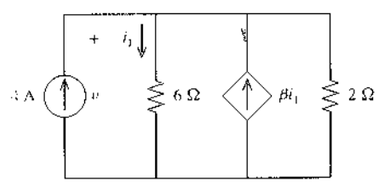
$$v_1 = 2(-i) = -2i \tag{3.2}$$

Using (3.2), we may eliminate v_1 from (3.1), resulting in

$$2(-2i) + 6i = 6$$

or $i = 3$ A. Thus the dependent source has complicated matters only to the extent of requiring the extra equation (3.2).

Let's use this example to consider further the significance of the voltage gain μ. With voltage gain of 3, we have determined the current i to be 3 A. If this gain were doubled, we would just replace $3v_1$ by $6v_1$ in (3.1) and repeat the subsequent steps to get $i = -1.5$ A. It was stated previously that the controlled source parameter (μ, β, r, or g) scales the size of the controlled variable per unit controlling variable. This does not imply that the circuit currents and voltages scale by the same ratio. Doubling the voltage gain in this example changed i from 3 A to −1.5 A, while the controlled voltage (VCVS terminal voltage) was reversed in sign but unchanged in magnitude. What has doubled is only the *ratio* of the controlled voltage per unit controlling voltage, not any particular current or voltage.



3.2.1. In the circuit of Fig. 3.4, add an 18-Ω resistor in parallel with the given 6-Ω resistor and find i, v_1, and the resistance seen by the independent source (i.e., the resistance that when attached across the independent source in place of the rest of the circuit would yield the same response i).

12 A; −24 V; 0.5 Ω

3.2.2. Find the current gain β that results in $v = 8$ V.

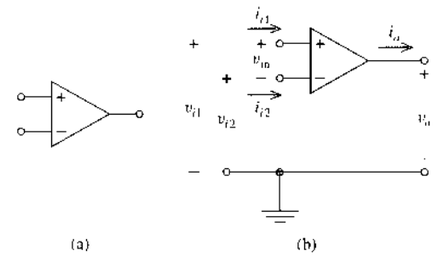
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3.2.3. In the circuit of Exercise 3.2.2, change the 2-Ω resistor to 12 Ω and find the resistance seen by the independent source for $\beta = 2$ (see Exercise 3.2.1 for a definition of "resistance seen"). Note that a negative resistance is possible only when a dependent source is present.

−12 Ω

3.3 OPERATIONAL AMPLIFIERS

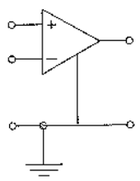
A dependent source has the important capability of generating a current or voltage that is an *amplified* version of its controlling variable. One practical circuit element embodying this capacity is the *operational amp (op amp)*, defined as *an electronic device that under proper circumstances behaves like a voltage-controlled voltage source (VCVS) with very high gain.* The circuit symbol for the op amp is shown in Fig. 3.5(a). The terminal



Operational amplifier (op amp): (a) circuit symbol; (b) currents and voltages.

marked by a plus sign in the circuit symbol is called the *noninverting input*, that by the minus sign the *inverting input*, and the unmarked terminal is the *output*. The voltages at these three nodes, v_{i1}, v_{i2}, and v_o, respectively, are all defined relative to *ground*, a node that is usually omitted from the op amp circuit symbol but is part of every op amp circuit. The op amp input voltage, or voltage across the op amp input terminals, is

$$v_{\text{in}} = v_{i1} - v_{i2}$$

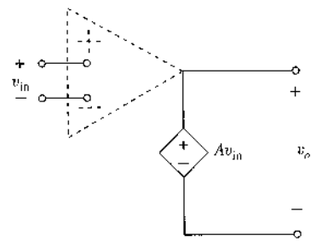


Alternative circuit symbol for the op amp.

Sometimes a fourth wire is added to the circuit symbol as shown in Fig. 3.6 to show the connection to the ground node explicitly. We will adopt the three-wire convention of Fig. 3.5(a) when drawing op amps in circuits.

Since the op amp is an electronic device, a power supply of the right ratings must be correctly connected to the op amp in order to energize it. This is one of the "proper circumstances" required in the definition to achieve useful op amp behavior, and these considerations will be briefly discussed in Section 3.7. Although well-behaved op amp circuits cannot be physically built and operated without due consideration of how these proper circumstances can be assured, the behavior of op amps as linear circuit elements may be analyzed assuming that this has been done. A satisfactory discussion of these requirements and how to meet them must be deferred to an electronics course, which often follows basic circuits courses such as this. For the present purpose of understanding how op amps behave in linear circuits, *we will assume that the proper environment for this behavior has been established in all op amp networks we will analyze.* In particular, since power supplies help establish these circumstances but otherwise do not influence linear op amp behavior, they will generally be omitted from op amp circuit diagrams.

Under the proper operating conditions, an op amp presumably behaves like a VCVS with high gain. The simplest circuit model for an op amp, called the *ideal voltage amplifier* model and shown in Fig. 3.7, consists of nothing more than a VCVS and some connecting wire. Note that this model shows that v_o, the op amp output voltage, is controlled by v_{in}, the difference between the op amp input terminal voltages. The VCVS voltage gain A dictates the degree of amplification of the input voltage in passage through the op amp (ratio of v_o to v_{in}). A is frequently referred to as the *open-loop gain* of the op amp. This nomenclature is related to the concept of feedback discussed in the next section. As we shall see, op amps are frequently used in configurations in which the op amp output is connected back to its input. Such circuits are called *closed loop* and



Ideal voltage amplifier op amp circuit model.

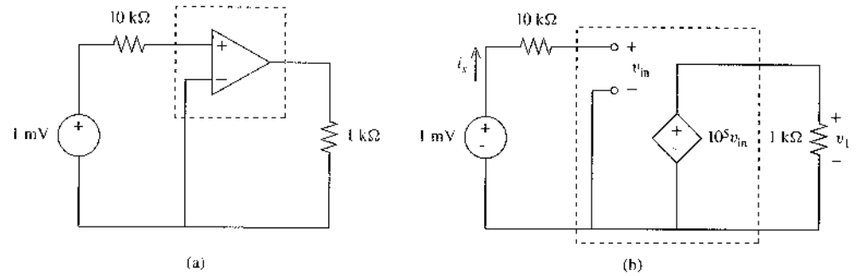
in general have different gain (ratio of output to input) than the open-loop op amp (no feedback), whose open-loop gain is simply the VCVS voltage gain A.

To use this model, we simply replace the op amp circuit symbol by the model and apply the methods of Section 3.2 to analyze the resulting circuit.

Example 3.2

Suppose that we connect an independent source across the input terminals of an op amp, and a load resistor between output and ground. The source has a Thevenin equivalent voltage of 1 mV and Thevenin equivalent resistance of 10 kΩ. The load resistance is 1 kΩ, and the op amp has an open-loop gain of 100,000. Find the source current i_s and load voltage v_l.

Examining Fig. 3.8(b), there can be no current ($i_s = 0$ A) through the 10-kΩ resistor since it is in series with an open circuit. Then, by KVL, $v_{\text{in}} = 0.001$ V. Thus the voltage of the VCVS is $100{,}000 \times 0.001 = 100$ V, which by KVL is also v_l. We see that the source voltage has been amplified by the open-loop gain and transferred across to the load. Note also that the power delivered to the load did not come from the independent source, which produces no power since $i_s = 0$ A. It must have come from the op amp or, more specifically, from the power supply energizing the op amp, even though this necessary source of power is not shown explicitly.

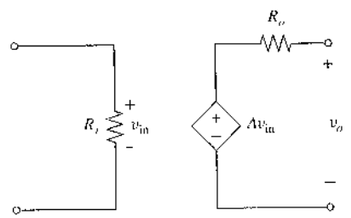


Circuit for Example 3.2: (a) circuit modules; (b) ideal voltage amp model substitution.

Reflection upon the ideal voltage amplifier model of Fig. 3.7 reveals some apparent shortcomings. First, the model predicts that an op amp draws no current at all from the input source (the input terminals are open circuited) and thus no power, yet performs its job of amplification. But basic physics suggests that any amplifier must draw at least a little power from its source in order to sense the signal it is supposed to amplify. Communication always requires some minimal expenditure of power by the source in order to transmit the message, yet the model predicts that no power is needed. Second, an equally questionable prediction is apparent on the output side of the model. If a load resistor is applied across the output, by decreasing its resistance further and further the current and thus the power delivered to the load by the op amp may be made arbitrarily large (the load voltage is fixed by the input voltage and open-loop gain, while power to

the load is v_1^2/R). Yet every device in the real world has some finite maximum power capability.

An *improved op amp model*, which remedies these shortcomings by the addition of two resistors to the ideal voltage amplifier model, is shown in Fig. 3.9. R_i is called the *input resistance* and R_o the *output resistance*. The ideal voltage amplifier model may be seen as a special case of the improved model with R_i = infinity and $R_o = 0$. By requiring these values to be both nonzero and finite in the improved model, we guarantee that some power will be expended by the source and that a finite maximum current and power will be developed across the load, no matter how small the load resistance is. Typical values for common op amps are $R_i = 1$ MΩ and $R_o = 30$ Ω. Just as with the previous model, the improved model is used by substituting it for the op amp circuit symbol in the circuit to be analyzed.



Improved op amp model.

Example 3.3

We will use the improved op amp model with $R_i = 1$ MΩ, $R_o = 30$ Ω, and $A = 10^5$ to determine v_2 in the circuit shown in Fig. 3.10(a). After replacement, the circuit is as shown in Fig. 3.10(b). To simplify analysis, everything to the left of the dashed line is first converted to its Thevenin equivalent. The open-circuit voltage at terminals a and b is, by voltage division,

$$v_{oc} = v_T = \frac{60 \times 10^3}{90 \times 10^3} v_1 = \tfrac{2}{3} v_1$$

and the resistance looking into terminals ab with source v_1 killed is the parallel equivalent of the 60-kΩ and 30-kΩ resistors:

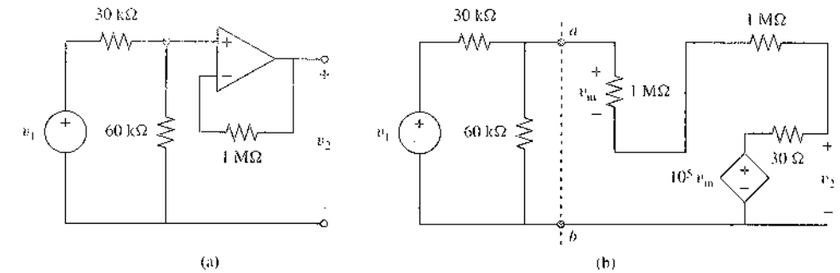
$$R_T = \frac{(30 \times 10^3)(60 \times 10^3)}{90 \times 10^3} = 20 \text{ kΩ}$$

Substituting the Thevenin equivalent circuit, we have the circuit of Fig. 3.11. Calling the loop current i, by KVL around the loop

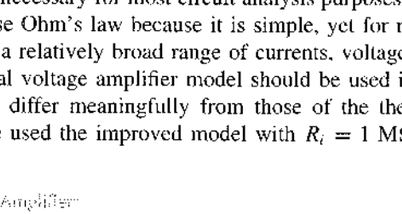
$$2.02 \times 10^6\, i + 10^5 v_{in} - \tfrac{2}{3} v_1 = 0$$

But since $i = v_{in}/10^6$, this may be solved for v_{in}:

$$v_{in} = \tfrac{2}{3} \times 10^{-5} v_1$$



(a) Circuit for Example 3.3; (b) after replacement of the op amp by the improved op amp model.



Circuit of Fig. 3.10 after Thevenin equivalent substitution.

Finally, since

$$v_2 = 10^5 v_{in} + 30i$$

we have the desired result:

$$v_2 = \tfrac{2}{3} v_1$$

Which op amp model should be used in a given problem? As in any model-based analysis, there is always some distance between physical objects and each of their models. An unending sequence of ever more accurate and more complex models may be invoked to represent a given object. The best model to use is not necessarily the most accurate, but rather the simplest one with the accuracy required for a given purpose. For instance, Ohm's law itself is only a simple, low-order model for the current–voltage behavior of a typical physical resistor. No physical resistor obeys Ohm's law exactly, and thus the predictions of Ohm's laws can be improved by using more complex models. More accurate models are available over specific ranges of current and voltage for specific types of resistors, but are unnecessary for most circuit analysis purposes and certainly are more difficult to use. We use Ohm's law because it is simple, yet for many physical resistors is quite accurate over a relatively broad range of currents, voltages, and temperatures.

The simpler ideal voltage amplifier model should be used in those settings where its predictions do not differ meaningfully from those of the the improved model. In Example 3.2, had we used the improved model with $R_i = 1$ MΩ, $R_o = 30$ Ω and the

same $A = 100{,}000$ as before, by a voltage divider calculation in the input loop the input voltage is

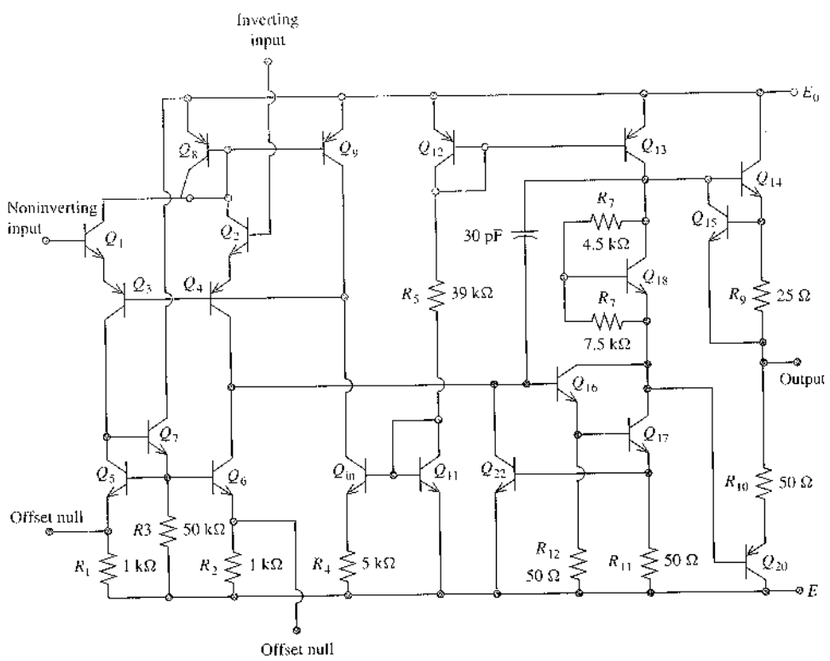
$$v_{\text{in}} = \frac{0.001R_i}{R_i + 10{,}000} = 0.00099 \text{ V}$$

compared to 0.001 V previously, while a second voltage divider applied to the output loop shows that

$$v_1 = \frac{(99.0)50000}{50{,}000 + R_o} = 98.9 \text{ V}$$

No current or voltage has changed by more than a fraction of 1%, which for most purposes means that the simple model is accurate enough. As we progress in our knowledge of op amp circuit analysis, we will develop guidelines for model selection. For now, the model to use will always be specified in each problem requiring analysis of an op amp circuit.

The two models introduced in this section are simple to use and in many circumstances predict the actual behavior of physical op amps quite accurately. We should never forget, though, that a model is different from the device it is modeling. A physical op amp is constructed out of many transistors, resistors, and other elements. Figure 3.12 is

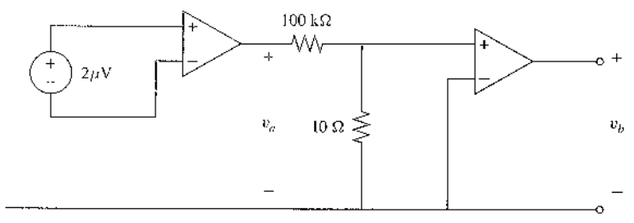


741 op amp. (Courtesy of Fairchild Semiconductor Corp.)

the circuit diagram for the 741 op amp, one of the most popular inexpensive low-power op amps. It is shown to illustrate the distance between a simple model, such as Figs. 3.7 and 3.9, and the circuit being modeled. This circuit diagram is complicated because it is not a simple matter to create a device that under the proper circumstances behaves like a voltage-controlled voltage source of high gain. Considerable circuit complexity is required to design a device calculated to behave so simply.

3.3.1. Determine the voltages v_a and v_b. Use the ideal voltage amplifier model with open-loop gain $A = 100{,}000$.

0.200 V; 2.00 V



3.3.2. Repeat Exercise 3.3.1 with the improved op amp model using $R_i = 1$ MΩ, $R_o = 30$ Ω. Are the voltages within 1% of Exercise 3.3.1? Within 0.1%?

0.200 V; 2.00 V; yes; yes

3.3.3. Using the circuit shown in Exercise 3.31 and the improved op amp model of Exercise 3.3.2, suppose that a load resistor R_1 was connected across v_b. For what value of R_1 would the current delivered to R_1 be a maximum? What would that maximum current be?

0 Ω; $\frac{1}{15}$ A

3.3.4. Redo Example 3.3 using the ideal voltage amplifier model for the op amp. Do the results agree with the example in which the improved op amp model was used?

$v_2 = \frac{2}{3}v_1$; yes

3.4 ROLE OF NEGATIVE FEEDBACK

Typical op amps may have open-loop gains in the range of 10^5 to 10^6 or more, yet relatively few circuits require such large voltage amplification. Why is the op amp required to behave like a VCVS of very high gain, since this gain is seldom used to create such extreme voltage amplification? To understand how open-loop gain can be converted

to other benefits no less important to successful circuit design than raw amplification, we will consider the concept of negative feedback.

Negative feedback is said to exist within an electrical circuit, or dynamic system of any kind, if *any change of the output has the algebraically opposite effect on the input*. In a negative feedback system, an increase in the algebraic value of the output (more positive or less negative) yields a decrease in the algebraic value of the input (less positive or more negative), and vice versa. Consider an athlete running a marathon. Define as input the concentration of oxygen in the blood and as output the athlete's resulting running speed. If at some point it should be decided to pick up the pace (increased output), oxygen will be consumed more rapidly and blood oxygen levels will decrease in turn (decreased input). If, instead, the pace is slowed, less oxygen is burned per unit time and its blood concentration will increase (increased input) due to the decreased pace (decreased output). This is an example of a negative feedback system. Changes in the output of a negative feedback system lead to opposite input changes, which tend to return the output toward its initial value. By this process, negative feedback systems tend to resist change and thereby to *stabilize* their own outputs.

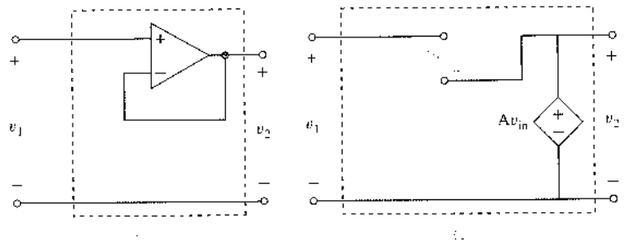
Negative feedback may be achieved in an op amp circuit by supplying a current path between the output terminal and the inverting input terminal. Consider the circuit of Fig. 3.13(a), called a *voltage follower circuit* for reasons that will soon be apparent. Replacing the op amp by its ideal voltage amplifier model as in Fig. 3.8(b), KVL around the left loop yields

$$-v_1 + v_{in} + A v_{in} = 0$$

or $v_{in} = v_1/(A+1)$. The right loop shows that $v_2 = A v_{in}$. Eliminating v_{in} from these last two equations and solving for v_2 yields the result

$$v_2 = \frac{A}{A+1} v_1$$

First let us verify that this is truly a negative feedback circuit. If the op amp output voltage v_2 were somehow to increase slightly, the voltage at the inverting input terminal v_{i2} would be increased by the same amount, while the voltage at the noninverting input



Voltage follower: (a) circuit diagram; (b) using ideal voltage amplifier model.

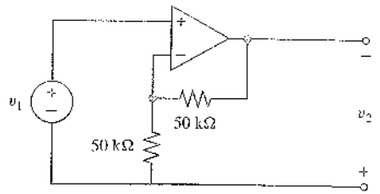
would be unaffected (it is determined solely by v_1). Since $v_{in} = v_{i1} - v_{i2}$, the increase in the op amp output v_2 will indeed result in an algebraic decrease in the op amp input v_{in}, the definitive sign of negative feedback.

Next we compare the behavior summarized in (3.6) with the op amp circuits of Section 3.3 in which negative feedback was not applied. When the voltage v_1 was applied directly across the op amp input terminals with no feedback, the op amp output voltage v_2 was Av_1. With negative feedback the ratio of output v_2 to circuit input v_1, or the *voltage transfer ratio*, has dropped from A to $A/(A+1)$. For $A = 100{,}000$, this is a drop from a voltage transfer ratio of 100,000 to just a bit less than unity. Indeed, for any reasonably large value of A, the voltage transfer ratio is quite close to unity, whence the name *voltage follower* (the output v_2 "follows" or matches the input v_1 without amplification or any other changes).

By making the negative feedback connection, it appears that we have thrown away the $A = 100{,}000$ open-loop op amp gain, instead settling for a voltage transfer ratio (with feedback) of approximately unity. What have we gained in compensation for the tremendous reduction in available amplification? Suppose that over time, as the circuit ages, the open-loop gain of our op amp declines 10%, from 100,000 to 90,000. In the circuit lacking negative feedback, the voltage transfer ratio will change between the same two values, or change 1 part in 10. In our voltage follower, $A/(A+1)$ will go from 0.999990 to 0.999989 as A declines from 100,000 to 90,000, a change of 1 part in 10^6. Thus, at the cost of a five-order-of-magnitude reduction in amplification, we have gained a five-order-of-magnitude increase in the constancy (decrease in the sensitivity) of the circuit's amplification factor.

Constancy of gain is essential in many applications and useful in most others. Using modern integrated-circuit technology, it is easy to build an op amp with high open-loop gain and do so inexpensively, but very difficult to regulate the value of the gain with high precision. Suppose that two op amps coming off the production line have open-loop gains that differ by 10%. If otherwise identical circuits in which these op amps are used have outputs that differ by 10% also, the performance of the resulting electronic end product could not be guaranteed to any degree of precision. By using op amps only in stabilizing negative feedback configurations, the copy-to-copy end circuit variations can be kept very low even when the copy-to-copy op amp variations are rather high. Other gain variations compensated by negative feedback include those due to temperature, aging of other circuit elements connected to the op amp, and variable load resistance. Negative feedback is also essential to decrease undesirable variations of gain with frequency, that is, increase the stage bandwidth, as discussed in Chapter 14.

In summary, *using negative feedback trades raw gain for constancy of gain*. The resulting circuits are highly insensitive to the numerical value of the open-loop op amp gain A, provided only that A can be counted on to be reasonably large. The voltage transfer ratios across these circuits may then be set by selecting the circuit type and the values of any external elements (e.g., resistors as in Fig. 3.15), matters much more easily controlled by the circuit designer than the open-loop op amp gains. Negative feedback configurations are sometimes referred to as *closed-loop circuits*, since the feedforward path through the op amp and the feedback path together form a closed loop. Because of its extreme usefulness in creating reliable circuit designs, all op amp circuits analyzed in the remainder of this book are of the negative feedback type.



3.4.1. Determine the voltage transfer ratio v_2/v_1 of a voltage follower using the improved op amp model with $A = 100{,}000$, $R_i = 1$ MΩ, and $R_o = 30$ Ω and with a load resistor of $R_1 = 100$ Ω connected across v_2. Is it within 1% of that computed using the ideal voltage amplifier model?

1.00; yes

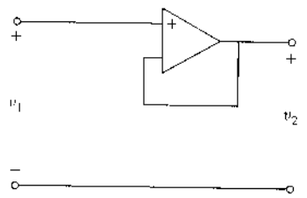
3.4.2. Determine the voltage transfer ratio v_2/v_1. Use the ideal voltage amplifier model.

$2A/(A + 2)$, or in the high open-loop gain limit, $v_2/v_1 = 2$

3.5 OP AMP BUILDING BLOCK CIRCUITS

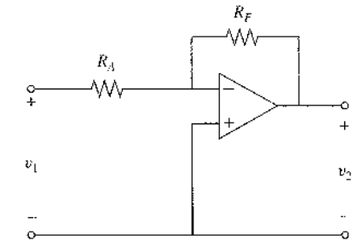
In this section we present several simple op amp circuits that are frequently used as modules or *building blocks* in the design of more complex circuits. Each building block performs one basic operation, such as the amplification of its input or the addition of two or more inputs. By interconnecting these building blocks, more complex results may be built up, in the same way that an arbitrarily complicated mathematical expression can be evaluated by the simple step-by-step application of add, subtract, multiply, and divide operations or a wall built by properly stacking building blocks made of brick or stone.

The first building block circuit is the *voltage follower* shown in Fig. 3.14. For this and the other building blocks to be described in this section, what the circuit "does" is summarized in its voltage transfer equation. *The voltage transfer equation specifies the output voltage in terms of the input voltage (or voltages) when the output port is open circuited.* The voltage-follower circuit was analyzed in Section 3.4, where the voltage transfer equation was found to be simply $v_2 = v_1$. Thus what the voltage follower does is produce an output voltage that equals (follows) its input voltage. This may seem like a rather modest achievement, but as we will see in Section 3.6, voltage followers are of the utmost practical benefit. Note that the input voltage to this circuit, v_1, should not be confused with the input voltage to the op amp, v_{in}. The voltage transfer equation and corresponding voltage transfer ratio (ratio of output voltage to input voltage, v_2/v_1



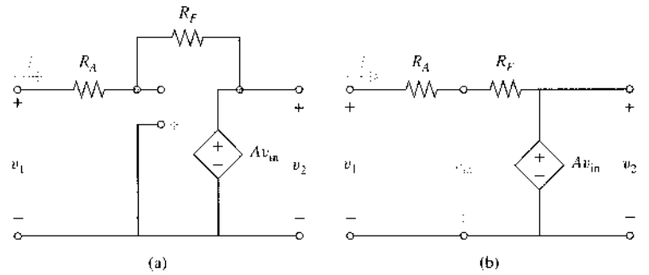
Voltage transfer equation: $v_2 = v_1$

Voltage follower.



Voltage transfer equation: $v_2 = -R_F/R_A v_1$

Inverting amplifier.



Analyzing the inverting amplifier: (a) using the ideal voltage amplifier model; (b) redrawn for clarity.

in Fig. 3.14) are defined in terms of the input to and output from the circuit, not the input to the op amp that is contained in the circuit.

The voltage transfer equation for the voltage follower was derived using the ideal voltage amplifier model. The same result may also be derived using the improved op amp model of Fig. 3.9, as shown in Exercise 3.5.1. For each building block, use of the ideal voltage amplifier model and use of the improved op amp model leads to the same voltage transfer equation in the limit as A goes to infinity (which we use to define the voltage transfer function since we know that A will be very high for practical op amps). Because analysis is a bit more laborious using the improved op amp model with its two additional resistors, but the results do not differ in the high-op-amp-gain limit from those derived using the simpler ideal voltage amplifier model, we will use the latter to study each building block circuit.

The second building block is the *inverting amplifier* of Fig. 3.15. Note that this circuit incorporates negative feedback, as evidenced by the current path from the output back to the inverting input. Indeed, all the building blocks will incorporate negative feedback. Replacing the op amp by its ideal voltage amplifier model results in the circuit of Fig. 3.16(a).

Examining Fig. 3.16(b), we see that

$$v_{in} = -Av_{in} - R_F i$$

Solving for i and using the fact that $v_2 = Av_{in}$ yields

$$i = \frac{-(A+1)}{R_F} v_{in} = \frac{-(A+1)}{R_F} \frac{v_2}{A}$$

Applying KVL around the outer loop gives

$$v_2 = v_1 - (R_A + R_F)i \tag{3.10}$$

Substituting (3.9) into (3.10) to eliminate i yields

$$v_2 = v_1 + \frac{R_A + R_F}{R_F} \frac{A+1}{A} v_2$$

Gathering v_2 terms and solving for v_2 gives the result

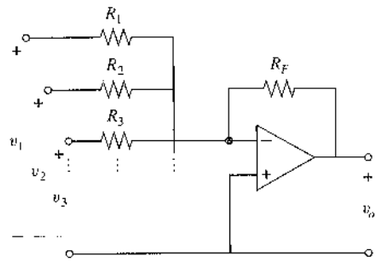
$$v_2 = \frac{-R_F}{R_A + 1/A(R_A + R_F)} v_1$$

In the high-op-amp-gain limit as A becomes arbitrarily large, we have the voltage transfer equation for the inverting amplifier:

$$v_2 = \frac{-R_F}{R_A} v_1$$

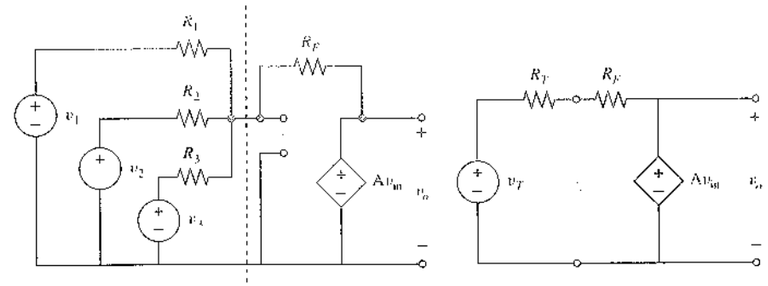
From (3.13) we see that the voltage transfer ratio is always negative for the inverting amplifier, whence its name. This building block circuit functions to create an inverted, scaled copy of the input with the scale (amplification) factor set by the designer's choice of external resistances R_F and R_A. Just as with all other building blocks, the voltage transfer equation shows no dependence on open-loop gain A, a consequence of our insistence on negative feedback.

The next building block circuit is the *inverting summer* of Fig. 3.17. It differs from the inverting amplifier by having more than one input. To analyze this circuit, we first convert it to an equivalent inverting amplifier and then use the results just derived for this circuit. The Thevenin equivalent of the circuit to the left of the dashed line in Fig. 3.18(a) is conveniently evaluated by first replacing each series combination of voltage source and resistor by its Norton equivalent and then adding current sources and combining parallel resistors. The result is shown in Fig. 3.18(b).



Voltage transfer equation: $v_o = \frac{-R_F}{R_1} v_1 + \frac{-R_F}{R_2} v_2 + \frac{-R_F}{R_3} v_3 + \cdots$ (3.14)

Inverting summer.



Analyzing the inverting summer: (a) using ideal voltage amplifier model; (b) after Thevenin transformation.

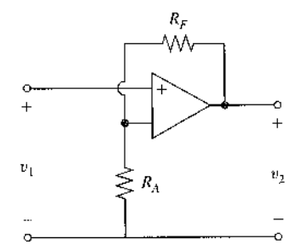
In Fig. 3.18(b),

$$R_T = R_1 \| R_2 \| R_3 = \frac{1}{1/R_1 + 1/R_2 + 1/R_3}$$

$$v_T = R_T \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} \right)$$

To fix ideas, exactly three inputs have been shown in Fig. 3.18(a) and used in the calculations above. Figure 3.18(b) contains a single-input inverting amplifier with input voltage v_T and voltage transfer ratio $-R_F/R_T$. Applying voltage transfer equation (3.13) for this circuit with input $v_1 = v_T$ and input resistance $R_A = R_T$ yields the voltage transfer equation for the three-input case, or in general (by adding the ellipsis, "...") the voltage transfer equation (3.14). The inverting summer output is the sum of its inverted and scaled inputs.

If it is desired to amplify without sign change, the *noninverting amplifier* shown in Fig. 3.19 may be used. Replacing the op amp with its ideal voltage amplifier model results in Fig. 3.20. Examining the output circuit, R_F and R_A are in series across v_2, or



Voltage transfer equation: $v_2 = \left(1 + \frac{R_F}{R_A}\right) v_1$ (3.15)

Noninverting amplifier.

by voltage division

$$v_{R_A} = \frac{R_A}{R_A + R_F} v_2$$

KVL around the left loop yields

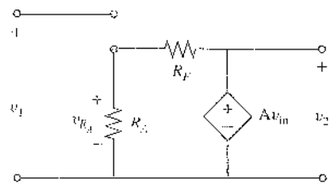
$$v_1 = v_{\text{in}} + v_{R_A}$$

Substituting v_2/A for v_{in} and (3.18) for v_{R1} gives

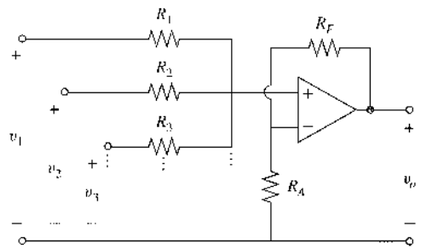
$$v_1 = \frac{v_2}{A} + \frac{R_A}{R_A + R_F} v_2$$

The first term goes to zero in the high-op-amp-gain limit, leaving the voltage transfer equation (3.17). For instance, $R_F = 10$ kΩ, $R_A = 5$ kΩ results in a voltage transfer ratio of +3. Note that whereas the inverting amplifier has no restriction on voltage transfer ratio magnitude, the noninverting voltage transfer ratio must be greater that unity (for passive resistors with nonnegative resistance). To achieve values less than unity, a voltage should first be divided (see Exercise 3.5.2) before being input to a noninverting amplifier.

The *noninverting summer* of Fig. 3.21 may be analyzed exactly as the previous (inverting) summer. First convert to an equivalent single-input circuit by Thevenin



Substituting the ideal voltage amplifier model.



Voltage transfer equation: $v_o = \left(1 + \frac{R_F}{R_A}\right)\left(\frac{R_T}{R_1}v_1 + \frac{R_T}{R_2}v_2 + \frac{R_T}{R_3}v_3 + \cdots\right)$

where

$$R_T = R_1 \| R_2 \| R_3 \| \cdots$$

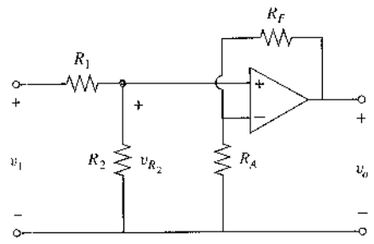
Noninverting summer.

transformation and then apply the voltage transfer equation for the single-input case already determined. By (3.17), the voltage transfer equation for the noninverting summer is

$$v_o = \left(1 + \frac{R_F}{R_A}\right) v_T$$

with v_T given by (3.16). While (3.16) is specific for three inputs, the ellipsis $\cdots$ in (3.20) once again indicates the immediate generalization to any number of inputs v_1, v_2, ... This building block has the effect of scaling and summing its inputs (without sign inversion).

These five circuits constitute the bulk of the building blocks we will use. One additional one will be introduced later. It is somewhat surprising, given the simplicity of these circuits, that they are quite sufficient to build up many circuits of considerable practical significance. We need only consider how they may be interconnected, as in Section 3.6, and allow inductors and capacitors in the building blocks to be ready to understand and to design many interesting circuits such as filters and analog computers.



3.5.1. Using the improved op amp model (Fig. 3.9) with $R_i = 1$ MΩ and $R_o = 30$ Ω, first determine the output v_2 of the voltage follower (Fig. 3.14) in terms of op amp gain A and then in the high-op-amp-gain limit. Does the voltage transfer equation agree with that calculated in (3.7)?

$\frac{A + 0.00003}{A + 1.00003} v_1$; v_1; yes

3.5.2. This exercise shows how to get a positive voltage transfer ratio of less than 1. Using the ideal op amp model, determine the voltage v_{R2}. Then determine v_o, noting that v_{R2} is the input to the noninverting amplifier. Specify values for the resistances yielding an overall voltage transfer ratio of $\frac{1}{3}$ (answer not unique).

$\frac{R_2}{R_1 + R_2} v_1$; $\frac{R_2}{R_1 + R_2}\left(1 + \frac{R_F}{R_A}\right) v_1$; $R_F = R_A$; $R_1 = 5R_2$

3.5.3. Design a noninverting summer whose voltage transfer equation is $v_o = 2v_1 + 5v_2$.

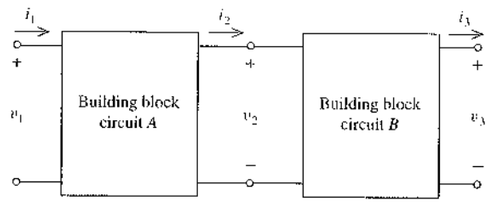
In Fig. 3.21, pick any R_A and R_2; then set $R_F = 6R_A$ and $R_1 = 2.5R_2$.

3.6 INTERCONNECTING OP AMP BUILDING BLOCKS

Op amp building blocks are simple modules that may be connected to produce more complex circuit behaviors. In this section we consider one particularly useful way to interconnect these modules and how to predict the behavior of the interconnected circuit from that of the modules that are its parts.

A *port* in a circuit is a pair of wires to which another subcircuit may be attached. Op amp building block circuits each have one output port and at least one input port. Circuits containing a single input and a single output port, for example the voltage follower and inverting and noninverting amplifiers of Section 3.5, are called *two-port circuits*. Those with additional ports, such as the inverting and noninverting summers, are *n-port circuits*.

The most direct way to connect a pair of two-port circuits is to tie the output port of one to the input port of the other. This is called the *cascade interconnection* of two-ports. Cascading two (or more) two-ports results in an interconnected circuit that is also a two-port, as shown in Fig. 3.22. Here v_1 is the input voltage to the overall cascaded circuit and v_3 the output voltage.



Cascade interconnection of two op amp building block units.

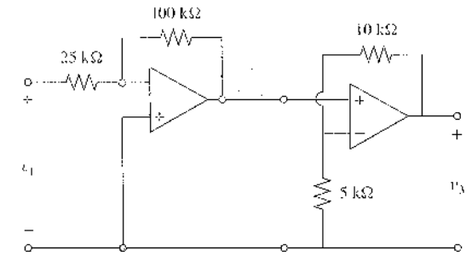
Every op amp building block is characterized by its voltage transfer equation, which specifies the output voltage as a function of the input voltage or voltages under the condition that the output port is open circuited. When op amp building blocks are interconnected, it is the voltage transfer function across the entire interconnected circuit that describes its behavior. Thus for two-port op amp building blocks in cascade, we will determine the overall voltage transfer equation.

Referring to Fig. 3.22, take the output port (v_3) to be open circuited as required by definition of the voltage transfer equation. Applying the voltage transfer equation for building block B, $v_3 = K_B v_2$, where K_B is the voltage transfer ratio. Then, to find the voltage transfer equation $v_3 = K_C v_1$ for the overall cascade, we need only express v_2 in terms of v_1. If $i_2 = 0$, the output of building block A is also open circuited, and v_2 is found by applying A's voltage transfer equation $v_2 = K_A v_1$. The overall result is then $v_3 = K_A K_B v_1$; that is, $K_C = K_A K_B$.

Example 3.4

Find the voltage transfer equation (input v_1, output v_3) for the circuit of Fig. 3.23. Use the ideal voltage amplifier model for the op amps.

The rightmost building block circuit is a noninverting amplifier with $v_3 = (1 + R_F/R_A)v_2$ or $v_3 = 3v_2$. $i_2 = 0$ since the the ideal voltage amplifier model for its op amp has an open-circuited input. The leftmost building block is an inverting amplifier with voltage transfer ratio $-R_F/R_A = -4$. Since its output port is open circuited ($i_2 = 0$), v_2 may be exactly evaluated using the voltage transfer equation for this building block, $v_2 = -4v_1$. Then, combining, $v_3 = 3(-4v_1) = -12v_1$. The voltage transfer ratio for the cascade is the product of the individual voltage transfer ratios.



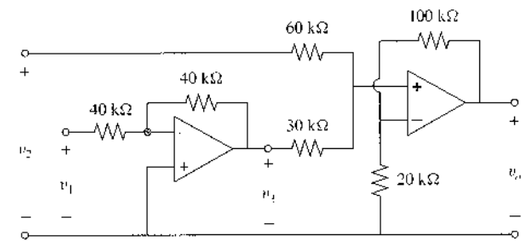
Circuit for Example 3.4.

Suppose that i_2 is not equal to zero in Fig. 3.22. The ideal voltage amplifier model of the op amp predicts that the output voltage of each building block must be independent of i_2 since it is just the voltage across a VCVS (see Fig. 3.7), whose source function does not depend on the output current i_2. So even though open-circuit output is assumed in its definition, the voltage transfer equation continues to apply in computing the output voltage for a building block circuit even when the output is not open circuited. This analysis using the ideal voltage amplifier model suggests that connecting a load, such as another op amp building block, to the output port of a previous building block will not affect its output voltage.

Thus *for building blocks in cascade, the overall voltage transfer ratio is simply the product of the individual voltage transfer ratios*, provided only that the ideal voltage amplifier model of the op amps may be accurately used. The same principle applies to n-port op amp building blocks as well, as illustrated in the next example.

Example 3.5

Find the voltage transfer equation for the circuit of Fig. 3.24. Use the ideal voltage amplifier model for the op amps.



Circuit for Example 3.5.

The rightmost building block is a two-input noninverting summer with input voltages v_2 and v_3 and output v_o. Using the voltage transfer equation (3.17) for this building block, we have

$$v_o = \left(1 + \frac{100\text{K}}{20\text{K}}\right)(60\text{K}\|30\text{K})\left(\frac{v_2}{60\text{K}} + \frac{v_3}{30\text{K}}\right)$$

where K stands for multiplication by 10^3 and || is the parallel equivalent operator.

$$60\text{K}||30\text{K} = \frac{1}{1/60\text{K} + 1/30\text{K}} = 20\text{K}$$

Then

$$v_a = 2v_2 + 4v_3$$

The remaining building block is an inverting amplifier with

$$v_3 = \frac{-40\text{K}}{40\text{K}} v_1 = -v_1$$

Combining these two results, the overall voltage transfer equation is

$$v_o = 2v_2 + 4(-v_1) = 2v_2 - 4v_1$$

By combining building blocks of different types, we may achieve results beyond what would be possible using any one alone. This example shows how a *difference*, as opposed to an inverting or noninverting sum, of voltages may be generated by combining inverting and noninverting building blocks.

The main result of this section, that when op amp building blocks are cascaded the overall voltage transfer equation may be found simply by combining the voltage transfer equations of each individual building block, was justified using the ideal voltage amplifier model for each op amp. The result followed from the lack of dependence of the output voltage on the output current. The improved op amp model of Fig. 3.9 contains an additional resistor R_o in its output loop. The current produced by its VCVS, which supplies both the output current (i_2 in Fig. 3.22) and the current to the feedback path present in every op amp building block, must all pass through R_o. By KVL, the improved op amp model thus predicts that the output voltage will be reduced as the output current increases (their sum equals Av_{in}, which does not depend on this current). The two models seem to disagree as to whether the output voltage will remain constant, as the ideal voltage amplifier model predicts, or will in fact decrease as the output current increases as the improved model suggests.

Which model gives the right prediction? As might be anticipated in calling it an "improved" op amp model, direct measurement verifies that as the output current of any building block circuit increases, its output voltage declines as predicted by the improved model. The decline of load (output) voltage with load current is called *loading* and is quite a general phenomenon, not restricted to op amp building blocks. As explored in Exercise 3.6.1, loading is the consequence of voltage division between internal resistance and load resistance. Some degree of loading must occur whenever a source has nonzero Thevenin equivalent resistance and is connected across a finite load resistance.

Must we then abandon our main results, derived using the ideal voltage amplifier model, which has no internal resistance and hence manifests no loading? For sufficiently low levels of output current the degree of loading, that is, reduction of output voltage, is negligible and our result holds with excellent accuracy. Current levels may be kept low by keeping the resistances in the building block circuits high, which is why values of 5 kΩ and above are used in the examples and problems.

In summary, to a good approximation a cascade of op amp building blocks has a voltage transfer equation that may be computed simply by combining the individual voltage transfer equations. *For two-ports in cascade, the overall voltage transfer ratio is the product of the constituent two-port voltage transfer ratios.* Using this combining rule, circuits involving multiple building blocks may be designed to accomplish relatively complex goals and yet be analyzed easily by multiplying voltage transfer ratios. The range and versatility of the building block approach to op amp circuit design, suggested by the results of this section, will be fully realized with the introduction of capacitors and inductors into the building blocks in Chapter 7.

3.6.1. Consider a source with Thevenin equivalent resistance R_T and Thevenin equivalent voltage v_T. If a load of R_L ohms is placed across the source terminals as in part (a) of the figure, (a) what is the open-circuit load voltage (voltage across R_L when R_L is infinite)? (b) Find the value for resistance R_L at which loading has reduced v_L to half of its maximum (open circuit) value. (c) If a voltage follower is inserted between source and load as shown in part (b) of the figure, show that loading is eliminated. *Buffering* of a source from a load to eliminate loading is one of the main uses of a voltage follower.

Answer: v_T, R_T, V_L (no buffer) $= (R_L + V_T)/(R_L + R_T)$, but V_L (with buffer) $= V_T$. Loading reduces the output voltage by a factor equal to the voltage-divider ratio.

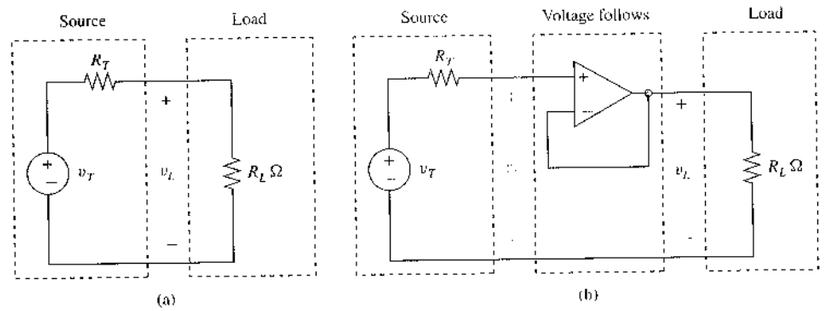
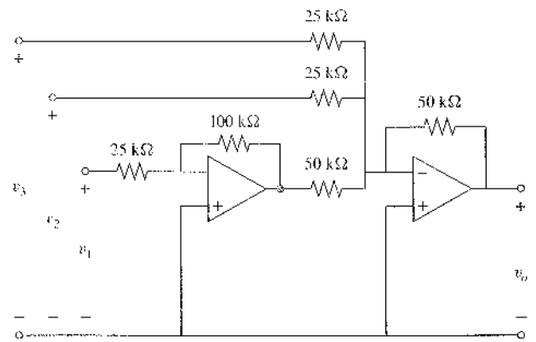


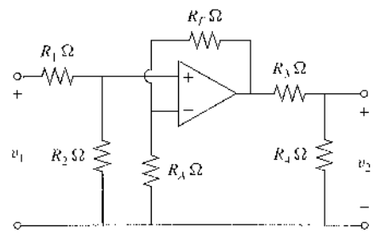
Figure 3.23 Loading with and without a buffer between source and load: (a) direct connection; (b) buffered connection using a voltage follower.

3.6.2. Design a circuit with the voltage transfer equation $v_o = 8v_1 - 2v_2 - v_3$. Use only inverting building blocks and select all resistances to be in the range 5 to 500 kΩ.

Answer: Express v_o as $v_o = -2(-4v_1) - 2v_2 - v_3$. Get the term in parentheses as the output of an inverting amplifier with input v_1 and voltage transfer ratio −4. Then route this voltage, along with the voltages

v_2 and v_3, to an inverting summer with appropriate voltage transfer ratios (-2, -2, and -1).





3.6.3. Determine the voltage transfer ratio v_2/v_1.

$$\frac{R_2}{R_1 + R_2}\left(1 + \frac{R_f}{R_A}\right)\frac{R_4}{R_3 + R_4}$$

3.7 PRACTICAL OP AMPS

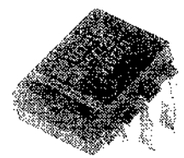
Op amps are not merely schematic diagrams to which the laws of circuit analysis may be applied but physical devices with size and weight requiring power supplies and attention to other practical considerations. Here the issues of physical packaging and limits of performance and environment are introduced very briefly. These topics are taken up at a more satisfying level of detail in courses in electronics circuits, which typically follow a first course in circuit analysis.

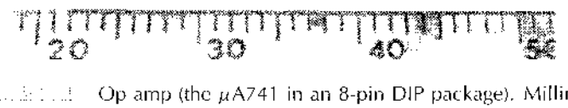
Physical Packaging

The first few generations of commercial op amps, which made their appearance more than 50 years ago, were large, heavy devices fabricated from vacuum tubes and discrete *RLC* components. Size, cost, and appetite for power limited their use to the most critical applications, particularly the performance of key mathematical operations in predigital computer fire-control units for air defense batteries (hence the name *operational amplifier*). In 1962, Fairchild Electronics introduced the μA709, the first monolithic op amp, miniaturized onto an eggshell-thin chip of silicon just a fraction of an inch square. The sharp reductions in size and cost created by this use of integrated-circuit technology opened the door to the extensive use of op amps in a broad range of industrial and consumer electronics. Successive generations of op amps have taken advantage of advances

in microelectronics, resulting in steadily improved integration, performance, and reliability, all at prices less than that of a postage stamp. Today's op amp is truly a wonder of invention and refinement and is one of the prime workhorses of modern electronic design.

Op amps are currently available in a variety of convenient packages and ratings. The popular general-purpose μA741 op amp in an 8-pin dual-in-line (DIP) package is shown in Fig. 3.25. Pins 4 and 7 are assigned to be connected to the power supply. Without the proper dc bias currents supplied by the power supply at the rated voltage levels, the transistors contained in the op amp circuit will not act to produce the VCVS with high-gain behavior that characterizes all functional op amps.





Op amp (the μA741 in an 8-pin DIP package). Millimeter scale.

Besides their packaging as separate chips, op amps are frequently deployed as modules contained in other integrated-circuit chips. Voltage followers are commonly used as buffers at the input and output pins of integrated circuits to prevent loading. Op amps form essential parts of analog-to-digital (AD) and digital-to-analog (DA) converters, instrumentation amplifiers, active filters, and numerous other chip-level electronic modules. It is not uncommon for dozens of op amp modules to be designed into a single very large scale integrated (VLSI) circuit chip.

Performance and Environmental Limits

The op amp was defined in Section 3.3 as an electronic device that under certain circumstances behaves like a VCVS with high gain. The circumstances that need to be established include proper environmental conditions and operation within a certain "performance envelope" or permissible range of the circuit variables. With careful attention to these constraints, the op amp, despite all its physical complexity, will indeed behave very much as predicted by its simple ideal voltage amplifier model.

Environmental conditions that need to be maintained include use of a power supply developing the right voltages and free from spikes and surges, adequate shielding from

other electronic and magnetic signals, and operation in the proper temperature range. Microminiaturization, with all its benefits, puts many elements on a very small piece of real estate and may result in overheating and circuit failure. Adequate cooling of densely populated circuit boards is an important consideration in designing modern electronic systems. For very dense circuits, such as the central processing unit of supercomputers, elaborate refrigeration systems are required to carry off the heat produced.

Performance limitations include voltage, current, power, and frequency. All voltages and currents must be kept within the linear regime of operation. Typically, this means that voltages cannot exceed the power supply values; and currents must be strictly limited by rated values. If called upon to produce too much output voltage, the op amp will *clip* near its power supply values; that is, the output will be limited in amplitude by the positive and negative power supply values. If too much power is demanded, the circuit will fail to conform to its predicted linear behavior or, more ominously, may overheat and ultimately burn out. Indeed, current-limiting circuits protecting against this are built into most popular op amp designs. Op amps are also limited in the frequency band over which they may act. While more broadband specialty op amps are available, most op amp circuits are designed to operate over frequencies well below 1 MHz. The frequency dependence of op amps will be taken up in Chapter 14.

The experienced op amp circuit designer will take other practical factors into consideration as well: shock and vibration, offset currents and voltages, and slew rate limits (maximum rate of change in volts/second, for example), and it is not surprising that an electronic device such as an op amp can be made to perform best in the hands of a skilled circuit designer. What is perhaps more surprising is that well-behaved op amp circuits can be made rather easily by those with less than expert status. Using today's highly integrated, miniaturized op amps, a few inexpensive resistors and batteries are all that is really required to construct an op amp circuit that behaves with a high degree of accuracy as the ideal voltage amplifier model of the op amp predicts.

3.8 DESIGN OF SIMPLE OP AMP CIRCUITS

In Chapter 2 we considered strictly resistive circuits, those containing only resistors and independent sources. In the design section of that chapter it was noted that the versatility of resistive circuits, their range of useful applications, is limited. The addition of just one more element, the op amp, greatly expands the range of potential applications, as we shall see below. The examples presented here point to the design of electrocardiogram amplifiers for biomedical application, analog-to-digital converters in computer engineering, and an averaging circuit for signal processing applications. Many more examples could be offered as well, but it is hoped that these are persuasive of the range and versatility of the op amp in practical applications.

The design examples here illustrate uses of op amp circuits constructed using the straightforward building block cascade approach described in Sections 3.5 and 3.6. Other interconnections of op amp stages will be considered in later chapters, and elements other than op amps and resistors, which will further enhance the variety of electronic circuits that can be designed.

The electrical potential difference $v_a(t) - v_b(t)$ measured between two electrodes secured to points a and b on the body's wall contains information on the state of health of the heart. Each time the heart beats, its mechanical contraction is triggered by electrical currents flowing through the heart. The strength and sequence of these crucial cardiac currents are encoded in the electrocardiogram (ECG) signal $v_o(t)$. We wish to design an ECG amplifier that will amplify this signal from its body surface value of approximately 2 mV peak to peak to the level of 1 V peak to peak needed to drive a stripchart recorder. The accompanying common-mode signal $v_{cm}(t) = \frac{1}{2}[v_a(t) + v_b(t)]$, which contains only noise and irrelevant signals from other body sources, should be small or nonexistent in the output.

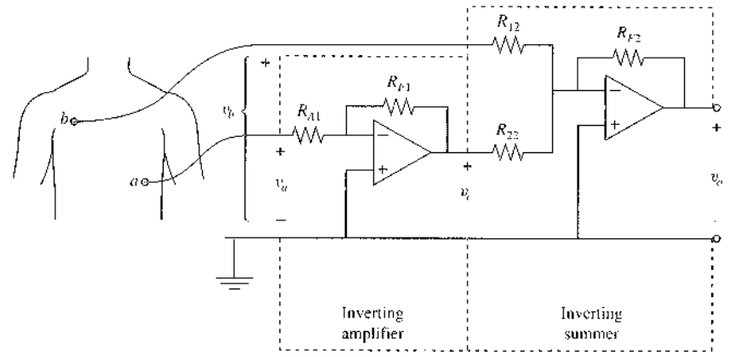
We begin our design by noting that the desired overall voltage transfer equation for our circuit is $v_o = 500(v_a - v_b)$. A circuit realizing this voltage transfer equation will scale a 2-mV difference signal to 1 V, as required. Moreover, the addition of a constant to both v_a and v_b does not change the output of this circuit, which is sensitive only to their potential difference. Thus the common-mode signal $v_{cm}(t)$ is effectively rejected by a circuit with this difference-voltage transfer equation.

We will use a two-stage design incorporating an inverting amplifier and an inverting summer as shown in Fig. 3.26. First an inverting amplifier will create

$$v_c = \mu_1 v_a$$

where $\mu_1 = R_{F1}/R_{A1}$ is the voltage gain for this stage. Then this output, along with v_a, will be input to an inverting summer whose voltage transfer equation will be

$$v_o = -\mu_{2c} v_c - \mu_{2b} v_b$$



ECG amplifier.

where $\mu_{2b} = R_{F2}/R_{12}$ and $\mu_{2c} = R_{F2}/R_{22}$. Combining these two equations gives

$$v_o = \mu_1\mu_{2c}v_a - \mu_{2b}v_b$$

Thus if we can set $\mu_1\mu_{2c} = \mu_{2b} = 500$, we will have our design. First setting

$$\mu_{2b} = \frac{R_{F2}}{R_{12}} = 500$$

we select $R_{12} = 1$ kΩ and $R_{F2} = 500$ kΩ. Now we may divide the desired gain $\mu_1\mu_{2c} = 500$ between the two stages any way we wish. It is helpful to balance the gains so we set $\mu_1 = \mu_{2c} = \sqrt{500}$. Then since

$$\mu_{2c} = \sqrt{500} = \frac{R_{F2}}{R_{22}} = \frac{5 \times 10^5}{R_{22}}$$

we have that $R_{22} = 22.4$ kΩ. Finally, $\mu_1 = R_{F1}/R_{A1} = \sqrt{500}$, so selecting $R_{F1} = 100$ kΩ, we have fixed the value of the final resistance:

$$R_{A1} = \frac{R_{F1}}{\sqrt{500}} = 4.47 \text{ kΩ}$$

which completes the design.

As with almost all designs, this solution is not unique. Indeed, more attractive solutions can be found, solutions that use only a single op amp, have less sensitivity to parameters, are easier to adjust, or possess a broader range of operating frequencies. But the circuit shown in Fig. 3.26 is quite capable of satisfying the simple ECG amplifier design specifications as stated.

Our goal is to design a digital-to-analog converter (DAC). Assume that we have an 8-bit digital input signal with transistor-to-transistor (TTL) logic levels +5 V and 0 V, and that the full-scale analog output should range from −10 V when the digital input is all zeros (0 V at each input pin, digital value 00000000_2) to 0 V for the highest input value (+5 V at each input pin, digital value 11111111_2).

Since each pin represents a power of 2, the formula by which a digital value is converted to its analog value will be

$$v_a = S\left(\sum_{k=0}^{7} 2^k v_k\right) + v_B \qquad (3.24)$$

where v_B is a bias voltage that sets the zero output value and S is a scale factor. In our case we wish $v_0 = v_1 = \cdots = v_7 = 0$ V to map into $v_a = -10$ V, or

$$-10 = v_B$$

so $v_B = -10$ V. We also want $v_0 = v_1 = \cdots = v_7 = +5$ V to map into $v_a = +10$ V, or

$$0 = S(255)(5) + v_B$$

which yields $S = 2/255$. Thus our voltage transfer equation is, by (3.24),

$$v_a = \left(\sum_{k=0}^{7} \frac{2^{k+1}}{255} v_k\right) - 10 \qquad (3.25)$$

This can be realized with a nine-input noninverting summer as shown in Fig. 3.27. Examining the form of the voltage transfer equation for this building block which is given in Fig. 3.21, in order that the gains on the eight successive single-bit inputs $v_7, v_6, \ldots, v_0$ each be in the ratio of 2, we must have $R_6 = 2R_7$, $R_5 = 2R_6 = 4R_7$, and so on. Then with $G_T = 1/R_T$,

$$\begin{aligned} G_T &= G_7 + G_6 + \cdots + G_0 + G_B \\ &= G_7(1 + \tfrac{1}{2} + \tfrac{1}{4} + \cdots + \tfrac{1}{128}) + G_B \\ &= \frac{255G_7}{128} + G_B \end{aligned}$$

Now to keep the numbers simple let us select $G_B = 129/128\ G_7$ so that

$$G_T = 3G_7$$

Then $R_T = \frac{1}{3}R_7$, and with the resistive ratios already defined we have, by the voltage transfer equation in Fig. 3.21,

$$v_a = \left(1 + \frac{R_F}{R_A}\right)\left(\frac{1}{3}\right)\left(\frac{v_0}{128} + \frac{v_1}{64} + \cdots + \frac{v_7}{1} + \frac{129v_B}{128}\right) \qquad (3.26)$$

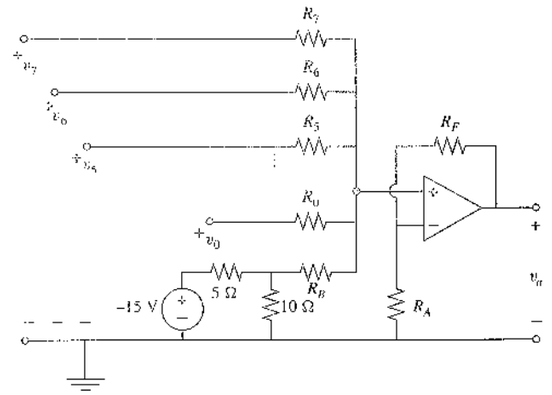
Comparing (3.26) and (3.25), the v_0 terms will be identical if

$$\left(1 + \frac{R_F}{R_A}\right)\left(\frac{1}{3}\right)\left(\frac{1}{128}\right) = \frac{2}{255}$$

Solving the last for R_F yields

$$R_F = 2.01R_A$$

Thus in Fig. 3.27 we set $R_A = 10$ kΩ, $R_F = 20.1$ kΩ. Since the v_0 terms in (3.25) and (3.26) are now identical and the ratios of these terms to the $v_1, \ldots, v_7$ terms in the two equations are identical, all eight of these terms must agree in the two equations. In Fig. 3.27 we show the bias voltage being produced by voltage division of a −15-V source (this is a common power supply voltage). Finally, each of the conductances $G_0, G_1, \ldots, G_6$, and G_B have been determined only as ratios compared to G_7. We may choose any convenient value for G_7 (or R_7) and the rest follow. If we select $R_7 = 1$ kΩ, then $R_6 = 2$ kΩ, $R_5 = 4$ kΩ, ..., $R_0 = 128$ kΩ, and $R_B = 128/129\,R_7 = 992$ Ω.



Eight-bit DAC.

The resulting design will convert TTL digital inputs to analog outputs in the range ±10 V. DACs are used wherever computers must be interfaced to nondigital circuitry, for instance in computer displays such as cathode ray tubes and computer-controlled electromechanical devices such as industrial robots.

For our final design example, we wish a circuit that will produce an output voltage which is the average of the outputs of several input voltages. Signal averagers are widely used in signal processing circuits; for instance, at the outputs of noisy transducers averaging is used to minimize the effects of random noise on repeated samples of the same input signal.

For N inputs $v_1, v_2, \ldots, v_N$, the average is

$$v_0 = \frac{1}{N}(v_1 + v_2 + \cdots + v_N)$$

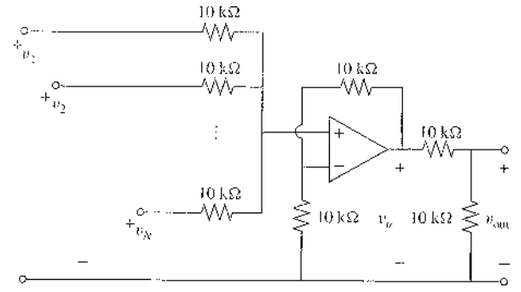
This is the voltage transfer equation for a noninverting summer, each of whose gains is $1/N$. Referring to the circuit diagram for this building block shown in Fig. 3.21, in order for the gains to be all equal we require that $R_1 = R_2 = \cdots = R_N = R$, so

$$R_T = R_1 || R_2 || \cdots || R_N = \frac{R}{N}$$

Then from that figure,

$$v_o = \left(1 + \frac{R_F}{R_A}\right)\frac{1}{N}(v_1 + v_2 + \cdots + v_N)$$

Note that for any finite values for R_F and R_A, the gains on each input signal will be too large, each by the same factor. An easy way to remedy this problem is to set R_F and R_A equal, so that



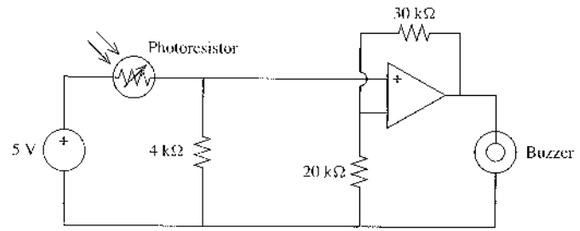
Signal averager circuit.

$R_F/R_A = 1$. Then v_o is too large by a factor of 2, so we voltage divide the output v_o by 2. The resulting circuit with all resistors set to a common value of 10 kΩ is shown in Fig. 3.28.

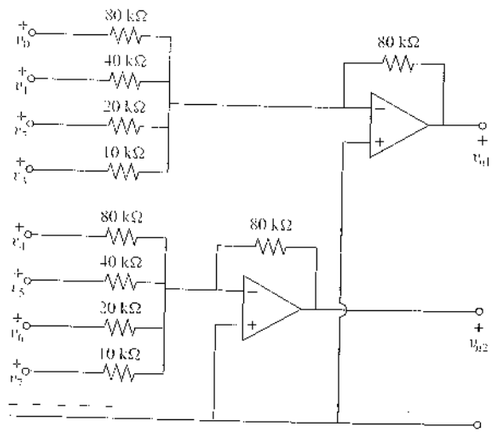
One limitation of this solution is that the output impedance of this circuit is larger than most of our op amp circuits due to the output voltage divider. If we are using the output of this circuit to drive a low-input-impedance circuit, such as a stripchart recorder, it would probably be necessary to buffer the connection with, say, a voltage follower. There are many other signal averaging circuits with features beyond those offered by our solution, but Fig. 3.28 does the job of signal averaging simply and accurately.

3.8.1. A certain photoresistor has resistance that is greater than 1 kΩ in the dark, and drops below 1 kΩ in the presence of moderate light. Design a light alarm using this photoresistor as the sensor. As the indicator of an alarm condition, use a buzzer that sounds when the voltage across it exceeds 10 V dc.

One of many possible solutions



3.8.2. Produce a DAC circuit with two outputs, each the analog-converted value of one "nibble," the top or bottom 4 bits of the 8-bit input. Assume TTL values and scale each of the two analog outputs to the inverted single-ended range 0 V (all 4 bits 0 V) to −75 V (all 4 bits 15 V).



v_o is the least significant bit, v is the most significant bit of the input byte.

SUMMARY

Dependent sources, those whose source functions depend on the value of a current or voltage elsewhere in the circuit, are used to model the behavior of active devices such as transistors and op amps. An op amp is a device which, under proper conditions, behaves like a voltage-controlled voltage source with very high gain. Op amps are among the most versatile of active devices and will be treated as a basic circuit element for the remainder of this book.

- Circuits containing dependent sources may be analyzed by treating the sources as if they were independent, then substituting the controlling variable into the dependent source function.
- The ideal voltage amplifier model for the op amp consists in an open-circuit input and voltage-controlled voltage source output.
- The improved op amp model replaces the open-circuit input with a high but finite (e.g., 1 MΩ) input resistance, and adds a low (e.g., 30 Ω) resistance in series in the output circuit. This model is more accurate but contains more elements.
- An op amp is in a negative feedback configuration when an incremental increase in its output voltage leads to a decrease in its input voltage.

- Negative feedback is necessary for useful op amp behavior in all linear circuit applications, such as the op amp building block circuits: inverting and noninverting amplifiers and summers, and the voltage follower.
- The voltage transfer equation is a linear equation relating the input voltage or voltages of an op amp circuit to its output voltage. The output-to-input voltage ratio is called the voltage transfer ratio.
- The voltage transfer ratio of two op amp building blocks connected in cascade is the product of their individual voltage transfer ratios.

The circuit diagrams for physical op amps are complicated, containing many transistors and other elements. But when properly configured, op amps can be modeled very simply and are an essential element in modern circuit design.

3.1. Draw the circuit symbols for two distinct controlled sources, each of which is equivalent to a 1-kΩ resistor.

3.2. Give examples of an active controlled source and a passive controlled source (see Problem 3.1).

3.3. For what value of transconductance of the dependent source in Problem 3.8 does a mathematical inconsistency arise?

3.4. Find v_1 and the power delivered to the 10-Ω resistor.

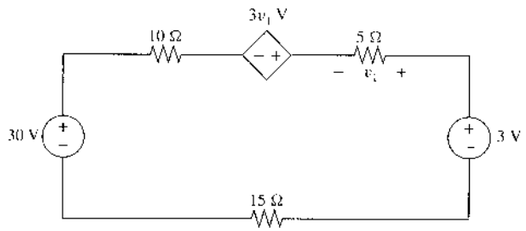


FIGURE P3.4

3.5. Find i_1.

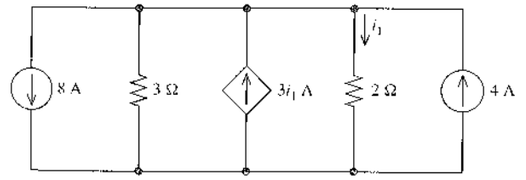


FIGURE P3.5

3.6. Find i.

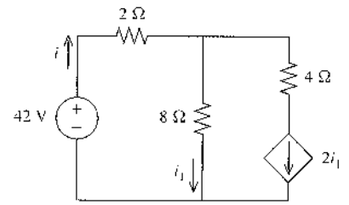


FIGURE P3.6

3.7. Find v_1.

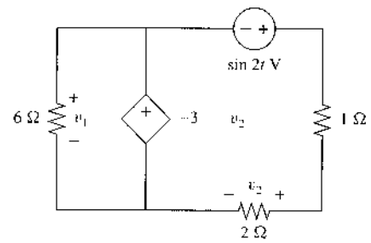
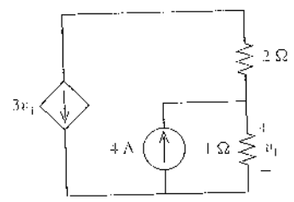
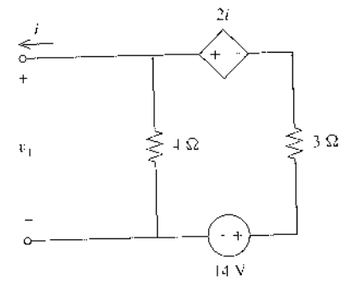


FIGURE P3.7

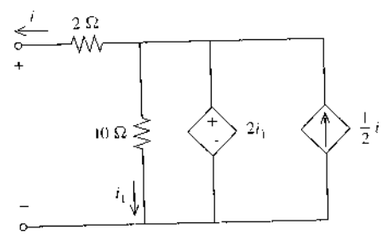
3.8. Which supplies more power, the independent source or the dependent source?



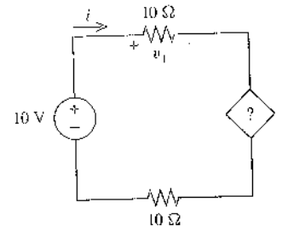
3.9. Find the Thevenin equivalent circuit.



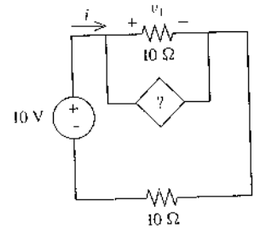
3.10. What single resistor is this circuit equivalent to?



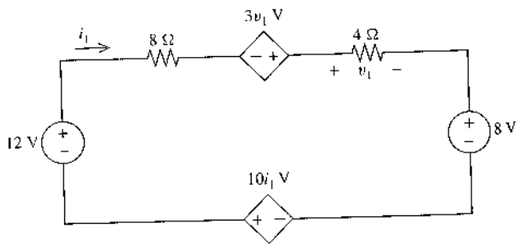
3.11. We wish to make $i = 1$ A. Find two different kinds of dependent sources that can do this, specifying their source functions.



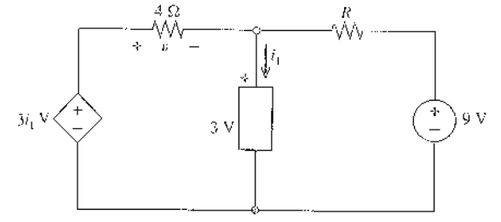
3.12. We wish to make $i = -1$ A. Find two different kinds of dependent sources that can do this, specifying their source functions.



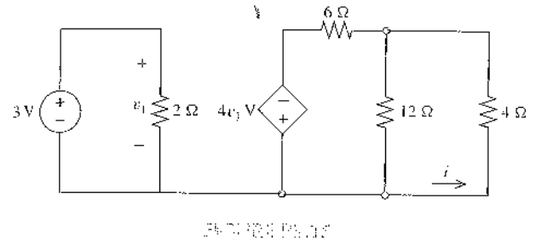
3.13. Find v_1 and i_1.



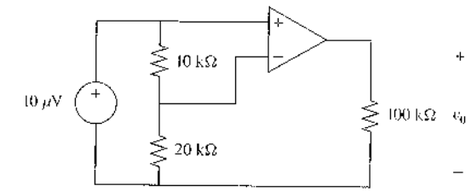
3.14. Find i_1 and v if (a) $R = 4\ \Omega$ and (b) $R = 12\ \Omega$.



3.15. Find i.



3.16. Find v_o. Use the ideal op amp model with $A = 100{,}000$.



3.17. Repeat Problem 3.16, but use the improved op amp model with parameters $A = 100{,}000$, $R_i = 1$ MΩ, and $R_o = 30\ \Omega$.

3.18. Show that the improved op amp model of Fig. 3.9 can be redrawn using a CCCS rather than a VCVS.

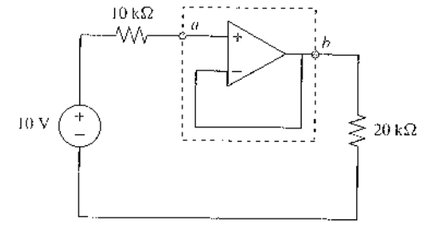
3.19. Using an inverting amplifier, design a circuit with $v_2 = -9v_1$. Keep resistances in the range 5 to 500 kΩ.

3.20. Using an inverting amplifier, design a circuit with $v_2 = -\frac{1}{9}v_1$. Keep resistances in the range 5 to 500 kΩ.

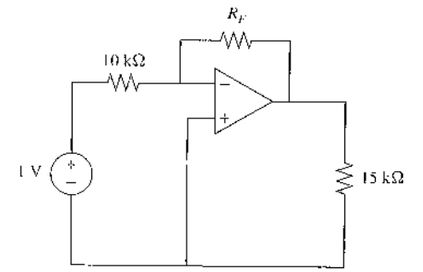
3.21. Using a noninverting amplifier, design a circuit with $v_2 = 7v_1$. Keep all resistors in the range 5 to 500 kΩ.

3.22. If all the external resistors in any one of the building blocks were scaled by the same constant, what effect would this have on the voltage transfer function? On the currents through the external resistor?

3.23. Compare the power delivered to the 20-kΩ resistor in this buffered circuit with that if the voltage follower were removed (a and b shorted together).

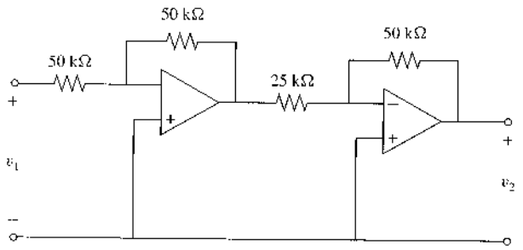


3.24. If we wish to supply $\frac{1}{2}$ mW of power to the 15-kΩ load, $R_F = ?$

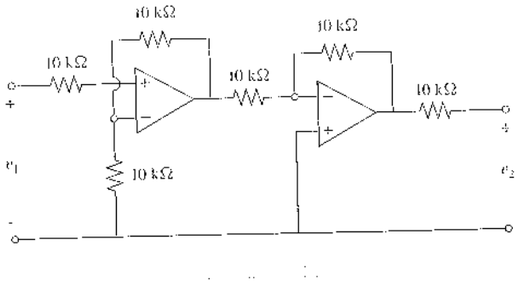


3.25. Using a voltage divider followed by a noninverting amplifier, design a circuit with $v_2 = +\frac{1}{7}v_1$. Can we reverse the order of these two subcircuits and get the same result?

3.26. Find k in the voltage transfer function $v_2 = kv_1$.



3.27. Find k in the voltage transfer function $v_2 = kv_1$.

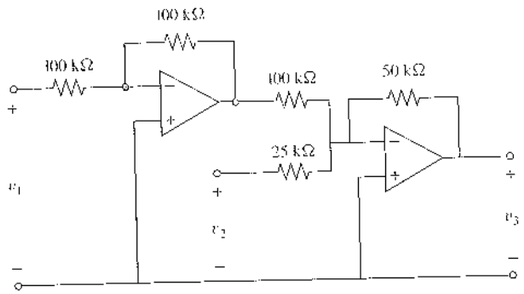


3.28. Show how you can create a circuit with $v_2 = 1000v_1$ by cascading three *identical* amplifier stages. Keep all R's in the range 5 to 500 kΩ.

3.29. Design a circuit with $v_3 = -2v_1 - 5v_2$. Keep all resistors in the range 5 to 500 kΩ.

3.30. Design a circuit with $v_5 = -v_1 - 2v_2 - 3v_3 + 4v_4$. Keep all resistors in the range 5 to 500 kΩ.

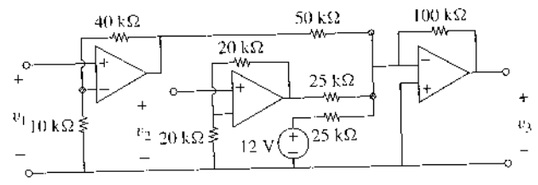
3.31. Find the voltage transfer function $v_3 = k_1v_1 + k_2v_2$.



3.32. Design a circuit with $v_3 = 3v_1 + 2v_2$ using one noninverting summer.

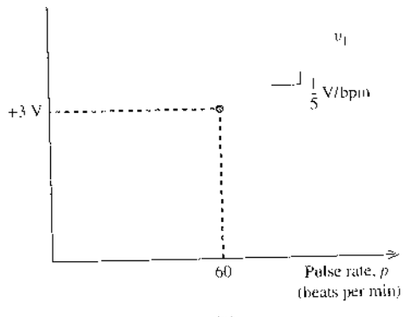
3.33. Repeat Problem 3.32 using two inverting stages instead of one noninverting.

3.34. Find the voltage transfer equation (inputs v_1, v_2; output v_3).

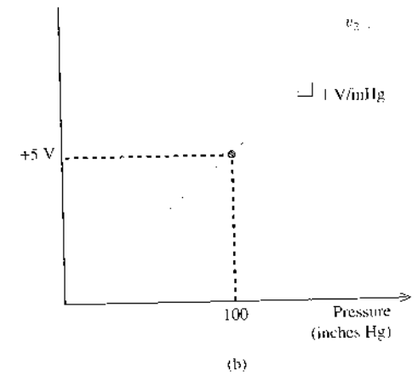


3.35. Design a circuit with $v_3 = 5v_1 + 4v_2$ two ways. (1) Use a single op amp, and (2) use op amps in inverting building blocks only.

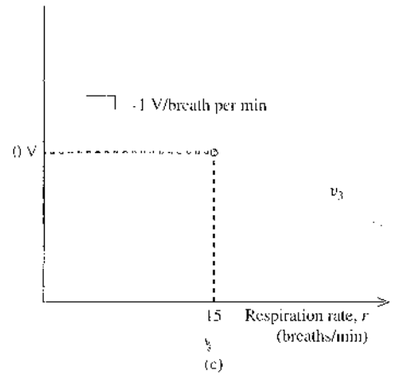
An intensive care unit patient monitoring system has three output voltages: $v_1(t)$ measures the pulse rate, $v_2(t)$ the average blood pressure, and $v_3(t)$ the respiration rate. The relationship between these voltages and the quantities they measure are shown below.



(a)



(b)



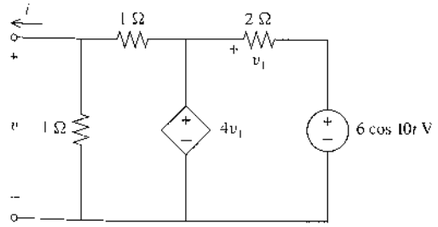
(c)

Design a circuit whose output is the alarm voltage $v_A = 2\Delta v_1 + 3\Delta v_2 + \Delta v_3$, where Δv_i is the fractional deviation of v_i from its nominal value shown. For instance, if the patient's pulse rate is 90 bpm, $\Delta v_1 = (90-60)/60 = +0.5$. The inputs to this circuit can be v_1, v_2, v_3, and ±15-V dc sources.

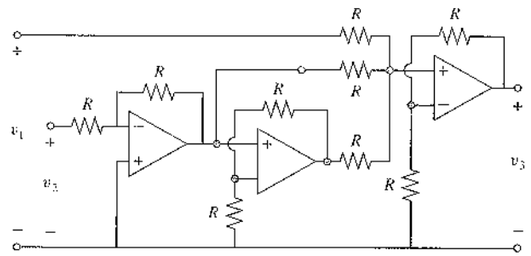
3.37. A photoresistor has a dark resistance of 1 kΩ, and its resistance drops 20 Ω/lux (a unit of light intensity). Design a circuit employing two such photoresistors that will output a +5-V signal when the light incident on one, the test photoresistor, is 1 lux brighter than the other or reference photoresistor, and −5 V when the reference photoresistor is 2 lux brighter.

3.38. Design a circuit whose voltage transfer equation is $v_{out} = 2500v_1 - 150v_2$. The voltage gain on any input to any single op amp stage should not exceed 8. Use as few op amps as you can.

3.39. Find the Thevenin and Norton equivalents.



Find the voltage transfer function $v_3 = k_1v_1 + k_2v_2$ when $R = 100$ kΩ. Repeat when $R = 10$ kΩ.



Design a circuit satisfying $v_5 = v_1 - v_2 + v_3 - v_4$. Keep all R's in the range 5 to 500 kΩ.

For what range of resistance R_1 do the values of v_2 computed using the ideal voltage amplifier model of Fig 3.7 and the improved model of Fig. 3.9 differ by at least 1%? Use $R_i = 1$ MΩ, $R_o = 30$ Ω.

For what range of resistance R_L do the values of v_2 computed using the ideal voltage amplifier model of Fig. 3.7 and the improved model of Fig. 3.9 differ by at least 1%? Use $R_i = 1$ MΩ, $R_o = 30$ Ω.

4

Analysis Methods



Gustav Robert Kirchhoff
1824–1887

There must be a fundamental story here [on his research with Bunsen].

Gustav Robert Kirchhoff

Ohm's law is fundamental to electric circuits, but to analyze even the simplest circuit requires two additional laws formulated in 1847 by the German physicist Gustav Robert Kirchhoff. These laws—Kirchhoff's current law and Kirchhoff's voltage law—are the more remarkable when we consider that Kirchhoff's principal interest was in his pioneering work in spectroscopy with the noted German chemist Robert Bunsen, to whom we owe the Bunsen burner. In that field there is another law of Kirchhoff: Kirchhoff's law of radiation.

Kirchhoff was born in Königsberg, East Prussia, the son of a lawyer. He entered the University of Königsberg at age 18 and graduated with his doctorate five years later. Upon graduation in 1847 he married the daughter of Friedrich Richelot, one of his famous mathematics teachers, and at the same time received a rare travel grant for further study in Paris. The political unrest that led to the 1848 wave of revolutions in Europe forced him to change his plans, and he became a lecturer in Berlin. Two years later he met Bunsen and the two began their famous collaboration. Kirchhoff's great success in spectroscopy drew attention away from his contributions in other branches of physics, but without his electrical laws there would be no circuit theory.

4.1 Linearity and Proportionality
4.2 Superposition
4.3 Nodal Analysis
4.4 Circuits Containing Voltage Sources
4.5 Mesh Analysis
4.6 Circuits Containing Current Sources
4.7 Duality
4.8 Virtual Short Principle for Op Amps
Summary
Problems

In Chapter 2 we considered methods of analyzing simple circuits, which we recall are those that may be described by a single equation involving a single unknown current or voltage. More general circuits must be described by a set of simultaneous equations involving several unknown circuit variables.

In this chapter we meet the challenge of these more general circuits by developing systematic ways of formulating and solving sets of equations that permit *complete analysis of any linear circuit*. We shall consider two methods, one based primarily on Kirchhoff's current law (the method called nodal analysis) and one on Kirchhoff's voltage law (mesh analysis).

Each of these methods results in a set of equations that contains only a small subset of all the currents and voltages in the circuit. It should be evident from our work in previous chapters that a complete analysis of a circuit can be performed by first *breaking the circuit*, that is, solving for the values of a few key circuit variables. Once these are known, the rest then follow easily. For example, in a simple circuit consisting of a single loop, the key variable is the loop current. Once we have broken the circuit by solving for this current, we may find every voltage in the circuit by applying the individual element laws with known current (and of course, every element current is just equal to the loop current).

We begin in Sections 4.1 and 4.2 by showing how the principles of proportionality and superposition can be used to divide a linear circuit problem involving several sources into component problems each involving a single variable or single source. This may sometimes be used to make more general circuits simple and thus solvable by those methods introduced previously, which are generally applicable only to simple circuits. We next develop the methods of nodal analysis and mesh analysis, a pair of techniques that proves indispensable in the practice of circuit analysis and design. The virtual short and open principles presented in Section 4.8 facilitate analysis of circuits containing op amps. The chapter concludes with a first look at the computer-based circuit simulation and analysis program called SPICE. We will make frequent use of SPICE in the course of our study to check our numerical results and to gain hands-on experience with a wider range of circuits than could be comfortably solved without the level of computational support provided by a digital computer. Solving, for instance, 10 simultaneous circuit analysis equations in 10 unknowns definitely loses its charm after the first few hours of hand calculation, and SPICE relieves us of that task.

4.1 LINEARITY AND PROPORTIONALITY

In Chapter 2 we defined a linear resistor as one that satisfied Ohm's law:

$$v = Ri$$

and we considered circuits that were made up of linear resistors and independent sources. We defined dependent sources in Chapter 3 and analyzed circuits containing both independent and dependent sources. The dependent sources that we considered all had source functions of the form

$$y = kx \tag{4.1}$$

where k is a constant and the variables x and y were circuit variables (voltages or currents). Clearly, Ohm's law is a special case of (4.1). In (4.1) the variable y is proportional to the variable x, and the graph of y versus x is a straight line passing through the origin. For this reason, some authors refer to elements that are characterized by (4.1) as *linear elements*. For our purposes, we shall define a linear element in a more general way, which includes (4.1) as a special case. If x and y are circuit variables associated with a two-terminal element, we shall say that the element is *linear* if multiplying x by a constant K results in the multiplication of y by the same constant K. This is called the *proportionality property* and evidently holds for all elements obeying (4.1) since, after multiplying both sides by K,

$$(Ky) = k(Kx)$$

Thus an element described by (4.1) is linear. In addition, elements with terminal laws of the forms

$$y = k\frac{dx}{dt}, \qquad y = k\int x(t)\,dt \tag{4.2}$$

are also linear, since these laws also imply that

$$(Ky) = k\left(\frac{d}{dt}Kx\right), \qquad (Ky) = k\int (Kx)\,dt$$

The op amp is a multiterminal element and is described by more than one equation. However, we shall use the op amp only in a feedback mode, as stated in Chapter 3, and in this case the op amp model circuits that result consist of linear elements. In this case the models introduced in Chapter 3, consisting of linear dependent sources and linear resistors, may be used in place of the op amp. Therefore, we may add the op amp to our list of linear elements.

We shall define a *linear circuit* as any circuit containing nothing but linear elements and independent sources. As examples, almost all the circuits we have considered thus far are linear circuits. Indeed, our attention throughout this book is limited exclusively to

linear circuits. The reasons for the focus on linear circuits are twofold: first, a great many interesting circuits, such as amplifiers and filters, are linear, and second, the techniques for analyzing linear circuits are very powerful and form the basis for all circuit analysis, whether linear or nonlinear.

The analysis equations of a linear circuit are obtained by applying KVL, KCL, and the element laws for each element in the circuit. The equations that result can be generally expressed as

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = y \tag{4.3a}$$

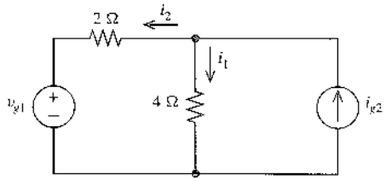
where the x_i's are circuit variables (currents and/or voltages) and y is a net algebraic sum of independent sources. For instance, if (4.3a) is a KVL equation around a loop consisting of resistors and independent voltage sources, each x_i may be a current, each a_i (plus or minus) a resistance value, and y the algebraic sum of voltage source functions. If (4.3a) is a KCL equation, the x_i's may be currents, a_i's plus or minus conductances, and y the net current into the node. All circuit variables in a given equation need not be of the same type, some of the unknown x_i's might in fact be currents and others voltages.

A sum of scaled variables, as on the left side of (4.3a), is referred to as a *linear combination* of x_1 through x_n. Note that linear combinations of linear elements also satisfy the *proportionality property*; that is, *scaling each source by K will scale each circuit variable by K*, since (4.3a) implies that

$$a_1(Kx_1) + a_2(Kx_2) + \cdots + a_n(Kx_n) = Ky \tag{4.3b}$$

Thus if $x_1, \ldots, x_n$ were the circuit variable values when y was the source, $Kx_1, \ldots, Kx_n$ will be their values with the net source y scaled by K.

Example 4.1



Linear circuit with two sources.

We seek i_1 and i_2 in Fig. 4.1. Applying KVL clockwise around the left loop,

$$-2i_2 + 4i_1 = v_{g1} \tag{4.4a}$$

we see that the source v_{g1} equals a linear combination of i_1 and i_2. By KCL at the top right node,

$$i_1 + i_2 = i_{g2} \tag{4.4b}$$

which, like all KCL and KVL equations in a linear circuit, also equates a linear combination of circuit variables to a net source value. Solving the last for i_2 and using this to eliminate i_2 from (4.4a), the current i_1 is found to be

$$i_1 = \tfrac{1}{6}v_{g1} + \tfrac{1}{3}i_{g2} \tag{4.5a}$$

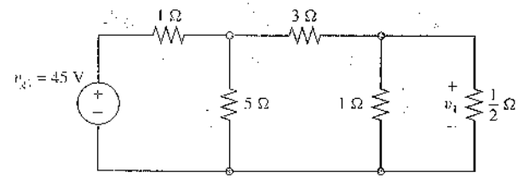
and substituting this value for i_1 into (4.4b),

$$i_2 = -\tfrac{1}{6}v_{g1} + \tfrac{2}{3}i_{g2} \tag{4.5b}$$

The solution expresses the circuit variables as linear combinations of the sources. By the proportionality principle, scaling each source by 2 should double all circuit variables. Examining (4.5), if v_{g1} and i_{g2} are doubled, the right sides of these equations are doubled and thus so are i_1 and i_2.

Example 4.2

We shall use the proportionality principle to assist in calculation of v_1 in the circuit of Fig. 4.2. Such a circuit is sometimes called a *ladder network*, due to its ladderlike configuration. Note that the unknown is far away from the only circuit variable we initially know, the value of the source. Thus several equations involving several unknowns can be expected if we begin writing equations at the source end of the ladder.



Ladder network example.

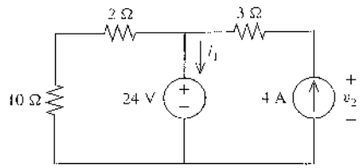
Instead, let us *guess a solution*, $v_1 = 1$ V, and see where this assumption leads. If $v_1 = 1$ V, surely $i_1 = 2$ A by Ohm's law, and $v_2 = 1$ V since parallel elements have the same voltage drop. Thus $i_2 = 1$ A, and at the top right node, $i_3 = i_2 + i_1 = 3$ A. Since i_3 flows through a 3-Ω resistor, $v_3 = 9$ V. Then, by KVL, $v_4 = v_3 + v_2 = 10$ V. Continuing right to left in this manner, $i_4 = 2$ A, $i_5 = i_3 + i_4 = 5$ A, $v_5 = 5$ V, and finally, if our guess ($v_1 = 1$ V) were right, we would have $v_{g1} = v_4 + v_5 = 15$ V.

This is not correct, since actually $v_{g1} = 45$ V. But after all, this was no more than a guess to get us started. By the proportionality relation, if a 15-V source gives an output $v_1 = 1$ V, as we have discovered, our 45-V source will give three times as much, so the correct answer must be

$$v_1 = \frac{45}{15}(1) = 3 \text{ V}$$

This method of assuming an answer for the output, working backward to obtain the corresponding input and finally adjusting the assumed output to be consistent with the actual input by means of the proportionality relation, is particularly easy to apply to the ladder network, but may be applied to other circuits as well.

A nonlinear circuit is, of course, one that has at least one element whose terminal relation is not of the form (4.1) or (4.2). An example is given in Exercise 4.1.4, for which it is seen that the proportionality property does not apply.



4.1.1. Find i_1 and v_2, with (a) the source values as shown, (b) the source values divided by 2, and (c) the source values multiplied by -2. Note how the principle of proportionality applies in (b) and (c).

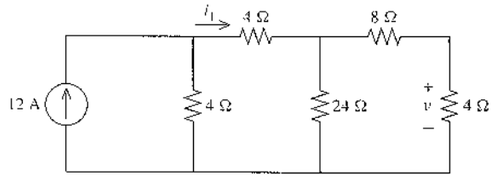
Answer: (a) 2 A, 36 V; (b) 1 A, 18 V; (c) -4 A, -72 V

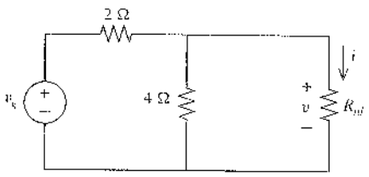
4.1.2. Suppose that $v_{g1} = 90$ V in Fig. 4.2. Find v_1, i_1, v_2, i_2, v_3, i_3, v_4, i_4, v_5, and i_5.

Answer: 6 V; 12 A; 6 V; 6 A; 54 V; 18 A; 60 V; 12 A; 30 V; 30 A

4.1.3. Find v and i using the principle of proportionality.

Answer: 8 V; 3 A





4.1.4. In this circuit R_{nl} is a nonlinear resistor with terminal law $v = 2i^3$. Find the source function v_g that supports $i = 1$ A and then that for $i = 2$ A, and show that the principle of proportionality is *not* satisfied. Why?

Answer: 5 V; 28 V. Violation of the principle of proportionality is that these answers are not in the proportion 1 : 2 (this is a nonlinear circuit).

4.2 SUPERPOSITION

In this section we consider linear circuits with more than one independent source. The linearity property makes it possible, as we shall see, to analyze these circuits by adding together the responses due to each source separately.

Recall from (4.3) that each circuit equation may be expressed as

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = y$$

where the x_i's are circuit variables and y the net source value in this KCL or KVL equation. To be specific, suppose that there are exactly two sources, y_a and y_b:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = y_a + y_b \qquad (4.6)$$

Now suppose that we "kill" the source b; that is, we set its source function y_b to zero but otherwise leave the circuit unchanged. Then, in this new circuit, we would have an equation corresponding to the one above:

$$a_1x_1^a + a_2x_2^a + \cdots + a_nx_n^a = y_a \qquad (4.7a)$$

In this equation, the left-hand side would be the same linear combination as before (the a_i's are unchanged), since only the value of source y_b was changed, not any of the linear elements or their placement in the circuit. The values of the circuit variables will be different due to the presence of just the single source a alone and are designated with an a superscript. x_i^a is thus the response to source a alone. If, instead, we kill the other source, source a, the circuit equation (4.6) becomes

$$a_1x_1^b + a_2x_2^b + \cdots + a_nx_n^b = y_b \qquad (4.7b)$$

As before, the a_i's in (4.7b) will be identical to those in (4.6), and the x_i^b's are the response to source b alone. Adding (4.7a) and (4.7b) together gives

$$a_1(x_1^a + x_1^b) + a_2(x_2^a + x_2^b) + \cdots + a_n(x_n^a + x_n^b) = y_a + y_b$$

Comparing (4.6) with the preceding equation, we arrive at our central result. Thinking of any of the x_i circuit variables as the response of this circuit to the sources, we have

$$x_i = x_i^a + x_i^b$$

or *the overall response of a circuit containing several sources is the sum of the responses to each individual source with the other sources killed.* This is the *principle of superposition.*

Our justification above assumed exactly two sources, but clearly can be applied, after slight generalization, to the case of several sources. Remember from Chapter 2 that current sources are killed, or zeroed out, by replacing them by open circuits and voltage sources by short circuits. Note also that *superposition only holds in general for linear circuits,* those for which (4.6) may be assumed, just as the proportionality principle is guaranteed only for linear circuits.

Example 4.3

First consider the circuit of Fig. 4.1, which was analyzed in Section 4.1. The circuit variables i_1 and i_2 were found in (4.5). Let us now use superposition to find these responses. First we kill the current source, resulting in the modified circuit of Fig. 4.3(a), and determine i_1^a and i_2^a. This is the component a problem, and i_1^a and i_2^a are the components of the responses i_1 and i_2 due to source a (the voltage source). This is a single-loop circuit, and

$$i_1^a = \frac{v_{g1}}{6}$$

$$i_2^a = -\frac{v_{g1}}{6}$$

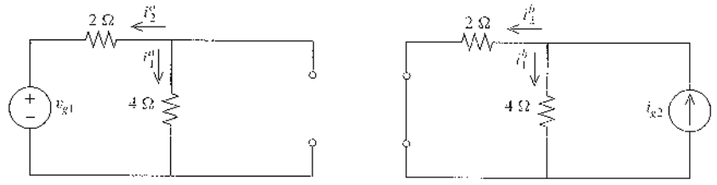
Next we kill instead the voltage source, resulting in the component b problem shown in Fig. 4.3(b). By current division,

$$i_1^b = \tfrac{1}{3} i_{g2}$$
$$i_2^b = \tfrac{2}{3} i_{g2}$$

By the superposition principle, each response is the sum of its component responses, or

$$i_1 = i_1^a + i_1^b = \tfrac{1}{6} v_{g1} + \tfrac{1}{3} i_{g2}$$
$$i_2 = i_2^a + i_2^b = -\tfrac{1}{6} v_{g1} + \tfrac{2}{3} i_{g2}$$

These results indeed agree with our previous calculations (4.5).



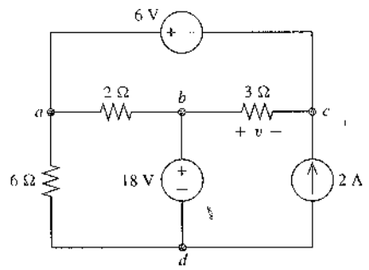
(a) Component a problem; (b) component b problem.

Example 4.3 demonstrates that a trade-off exists in the use of superposition. On the one hand, there are two (or more) component problems to solve instead of just the one we were given originally. On the other hand, each component problem is simpler than the original, both because it contains only one source and because killing sources reduces the number of nodes and/or loops in the circuit. Indeed, both component problems in Example 4.3 involve simple circuits (single loop in component a, single pair of nodes in component b), whereas the original circuit was not simple and could not be analyzed without solving simultaneous equations in two unknowns [(4.4a) and (4.4b)].

Example 4.4

As a second example, let us find the voltage v in the circuit with three independent sources shown in Fig. 4.4. Using superposition, we will find the components of v due to each source separately, referring to the 6-V source as source a, the 18-V source as source b, and the 2-A source as source c.

The component a problem is shown in Fig. 4.5(a), in which the b and c sources have been killed. Killing the 18-V source shorts nodes b and d together into a single node. The circuit has been drawn



Example 4.4 Example with three independent sources.

to show this. The parallel equivalent of the 2- and 6-Ω resistors is $\frac{3}{2}$ Ω, and v^a can be found by voltage division:

$$v^a = \frac{3(6)}{3 + \frac{3}{2}} = 4 \text{ V}$$

The component b problem is shown in Fig. 4.5(b). Killing the 6-V source has drawn nodes a and c together as shown. Once again, after replacing the 2- and 3-Ω resistors by their parallel equivalent $\frac{6}{5}$ Ω, v^b can be found by voltage division:

$$v^b = \frac{\frac{6}{5}}{\frac{6}{5} + 6}(18) = 3 \text{ V}$$

The component c problem [Fig. 4.5(c)] has three resistors in parallel:

$$v^c = -\frac{1}{\frac{1}{6} + \frac{1}{2} + \frac{1}{3}}(2) = -2 \text{ V}$$

By superposition, the value of v is the sum of its components:

$$v = v^a + v^b + v^c = 4 + 3 - 2 = 5 \text{ V}$$

Note that we have solved a set of simple circuit problems in place of an original problem that, if not for superposition, would have required simultaneous equations.

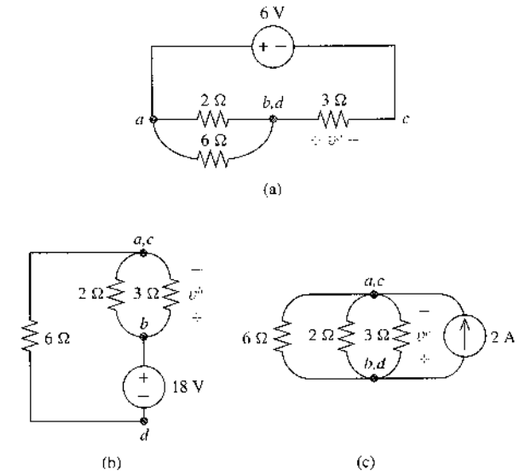
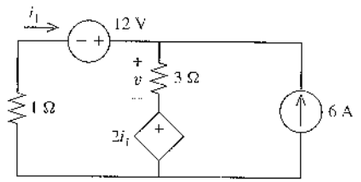


FIGURE 4.5 (a) Component a; (b) component b; (c) component c.

Superposition is a powerful tool that allows us to solve for currents and voltages in *linear* circuits containing several sources by adding (superposing) component currents and voltages. In general, superposition will not work in nonlinear circuits (those that contain one or more nonlinear elements). Its validity is also limited, even in linear circuits, to calculation of currents and voltages. In particular, the total power dissipated by an element in a linear circuit is *not* in general the sum of separate powers due to each source. This is because power is a *nonlinear* function of the circuit variables current and voltage. The use of superposition is limited to calculating currents and voltages in linear circuits.

Example 4.5

This example illustrates the proper use of superposition when power is to be calculated and also its use when there is a dependent source present. We seek the voltage v and the power dissipated by the 3-Ω resistor in the circuit of Fig. 4.6. To find v, we shall use superposition. Only *independent sources* generate components, so we decompose into two component problems a and b, as shown in Fig. 4.7. For component a, we kill the current source. Figure 4.7(a) is a single-loop circuit with the KVL equation



Circuit with a dependent source.

$$-12 + 3i_1^a + 2i_1^a + i_1^a = 0$$

or $i_1^a = 2$ A, and

$$v^a = 3i_1^a = 6 \text{ V}$$

In the component b problem, with the voltage source killed, KCL at the top node yields a current downward through the middle branch of $6 + i_1^b$. KVL around the left loop is then

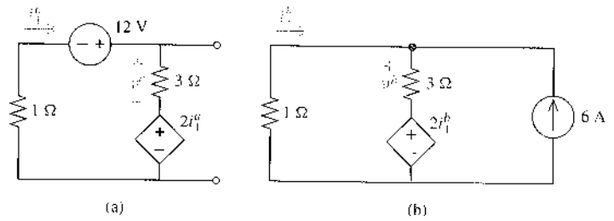
$$i_1^b + 3(6 + i_1^b) + 2i^b = 0$$

or $i_1^b = -3$ A, and

$$v^b = 3(6 + i_1^b) = 9 \text{ V}$$

Then, by superposition of component voltages,

$$v = v^a + v^b = 6 + 9 = 15 \text{ V}$$

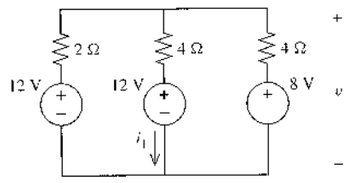


(a) Component a; (b) component b.

We also need the power through the 3-Ω resistor. Since we know its voltage v,

$$p = \frac{v^2}{R} = \frac{15^2}{3} = 75 \text{ W}$$

Note that we found v by superposition and then used the total voltage v *after* superposition of voltage components to compute the power. Had we computed the power through the resistor in the component problems separately and tried to superpose them, this would have given us a different and erroneous result, since the sum of component powers, $(v^{a2} + v^{b2})/R$, is not the same as the power due to the sum of components, $v^2/R = (v^a + v^b)^2/R$. Even in linear circuits, power does not superpose; only voltage and current do.



4.2.1. Solve Exercise 4.1.1 using superposition.

4.2.2. Find v and i_1 by superposition. Check by Thevenin-Norton transformations.

Answer: 11 V; $-\frac{1}{4}$ A

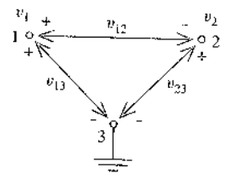
4.2.3. Replace the 8 V by $4i_1$ in Exercise 4.2.2, converting an independent source to a dependent source. Find v and i_1 by superposition.

Answer: 8 V; −1 A

4.3 NODAL ANALYSIS

In this section we develop a general method of circuit analysis in which voltages are the unknowns to be found. A convenient choice of voltages for many networks is the set of *node voltages*. Since a voltage is defined as existing between two nodes, it is convenient to select one node in the network to be a *reference node* and then associate a voltage at each of the other nodes. *The voltage of each of the nonreference nodes with respect to the reference node is defined to be a node voltage.* It is common practice to select reference directions for these voltages so that the plus ends are all at the nonreference nodes and the minus ends all at the reference node. For a circuit containing N nodes, there will be $N - 1$ nonreference nodes and thus $N - 1$ node voltages. Nodal analysis is a method in which we will break the circuit, that is, solve for a key set of circuit variables, by finding the node voltages themselves. Any other current or voltage will follow easily once the circuit is broken.

The reference node is often chosen to be the node to which the largest number of branches are connected. Many practical circuits are built on a metallic base or chassis, and usually a number of elements are connected to the chassis, which becomes a logical choice for the reference node. In many cases, such as in electric power systems, the chassis is shorted to the earth itself, becoming part of a single chassis–earth node. For



Reference and nonreference nodes.

this reason, the reference node is frequently referred to as *ground* or the *ground node*. The reference node is thus at ground potential or zero potential, and each other node may be considered to be at some potential above or below zero specified by the value of its node voltage.

The equations of nodal analysis are obtained by applying KCL at the nodes. Recall that each term in a KCL equation is an element current. For a resistor, this current is proportional to its voltage. This voltage, like any element voltage, is equal to a node voltage (if one end of the element is tied to the reference node) or the difference of two node voltages (if both ends are tied to nonreference nodes). For example, in Fig. 4.8 the reference node is node 3 with zero or ground potential. The symbol shown attached to node 3 is the standard symbol for ground, as noted in Chapter 3. The nonreference nodes 1 and 2 have node voltages v_1 and v_2. Thus the element voltage v_{12} with the polarity shown is

$$v_{12} = v_1 - v_2$$

The other element voltages shown are

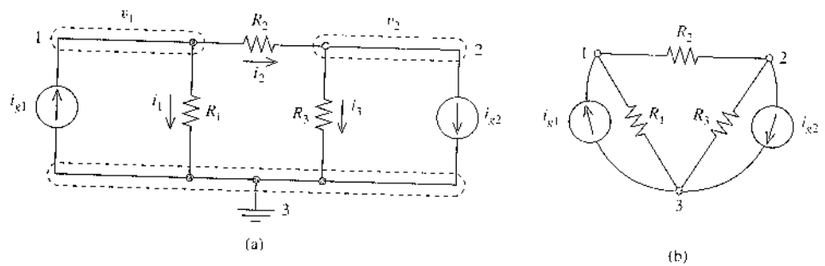
$$v_{13} = v_1 - 0 = v_1$$

$$v_{23} = v_2 - 0 = v_2$$

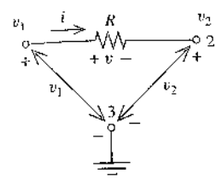
These equations may be established by applying KVL around the loops (real or imagined). Evidently, if we know all the node voltages, we may find all the element voltages and thence all the element currents.

The application of KCL at a node, expressing each unknown current in terms of the node voltages, results in a *node equation*. Clearly, simplification in writing the resulting equations is possible when the reference node is chosen to be a node with a large number of elements connected to it. As we shall see, however, this is not the only criterion for selecting the reference node, although it is frequently the overriding one. Since we are going to apply KCL systematically at circuit nodes, the most straightforward case to consider is that of circuits whose only sources are independent current sources. We begin with examples of this type.

In the network shown in Fig. 4.9(a), there are three nodes, dashed and numbered as shown. [This may be easier to see in the redrawn version of Fig. 4.9(b).] Since there are four elements connected to node 3, we select it as the reference node, identifying it by the ground symbol shown.



Circuit containing independent current sources.



Single element.

Before writing the node equations, consider the element shown in Fig. 4.10, where v_1 and v_2 are node voltages. The element voltage v is given by

$$v = v_1 - v_2$$

and thus by Ohm's law we have

$$i = \frac{v}{R} = \frac{v_1 - v_2}{R}$$

or

$$i = G(v_1 - v_2)$$

where $G = 1/R$ is the conductance. That is, *the current from node 1 to node 2 through a resistor is the difference of the node voltage at node 1 and the node voltage at node 2 divided by the resistance R, or multiplied by the conductance G.* This relation will allow us to write the node equations rapidly by inspection directly in terms of the node voltages.

Now returning to the circuit of Fig. 4.9, the sum of the currents leaving node 1 must be zero, or

$$i_1 + i_2 - i_{g1} = 0$$

In terms of the node voltages, this equation becomes

$$G_1 v_1 + G_2(v_1 - v_2) - i_{g1} = 0$$

We could have obtained this equation directly using the procedure of the preceding paragraph. Applying KCL at node 2 in a similar manner, we obtain

$$-i_2 + i_3 + i_{g2} = 0$$

or

$$G_2(v_2 - v_1) + G_3 v_2 + i_{g2} = 0$$

Instead of summing currents leaving the node to zero, we could have used the form of KCL that equates the sum of currents leaving the node to the sum of currents entering the node. Had we done so, the terms i_{g1} and i_{g2} would have appeared on the right-hand side:

$$G_1 v_1 + G_2(v_1 - v_2) = i_{g1}$$

$$G_2(v_2 - v_1) + G_3 v_2 = -i_{g2}$$

Rearranging these two equations results in

$$(G_1 + G_2)v_1 - G_2 v_2 = i_{g1} \tag{4.8a}$$

$$-G_2 v_1 + (G_2 + G_3)v_2 = -i_{g2} \tag{4.8b}$$

These equations exhibit a symmetry that may be used to write the equations in the rearranged form (4.8) directly by inspection of the circuit diagram. In (4.8a) the coefficient of v_1 is the sum of conductances of the elements connected to node 1, while the coefficient of v_2 is the negative of the conductance of the element connecting node 1 to node 2. The same statement holds for (4.8b) if the numbers 1 and 2 are interchanged. Thus node 2 plays the role in (4.8b) of node 1 in (4.8a). That is, it is the node at which KCL is applied. In each equation the right-hand side is the current from the current sources that enters the corresponding node.

In general, in networks containing only conductances and current sources, KCL applied at the kth node, with node voltage v_k, may be written as follows. *On the left*

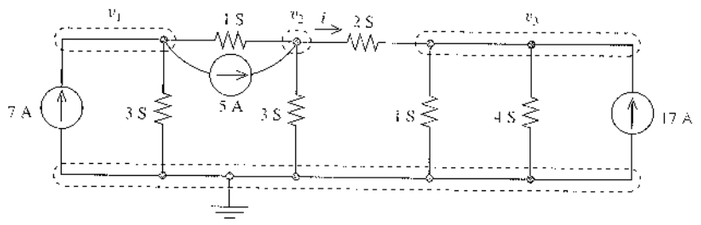
*side of the node k equation, the coefficient of the k*th*-node voltage is the sum of the conductances connected to node k, and the coefficients of the other node voltages are the negatives of the conductances between those nodes and node k. The right side of this equation consists of the net current flowing into node k due to current sources.*

This predictable pattern makes it easy to write down the node equations. Note that the signs, positive on the left-hand side for v_k terms and negative for other node voltage terms, and positive on the right-hand side for current sources flowing into node k, are a consequence of the form of KCL chosen. While other forms could be used quite as correctly, we advocate sticking to the form recommended, with the payoff that the terms will always fall in this pattern. It helps to make the pattern of signs fixed and predictable, so we can focus our attention on the larger issues when analyzing a circuit.

Nodal analysis consists in writing KCL node equations described above at all non-reference nodes in the circuit. This yields $N - 1$ linear equations in a similar number of unknowns (the node voltages). As discussed in Appendix C, these equations are linearly independent and thus are guaranteed to possess a unique solution. The node voltages may be found by a variety of means, including Gauss elimination, Cramer's rule, and matrix inversion.

Example 4.6

Consider the circuit of Fig. 4.11. The bottom node has been selected as the reference node since so many elements connect to it. The resistors are labeled according to their conductances.



Circuit for Example 4.6.

Since there are three nonreference nodes, there will be three equations in three unknown node voltages. At node v_1, we note that the sum of conductances is $3 + 1 = 4$, the negative of the conductance connecting node v_2 to node v_1 is -1, and the net source current entering node 1 is $7 - 5 = 2$. Thus the first node equation is

$$4v_1 - v_2 = 2 \tag{4.9}$$

Similarly, at nodes v_2 and v_3, we have

$$-v_1 + 6v_2 - 2v_3 = 5 \tag{4.10}$$

$$-2v_2 + 7v_3 = 17 \tag{4.11}$$

We may solve (4.9) and (4.10) for the node voltages using any one of a variety of methods for solving simultaneous equations. Three such methods are matrix inversion, Cramer's rule, and Gaussian elimination. For the reader who is not familiar with these methods, a discussion is given in Appendix A. Selecting Cramer's rule, first find the determinant of the coefficient matrix, given by

$$\Delta = \begin{vmatrix} 4 & -1 & 0 \\ -1 & 6 & -2 \\ 0 & -2 & 7 \end{vmatrix} = 145 \tag{4.12}$$

To determine v_1, we replace the first column of the coefficient matrix by the vector of constants on the right-hand side of (4.9)–(4.10), compute its determinant, and divide by the determinant of the coefficient matrix already found.

$$v_1 = \frac{\begin{vmatrix} 2 & -1 & 0 \\ 5 & 6 & -2 \\ 17 & -2 & 7 \end{vmatrix}}{\Delta} = 1 \text{ V}$$

v_2 is found by replacing the second and v_3 the third column of the coefficient matrix and calculating as above, yielding $v_2 = 2$ V and $v_3 = 3$ V.

Now that we have broken the circuit by finding the node voltages, we may easily find any other voltage or current. For example, if we want the current i in the 2-S element, it is given by

$$i = 2(v_2 - v_3) = 2(2 - 3) = -2 \text{ A}$$

Note that the coefficient matrix shown in (4.11) is symmetric [the (i, j) and (j, i) elements are equal]. This follows from the fact that the conductance between nodes i and j is that between nodes j and i. Symmetry further simplifies writing the node equations. While symmetry will hold as a general rule for all circuits not containing dependent sources, symmetry of the coefficient matrix cannot be counted on in that case, as we shall see in the next example.

Example 4.7

Consider the circuit of Fig. 4.12, which contains dependent current sources. We will begin by writing the node equations exactly as if the sources were independent. At node 1,

$$(1)(v_1) + (1)(v_1) + (2)(v_1 - v_2) = 5 - 5i$$

and at node 2,

$$\tfrac{1}{2}(v_2) + (2)(v_2 - v_1) = 5i + 2v$$

We next express the controlling variables for the dependent sources, i and v in these equations, in terms of the node voltages. By Ohm's law,

$$i = v_1$$

and by inspection

$$v = v_1 - v_2$$

Substituting the last two equations into the preceding two,

$$(1)(v_1) + (1)(v_1) + (2)(v_1 - v_2) = 5 - 5v_1$$

$$\tfrac{1}{2}(v_2) + (2)(v_2 - v_1) = 5v_1 + 2(v_1 - v_2)$$

These two equations in two unknowns can be solved by Cramer's rule, matrix inversion, or Gauss elimination, as desired. Selecting matrix inversion, we first rewrite as

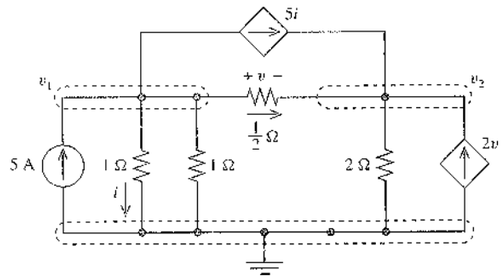
$$\begin{bmatrix} 9 & -2 \\ -9 & \frac{9}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

The determinant of the coefficient matrix is $(9)\left(\frac{9}{2}\right) - (-9)(-2) = 45/2$ and the inverse is

$$\frac{2}{45}\begin{bmatrix} \frac{9}{2} & 2 \\ 9 & 9 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & \frac{4}{45} \\ \frac{2}{5} & \frac{2}{5} \end{bmatrix}$$

Then

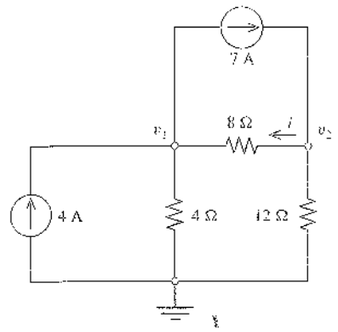
$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & \frac{4}{45} \\ \frac{2}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



Circuit containing dependent sources.

From Example 4.7 we see that the presence of dependent sources destroys the symmetry in the coefficient matrix [see (4.12)] and that in such circuits the elements of this matrix may no longer simply be interpreted as sums of conductances, since the dependent sources also contribute. On the other hand, the presence of dependent sources has not significantly complicated nodal analysis, requiring only an additional substitution step, replacing controlling variables by node voltages.

4.3.1. Take all resistors in Fig. 4.9 to be 1 Ω and both current source functions to be 1 A. Using nodal analysis, find the node voltages and the three labeled currents.



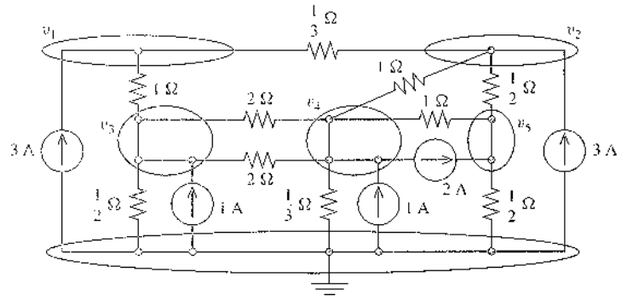
$v_1 = 1$ V; $v_2 = -1$ V; $i_1 = 1$ A; $i_2 = 2$ A; $i_3 = -1$ A

4.3.2. Using nodal analysis, find v_1, v_2, and i.

4 V; 36 V; 4 A

4.3.3. Write the nodal equations directly in vector-matrix form. Do not solve.

$$\begin{bmatrix} 4 & -3 & -1 & 0 & 0 \\ -3 & 6 & 0 & -1 & -2 \\ -1 & 0 & 4 & -1 & 0 \\ 0 & -1 & -1 & 6 & -1 \\ 0 & -2 & 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 1 \\ 3 \\ -2 \end{bmatrix}$$



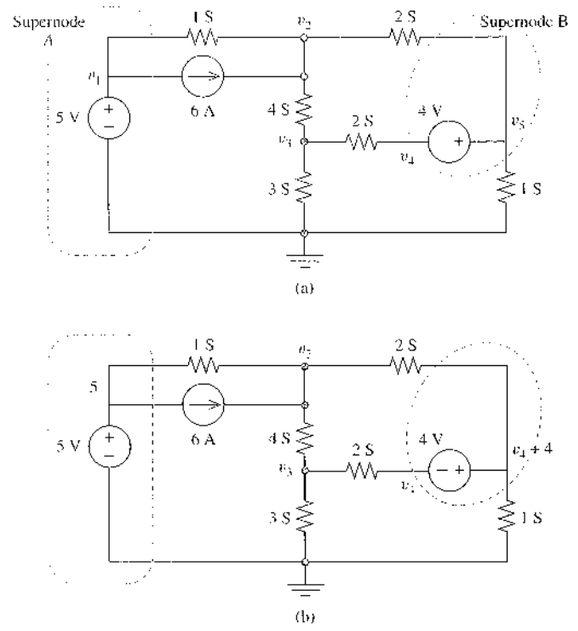
4.4 CIRCUITS CONTAINING VOLTAGE SOURCES

At first glance it may seem that the presence of voltage sources in a circuit complicates nodal analysis. We can no longer write the KCL node equations, since there is no way to express the currents through these circuit elements in terms of their node voltages. As discussed in Chapter 2, the element law for a voltage source does not relate its current to its voltage, so we cannot use it to replace a current unknown by a voltage unknown in the node equation.

However, as we shall see, nodal analysis in the presence of voltage sources proves no more complicated, requiring only a small modification to the basic method for writing the equations of nodal analysis presented in Section 4.3. In fact, we will come to welcome voltage sources, since they reduce the number of simultaneous node equations that must be solved, yielding one less equation per voltage source.

Example 4.8

To illustrate the procedure, let us consider the circuit of Fig. 4.13. For convenience we have labeled the resistors by their conductances. Note that we have enclosed voltage sources in separate regions indicated by dashed lines. Recalling that the generalized form of KCL states that all currents entering a closed region must sum to zero



(a) Circuit containing voltage sources; (b) node voltages in supernodes relabeled.

(just as they do for ordinary nodes), we will refer to each of these regions as a *supernode*.

Each supernode contains two nodes, one a nonreference node and another node that may be a second nonreference node (e.g., supernode B in Fig. 4.13) or the reference node (e.g., supernode A). Supernodes containing the reference node have one node voltage variable, such as v_1 in Fig. 4.13(a). But since there is an element with known voltage drop connecting v_1 to a known node voltage (zero since it is the reference node), v_1 is not in fact an unknown and may be determined immediately:

$$v_1 = 5 + 0 = 5 \text{ V}$$

Similarly, supernodes not containing the reference node have two node voltage variables [v_4 and v_5 in Fig. 4.13(a)]. However, since their difference is known,

$$v_5 - v_4 = 4$$

we may consider just one of these node voltages as an unknown, say v_4, and the other known in terms of v_4 as

$$v_5 = v_4 + 4$$

The number of node voltage unknowns is thus reduced by the number of independent voltage sources. This is shown in Fig. 4.13(b) by relabeling each supernode so that one of its node voltages is expressed as the algebraic sum of the node voltage at the other plus the known voltage drop across the voltage source.

To complete the formulation of the nodal analysis equations, let us *apply KCL to all supernodes not containing the reference node and to all other nonreference nodes*. This has the effect of reducing the number of node equations by one per voltage source. *The presence of voltage sources thus reduces the number of equations and the number of unknowns in nodal analysis by one per voltage source.* For instance, in Fig. 4.13(a) there are five nonreference nodes, and if the voltage sources were all current sources, five equations in five unknowns would be required. Because there are two voltage sources, however, we may break the circuit by using just three equations in three unknowns, as we now demonstrate.

We do not require a KCL equation for supernode A because it contains the reference node. Referring to Fig. 4.13(b), equating currents out of node 2 through resistors to net current into node 2 via current sources in the usual way,

$$(1)(v_2 - 5) + (4)(v_2 - v_3) + (2)[v_2 - (v_4 + 4)] = 6$$

while at node 3

$$(4)(v_3 - v_2) + (3)(v_3 - 0) + (2)(v_3 - v_4) = 0$$

and at supernode B,

$$(2)(v_4 - v_3) + (2)[(v_4 + 4) - v_2] + (1)[(v_4 + 4) - 0] = 0$$

Rearranging these equations in vector-matrix form yields

$$\begin{bmatrix} 7 & -4 & -2 \\ -4 & 9 & -2 \\ -2 & -2 & 5 \end{bmatrix} \begin{bmatrix} v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -12 \end{bmatrix}$$

The coefficient matrix has all the same properties it did in the absence of voltage sources: the on-diagonal elements are the sum of all conductances into the node or supernode at which that equation was written, the (i, j) element is the negative sum of conductances connecting the ith equation's node or supernode with the jth equation's, and the matrix is symmetric. The source vector on the right-hand side, however, is no longer simply the net current in due to current sources, since it now contains terms due to the voltage sources as well.

Example 4.9

As a second example, let us find v and i in the circuit of Fig. 4.14. Assigning node voltages of 0 V and $0+20 = 20$ V to one supernode and v and $v+3$ to the other, there is one supernode not containing the reference node, and there we must write a KCL equation. Summing

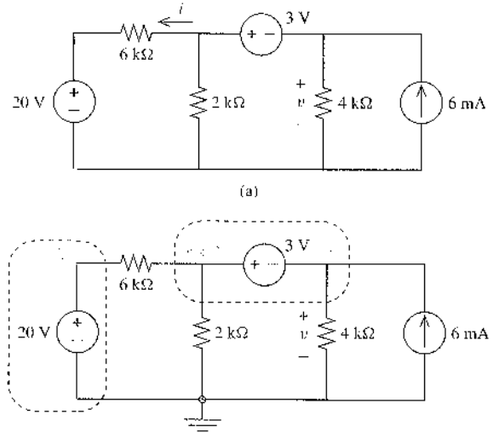
currents into this region in the usual way gives

$$\frac{1}{6\times10^3}[(v+3)-20]+\frac{1}{2\times10^3}[(v+3)-0]+\frac{1}{4\times10^3}(v-0)=6\times10^{-3}$$

Solving the equation yields $v = 8$ V. Having broken the circuit, we may now easily find any other circuit variable, such as i. This current is

$$i=\frac{1}{6\times10^3}[(v+3)-20]=\frac{1}{6\times10^3}(-9)=-\tfrac{3}{2}\text{ mA}$$

Note that while there are three nonreference nodes in this problem, we had to solve only a single equation in one unknown, not three in three, because of the welcome presence of the two voltage sources.



(a) Circuit for Example 4.9; (b) redrawn with supernodes and node voltages labeled.

Example 4.10

Next consider an example containing a dependent voltage source as shown in Fig. 4.15. In Example 4.7 we initially treated dependent current sources as if they were independent, for the purpose of writing the nodal equations, and then replaced their controlling variables by node voltages. We proceed in the same fashion here.

Equations are needed at nodes v_1 and v_2, but not at either supernode since they both contain the reference node. The two equations required are

$$4v_1+2(v_1-v_2)=10 \qquad (4.13a)$$

$$2(v_2-v_1)+9(v_2-10v)=-6 \qquad (4.13b)$$

In writing these equations, we proceeded as if the dependent voltage source were an independent source of value $10v$. Next we replace the controlling variable by node voltages. Since v is the element voltage between nodes with voltages given as $10v$ and v_1,

$$v=10v-v_1$$

or $v=\frac{1}{9}v_1$. Substituting this into (4.13b), (4.13a) becomes

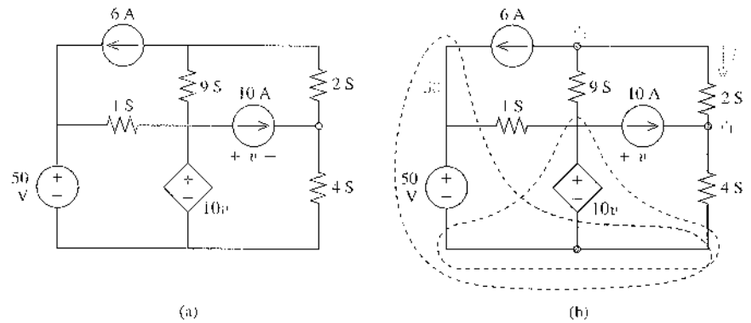
$$6v_1-2v_2=10$$
$$-12v_1+11v_2=-6$$

Solving by Gaussian elimination, we multiply the second equation by $\frac{1}{2}$ and add it to the first, yielding

$$v_2=2\text{ V}$$

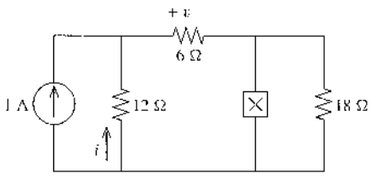
Substituting this back into the first equation yields $v_1=\frac{7}{3}$ V. With the circuit broken, analysis may be completed by inspection. For instance, the current i is

$$i=2(v_2-v_1)=2(2-\tfrac{7}{3})=-\tfrac{2}{3}\text{ A}$$



(a) Circuit containing dependent voltage sources; (b) redrawn to show supernodes.

EXERCISES



EXERCISE 4.4.1

4.4.1. Using nodal analysis, find v if element x is a 3-V independent voltage source with the positive terminal at the top.
Answer 3 V

4.4.2. Find v in Exercise 4.4.1 if element x is a 2-A independent current source directed downward.
Answer 8 V

4.4.3. Find v in Exercise 4.4.1 if element x is a dependent voltage source of $12i$ V with the positive terminal at the bottom.
Answer 12 V

4.4.4. With x an independent voltage source as in Exercise 4.4.1, suppose that we desire a value of +5 V for v. What should the source function for x be?

Ans. −3 V

4.5 MESH ANALYSIS

In the nodal analysis of the preceding sections we applied KCL at the nonreference nodes of the circuit. We now consider a related method known as *mesh analysis*, in which KVL is applied around certain loops in the circuit. As we shall see, in this case the unknowns in the resulting equations will be currents.

We restrict attention in this section to *planar circuits*, circuits that can be drawn on a plane surface (like a piece of paper) so that no elements or connecting wires cross. In this case the plane is divided by the circuit diagram into distinct areas in the same fashion that the solid framework in a window separates and outlines each individual windowpane. The closed boundary of each such windowpane area is called a *mesh* of the circuit. Thus a mesh is a special case of a *loop*, which we consider to be any closed path of elements in the circuit passing through no node or element more than once. In other words, a mesh is a loop that contains no elements within it.

Example 4.11

The circuit of Fig. 4.16 is planar and contains three meshes identified by the arrows. Mesh 1 contains the elements R_1, R_2, R_3, and v_{g1}; mesh 2 contains R_2, R_4, v_{g2}, and R_5; and mesh 3 contains R_5, v_{g2}, R_6, and R_3.

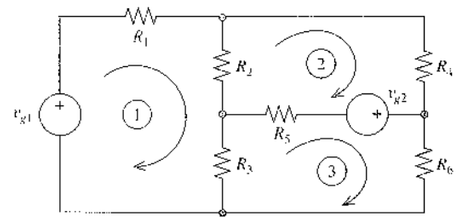


FIGURE 4.16 Planar circuit with three meshes.

In the case of *nonplanar circuits*, we cannot define meshes and mesh analysis cannot be performed. Mesh analysis is thus not as general as nodal analysis, which had no such topological restriction. A KVL-based technique may still be employed, however, even for nonplanar circuits. The procedure in this case is similar to mesh analysis, but the equations are not as easily formulated. This technique, *generalized loop analysis*, is summarized briefly in Appendix C. Fortunately, the large majority of electric circuits we shall have need to analyze in practice are planar, and for those we may choose nodal or mesh analysis. For nonplanar circuits, the only general method developed in this book is nodal analysis.

We define a *mesh current* as the current that circulates around a mesh. If an element is located on a single mesh such as v_{g1} or R_2 in Fig. 4.16, it carries an element current just equal to this mesh current (mesh current i_1 for v_{g1}, i_2 for R_4). If an element is located on the boundary between two meshes, such as R_2 in the same figure, its element current is the algebraic sum of the two mesh currents circulating through it ($i_1 - i_2$ for element R_2). Labeling element currents with capital letters and mesh currents with lowercase letters, both R_1 and v_{g1} have the same element current I_1, where

$$I_1 = i_1$$

The element current I_3 defined positive downward through R_3 is

$$I_3 = i_1 - i_2$$

Mesh analysis consists in writing KVL around each mesh in the circuit, using mesh currents as the unknowns. The resulting system of equations is guaranteed to be linearly independent, as shown in Appendix C, and thus possesses an unique solution. As with nodal equations, the solution may be found by any convenient method such as Gauss elimination, Cramer's rule, or matrix inversion.

Example 4.12

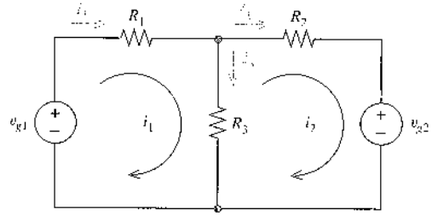


FIGURE 4.17 Circuit with two meshes.

Consider the circuit of Fig. 4.17. Writing KVL clockwise around the first mesh,

$$R_1 i_1 + R_3(i_1 - i_2) = v_{g1}$$

and clockwise around the second mesh,

$$R_2 i_2 + R_3(i_2 - i_1) = -v_{g2}$$

Note that in the second equation we have expressed the element current through R_3 as $i_2 - i_1$, so its reference direction arrow points in the direction in which we are traversing the loop (clockwise). This way the voltage drop is positive in this KVL equation. Had we used $i_1 - i_2$ as the element current, as we did in the first equation, we would have had a minus sign in front of the corresponding term, $-R_3(i_1 - i_2)$ in the second equation. We will always use implied element current reference directions that point in the direction in which we are traversing the mesh, thus keeping the signs for all resistive voltage drop terms positive. This is for convenience, of course; it is never incorrect to replace a term like $+R_3(i_2 - i_1)$ in any equation by $-R_3(i_1 - i_2)$.

There is also a *shortcut method* of writing mesh equations that is similar to the shortcut nodal method of Section 4.2. Rearranging the two equations of Example 4.12,

$$(R_1 + R_3)i_1 - R_3 i_2 = v_{g1} \tag{4.14a}$$

$$-R_3 i_1 + (R_2 + R_3)i_2 = -v_{g2} \tag{4.14b}$$

we note that in the first equation, corresponding to the first mesh, the coefficient of the first current is the sum of the resistances associated with the first mesh, and the coefficient of the other mesh current is the negative of the resistance common to that mesh and the

first mesh. The right side of the first equation is the net voltage rise in the direction of traversal due to voltage sources in the first mesh. The same pattern applies to each equation. *On the left side of the mesh k equation, the coefficient of the mesh k current is the sum of the resistances of mesh k, and the coefficients of the other mesh currents are the negatives of the resistances on the boundary between their meshes and mesh k. The right side of the mesh k equation consists of the net voltage rise in this mesh due to voltage sources.* This shortcut procedure is a consequence of selecting all the mesh currents in the same direction (clockwise in Fig. 4.17) and writing KVL as each mesh is traversed in the direction of its mesh current. Of course, the shortcut method applies only when no sources are present except independent voltage sources.

Example 4.13

In Fig. 4.16 define i_1, i_2, and i_3 as the mesh currents circulating clockwise in meshes 1, 2, and 3, respectively. Applying the shortcut method to mesh 1, we have

$$(R_1 + R_2 + R_3)i_1 - R_2 i_2 - R_3 i_3 = v_{g1}$$

This result may be checked by applying KVL to mesh 1, resulting in

$$R_1 i_1 + R_2(i_1 - i_2) + R_3(i_1 - i_3) = v_{g1}$$

The two results are evidently the same.

Applying KVL to meshes 2 and 3 yields, in the same manner,

$$-R_2 i_1 + (R_2 + R_4 + R_5)i_2 - R_5 i_3 = -v_{g2}$$

$$-R_3 i_1 - R_5 i_2 + (R_3 + R_5 + R_6)i_3 = v_{g2}$$

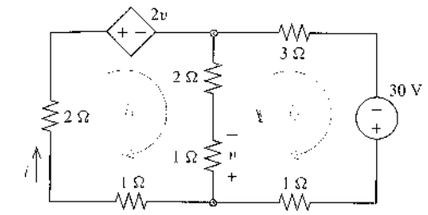
The analysis is completed by solving these three mesh equations for the three unknown mesh currents.

The same symmetry is present in the mesh equations as was noted in the nodal equations. Rewriting the three mesh equations (4.15) as a single vector matrix equation, we have

$$\begin{bmatrix} R_1 + R_2 + R_3 & -R_2 & -R_3 \\ -R_2 & R_2 + R_4 + R_5 & -R_5 \\ -R_3 & -R_5 & R_3 + R_5 + R_6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} v_{g1} \\ -v_{g2} \\ v_{g2} \end{bmatrix}$$

The diagonal elements are the sums of the resistances in the meshes, and the off-diagonal elements are the negatives of the resistances common to the meshes corresponding to the row and column of the element. That is, $-R_2$ in row 1, column 2 or in row 2, column 1 is the negative of the resistance common to meshes 1 and 2, and so on. Thus the matrix is symmetric. As was the case with nodal analysis, this symmetry is not preserved if there are dependent sources present.

Example 4.14



Example with a dependent source.

We seek the current i in Fig. 4.18. This circuit contains a dependent voltage source. As with nodal analysis, we work with dependent sources by first writing the analysis equations as if the source were independent; we then replace the controlling variable by the desired variables of the analysis method (in this case the mesh currents). Using the shortcut method yields

$$6i_1 - 3i_2 = -2v$$

$$-3i_1 + 7i_2 = 30$$

In (4.16), 6 Ω is the resistance around mesh 1, 7 Ω is the resistance around mesh 2, −3 Ω is the negative of the net resistance on the border between the meshes, $-2v$ is the clockwise voltage rise due to voltage sources in mesh 1, and 30 is the rise due to sources in mesh 2. Replacing the controlling variable v by $1(i_2 - i_1)$ via Ohm's law gives us

$$\begin{bmatrix} 4 & -1 \\ -3 & 7 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 30 \end{bmatrix}$$

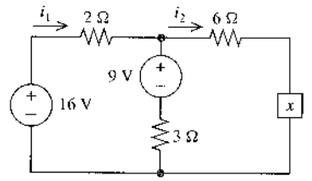
We will solve for i_1, which is the desired current i, using Cramer's rule. The determinant of the coefficient matrix is

$$\Delta = (4)(7) - (-1)(-3) = 25$$

Replacing the first column of the coefficient matrix by the vector of right-hand sides and calculating the determinant yields

$$i_1 = \frac{\begin{vmatrix} 0 & -1 \\ 30 & 7 \end{vmatrix}}{\Delta} = \frac{6}{5} \text{ A}$$

Exercises



Exercise 4.5.2

4.5.1. Using mesh analysis, find i_1 and i_2 in Fig. 4.17 if $R_1 = 2$ Ω, $R_2 = 4$ Ω, $R_3 = 3$ Ω, $v_{g1} = 9$ V, and $v_{g2} = -5$ V.

Answer: 3 A; 2 A

4.5.2. Repeat Exercise 4.5.1 with $R_1 = 1$ Ω, $R_2 = 2$ Ω, $R_3 = 4$ Ω, $v_{g1} = 21$ V, and $v_{g2} = 0$. Check by using equivalent resistance and current division.

Answer: 9 A; 6 A

4.5.3. Using mesh analysis, find i_1 and i_2 if element x is a 6-V independent voltage source with the positive terminal at the top.

Answer: 2 A; 1 A

4.5.4. Repeat Exercise 4.5.3 if element x is a dependent voltage source of $6i_1$ V with the positive terminal at the bottom.

Answer: 5 A; 6 A

4.6 CIRCUITS CONTAINING CURRENT SOURCES

As in the case of nodal analysis with voltage sources, mesh analysis results in fewer equations if current sources are present. To illustrate this point, let us consider the circuit of Fig. 4.19(a), which has two current sources and a voltage source.

Circuits with three meshes will, in the absence of current sources, have three unknown mesh currents. Note, however, that each current source bordering only one mesh, such as i_{g1} in Fig. 4.19(a), results in one less *unknown* mesh current, since the mesh current must agree with the given current source function. Thus we relabel the mesh current i_3 in Fig. 4.19(b) by its known value,

$$i_3 = -i_{g1}$$

Similarly, for each current source on the boundary between two meshes, one of the two mesh currents may be expressed in terms of the other. The current source between meshes 1 and 2 permits us to relabel i_2 in terms of i_1, since by examination of Fig. 4.19(a),

$$i_{g2} = i_2 - i_1$$

or

$$i_2 = i_{g2} + i_1$$

as shown in Fig. 4.19(b). *The presence of current sources thus reduces the number of unknowns in mesh analysis by one per current source.*

We next consider the selection of a set of KVL equations to match this reduced number of unknown currents. Imagine the circuit that results if we were to kill all current sources by replacing them by open circuits. Some of its meshes correspond to meshes of the original circuit, others to loops in the original circuit that are not meshes in the original circuit since they are not empty (they had current sources inside them). These are called *supermeshes*. This new circuit has one less mesh per current source than the original. *KVL around loops in the circuit defined as the set of meshes that result when all current sources are removed yields the complete set of mesh analysis equations for circuits containing current sources.* In Fig. 4.19(b), if the current sources were killed, there would remain only a single mesh traversing the three resistors and the voltage source. This is not a mesh in the original circuit, since it encloses a current source. Writing KVL around

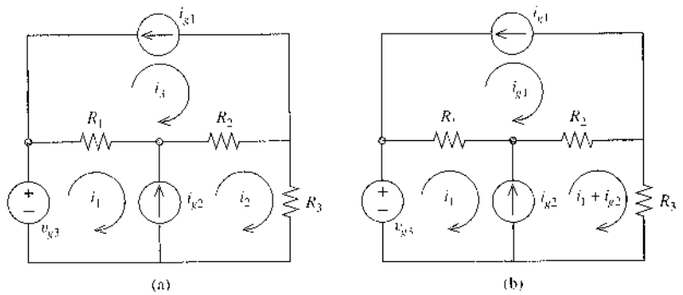


FIGURE 4.19 (a) Circuit with current sources; (b) mesh currents relabeled.

this supermesh of the original circuit gives

$$R_1(i_1 + i_{g1}) + R_2(i_1 + i_{g2} + i_{g1}) + R_3(i_1 + i_{g2}) = v_{g3}$$

This is a single equation in a single unknown mesh current i_1. Solving yields

$$i_1 = \frac{v_{g3} - R_1 i_{g1} - R_2(i_{g1} + i_{g2}) - R_3 i_{g2}}{R_1 + R_2 + R_3}$$

This solution is considerably easier than the three equations in three unknown mesh currents we would be left with if the current sources were voltage sources. We welcome current sources in mesh analysis for this reason.

Example 4.15

Let us apply mesh analysis to the circuit of Fig. 4.20(a). Examining this figure, the current sources are related to the mesh currents by

$$9 = i_1 - i_3 \tag{4.17a}$$

$$2v_x = i_4 - i_3 \tag{4.17b}$$

Solving (4.17a) for i_3, we have

$$i_3 = i_1 - 9$$

and (4.17b) for i_4 gives

$$i_4 = 2v_x + i_3 = 2v_x + i_1 - 9$$

These equations permit us to relabel the four mesh currents in terms of only two unknown mesh currents, i_1 and i_2. This is shown in Fig. 4.20(b).

The two equations required to find these two unknown mesh currents are found by imagining the current sources killed. In this modified circuit only two meshes would exist, one corresponding to mesh 2 of the original circuit and the other to a supermesh consisting of the union of the old meshes 1, 3, and 4 from Fig. 4.20(a). Note that this supermesh is a loop but *not* a mesh in our actual circuit of

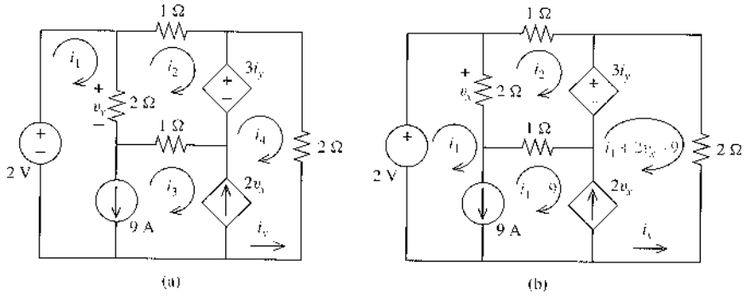


FIGURE 4.20 (a) More complicated circuit; (b) mesh currents relabeled.

Fig. 4.20(a) or (b), since the current sources lie inside its boundary (hence its designation as a supermesh). Writing KVL around mesh 2 in Fig. 4.20(b) gives

$$1i_2 + 3i_y + 1(i_2 - i_1 + 9) + 2(i_2 - i_1) = 0$$

and around the supermesh,

$$2(i_1 - i_2) + 1(i_1 - 9 - i_2) - 3i_y + 2(i_1 + 2v_x - 9) - 2 = 0$$

Gathering common terms yields

$$-3i_1 + 4i_2 = -9 - 3i_y$$

$$5i_1 - 3i_2 = 29 - 4v_x + 3i_y$$

Thus far we have ignored the fact that two of the sources are, in fact, dependent. This is the same two-step strategy we employed with nodal analysis in the presence of dependent sources. To complete the analysis, the controlling variables v_x and i_y will be replaced by mesh currents. Inspecting the circuit diagram Fig. 4.20(b), this is readily accomplished.

$$v_x = 2(i_1 - i_2)$$

and

$$i_y = -(i_1 + 2v_x - 9)$$

$$= -(i_1 + 2[2(i_1 - i_2)] - 9$$

or

$$i_y = -5i_1 + 4i_2 + 9$$

Using (4.20) to eliminate v_x and i_y from (4.19) yields

$$-18i_1 + 16i_2 = -36$$

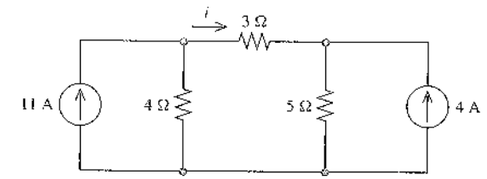
$$28i_1 - 23i_2 = 56$$

Solving, we find that $i_1 = 2$ A and $i_2 = 0$ A. This result may be confirmed by use of any of our linear algebraic methods (Gauss, Cramer, or matrix inversion) or by back substitution of i_1 and i_2 into these two equations.

With the circuit now broken, any other variable needed may be easily recovered. The other two mesh currents may be found from their labels in Fig. 4.20(b) in terms of i_1 and i_2. All resistive voltages follow from Ohm's law.

In general, before analyzing any circuit, we should note how many equations are required in nodal analysis and in mesh analysis and choose the method with fewer equations. The number of nodal equations is the number of nodes (less one, the reference node) minus the number of voltage sources. The number of mesh equations is the number of meshes minus the number of current sources. It is further assumed that series and parallel equivalents have been substituted where possible before performing either nodal or mesh analysis. Series elements multiply the number of nodes unnecessarily; similarly, parallel elements cause meshes to proliferate.

4.6.1. Using mesh analysis, find i.



Answer: 2 A

4.6.2. Using mesh analysis, find v in Fig. 4.6. This was found in Example 4.5 using superposition.

Answer: 15 V

4.6.3. In Fig. 4.19(a), let $R_1 = 4\ \Omega$, $R_2 = 6\ \Omega$, $R_3 = 2\ \Omega$, $i_{g1} = 4$A, $i_{g2} = 6$A, and $v_{g3} = 52$ V. Find the power delivered to R_3.

Answer: 18 W

4.7 DUALITY

There is an interesting symmetry of opposites exhibited by pairs of network equations which we have encountered. For example, Ohm's law may be stated in v form as

$$v = Ri \tag{4.21a}$$

or in i form as

$$i = Gv \tag{4.21b}$$

Equation (4.21b) may be derived from (4.21a) by solving for i and replacing $\frac{1}{R}$ by G, of course. But note that either equation may be formed from the other by *replacing each symbol by its paired opposite:* i by v, v by i, R by G, G by R.

Similarly, in the case of series resistances, $R_1, R_2, \ldots, R_n$, the equivalent resistance was shown in Section 2.3 to follow the "series resistance adds" rule:

$$R_s = R_1 + R_2 + \cdots + R_n$$

Replacing R by G and *series* by *parallel*,

$$G_p = G_1 + G_2 + \cdots + G_n$$

which is the "parallel conductances add" rule of Section 2.4. It is clear that one of these equations may be obtained from the other by interchanging resistances and conductances and the subscripts s and p (i.e., series and parallel).

There is a systematic symmetry of opposites, or *duality*, between resistance and conductance, current and voltage, and series and parallel. We acknowledge this by

Some Dual Quantities

v	i	Series	Parallel
R	G	Mesh	Node
Open circuit	Short circuit	KVL	KCL
Ind. voltage source	Ind. current source	v form	i form
VCVS	CCCS	Thevenin form	Norton form
CCVS	VCCS	Voltage divider	Current divider

defining these quantities as *duals* of each other. That is, R is the dual of G, i is the dual of v, series is the dual of parallel, and vice versa in each case.

Another simple case of dual equations is

$$v = 0$$

and its dual

$$i = 0$$

In the general case, an element described by $v = 0$ is a *short circuit*, and one described by $i = 0$ is an *open circuit*. Thus short circuits and open circuits are duals. Table 4.1 is a table of duals.

Note that *dual* quantities are not *equivalents*; they do not even fit in the same circuit. For instance, R and G are duals, but an 8-Ω resistor and 8-S conductance are not equivalent. Similarly, the dual to a Thevenin form (series voltage source and resistor) is a Norton form (parallel current source and conductance), but they are not equivalent since they do not in general satisfy $V_T = R_T I_N$ or $R_N = R_T$ as required for Thevenin-Norton equivalents.

Example 4.16

Consider the circuit of Fig. 4.21(a). The mesh equations are given by

$$(R_1 + R_2)i_1 - R_2 i_2 = v_g$$
$$-R_2 i_1 + (R_2 + R_3)i_2 = 0 \qquad (4.22)$$

To obtain the dual of (4.22), we simply replace the R's by G's, the i's by v's, and v by i. The result is

$$(G_1 + G_2)v_1 - G_2 v_2 = i_g$$
$$-G_2 v_1 + (G_2 + G_3)v_2 = 0 \qquad (4.23)$$

These are the nodal equations of a circuit having two nonreference node voltages, v_1 and v_2, three conductances, and an independent current source i_g. From our shortcut procedure for nodal equations we see that G_1 and G_2 are connected to the first node, G_2 is common to the two nodes, G_2 and G_3 are connected to the second node, and i_g enters the first node. Since G_1 and i_g are not connected to the second node and G_3 is not connected to the first node, these elements

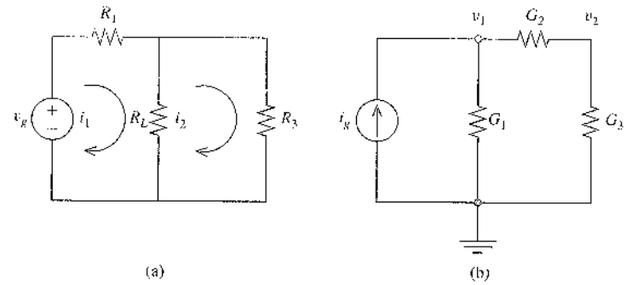


Figure 4.21 (a) Circuit; (b) dual circuit.

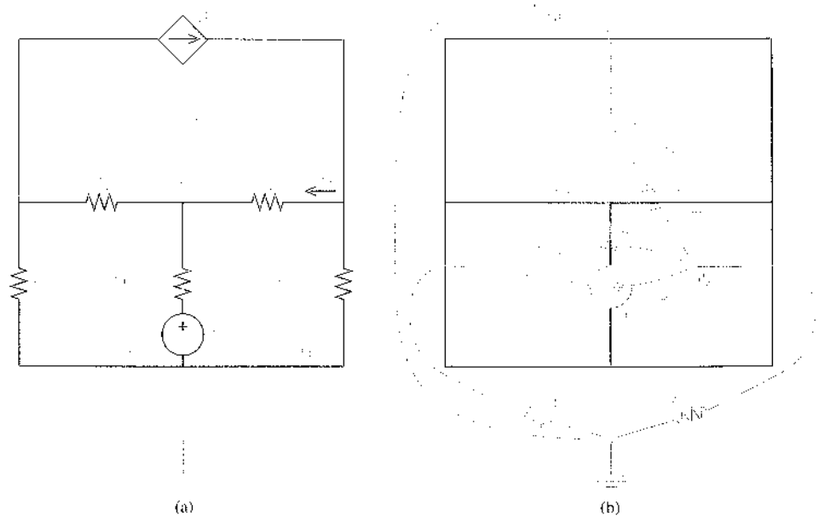
are connected to the reference node. The primal (original) circuit and its dual are shown in Fig. 4.21.

Figure 4.21(a) may be described as v_g in series with R_1 and the parallel combination of R_2 and R_3. Replacing the quantities in this statement by their duals, we see correctly that Fig. 4.21(b), its dual circuit, is i_g in parallel with G_1 and the series combination of G_2 and G_3.

Next consider systematic construction of the dual circuit from the circuit diagram of the original, or *primal, circuit*. As illustrated in Example 4.16, meshes in the original (primal) circuit are dual to nodes in the dual circuit. Each mesh current in the primal became a node voltage in the dual, thus each mesh in the primal corresponds to a node in the dual. This suggests that we begin constructing the dual circuit diagram by placing a node in the center of each mesh of the primal as shown in Fig. 4.22(a). One additional node is needed in the dual circuit, the reference node, for which there is no node voltage variable or corresponding mesh in the primal circuit. Elements on the outer periphery of the primal circuit have only one mesh current passing through them, so they will be dual to elements with only one node voltage variable across them. These must be elements connected to the reference node in the dual circuit, since all others will have two node voltages across them. Thus we add one last node to our dual circuit, the reference node, placing it somewhere outside the primal circuit as shown in Fig. 4.22(a).

To complete the dual circuit we must add its elements. Note that all terms in the first primal mesh equation will be dual to terms in the first dual node equation. So we attach one element at node 1 of the dual through each of the primal circuit elements surrounding this node. The element we attach will be the dual element, of course. The completed dual circuit is shown in Fig. 4.22(b). The dual circuit has the same number of elements as the primal and one more node than the primal has meshes. Note that the controlled source, a current-controlled current source (CCCS), has gone over to its dual controlled source, a voltage-controlled voltage source (VCVS). The controlling current $i_c = i_3 - i_2$ is replaced by its dual, the controlling voltage $v_c = v_3 - v_2$. Since the reference arrows for the primal source and its mesh current pointed in the same direction, the reference +/− signs for the dual source agree with the dual variable, v_3 (plus at v_3).

Duality has many uses in circuit theory. It may be used to generalize a theoretical result immediately to its dual without further proof. For instance, we could have invoked



(a) (b)

Construction of the dual circuit: (a) node placement; (b) dual circuit.

duality to write down the current-divider law for conductances in parallel from its dual, the voltage-divider law for resistances in series. Duality can also be used to produce circuits whose *currents* have certain desirable properties from other circuits whose *voltages* have those properties, or vice versa. Since nodal analysis on the dual is identical to mesh analysis on the primal, duality further permits us to use just one general method, say nodal analysis, whichever way is most numerically efficient. For instance, a circuit with fewer meshes than node voltages can be dualized and then nodal analysis done with the same benefits as mesh analysis on the primal. This is particularly useful in the production of computer-based tools, which by this duality-based approach only need to "know" one method, either nodal or mesh analysis, both need not be programmed.

Example 4.17

Suppose we are required to use mesh analysis to find the current i_L of Fig. 4.23(a). If we were to apply mesh analysis to this circuit directly, there would be three simultaneous equations in three unknown mesh currents. The dual circuit is shown in Fig. 4.23(b), and while there are four meshes, the presence of three current sources permits the circuit to be broken by solving a single equation in one unknown. Around the supermesh formed by the resistances,

$$(1)i_1 + \tfrac{1}{2}i_2 + \tfrac{1}{2}i_3 + \tfrac{1}{4}i_4 = 0 \qquad (4.24)$$

We may eliminate i_2, i_3, and i_4 by noting from the circuit diagram that the currents through the current sources are $i_4 - i_1 = 6$, $i_4 - i_2 = 2$, and $i_4 - i_3 = 3$, so that in terms of i_4, $i_1 = i_4 - 6$, $i_2 = i_4 - 2$,

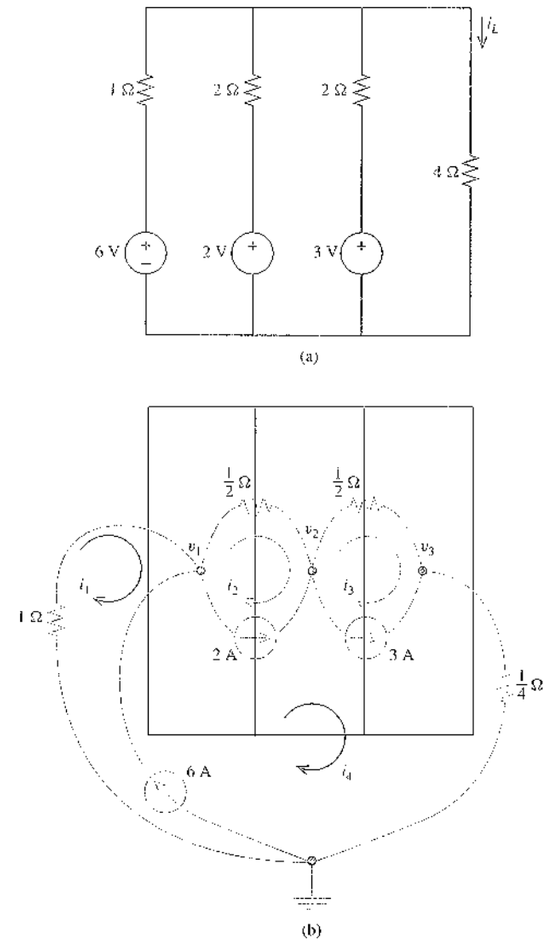


FIGURE 4.23 (a) Circuit for Example 4.17; (b) dual circuit.

and $i_3 = i_4 - 3$. Substituting these into (4.24) gives us

$$(1)(i_4 - 6) + \tfrac{1}{2}(i_4 - 2) + \tfrac{1}{2}(i_4 - 3) + \tfrac{1}{4}i_4 = 0$$

Solving, we have $i_4 = \frac{34}{9}$ A. The desired current i_L is the mesh current in the third mesh in the primal circuit. This current is dual to the node voltage v_3 in the dual circuit, which is

$$v_3 = \frac{i_4}{4} = \frac{17}{18}\ \text{V}$$

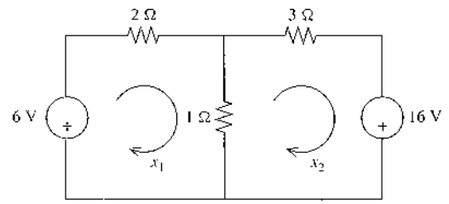
so $i_L = \frac{17}{18}$ A.

4.7.1. A circuit satisfies the equations

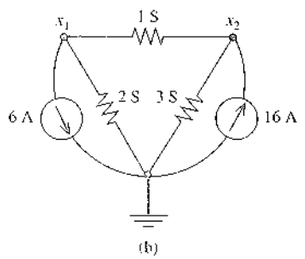
$$3x_1 - x_2 = -6$$

$$-x_1 + 4x_2 = 16$$

where x_1 and x_2 are both node voltages or both mesh currents. Sketch the primal and dual circuits.

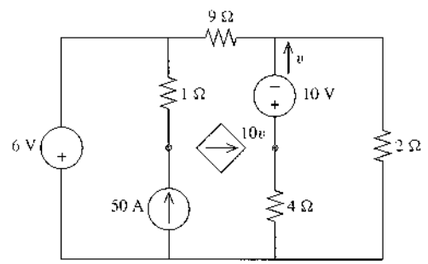


(a)



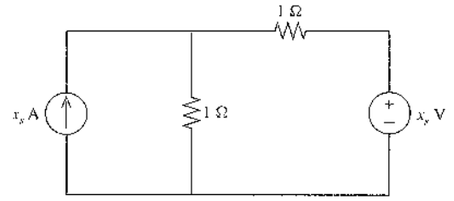
(b)

4.7.2. Find the dual of Fig. 4.15.



4.7.3. Find a circuit that is *self-dual*; that is, its dual is identical to itself.

There are many self-dual circuits. Here is one.



4.8 VIRTUAL SHORT PRINCIPLE FOR OP AMPS

A general approach to the analysis of circuits containing op amps was developed in Chapter 3. Each op amp in the circuit diagram is replaced by a suitable model, for instance, the ideal voltage amplifier model of Fig. 3.7 or the improved model of Fig. 3.9. Since these models contain only resistors and dependent sources, the resulting circuit may be analyzed just as any other linear circuit would be.

There is another approach to analyzing op amp circuits that, where applicable, is far simpler to carry out. It does not require replacing op amps by models but is applied to the original circuit diagram directly. For each op amp in the circuit (Fig. 4.24), two rules are used:

1. The current into the op amp input terminals is zero: $i_{i1} = i_{i2} = 0$.
2. The voltage across the op amp input terminals is zero: $v_{in} = 0$.

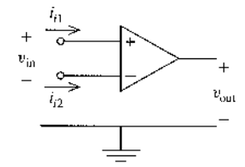
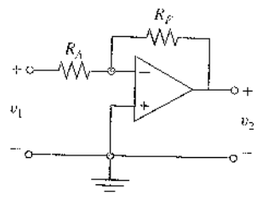


FIGURE 4.24 Virtual open and short principles.

The first of these rules asserts that the op amp input behaves like an open circuit. Since the ideal voltage amplifier model of Fig. 3.7 employs an open-circuited input, this rule, the *virtual open principle*, is not hard to justify. The second rule asserts that the op amp input also behaves like a short circuit (no voltage drop across its terminals). This rule, the *virtual short principle*, requires a bit more reasoning. We first give examples of the use of these rules and then consider the justification of the virtual short principle and the key assumption under which it is valid.

Example 4.18



Inverting amplifier.

In Section 3.5 the inverting amplifier circuit shown in Fig. 4.25 was analyzed. Replacing the op amp by its ideal voltage amplifier model with open-loop gain A, it was found that the output voltage v_2 is given by

$$v_2 = \frac{-R_F}{R_A + \frac{1}{A}(R_A + R_F)} v_1$$

In the high-op-amp-gain limit (as A becomes arbitrarily large), we have the voltage transfer relation for the inverting amplifier (3.8) repeated here:

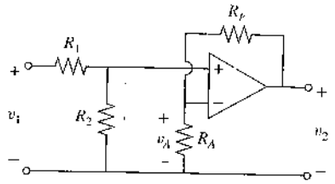
$$v_2 = -\frac{R_F}{R_A} v_1$$

We now show how (4.25b) can be derived from the virtual open and virtual short principles *without resort to models or passing to limits*. Since the noninverting input terminal in Fig. 4.25 is at ground, by the virtual short principle, so is the inverting input terminal. Then the node connecting the two resistors is at ground potential, implying that the full voltage v_1 is across R_A and v_2 is across R_F. Summing the currents into this node gives

$$\frac{v_1}{R_A} + \frac{v_2}{R_F} = 0$$

In this equation the current into the op amp at this node is zero by the virtual open principle. Rearranging (4.26) slightly yields the voltage transfer relation (4.25b).

Example 4.19



Circuit for Example 4.19.

Find v_2 in Fig. 4.26.

Since no current flows into the op amp at the inverting input terminal (the virtual open principle), R_F and R_A are in series, and thus they divide the voltage v_2, with

$$v_A = \frac{R_A}{R_A + R_F} v_2$$

Since no voltage is dropped across the op amp input (the virtual short principle), this is also the voltage at the noninverting input terminal, or

$$v = v_A$$

Looking at the left loop, R_1 and R_2 are in series across v_1 (again, no current into the op amp due to the virtual open principle). Thus by voltage division of v_1,

$$v = \frac{R_2}{R_1 + R_2} v_1$$

Combining these last three equations, it must be that

$$\frac{R_A}{R_A + R_F} v_2 = \frac{R_2}{R_1 + R_2} v_1$$

or

$$v_2 = \left(1 + \frac{R_F}{R_A}\right) \frac{R_2}{R_1 + R_2} v_1$$

To justify the virtual short principle, we reason as follows. Assume, in a given op amp circuit, that the op amp output voltage remains finite in the high-op-amp-gain limit as the op amp gain A becomes arbitrarily large. Since the output voltage is A times the input, *as A becomes arbitrarily large the op amp output voltage will remain finite only if the input voltage becomes arbitrarily small.* Passing to the high-op-amp-gain limit, the input must be exactly zero, or we would have the product of a nonzero input times an infinite gain producing an infinite output voltage, contradicting the assumption that the output remain finite.

Thus, for circuits whose op amp output voltages remain finite in the high-op-amp-gain limit, the virtual short principle follows. What class of circuits is this? Certainly, the circuit described by (4.25a) is in this class. Note that as A gets larger and larger, the op amp output voltage v_2 tends not to infinity, but rather, to the finite value $(-R_F/R_A)v_1$ specified in (4.25b).

Indeed, any circuit with negative feedback, such as all the op amp building block circuits described in Chapter 3, will be "well behaved" enough to tend to a finite output in the high-op-amp-gain limit. Thus *we may use the virtual short and open principles (4.25) to analyze any op amp circuit that employs negative feedback.* As has been noted previously, only such circuits are practical linear op amp circuits anyway, since the rest will have op amp output voltages that "blow up," exceeding their physical output voltage ratings when plugged in. This violation of safe limits will result in behavior certainly nonlinear and perhaps rapidly destructive. So our virtual open and short rules will be valid for any useful linear op amp circuit.

Example 4.20

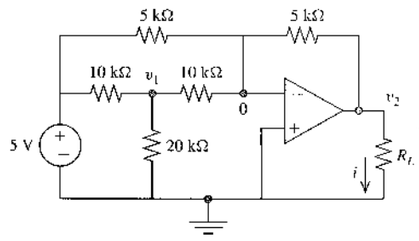


FIGURE 4.27 More complicated op amp circuit.

The op amp circuits analyzed so far have been of the simple type, that is, reducible to a single equation. This example demonstrates how to to solve a general op amp circuit problem. We will use nodal analysis together with the virtual short and open principles. We seek the load current i and the voltage v.

The circuit of Fig. 4.27 has two unknown node voltages, labeled v_1 and v_2. By the virtual short principle, the inverting input terminal is at ground potential, and we have labeled its node voltage 0 V. Thus we need two node equations. For circuits not containing op amps, we write KCL at each node or supernode containing an unknown node voltage. Where op amps are present, we modify that strategy by *writing KCL at the inverting input node rather than the op amp output node.* At nodes 1 and the noninverting input node, respectively,

$$\frac{1}{10 \times 10^3} + \frac{1}{20 \times 10^3} v_1 + \frac{1}{10 \times 10^3}(v_1 - 5) = 0$$

$$\frac{1}{10 \times 10^3}(0 - v_1) + \frac{1}{5 \times 10^3}(0 - 5) + \frac{1}{5 \times 10^3}(0 - v_2) = 0$$

The virtual open principle was used in (4.27b). Solving (4.27a) for v_1 yields

$$v_1 = 2 \text{ V}$$

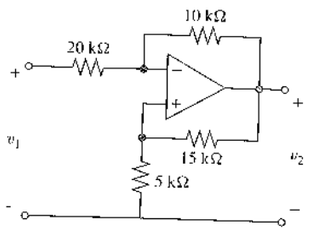
and substituting into (4.27b) gives

$$v_2 = -\frac{1.2}{-0.2} = -6 \text{ V}$$

Then the desired unknowns are $i = -6/1000 = -6$ mA and $v = 5 - v_1 = 3$ V.

We will consistently choose not to write node equations at op amp output nodes to avoid complications due to the incomplete nature of the op amp circuit diagram. As mentioned in Chapter 3, there are several connections to the op amp not shown in the circuit diagram. In particular, the lead connecting the op amp output to ground is usually omitted (see Fig. 3.6). Indeed, drawing the op amp as a three-terminal device appears to violate the generalized KCL, since each of the two input currents is zero (the virtual open principle), while the op amp output current in general is not zero. The missing output lead to ground makes it difficult to determine the op amp output current directly and thus hard to write a node equation there. Instead, we choose to write KCL at the inverting input node, where all currents can be easily determined. Once the circuit is broken, the op amp output current can be found easily if so desired.

The virtual short and open principles rules are designated "virtual" because in practice the input current and input voltage to physical op amps being operated in negative feedback circuits are small enough to be virtually, although not exactly, zero. This is exactly what we would expect if the op amp input had high but not infinite resistance, and the op amp gain A were large but finite (conditions characteristic of physical op amps). The virtual open and short principles are excellent approximations and may be used to simplify analysis wherever op amps are operated in the linear, negative feedback mode. Note, however, that use of the virtual short principle applies exactly in the high-op-amp-gain limit only. If the circuit variables for a specific finite gain A are desired, the less convenient model-substitution method developed in Chapter 3 should be employed.



4.8.1. Use the virtual short and open (* 11 12) principles to find the voltage transfer ratio v_2/v_1 for the noninverting amplifier of Fig. 3.19.

Answer: $1 + R_F/R_A$

4.8.2. Use the virtual short and open principles to find the voltage transfer ratio v_2/v_1 for the voltage follower of Fig. 3.14.

Answer: 1

4.8.3. Use nodal analysis and the virtual short and open principles to find the voltage transfer ratio v_2/v_1.

Answer: -4/5

We have seen in this chapter that general linear resistive circuits can be analyzed efficiently by writing and then solving a set of simultaneous nodal or mesh equations. The computations required for hand solution of a simple circuit are minimal, and one with just two or three unknowns (node voltages or mesh currents) is still reasonable. But solving perhaps six equations in six unknowns or 20 equations in 20 unknowns by hand takes quite a long time. Moreover, the great number of steps makes the hand solution process error prone, demanding even more time for careful step-by-step checking. Most of this extensive work time is occupied doing repetitive calculations, with little payoff in terms of learning or new insights.

For this reason, generations of students of electrical engineering had to content themselves with working with a limited number of circuits each of very modest size. Valuable experience that could be gained through wider-ranging analysis exploring somewhat bigger circuits was precluded by the daunting prospect of the price demanded, hours of repetitive hand "cranking." Fortunately, this is no longer the case. Modern technology offers the digital computer, a splendid tool capable of relieving us of the tedious and inefficient use of time inherent in lengthy hand calculations. Over the last four decades the computer has revolutionized how every kind of engineering is done, whether the design of a space station, the improvement of automobile emission control devices, or the analysis of electrical circuits. With computer support for repetitive calculations, larger and more realistic problems can be solved and more experience with more complex systems can be gained. Both the student and the engineering practitioner benefit by the opportunity for more advanced, more realistic problem-solving experience afforded by the assistance of a computer equipped with the right software programs. While many useful computer programs are available to aid in the analysis of electrical circuits, we shall settle on one that is widely available on a variety of computer platforms. This software program is called SPICE (Simulation Program with Integrated Circuit Emphasis).

We shall make frequent use of this tool. One common use will be to check hand results (many of the problems in subsequent chapters ask the student to solve a circuit problem by hand and then "check using SPICE"). A second will be to solve problems involving many nodes and/or meshes. A third important use of SPICE will be to explore the effect of changing the value of a single circuit element, such as a specific resistance within a circuit. This is especially important for "cut-and-try" design, where we may try several values of a circuit element before settling on one resulting in the best circuit performance. Typically, the entire numerical analysis needs to be repeated for each value tried, and the effort involved effectively eliminates hand calculation as a realistic possibility in many design studies. The alternative to repeated trials is to write down an equation explicitly describing the functional dependence of a selected measure of circuit performance on the parameter in question and then somehow solve this equation for the optimal value. With complex circuits this is seldom a practical alternative to the repetitive process of cut-and-try.

SPICE will prove helpful in each of these ways, but it remains only a computational tool. One way in which we cannot use SPICE is to substitute for the intellectually challenging, sometimes painful process of acquiring a mastery of circuits concepts and

methods. Problem-solving skills deepen only through concentrated thought and repeated experience. Just as ownership of a good desk calculator does not make one a skilled accountant or access to a digital music synthesizer does not make one a noted composer, so in electrical circuit work the computer can really only save us a bit of time. But since time is a precious commodity to every busy student or professional, we shall take maximum advantage of this time-saving tool where we can.

SPICE, developed with the support of the National Science Foundation by students and faculty at the University of California at Berkeley, is a program that has become the de facto standard for analog circuit simulation. SPICE's success is due to a combination of virtues: the program is powerful, easy to operate, and inexpensive, is supported on most computers of engineering significance, and is well maintained. This is not to suggest that SPICE is the absolute best product available for circuit analysis; indeed, there are more powerful and more comprehensive software packages available commercially. But SPICE will do what is needed for our purposes and is more widely and inexpensively available than other choices.

PSpice, a version of SPICE rewritten for personal computers such as the IBM PC family, the Apple Macintosh, and many other smaller computer platforms, is often used for illustration in this book. Although they differ in some particulars, where possible we shall stick to features common to both the SPICE and PSpice "dialects." Indeed, when we refer to SPICE, we will generically mean both SPICE, as written at the University of California at Berkeley for mainframe computers, and PSpice, as written by the MicroSim Corporation for various personal computers and workstations. Where necessary to distinguish them, we will explicitly do so. Other variants of SPICE have been developed, such as HSPICE and SPICE-Plus, which bear considerable similarity as well.

The student unfamiliar with SPICE is encouraged to review Appendix D. The remainder of this section is devoted to an introduction to those SPICE formats and commands necessary for the class of circuit analysis problems we have studied thus far, linear resistive circuits with constant (dc) independent sources.

Before the SPICE program can be run, the first step is the creation of a SPICE *input file*. The purpose of the SPICE input file is threefold: to specify completely the circuit we wish to analyze, to alert the SPICE program to which type of analysis we wish done, and to identify the desired output variables. The SPICE input file is formatted as an ordinary text file (ASCII file) and is typically created with a utility software program called a *text editor* available on the computer system being used.*

The SPICE input file is organized as a set of statements, each of which terminates with a carriage return. The first statement is a *title statement*, identifying the problem under study. There is no particular format to this statement, except that the first column must not be blank. The last statement is the *.END statement*. All other statements come between these two.

The circuit is identified by a set of *element statements*. Each statement specifies the element type, name, location in the circuit, and its element value. The format for a

*It is assumed the reader is familiar with the operating system, a text editor, and a command to print a text file to screen or hard copy on the computer system on which SPICE will be run.

resistor element statement is

```
RXXXXX      N1      N2      VALUE
```

where R indicates it is a resistor, XXXX is its name, N1 and N2 are the two nodes at its terminals, and VALUE its resistance in ohms. Note that symbols from SPICE input or output files will always be distinguished by the use of the boldface fixed-width font shown.

The format for an independent dc voltage source is

```
VXXXXX      N1    N2    DC     VALUE
```

and the independent dc current source is

```
IXXXXX      N1    N2    DC     VALUE
```

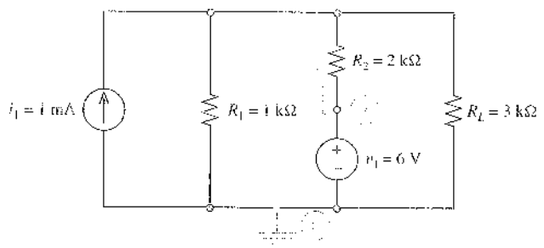
Although the order in which the nodes are listed does not matter for a resistor, for independent sources order is used to specify the reference direction. The plus sign of the voltage source reference direction is taken to be at the node N1, and the arrow of the current source reference direction is taken to point from N1 toward N2. VALUE is the source function in volts (for the voltage source) or amperes (for the current source).

The SPICE input file will also contain *control statements* that specify the type of analysis to be done and the outputs to be viewed. The default mode of analysis is dc; thus, in the absence of statements specifying other modes of analysis (such as .AC or .TRAN statements used in later chapters), we will get the dc solution we are presently interested in generating. The output may be a list of numbers, or it may be plotted. For dc circuits, whose circuit variables do not vary over time, only the numerical output needs to be considered. For outputting a list of numbers, the control statement is

```
.PRINT     DC CVLIST
```

CVLIST is a list of the circuit variables we wish printed out. Each voltage in the list is formatted V(I,J), the node voltage at node I minus that at node J (if $J = 0$, the reference node, the second argument may be omitted). Each current in CVLIST is formatted I(NXXXX). This is the current I through element NXXXX, with the reference arrow for I pointing from the first node to the second node appearing in the element statement for NXXXX. For instance, V(2,5) is the voltage at node 2 minus the voltage at node 5, V(3) is the voltage at node 3 relative to node 0 (ground), and I(VREF) is the current through the voltage source VREF, with current reference direction pointing from the plus end of VREF (the first node on the VREF element statement) toward the minus end.

As the first SPICE example, consider the circuit of Fig. 4.28, where the node designations have been circled for clarity. Note that the reference node is designated as node 0. The SPICE input file that



Circuit for first SPICE example.

will result in a dc analysis and output the values of node voltage 1 and the current i through the 6 V source, is given next.

```
SPICE input file for first example
*That was the title  statement.
*Here are the element  statements:
IS1   0   1   DC    1MA
VS1   2   0   DC    6
R1    1   0   1KILOHM
R2    1   2   2K
RL    1   0   3K
*Here are the control  statements:
.PRINT    DC   V(1)   I(VS1)
*Type of analysis defaults to dc
.END
```

Note that the element statements fully describe the circuit (which elements are where and what the source reference directions are). *The element statements comprise a complete representation of the circuit diagram.*

The default units for the element values are volts, amperes, and ohms unless otherwise specified. Thus the value of 6 in the voltage source element statement means 6 V. The units may be scaled by a power of 10 if an appropriate suffix is attached to the numerical value: M means 10^{-3}, K means 10^{3}, and so on (see Appendix D for the complete list). Other suffix letters are ignored (such as the A in the IS1 element statement added only for clarity). Any line beginning with an asterisk * is a comment statement and is ignored by the SPICE program.

Suppose that this SPICE input file is typed into the computer and stored under the filename `EX4-2A.CIR`. The SPICE circuit analysis program itself may now be run, using a command syntax that will vary with the variant of SPICE and operating system being used. Upon invoking the SPICE program, the user specifies `EX4-2A.CIR` as the input file to be analyzed and declares a name for the output file to be created (typically, `EX4-2A.OUT`). After execution is complete, examination of this output file on the terminal screen or as hard copy will reveal (among other material, such as an echo of the SPICE input file) the lines

```
NODE VOLTAGE     NODE VOLTAGE
 (1)5.1429         (2)6.0000

VOLTAGE SOURCE CURRENTS
NAME CURRENT
VS1 -7.143E-04
```

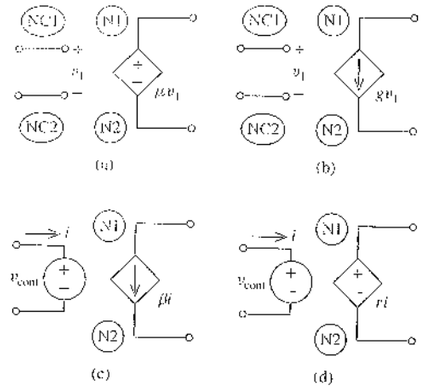
Thus the values for node voltage 1 and current i are 5.1429 V and −0.7143 mA, respectively.

A somewhat peculiar limitation in some versions of SPICE is that only currents through independent voltage sources may be included in the output statement as part of `CVLIST`. This limitation is found, for instance, in SPICE versions 1 and 2 and also in SPICE version 3 when using "batch mode." To output currents that do not happen to flow through independent voltage sources, we must add a *dummy voltage source*, a +0 V voltage source in series with the desired current. This source will have no other effect on the circuit, since no current or voltage will be changed by its inclusion. Most SPICE variants, such as PSpice, permit the more general form `I(NXXXX)` in `CVLIST`, and dummy voltage sources are not needed to `.PRINT` currents. The documentation for the specific variant and version of SPICE should be consulted to determine whether only currents through independent voltage sources are permitted.

Along with the resistors and independent sources discussed above, resistive circuits may also contain dependent sources. These come in the four "flavors" illustrated in Fig. 3.1. In SPICE, the four types of dependent sources have element statements with the following formats:

```
VCVS:    EXXXXX   N1   N2   NC1    NC2    MU
VCCS:    GXXXXX   N1   N2   NC1    NC2    G
CCCS:    FXXXXX   N1   N2   VCONT         BETA
CCVS:    HXXXXX   N1   N2   VCONT         R
```

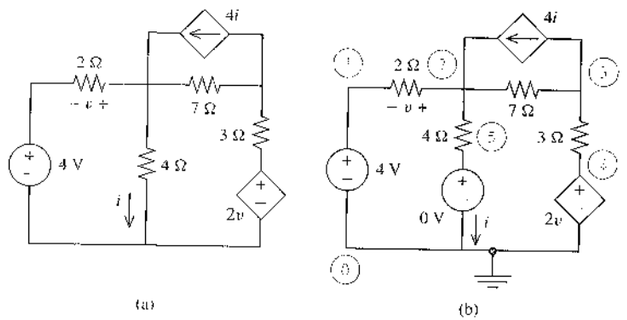
The conventions for these elements are illustrated in Fig. 4.29. In each case, `N1` and `N2` are the source nodes. For the voltage sources (VCVS and CCVS), `N1` is at the positive end and `N2` at the negative end of the voltage reference direction, just as was the convention for independent voltage source element statements. For the current sources (VCCS and CCCS), the arrow, indicating current reference direction, points from `N1` to `N2`, once again consistent with the case of independent sources. `NC1` and `NC2` are the nodes of the controlling voltage, with `NC1` at its plus end. `VCONT` is the name of an independent voltage source through which the controlling current flows. This current is defined to have its reference direction arrow point from the positive node of `VCONT` to the negative. If we wish to use a controlling current that does not flow through an independent voltage source, we need to insert a dummy voltage source in series with this current.



Dependent source conventions in SPICE.

The final field in each of these statements is the constant that multiplies the controlling variable to give the dependent source function. These were defined in Chapter 3 as voltage gain μ for the VCVS, transconductance g for the VCCS, current gain β for the CCCS, and transresistance r for the CCVS.

This example contains two controlled sources, a CCCS and a VCVS. Since the controlling current i does not flow through a voltage source, we must insert a dummy voltage source in series to permit this current to be used. We wish to determine the voltage v and also the power dissipated by the 2-Ω resistor (see Fig. 4.30). SPICE always outputs the power delivered by each independent source in a dc problem without even being asked. Power delivered by or dissipated by other elements, though, must be computed by hand from current and voltage. So we will print out v and the current through the 4-V source, which happens to be in series and thus carries the



(a) Circuit; (b) prepared for input as SPICE input file.

same current as the desired current through the 2-Ω resistor. The SPICE input file follows.

```
SPICE input file for the second example
*
V1      1       0       DC      4
R1      1       2       2
R2      2       5       4
*Here comes the dummy voltage source:
VD      5       0       0
R3      2       3       7
*Here is the CCCS with current gain 4:
F1      3       2       VD      4
R4      3       4       3
*And the VCVS with voltage gain 2:
E1      4       0       2       1       2
.PRINT  DC  V(2,1)   I(V1)
.END
```

The resulting SPICE output shows us that the 4-V source current is −1.4 A and the voltage v is −28.0 V. Then the power dissipated by the 2-Ω resistor is $(-1.4)(-28.0) = 39.2$ W.

SPICE has many additional element statements, modes of analysis, and other features that we consider as the need for them arises in later chapters. For the present, contenting ourselves with dc analysis of linear resistive circuits, we introduce just two additional capabilities of SPICE: the dc sweep and the subcircuit.

A dc sweep is performed when we wish to step through several values of a dc source, each time performing a dc analysis and printing the outputs. This mode of analysis is useful, for instance, in determining whether an output variable will exceed specified limits over an anticipated range of input (dc source) values. A dc sweep mode of analysis is triggered by including a `.DC` control statement in the SPICE input file:

```
.DC  SNAME  START  STOP  DELTA
```

`SNAME` is the name of the independent source to be swept (current or voltage), `START` and `STOP` are the initial and final source values desired, and `DELTA` is the increment between successive source values within the sweep. The next example will include a dc sweep.

A powerful feature of SPICE is the ability to define subcircuits, which can be typed once in a SPICE input file and then referenced as often as needed without repeating its element statements. Each time the subcircuit is needed, it is "called," similar to a subroutine call in FORTRAN or a function invocation in C. The subcircuit is defined by including a set of lines beginning with a `.SUBCKT` statement and ending with an

.ENDS (end subcircuit) statement. The format is

```
.SUBCKT  SUBNAME NA      NB      NC...
*Subcircuit element  statements go here
...
...
.ENDS
```

SUBNAME is a name given to the subcircuit. The remaining fields on the .SUBCKT statement are the numbers of the nodes of the subcircuit at which the subcircuit will be connected to the main circuit. The subcircuit is called from elsewhere in the SPICE input file by a statement of the form

```
XYYYYY  N1  N2  N3...    SUBNAME
```

Here X indicates a subcircuit call, YYYYY is an arbitrary identifier for this particular subcircuit invocation, N1, N2, N3, ... are the node numbers within the main circuit at which the subroutine is connected, and SUBNAME is the name of the subcircuit to be used.

SPICE is informed which subcircuit node to connect to which main circuit node by the order in which nodes appear. *The first node on the* .SUBCKT *statement will be connected to the first node on the* X *subroutine call statement, the second to the second, and so on.* The node numbers within a subcircuit definition are independent of the node numbers in the main part of the SPICE input file (with the exception only of the reference node or common ground, node 0, which always refers to the same node wherever it may appear). This independent numbering of nodes corresponds to the independent use of "formal" variables in a FORTRAN subroutine or C function and "actual" variables in the main program. Independent node numbering is what permits us to insert the same subroutine in several different places in the main circuit without confusion.

Example 4.23

As a final example, let us consider the circuit containing two op amps shown in Fig. 4.31. We will replace each op amp by the improved circuit model of Fig. 3.9 (repeated in Fig. 4.32), defining this op amp model as a .SUBCKT to avoid entering its element statements twice. The type of analysis we will do is a dc sweep of the independent dc source between the limits of −5 and +5 V, printing out the op amp output voltages. Note that same node numbers are used independently in the main circuit and the subcircuit. The SPICE input file used is:

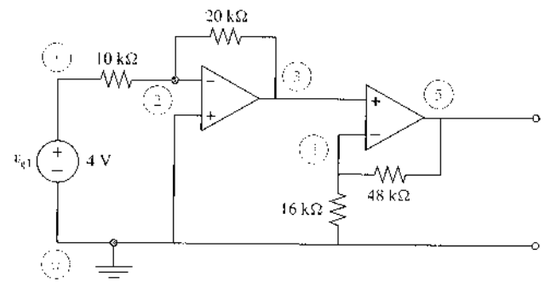


FIGURE 4.31 Circuit for Example 4.23.

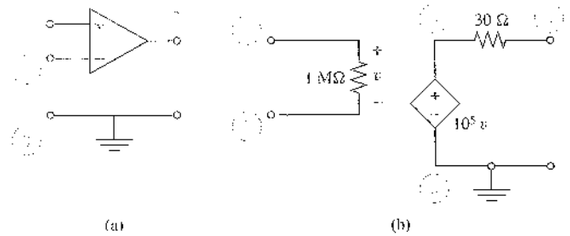


FIGURE 4.32 Subcircuit for Example 4.23.

```
Ex4-23: circuit with two op amps
*
*Main circuit:
VG1     1       0       DC          1
R1      1       2       10K
R2      2       3       20K
X1      0       2         3     OPAMP
*Order on nodes on above  statement matches
*order on .subckt  statement: + in, - in, out.
R3      4       0       16K
R4      4       5       48K
X2      3       4         5     OPAMP
*Here comes the op amp subcircuit:
.SUBCKT  OPAMP  1       2           3
*Node 1 is the +in, 2 the -in, and 3 the output node
RIN     1       2      1MEG
E1      4       0         1         2   100K
R0      4       3        30
.ENDS
*Finally the analysis mode and output statments:
.DC   VG1       -5        +5          1
.PRINT DC     V(3)      V(5)
.END
```

After running SPICE with this input file, the output contains the following lines:

```
      VG1             V(3)            V(5)
-5.00E+00       1.00E+01        4.00E+01
-4.00E+00       8.00E+00        3.20E+01
-3.00E+00       6.00E+00        2.40E+01
-2.00E+00       4.00E+00        1.60E+01
-1.00E+00       2.00E+00        7.99E+00
 0.00E+00       0.00E+00        0.00E+00
 1.00E+00      -2.00E+00       -7.99E+00
 2.00E+00      -4.00E+00       -1.60E+01
 3.00E+00      -6.00E+00       -2.40E+01
 4.00E+00      -8.00E+00       -3.20E+01
 5.00E+00      -1.00E+01       -4.00E+01
```

These are the op amp outputs for input voltage VG1 swept from -5 to $+5$ V by increments of $+1$ V. Note that in each row $V(3) = -2$ VG1, confirming the observation that the first op amp stage is an inverting amplifier with gain of $-R_2/R_1 = -2$. Also, $V(5) = 4$ V(3), since V(3) is input into a noninverting amplifier with gain of $(1 + R_4/R_3) = +4$. Note also that the column of V(3) and of V(5) values both scale with VG1, a consequence of the principle of proportionality. For instance, the values of these variables for VG1 $= 2$ V are double those for VG1 $= 1$ V.

For all its power, SPICE is just a numerical algorithm and will have round-off errors. For instance, V(5) is exactly four times V(3), as expected, for almost all values of V(3). But the value V(3) $= 2.0000$ V yields V(5) $= 7.9999$ V, not 8.000 V. SPICE is not an algorithm endowed with artificial intelligence. It does strictly numerical calculations, not symbolic ones. With a hand calculator we may be surprised that division by a large number followed by multiplication by that same number sometimes will not return the original value to the display, so we cannot expect SPICE to give exact values, just close numerical approximations.

4.9.1. Write a SPICE input file for the circuit shown in Exercise 4.1.1.


```
Exercise 4.9.1 SPICE input file
R11     0       1       10
R2      1       2       2
V1      2       0       DC      24
R3      2       3       3
I2      0       3       DC      4
.PRINT  DC   I(V1)   V(3,0)
.END
```

4.9.2. Write a SPICE input file for Fig. 4.6.

```
Exercise 4.9.2 SPICE input file
R1      1       0       1
V1      2       1       DC      12
R2      2       3       3
H1      3       0       V1      -2
I2      0       2       DC      6
.PRINT DC    I(V1)  V(2,3)
.END
```

The *negative* of the value I(V1) will correspond to the desired output variable i in the figure. This sign change could be avoided, as well as the minus sign on the transresistance of the controlled source (see the H statement), by putting a dummy voltage source in series with the 12-V source but with opposite reference direction as the 12-V source.

SUMMARY

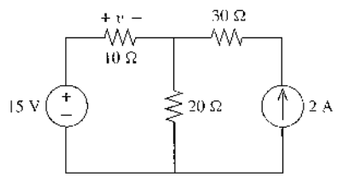
In this chapter tools of great generality for the analysis of linear circuits are introduced: the principle of superposition, the methods of nodal and mesh analysis, and a computer program capable of simulating a very broad range of circuits called SPICE.

- Linear circuits satisfy the principle of proportionality: scaling all independent sources by the same factor scales all responses similarly.
- They also satisfy the principle of superposition: the total response is the superposition, or sum, of responses to each independent source with the others killed.
- In nodal analysis the key variables are the node voltages, and KCL is written at all nodes but the reference node.
- In nodal analysis, each voltage source defines a supernode containing only one unknown node voltage and KCL is applied to the supernode.
- In mesh analysis the key variables are the mesh currents, and KVL is written around all meshes.

- In mesh analysis each current source defines a supermesh containing only one unknown loop current and KVL is applied around the supermesh.
- Once a circuit is broken, it is an easy matter to express any desired current or voltage in terms of the known key variables (node voltages or mesh currents).
- Dual circuits are constructed by placing dual node i inside primal mesh i for each primal mesh, and a reference node outside. For each element in primal meshes i and j, its dual is connected between the nodes i and j of the dual circuit.
- The virtual short and virtual open principles for analyzing op amp circuits in negative feedback configurations state that the voltage across, and the current into, the input terminals of the op amp both equal zero.
- Nodal analysis is recommended for op amp circuits using the virtual short and open principles. Do not write nodal KCL equations at op amp output nodes.
- SPICE requires an input file which details the circuit, establishes the desired mode of operation and indicates the desired outputs variables and formats.

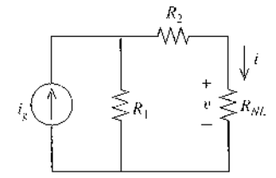
The examples in this chapter apply the principle of superposition, and the methods of mesh and nodal analysis, strictly to resistive circuits studied in the time domain. This is not a limitation of these methods but reflects the limited experience we have gained so far. Superposition, mesh and nodal analysis will continue to apply, to be our most basic tools, throughout the book: in the analysis of linear circuits containing storage elements (Chapters 5 to 7), those studied in the phasor domain (Chapters 8 to 11), the s-domain (Chapters 12 to 16), and the Fourier domain (Chapter 17).

4.1. By what factor must we scale *both* source functions in order to get $v = +1$ V across the 10-Ω resistor?

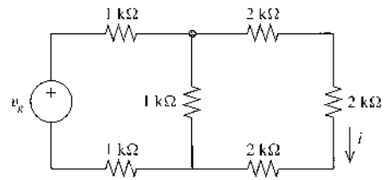


4.2. What does the principle of proportionality predict will happen to a linear circuit if all independent sources are scaled by zero?

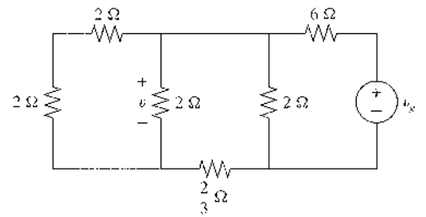
4.3. R_1 and R_2 are linear resistors, but R_{NL} satisfies $i = e^{-v}$. Demonstrate that this circuit violates the proportionality principle.



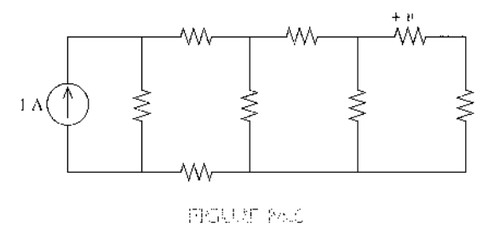
4.4. If $i = 1$ A, what must v_g be? Use this to determine i if $v_g = 1$ kV.



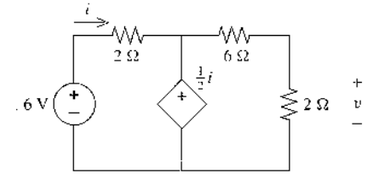
4.5. Use proportionality to find the source function v_g that makes $v = 6$ V.



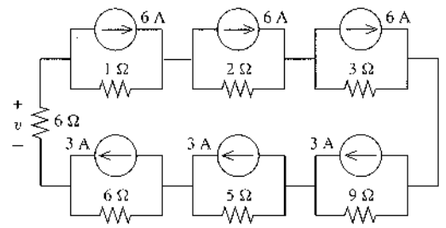
4.6. Find v using the proportionality principle. All resistors are 10 Ω.



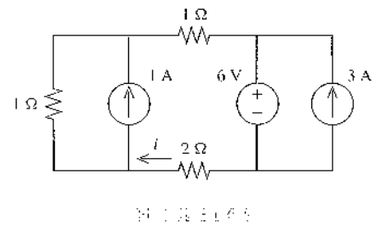
4.7. Find v using the proportionality principle.



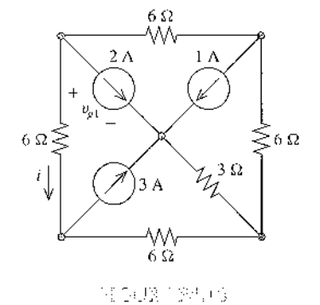
4.8. Find v by superposition. Check using Thevenin–Norton transformations.



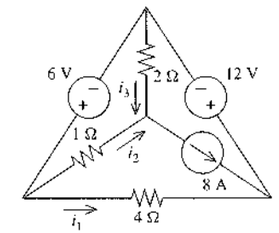
4.9. Solve for i by superposition.



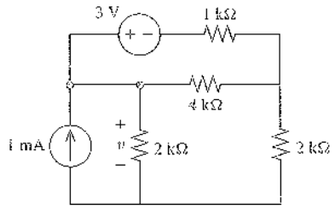
4.10. Use superposition to find v_{g1} and i.



4.11. Use superposition to find the three resistive currents shown.

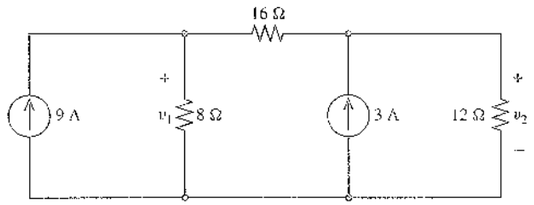


4.12. Use superposition to find v. Check by using Thevenin–Norton transformations.

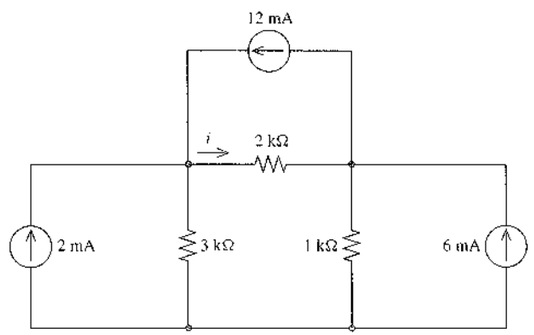


4.13. What should we replace the 6-A source function by in Problem 4.18 in order that $v_1 = 0$?

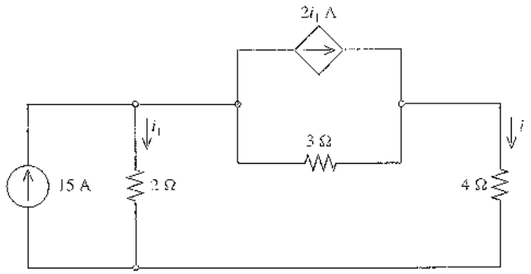
4.14. Using nodal analysis, find v_1 and v_2.



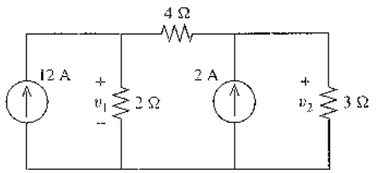
4.15. Using nodal analysis, find i.



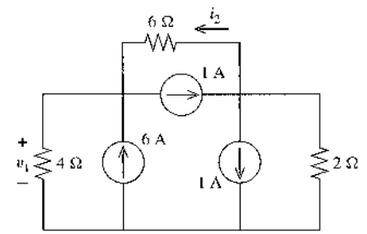
4.16. Using nodal analysis, find i and i_1.



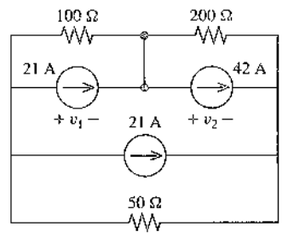
4.17. Use nodal analysis to find the indicated variables. Check using superposition.



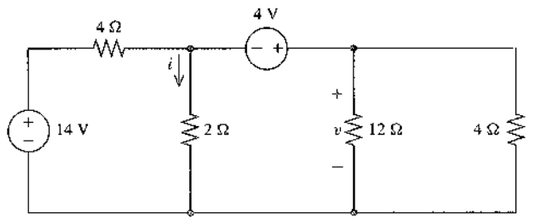
4.18. Repeat Problem 4.17 for the circuit shown.



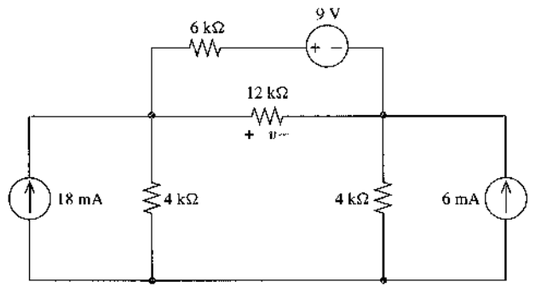
4.19. Repeat Problem 4.17 for the circuit shown.



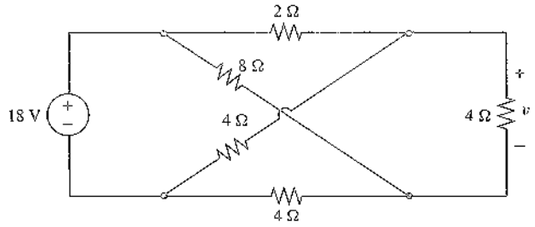
4.20. Find v and i using nodal analysis.



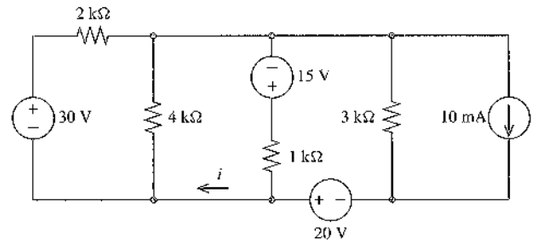
4.21. Using nodal analysis, find v.



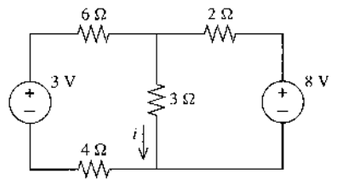
4.22. Using nodal analysis, find v.



4.23. Using nodal analysis, find i.



4.24. Repeat Problem 4.17 for the circuit shown.

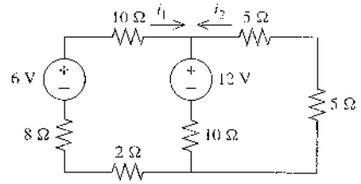


4.25. Replace the 8-V independent source in Problem 4.24 with an $8i$ CCVS (same reference direction as shown). Find i.

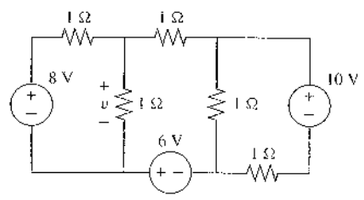
4.26. Show that by the very definition of node voltages, KVL around every loop is satisfied automatically.

4.27. Show that by the very definition of mesh currents, KCL at each node is satisfied automatically.

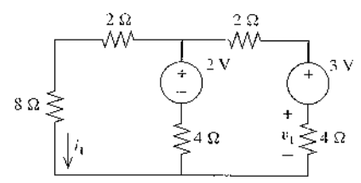
4.28. Use mesh analysis to find the indicated variables. Check using superposition.



4.29. Repeat Problem 4.28 for the circuit given.



4.30. Use mesh analysis to find the indicated variables. Solve the mesh equations by Cramer's rule.



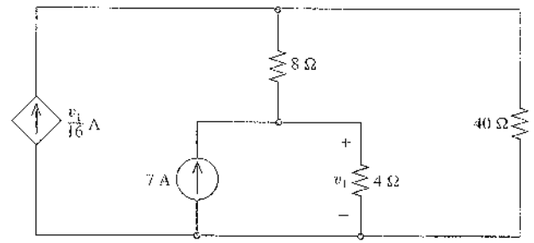
4.31. Repeat Problem 4.30 using matrix inversion.

4.32. Repeat Problem 4.30 using Gauss elimination.

4.33. Repeat Problem 4.30 if the 2-V independent source is replaced by a VCVS (same reference direction) with source function $-2v_1$.

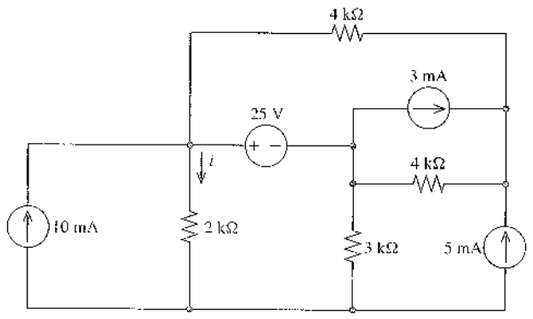
4.34. Solve Problem 4.3 using mesh analysis.

4.35. Using mesh or loop analysis, find the power delivered to the 8-Ω resistor.

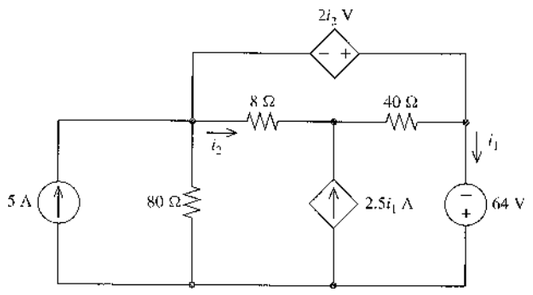


4.36. Solve Problem 4.15 using mesh analysis.

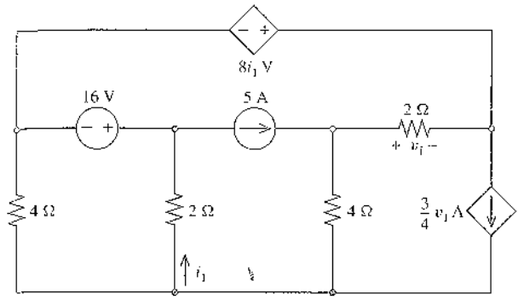
4.37. Find i using both nodal and loop analysis.



4.38. Find i_1 and i_2 using loop or mesh analysis.



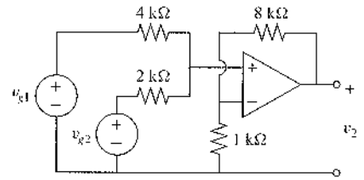
4.39. Find v_1.



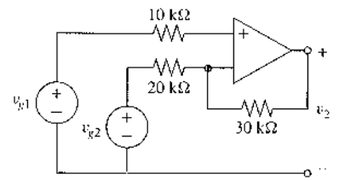
4.40. Solve Problem 4.56 by mesh analysis.

4.41. Solve Problem 4.23 using nodal analysis, and solve again using mesh analysis.

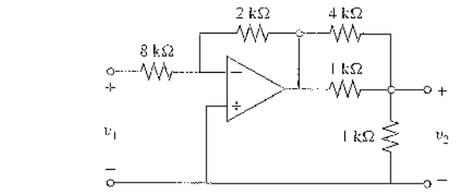
4.42. Using the virtual short and open principles, find v_2 in terms of v_{g1} and v_{g2}.



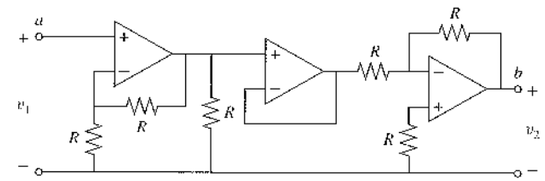
4.43. Repeat Problem 4.42 for the circuit shown.



4.44. Use the virtual short and open principles to find the voltage transfer ratio v_2/v_1.



4.45. Repeat Problem 4.44 for this circuit. $R = 10$ kΩ.



4.46. Repeat Problem 4.45, but add a 10-kΩ resistor between a and b.

4.47. Use SPICE to solve Problem 4.53.

4.48. Use SPICE to solve Problem 4.23.

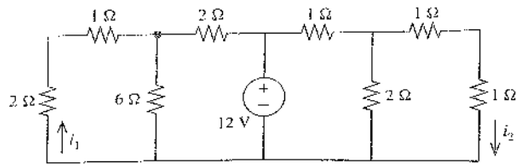
4.49. Use SPICE to solve Problem 4.28. Do a dc sweep from 0 to 12 V on the 6-V source, with 1-V increments.

4.50. Use SPICE to solve Problem 4.44. Replace the op amp by the improved op amp model of Fig. 3.9, with $R_i = 1$ MΩ, $R_o = 30$ Ω, and $A = 10^5$.

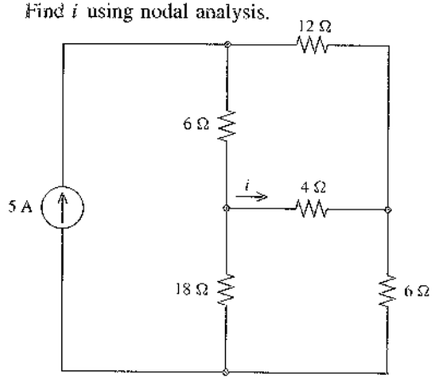
4.51. Use SPICE to solve Problem 4.46. Replace the op amps as directed in Problem 4.50.

4.52. This problem is intended to check how close the input current and voltage to an op amp are to zero (as the virtual open and short principles suggest). In the circuit of Problem 4.44, replace each op amp by its improved model (Fig. 3.9) with $R_i = 1$ MΩ, $R_o = 30$ Ω, and $A = 10^5$. Let $v_1 = 1$ V. Use SPICE to determine the currents into each op amp and the voltages across these inputs.

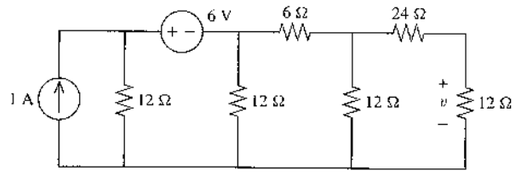
4.53. Use proportionality to find i_1 and i_2.



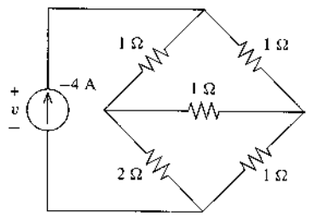
Find i using nodal analysis.



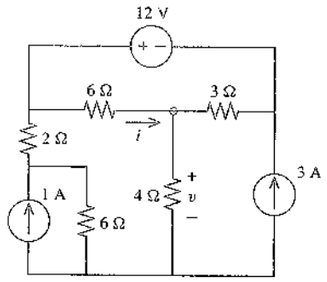
Use superposition to find v. For each component problem, use the proportionality principle; that is, assume that the component $v = 1$ V.



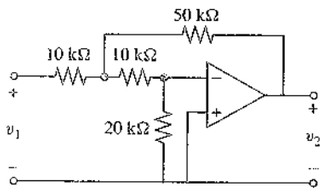
Use nodal analysis to find v for the circuit shown.



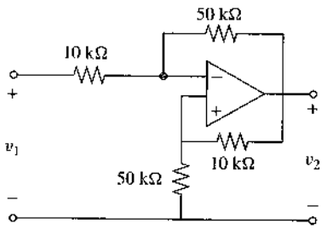
Solve by nodal analysis; then solve again using mesh analysis. Find all indicated variables.



Use the virtual short and open principles to find the voltage transfer ratio v_2/v_1.



Use the virtual short and open principles to find the voltage transfer ratio v_2/v_1.



5

Energy-Storage Elements



Michael Faraday
1791–1867

My greatest discovery was Michael Faraday.

Sir Humphry Davy

On August 29, 1831, Michael Faraday, the great English chemist and physicist, discovered electromagnetic induction, when he found that moving a magnet through a coil of copper wire caused an electric current to flow in the wire. Since the electric motor and generator are based on this principle, Faraday's discovery profoundly changed the course of world history. When asked by the British prime minister years later what use could be made of his discoveries, Faraday quipped, "Some day it might be possible to tax them."

Faraday, one of 10 children of a blacksmith, was born near London. He was first apprenticed to a bookbinder, but at age 22 he realized his boyhood dream by becoming assistant at the Royal Institution to his idol, the great chemist Sir Humphry Davy. He remained at the Institution for 54 years, taking over Davy's position when Davy retired. Faraday was perhaps the greatest experimentalist who ever lived, with achievements to his credit in nearly all the areas of physical science under investigation in his time. To describe the phenomena he investigated, he and a science-philosopher friend invented new words such as electrolysis, electrolyte, ion, anode, and cathode. To honor him, the unit of capacitance is named the *farad*.

5.1 Capacitors
5.2 Energy Storage in Capacitors
5.3 Series and Parallel Capacitors
5.4 Inductors
5.5 Energy Storage in Inductors
5.6 Series and Parallel Inductors
5.7 DC Steady State
5.8 Practical Capacitors and Inductors
5.9 Singular Circuits
Summary
Problems

Up to now we have considered only resistive circuits, that is, circuits containing resistors and sources (and op amps, which can be replaced by models consisting of resistors and sources). The terminal characteristics of these elements are simple algebraic equations that lead to circuit equations that are algebraic. In this chapter we introduce two important dynamic circuit elements, the capacitor and the inductor, whose terminal equations are integrodifferential equations (involve derivatives and/or integrals) rather than algebraic equations. These elements are referred to as *dynamic* because they store energy that can be retrieved at some later time. Another term used, for this reason, is *storage elements*. We first describe the property of capacitance and discuss the mathematical model of an ideal capacitor. The terminal characteristics and energy relations are then given, followed by equivalents for parallel and series connections of two or more capacitors. We then repeat this procedure for the inductor. The chapter concludes with a discussion of practical capacitors and inductors and their equivalent circuits.

5.1 CAPACITORS

A capacitor is a two-terminal device that consists of two conducting bodies that are separated by a nonconducting material. Such a nonconducting material is known as an *insulator* or a *dielectric*. Because of the dielectric, charges cannot move from one conducting body to the other within the device. They must therefore be transported between the conducting bodies via external circuitry connected to the terminals of the capacitor. One very simple type, the parallel-plate capacitor, is shown in Fig. 5.1. The conducting bodies are flat, rectangular conductors that are separated by the dielectric material.

Starting with a capacitor that is initially uncharged, suppose that by means of some external circuit we have transferred q coulombs of charge from one plate to the other. To determine the potential difference which results, we imagine a +1-C test charge subsequently being moved between points b and a in Fig. 5.1. Since the test charge is positive and there is $+q$ charge on the plate toward which we are moving, work must be done to move it. Equal work must be done on it due to attractive force of the $-q$ charge on the

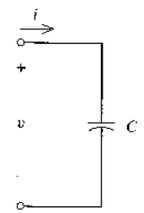
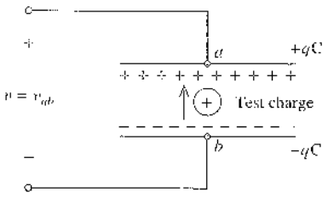


FIGURE 5.2 Circuit symbol for a capacitor.



Parallel-plate capacitor.

plate we are leaving. Thus, from the definition of potential difference between two points in space in Chapter 2 as the net work done in transporting a +1-C test charge between the points, the plate with the $+q$ charge is at a higher potential than the other one; that is,

$$v_{ab} = f(q)$$

where apparently the function f is an increasing one, since the larger the separated charge q, the greater the work done on the test charge.

For many physical capacitors, the general charge–voltage relationship (5.1a) is linear. A *linear capacitor* is defined as a two-terminal device whose charge–voltage relationship is a straight line intersecting the origin, $v_{ab} = f(q) = Kq$, or

$$q = Cv$$

Here $v = v_{ab}$ is the voltage between the plates, and $C = 1/K$ is the slope of the graph plotting q as a function of v. C is known as the *capacitance* of the device, with the dimension of coulombs per volt. The unit of capacitance is known as the *farad* (F), named for the famous British physicist Michael Faraday (1791–1867). A 1-farad capacitor will separate 1 coulomb of charge for each volt of potential difference between its plates.

It is interesting to note in the example above that the net charge within the capacitor is always zero. Charges removed from one plate always appear on the other, so the total charge remains zero. We should also observe that charges leaving one terminal enter the other. This fact is consistent with the requirement that current entering one terminal must exit the other in a two-terminal device, a consequence of KCL. Since the current is defined as the time rate of change of charge, differentiating (5.1b), we find that

$$i = C\frac{dv}{dt} \tag{5.2}$$

which is the current–voltage relation or terminal law for a capacitor.

The circuit symbol for the capacitor and the current–voltage convention that satisfies (5.2) are shown in Fig. 5.2. It is apparent that moving a charge of $+q$ in Fig. 5.1 from the lower to the upper plate represents a current flowing into the upper terminal and thence onto the upper plate. The movement of this charge causes the upper terminal to become more positive than the lower one by an amount v. Hence the current–voltage convention of Fig. 5.2 is satisfied. Thus, in order that the terminal law (5.2) be satisfied, the current and voltage reference directions must be related as shown in Fig. 5.2, with

the arrow pointing into the positive end of the voltage reference direction. This is the passive sign convention we required for use with resistors earlier. If the passive sign convention is violated by reversing either the voltage or the current reference direction, the current entering the positive terminal is $-i$ and (5.2) becomes

$$i = -C\frac{dv}{dt}$$

We recall that for this case a minus sign was also required in Ohm's law.

Example 5.1

As an example, suppose that the voltage on a 1-μF capacitor is $v = 6\cos 2t$ V. Then, with terminal variables satisfying the passive sign convention as in Fig. 5.2,

$$\begin{aligned} i &= C\frac{dv}{dt} = (10^{-6})(-12\sin 2t) \\ &= -12\sin 2t \ \mu\text{A} \end{aligned}$$

In (5.2) we see that if v is constant, the current i is zero. Therefore, *a capacitor acts like an open circuit to a dc voltage*. On the other hand, the more rapidly v changes, the larger the current flowing through its terminals.

Example 5.2

Consider a voltage that increases linearly from 0 to 1 V in a^{-1} s, given by

$$v = \begin{cases} 0, & t < 0 \\ at, & 0 \le t \le a^{-1} \\ 1, & t > a^{-1} \end{cases}$$

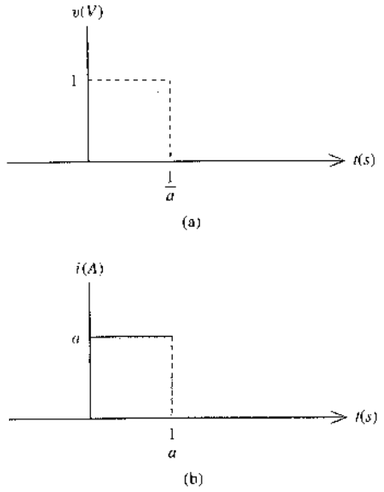


FIGURE 5.3 Voltage and current waveforms in a 1-F capacitor.

If this voltage is applied to the terminals of a 1-F capacitor (an unusually large value chosen for numerical convenience), the resulting current (in amperes) is

$$i = \begin{cases} 0, & t < 0 \\ a, & 0 \le t \le a^{-1} \\ 0, & t > a^{-1} \end{cases}$$

Plots of v and i are shown in Fig. 5.3. We see that i is zero when v is constant and that it is equal to a constant when v increases linearly. If the slope a is made larger, i must increase. It is apparent that if $a^{-1} = 0$ (a is infinite), v changes abruptly (in zero time) from 0 to 1 V.

In general, any abrupt or instantaneous changes in voltage, such as in the example above, require an infinite current flow through the capacitor. An infinite current, however, is a physical impossibility. Thus abrupt or instantaneous changes in the voltage across a capacitor are not possible, and *the voltage across a capacitor is always continuous* even though the current may be discontinuous. This is called the *continuity principle*. (Circuits that appear to violate this principle, but themselves prove to contain mathematical contradictions, are taken up in Section 5.9.)

If voltage is continuous in time, then, by (5.1b), charge must be also. Thus an alternative statement of the continuity principle, particularly useful in circuits with several capacitors, is that the *total charge cannot change instantaneously* (conservation of charge).

Let us now find $v(t)$ in terms of $i(t)$ by integrating both sides of (5.2) between times t_0 and t. The result is

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0)$$

where $v(t_0) = q(t_0)/C$ is the voltage on C at time t_0. In this equation, the integral term represents the voltage that accumulates on the capacitor in the interval from t_0 to t, whereas $v(t_0)$ is that which accumulates from $-\infty$ to t_0. The voltage $v(-\infty)$ is taken to be zero. Thus an alternative form of (5.3a) is

$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\,d\tau$$

Example 5.3

In applying (5.3b), we are obtaining the area associated with a plot of i from $-\infty$ to t. In Fig. 5.3, for example, since $v(-\infty) = 0$ and $C = 1$ F, we have

$$v(t) = \frac{1}{1}\int_{\infty}^{t} (0)\,d\tau + v(-\infty) = 0, \qquad t \le 0$$

Therefore, $v(0) = 0$, and

$$v(t) = \int_{0}^{t} a\,d\tau + v(0) = at, \qquad 0 \le t \le a^{-1}$$

This last equation shows that $v(1/a) = 1$, so

$$v(t) = \frac{1}{1}\int_{1/a}^{t} (0)\,d\tau + v\frac{1}{a} = 1, \qquad t \ge a^{-1}$$

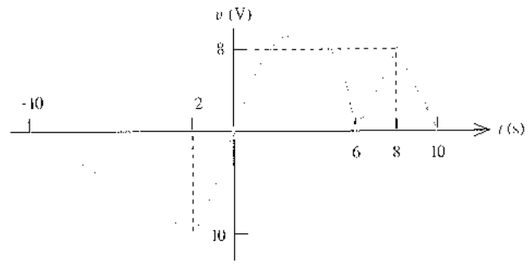
which agrees with $v(t)$ in Fig. 5.3. In this example we see that $v(t)$ and $i(t)$ do not necessarily have the same shape. Specifically, their maxima do not necessarily occur at the same time, nor their minima, unlike the case for the resistor. In fact, inspection of Fig. 5.3 reveals that the current may be discontinuous, even though the voltage must be continuous, as stated previously.

5.1.1. A 100-nF capacitor has a voltage $v = 10 \sin 1000t$ V. Find its current.

[illegible] $\cos 1000t$ mA

5.1.2. A 400-μF capacitor has voltage as shown. It is piecewise linear except for $0 \le t \le 6$, $v(t) = (t)(6 - t)$. Find its current i at $t = -4$, -1, 1, 5, and 9 s.

[illegible] -0.5, 2, 1.6, -1.6, and -1.6 mA



5.1.3. A constant current of 10 mA is charging a 10-μF capacitor (entering its positive voltage terminal). If the capacitor was initially charged to 5 V, find the charge and voltage on it after 20 ms.

[illegible] 0.25 mC; 25 V

5.1.4. Let the graph of Exercise 5.1.2 be the graph of the current in milliamperes versus time in milliseconds in a 0.25-μF capacitor. Find the voltage at $t = -4$, -1, 6, and 9 ms.

[illegible] -90, -190, -56, and 0 V

5.2 ENERGY STORAGE IN CAPACITORS

The terminal voltage across a capacitor is accompanied by a separation of charges between the capacitor plates. These charges have electrical forces acting on them. An *electric field*, a basic quantity in electromagnetic theory, is defined as the position-dependent force acting on a unit positive charge. Thus the forces acting on the charges within the capacitor can be considered to result from an electric field. It is for this reason that the energy stored or accumulated in a capacitor is said to be stored in the electric field located between its plates. Quantitatively, the energy stored in a capacitor, from (1.6) and (5.2), is given by

$$w_C(t) = \int_{\infty}^{t} vi \, d\tau = \int_{-\infty}^{t} v\left(C\frac{dv}{d\tau}\right) d\tau$$

$$w_C(t) = \int_{-\infty}^{t} Cv \, dv = \tfrac{1}{2}Cv^2(t)$$

The last equality follows from $v(-\infty) = 0$.

From this result we see that $w_c(t) \ge 0$. Therefore, by (1.7), *the capacitor is a passive circuit element*. The only other passive element we have encountered thus far was the resistor. Resistors dissipate energy, as we have seen. Unlike the resistor, the ideal capacitor cannot dissipate any energy. The net energy $w_C(t)$ supplied to it by the external circuit is stored in the electric field within the device and can be fully recovered.

Consider, for instance, a 1-F capacitor that has a voltage of 10 V. The energy stored is

$$w_c = \tfrac{1}{2}Cv^2 = 50 \text{ J}$$

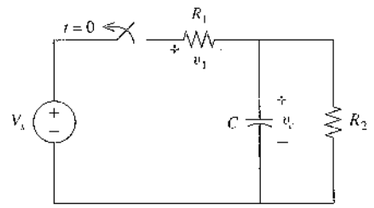
Suppose that the capacitor is disconnected from any external circuit so that no additional current can flow, and the charge, voltage, and energy all remain constant. If we subsequently connect a resistor across the capacitor, a current flows through it as the charges separated on the plates seek one another, until all the stored energy (50 J) has been dissipated in the form of heat by the resistor and the voltage across the RC combination settles to zero. Systematic analysis of such circuits is found in Chapter 6.

As was pointed out earlier, the voltage on a capacitor is a continuous function of time. Thus, by (5.4), we see that *the energy stored in a capacitor is also continuous*. This is not surprising, since otherwise, energy would have to be transported from one place to another in zero time. Since power is the rate of energy change per unit time, instantaneous change of energy demands infinite power, a physical impossibility.

To illustrate continuity of capacitor voltage, let us consider Fig. 5.4, which contains an *ideal switch* that is opened at $t = 0$, as indicated. An ideal switch is defined to be a two-terminal element that switches between an open circuit and a short circuit instantaneously. Unless otherwise specified, all switches used in this book are considered to be ideal.

To discuss the effect of the switching action, we first need to consider two times near $t = 0$. We shall denote as $t = 0-$ the time just before the switching action and $t = 0+$ as the time just after the switching action. Theoretically, no time has elapsed between $0-$ and $0+$, but the two times represent radically different states of the circuit. $v_c(0-)$ is the voltage on the capacitor just before the switch acts, and $v_c(0+)$ is the voltage immediately after switching. Similarly, $i_C(0-)$ and $i_C(0+)$ are the capacitive current just before and just after the switch changes state. More formally, $v_c(0-)$ is the limit of $v_c(t)$ as t approaches zero from the left [through negative ($t < 0$) values], and $v_c(0+)$ is the limit as t approaches zero from the right [through positive ($t > 0$) values]. Indeed, mathematically, the times $0-$ and $0+$ are convenient fictions without precise definition (when is the first instant greater than zero?), although the values of circuit variables such as $v_C(0-)$ and $v_C(0+)$ are defined precisely by the limits from the left and right.

Example 5.4



Circuit illustrating continuity of capacitor voltage.

Suppose that in Fig. 5.4 we have $V_s = 6$ V and $v_c(0) = 4$ V. Just prior to the switching action ($t = 0-$), we have $v_1(0-) = V_s - v_c(0-) = 2$ V. Immediately after the switch is opened, we have $v_1(0+) = 0$, since no current is flowing in R_1 (it is in series with an open circuit). However, since v_c, a capacitive voltage, must be continuous, we have

$$v_c(0+) = v_c(0-) = 4 \text{ V}$$

Thus the voltage on resistor R_1 has changed abruptly, but that on the capacitor has not. The voltage on R_2 is the same as the capacitor voltage and thus has not changed abruptly either.

Although apparently prohibited by the continuity principle, circuit diagrams can be drawn for which capacitor voltages seem forced to change abruptly. For example, consider two capacitors and an ideal switch forming a closed loop. If the capacitors have unequal voltages just as the switch closes, by continuity their values must remain unequal at $t = 0+$, while by KVL their voltages must be equal at $t = 0+$. Apparently, we must either concede an infinite power flow (by accepting violation of the continuity principle) or abandon KVL for these circuits. We shall consider such *singular circuits* in more detail in Section 5.9. Singular circuits are similar to the circuits discussed in Chapter 2 that contain unequal voltage sources in parallel or current sources in series. Such circuits lead to mathematical contradictions and cannot be analyzed consistently. In practice, the contradiction is removed by replacing ideal elements by more accurate physical models of them, including Thevenin equivalent resistance in the case of the sources of Chapter 2 and parallel leakage resistance in the present case of capacitors. In the real world, continuity and the Kirchhoff laws are both valid in lumped physical circuits.

5.2.1. A 0.2-μF capacitor has a charge of 20 μC. Find its voltage and energy.

Answer: 100 V; 1 mJ

5.2.2. If the energy stored in a 0.5-F capacitor is 25 J, find the voltage and charge.

Answer: 10 V; 5 C

5.2.3. In Fig. 5.4, let $C = \frac{1}{4}$ F, $R_1 = R_2 = 4\ \Omega$, and $V = 20$ V. If the current in R_2 at $t = 0-$ is 2 A directed downward, find at $t = 0-$ and at $t = 0+$ (a) the charge on the capacitor, (b) the current in R_1 directed to the right, (c) the current in C directed downward, and (d) dv_C/dt.

Answer: (a) 2, 2 C; (b) 3, 0 A; (c) 1, −2 A; (d) 4, −8 V/s

5.3 SERIES AND PARALLEL CAPACITORS

In this section we determine the equivalents for series and parallel connections of capacitors. Recall that two subcircuits are equivalent if they have the same terminal law. We shall match the terminal law for series-connected capacitors with that for a single capacitance, the series equivalent capacitance, and then repeat for the parallel case.

Let us first consider the series connection of N capacitors, as shown in Fig. 5.5(a). Applying KVL, we find

$$v = v_1 + v_2 + \cdots + v_N$$

Using the terminal law (5.3a) gives

$$\begin{aligned} v(t) &= \left(\frac{1}{C_1}\int_{t_0}^{t} i\,d\tau + v_1(t_0)\right) \\ &\quad + \left(\frac{1}{C_2}\int_{t_0}^{t} i\,d\tau + v_2(t_0)\right) + \cdots + \left(\frac{1}{C_N}\int_{t_0}^{t} i\,d\tau + v_N(t_0)\right) \\ &= \left(\sum_{n=1}^{N}\frac{1}{C_n}\right)\int_{t_0}^{t} i\,d\tau + v(t_0) \end{aligned}$$

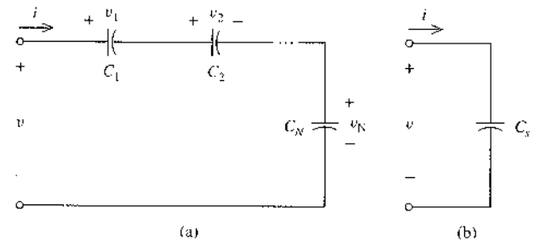
where in the last we have used (5.5). This is the desired terminal law for the series-connected capacitors. The terminal law for Fig. 5.5(b) is simply

$$v(t) = \frac{1}{C_s}\int_{t_0}^{t} i\,d\tau + v(t_0)$$

The two terminal laws are identical, hence the circuits are equivalent, if

$$\frac{1}{C_s} = \sum_{n=1}^{N}\frac{1}{C_n}$$

The equivalent of a chain of series-connected capacitors is a single capacitor whose inverse capacitance is the sum of inverses of the series capacitances.



(a) Series capacitors; (b) equivalent circuit.

In the case of just two capacitors in series, this general result simplifies to

$$C_s = \frac{C_1 C_2}{C_1 + C_2}$$

For two capacitors in series, the equivalent capacitance is the product over the sum of the two individual capacitances. The product-by-sum equivalence rule was introduced in (2.15) for the case of two resistors in parallel. As with parallel resistors, this rule is limited to the special case of exactly two such elements. For three or more capacitors in series, the general result (5.6) should be used.

We next turn to the parallel connection of N capacitors, as shown in Fig. 5.6(a). KCL at the top node yields

$$i = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + \cdots + C_N \frac{dv}{dt}$$

or

$$i = \left(\sum_{n=1}^{N} C_n \right) \frac{dv}{dt}$$

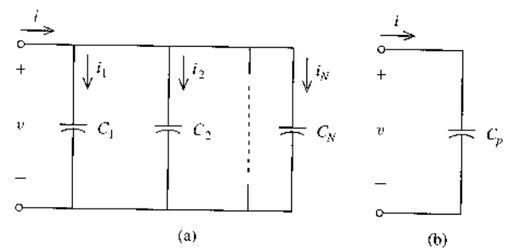
The corresponding terminal law for Fig. 5.6(b) is

$$i = C_p \frac{dv}{dt}$$

Comparing the last two equations, the terminal laws are identical and the circuits equivalent if

$$C_p = \sum_{n=1}^{N} C_n$$

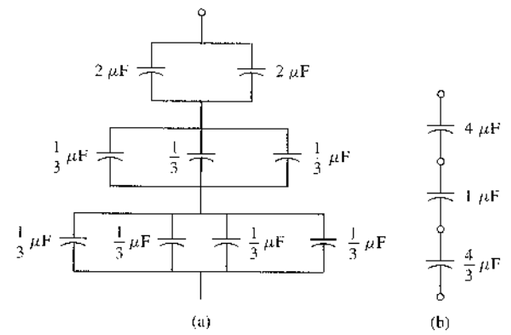
The equivalent of parallel capacitors is a single capacitor whose capacitance is the sum of capacitances of the parallel capacitors. More simply put, parallel capacitance adds.



(a) Parallel capacitors; (b) equivalent circuit.

Example 5.5

Determine the equivalent capacitance C_{eq} of the circuit shown in Fig. 5.7(a).



Circuit for Example 5.5.

Since parallel capacitances add, Fig. 5.7(a) may be immediately simplified to its equivalent, Fig. 5.7(b). Then, using the series equivalent rule (5.6), with μF units,

$$\frac{1}{C_{eq}} = \frac{1}{4} + \frac{1}{1} + \frac{1}{\frac{4}{3}} = 2$$

or $C_{eq} = \frac{1}{2}\ \mu F$.

Example 5.6

Determine the voltage $v_1(t)$ for $t \geq 0$ in Fig. 5.8(a) if $v_1(0) = 10$ V. The equivalent of the series-connected capacitors is shown in Fig. 5.8(b),

$$\frac{1}{C_s} = \frac{1}{1} + \frac{1}{2} + \frac{1}{2}$$

or $C_s = \frac{1}{2}$ F. Then

$$i = C_s \frac{d}{dt}(3 \sin 2t) = \frac{1}{2}(6 \cos 2t) = 3 \cos 2t \text{ A}$$

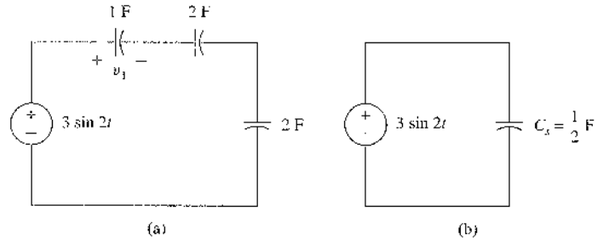
The voltage across the 1-F capacitor is then given by (5.3), or

$$v_1(t) = \frac{1}{C_1} \int_{t_0}^{t} i(\tau)\, d\tau + v_1(t_0) = \int_0^t (3 \cos 2\tau)\, d\tau + 10$$

which yields $v_1(t) = \frac{3}{2} \sin 2t + 10$ V. Note that evaluating this result at $t = 0$ gives the correct initial value for $v_1(t)$. We may also check that

$$i = C_1 \frac{dv_1}{dt} = 1 \frac{d}{dt}\left(\frac{3}{2} \sin 2t + 10 \right) = 3 \cos 2t \text{ A}$$

which is consistent with our computed value for $i(t)$.



Circuit for Example 5.6.

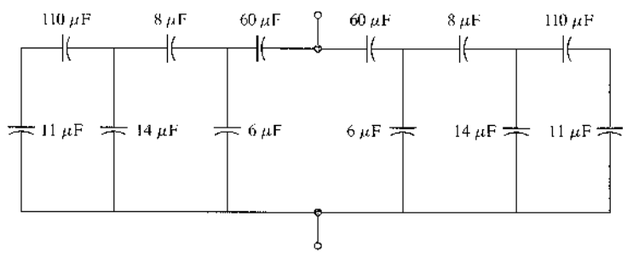
It is interesting to note that the equivalent capacitance of series and parallel capacitors follows the same rules as equivalent conductance of series and parallel conductances. That is, for the parallel case the equivalent parameter (conductance G_P or capacitance C_P) is the sum of the individual parameters, while in series the equivalent parameter (G_S or C_S) is the inverse of the sum of inverses of the individual parameters. This correspondence follows from the fact that the terminal laws for the conductance and capacitance are of the same form: the parameter (G or C) multiplies the terminal voltage (or derivative of terminal voltage) to yield terminal current. This functional correspondence makes the foregoing derivations of capacitive equivalents correspond line for line to those for conductance equivalents presented earlier in Chapter 2.

5.3.1. Find the maximum and minimum values of capacitance that can be obtained from ten 1-μF capacitors. How should they be connected?

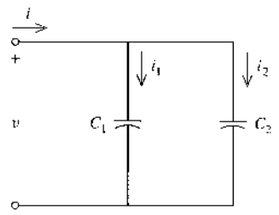
Answer: 10 μF (parallel); 0.1 μF (series connected)

5.3.2. Find the equivalent capacitance.

Answer: 20 μF



Circuit for Exercise 5.3.2.



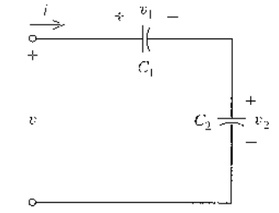
Circuit for Exercise 5.3.3.

5.3.3. Derive an equation for current division between two parallel capacitors by finding i_1 and i_2.

Answer: $C_1 i/(C_1 + C_2)$; $C_2 i/(C_1 + C_2)$

5.3.4. Derive an equation for voltage division between two initially uncharged series capacitors by finding v_1 and v_2.

Answer: $C_2 v/(C_1 + C_2)$; $C_1 v/(C_1 + C_2)$



5.4 INDUCTORS

In the preceding sections we found that the terminal characteristics of the capacitor are the result of forces that exist between electric charges due to their separation in space. Just as these charges exert position-dependent or *electrostatic* force upon one another, so moving charges, or currents, exert *electrodynamic* force.

The force that is exchanged by two neighboring current-carrying wires was determined experimentally by the French scientist André Marie Ampère (1775–1836) in the early nineteenth century. This force can be characterized by postulating the existence of a *magnetic field*. The magnetic field, in turn, can be thought of in terms of a *magnetic flux* that forms closed loops about electric currents. The cause of the flux is motion of charge in the wires, that is, the current. The study of magnetic fields, like that of electric fields mentioned in Section 5.3, comes in a course devoted to electromagnetic theory. Here we are interested primarily in the role of magnetic fields in linear lumped circuits, most prominently in the circuit element called an inductor.

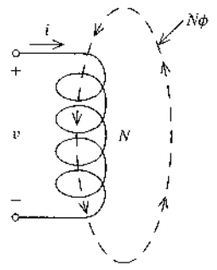


Figure 5.9 Simple model of an inductor.

An *inductor* is a two-terminal device that consists of a coiled conducting wire wound around a core. A current flowing through the device produces a magnetic flux ϕ that forms closed loops threading its coils, as shown in Fig. 5.9. Suppose that the coil contains N turns and that the flux ϕ passes through each turn. In this case the total flux linked by the N turns of the coil, denoted by λ, is

$$\lambda = N\phi$$

This total flux is commonly referred to as the *flux linkage*. The unit of magnetic flux is the weber (Wb), named for the German physicist Wilhelm Weber (1804–1891). In a linear inductor, the flux linkage is directly proportional to the current flowing through the device. For linear inductors, therefore, we may write

$$\lambda = Li$$

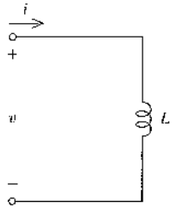
where L, the constant of proportionality, is the inductance in webers per ampere. The unit of 1 Wb/A is known as the *henry* (H), named for the American physicist Joseph Henry (1797–1878).

In (5.8) we see that an increase in i produces a corresponding increase in λ. This increase in λ induces a voltage in the N-turn coil. The fact that voltages are induced by time-varying magnetic flux was first discovered by Henry. Henry, however, repeating the mistake of Cavendish with the resistor, failed to publish his findings. As a result, Faraday is credited with discovering the law of electromagnetic induction. This law states that the induced voltage is equal to the time rate of change of the total magnetic flux. In mathematical form, the law is

$$v = \frac{d\lambda}{dt}$$

which with (5.8) yields the terminal law for an inductor:

$$v = L\frac{di}{dt} \tag{5.9}$$



Circuit symbol for an inductor.

Clearly, as i increases, a voltage is developed across the terminals of the inductor, the polarity of which is shown in Fig. 5.10. This voltage opposes any additional increase in i, for if this were not the case, that is, if the polarity were reversed, the induced voltage would "aid" the current. This physically cannot be true because the current and voltage would then both increase indefinitely.

The circuit symbol for the inductor whose terminal law is given in (5.9) and shown in Fig. 5.10. Just as in the cases of the resistor and the capacitor, the voltage and current reference directions are assumed to satisfy the passive sign convention. If either reference direction (but not both) is reversed, a negative sign must be introduced into the terminal law, yielding $v = -L(di/dt)$.

The terminal law of the inductor, (5.9), shows that if i is constant, the voltage v is zero. Therefore, *an inductor acts like a short circuit to a dc current*. On the other hand, the more rapidly i changes, the greater is the voltage that appears across its terminals. This is due to the more rapidly time-varying magnetic flux i produced, which is the cause of the induced voltage v across the terminals of the inductor.

Example 5.7

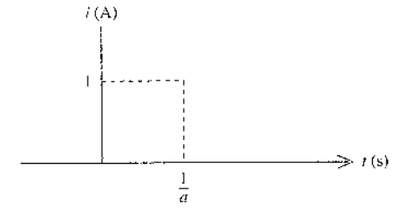
Consider a current that increases linearly from 0 to 1 V in a^{-1} s, given by

$$i = \begin{cases} 0, & t < 0 \\ at, & 0 \le t \le a^{-1} \\ 1, & t > a^{-1} \end{cases}$$

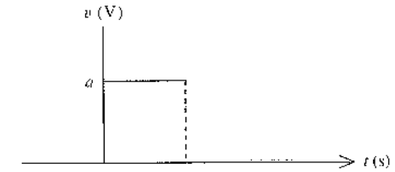
If this current is applied to the terminals of a 1-H inductor, the resulting voltage is

$$v = \begin{cases} 0, & t < 0 \\ a, & 0 \le t \le a^{-1} \\ 0, & t > a^{-1} \end{cases}$$

Plots of i and v are shown in Fig. 5.11. We see that v is zero when i is constant and that it is equal to a when i increases linearly. If a



(a)



(b)

Current and voltage waveforms in a 1-H inductor.

is made larger, v changes more rapidly and v increases. It is apparent that if $a^{-1} = 0$ (a is infinite) i changes abruptly (in zero time) from 0 to 1 V.

In general, abrupt changes in the current through an inductor, such as in Example 5.7, require that an infinite voltage appear across the terminals of the inductor. As observed in the case of the capacitor, infinite current or voltage is a physical impossibility. Thus instantaneous changes in the current through an inductor are not possible. We observe that *the current through an inductor is always continuous*. This is the other half of the *continuity principle*, shown previously to apply to capacitive voltage. Inductive current is also continuous, even though the voltage across an inductor may change discontinuously, as may capacitive current.

Another interesting parallel to our previous discussion of the capacitor is revealed by comparing Example 5.7 with Example 5.2. We see that by reversing the roles of i and v the same terminal pair is produced in the inductor as in the capacitor. This is less surprising when we note that the terminal laws for these two elements, $i = C(dv/dt)$ and $v = L(di/dt)$, are of the same form with only the current and voltage interchanged. This *duality* of capacitor and inductor will make the remaining results for the inductor strongly reminiscent of the previous ones for the capacitor.

Let us find the current $i(t)$ in terms of the voltage $v(t)$ for the inductor. Integrating (5.9) from time t_0 to t and solving for $i(t)$, we have

$$i(t) = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + i(t_0) \tag{5.10}$$

In this equation the integral term represents the current buildup from time t_0 to t, whereas $i(t_0)$ is the current at t_0. Obviously, $i(t_0)$ is the current that accumulates from $t = -\infty$ to t_0, where $i(-\infty) = 0$. Thus an alternative to (5.10) is

$$i(t) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)\,d\tau$$

Example 5.8

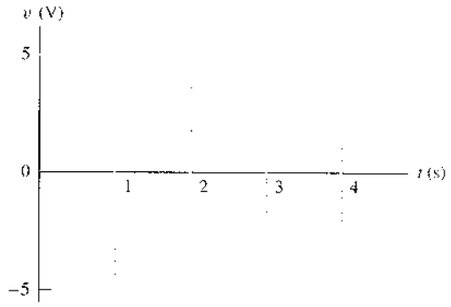
Consider a 1-H inductor. In the application of (5.10) we are obtaining the net area under the graph of v from $-\infty$ to t, since $i(t_0)$ represents the area from $-\infty$ to t_0 . Setting $t_0 = 0$ in Fig. 5.11, for instance, we can compute the current at $t = 1/a$ to be

$$i\left(\frac{1}{a}\right) = \frac{1}{1}\int_{0}^{1/a} v(\tau)\,d\tau + 0 = \int_{0}^{1/a} a\,d\tau = 1 \text{ A}$$

If we prefer, we may set $t_0 = \frac{1}{2}a$, a time at which $i(t_0) = \frac{1}{2}$, and recompute

$$i\left(\frac{1}{a}\right) = \frac{1}{1}\int_{1/2a}^{1/a} v(\tau)\,d\tau + v\frac{1}{2a} = \int_{1/2a}^{1/a} a\,d\tau + \frac{1}{2} = 1 \text{ A}$$

In this example we see that v and i, just as in the case of the capacitor, do not necessarily have the same variation in time. Inspection of Fig. 5.11, for example, shows that the voltage can be discontinuous even though the current is always continuous.



5.4.1. A 10-mH inductor has a current of $50\cos 1000t$ mA. Find its voltage and its flux linkage. Assume that the passive sign convention is satisfied.

$-0.5\sin 1000t$ V; $0.5\cos 1000t$ mWb

5.4.2. Find the current $i(t)$ for $t > 0$ in a 20-mH inductor having a voltage of $-5\sin 50t$ V if $i(0) = 5$ A. Assume that the passive sign convention is satisfied.

$5\cos 50t$ A

5.4.3. Find the current in a 0.5-H inductor, for $0 \le t \le 2$ s, if $i(0) = 0$ and the voltage is as shown.

$10t$ A, $0 \le t \le 1$; $10(2 - t)$ A, $1 < t \le 2$

5.5 ENERGY STORAGE IN INDUCTORS

A current i flowing through an inductor causes a total flux linkage λ to be produced, which passes through the turns of the inductor's coil. Just as work was performed in moving charges between the plates of a capacitor, work must be done by the external

circuit to establish the flux ϕ in the inductor. This work or energy is said to be stored in the magnetic field. Just as was the case for energy stored in the electric field of a capacitor, it can be reclaimed later by the external circuit. Using (1.6) and (5.9), the energy stored in an inductor is given by

$$w_L(t) = \int_{-\infty}^{t} vi\,d\tau = \int_{-\infty}^{t} \left(L\frac{di}{d\tau}\right) i\,d\tau$$

$$= \int_{i(-\infty)}^{i(t)} Li\,di = \tfrac{1}{2}Li^2(t)$$

Inspection of the result reveals that $w_L(t) \ge 0$. Therefore, by (1.7), we see that *the inductor is a passive circuit element.* Like the ideal capacitor, the ideal inductor does not dissipate any power. Consider, for example, a 2-H inductor that is carrying a current of 5 A. The energy stored is

$$w_L = \tfrac{1}{2}Li^2 = 25 \text{ J}$$

Suppose that this inductor is disconnected instantaneously from the external circuit that caused the 5-A current to flow and simultaneously is connected across a resistor. Current will flow through the inductor–resistor combination until all the energy previously stored in the inductor (25 J) has been dissipated by the resistor and the current settles to zero. Complete analysis of circuits of this type is found in Chapter 6.

Example 5.9

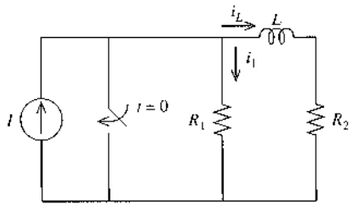


FIGURE 5.12 Circuit for Example 5.9.

Since inductive currents are continuous, it follows that the energy stored in an inductor, like that stored in a capacitor, is also continuous. To illustrate this, consider the circuit of Fig. 5.12, which contains a switch that is closed at $t = 0$, as indicated. Suppose that $i_L(0-) = 2$ A and $I = 3$ A. Then $i_1(0-) = 3 - 2 = 1$ A. Just after the switch is closed, we have $i_1(0+) = 0$ since a short circuit is placed across R_1. However, we have, by continuity,

$$i_L(0+) = i_L(0-) = 2 \text{ A}$$

Thus the resistor current has changed abruptly, but the inductive current has not.

An example of a singular circuit for which inductive currents appear to be discontinuous is given in Section 5.9. As in the case of singular capacitive circuits, the apparent discontinuity in the energy stored in the inductors cannot be accounted for in the ideal lumped-circuit model. Physical circuits having inductors contain associated resistance that does not permit the infinite inductor voltages that must accompany abrupt changes in inductor currents. We shall be concerned primarily with circuits of this type.

5.5.1. Derive an expression for the energy stored in an inductor in terms of the flux linkage λ and the inductance L.

$\lambda^2/2L$

5.5.2. A 40-mH inductor has a current $i = 100\cos 10\pi t$ mA. Find the flux linkage, voltage, and the energy at $t = 1/30$ s.

2 mWb; $-20\sqrt{3}\pi$ mV; 50 μJ

5.5.3. A 2-mH inductor has a voltage $v = 2\cos 1000t$ V with $i(0) = 1.5$ A. (a) Find the energy stored in the inductor at $t = \pi/6$ ms. (b) What is the maximum energy stored and at what time(s)? (c) What is the minimum energy stored and at what time(s)?

(a) 4 mJ; (b) $\frac{25}{4}$ mJ at $t = 2\pi n + \pi/2$; (c) $\frac{1}{4}$ mJ at $t = (2n+1)\pi + \pi/2$, $n = 0, 1, \ldots$

5.5.4. In Fig. 5.12, let $I = 5$ A, $R_1 = 6\ \Omega$, $R_2 = 4\ \Omega$, $L = 2$ H, and $i_1(0-) = 2$ A. If the switch closes at time $t = 0$, find $i_L(0-)$, $i_L(0+)$, $i_1(0+)$, and $di_L(0+)/dt$.

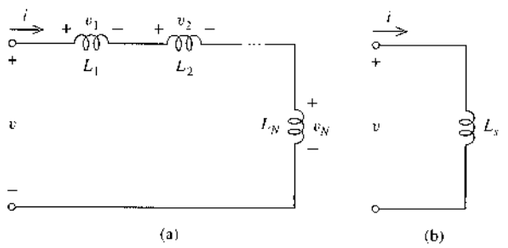
3 A; 3 A; 0; -6 A/s

5.6 SERIES AND PARALLEL INDUCTORS

In this section we determine the equivalent inductance for series and parallel connections of inductors. Just as we found the series and parallel capacitive equivalents to follow the rules derived for conductances, we will now see that series and parallel inductors have equivalents whose values follow the same rules we derived for resistors.

First consider the series connection of N inductors shown in Fig. 5.13(a). Summing voltage drops around the loop and using the terminal law $v = L(di/dt)$ gives

$$v = L_1\frac{di}{dt} + L_2\frac{di}{dt} + \cdots + L_N\frac{di}{dt}$$
$$= \left(\sum_{n=1}^{N} L_N\right)\frac{di}{dt}$$



(a) Series inductors; (b) equivalent circuit.

Comparing the terminal law from Fig. 5.13(b), $v = L_S(di/dt)$, the two circuits are equivalent if

$$L_S = \sum_{n=1}^{N} L_N$$

The equivalent of a chain of series inductors is a single inductor whose inductance is the sum of the individual inductances. Less formally, series inductance adds.

The parallel case is illustrated in Fig. 5.14(a). Summing currents at the top node and using the terminal law in the form (5.10),

$$i(t) = \left(\frac{1}{L_1}\int_{t_0}^{t} v\,d\tau + i_1(t_0)\right) + \left(\frac{1}{L_2}\int_{t_0}^{t} v\,d\tau + i_2(t_0)\right) + \cdots$$
$$+ \left(\frac{1}{L_N}\int_{t_0}^{t} v\,d\tau + i_N(t_0)\right)$$

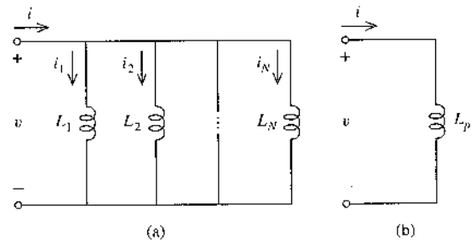
Gathering like terms and noting that $i = i_1 + i_2 + \cdots + i_N$, the terminal law is

$$i(t) = \left(\sum_{n=1}^{N}\frac{1}{L_n}\right)\int_{t_0}^{t} v\,d\tau + i(t_0)$$

Comparing the terminal law of Fig. 5.14(b) given in (5.10), they match if

$$\frac{1}{L_P} = \sum_{n=1}^{N}\frac{1}{L_n}$$

The equivalent of parallel inductors is a single inductor whose inverse inductance is the sum of the inverses of the parallel inductances.



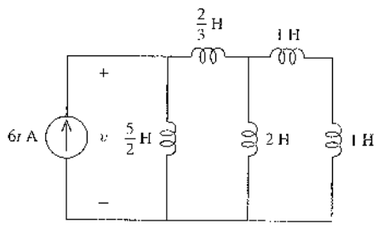
(a) Parallel inductors; (b) equivalent circuit.

In the case of two parallel inductors L_1 and L_2, (5.13) may be simplified to

$$L_p = \frac{L_1 L_2}{L_1 + L_2}$$

which is directly analogous to the equivalence of two parallel resistances (2.15). As mentioned previously in that case, the product-by-sum rule only holds for exactly two elements. For three or more, the general result (5.13) should be applied.

Example 5.10



Circuit for Example 5.10.

We wish to find v in Fig. 5.15. We will first find an equivalent inductor. The two 1-H inductors in series are equivalent to a 2-H inductor, which is in parallel with the 2-H inductor shown in the figure. Their parallel equivalent is

$$L_{eq1} = \frac{(2)(2)}{2+2} = 1 \text{ H}$$

L_{eq1} is in turn in series with $\frac{2}{3}$ H, yielding a series equivalent of $\frac{5}{3}$ H. Finally, this is in parallel with the $\frac{5}{2}$-H inductor for an overall equivalent L_{eq}:

$$\frac{1}{L_{eq}} = \frac{1}{\frac{5}{3}} + \frac{1}{\frac{5}{2}} = \frac{3}{5} + \frac{2}{5} = 1$$

and $L_{eq} = 1$ H. Then, for this circuit,

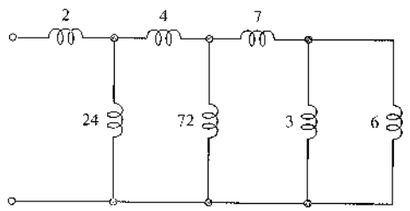
$$v = 1\frac{d}{dt}(6t) = 6 \text{ V}$$

5.6.1. Find the maximum and minimum values of inductance that can be obtained using five 20-mH inductors and ten 10-mH inductors.

Answer: 200 mH; 800 μH

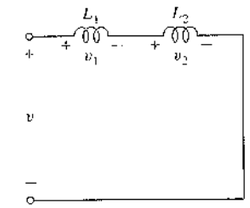
5.6.2. Find the equivalent inductance. Inductor values shown are in millihenries.

Answer: 10 mH



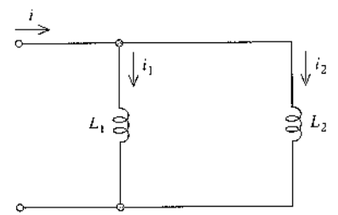
5.6.3. Derive an equation for voltage division between two series inductors by finding v_1 and v_2.

Answer: $L_1 v/(L_1 + L_2)$; $L_2 v/(L_1 + L_2)$



5.6.4. Derive an equation for current division between two parallel inductors with no initial current by finding i_1 and i_2.

Answer: $\frac{L_2 i}{L_1 + L_2}$; $\frac{L_1 i}{L_1 + L_2}$



5.7 DC STEADY STATE

Before introduction of capacitors and inductors, we focused on resistive circuits, those containing only resistors and sources. In resistive circuits whose independent sources are all dc (constant values), it is significant that all currents and voltages will also be dc. This follows from the fact that nodal or mesh analysis equations will contain no time-varying quantities. The solutions of these equations will clearly be constants, that is, dc values. Substituting these constant node voltages or mesh currents into Ohm's law to find the remaining circuit variables will produce only more dc currents and voltages.

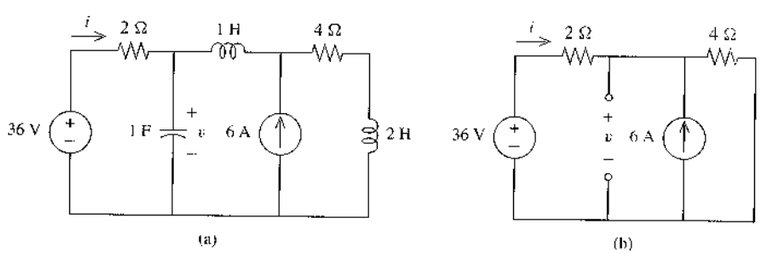
A circuit is said to be in *dc steady state* if all currents and voltages in that circuit are constant. Clearly, only circuits whose independent sources are all dc can possibly attain dc steady state, for if some source function were not constant, that current or voltage would not be dc. From Section 5.6, resistive circuits containing only dc independent sources are always in dc steady state. However, the introduction of derivatives and integrals into the analysis equations by inductors and capacitors makes time-varying solutions a distinct

possibility even when sources are constant. The relationship of dc steady state to circuits containing inductors and capacitors is considered next.

It was noted earlier in this chapter that since $i = C\,dv/dt$ is the terminal law for a capacitor, if v is a constant, then $i = 0$; that is, the capacitor is equivalent to an open circuit. Similarly, if i is a constant, $v = L(di/dt)$ suggests that the inductor is equivalent to a short circuit since its voltage is $v = 0$. Thus *in dc steady state, where all currents and voltages are constants, we may analyze the circuit by replacing inductors by short circuits and capacitors by open circuits.*

Example 5.11

Assume that the circuit of Fig. 5.16(a) is in dc steady state. Find i and v.



(a) Circuit; (b) dc steady-state equivalent.

In dc steady state we may replace inductors by shorts and capacitors by open circuits, as shown in Fig. 5.16(b). Applying mesh analysis, this is a two-mesh circuit. The presence of the current source reduces the analysis to the single equation around the supermesh consisting of the outer loop,

$$2i + 4(i + 6) = 36$$

or $i = 2$ A. Then, by KVL,

$$v = 4(i + 6) = 4(8) = 32 \text{ V}$$

The assumption of dc steady state in *RLC circuits* (those with resistors, inductors, and capacitors), which contain dc sources only, thus leads to self-consistent results. Replacing inductors and capacitors by shorts and opens, we are left with a resistive circuit to analyze and are thus guaranteed a dc steady-state outcome of the analysis. Self-consistency, however, only proves that dc steady state is possible with such circuits, not that it is inevitable. As discussed in Chapter 6, most *RLC* circuits with dc sources eventually converge to dc steady state. But some never do; and even for those that do, the next example shows that they typically do not "start" in dc steady state.

Example 5.12

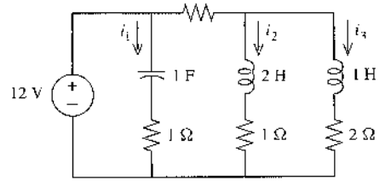
Suppose that before time $t = t_0$ the capacitor in Fig. 5.16 is charged up to $v = 100$ V and at t_0 is switched into the position shown in the circuit. Then, by continuity of capacitive voltages, its value

at $t = t_0+$ will be 100 V as well. But the dc steady-state value for this voltage is 32 V, as shown in Example 5.11. We must conclude that this circuit is not in dc steady state at time $t = t_0+$, since for $t > t_0$ at least one current or voltage in this circuit will not be dc.

Dc steady state is an important concept that we will meet again at several points in our investigation of linear circuits. It is particularly helpful in determining initial conditions for many circuits with time-varying responses. Fortunately, performing a dc steady-state analysis on an *RLC* circuit is simplified by the replacement of inductors and capacitors by shorts and opens, thus becoming the straightforward job of solving a reduced resistive, rather than *RLC*, circuit.

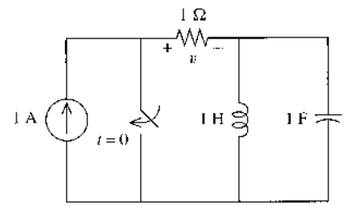
5.7.1. Find the dc steady-state value of i_1, i_2, and i_3.

0; 3 A; $\frac{3}{2}$ A



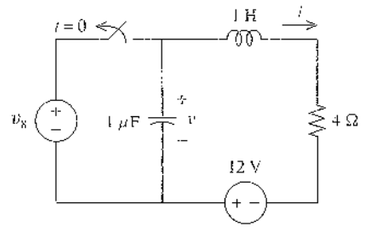
5.7.2. Assume that this circuit is in dc steady state just before $t = 0$. Find $v(0-)$ and $v(0+)$. Is the circuit in dc steady state just after $t = 0$?

1 V; 0 V; no



5.7.3. Find a value for the source function v_g so that $i(0+) = 2$ A. For this v_g, find $v(0-)$ and $v(0+)$.

−4 V; −4 V; −4 V



5.8 PRACTICAL CAPACITORS AND INDUCTORS

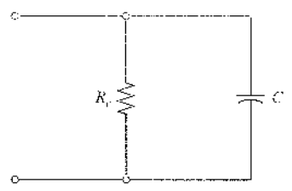
Commercially important capacitors are produced either as discrete elements or are deposited onto integrated-circuit (IC) substrates during the IC manufacturing process. Discrete capacitors are available in a wide variety of types, values, and voltage ratings. The capacitor type is generally classified by the kind of dielectric used, and its capacitance is determined by the type of dielectric and the physical geometry of the device. The voltage rating, or working voltage, is the maximum voltage that can safely be applied to the capacitor. Voltages exceeding this value may permanently damage the device by destroying or breaking down the dielectric.

Simple discrete capacitors may be constructed using two strips of metal foil separated by a dielectric insulating material. The foil and dielectric are pressed together into sheets that are then rolled or folded into a compact package. Electrical conductors attached to each metal-foil sheet constitute the terminals of the capacitor.

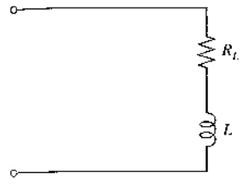
Unlike ideal capacitors, practical capacitors dissipate small but nonzero amounts of power. This is due primarily to *leakage currents* that flow within the dielectric material in the device. Practical dielectrics have a nonzero conductance that allows charge to leak directly from one capacitor plate back to the other (internally, in addition to the nonleakage current moving through the external circuit path). Leakage current may be included in a circuit model for a practical capacitor by placing a resistance in parallel with an ideal capacitor, as shown in Fig. 5.17.

Common types of discrete capacitors, designated by their dielectric, include ceramic, Mylar, Teflon, and polystyrene. In addition, larger capacitive values are usually of the electrolytic type requiring polarized use (one terminal of the capacitor must be kept at a higher potential than the other). The materials and structures of integrated-circuit capacitors depend on the IC technology being used, frequently consisting of a thin layer of an insulating oxide dielectric between layers of doped semiconductor materials acting as the conducting plates.

Practical inductors are available only as discrete elements (or preassembled into electronic packages containing many discrete elements), not as constituents of integrated



Circuit model for a practical capacitor.



Circuit model for a practical inductor.

circuits. Like practical capacitors, they dissipate a small but nonzero amount of power. This dissipation results from losses associated with the fact that the wire making up the inductor coil has some resistance and also from core losses (losses due to currents induced in the core by the strong magnetic field present there). The magnetic flux is best focused in the inductor core by the use of ferromagnetic materials, but this also increases the core losses. These losses may be modeled by inserting a series resistor into the circuit model for a practical inductor as shown in Fig. 5.18.

The unavailability of IC inductors, a consequence of the difficulty of reproducing coils on flat surfaces and also of the poor magnetic properties of most IC materials, has limited the use of inductors in modern miniaturized circuit design. Much effort has been devoted to finding inductorless designs for IC applications. There are still many electronic applications, however, where inductors are commonplace, ranging from telephone circuits and radio receivers to power supplies and electric motors.

5.8.1. Suppose that we wish to construct a 1-F capacitor by putting 10-μF capacitors in parallel. How many do we need? If the leakage resistance is 10 MΩ for each 10-μF capacitor, what will the total leakage resistance and current be if the 1-F capacitor is put across a 100-V dc source?

Answer: 100,000; 100 Ω; 1 A

5.8.2. Suppose that we create a 50-mH inductor by winding 10,000 turns of thin wire around a 1-cm-diameter form. If the wire has a resistance of 0.01 Ω/cm, what power will be dissipated by the inductor if it carries a current of 20 mA dc? What would it be if this practical inductor was an ideal inductor?

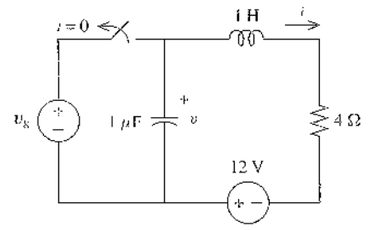
Answer: 40π mW; 0 mW

5.9 SINGULAR CIRCUITS

A circuit in which a switching action takes place that *appears* to violate the continuity principle by producing discontinuities in capacitor voltages or inductor currents is called a *singular circuit*. In this section we consider such circuits and show how the apparent violation may be consistently interpreted and resolved.

5.7.3. Find a value for the source function v_g so that $i(0+) = 2$ A. For this v_g, find $v(0-)$ and $v(0+)$.

−4 V; −4 V; −4 V



5.8 PRACTICAL CAPACITORS AND INDUCTORS

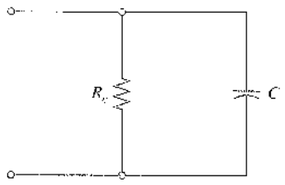
Commercially important capacitors are produced either as discrete elements or are deposited onto integrated-circuit (IC) substrates during the IC manufacturing process. Discrete capacitors are available in a wide variety of types, values, and voltage ratings. The capacitor type is generally classified by the kind of dielectric used, and its capacitance is determined by the type of dielectric and the physical geometry of the device. The voltage rating, or working voltage, is the maximum voltage that can safely be applied to the capacitor. Voltages exceeding this value may permanently damage the device by destroying or breaking down the dielectric.

Simple discrete capacitors may be constructed using two strips of metal foil separated by a dielectric insulating material. The foil and dielectric are pressed together into sheets that are then rolled or folded into a compact package. Electrical conductors attached to each metal-foil sheet constitute the terminals of the capacitor.

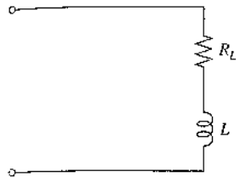
Unlike ideal capacitors, practical capacitors dissipate small but nonzero amounts of power. This is due primarily to *leakage currents* that flow within the dielectric material in the device. Practical dielectrics have a nonzero conductance that allows charge to leak directly from one capacitor plate back to the other (internally, in addition to the nonleakage current moving through the external circuit path). Leakage current may be included in a circuit model for a practical capacitor by placing a resistance in parallel with an ideal capacitor, as shown in Fig. 5.17.

Common types of discrete capacitors, designated by their dielectric, include ceramic, Mylar, Teflon, and polystyrene. In addition, larger capacitive values are usually of the electrolytic type requiring polarized use (one terminal of the capacitor must be kept at a higher potential than the other). The materials and structures of integrated-circuit capacitors depend on the IC technology being used, frequently consisting of a thin layer of an insulating oxide dielectric between layers of doped semiconductor materials acting as the conducting plates.

Practical inductors are available only as discrete elements (or preassembled into electronic packages containing many discrete elements), not as constituents of integrated



Circuit model for a practical capacitor.



Circuit model for a practical inductor.

circuits. Like practical capacitors, they dissipate a small but nonzero amount of power. This dissipation results from losses associated with the fact that the wire making up the inductor coil has some resistance and also from core losses (losses due to currents induced in the core by the strong magnetic field present there). The magnetic flux is best focused in the inductor core by the use of ferromagnetic materials, but this also increases the core losses. These losses may be modeled by inserting a series resistor into the circuit model for a practical inductor as shown in Fig. 5.18.

The unavailability of IC inductors, a consequence of the difficulty of reproducing coils on flat surfaces and also of the poor magnetic properties of most IC materials, has limited the use of inductors in modern miniaturized circuit design. Much effort has been devoted to finding inductorless designs for IC applications. There are still many electronic applications, however, where inductors are commonplace, ranging from telephone circuits and radio receivers to power supplies and electric motors.

5.8.1. Suppose that we wish to construct a 1-F capacitor by putting 10-μF capacitors in parallel. How many do we need? If the leakage resistance is 10 MΩ for each 10-μF capacitor, what will the total leakage resistance and current be if the 1-F capacitor is put across a 100-V dc source?

Answer: 100,000; 100 Ω; 1 A

5.8.2. Suppose that we create a 50-mH inductor by winding 10,000 turns of thin wire around a 1-cm-diameter form. If the wire has a resistance of 0.01 Ω/cm, what power will be dissipated by the inductor if it carries a current of 20 mA dc? What would it be if this practical inductor was an ideal inductor?

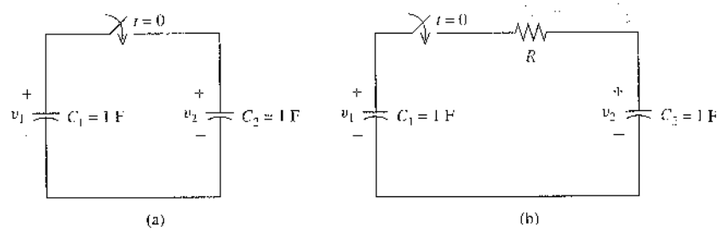
Answer: 40π mW; 0 mW

5.9 SINGULAR CIRCUITS

A circuit in which a switching action takes place that *appears* to violate the continuity principle by producing discontinuities in capacitor voltages or inductor currents is called a *singular circuit*. In this section we consider such circuits and show how the apparent violation may be consistently interpreted and resolved.

Example 5.13

Let us consider first Fig. 5.19(a), where the 1-F capacitors C_1 and C_2 have voltages of 1 V and 0 V, respectively, just prior to the closing of the switch. That is, $v_1(0-) = 1$ V and $v_2(0-) = 0$ V. Just after the switch closes, continuity requires that $v_1(0+) = 1$ and $v_2(0+) = 0$. But KVL applied at time 0+ demands that $v_1(0+) + v_2(0+) = 0$. Either continuity or KVL is apparently violated. Which rule should give way?



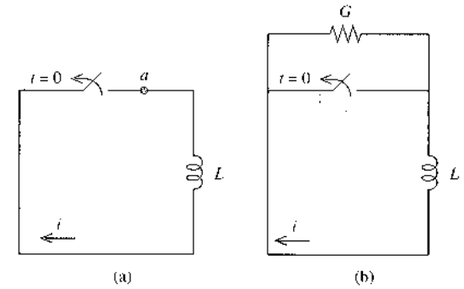
(a) Singular circuit with capacitors; (b) improved circuit model.

Recall that our justification of continuity was based on the assertion that infinite currents and voltages are not possible in *practical* circuits. In the realm of pure mathematics, there is nothing wrong with the concept of infinity. It serves us well in imagining what the function $f(x) = 1/x$ approaches as x goes to zero, or what the sum of all positive integers up to N goes to as N gets bigger and bigger.

What goes wrong in singular circuits is that they are not useful models for *practical* circuits, those that can be approximated in practice in the real world by physical means. They are similar to other impractical circuits we have identified previously, that is, parallel voltage sources, or series current sources, with different source functions. We earlier concluded that the terminal law for an ideal independent source, so useful in the context of most circuits, clashes irreconcilably with the Kirchhoff laws for these peculiar circuits. We agreed in that case to *resolve the apparent contradictions when they arise by use of a better model.* In the case of the sources, this was done by adding a finite, nonzero Thevenin or Norton equivalent resistance to the ideal source model, effectively eliminating the mathematical conflict. If we stubbornly insist on the original, inadequate source models in these circuits, we simply have to admit an inherent mathematical contradiction and refuse to analyze the self-contradictory circuit further.

In the case of singular circuits such as Fig. 5.19(a), the apparent contradiction between continuity and the Kirchhoff laws is also removed by using a better circuit model. One such model takes the nonzero resistance of any practical wire into account by adding a series resistance into the loop, as shown in Fig. 5.19(b). Then we may have continuity satisfied, that is, $v_1(0+) = 1$ and $v_2(0+) = 0$, and at the same time have KVL satisfied, since the difference of these voltages will now be accounted for as a voltage drop across the wire v_R, where $v_R(0+) = v_1(0+) - v_2(0+) = 1$ V. Note that if the wire resistance R is small, as we might expect in practice, then by Ohm's law the current in the wire, $i(0+) = v/R = 1/R$, will be quite large. Indeed, in the limit as R gets smaller and smaller, the current goes to infinity, the "impractical" value demanded by the circuit of Fig. 5.19(a). Normally, of course, we need not model the wire resistance since this is not required to limit the current to a finite value.

The second example of a singular circuit is shown in Fig. 5.20(a). Once again a contradiction between continuity and a Kirchhoff law is apparent at time $t = 0+$, the time continuity of inductive current i and Kirchhoff's current law applied at node a. Once again, resolution of the conflict requires use of a more practical model. Suppose that we were to build this circuit physically. What would happen when we threw the switch open? The power stored in the inductor's magnetic field would not instantaneously drop to zero, as it would be required to do if the current instantaneously dropped to zero. The current would seek another path. Indeed, we would probably be greeted with a quite visible spark arcing across the air gap produced as the switch is opened. A model for the conductivity of the air gap is included in Fig. 5.20(b). If conductance G is small, as expected in practice (air is a poor conductor), then with $i(0+) = 1$ A by continuity, we have $v(0+) = 1/G$, a large value. Indeed, this circuit suggests the design of automobile breaker point ignition systems, in which the goal is to produce a very high voltage by interrupting the current through the ignition coil or inductor with a pair of moving breaker points. The inevitable arcing predicted by our analysis is what causes these points to corrode and eventually fail.



(a) Singular circuit with inductor; (b) improved circuit model.

5.9.1. Suppose that $i = \frac{1}{2}$ A in Fig. 5.20(b). If the air gap conductance $G = 10$ μS between the switch terminals, how much voltage is produced across the air gap immediately after the switch is thrown open?

50 kV

5.9.2. If an ideal 100-μF capacitor is charged to 100 V and then suddenly shorted with an ideal connecting wire, assuming that KVL is not violated, how much energy must be instantaneously shed by the capacitor? In the same situation, using a practical model, we include 0.01-Ω

wire resistance in series with the ideal capacitor. What is the current flow through the wire just after it is connected? What is its dc steady-state value?

1/2 J; 10 kA; 0 A

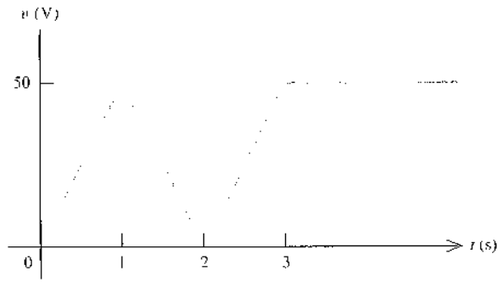
SUMMARY

This chapter adds inductors and capacitors to the set of circuit elements we employ, thereby completing the list. Their i–v laws contain derivatives and integrals, thus bringing calculus, and most particularly differential equations, front and center. Although not capable of generating energy, these elements store energy drawn from the rest of the circuit and later return it. This produces more complex behavior than resistors are capable of, which is explored in Chapters 6 and 7.

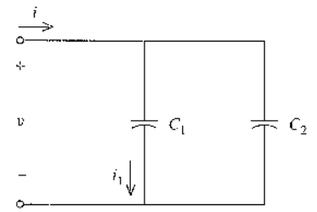
- Capacitors obey the i–v law $i = C\,dv/dt$, where C is the capacitance in farads and i and v are assumed to satisfy the passive sign convention.
- Energy is stored by a capacitor in the electrical field between its plates. Its value is $\frac{1}{2}Cv^2$.
- Parallel capacitance adds, series capacitance obeys the reciprocal–reciprocal law.
- Inductors obey the i–v law $v = L\,di/dt$, where L is the inductance in henries and i and v are assumed to satisfy the passive sign convention.
- Energy is stored by an inductor in the magnetic field threading its turns. Its value is $\frac{1}{2}Li^2$.
- Series inductance adds, parallel inductance obeys the reciprocal–reciprocal law.
- In dc steady state $dv/dt = di/dt = 0$, so capacitors behave like open circuits, inductors like short circuits.
- The continuity principle states that inductive currents and capacitive voltages cannot jump discontinuously. To do so requires infinite power.
- Singular circuits are those in which either the Kirchhoff laws or the continuity principle must be violated. They can be avoided with the use of more realistic circuit models.

5.1. Find the current i through a 20-μF capacitor whose voltage is $v(t) = 2t\cos 6t$ V. Assume that the passive sign convention is satisfied.

5.2. Find the current i through a 1-F capacitor whose voltage v is shown, (a) assuming that i and v satisfy the passive sign conventions and (b) assuming that they do not.



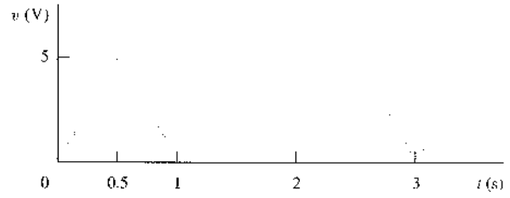
5.3. If this subcircuit has terminal law $i = 12(dv/dt)$ and i_1 is found to be $20\cos 2t$ A when v is $4\sin 2t$ V, find C_1 and C_2.



5.4. A 3-pF capacitor is in series with a current source of value $i = 5\sin 1000\pi t$ μA. If $v(0) = 0$ V, what is its voltage v at $t = 1$ ms? At $t = 1$ s? Assume that v and i satisfy the passive sign convention.

5.5. Find the voltage v across a 1-F capacitor whose current in milliamperes is given in the graph for Problem 5.2.

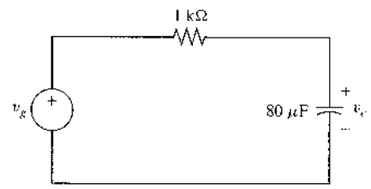
5.6. The voltage across a 0.2-μF capacitor is the triangular wave shown. Find the current and power for $0 < t < 3$ s.



5.7. What constant current is required to deliver a charge of 40 μC to a 1-μF capacitor in 4 ms?

5.8. How long will it take for a constant 20-mA current to deliver a charge of 80 μC to a 1-μF capacitor? If the same charge resides on a 20-μF capacitor, what is the voltage?

5.9. Find v_g if $v_C = 25e^{-10t}$ V.

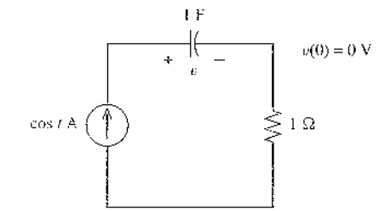


5.10. Find the energy stored in the capacitor in Problem 5.1 versus time.

5.11. A capacitor with an equal and opposite charge of 4 μC on each plate is found to have a voltage of 80 V. Find its capacitance and the energy stored.

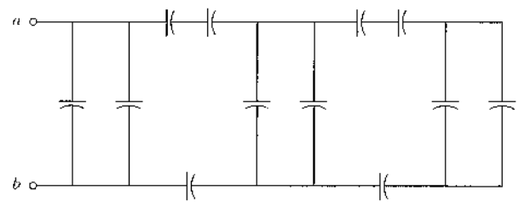
5.12. A 100-μF capacitor is placed in parallel with an independent voltage source of te^{-t} V at $t = 0$. At what time instant will the power transferred by the source to the capacitor be maximum? What will this maximum power be?

5.13. Determine the energy stored in the capacitor and the energy dissipated by the resistor as functions of t for $t \geq 0$. As t gets larger and larger, which element is the larger net energy "customer" for the source?

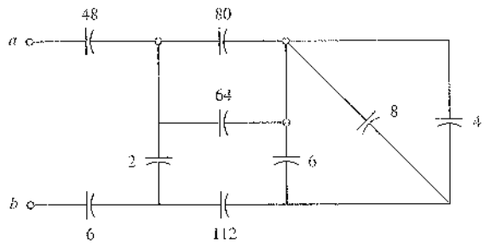


5.14. The current through a 0.01-F capacitor is $5\cos 25t$ A. Find the voltage v and the power p for $t > 0$ if $v(0) = 0$. Find the maximum value of p and the smallest value of t for which it occurs.

5.15. All capacitors are 1 μF. Find the equivalent capacitance at a–b.



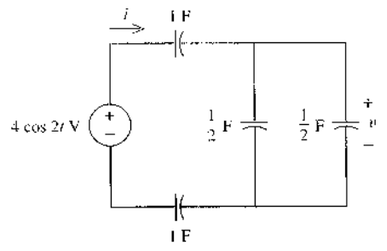
5.16. The capacitances are in picofarads. Find the equivalent capacitance at a–b.



5.17. What is the equivalent capacitance of the series interconnection of m subcircuits each consisting of n identical C-farad capacitors in parallel? Use this result to design a $\frac{41}{42}$-μF capacitor using only $\frac{1}{3}$-F capacitors.

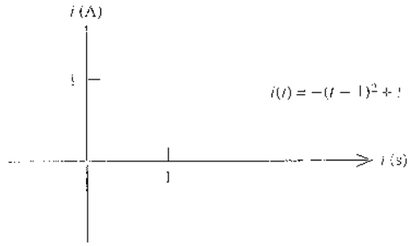
5.18. We wish to construct a 12.35-μF capacitor out of 10-μF capacitors. Show how this can be done. How many 10-μF capacitors do you need?

5.19. Find i and v for $t \geq 0$ if all the capacitors are initially discharged at $t = 0$.



5.20. Find the voltage v across a 20-mH inductor whose current is $i(t) = 4 \sin 10\pi t + 1$ A. Assume that the passive sign convention is satisfied.

5.21. Find the voltage v across a 1-H inductor whose current i is shown. Assume that the passive sign convention is satisfied.



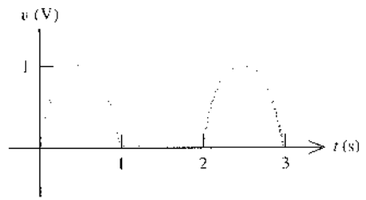
5.22. If an inductor with current $i(t) = 20 \cos 100t$ mA has a voltage $v(t) = 5 \sin 100t$ V: (a) What is its inductance? (b) Do i and v together satisfy the passive sign convention?

5.23. For what value of ω will the amplitude of the voltage across a 2-mH inductor with current $i = 18 \cos \omega t$ A be +12 V?

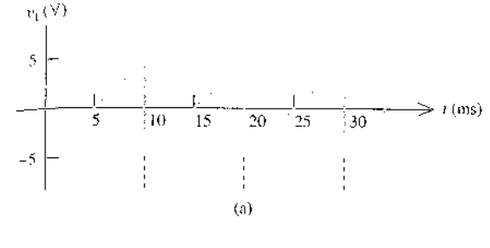
5.24. Solve Problem 5.2 if v is the voltage across a 50-mH inductor, and $i(0) = 0$.

5.25. What class of voltage waveforms will produce piecewise linear current waveforms through an inductor? Justify.

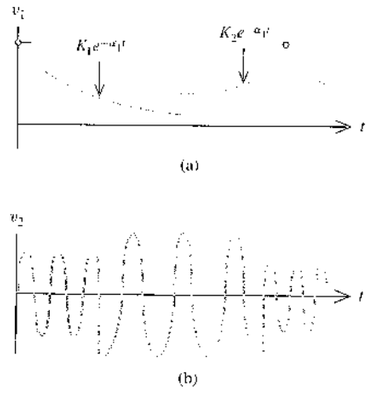
5.26. Let $v(t) = \sin \pi t$ V for $0 \leq t \leq 1$ and $2 \leq t \leq 3$, but otherwise $v(t) = 0$. If this is the voltage across an inductor, find the current through it for $t \geq 0$. Assume that $i(0) = 1$ A and that i and v together satisfy the passive sign convention.



5.27. If $i(0) = 10$ A is the initial current in a 1-H inductor, find the currents for $t > 0$ for the two voltages shown. Assume that the passive sign convention is satisfied.

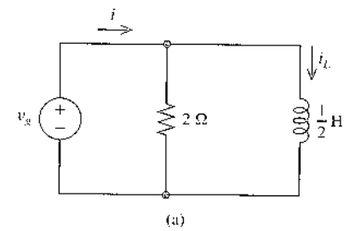


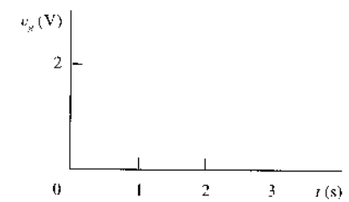
5.28. What type of current waveforms in inductors will produce piecewise exponential voltage waveforms such as those shown in (a)? Piecewise sinusoidal as in (b)?



5.29. Find the terminal voltage of a 10-mH inductor if the current is (a) 2 A, (b) $20t$ A, (c) $10 \sin 100t$ A, and (d) $10(1 - e^{-5t})$ A.

5.30. For the circuit shown in (a), the source voltage is given in (b). Find the current i if $i_L(0) = -1$ A for (a) $0 < t < 1$ s and (b) $1 < t < 2$ s.





5.31. The current through a 0.1-H inductor is $i = 10 \cos 10t$ mA. Find (a) the terminal voltage, (b) the power, (c) the stored energy, and (d) the maximum value of the power being absorbed.

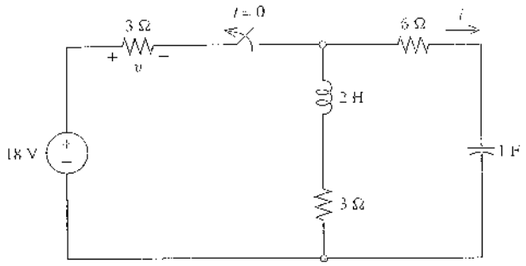
5.32. Find the energy stored versus t for the inductor of Problem 5.20.

5.33. Repeat Problem 5.32 for the inductor of Problem 5.21.

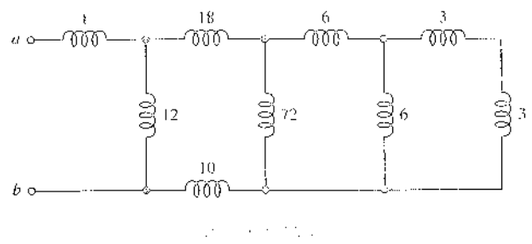
5.34. What is the maximum power that must be generated by a voltage source of $10 \cos 2t$ V that is placed in parallel with a 10-mH inductor at $t = 0$. Assume that the inductive current at $t = 0-$ is zero.

5.35. Find w_2/w_1, the ratio of energy stored in two series inductors if both start at $t = 0$ with $i(0) = 0$.

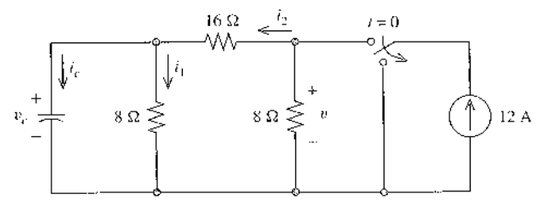
5.36. If $v(0-) = 9$ V, $i(0-) = 1$ A, and the switch is opened at $t = 0$, find $i(0+)$ and $di(0+)/dt$.



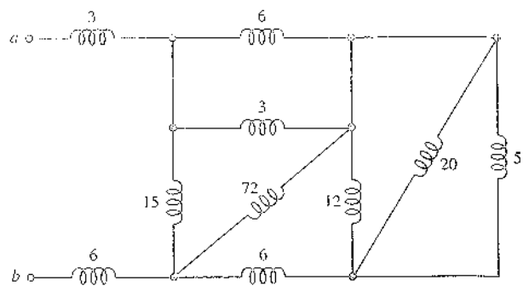
5.37. The inductances shown are all in mH. Find the equivalent inductance seen at terminals *a–b*.



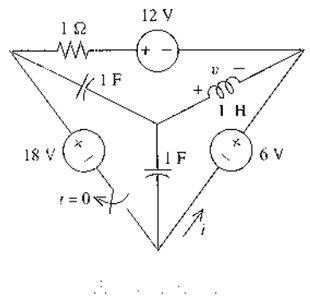
5.38. If $v(0-) = 72$ V, find i_1, i_2, i_c, v_c, and v at $t = 0-$ and at $t = 0+$.



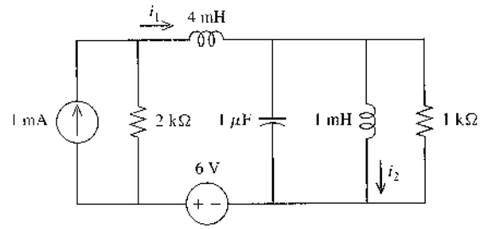
5.39. If inductances are all in mH, find the equivalent inductance seen at terminals *a–b*.



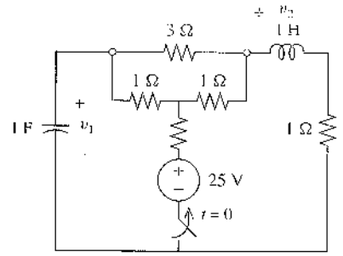
5.40. This circuit is in dc steady state at $t = 0-$. Find $v(0-)$, $v(0+)$, $i(0-)$, and $i(0+)$.



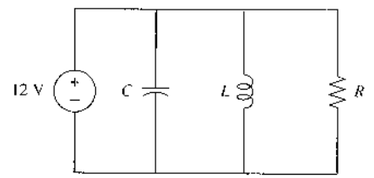
5.41. Find the dc steady-state currents i_1 and i_2.



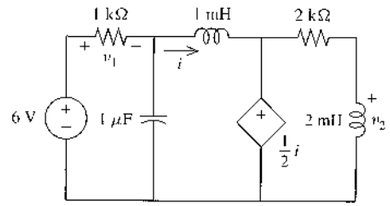
5.42. If the circuit is in dc steady state at $t = 0-$, find v_1 and v_2 at $t = 0-$ and $t = 0+$.



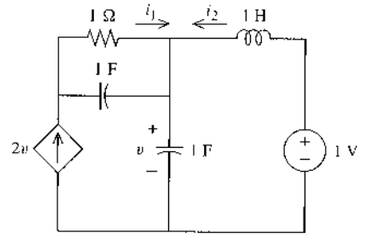
5.43. Show that this circuit does not have a dc steady state. Why?



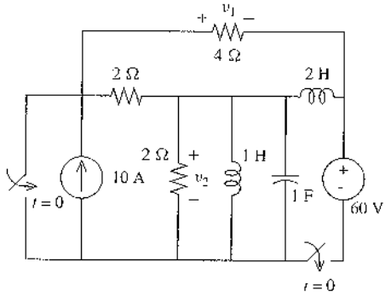
5.44. Find i, v_1, and v_2 in dc steady state.



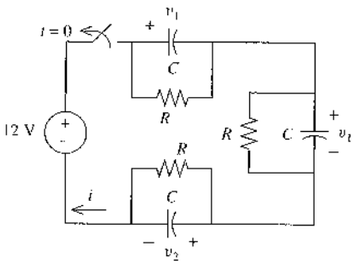
5.45. Repeat Problem 5.41 for this circuit.



5.46. Repeat Problem 5.42 for this circuit. Note there are two switches, both acting at the same time $t = 0$.



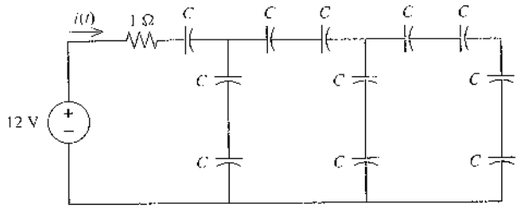
5.47. Assume that the circuit is in dc steady state at $t = 0-$. Find $i(0-)$ and $i(0+)$, in terms of R and C.



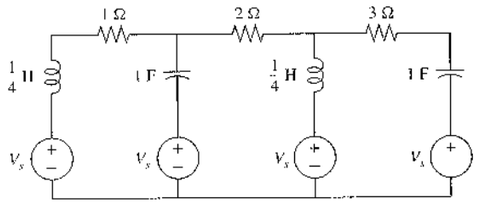
5.48. Repeat Problem 5.43 using practical inductors and capacitors in place of the ideal ones shown. Does the same conclusion hold?

5.49. Sketch a subcircuit containing five capacitors and two marked terminals *a–b* such that none of the capacitors are in series or parallel. Is this possible with a four-capacitor subcircuit? If so, sketch.

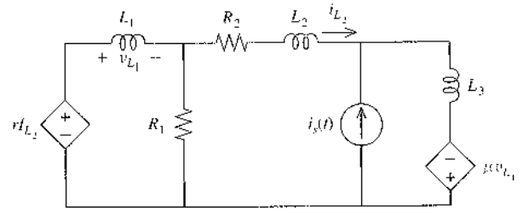
5.50. Find C so that $i(t) = 12e^{-4t}$ A, $t > 0$. Assume that all capacitive voltages are zero at $t = 0-$.



Find the dc steady-state power dissipated by each of the three resistors, and produced by each of the four sources.



Write (do not solve) the mesh equations. The only unknowns should be mesh currents. Then repeat for nodal equations.





Joseph Henry
1797–1878

Blot out these two names [Joseph Henry and Michael Faraday] and the civilization of the present world would become impossible.

H. S. Carhart

6

First-Order Circuits

Michael Faraday's great discovery in 1831 of electromagnetic induction was being independently duplicated at about the same time by American physicist Joseph Henry, but Faraday was credited with the discovery because his results were published first. Henry became famous, however, as the discoverer of the inductance (called self-inductance) of a coil and as the developer of powerful electromagnets capable of lifting thousands of pounds of weight. He was also America's foremost nineteenth-century physicist and the first secretary of the newly formed Smithsonian Institution.

Henry was born near Albany, New York, and his early years were spent in poverty. His ambition was to become an actor until by chance at age 16 he happened upon a book of science, which caused him to devote his life to the acquisition of knowledge. He enrolled in the Albany Academy and upon graduation became a teacher there. In 1832 he joined the faculty of the College of New Jersey, now Princeton, and in 1846 joined the Smithsonian Institution. In his honor the unit of inductance was given the name *henry* 12 years after his death.

6.1 *Simple RC and RL Circuits without Sources*
6.2 *Time Constants*
6.3 *General First-Order Circuits without Sources*
6.4 *Circuits with DC Sources*
6.5 *Superposition in First-Order Circuits*
6.6 *Unit Step Function*
6.7 *Step and Pulse Responses*
6.9 *Design of First-Order Circuits*
Summary
Problems

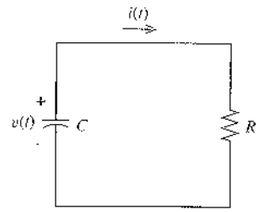
In Chapter 4 resistive circuits were analyzed, and in Chapter 5 the two energy storage elements, capacitors and inductors, were introduced. In this chapter we analyze the class of circuits that result when we add to resistive circuits a single storage element. Such a circuit is called a *first-order circuit* because, as we shall see, the storage element results in a first-order differential equation characterizing the circuit.

We concern ourselves first with simple RC and RL circuits that contain no independent sources. Lacking such sources, any circuit response can only be the result of initial capacitive or inductive energy stored within the circuit and is thus dictated by the nature of the circuit itself. For this reason the response is known as the *natural response* of the circuit. The natural response is characterized by a single time constant or rate of exponential decay.

Following our study of circuits without sources, we shall consider the total response of first-order circuits containing sources. We shall find that it consists of two additive parts, a natural response, identical in form to the response in the absence of sources, and a *forced response*, governed by the form of the forcing functions (source functions of the independent sources). The case in which all independent sources are dc, that is, constant, is considered in detail. Step and pulse sources and their responses are next discussed. The chapter concludes with a look at how SPICE can be used to determine responses for these circuits.

6.1 SIMPLE *RC* AND *RL* CIRCUITS WITHOUT SOURCES

We begin our study of first-order circuits by considering the simple one-loop circuit containing just a capacitor and a resistor, as shown in Fig. 6.1. Suppose that the capacitor is charged to a voltage of V_0 volts at an *initial time*, which we shall take as $t = 0$. By initial time we mean the beginning of the time period we are interested in, not the time of the circuit's construction (the circuit must have a history extending beyond our initial time, since the value of V_0 volts must have been established by external circuitry before this time).



Source-free *RC* circuit.

Since there are no current or voltage sources in the network, the circuit response (v or i) is due entirely to the energy that is stored initially in the capacitor. The energy at the initial time $t = 0$ is, by (5.4),

$$w_C(0) = \tfrac{1}{2}CV_0^2$$

We seek to determine $v(t)$ for $t > 0$. Applying KCL at the top node yields

$$C\frac{dv}{dt} + \frac{v}{R} = 0$$

or

$$\frac{dv}{dt} + \frac{1}{RC}v = 0$$

which is a *first-order differential equation*. (The *order* of a differential equation is defined as the order of the highest-order unknown derivative contained in the equation.)

Several methods are available for solving first-order differential equations. One straightforward method is to rearrange the terms in the equation so as to separate the variables v and t. These variables may be separated by rewriting (6.2) as

$$\frac{dv}{v} = -\frac{1}{RC}dt$$

Then, taking the indefinite integral of each side, we have

$$\int \frac{dv}{v} = -\frac{1}{RC}\int dt$$

or

$$\ln(v) = -\frac{t}{RC} + k$$

with k the constant of integration. Exponentiating both sides, we have

$$v = Ke^{-(t/RC)}$$

where e is the base of the natural logarithms, $e = 2.71828\ldots$, and $K = e^{+k}$.

For this solution to be valid in the interval of interest $t > 0$, continuity requires that $v(0)$ must match the specified initial condition $v(0) = V_0$. Thus while $v(t)$ in (6.4) will satisfy the differential equation for any K, for only a single value of K will this solution satisfy both differential equation and initial condition. This is found by forcing $v(0)$ in (6.4) to the required value of V_0:

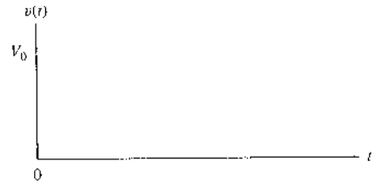
$$v(0) = Ke^0 = K = V_0$$

whence

$$v(t) = V_0e^{-(t/RC)}$$

Note that all other circuit variables follow immediately from (6.5), although for the moment we will keep our focus on the capacitive voltage $v(t)$.

A graph of the circuit response $v(t)$ is shown in Fig. 6.2. The voltage begins at V_0, as required, and then decays exponentially to zero with time. The rate at which it decays to zero is given by the RC product of the circuit. Since this response is governed by the circuit elements themselves and not by some independent source "forcing" a different



Voltage $v(t)$ in the circuit of Fig. 6.1.

behavior during $t \geq 0$, this response is called the *natural response* of the circuit. The natural response is synonymous with the response in the absence of independent sources.

At any time t, the energy stored in the capacitor $w_C(t)$ is given by $\frac{1}{2}Cv^2(t)$. The resistor stores no energy; thus the total energy stored in the circuit at time t is, by (6.5),

$$w_C(t) = \tfrac{1}{2}C\left[V_0 e^{-(t/RC)}\right]^2 = \tfrac{1}{2}CV_0^2 e^{-2t/RC}$$

Comparing with the initial stored energy given by (6.1) gives

$$w_C(t) = w_C(0)e^{-2t/RC} \tag{6.6}$$

Note that the stored energy is exponentially decaying to zero from its initial value $w(0)$. Where does this lost energy go? We know that the power dissipated by a resistor is

$$p_R(t) = \frac{v_R^2}{R} = \frac{\left[V_0 e^{-(t/RC)}\right]^2}{R}$$

Integrating from the initial time 0 to time t, we have the total energy dissipated by the resistor from time 0 to t:

$$w_R(t) = \frac{V_0^2}{R}\int_0^t e^{-2\tau/RC}\,d\tau$$

or

$$w_R(t) = \frac{1}{2}CV_0^2\left(1 - e^{-2t/RC}\right) = w_C(0)\left(1 - e^{-2t/RC}\right)$$

Comparing the last to (6.6), we see where the missing energy has gone.

$$w_C(0) - w_C(t) = w_R(t)$$

At each instant of time, the decrease in energy stored in the capacitor since $t = 0$ exactly equals the total energy dissipated by the resistor up to that time. As time increases, less energy remains stored in the circuit; and in the limit as t gets very large, all the initial stored energy $w_C(0)$ has been dissipated as heat in the resistor and no stored energy remains. As this occurs, $v(t)$ and all other voltages and currents in the circuit go to zero as well. Clearly, it is the internal energy stored at the initial time that is driving this "natural" response.

Example 6.1

Consider the circuit of Fig. 6.1 with $C = 1\ \mu$F, $R = 1$ kΩ, and an initial voltage of $v(0) = 5$ V across the capacitor. We wish to know the current $i(t)$ and the maximum power dissipated by the resistor.

From (6.5), the capacitive voltage is

$$v(t) = V_0 e^{-t/RC} = 5e^{-1000t}\ \text{V}$$

Then, since $v(t)$ is also the voltage across the resistor,

$$i(t) = \frac{v(t)}{R} = 0.005e^{-1000t}\ \text{A}$$

The power dissipated by the resistor is

$$p_R(t) = \frac{v^2(t)}{R} = 0.025e^{-2000t}\ \text{W}$$

Thus the maximum power of 25 mW is dissipated right at the initial time $t = 0$. This is confirmed by the graph in Fig. 6.2, which shows that the sharpest reduction in capacitive voltage, and thus stored power, occurs at the initial time.

Example 6.1 used elements with values commonly found in low-power circuits: resistance in kilohms and capacitance on the scale of microfarads. This is a departure from most other examples, which have used elements on the scale of 1 Ω and 1 F. While convenient for calculation, this scale of R and C values is not often seen in practice. Resistors of the scale of 1 Ω lead to high current and power dissipation at common voltage levels of 1.5 to 150 V, and capacitors on the scale of 1 F would be massively large and expensive.

For circuits with elements in the far more common kΩ–μF ranges, it is often convenient to use a *scaled* metric system to write equations. Carrying cumbersome powers of 10 through most calculations can be avoided. Suppose that we use the scaled units listed in Table 6.1. The first three are basic SI quantities, the rest are derived from them. Since $R = E/I$ (we are using E to avoid confusing voltages or electrical potential difference, with the unit of volts), then V/mA = kΩ and the scaled resistance unit is the kilohm. Similarly, the scaled unit of capacitance is mA–ms/V or μF. In this system inductance is measured as V–ms/mA = V–s/A, which is the basic SI unit of henries (H). *The scaled SI unit system V–mA–ms–kΩ–μF–H is consistent without the need to carry powers of 10.* Had we used this scaled system in Example 6.1 with $R = 1$ kΩ, $C = 1\ \mu$F, and $V_o = 1$ V, applying (6.5) we would have written

$$v(t) = 5e^{-t/(1)(1)} = 5e^{-t}\ \text{V}$$

Table 6.1 Standard and Scaled SI Unit Systems

	Quantity	Standard SI Unit	Scaled SI Unit
Basic:	Voltage (E)	V	V
	Current (I)	A	mA
	Time (t)	s	ms
Derived:	Resistance (E/I)	Ω	kΩ
	Capacitance ($I - t/E$)	F	μF
	Inductance ($V - t/I$)	H	H
	Power ($E - I$)	W	mW

and the time constant is $\tau = 1$, with units of milliseconds. Continuing, we find that

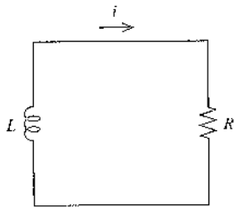
$$i(t) = \frac{v(t)}{R} = \frac{5e^{-t}}{1} = 5e^{-t} \text{ mA}$$

$$p_R(t) = \frac{v^2(t)}{R} = \frac{(5e^{-t})^2}{1} = 25e^{-2t} \text{ mW}$$

Comparing Example 6.1, the scaled system results in more compact equations. The reader is encouraged to use this system on problems that fit these scaled units. Many such will be found in the exercises and in the end-of-chapter problems in the remainder of this book.

We next turn to the simple RL circuit shown in Fig. 6.3. As with the previous RC circuit, there are no external sources and the response will be driven by initial stored energy, in this case proportional to the square of the initial current $i(0) = I_0$ through the inductor. The energy stored at the initial time is, by (5.11),

$$w_L(0) = \tfrac{1}{2} L I_0^2$$



Source-free *RL* circuit.

Applying KVL yields

$$L\frac{di}{dt} + Ri = 0$$

or

$$\frac{di}{dt} + \frac{R}{L}i = 0$$

This equation is of the same form as that of (6.2) for the RC circuit. We may therefore solve it by the same method, separation of variables.

Instead, let us observe that since the present equation is of the same form as (6.2), we expect its solution will be of the same form as well. We know that the form of this previous solution, given in (6.5), is a constant times an exponential function of time. Guided by this experience, let us introduce a *trial natural solution*,

$$i(t) = Ke^{st}$$

where K and s are constants to be determined. If the trial natural solution truly solves the differential equation, we must be able to substitute it successfully into this equation, or

$$\frac{d}{dt}(Ke^{st}) + \frac{R}{L}(Ke^{st}) = 0$$

from which we see that

$$\left(s + \frac{R}{L}\right)(Ke^{st}) = 0$$

If our trial natural solution is to work, this equation must be valid for all $t \geq 0$. However, the factor Ke^{st} cannot be zero for $t \geq 0$, since that would require $K = 0$; hence $i(t) = 0$ for all $t \geq 0$ in contradiction to the initial value $i(0) = I_0$. We conclude that if the trial natural solution is valid, it must be that the other factor is zero:

$$s + \frac{R}{L} = 0$$

This is called the *characteristic equation* for this differential equation, and its solution s specifies the exponent in the solution of the differential equation, in this case

$$s = -\frac{R}{L}$$

We conclude that a valid natural solution must be of the somewhat more specific form

$$i(t) = Ke^{-(R/L)t}$$

Any $i(t)$ of this form satisfies the differential equation (6.7). The only other test a solution must pass is that it satisfy the given initial condition $i(0) = I_0$. Applying this requirement,

$$i(0) = I_0 = Ke^0 = K$$

The only trial natural solution satisfying both the initial condition and the differential equation is the one with specified constants s and K, or

$$i(t) = I_0 e^{-(R/L)t}$$

which is the desired solution. We may verify this solution by substituting $i(t)$ given by (6.10) into the original differential equation we set out to solve (6.7) and confirming that it matches the required initial condition as well. It passes these tests.

Let us review our reasoning. We guessed at an exponential form for the solution and then found values for its parameters s and K that made our guess demonstrably correct. There is no doubt the result is the desired solution, since it solves the differential equation and has the right initial value, and we ask no more of a solution than these two requirements.

Example 6.2

To demonstrate the method of characteristic equations in the case that the initial time is not zero, consider the circuit of Fig. 6.3 with $R = 2\ \Omega$, $L = 1$ H, and initial current $i(t_0) = 3$ A at time $t_0 = 10$ s. By (6.7) we have

$$\frac{di}{dt} + 2i = 0$$

The characteristic equation is $s + 2 = 0$ or $s = -2$. Then the trial natural solution is

$$i(t) = Ke^{-2t}$$

and evaluating at $t_0 = 10$ in order to match initial conditions,

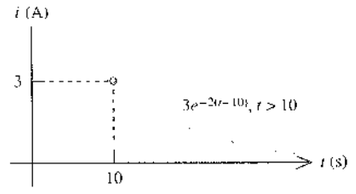
$$i(t_0) = 3 = Ke^{-2(10)}$$

which yields $K = 3e^{20}$. Thus the desired current is

$$i(t) = 3e^{20}e^{-2t} \text{ A}$$

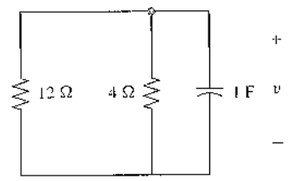
or $$i(t) = 3e^{-2(t-10)} \text{ A}$$

A plot of this current is shown in Fig. 6.4.



$i(t)$ in Example 6.2 for $t \geq 10$.

Examining (6.5) and (6.10), we see that the capacitive voltage and inductive current in these single-loop, source-free circuits are both decaying exponentials. Since the derivative of an exponential is also an exponential, the current through the capacitor $i = C(dv/dt)$ and the voltage across the inductor $v = L(di/dt)$ in these circuits are also decaying exponentials with the same exponents. The significance of the constants $1/RC$ and R/L in these exponentials will be taken up in Section 6.2.



6.1.1. In Fig. 6.1, let $t_0 = 0$, $V_0 = 10$ V, $R = 1$ kΩ, $C = 1$ μF. Find v and i at $t = 1$ ms, $t = 2$ ms and $t = 5$ ms.

Answer: 3.68 V, 3.68 mA; 1.35 V, 1.35 mA; 0.067 V, 0.067 mA

6.1.2. If $v(0) = +10$ V, at what time will $v(t)$ equal +1 V?

Answer: 6.9 s

6.1.3. If $w(t)$ is the energy stored in the capacitor in Exercise 6.1.2, find a formula for $w(t)$, $t > 0$. At what time does half the original stored energy remain?

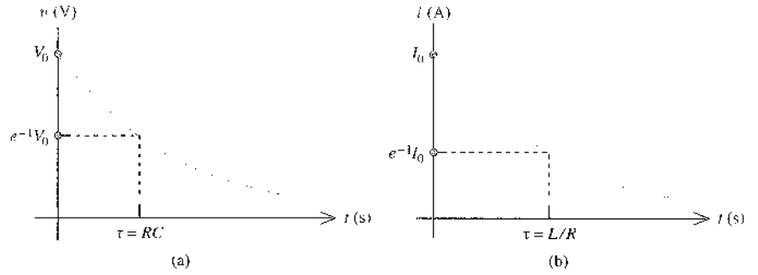
Answer: $50e^{-2t/3}$ J; 1.04 s

6.1.4. In Fig. 6.3, $R = 1$ kΩ. If the voltage across the inductor, defined to satisfy the passive sign convention together with i, is $v_L(t) = 10e^{-200t}$ for $t > 0$ s, find L and the initial current $i(0)$.

Answer: 5 H; 10 mA

6.2 TIME CONSTANTS

In networks that contain energy-storage elements it is very useful to characterize with a single number the rate at which the natural response decays to zero. This role is played by a quantity called the time constant of the circuit.



(a) RC circuit response; (b) RL circuit response.

Consider the graph shown in Fig. 6.5(a), which illustrates a natural response (the capacitive voltage) of the simple RC circuit discussed in Section 6.1. The time required for a natural response to decay by a factor of $1/e$ is defined as the *time constant* of the circuit, which we shall denote by τ. For the given RC circuit, the natural response is

$$v(t) = V_0 e^{-t/RC}$$

By definition of the time constant τ, we require that τ satisfy

$$\frac{v(t+\tau)}{v(t)} = \frac{1}{e}$$

or $$v(t+\tau) = e^{-1}v(t)$$

Substituting for the voltage v, this equation becomes

$$V_0 e^{-(t+\tau)/RC} = e^{-1}V_0 e^{-t/RC}$$

or, after canceling common factors, we have

$$\tau = RC$$

The units of τ are $\Omega \cdot \text{F} = (\text{V/A})(\text{C/V}) = \text{C/A} = \text{s}$. In terms of the time constant τ, the natural response is

$$v(t) = V_0 e^{-t/\tau}$$

The time constant for the RL circuit is computed similarly from Fig. 6.5(b), with

$$i(t) = I_0 e^{-(R/L)t}$$

In this case we have

$$I_0 e^{-(R/L)(t+\tau)} = e^{-1} I_0 e^{-(R/L)t}$$

which, after cancellation, identifies the time constant τ as

$$\tau = \frac{L}{R}$$

and we may rewrite the response in terms of τ as

$$i(t) = I_0 e^{-t/\tau}$$

The time constant τ units are $(H/\Omega) = [V/(A/s)]/(V/A) = s$, once again.

In both RC and RL circuits, each τ seconds the natural response is reduced by a factor of $1/e$ relative to its value at the beginning of that one-time-constant interval. The response at the end of one time constant is reduced to $e^{-1} = 0.368$ of its initial value. At the end of two time constants, it is equal to $e^{-2} = 0.135$ of its initial value, and at the end of five time constants it has become $e^{-5} = 0.0067$ of its initial value. After four or five time constants, the response is essentially zero.

For the RC circuit of Fig. 6.1, we see in Fig. 6.2 that the capacitive voltage $v(t)$ is exponentially decaying to zero with time constant $\tau = RC$. Let us complete our analysis of this circuit by considering the other circuit variables. The resistive voltage just equals $v(t)$ by KVL. The mesh current can be found by dividing the resistive voltage by R and is thus also an exponential decay with $\tau = RC$. *All currents and voltages are exponential decays with the same time constant* τ. Examination of the RL circuit of Fig. 6.3 shows the same statement is true there as well. The behavior of all circuit responses is fixed by a single time constant, $\tau = RC$ in the RC circuit and $\tau = L/R$ in the RL circuit. This is why we refer to τ as the time constant of the *circuit*, rather than the time constant of any specific current or voltage within the circuit.

An interesting property of exponential functions is shown in Fig. 6.6. A tangent to the curve at $t = 0$ intersects the time axis at $t = \tau$. This is easily verified by considering the equation of a straight line tangent to the curve at $t = 0$, given by

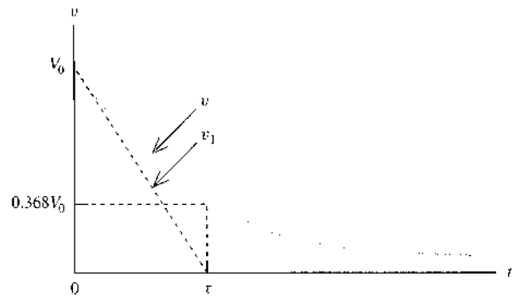
$$v_1(t) = mt + V_0$$

where m is the slope of the line. Differentiating v, we have

$$\frac{dv}{dt} = \frac{-V_0}{\tau} e^{-(t/\tau)}$$

Thus the slope of v at $t = 0$ is $-V_0/\tau$. Then, with $m = -V_0/\tau$,

$$v_1(t) = \frac{-V_0}{\tau} t + V_0$$



and

$$v_1(\tau) = 0$$

as asserted. Similarly, a tangent to the curve at a time t_1 intersects the time axis at $t_1 + \tau$ (see Problem 6.9). This fact is often useful in sketching the exponential function.

From Fig. 6.6 we see that an alternative definition for the time constant is the time required for the natural response to become zero if it decreased at a constant rate equal to the initial rate of decay. Of course, it does not decrease at a constant rate, since its decay rate is everywhere proportional to its value, which is steadily approaching zero.

Knowledge of the time constant allows us to predict the general form of the response, (6.13) or (6.15), but to complete the solution we must know the initial value V_0 or I_0. Frequently, this must be determined in a circuit with switches. Since capacitive voltages and inductive currents are continuous (as discussed in Chapter 5), the desired values just after the switch acts at 0+ can be determined from the state of the circuit just before, at $t = 0-$.

Example 6.3

To illustrate the procedure, let us find the capacitor voltage $v(t)$ in Fig. 6.7(a), given that the circuit was in dc steady state just before the opening of the switch. In dc steady state the capacitor may be replaced by (is equivalent to) an open circuit. The resistance to the left of this open circuit is

$$R_{eq} = 8 + \frac{(3)(6)}{3+6} = 10\ \Omega$$

and, by voltage division,

$$v(0-) = \frac{100 R_{eq}}{R_{eq} + 15} = 40\ \text{V}$$

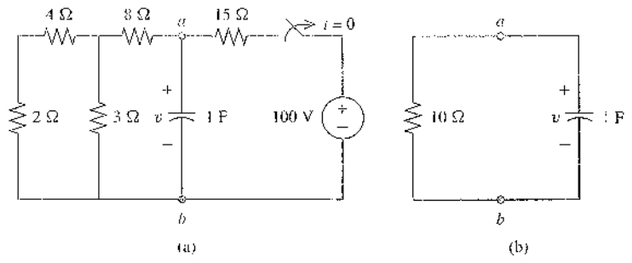
At $t = 0$ the switch opens. By continuity of capacitive voltage, $v(0+) = v(0-) = 40$ V. For $t > 0$, we have a sourceless circuit, as shown in Fig. 6.7(b), where resistors to the left of the capacitor have been replaced by their equivalent $R_{eq} = 10\ \Omega$. The time constant for this RC network is simply the product of the capacitance and the equivalent resistance, given by

$$\tau = R_{eq} C = (10)(1) = 10\ \text{s}$$

Therefore, by (6.13) the voltage is

$$v(t) = 40 e^{-\frac{t}{10}}\ \text{V}$$

Every 10 s, the voltage will decline by a factor of $1/e$, or 0.368. After 50 s, the voltage has declined to $40e^{-5} = 0.27$ V, illustrating that after 5 time constants the response has been reduced by more than two orders of magnitude. Thus after 10 time constants there is a reduction of more than four orders of magnitude, after 15 time constants a reduction of more than six orders of magnitude, and so on.



(a) Circuit for Example 6.3; (b) equivalent for $t \geq 0$.

6.2.1. In an RC circuit, determine (a) τ for $R = 2$ kΩ and $C = 10$ μF, (b) C for $R = 10$ kΩ and $t = 20$ μs, and (c) R for $v(t)$ on a 2-μF capacitor to halve every 20 ms.

(a) 20 ms; (b) 2 nF; (c) 14.43 kΩ

6.2.2. An RC circuit consists of a 20-kΩ resistor and a 0.05-μF capacitor. It is desired to decrease the current in the network by a factor of 5 without changing the capacitor voltage. Find the necessary values of R and C.

100 kΩ; 0.01 μF

6.2.3. In a single-loop RL circuit, the current is determined to be 2 mA at $t = 10$ ms and 100 μA at $t = 46$ ms. Find the time constant τ and the initial current at time $t = 0$.

12 ms; 4.6 mA

6.3 GENERAL FIRST-ORDER CIRCUITS WITHOUT SOURCES

The simple first-order RC and RL circuits examined in Section 6.2 were seen to have decaying responses governed by a single time constant. Generalizing, we must expect the same behavior from any circuit that is equivalent, whether it is simple (solvable from a single equation) or not.

Example 6.4

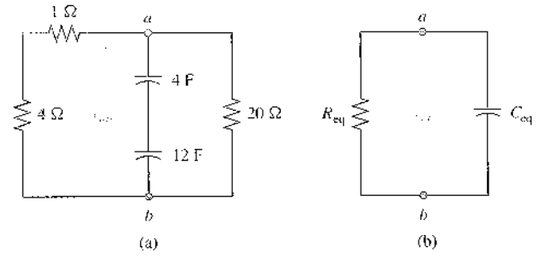
We wish to determine the time constant of the circuit of Fig. 6.8(a). We first find an equivalent circuit of the simple form of Fig. 6.8(b).

$$C_{eq} = \frac{(4)(12)}{4 + 12} = 3 \text{ F}$$

$$R_{eq} = \frac{(20)(5)}{20 + 5} = 4 \ \Omega$$

The time constant τ in Fig. 6.8(b) is

$$\tau = R_{eq}C_{eq} = (4)(3) = 12 \text{ s}$$



(a) RC circuit; (b) equivalent circuit.

Since Figs. 6.8(a) and (b) are equivalent circuits, the voltage $v_{ab}(t)$ must be the same. This voltage has time constant 12 s in (b), so it must also in (a). Indeed, all currents and voltages in both Figs. 6.8(a) and (b) decay exponentially to zero with $\tau = 12$ s.

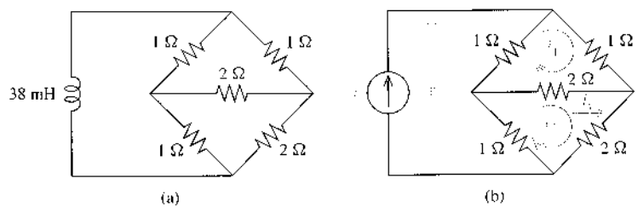
From Example 6.4 we see that RC circuits containing a single equivalent R_{eq} and C_{eq} can be analyzed by reduction to these equivalents, with $\tau = R_{eq}C_{eq}$. Clearly, the same statement applies to RL circuits with single equivalent inductance and resistance, with $\tau = L_{eq}/R_{eq}$. Note that it may not always be possible to use series and parallel rules to find these equivalents, however. The next example shows the general approach in this case.

Example 6.5

We wish to find the time constant of the circuit of Fig. 6.9(a). The circuit to the right of the inductor is purely resistive, and we seek its equivalent resistance R_{eq}. Since none of the resistors is in series or parallel, we cannot use series–parallel equivalents. To find R_{eq}, we will force i to flow into the circuit, as shown in Fig. 6.9(b), and determine the resulting voltage v. Since the terminal law for the circuit is $v = R_{eq}i$, the ratio between v and i is the desired R_{eq}.

Figure 6.9(b) is a three-mesh circuit with one current source, so two mesh equations are required. The mesh equations are

$$\begin{bmatrix} 4 & -2 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} i \\ i \end{bmatrix}$$



(a) RL circuit; (b) circuit to determine R_{eq}.

Inverting the matrix, the determinant is $(4)(5) - (-2)(-2) = 16$, and

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \frac{1}{16}\begin{bmatrix} 5 & 2 \\ 2 & 4 \end{bmatrix} = \frac{1}{16}\begin{bmatrix} 7i \\ 6i \end{bmatrix}$$

By KVL around the outer loop,

$$\begin{aligned} v = v_{ab} &= 1i_1 + 2i_2 \\ &= \tfrac{1}{16}(7i + 12i) \\ &= \tfrac{19}{16}i \end{aligned}$$

Thus $R_{eq} = \frac{19}{16}\ \Omega$ and

$$\tau = \frac{L}{R_{eq}} = \frac{0.038}{19/16} = 0.032\ \text{s} = 32\ \text{ms}$$

Op amp circuits will also exhibit first-order behavior when they contain one storage element or when all storage elements can be combined into a single equivalent. They can be analyzed efficiently using nodal analysis and the virtual short- and open-circuit principles as discussed in Section 4.8.

Example 6.6

Suppose that at $t = 0$, the output in Fig. 6.10 is $v_2(0) = V_0$ and that $v_g(t) = 0$ for $t > 0$. We wish to find $v_2(t)$ for $t > 0$. By the virtual short principle, node 1 is at ground potential. Summing currents into this node gives

$$\frac{v_g - 0}{R_A} + \frac{v_2 - 0}{R_F} + C\frac{d(v_2 - 0)}{dt} = 0$$

or, with $v_g = 0$ for $t > 0$,

$$\frac{dv_2}{dt} + \frac{1}{R_F C}v_2 = 0$$

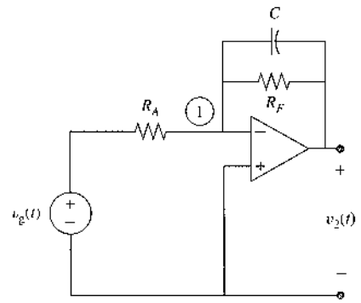


FIGURE 6.10 Circuit for Example 6.6.

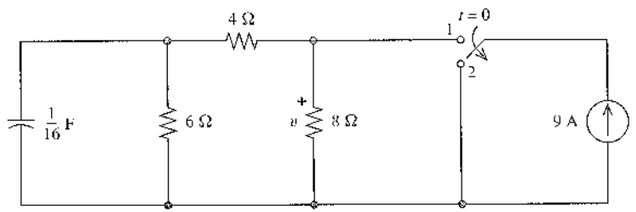
This is our standard unforced first-order differential equation with the solution

$$v_2(t) = v_2(0)e^{-t/R_F C} = V_0 e^{-t/R_F C}$$

In summary, a sourceless circuit containing a single storage element, or the equivalent of a single storage element, will have all responses decay to zero with time constant $\tau = RC$ in the RC case or $\tau = L/R$ in the RL case. Note, however, that circuits with two or more storage elements may well not have a single equivalent storage element and thus will not be broken through the solution of a single first-order differential equation. These are not first-order circuits, and their responses cannot be characterized by a single time constant. Examples include any circuit with both capacitors and inductors and the RC circuit of Problem 6.16. Such circuits are taken up in Chapter 7.

6.3.1. The circuit is in steady state at $t = 0-$ and the switch is moved from position 1 to position 2 at $t = 0$. Find the response $v(t)$ for $t > 0$. What is the time constant τ?

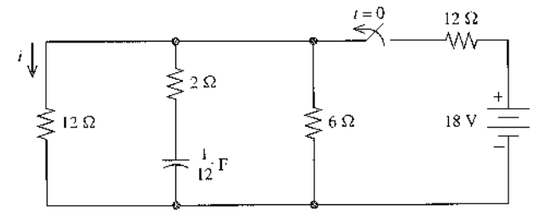
Answer: $16e^{-4t}$ V; $\frac{1}{4}$ s



EXERCISE 6.3.1

6.3.2. Find i for $t > 0$ if the circuit is in steady state at $t = 0-$.

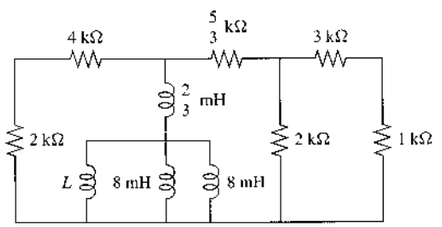
Answer: $\frac{1}{4}e^{-2t}$ A



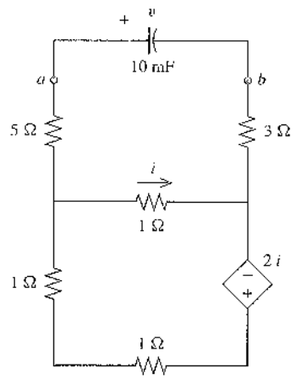
EXERCISE 6.3.2

6.3.3. For what value of L will the time constant τ be 1 μs?

Answer: 2 mH



EXERCISE 6.3.3



6.3.4. (a) Find the equivalent resistance of the circuit below a–b. (b) Use the result to find $v(t)$ for $t > -1$ if $v(-1) = 20$ V.

(a) 10 Ω; (b) $20e^{-10(t+1)}$

6.4 CIRCUITS WITH DC SOURCES

In the preceding sections we have considered source-free circuits whose responses have been the result of initial energies stored in capacitors and inductors. All independent current or voltage sources were removed or switched out of the circuits prior to finding the natural responses. It was shown that these responses, when arising in circuits containing a single equivalent capacitor or inductor, decay to zero exponentially over time.

In this section we examine circuits that in addition to having initial stored energies are driven by constant (dc) independent current or voltage sources. For these circuits we shall obtain solutions that are the result of both initial stored energy and that continually supplied by the sources. We shall find that the responses in this case consist of two parts, one of which is, like the sources themselves, a constant.

Let us begin by considering the circuit of Fig. 6.11. The network consists of the parallel connection of a constant current source and a resistor that is switched at time $t = 0$ across a capacitor having a voltage $v(0-) = V_0$ V. For $t > 0$, the switch is closed and a nodal equation at the upper node is given by

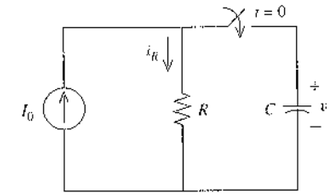
$$C\frac{dv}{dt} + \frac{v}{R} = I_0$$

or

$$\frac{dv}{dt} + \frac{1}{RC}v = \frac{I_0}{C} \qquad (6.\ \ )$$

This first-order differential equation is identical to that for the source-free RC case (6.2), except for the addition of a *forcing term* on the right-hand side. A forcing term in a differential equation is a term $f(t)$ independent of the unknown time functions.

As in the unforced case, this first-order differential equation may be solved either by separation of variables or by a trial solution approach. We choose the latter, since this method extends to higher-order circuits where separation of variables cannot be used.



RC network with dc source.

Examining (6.17) we see that the left-hand side is a linear combination of v and its derivative, while the right-hand side is just a constant. Thus $v(t)$ must be some time function with the property that a linear combination of v and its derivative is constant for all $t > 0$. What kinds of functions have this property? Certainly, $v(t)$ itself equal to a constant is one such class of functions, for the derivative of a constant is zero; so every linear combination of a constant and its derivative is, as required, also just a constant.

Guided by this observation, let us use a *trial forced solution* of the form

$$v_f(t) = A$$

where A is a constant to be determined. As we did with the trial natural solution employed earlier, we will substitute it into the differential equation that it must satisfy to test its validity. Replacing v by $v_f(t) = A$ in (6.17) yields

$$\frac{d}{dt}A + \frac{A}{RC} = \frac{I_0}{C}$$

or

$$A = I_0R$$

Thus we have found a (constant) time function $v_f(t)$ that satisfies the differential equation:

$$v_f(t) = I_0R$$

Any such time function satisfying the forced differential equation is called a *forced solution*. We have found a forced solution to (6.17).

There is a second requirement that any proposed solution to our original problem must satisfy, however. It is not enough to satisfy the differential equation, as $v_f(t)$ has been shown to do. A valid solution must also satisfy the initial condition $v(0-) = v(0+) = V_0$ V (we have used continuity of capacitive voltages to project V_0 across the switch time $t = 0$). Evaluating our forced solution $v_f(t)$ at $t = 0+$, we have $v_f(0+) = I_0R$. So the forced solution agrees with the given initial condition if and only if $V_0 = I_0R$. Examining Fig. 6.11, V_0 and I_0 are clearly the result of quite separate causes; V_0 is an initial condition established before $t = 0$, and I_0 is an independent source function. Thus, while $V_0 = I_0R$ may be satisfied for particular choices of initial conditions and source functions, there is no reason in general for this equation to be true.

Thus, if our forced solution $v_f(t)$ happens to satisfy the given initial condition, we have found the solution; it is $v(t) = v_f(t)$. In most cases, $v_f(t)$ will not satisfy the initial conditions. We are apparently stuck, for there are no additional free parameters in our forced solution $v_f(t)$ to use to satisfy the initial condition.

Let $v_n(t)$ denote any solution to the *unforced* version of our differential equation; that is, $v_n(t)$ satisfies

$$\frac{dv_n}{dt} + \frac{1}{RC}v_n = 0 \tag{6.18}$$

The unforced version is formed by setting the forcing term to zero. From previous work we know that the trial natural solution for v_n is of the form

$$v_n = Ke^{st}$$

with characteristic equation

$$s + \frac{1}{RC} = 0$$

or $s = -1/RC$, so

$$v_n = Ke^{-(1/RC)t} \tag{6.19}$$

We have thus far discovered a v_f that solves the forced differential equation (6.17), and a v_n that, for any value of its free parameter K, satisfies the associated unforced differential equation.

Consider finally a *sum of forced and natural solutions*, $v = v_n + v_f$. Substituting this sum for v on the left side of (6.17),

$$\frac{d}{dt}(v_n + v_f) + \frac{1}{RC}(v_n + v_f) = \left(\frac{dv_n}{dt} + \frac{1}{RC}v_n\right) + \left(\frac{dv_f}{dt} + \frac{1}{RC}v_f\right)$$

$$= 0 + \frac{I_0}{C}$$

where the first term in parentheses on the right evaluates to zero since v_n satisfies the unforced differential equation, and the second evaluates to I_0/C since v_f satisfies the forced version. Thus we see that $v = v_n + v_f$ satisfies the forced differential equation *regardless of the value of the free parameter K in the natural solution* v_n. Returning to the requirement that not only the forced differential equation, but also the initial condition, be satisfied,

$$v(t) = v_n(t) + v_f(t) = Ke^{-t/RC} + RI_0 \tag{6.20}$$

Evaluating at $t = 0$, the initial condition is matched if

$$v(0+) = V_0 = K(1) + RI_0$$

which requires that

$$K = V_0 - RI_0$$

Inserting this into (6.20), we have found our desired solution to the forced differential equation (6.17) with given initial condition:

$$v(t) = (V_0 - RI_0)e^{-(1/RC)t} + RI_0$$

This solution may be tested by substitution into (6.17) and noting that evaluation at $t = 0$ does yield the specified initial value V_0.

Summarizing this method, the following steps solve the forced differential equation with constant forcing term and given initial conditions:

1. Using a trial forced solution equal to the unknown constant A, substitute into the forced differential equation to find the forced solution.
2. Using a trial natural solution equal to Ke^{st}, substitute into the unforced differential equation, solving the characteristic equation for s.
3. The total solution is the sum of natural and forced solutions. Evaluate the sum at the initial time and set equal to the required initial value to find K.

Example 6.7

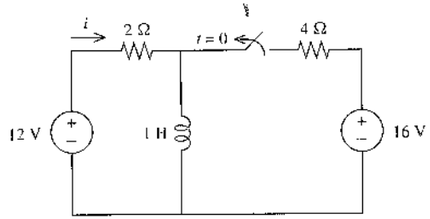


FIGURE 6.13 Circuit for Example 6.7.

We wish to find $i(t)$ for $t > 0$ in the circuit of Fig. 6.12. Before $t = 0$, the circuit is as shown in Fig. 6.13(a). Assuming that this circuit has had sufficient time to go to dc steady state prior to $t = 0$, the inductor acts like a short circuit and

$$i_L = i + i_1 = \frac{12}{2} + \frac{16}{4} = 10 \text{ A}$$

at $t = 0-$. At $t = 0$ the switch opens, and for $t > 0$ the circuit is redrawn as in Fig. 6.13(b). In this circuit, by KVL,

$$\frac{di}{dt} + 2i = 12 \tag{6.21}$$

and $i(0+) = i(0-) = 10$ A by continuity of inductive currents. The trial forced solution is $i_f = A$, and inserting this into the forced differential equation (6.21) gives

$$0 + 2A = 12$$

or $A = 6$. The unforced version of this equation is

$$\frac{di_n}{dt} + 2i_n = 0$$

with characteristic equation

$$s + 2 = 0$$

or $s = -2$. Putting these results together yields

$$i(t) = i_n(t) + i_f(t) = Ke^{-2t} + 6$$

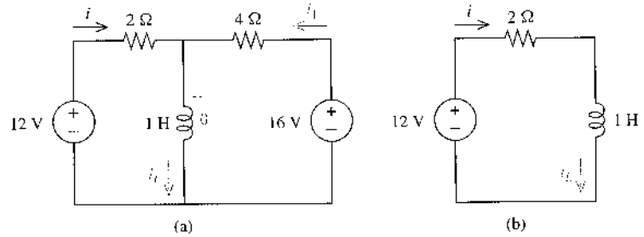


FIGURE 6.14 (a) Circuit before $t = 0$; (b) circuit after $t = 0$.

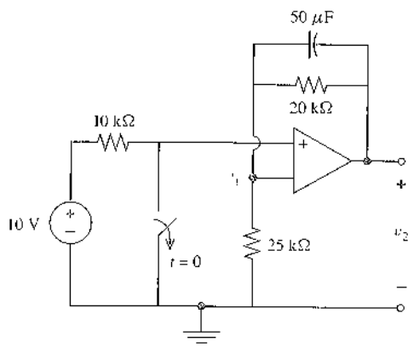
and matching initial conditions, we have

$$i(0+) = 10 = K(1) + 6$$

We conclude that $K = 4$ and the total solution is

$$i(t) = 4e^{-2t} + 6 \text{ A}$$

Example 6.8



Circuit for Example 6.8.

This example, shown in Fig. 6.14, demonstrates application of these ideas to op amp circuits. The circuit is in dc steady state just before the switch acts at $t = 0$, and we wish to find $v_2(t)$ for all times t. With the op amp input shorted to ground for $t < 0$, v_1 and v_2 are zero for $t < 0$. For $t > 0$, the voltage at node 1 is 10 V, since the voltage at the noninverting input is 10 V by the virtual open principle, and v_1 equals this value by the virtual short principle. The node equation at the inverting input node is then

$$\left(\frac{1}{25 \times 10^3}\right)(10 - 0) + (50 \times 10^{-6})\frac{d}{dt}(10 - v_2) + \left(\frac{1}{20 \times 10^3}\right)(10 - v_2) = 0$$

or

$$\frac{dv_2}{dt} + v_2 = 18 \qquad (6.$$

The trial forced solution is $v_{2f} = A$, and plugging it into (6.22), $A = 18$. The characteristic equation of the unforced version of (6.22) is $s + 1 = 0$, or

$$v_{2n}(t) = Ke^{-t}$$

Then the total solution is

$$v_2(t) = v_{2n}(t) + v_{2f}(t) = Ke^{-t} + 18$$

At $t = 0+$, there is no current flow through the 10-kΩ resistor, so the voltage at the noninverting input is 10 V. Since no voltage drop occurs across the op amp input terminals (the virtual short principle), $v_1(0+) = 10$ V also. Continuity of the capacitive voltage requires that

$$v_2(0+) - v_1(0+) = v_2(0-) - v_1(0-) = 0 - 0 = 0$$

Thus $v_2(0+) = 10$ V also, and evaluating the constant K, we obtain

$$v_2(0+) = K(1) + 18 = 10 \text{ V}$$

or $K = -8$. Thus

$$v_2(t) = 18 - 8e^{-t} \text{ V}, \qquad t \geq 0 \qquad (6.$$

which, combined with the earlier observation that $v_2(t) = 0$ for $t < 0$, completes the solution for all t.

The presence of the op amp has not altered the fact that a single storage element will yield a single first-order differential equation (6.22). As recommended in Chapter 4, we used nodal analysis with op amp circuits and wrote our node equation at an op amp input node rather than the output node.

Example 6.9

Our final example, with circuit shown in Fig. 6.15, illustrates a general first-order circuit containing several independent and dependent sources. In this circuit we seek the current $i(t)$ through the inductor for $t > 0$, where $i(0-) = 2$ A.

Such problems can be approached several different ways. A basic approach would be to perform mesh or nodal analysis on the circuit. This would lead to coupled first-order differential equations, which could then be reduced to a single first-order differential equation by substitution. The reduction is not always straightforward, and here we recommend other routes to securing this single first-order differential equation. In this example we employ the strategy of simplification by Thevenin–Norton transformation. Another method, based on superposition, will be developed in Section 6.5.

We will find the Thevenin transform of the entire circuit less its storage element. Then, replacing all but the storage element by the two-element equivalent, we will be left with a simple one-loop circuit to analyze. To do so, we will find the open-circuit voltage v_{oc} and short-circuit current i_{sc} defined in Fig. 6.16 at the inductor terminals ab. v_{oc} is found from Fig. 6.16(a). By KVL,

$$1i_1 + 2i_1 + 2(i_1 + 10) = 80$$

or $i_1 = 12$. Then around the left loop, again by KVL,

$$1(12) + v_{oc} + 3(12) = 80$$

so $v_{oc} = 32$ V. To find i_{sc} in Fig. 6.16(b), we write the two required mesh analysis equations,

$$1i_1 = -3v + 80$$

$$2i_2 + 2(i_2 + 10) = 3v$$

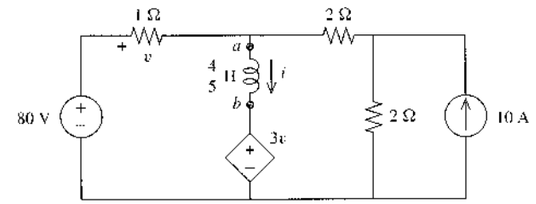
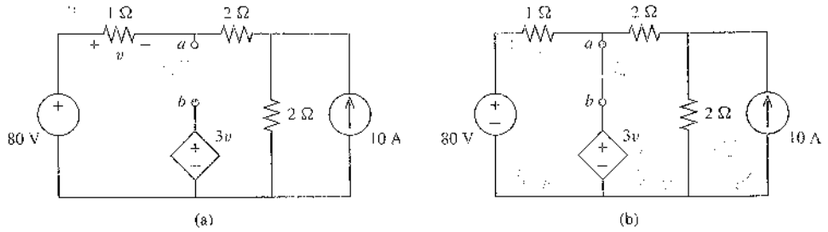


FIGURE 6.15 Circuit for Example 6.9.



(a) Circuit for v_{oc}; (b) circuit for i_{sc}.

Substituting $v = i_1$, by the first mesh equation we have $i_1 = 20$. Using this in the second, we have

$$2i_2 + 2(i_2 + 10) = 3(20) = 60$$

or $i_2 = 10$. Then

$$i_{sc} = i_1 - i_2 = 20 - 10 = 10 \text{ A}$$

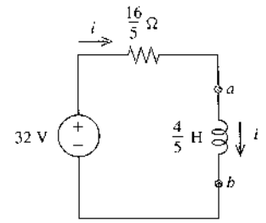
The Thevenin equivalent is shown in Fig. 6.17. We have reduced the problem to a familiar one. By KVL, around this simple RL circuit,

$$\frac{di}{dt} + 4i = 40$$

with initial condition $i(0-) = i(0+) = 2$ A. We have solved this first-order differential equation several times before, and

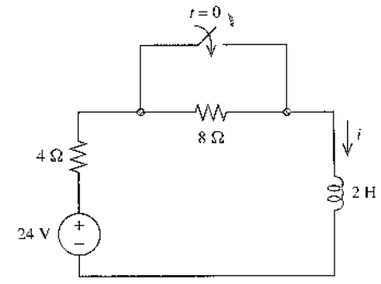
$$i(t) = Ke^{-4t} + A = 10 - 8e^{-4t} \text{ A}$$

with $A = 10$ found by substitution of $i_f = A$ into (6.24) and K by then matching the initial condition.

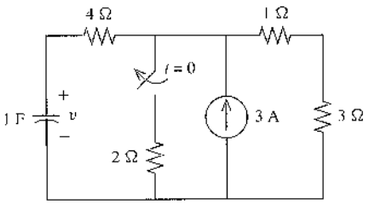


Circuit equivalent of Fig. 6.15.

The solutions that we have encountered so far in this chapter are often referred to in other more descriptive terms. Two such terms that are widely used are the *transient response* and the *steady-state response*. The transient response is the transitory portion of the total response, which approaches zero as time increases. The steady-state response, on the other hand, is that part of the total response that remains after the transient response has become zero. In the case of dc sources, the steady-state response is constant and is the dc steady state discussed in Section 5.7. In Example 6.9, for the response variable $i(t)$ given in (6.25), the transient response is $-8e^{-4t}$ A and the dc steady-state response is 10 A.



6.4.1. Find v for $t > 0$ if the circuit is in steady state at $t = 0-$.
Answer: $12 - 8e^{-t/8}$ V



6.4.2. The circuit is in steady state at $t = 0-$. Find i for $t > 0$.
Answer: $6 - 4e^{-2t}$ A

6.4.3. Resolve Exercise 6.4.1 if the switch closes at $t = 0$ rather than opens.
Answer: $4 + 8e^{-3/16t}$ V

6.5 SUPERPOSITION IN FIRST-ORDER CIRCUITS

In our discussion of resistive circuits, we saw that superposition could be applied to the analysis of circuits with more than one source. The overall response is computed as a superposition (i.e., a sum) of responses to each source individually, with all other sources killed.

To see how superposition applies in the case of first-order circuits, consider the general first-order differential equation governing all such circuits,

$$\frac{dx}{dt} + \frac{1}{\tau}x = f(t), \qquad x(t_0) = x_0 \tag{6.45}$$

In this equation, $x(t)$ is the circuit variable (current or voltage) and $f(t)$ the forcing function. If we kill all independent sources in the circuit, we will have the unforced equation, that is, $f(t)$ replaced by 0, but otherwise the equation is unchanged. Let $x_{ic}(t)$ solve the problem with independent sources killed; that is,

$$\frac{dx_{ic}}{dt} + \frac{1}{\tau}x_{ic} = 0, \qquad x_{ic}(t_0) = x_0$$

Here $x_{ic}(t)$ is the response due to the initial conditions. Now suppose we return the sources into the circuit, but instead, kill (zero out) the initial condition. The response in

this case we label $x_s(t)$, the response due to the independent sources, satisfying

$$\frac{dx_s}{dt} + \frac{1}{\tau}x_s = f(t), \qquad x_s(t_0) = 0$$

It is easy to see that the superposition of these two responses $x(t) = x_{ic}(t) + x_s(t)$ satisfies the original forced differential equation, since

$$\begin{aligned}\frac{dx}{dt} + \frac{1}{\tau}x &= \frac{d}{dt}(x_{ic} + x_s) + \frac{1}{\tau}(x_{ic} + x_s) \\ &= \left(\frac{d}{dt}x_{ic} + \frac{1}{\tau}x_{ic}\right) + \left(\frac{d}{dt}x_s + \frac{1}{\tau}x_s\right) = 0 + f(t)\end{aligned}$$

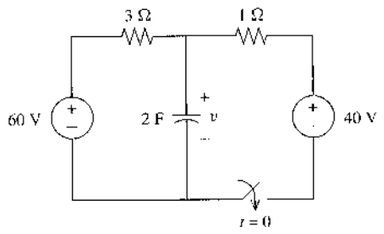
and that it satisfies the initial condition,

$$\begin{aligned}x(t_0) &= x_{ic}(t_0) + x_s(t_0) \\ &= x_0 + 0\end{aligned}$$

This shows that *the total response is a superposition of the initial condition response with all independent sources killed and the response to independent sources with the initial condition killed.* The initial condition may be thought of as another source, along with the independent sources. This makes physical sense since the initial condition is indeed a source of energy for the natural response. $x_{ic}(t)$ is the natural response to this initial condition "source" with the other (independent) sources killed and must be superposed with the forced response $x_s(t)$ driven by those independent sources, for which the initial condition source is in turn killed.

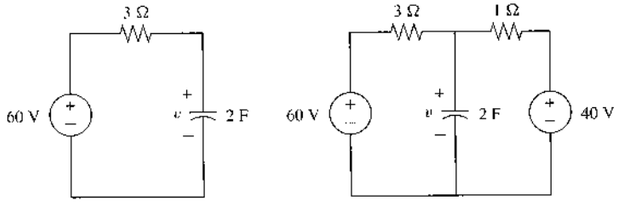
In the case that there is more than one independent source, superposition may further be applied to compute $x_s(t)$, the forced response with initial conditions killed. That is, *the forced response $x_s(t)$ is the superposition of the responses to each of the independent sources with all others killed (including the initial conditions).*

Example 6.10



Circuit for Example 6.10.

Consider the circuit of Fig. 6.18, in which we wish to find v. Prior to $t = 0$ the switch is open, and, as shown in Fig. 6.19(a), in dc steady state $v(0-) = 60$ V. At $t = 0$ the switch closes, and for $t > 0$ we have the circuit of Fig. 6.19(b). We will compute v for $t > 0$ as the superposition of three components: forced components due to the two independent sources and a natural component due to the initial condition $v(0+) = v(0-) = 60$ V.



(a) Equivalent for $t < 0$; (b) for $t > 0$.

Denote the component due to the 60-V source as v_{1s}. Killing the other source, the node equation at the top center node is

$$\frac{1}{3}(v_{1s} - 60) + \frac{1}{1}(v_{1s} - 0) + \frac{2dv_{1s}}{dt} = 0$$

or

$$\frac{dv_{1s}}{dt} + \frac{2}{3}v_{1s} = 10$$

The initial condition imposed on v_{1s} is $v_{1s}(0+) = 0$, since we are required to kill the initial conditions when computing the forced components. The solution to (6.27) with the given initial conditions, using the method described in Section 6.4, may easily be verified to be

$$v_{1s}(t) = 15[1 - e^{-(2/3)t}]$$

The component due to the 40-V source, v_{2s} is found by killing the 60-V source (and the initial condition). The node equation becomes

$$\frac{1}{3}v_{2s} + \frac{1}{1}(v_{2s} - 40) + \frac{2dv_{2s}}{dt} = 0$$

or

$$\frac{dv_{2s}}{dt} + \frac{2}{3}v_{2s} = 20$$

with $v_{2s}(0+) = 0$, the solution is

$$v_{2s}(t) = 30[1 - e^{-(2/3)t}]$$

The final component is $v_{ic}(t)$, found by killing both independent sources and exciting a response by using the true initial condition. From Fig. 6.19(b), upon shorting the two voltage sources the two resistors are in parallel with an equivalent of $\frac{3}{4}$ Ω and $RC = (\frac{3}{4})(2) = \frac{3}{2}$ in this sourceless circuit, or

$$\frac{dv_{ic}}{dt} + \frac{2}{3}v_{ic} = 0$$

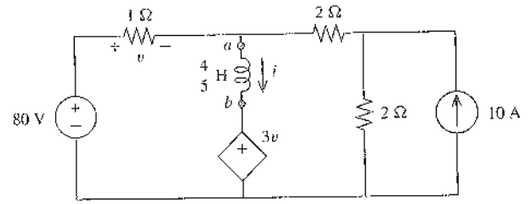
with initial condition $v_n(0+) = v_n(0-) = 60$. The solution is $v_{ic} = Ke^{-2/3t}$ with K computed to match the initial condition, or

$$v_{ic}(t) = 60e^{-(2/3)t} \text{ V}$$

We have found the responses driven by each source, the two independent sources, and the initial condition. It only remains to superpose the components:

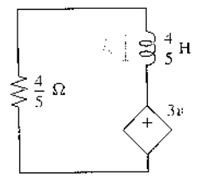
$$\begin{aligned}v(t) &= v_{ic}(t) + v_{1s}(t) + v_{2s}(t) \\ &= 15e^{-(2/3)t} + 45 \text{ V}\end{aligned}$$

This is the solution for $t > 0$, with $v(t) = 60$ V the solution for $t < 0$.



Circuit for Example 6.11.

Example 6.11



Circuit for component i_{ic}.

We shall redo Example 6.9 using superposition. The circuit diagram shown in Fig. 6.15 is repeated for convenience as Fig. 6.20. First we will compute i_{ic}, the component of i due to the initial conditions with all independent sources killed. In this case the resistors may be replaced by an equivalent of value $R_{eq} = \frac{4}{5}$, and we have a single-loop circuit as shown in Fig. 6.21. By KVL,

$$\frac{4}{5}\left(\frac{di_{ic}}{dt}\right) + 3v + \frac{4}{5}i_{ic} = 0$$

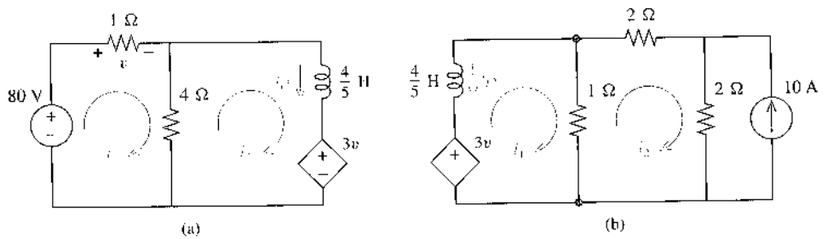
Using Ohm's law, $v = \frac{4}{5}i_{ic}$, and dividing by $\frac{4}{5}$, the above is

$$\frac{di_{ic}}{dt} + 4i_{ic} = 0$$

The given initial value is $i(0-) = i(0+) = 2$ A, leading to the component

$$i_{ic}(t) = 2e^{-4t} \text{ A} \tag{2}$$

Next we need the forced components i_{f1} due to the independent voltage source and i_{f2} due to the current source. To determine i_{f1} we kill the initial conditions and all other independent sources, in this case the current source, as shown in Fig. 6.22(a). Here the branch containing the inductor has been drawn on the right for convenience.



(a) Circuit for component i_{f1}; (b) circuit for i_{f2}.

The mesh equations for Fig. 6.22(a) are

$$i_1 + 4(i_1 - i_2) = 80$$

$$\frac{4}{5}\frac{d}{dt}i_2 + 3i_1 + 4(i_2 - i_1) = 0$$

Solving the first of these equations for i_1 and substituting into the second yields

$$\frac{d}{dt}i_2 + 4i_2 = 20, \qquad i_2(0+) = 0$$

with zero initial inductive current, since this is a forced response with initial conditions killed. The solution to the above is

$$i_2(t) = 5(1 - e^{-4t}) = i_{f1}(t)$$

The third and final component, $i_{2f}(t)$, is computed with the voltage source killed and the current source returned to the circuit, as shown in Fig. 6.22(b) (the leftmost two parallel branches have been drawn in reverse order for convenience). The mesh equations are

$$\frac{4}{5}\left(\frac{di_1}{dt}\right) + (i_1 - i_2) - 3(i_2 - i_1) = 0$$

$$2i_2 + 2(i_2 + 10) + (i_2 - i_1) = 0$$

Solving the second mesh equation for i_2 and substituting into the first gives us

$$\frac{di_1}{dt} + 4i_1 = -20, \qquad i_1(0+) = 0$$

once again using the zero initial inductor current to derive initial conditions. This solution is

$$i_1(t) = -5(1 - e^{-4t}) \text{ A}$$

From Fig. 6.22(b) we see that

$$i_{f2}(t) = -i_1(t) = 5(1 - e^{-4t}) \text{ A}$$

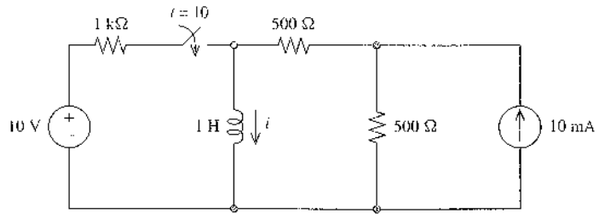
Using superposition, the total solution is the sum of components (6.32) to (6.34):

$$\begin{aligned} i(t) &= i_{ic}(t) + i_{f1}(t) + i_{f2}(t) \\ &= 2e^{-4t} + 10(1 - e^{-4t}) = 10 - 8e^{-4t} \text{ A} \end{aligned}$$

which agrees with our result from Example 6.9.

6.5.1. Use superposition to find i for $t > 10$ s. Assume that the circuit is in dc steady state at time $t = 10-$ s.

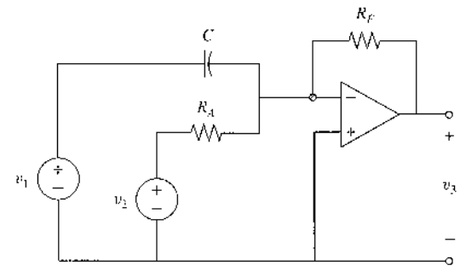
$5 + 10[1 - e^{-500(t-10)}]$ A



6.5.2. Repeat Exercise 6.5.1 using the method of Thevenin transform.

6.5.3. Use superposition to find v_3 in terms of v_1 and v_2.

$v_3 = -R_F C\dfrac{dv_1}{dt} - \dfrac{R_F}{R_A}v_2$



6.6 UNIT STEP FUNCTION

In the preceding sections we analyzed circuits in which independent sources have been suddenly inserted into the networks. At the instant these sources are applied, the voltages or currents at the points of application change abruptly. Forcing functions whose values change in this manner are called *singularity functions*, since they have one time instant at which they exhibit singular or unusual behavior.

There are many singularity functions that are useful in circuit analysis (and we defer their full description to Chapter 13). One of the most important is the *unit step function*, so named by the English engineer Oliver Heaviside (1850–1925). The unit step



Graph of the unit step function $u(t)$.

function is the function that is equal to zero for all negative values of its argument and that is equal to one for all positive values of its argument. If we denote the unit step function by the symbol $u(t)$, a mathematical description is

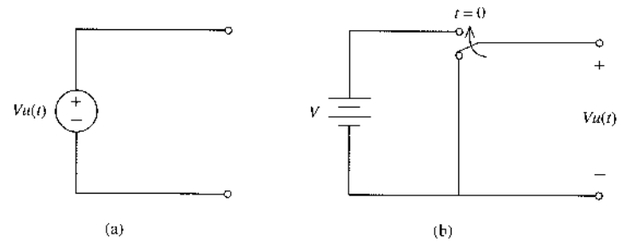
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

From its graph shown in Fig. 6.23 we see that at $t = 0$, $u(t)$ changes abruptly from 0 to 1. Nowhere else does its value change except at this "singular" moment in its behavior. Some authors define $u(0)$ to be 1; others prefer $u(0) = \frac{1}{2}$; but we are leaving $u(t)$ undefined at $t = 0$.

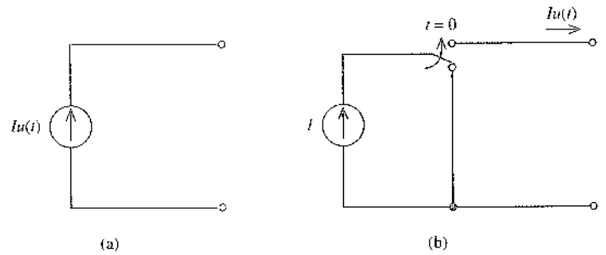
A voltage that steps from 0 to V volts at $t = 0$ may be represented by the product $Vu(t)$. Clearly, this voltage is 0 for $t < 0$ and V volts for $t > 0$. A voltage step source of V volts is shown in Fig. 6.24(a). A circuit that is equivalent to this source is shown in Fig. 6.24(b). A short circuit exists for $t < 0$, and the voltage is zero. For $t > 0$, a voltage V appears at the terminals of the equivalent circuit.

Equivalent circuits for a current step source of I amperes are shown in Fig. 6.25. An open circuit exists for $t < 0$, and the current is zero. For $t > 0$, the switching action causes a terminal current of I amperes to flow at the terminals of the equivalent circuit.

Unit step sources of the type shown in Figs. 6.24 and 6.25 will be frequently used in circuits involving sources that instantaneously switch in or out. Note in these



(a) V volt voltage step source; (b) equivalent circuit.



(a) I-ampere current step source; (b) equivalent circuit.

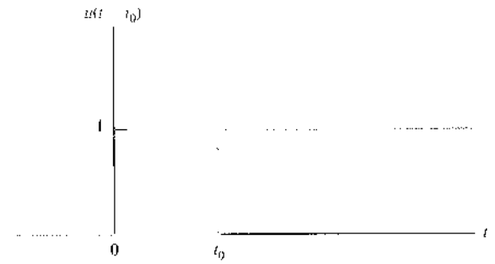
figures that the unit step forms Figs. 6.24(a) and 6.25(a) are simpler schematically than the combinations of dc sources and switches they replace in Figs. 6.24(b) and 6.25(b), resulting in simpler circuit diagrams.

This representation of switched variables by unit steps is not limited to the simple cases shown above. Just as we used the unit step to represent a voltage or current that jumps from 0 to another value at $t = 0$, we may use sums of scaled and time-shifted unit step functions to represent any "staircase" voltage or current, that is, any function with jump discontinuities that remains constant between jumps.

Returning to the definition of the unit step function given in (6.35), suppose that we replace t by $t - t_0$ in the three places that it occurs, which results in

$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases} \qquad (6.36)$$

The graph of this function is shown in Fig. 6.26. $u(t - t_0)$ is the unit step time-shifted by t_0 *to the right*, that is, delayed by t_0 seconds.



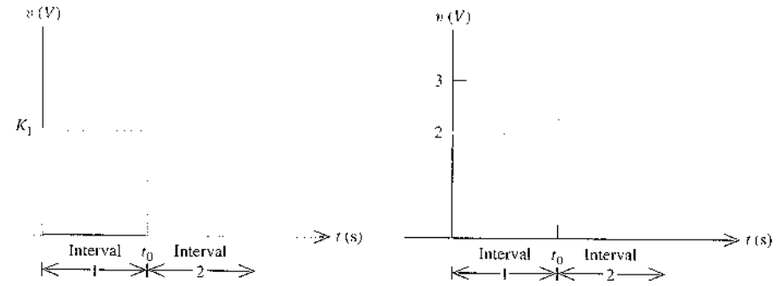
Time-shifted unit step.

Example 6.12

Consider the voltage $v(t)$ formed as a linear combination of a unit step with a second, time-shifted unit step:

$$v(t) = K_1 u(t) + K_2 u(t - t_0)$$

Suppose first that the second unit step is right-shifted (delayed); in other words, $t_0 > 0$. We will determine the value of $v(t)$ by use of (6.35) and (6.36). For $t < 0$, neither unit step has "switched on" yet



Two linear combinations of shifted unit steps.

and $v(t) = 0$. For t in the interval $0 < t < t_0$, the first unit step has switched on (its argument is positive), but the second has not yet (its argument is still negative), so $v(t) = K_1$. After t_0 they are both switched on and $v(t) = K_1 + K_2$. So

$$v(t) = \begin{cases} 0, & t < 0 \\ K_1, & 0 < t < t_0 \\ K_1 + K_2, & t > t_0 \end{cases}$$

By proper selection of the scaling factors K_1 and K_2, we can create any desired level value in the intervals marked 1 and 2 in Fig. 6.27. For instance, with $K_2 = -K_1$ we have the rectangular voltage pulse shown in Fig. 6.27(a), and with $K_1 = 2$ and $K_2 = 1$ we have a staircase function with the level values 2 and 3 shown in Fig. 6.27(b).

We may also *time-reverse* a unit step. Replacing t by $-t$ in (6.35),

$$u(-t) = \begin{cases} 1, & t < 0 \\ 0, & t > 0 \end{cases}$$

Time-reversed unit steps are useful in describing sources with nonzero values until the switch time, after which they are switched out of the circuit.

Example 6.13

Consider the circuit of Fig. 6.28(a). For all $t < 0$ the voltage source has a value of 10 V, as shown in Fig. 6.28(b), leading to a dc steady-state current of $i(0-) = 2$ A. After $t = 0$ the source may be replaced by a short circuit [since $u(-t) = 0$ for $t > 0$], leading to the circuit of Fig. 6.28(c). From our previous analysis of the simple sourceless RL circuit, we know that (6.10)

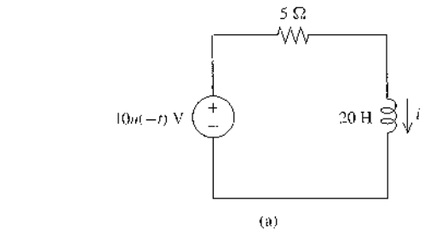
$$i(t) = I_0 e^{-t/\tau} \text{ A}$$

With $\tau = L/R = 4$ and I_0 determined by the initial condition,

$$i(0+) = i(0-) = 2 = I_0(1)$$

whence the total solution

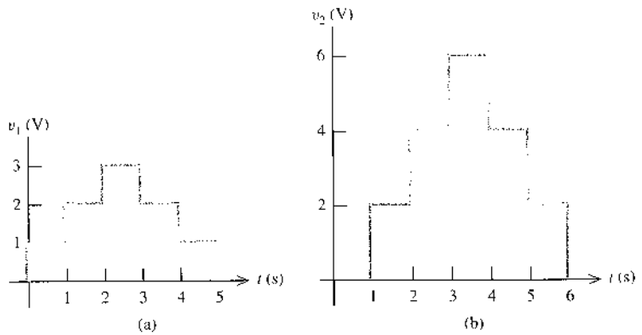
$$i(t) = 2e^{-t/4} \text{ A}$$



(a) Circuit with time-reversed step source; (b) equivalent for $t < 0$; (c) equivalent for $t > 0$.

6.6.1. Express $v_1(t)$ as a sum of scaled and time-shifted unit steps. How is $v_2(t)$ related to $v_1(t)$? Use these two to express $v_2(t)$ as a sum of scaled and time-shifted unit steps.

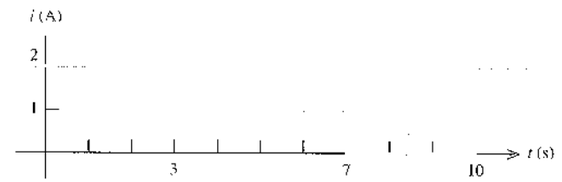
$u(t)+u(t-1)+u(t-2)-u(t-3)-u(t-4)-u(t-5)$; $v_2(t) = 2v_1(t-1)$; $2[u(t-1)+u(t-2)+u(t-3)-u(t-4)-u(t-5)-u(t-6)]$

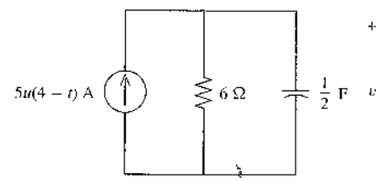


EXERCISE 6.6.2

6.6.2. Express $i(t)$ as a sum of two time-reversed and one normal unit step.

$u(3-t)+u(7-t)+2u(t-10)$





EXERCISE 6.6.3

6.6.3. Find $v(t)$ for $t > 4$. Assume that the circuit is in dc steady state just before the unit step jump time.

Answer: $30e^{-(t-4)/3}$ V

6.7 STEP AND PULSE RESPONSES

The step response is defined as the response of a circuit having only one independent source, which is a unit step function. The response and step source can each be either a current or a voltage. Since the unit step source presents zero source function until time $t = 0-$, the dc steady-state response by the circuit at $t = 0-$ must be zero. There can be no initial energy stored in the storage elements at the singular time when the source switches on because there has been no source for $t < 0$ to energize the storage elements. Thus the step response is the response to a unit step input with no initial energy stored in the circuit.

Example 6.14

As an example, let us find the step response $v(t)$ in the simple RC circuit of Fig. 6.29(a) having an input of $v_g(t) = u(t)$ V. The equivalent circuit is shown in Fig. 6.29(b). For $t < 0$, $v_g(t) = 0$ and so the dc steady state at $t = 0-$ is $v(0-) = 0$ V. By continuity of capacitive voltages, $v(0+) = 0$ V as well. Applying KCL for $t > 0$ yields

$$\frac{dv}{dt} + \frac{1}{RC}v = \frac{1}{RC} \tag{6.38}$$

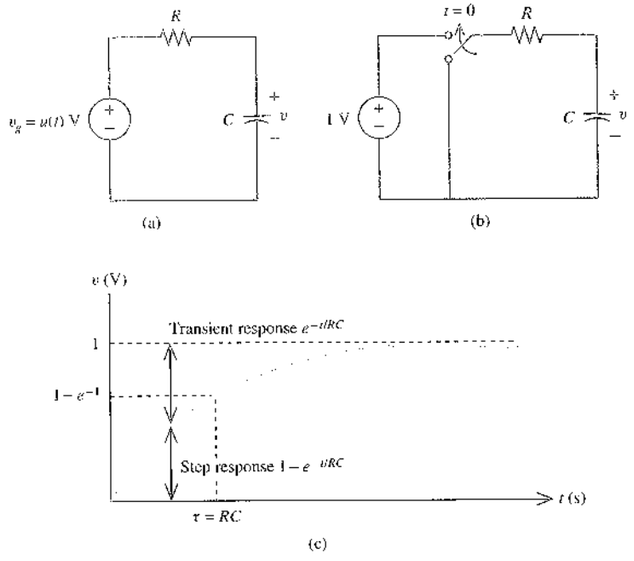
The characteristic equation is

$$s + \frac{1}{RC} = 0$$

leading to the natural response

$$v_n(t) = Ke^{-t/RC} \tag{6.39}$$

The trial forced solution is $v_f = A$, which when substituted into (6.38) yields $A = 1$. Combining this with the natural solution (6.39),



(a) *RC* circuit with a voltage step input; (b) equivalent circuit; (c) step response of *RC* circuit.

the total solution is

$$v(t) = Ke^{-t/RC} + 1$$

Finally, matching the initial condition $v(0+) = 0$ gives $K = -1$. The result is

$$v(t) = \begin{cases} 0, & t < 0 \\ 1 - e^{-t/RC}, & t > 0 \end{cases} \qquad (6.\ )$$

where we have used the fact that before the singular time all currents and voltages were zero. This can be written somewhat more concisely as

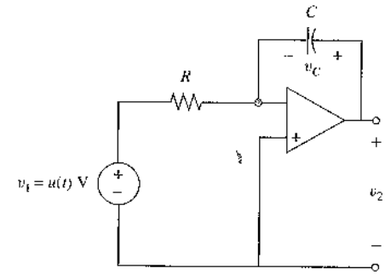
$$v(t) = (1 - e^{-t/RC})u(t) \qquad (6.\ )$$

avoiding the "laundry list" form of (6.40). The step response is shown in Fig. 6.29(c).

Once again it is the time constant that tells us the rate at which the transient response decays to zero. Each period of duration of one time constant, the transient reduces to e^{-1} times its value at the beginning of that period. Circuits with short time constants reach their steady-state value more quickly.

Note by (6.41) that multiplication of any other time function by a unit step forces the product to be zero prior to the singular time and to be equal to the other time function afterward. We will make frequent use of this convenient property of the unit step to simplify other equations arising in subsequent chapters.

Example 6.15



Circuit for Example 6.15.

As a second example, let us find $v_2(t)$ in the circuit of Fig. 6.30, consisting of a resistor, a capacitor, and an op amp, when the input is a unit step voltage source. The nodal equation at the inverting terminal of the op amp is, upon division by C,

$$\frac{dv_2}{dt} = -\frac{1}{RC}v_1$$

Integrating both sides of this equation between the limits of 0+ and t, we have

$$v_2(t) = -\frac{1}{RC}\int_{0+}^{t} v_1(\tau)\,d\tau + v_2(0+)$$

While our current interest is centered on the specific case $v_1(t) = u(t)$, note that by (6.42) the general effect of this circuit is to create an output voltage $v_2(t)$ that is proportional to the integral of the input $v_1(t)$ plus a constant. For this reason the circuit is called an *integrator*. Some authors prefer the term *inverting integrator* because of the negative sign in the voltage transfer equation (6.42).

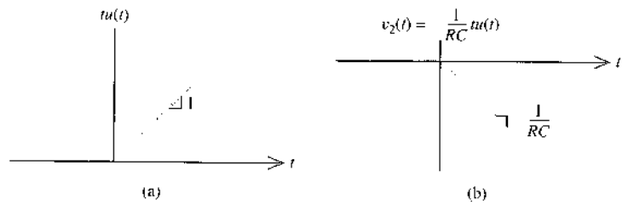
In the case of a unit step input $v_1(t) = u(t)$, all currents and voltages in the circuit are zero at 0− and, by continuity, $v_C(0+) = 0$ V as well. The virtual short principle tells us that the inverting terminal is at ground potential; thus $v_C = v_2$ and the initial condition term in (6.42) is $v_2(0+) = 0$ V. For $t > 0$, the integrand $v_1(t) = u(t) = 1$; thus

$$v_2(t) = -\frac{1}{RC}\int_{0+}^{t} d\tau + 0 = \left(-\frac{1}{RC}\right)t$$

The step response of our integrator circuit is 0 for $t < 0$ and the above for $t > 0$, or, using multiplication by $u(t)$ to make this equation concise, for all t

$$v_2(t) = -\frac{1}{RC}tu(t)$$

The function $tu(t)$ is called the *unit ramp* singularity function and is shown in Fig. 6.31(a). Our step response $v_2(t)$ is a negative-going ramp function, scaled by $-1/RC$ and is shown in Fig. 6.31(b).



(a) Unit ramp function; (b) step response of the circuit of Fig. 6.30.

Closely related to the step response is the pulse response. We saw in Section 6.6 how a linear combination of two unit steps could be used to produce a rectangular pulse, for instance Fig. 6.27(a). We now consider the response of a first-order circuit to a pulse input. As before, we assume no other independent sources are present in the circuit.

Prior to the leftmost jump, or *leading edge* of the pulse, there are no currents or voltages for the same reason mentioned for the step response: the dc steady-state response to a zero-value source must be zero. Nothing has turned on yet. During the pulse, the response must be identical to a step response, since the pulse and equal-amplitude step inputs will only differ after the second jump time, which has not yet occurred. Finally, after this second jump time, or *trailing edge* of the pulse, the source is back to zero. We need only compute the response of a sourceless circuit with the proper initial condition (applied at the trailing edge of the pulse).

Example 6.16

To illustrate the pulse response of a first-order circuit, consider Fig. 6.32(a), where the source is the current pulse

$$i_g(t) = 6(u(t) - u(t-1))\ \text{A}$$

This source is zero before $t = 0$, supplies 6 A in the interval $0 < t < 1$, and then shuts down again. It is sketched in Fig. 6.33(a). Replacing the source and 5-Ω resistor by its Thevenin equivalent, we have the single-loop circuit of Fig. 6.32(b). KVL around the loop and division by 5 yield

$$\frac{di}{dt} + i = i_g(t) \qquad (6.43)$$

Before $t = 0$ all currents and voltages are zero, since there is no source of energy to stimulate nonzero responses. Thus $i(0-) = i(0+) = 0$ A. For $0 < t < 1$, $i_g(t) = 6$, and for this interval of time we have

$$\frac{di}{dt} + i = 6, \quad i(0+) = 0 \qquad (6.44)$$

The characteristic equation is $s + 1 = 0$, and the natural response is Ke^{-t}. The trial forced solution is A, and plugging in we see that

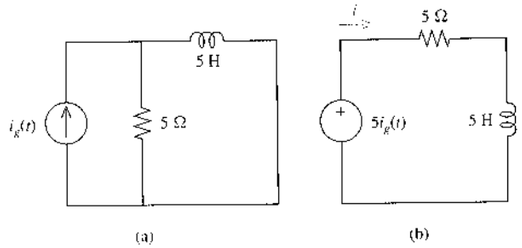


FIGURE 6.32 (a) Circuit for Example 6.16; (b) equivalent circuit.

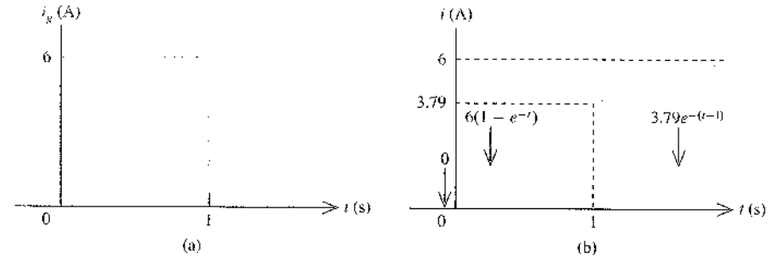


FIGURE 6.33 (a) Pulse current source; (b) pulse response.

$A = 6$. Matching the zero initial condition,

$$i(0+) = K(1) + 6$$

or $K = -6$. We have thus far that $i(t) = 0$ for $t < 0$, and

$$i(t) = 6(1 - e^{-t})\ \text{A}, \quad 0 < t < 1$$

This portion of the response is seen in Fig. 6.33(b). The dashed line shows that, if the source did not switch back off at $t = 1$, the response would approach the value of 6, characteristic of the step response for (6.44). This is only reasonable, since if the source did not switch back off it would be a step source.

Returning to (6.43), for $t > 1$ we have $i_g(t) = 0$ or

$$\frac{di}{dt} + i = 0, \quad t > 1$$

Thus in the interval $t > 1$ we need only solve an unforced first-order circuit. The time constant is $\tau = 1$, and the solution is

$$i(t) = Ke^{-t}, \quad t > 1$$

Since (6.45) is valid for $t > 1$, its initial condition must be at $t = 1+$, just after the trailing edge of the pulse. We have already determined the current just before this time, since by (6.45)

$$i(1-) = 6(1 - e^{-1}) = 3.79\ \text{A}$$

By continuity of inductive currents, $i(1+) = 3.79$ as well. To simplify the initial condition matching, we will rewrite the natural solution (6.47) as

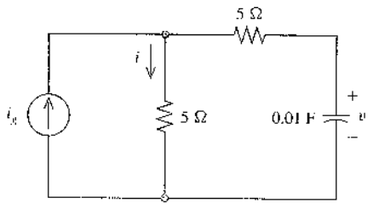
$$i(t) = K_1 e^{-(t-1)}, \quad t > 1$$

These two forms are equivalent, with the new unknown K_1 replacing K ($K = K_1 e^1$). Then, by (6.48) evaluated at time $1+$, $K_1 = 3.79$ and

$$i(t) = 3.79e^{-(t-1)}\ \text{A}, \quad t > 1$$

which completes the full time history of $i(t)$. A graph of this pulse response is shown in Fig. 6.33(b).

Comparing the pulse input to pulse response in Fig. 6.33, we see that the effect of this first-order circuit has been to smooth the input and slow down its abrupt transitions. Although the pulse jumps instantaneously to its full value at $t = 0$, its pulse response grows smoothly for $0 < t < 1$. And although the pulse returns instantaneously to zero at $t = 1$, its response decays to zero continuously with time constant $\tau = 1$ s. This smoothing effect is characteristic of a class of circuits known as low-pass filters, which will be taken up in Chapter 14.

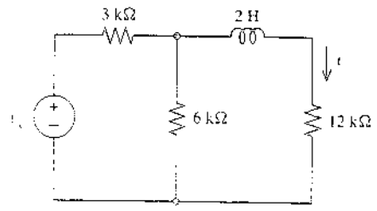


6.7.1. Find the step responses i and v [step response implies $i_g = u(t)$ A].

$(1 - \frac{1}{2}e^{-10t})u(t)$ A; $5(1 - e^{-10t})u(t)$ V

6.7.2. Find the response i to $v_g = 42u(t)$ V.

$2(1 - e^{-7t})u(t)$ mA (t in ms)



6.7.3. Use the answer from Exercise 6.7.2 to find the response i to $v_g = 210u(t)$ V.

By linearity, $10(1 - e^{-7t})u(t)$ A

SPICE was introduced earlier and used to analyze resistive circuits with dc sources. The program, of course, does much more than this. Here we discuss its use in obtaining transient responses.

The basic format of a SPICE input file and the element statements for resistors, dc sources, and linear controlled sources were introduced in Section 4.9. The format for storage element statements (capacitors and inductors) is

```
CXXXXX     N1     N2     VALUE     <IC = VOLTS>
LXXXXX     N1     N2     VALUE     <IC = AMPS>
```


Here C or L indicates capacitor or inductor, XXXXX is the name given this specific capacitor or inductor, N1 and N2 are the element nodes, and VALUE is the capacitance in farads or inductance in henrys. The final field is optional (angle brackets will be used to mark optional fields in SPICE statements), consisting of the initial conditions. The initial time at which these conditions are applied is always time zero, $t = 0$. The reference direction for the initial capacitive voltage of value VOLTS has its plus sign at the node N1, and that for the initial inductive current AMPS has its arrow pointing from N2 toward N1.

The control statement that will direct SPICE to perform a transient analysis is the .TRAN statement, whose basic format is

```
.TRAN      TSTEP      TSTOP      <UIC>
```

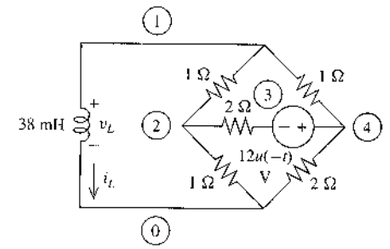
TSTEP is the time step between values to be printed or plotted (output statements for transient analysis will be described shortly). TSTOP is the end of the time interval, assumed to begin at time $t = 0$, during which analysis will be performed and output produced. Inclusion of the optional UIC field directs SPICE to use the initial conditions specified by the user on the storage element statements. If this field is absent, SPICE will compute the dc steady state at $t = 0-$ and use the steady-state inductor currents and capacitor voltages as the initial conditions.

The output of a transient analysis can either be printed as a list of numbers or plotted. The formats for these output control statements are

```
.PRINT     TRAN     CVLIST
.PLOT      TRAN     CVLIST
```

where CVLIST is the list of desired circuit variables.

This example illustrates the use of SPICE both to determine initial conditions and the subsequent transient analysis. We seek the current $i_L(t)$ in Fig. 6.34. Since the voltage source is $12u(-t)$ V, its value for $t < 0$ is 12 V. The initial condition $i_L(0+) = i_L(0-)$ may be found by replacing the inductor with a short circuit and finding the dc value for i_L with the 12-V source in place.



Circuit for Example 6.17.

```
Example 6.17 run 1: determining the IC
*PUT A DUMMY V SOURCE IN PLACE OF THE
*SHORTED L SO WE CAN PRINT IL=I(VD)
VD      1       0       DC      0
R1      1       2       1
R2      1       4       1
R3      2       3       2
*HERE IS THE 12 V SOURCE
VS      4       3       DC      12
R4      2       0       1
R5      4       0       2
.PRINT DC     I(VD)
.END
```

The printed output contains the lines

```
VOLTAGE SOURCE CURRENTS
NAME            CURRENT

VD              6.316E-01
VS             -3.789E+00
```

Thus the initial current is $i_L(0-) = i_L(0+) = 0.6316$ A. For $t > 0$ the circuit is source free, and the input file to plot $i_L(t)$ for $t \geq 0$ using this initial condition is

```
Example 6.17 run 2: the response for t>0
*ENTER THE IC CALCULATED IN RUN 1
L1      0       1       38M     IC=0.6316
R1      1       2       1
R2      1       4       1
R3      2       3       2
*THE FOLLOWING STATEMENT CAN BE OMITTED
*AND THE NODES RENUMBERED IF PREFERRED
VS      3       4       DC      0
R4      2       0       1
R5      4       0       2
.TRAN   0.2     10      UIC
*PUT IN A DUMMY V SOURCE IF YOUR VERSION
*OF SPICE DOESN'T PERMIT FORMS LIKE I(L1)
.PLOT         TRAN    I(L1)
.END
```

The output is shown in Fig. 6.35. The current $i_L(t)$ decays exponentially from its initial value, as any sourceless RC or RL circuit variable must. To determine τ, note that at the end of one time constant the response decays to $e^{-1}(0.6316) = 0.2324$ V. Examining Fig. 6.35, this occurs at 32 ms, so $\tau = 32$ ms. Note that this agrees with the time constant calculated in Example 6.5 using the same circuit, which was done without the help of SPICE.


```
**** 10/10/94 09:28:11 ********* Evaluation PSpice (January 1991) ************
 Ex 6.17 run 2: the response for t>0

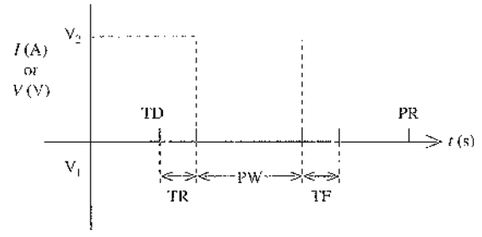
 ****     TRANSIENT ANALYSIS                TEMPERATURE =   27.000 DEG C

******************************************************************************


  TIME        I(L1)
(*)---------    0.0000E+00   2.0000E-01   4.0000E-01   6.0000E-01   8.0000E-01
  0.000E+00  6.316E-01 .                                          *           .
  4.000E-03  5.575E-01 .                                          *           .
  8.000E-03  4.923E-01 .                                        *             .
  1.200E-02  4.344E-01 .                                      *               .
  1.600E-02  3.838E-01 .                                    *                 .
  2.000E-02  3.382E-01 .                                  *                   .
  2.400E-02  2.984E-01 .                                *                     .
  2.800E-02  2.631E-01 .                              *                       .
  3.200E-02  2.324E-01 .                            *                         .
  3.600E-02  2.059E-01 .                          *                           .
  4.000E-02  1.808E-01 .                        *                             .
  4.400E-02  1.597E-01 .                      *                               .
  4.800E-02  1.409E-01 .                    *                                 .
  5.200E-02  1.242E-01 .                  *                                   .
  5.600E-02  1.097E-01 .                *                                     .
  6.000E-02  9.678E-02 .              *                                       .
  6.400E-02  8.534E-02 .              *                                       .
  6.800E-02  7.536E-02 .            *                                         .
  7.200E-02  6.649E-02 .          *                                           .
  7.600E-02  5.863E-02 .          *                                           .
  8.000E-02  5.178E-02 .        *                                             .
  8.400E-02  4.568E-02 .        *                                             .
  8.800E-02  4.028E-02 .      *                                               .
  9.200E-02  3.557E-02 .    *                                                 .
  9.600E-02  3.138E-02 .    *                                                 .
  1.000E-01  2.767E-02 .  *                                                   .
  1.040E-01  2.444E-02 .  *                                                   .
  1.080E-01  2.156E-02 .*                                                     .
  1.120E-01  1.901E-02 .*                                                     .
  1.160E-01  1.679E-02 .*                                                     .
  1.200E-01  1.481E-02 .*                                                     .
  1.240E-01  1.306E-02 .*                                                     .
  1.280E-01  1.154E-02 .*                                                     .
  1.320E-01  1.018E-02 .*                                                     .
  1.360E-01  8.974E-03 .*                                                     .
  1.400E-01  7.925E-03 .*                                                     .
  1.440E-01  6.992E-03 *                                                      .
  1.480E-01  6.165E-03 *                                                      .
  1.500E-01  5.791E-03 *                                                      .
```

SPICE transient analysis output.

SPICE may also be used to compute step and pulse responses. To determine the step response, we need only set all initial conditions to zero and include the independent step source as a dc source in the SPICE input file. Pulse responses are handled similarly, except we declare the source to be a *pulse source* through use of the element



Parameters of a SPICE pulse source.

statements

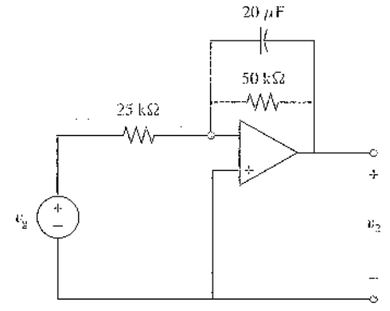
```
VXXXXX   N1   N2  PULSE(V1 V2  TD  TR  TF  PW  PR)
IXXXXX   N1   N2  PULSE(V1 V2  TD  TR  TF  PW  PR)
```

The seven numbers within the parentheses specify the parameters of the pulse source, as shown in Fig. 6.36. The rise and fall times of the pulse TR and TF cannot be set to zero. To model instantaneous switching, these values should be set much smaller than the time constant of the circuit.

We will compute the unit step and pulse response $v_2(t)$ for the op amp circuit shown in Fig. 6.37. We will use the improved op amp model (Fig. 3.9).

```
Example 6.18 (FIG. 6.38)
*FOR STEP RESPONSE WE'LL USE A DC SOURCE
VG      1       0       DC      1
RA      1       2       25K
RF      2       3       50K
CF      2       3       20U     IC=0
XOA     0       2       3       OPAMP
*HERE IS THE OP AMP SUBCIRCUIT
.SUBCKT OPAMP       1       2       3
*NODE 1 IS +IN, 2 -IN, 3 OUTPUT
RIN     1       2       1MEG
E1      4       0       1       2       100K
ROUT    4       3       30
.ENDS
*FINALLY, THE CONTROL STATEMENTS
.TRAN   0.1     10      UIC
.PLOT       TRAN    V(3)    V(1)
.END
```

The step response output is shown in Fig. 6.38. Based on Example 6.6, we would expect the step response to have a transient decaying to zero with time constant $\tau = R_F C_F$ or (50 kΩ)(20 μF)



Circuit for Example 6.18.

= 1000 ms, that is, 1 s. Noting from Fig. 6.38 that the steady-state value is −2 V, the initial transient must be +2 V and the value after one time constant should be $e^{-1}(2) - 2 = -1.26$. This occurs at $t = 1.00$ s as expected.

Continuing the example, we will find the pulse response to an input pulse of amplitude 1 V and duration 1 s. Replacing the VG element statement in the SPICE input file by

```
VG    1     0       PULSE ( 0  1  0  1M  1M  1  10 )
```

and rerunning SPICE results in the pulse response output shown in Fig. 6.39. The step and pulse responses are identical until the pulse trailing edge ($t = 1$ s), after which the pulse response exhibits a natural decay to zero with time constant $\tau = 1$ s, while the step response continues toward the forced dc steady state of −2 V. From Fig. 6.37, we see that in dc steady state, with the capacitor equivalent to an open circuit, this is an inverting amplifier (as discussed in Chapter 3) with voltage transfer ratio of −2. Thus with a unit step input, we should closely approach this steady state after 5 to 10 time constants (5 to 10 s). The data in Fig. 6.38 confirm this.

SPICE includes several other transient sources as well. Each is of the form

```
VXXXXX   N1   N2   TSPEC
IXXXXX   N1   N2   TSPEC
```

where N1 and N2 are the positive and negative source nodes, and TSPEC is the transient source specification. Three of the most useful transient sources are the piecewise linear PWL, exponential EXP, and damped sinusoidal SIN sketched in Figs. 6.39 and 6.40.

In addition to the coarse, line-printer style of graph afforded by the .PLOT statement, PSpice permits terminals equipped with graphics adapters to display plotted output

```
**** 10/10/94 09:26:24 ******** Evaluation PSpice (January 1991) ********

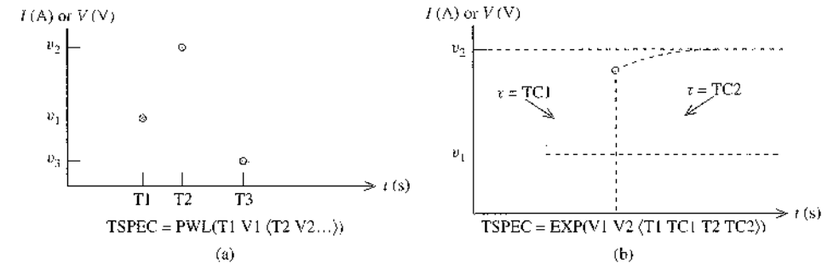
EXAMPLE 6.18: STEP RESPONSE

****     TRANSIENT ANALYSIS                TEMPERATURE =   27.000 DEG C

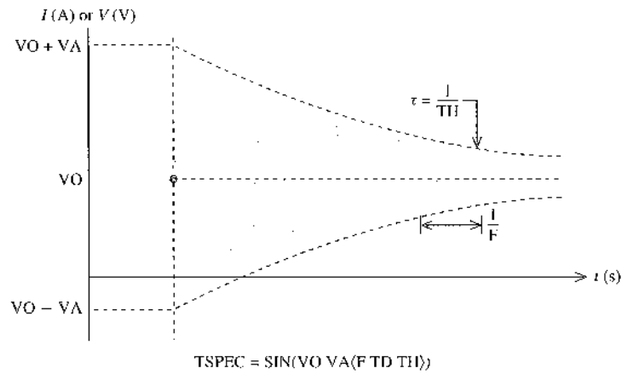
**********

  TIME          V(3)
(*)           -2.0000E+00  -1.5000E+00  -1.0000E+00  -5.0000E-01   0.0000E+00
  0.000E+00 -7.956E-06 .                .            .            .            .
  1.000E-02  1.894E-01 .                .            .            .            .
  2.000E-02 -5.594E-01 .                .            .            .            .
  3.000E-02 -9.140E-01 .                .            .            .            .
  4.000E-02  6.548E-01 .                .            .            .            .
  5.000E-02 -7.844E-01 .                .            .            .            .
  6.000E-02 -8.896E-01 .                .            .            .            .
  7.000E-02  2.006E+00 .                .            .            .            .
  8.000E-02 -2.100E+00 .                .            .            .            .
  9.000E-02 -1.787E+00 .                .            .            .            .
  1.000E+00  1.264E+00 .                .            .            .            .
  1.100E+00 -1.335E+00 .                .            .            .            .
  1.200E+00 -1.398E+00 .                .            .            .            .
  1.300E+00 -1.456E+00 .                .            .            .            .
  1.400E+00 -1.507E+00 .                .            .            .            .
  1.500E+00 -1.555E+00 .                .            .            .            .
  1.600E+00 -1.597E+00 .                .            .            .            .
  1.700E+00 -1.636E+00 .                .            .            .            .
  1.800E+00  1.670E+00 .                .            .            .            .
  1.900E+00 -1.702E+00 .                .            .            .            .
  2.000E+00 -1.730E+00 .                .            .            .            .
  2.100E+00  1.756E+00 .                .            .            .            .
  2.200E+00 -1.779E+00 .                .            .            .            .
  2.300E+00 -1.801E+00 .                .            .            .            .
  2.400E+00  1.820E+00 .                .            .            .            .
  2.500E+00 -1.837E+00 .                .            .            .            .
  2.600E+00  1.852E+00 .                .            .            .            .
  2.700E+00 -1.867E+00 .                .            .            .            .
  2.800E+00 -1.879E+00 .                .            .            .            .
  2.900E+00  1.891E+00 .                .            .            .            .
  3.000E+00 -1.901E+00 .                .            .            .            .
  3.100E+00 -1.911E+00 .                .            .            .            .
  3.200E+00  1.919E+00 .                .            .            .            .
  3.300E+00 -1.927E+00 .                .            .            .            .
  3.400E+00 -1.934E+00 .                .            .            .            .
  3.500E+00  1.940E+00 .                .            .            .            .
  3.600E+00 -1.946E+00 .                .            .            .            .
  3.700E+00 -1.951E+00 .                .            .            .            .
  3.800E+00  1.956E+00 .                .            .            .            .
  3.900E+00 -1.960E+00 .                .            .            .            .
  4.000E+00 -1.964E+00 .                .            .            .            .
  4.100E+00 -1.967E+00 .                .            .            .            .
  4.200E+00 -1.970E+00 .                .            .            .            .
  4.300E+00 -1.973E+00 .                .            .            .            .
  4.400E+00  1.976E+00 .                .            .            .            .
  4.500E+00 -1.978E+00 .                .            .            .            .
  4.600E+00 -1.980E+00 .                .            .            .            .
  4.700E+00  1.982E+00 *                .            .            .            .
  4.800E+00 -1.984E+00 *                .            .            .            .
  4.900E+00 -1.985E+00 *                .            .            .            .
  5.000E+00  1.987E+00 *                .            .            .            .
  5.100E+00 -1.988E+00 *                .            .            .            .
  5.200E+00 -1.989E+00 *                .            .            .            .
  5.300E+00  1.990E+00 *                .            .            .            .
  5.400E+00 -1.991E+00 *                .            .            .            .
  5.500E+00 -1.992E+00 *                .            .            .            .
  5.600E+00  1.993E+00 *                .            .            .            .
  5.700E+00 -1.993E+00 *                .            .            .            .
  5.800E+00 -1.994E+00 *                .            .            .            .
  5.900E+00  1.995E+00 *                .            .            .            .
  6.000E+00 -1.995E+00 *                .            .            .            .
  6.100E+00 -1.996E+00 *                .            .            .            .
  6.200E+00  1.996E+00 *                .            .            .            .
```

Step response for Example 6.18.



(a) Piecewise linear source; (b) exponential source.



TSPEC = SIN(VO VA(F TD TH))

Damped sinusoidal source.

at the maximum available screen resolution. The control statement

```
.PROBE
```

when added to the input file results in the creation of an auxiliary data file that supports the operation of PROBE, a graphics postprocessor software program bundled with PSpice. Upon execution, PROBE permits the circuit variables to be plotted and scales for the axes and other display characteristics to be selected and changed interactively. A PROBE plot of the same pulse response .PLOTted in Fig. 6.41 is shown in Fig. 6.42.*

Finally, a circuit with nonzero or multiple switch times can be analyzed with SPICE by shifting the switch time (or times) to zero. For instance, suppose that we have a circuit in dc steady state at $t = 0-$, at which time a switch acts, followed by a second switch action at $t = t_0 > 0$. To determine the conditions at $t = 0-$, we first do a dc analysis of the circuit with the switches in their $t < 0$ positions. The resulting values are used as initial conditions on the storage element statements for a second SPICE run, this time a transient analysis of the circuit with its switches in their $0 < t < t_0$ positions.

*Users of SPICE version 3, DESIGN CENTER, and other SPICE variants or supersets may also have access to high-resolution graphics. Consult your user's guide.

```
**** 10/ 3/94 09:37:06 ********* Evaluation SPICE (January 1991) ***********

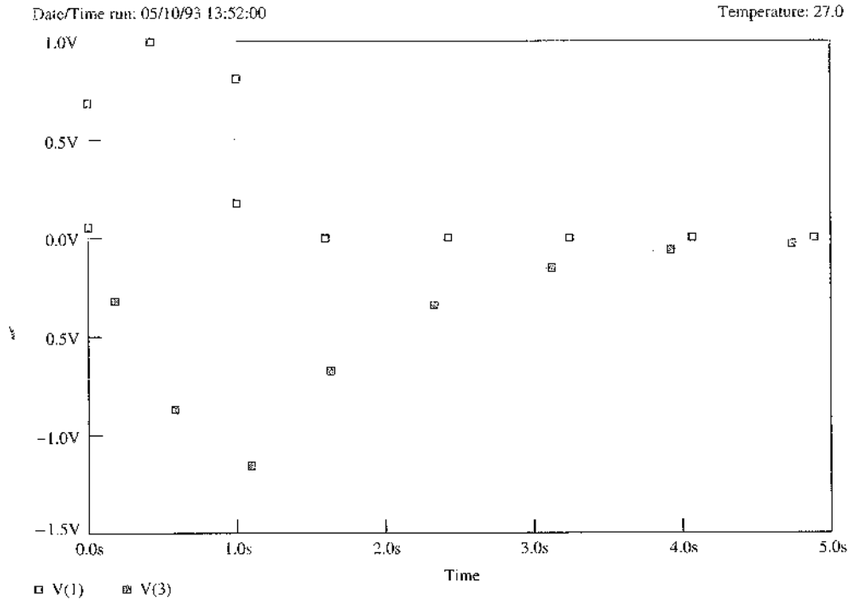
EXAMPLE 6.18: PULSE RESPONSE

****     TRANSIENT ANALYSIS              TEMPERATURE =   27.000 DEG C

******************************************************************************


  TIME         V(2)
(*) ---------   -1.5000E+00   -1.0000E-00  -5.0000E-01   0.0000E+00   5.0000E-01
                ------------  ------------  ------------  ------------  ----------
  0.0000E+00  1.6805E-08 .                              .             *
  1.000E-01 -1.890E-01 .                                .          *  .
  2.000E-01 -3.603E-01 .                                .       *     .
  3.000E-01 -5.113E-01 .                                .    *        .
  4.000E-01 -6.578E-01 .                                .  *           .
  5.000E-01  7.822E-01 .                                .*             .
  6.000E-01 -9.014E-01 .                               *.              .
  7.000E-01 -1.007E+00 .                              * .              .
  8.000E-01  1.106E+00 .                             *  .              .
  9.000E-01 -1.186E+00 .                           *    .              .
  1.000E-00 -1.266E+00 .                         *      .              .
  1.100E-00  1.149E+00 .                        *       .              .
  1.200E+00 -1.041E+00 .                      *         .              .
  1.300E+00 -9.419E-01 .                     *          .              .
  1.400E+00  8.541E-01 .                   *            .              .
  1.500E+00 -7.710E-01 .                  *             .              .
  1.600E+00 -6.986E-01 .                *               .              .
  1.700E+00  6.307E-01 .               *                .              .
  1.800E+00 -5.715E-01 .             *                  .              .
  1.900E+00 -5.159E-01 .            *                   .              .
  2.000E+00  4.674E-01 .          *                     .              .
  2.100E+00 -4.257E-01 .         *                      .              .
  2.200E+00 -3.913E-01 .       *                        .              .
  2.300E+00 -3.462E-01 .      *                         .              .
  2.400E+00 -3.127E-01 .    *                           .              .
  2.500E+00  2.822E-01 .   *                            .              .
  2.600E+00 -2.568E-01 .  *                             .              .
  2.700E+00  2.305E-01 . *                              .              .
  2.800E+00 -2.093E-01 .*                               .              .
  2.900E+00 -1.888E-01 .                                *              .
  3.000E+00  1.712E-01 .                                .              *
  3.100E+00 -1.545E-01 .                                .             *
  3.200E+00 -1.400E-01 .                                .           *
  3.300E+00  1.264E-01 .                                .          *
  3.400E+00 -1.145E-01 .                                .        *
  3.500E+00 -1.034E-01 .                                .      *
  3.600E+00  9.367E-02 .                                .    *
  3.700E+00 -8.457E-02 .                                .  *
  3.800E+00 -7.662E-02 .                                .*
  3.900E+00  6.917E-02 .                                *
  4.000E+00 -6.260E-02 .                               *.
  4.100E+00 -5.658E-02 .                              * .
  4.200E+00  5.127E-02 .                             *  .
  4.300E+00 -4.628E-02 .                           *    .
  4.400E+00 -4.183E-02 .                         *      .
  4.500E+00  3.788E-02 .                       *        .
  4.600E+00 -3.438E-02 .                     *          .
  4.700E+00 -3.087E-02 .                   *            .
  4.800E+00 -2.806E-02 .                 *              .
  4.900E+00 -2.533E-02 .               *                .
  5.000E+00 -2.295E-02 .             *                  .
  5.100E+00 -2.072E-02 .           *                    .
  5.200E+00 -1.877E-02 .         *                      .
  5.300E+00  1.695E-02 .       *                        .
  5.400E+00 -1.536E-02 .     *                          .
  5.500E+00 -1.386E-02 .   *                            .
  5.600E+00  1.256E-02 .  *                             .
  5.700E+00 -1.124E-02 . *                              .
  5.800E+00 -1.027E-02 .*                               .
  5.900E+00  9.275E-03 .                                *
  6.000E+00 -8.404E-03 .                               *.
  6.100E+00 -7.587E-03 .                              * .
  6.200E+00  6.874E-03 .                             *  .
```

Pulse response for Example 6.18 (.PLOT output).

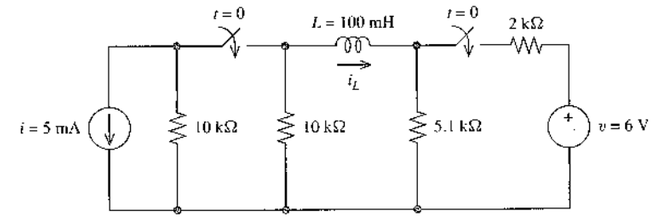


Pulse response for Example 6.18 (PROBE output).

We terminate this analysis at the time t_0, outputting the values of the capacitive voltages and inductive currents. These values are input as initial conditions for a final run, using a SPICE input file reflecting the circuit configuration that results by setting the switches in their $t > t_0$ positions. Since SPICE always begins a transient analysis at $t = 0$, we will remember to add t_0 to every time listed in the final output. An example with multiple switch times is given in Exercise 6.8.3.

6.8.1. Using SPICE, find (a) $i_L(0-)$ and (b) i_L (30 μs).

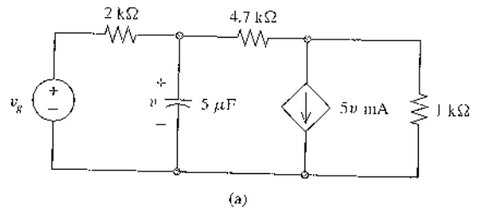
(a) -2.475 mA; (b) $+0.285$ mA



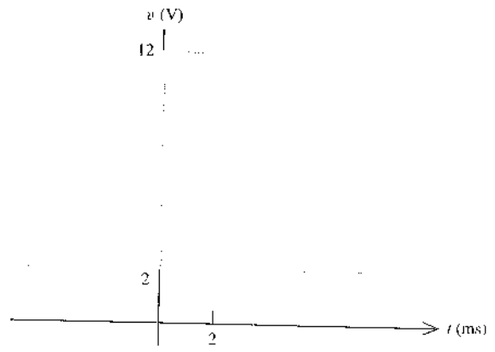
6.8.2. Repeat Exercise 6.8.1 for the case in which the rightmost switch closes at time $t = 20\ \mu s$ rather than $t = 0$.

(a) -2.475 mA; (b) $+0.218$ mA

6.8.3. Plot the response $v(t)$ for $0 < t < 10$ ms if v_g is a pulse of amplitude 10 V and duration 2 ms, and the initial value of v is $+2$ V.

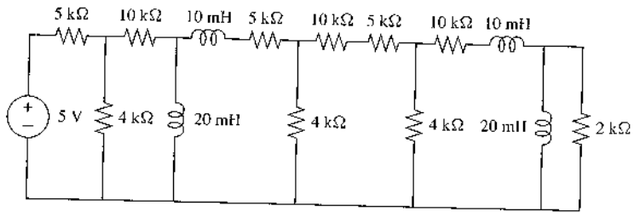


(a)



(b)

6.8.4. Write a SPICE input file for this circuit using two subcircuits. Assume that all initial currents are zero. Omit the .TRAN or .DC statement and output control statements.




```
Exercise 6.8.4
VS      1       0       DC      5
XR1     1       2       RSUBCKT
XL1     2       3       LSUBCKT
XR2     3       4       RSUBCKT
XR3     4       5       RSUBCKT
XL2     6       5       LSUBCKT
RL      6       0       2K
.SUBCKT         RSUBCKT                 1       3
R1      1       2       5K
R2      2       3       10K
R3      2       0       4K
.ENDS
.SUBCKT         LSUBCKT         1       2
L1      1       2       10M     IC=0
L2      1       0       20M     IC=0
.ENDS
.END
```

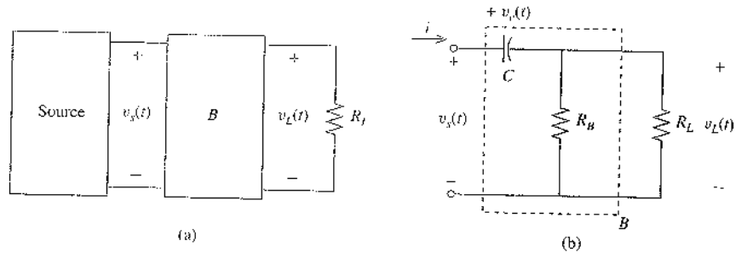
6.9 DESIGN OF FIRST-ORDER CIRCUITS

First-order circuits have two parameters that may be assigned. These may be specified as the resistance R and inductance L, resistance R and capacitance C, or the time constant τ and the gain of the circuit. First-order circuits may be either passive, designed using only RLC elements, or active, incorporating op amps or other active devices. What they have in common is that they are described by first-order differential equations.

In Design Example 6.1 we demonstrate one common use of first-order circuits: to block or trap dc. Often, a dc bias voltage is produced by a source circuit and it is desired to prevent this dc value from appearing across the terminals of the load. In this case a first-order circuit can be interposed between source and load, serving as a buffer between them.

Design a first-order buffer circuit B which blocks any dc source voltage component from appearing across the load R_L [see Fig. 6.43(a)]. In addition, the half-amplitude pulse width of the step response across the load should be no more than 1 ms for any load R_L.

The i–v law for a capacitor is $i = C\,dv/dt$. If the voltage $v(t)$ is a constant (dc), then $dv/dt = 0$ and no dc current will flow through the capacitor. Thus we can place any capacitor C in series with the source voltage to block dc. If the buffer circuit B in Fig. 6.43(b) was only the capacitor, in other words if the resistor labeled R_B in this figure were omitted, the time constant of the resulting RC circuit would be $\tau = R_L C$ and could be arbitrarily long for large R_L regardless of our choice for C. So we put a resistor R_B



(a) Block diagram; (b) buffer circuit design.

at the output of B as shown. Determination of appropriate values for C and R_B follow.

Let R denote the parallel equivalent of R_B and R_L. The voltage response $v_C(t)$ across the capacitor to a unit step input $v_S(t) = u(t)$ V is, repeating (6.41),

$$v_C(t) = (1 - e^{-t/RC})u(t)$$

and thus by KVL applied to Fig. 6.43(b),

$$v_L(t) = u(t) - (1 - e^{-t/RC})u(t) = e^{-t/RC}u(t)$$

This pulselike step response is sketched in Fig. 6.44. In order to have the response reduced to 0.5 V at $t = \tau_h = 0.001$ s, defined as the *half-amplitude pulse width*, we need

$$v_L(t)|_{t=0.001} = 0.5 = e^{-0.001/\tau}$$

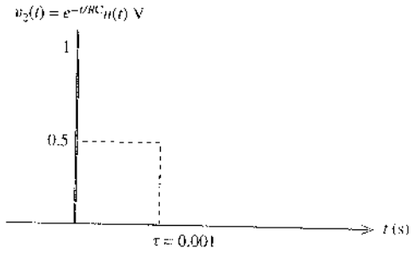
where $\tau = RC$ is the time constant. Taking logarithms gives

$$\frac{-0.001}{\tau} = \log 0.5 = -0.693$$

or $\tau = 1.44$ ms. Thus in order that the half-amplitude pulse width $\tau_h = 1$ ms or less, it is necessary that the circuit time constant $\tau = RC$ be 1.44 ms or less. If we choose $R_B = 100$ kΩ, then since R is the parallel equivalent of R_B and another resistance (R_L), R cannot exceed 100 kΩ. Then choosing C so that

$$RC = 0.00144 = (10^5)C$$

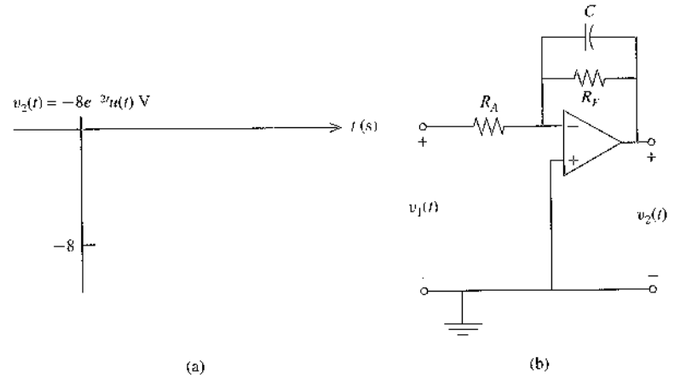
or $C = 14.4$ nF, the design is complete. For open-circuit loads (R_L infinite) the half-amplitude pulse width will be 1 ms, and for finite loads leading to shorter time constants the half-amplitude pulse width will be less than 1 ms, as required. Note that any selection of R_B and C such that R_BC is less than or equal to 1.44 ms will also satisfy the design requirements.



Step response of the load voltage $v_L(t)$.

If voltage gains of greater than unity are required in a first-order circuit, active design must be employed. An example based upon the inverting amplifier op amp building block follows.

Design a circuit whose unit step response is the voltage waveform shown in Fig. 6.45(a).



(a) Desired step response; (b) circuit design.

Examining the waveform, in dc steady state we need a voltage gain of -8, which can be achieved by an inverting amplifier introduced in Fig. 3.15 with resistance ratio $R_F/R_A = 8$. To achieve a single exponential transition to steady state as required, we need a first-order circuit, so we must add a storage element. Choosing a capacitor, it cannot be placed in series with either resistor since the dc-blocking property of a capacitor makes it an effective infinite resistance and the dc steady-state gain will certainly be affected. Trying C in parallel with R_F, we have the circuit of Fig. 6.45(b). Noting that by the virtual short principle the noninverting input is at ground and applying KCL at the noninverting input node, we obtain

$$\frac{v_1}{R_A} + \frac{v_2}{R_F} + C\frac{dv_2}{dt} = 0$$

or

$$\frac{dv_2}{dt} + \frac{1}{R_FC}v_2 = \frac{-1}{R_AC}v_1$$

Setting $v_1(t) = u(t)$ V to find the step response, the last is a first-order differential equation with dc source as discussed in Section 6.4. The forced response will be a constant A satisfying

$$\frac{dA}{dt} + \frac{A}{R_FC} = \frac{-1}{R_AC}$$

or $A = -R_F/R_A$. The natural response will be an exponential with $\tau = R_FC$, so the total response for $t > 0$ must be

$$v_2(t) = Ke^{-t/R_FC} - \frac{R_F}{R_A}$$

The step response is always measured using zero initial conditions,

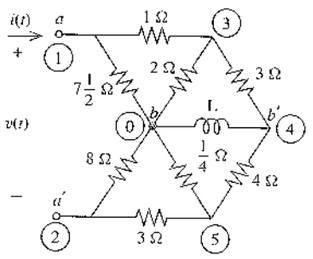
so

$$v_2(0) = K - \frac{R_F}{R_A} = 0$$

or

$$v_2(t) = \frac{-R_F}{R_A}(1 - e^{-t/R_FC})u(t)$$

Returning to Fig. 6.45(a), we need only pick the resistance ratio $R_F/R_A = 8$ and the time constant R_FC of the feedback network equal to $\frac{1}{2}$ s to complete the design. For instance, we may select $R_F = 80$ kΩ, $R_A = 10$ kΩ, and $C = 6.25$ μF.



Circuit for Design Example 6.3.

A power company is supplying electrical power to a load region modeled by the resistive network shown in Fig. 6.46. If an L-henry inductance is introduced between the pair of nodes b and b' shown, find a formula for the time constant of the resulting RL load circuit as seen from the source terminals a–a'.

This design problem is a bit different from the previous ones, in that we must design many circuits at the same time, those with all possible time constants τ. Also, we have rigid structural constraints on our design; the form of the circuit layout is given. There is no single format for design problems, they come in many forms and with a great variety of constraints.

Since there is only one inductor, this is a first-order circuit with time constant L/R_{eq}, where R_{eq} is the equivalent resistance. Rather than compute the time constant using exact analytic methods, which would be time consuming since the resistors are neither in series nor parallel, we will use SPICE and numerical approximation. Suppose that we set $L = 1$ H. If we determine $\tau = \tau_1$ for $L = 1$ H, we would know $R_{eq} = 1/\tau_1$ and hence the desired formula $\tau(L) = L/R_{eq}$. A SPICE input file for this design strategy is shown below.

```
DESIGN EXAMPLE 6.3
*
I       2       1       DC              1
R1      1       0         7.5
R2      2       0         8
R3      1       3         1
R4      2       5         3
R5      3       0         2
R6      5       0         0.25
R7      3       4         3
R8      4       5         4
L       4       0         1             IC=0
.TRAN   0.2     5       UIC
.PLOT TRAN V(1,2)
.END
```



```
**** 03/04/95 15:47:55 ********* Evaluation PSpice (January 1991) *************

 DESIGN EXAMPLE 6.3

 ****     TRANSIENT ANALYSIS                TEMPERATURE =   27.000 DEG C

******************************************************************************


  TIME         V(1,2)
(*)----------    4.0000E+00   4.0500E+00   4.1000E+00   4.1500E+00   4.2000E+00
  0.000E+00  4.163E+00 .                                            *
  2.000E-01  4.107E+00 .                                         *
  4.000E-01  4.070E+00 .                                      *
  6.000E-01  4.047E+00 .                                   *
  8.000E-01  4.032E+00 .                                *
  1.000E+00  4.022E+00 .                             *
  1.200E+00  4.016E+00 .                          *
  1.400E+00  4.012E+00 .                       *
  1.600E+00  4.009E+00 . *
  1.800E+00  4.008E+00 . *
  2.000E+00  4.007E+00 . *
  2.200E+00  4.006E+00 . *
  2.400E+00  4.005E+00 .*
  2.600E+00  4.005E+00 .*
  2.800E+00  4.005E+00 .*
  3.000E+00  4.005E+00 .*
  3.200E+00  4.005E+00 .*
  3.400E+00  4.005E+00 .*
  3.600E+00  4.005E+00 .*
  3.800E+00  4.005E+00 .*
  4.000E+00  4.005E+00 .*
  4.200E+00  4.005E+00 .*
  4.400E+00  4.005E+00 .*
  4.600E+00  4.005E+00 .*
  4.800E+00  4.005E+00 .*
  5.000E+00  4.005E+00 .*

 JOB CONCLUDED

 TOTAL JOB TIME            .33
```

FIGURE 6.47(a) SPICE output.

The resulting plot, shown as Fig. 6.47(a), shows that the steady-state voltage response is 4.005 V. Since the initial voltage was 4.163 V, the size of the transient response is $4.163 - 4.005 = 0.158$ V. The time constant τ is the time at which the transient is reduced to e^{-1} of its initial value, or

$$v_{12}(\tau) = 4.005 + e^{-1}(0.158) = 4.063$$

Figure 6.47(a) has too coarse time sampling to make out the time at which $v(1, 2)$ takes on this value with acceptable accuracy. Expanding the scale using the edited transient mode control statement

```
.TRAN   0.01    0.5     UIC
```

results in the plot shown in Fig. 6.47(b), from which we see that

```
**** 03/04/95 16:02:44 ******** Evaluation PSpice (January 1991) ************
DESIGN EXAMPLE 6.3

****     TRANSIENT ANALYSIS                TEMPERATURE =   27.000 DEG C
*******************************************************************************

  TIME         V(1,2)
(*) ---  ----      4.0500E+00   4.1000E+00   4.1500E+00   4.2000E+00   4.2500E+00
 2.000E-01  4.106E+00 .                                                        .
 2.100E-01  4.104E+00 .                                                        .
 2.200E-01  4.102E+00 .                                                        .
 2.300E-01  4.100E+00 .                                                        .
 2.400E-01  4.098E+00 .                                                        .
 2.500E-01  4.096E+00 .                                                        .
 2.600E-01  4.094E+00 .                                                        .
 2.700E-01  4.092E+00 .                                                        .
 2.800E-01  4.090E+00 .                                                        .
 2.900E-01  4.088E+00 .                                                        .
 3.000E-01  4.086E+00 .                                                        .
 3.100E-01  4.084E+00 .                                                        .
 3.200E-01  4.083E+00 .                                                        .
 3.300E-01  4.081E+00 .                                                        .
 3.400E-01  4.079E+00 .                                                        .
 3.500E-01  4.078E+00 .                                                        .
 3.600E-01  4.076E+00 .                                                        .
 3.700E-01  4.074E+00 .                                                        .
 3.800E-01  4.073E+00 .                                                        .
 3.900E-01  4.071E+00 .                                                        .
 4.000E-01  4.070E+00 .                                                        .
 4.100E-01  4.069E+00 .                                                        .
 4.200E-01  4.067E+00 .                                                        .
 4.300E-01  4.066E+00 .                                                        .
 4.400E-01  4.064E+00 .                                                        .
 4.500E-01  4.063E+00 .                                                        .
 4.600E-01  4.062E+00 .                                                        .
 4.700E-01  4.061E+00 .                                                        .
 4.800E-01  4.059E+00 .                                                        .
 4.900E-01  4.058E+00 .                                                        .
 5.000E-01  4.057E+00 .                                                        .

JOB CONCLUDED

TOTAL JOB TIME             .35
```

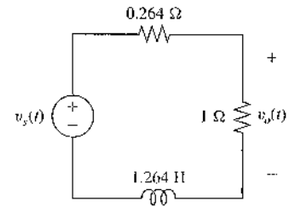
SPICE output with different sampling interval and limits.

$\tau = 0.45$ s. Then $1/R_{eq} = 0.45$ and our final design result is the expression

$$\tau(L) = \frac{L}{R_{eq}} = 0.45L$$

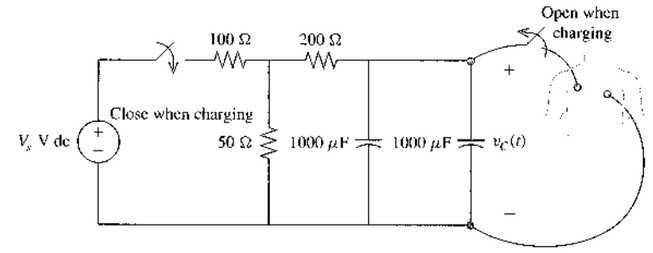
Design Example 6.3 was the first in which SPICE was used to save labor. We proceeded by evaluating a candidate circuit, editing the input file, and evaluating again in a cyclic manner until a design goal was reached. Sometimes it is the circuit element values or locations that are changed; other times, output parameters such as the transient sampling interval and duration in this example. In either case the tedium of repetitive, compute-intensive hand circuit evaluations are bypassed. Most of the remaining designs will use SPICE in this fashion.

6.9.1. Design a circuit with time constant $\tau = 1$ s whose response to the voltage pulse $v_s(t) = 2u(t) - 2u(t-1)$ just reaches, but never exceeds, 1 V.



6.9.2. A defibrillator is a bioelectric device for delivering a shock to stabilize the heart. If the defibrillator voltage is $v_C(t)$ in the figure, what is the minimum dc voltage source V_s that will charge the defibrillator from zero initial stored energy up to its working voltage of 400 V in 1 s?

1359.5 V



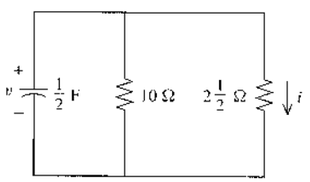
SUMMARY

First-order circuits are those characterized by a single first-order differential equation. They can be identified by the presence of a single equivalent storage element (inductor or capacitor). The behavior of first-order circuits may be studied by writing and solving their differential equations in the time domain, which is the focus of this chapter.

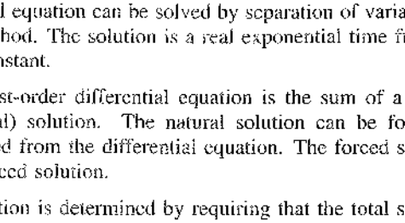
- An unforced first-order differential equation can be solved by separation of variables or by the characteristic equation method. The solution is a real exponential time function characterized by a single time constant.
- The total solution to a forced first-order differential equation is the sum of a forced solution and an unforced (natural) solution. The natural solution can be found as above, with forcing terms removed from the differential equation. The forced solution can be found by use of a trial forced solution.
- The multiplier of the natural solution is determined by requiring that the total solution match the initial condition.
- If more than one independent source is present, superposition may be used to determine the overall forced response. For each source, all other sources are killed, and the resulting forced responses superposed.
- The unit step function $u(t)$ is defined as zero for $t < 0$, one for $t > 0$. It models the function of a switch which turns on at $t = 0$. The unit step response of a circuit is a common way to characterize its behavior.
- Inductors and capacitors with arbitrary initial conditions are supported by SPICE. The .TRAN control statement is used to generate a transient analysis.

Unlike resistive circuits, first-order circuits contain responses that do not simply mimic the waveshapes of the sources exciting them. These are the simplest circuits that can create something new, something not already inherent in the circuit inputs, namely real exponential time functions. Circuits with more than one storage element can produce a much broader range of new responses and are the subject of the next chapter.

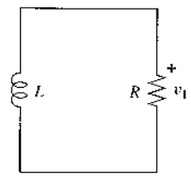
6.1. Find $v(t)$ and $i(t)$ for $t > 0$ if $v(0) = 2$ V.



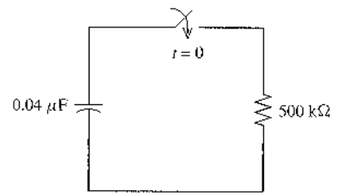
6.2. Find $v(t)$ for $t > 0$. Assume that the circuit is in dc steady state at $t = 0-$.



6.3. Determine $v_1(t)$ for $t > 0$ if $v_1(0-) = -12$ V.



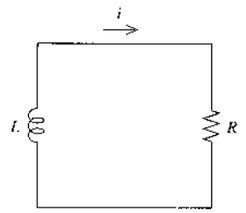
6.4. If the initial stored energy in the capacitor at time $t = 0-$ is 0.18 μJ, how much energy will remain stored at time $t = 20$ ms? At time $t = 200$ ms?



6.5. The energy $w = \frac{1}{2}Li^2$ stored in the inductor satisfies an enforced first-order differential equation

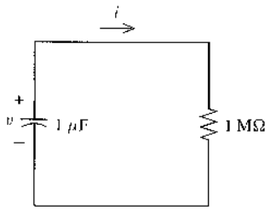
$$\frac{dw}{dt} + \alpha w = 0$$

Find α.



6.6. Design a sourceless RL circuit with a time constant of 10 ms using only 2-mH inductors and 1-kΩ resistors. Use the minimum number of circuit elements possible.

6.7. What is the time constant τ? Express $v(t)$ and $i(t)$ in terms of τ if $v(0) = +100$ V.



6.8. In terms of τ, how much time does it take a response in an unforced RC or RL circuit to decay by a factor of 2? By a factor of 10? By a factor of 1000?

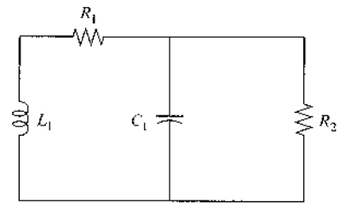
6.9. Sketch $w(t)$ versus t for the circuit of Problem 6.7. What is the time constant of this waveform? Why does it differ from $\tau = RC$?

6.10. From Fig. 6.6, the initial slope of a waveform with time constant linearly extrapolated hits $v(t) = 0$ at $t = \tau$. Where does a similar construction from the point $v(\tau)$ hit the horizontal axis?

6.11. Design a source-free circuit with $\tau = 1$ μs. If we have $w(0) = 1$ J in this circuit, how much energy will it dissipate by time $t = 1$ μs? $t = 5$ μs?

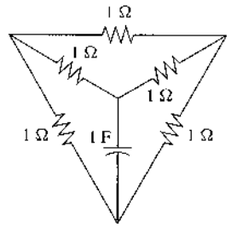
6.12. A sourceless RL circuit has 4-mA current at time $t = 2$ ms and 1 mA at time $t = 4$ ms. What was the initial current at time $t = 0$?

6.13. Show that this circuit is characterized by a second-order differential equation, so it is not a first-order circuit.

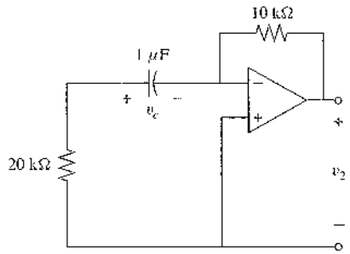


6.14. Repeat Problem 6.13 for the case in which C_1 is replaced by a second inductor L_2.

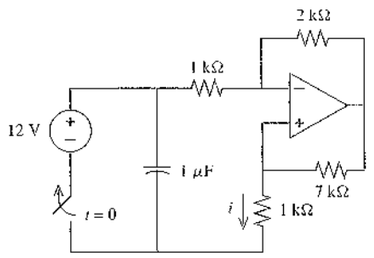
6.15. Find the time constant τ.



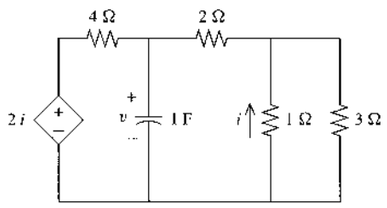
6.16. Find $v_2(t)$ for $t > 0$ if $v_c(0-) = +6$ V.



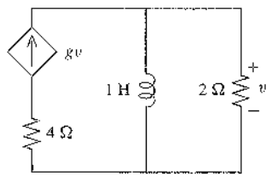
6.17. Find $i(t)$, $t > 0$, if the circuit is in dc steady state at time $t = 0-$.



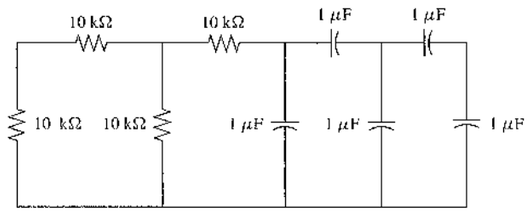
6.18. Find $i(t)$ for $t > 10$ s if $v(10-) = 2$ V. Repeat for $v(10-) = 0$ V.



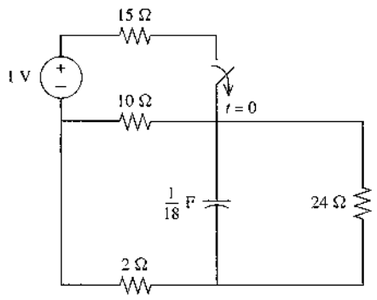
6.19. Find a value for g so that the time constant in this circuit is $\tau = 4$ s.



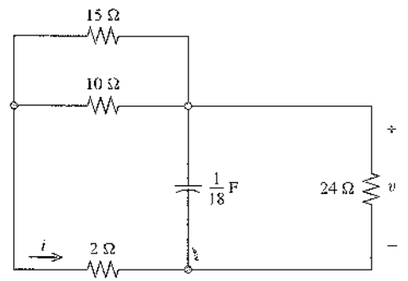
6.20. Find the time constant for this circuit.



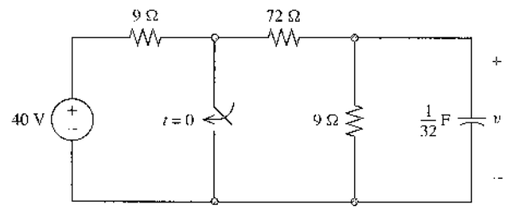
6.21. Find all currents and voltages for $t > 0$ in this circuit. Assume dc steady state at $t = 0-$.



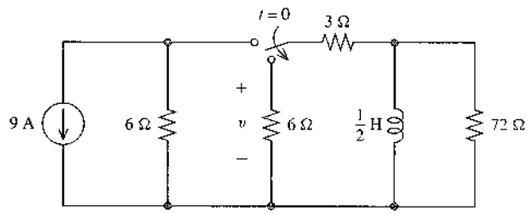
6.22. Find v and i for $t > 0$ if $i(0) = 1$ A.



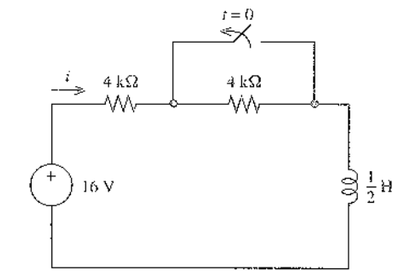
6.23. If the circuit is in steady state at $t = 0-$, find v for $t > 0$.



6.24. Find v for $t > 0$ if the circuit is in steady state at $t = 0-$.



6.25. Find i for $t > 0$ if the circuit is in steady state at $t = 0-$.



6.26. Solve the differential equation

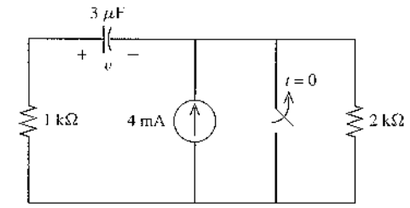
$$\frac{di}{dt} + 6i = 24, \qquad i(0) = 1$$

6.27. Solve the equation

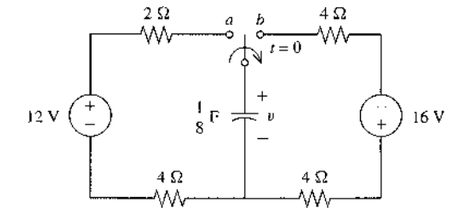
$$\int_0^t (i(\tau) + 2)\, d\tau + 3i(t) = 6$$

for $t > 0$ by converting it to a differential equation.

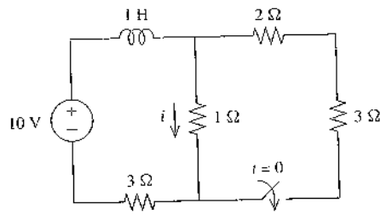
6.28. Find v for $t > 0$. Assume dc steady state at $t = 0$.



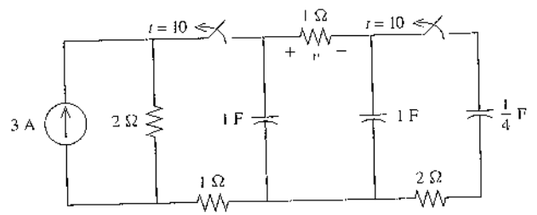
6.29. Find $v(t)$ for $t > 0$. Assume dc steady state at time $t = 0$.



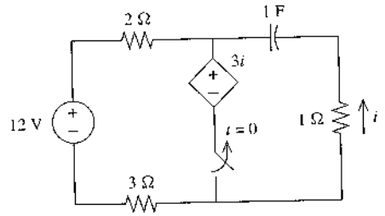
6.30. Find $i(0-)$ and $i(t)$ for $t > 0$. Assume dc steady state at $t = 0$.



6.31. Find v for $t > 10$. Assume dc steady state at $t = 10-$.

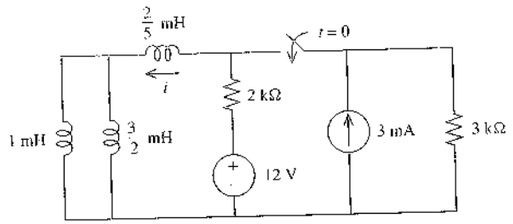


6.32. Find i for $t > 0$. Assume dc steady state at $t = 0-$. The controlled source has transresistance of 3 Ω.

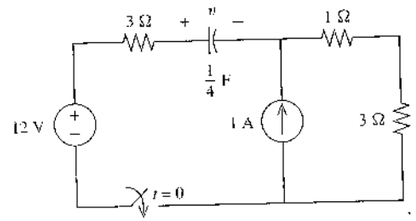


6.33. Repeat Problem 6.32 for the case in which the switch closes, rather than opens, at time $t = 0$.

6.34. Find i for $t > 0$. Assume dc steady state at $t = 0$.

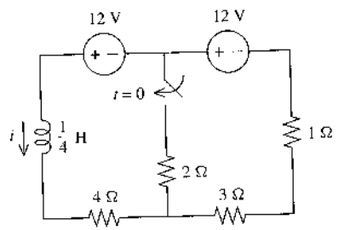


6.35. Find v for $t > 0$ using superposition. Assume the circuit is in dc steady state at $t = 0-$.

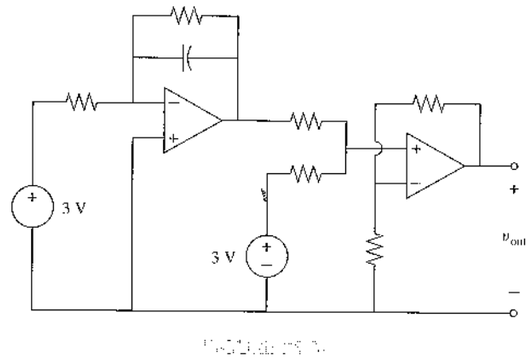


6.36. Repeat Problem 6.35 using the method of Thevenin transformation of all but the storage element.

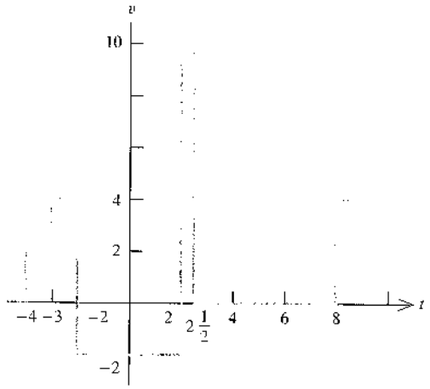
6.37. Find i for $t > 0$. Assume dc steady state at $t = 0-$.



6.38. Find $v_{out}(t)$ for $t > 0$. Assume that all resistors are 10 kΩ and the capacitor is 20 μF and has no voltage across it at time $t = 0-$.



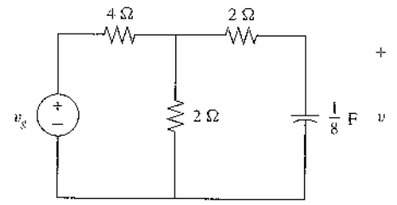
6.39. Express $v(t)$ in terms of step functions.



6.40. Sketch the voltage $v(t) = -3u(-t) + 3u(t + 2) + 2u(t) - 2u(t - 4)$.

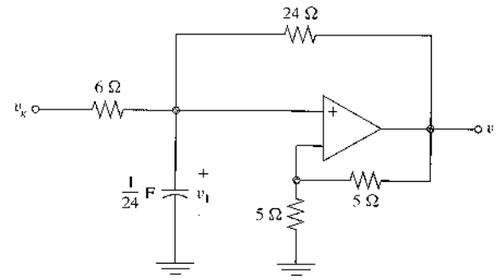
6.41. Express $\int_{-\infty}^{t} v(\tau)d\tau$, $v(t)$ from 6.39, in terms of ramp functions.

6.42. Find the step response $v(t)$.

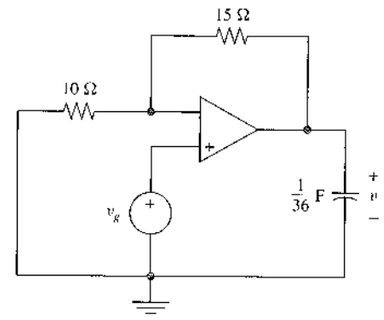


6.43. Repeat Problem 6.42 for the pulse response if $v_g(t) = u(t) - u(t - 1)$.

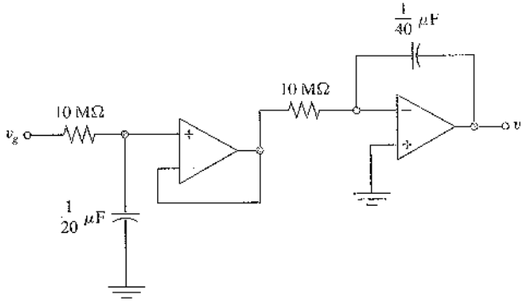
6.44. Find v for $t > 0$ if $v_g = 3u(t)$ V.



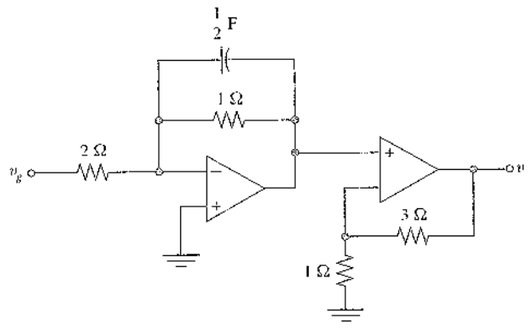
6.45. Find for $t > 0$ the current downward in the capacitor if $v(0) = 0$, and (a) $v_g = 4$ V, (b) $v_g = 2e^{-2t}$ V, and (c) $v_g = 2\cos 2t$ V.



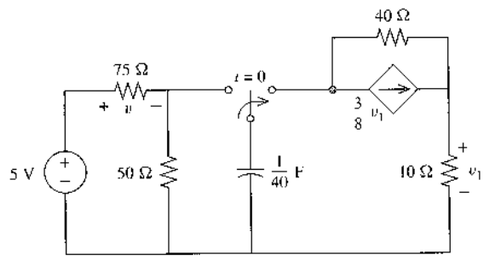
6.46. Find v for all time if $v_g = 2u(t)$ V.



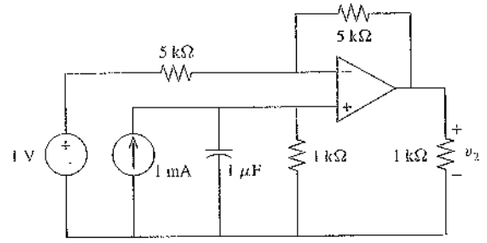
6.47. Find v for all t if (a) $v_g = 2u(t)$ V, and (b) $v_g = 2[u(t) - u(t-1)]$ V.



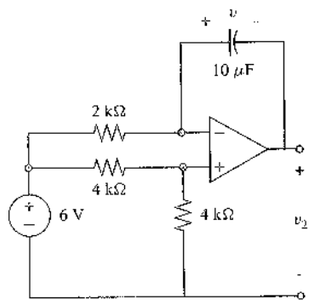
6.48. Find v and v_1 for $t > 0$. Assume dc steady state at $t = 0-$. Check using SPICE. The controlled source has transconductance $g = \frac{3}{8}$ S.



6.49. Find v_2 if the capacitive voltage at time $t = 0-$ is 0. Check using SPICE.

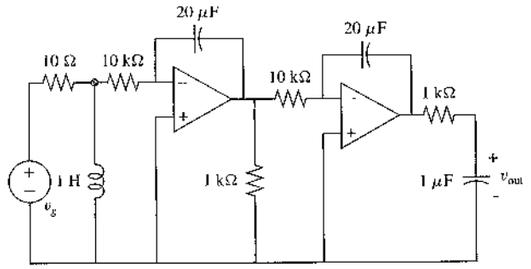


6.50. Find v_2 and v for $t > t_0$ if $v(t_0) = 4$ V. Check using SPICE.



6.51. Solve Problem 6.18 using SPICE.

6.52. Use SPICE to determine the output of the circuit to a unit step input. Plot the response from $t = 0$ to $t = 20$ ms. Use subcircuits to describe the op amps and the integrators.

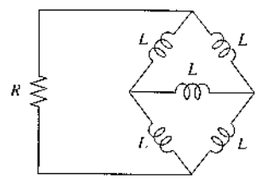


Interference from a car's ignition system induces 4000 identical equally spaced pulse voltage waveforms per second across the antenna wire when it is open circuited. The rise and fall times are $TR = TF = 1$ μs (see Fig. 6.36) and full-amplitude width is PW = 8 μs. The pulse has baseline V1 = 0 and amplitude V2 = 1 mV. If the radio is a 100-kΩ resistive load as seen by the antenna, determine an ideal inductor that can be placed in series with the radio input which reduces the peak-to-peak interference across the radio input to 0.1 mV ± 5%. This is called an *RF suppressor*. Model the antenna as having a 100-kΩ Thevenin equivalent resistance.

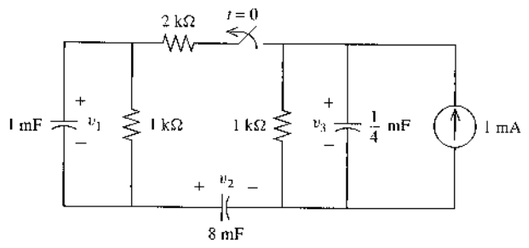
A certain loudspeaker is modeled as an 8-Ω resistor in series with a 2-mH inductor. Design a circuit to connect the loudspeaker to an independent voltage source V_g so that the time constant of the overall circuit is 1 ms and the power dissipated by the loudspeaker when $V_g = 10$ V dc is 1 W.

Design an RC circuit whose zero initial condition response to the current input $I_s(t) = u(t) - 2u(t-1)$ has a zero crossing of the capacitive voltage at time $t = 2$ s.

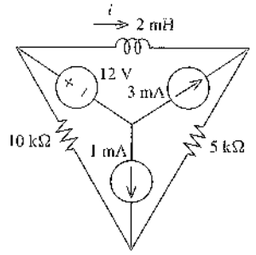
Find the time constant τ.



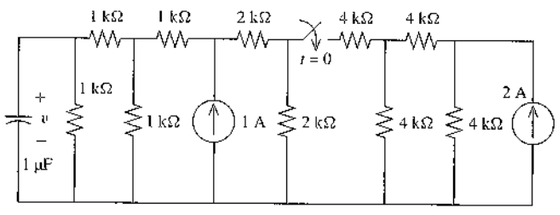
Find v_1, v_2, and v_3, for $t > 0$. Assume the circuit is in dc steady state at time $t = 0-$.



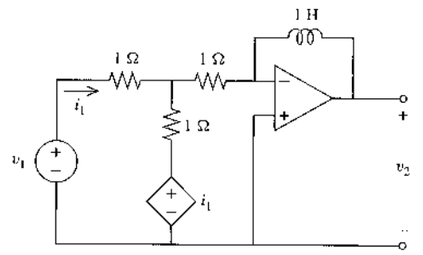
Find all currents and voltages in this circuit. Assume $i(0-) = 0$.



Find $v(t)$, $t > 0$ if the circuit is in dc steady state at $t = 0$.

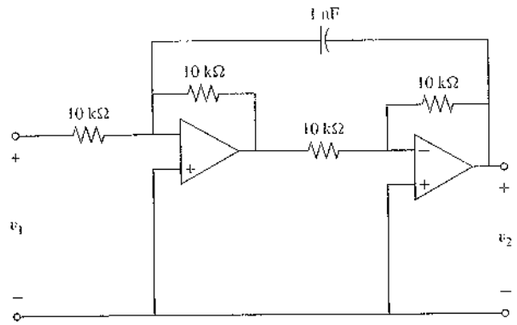


Find the voltage transfer equation for $v_2(t)$ in terms of $v_1(t)$.



Repeat Problem 6.29 for the case in which the switch moves $a \rightarrow b$ at $t = 0$ and then moves back to a at time $t = 3$ s.

Find the unit step response $v_2(t)$ to the input $v_1(t)$.



7

Second-Order Circuits



Samuel F. B. Morse
1791–1872

What hath God wrought!
[The famous message tapped out on the first telegraph]

Samuel F. B. Morse

The first practical application of electricity is said by many to be the telegraph, developed by Samuel F. B. Morse, an American portrait painter and inventor. Morse built on the ideas of the famous American physicist Joseph Henry, using the opening and closing of relays to produce the dots and dashes (or Morse code) that represent letters and numbers.

Morse was born in Charlestown, Massachusetts, the son of a minister and author. He studied to be an artist at Yale and the Royal Academy of Arts in London, and by 1815 he was considered to be moderately successful. In 1826 he helped found and became the first president of the National Academy of Design. But the previous year his wife had died, in 1826 his father died, and in 1828 his mother died. The following year the distressed Morse went to Europe to recover and study further. In 1832, while returning home on board the passenger ship *Sully*, he met an eccentric inventor and became intrigued with developing a telegraph, the principle of which had already been considered by Henry. By 1836 Morse had a working model, and in 1837 he acquired a partner, Alfred Vail, who financed the project. Their efforts were rewarded with a patent and the financing by Congress of a telegraph in 1844, over which Morse—on May 24, 1844—sent his now-famous message, "What hath God wrought!"

7.1 Circuits with Two Storage Elements
7.2 Second-Order Equations
7.3 Natural Response
7.4 Forced Response
7.5 Total Response
7.6 Unit Step Response
7.7 Design of Second-Order Circuits
Summary
Problems

The circuit analysis equations for linear circuits with energy storage elements may be expressed as linear differential equations, because the element laws are such that each term in the mesh or nodal equations is a derivative, integral, or multiple of the unknowns and the source variables. Evidently, a single differentiation will remove any integrals that it may contain, so in general the mesh or nodal analysis equations for a given circuit may be considered to be differential equations. The *describing equation*, a single equation whose only unknown is a selected output current or voltage, may be obtained by manipulation of these analysis equations.

The circuits containing storage elements that we have considered so far were first-order circuits. That is, they were described by first-order differential equations. This is always the case when there is only one storage element in the circuit, only one remains after series–parallel simplification, or when a switching action divides the circuit into two or more independent subcircuits each having at most one storage element.

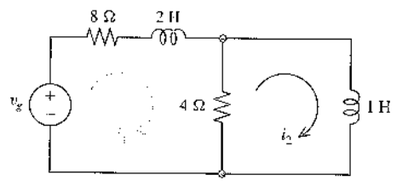
In this chapter we consider second-order circuits, which, as we shall see, contain two storage elements and have describing equations that are second-order differential equations. We show that the total response is the sum of a natural and forced response, as with first-order circuits. Examination of the characteristic equation shows the natural response of a *RLC* circuit is not limited to real exponential decays, as with *RC* or *RL* circuits, but may also include sinusoids, damped sinusoids, and t-multiplied forms. The forced response is determined from a trial form as with first-order circuits, and the total response is computed by requiring that two initial conditions, derived from the initial energies stored in the two storage elements, be satisfied.

In general, nth-order circuits, containing n storage elements, are described by nth-order differential equations. The results for first- and second-order circuits may readily be extended to the general case using the differential-equations-based methods of Chapters 5 and 6, but we shall not do so here. There is a more efficient method for analyzing these higher-order circuits based on Laplace transform analysis of the circuit, which is introduced in Chapter 12.

7.1 CIRCUITS WITH TWO STORAGE ELEMENTS

The circuits considered in Chapter 6 gave rise to first-order describing equations. If the circuit has two storage elements, one inductor and one capacitor, or two of the same type but not series or parallel equivalent to a single element, a second-order differential equation will describe the circuit.

Example 7.1



Second-order circuit with two inductors.

Consider the circuit of Fig. 7.1, in which we seek the mesh current i_2. The circuit contains two inductors. The mesh equations are

$$2\frac{di_1}{dt} + 12i_1 - 4i_2 = v_g \tag{7.1a}$$

$$-4i_1 + \frac{di_2}{dt} + 4i_2 = 0 \tag{7.1b}$$

From the second of these we have

$$i_1 = \frac{1}{4}\left(\frac{di_2}{dt} + 4i_2\right) \tag{7.2}$$

which differentiated results in

$$\frac{di_1}{dt} = \frac{1}{4}\left(\frac{d^2i_2}{dt^2} + 4\frac{di_2}{dt}\right) \tag{7.3}$$

Substituting (7.2) and (7.3) into (7.1a) to eliminate i_1, we have, after multiplying through by 2, the describing equation for i_2:

$$\frac{d^2i_2}{dt^2} + 10\frac{di_2}{dt} + 16i_2 = 2v_g \tag{7.4}$$

This is a second-order differential equation (one in which the highest order of differentiation of the unknown is 2). For this reason we refer to Fig. 7.1 as a *second-order circuit* and note that, typically, second-order circuits contain two storage elements.

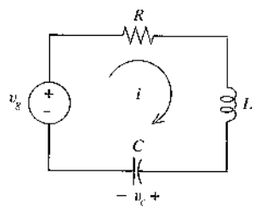
The two storage elements in a second-order circuit may be of the same type, as in Fig. 7.1, or one inductor and one capacitor, as in Fig. 7.2. This is the important *series RLC circuit*, which we return to throughout the chapter. By KVL,

$$Ri + \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + v_c(t_0) + L\frac{di}{dt} = v_g$$

Differentiating and dividing by L yields

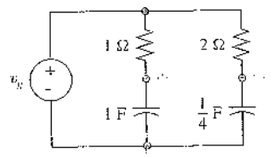
$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{1}{LC}i = \frac{dv_g}{dt}$$

which is a second-order differential equation; hence the series *RLC* circuit is also a second-order circuit.



Series *RLC* circuit.

There are exceptions to the rule that two-storage-element circuits have second-order describing equations. If two or more storage elements of the same kind (inductors or capacitors) can be replaced by a single equivalent, they count as a single storage element in determining the describing equation order. In other cases, the storage elements may not interact. For example, consider the circuit of



First-order circuit with two storage elements.

Fig. 7.3, which has two capacitors. Nodal analysis equations for this circuit are

$$\frac{dv_1}{dt} + v_1 = v_g$$

$$\frac{dv_2}{dt} + 2v_2 = 2v_g$$

These are *uncoupled* first-order differential equations; that is, neither unknown appears in both equations. Each may be solved using the methods of Chapter 6. Since the describing equations are first order, this is not a second-order circuit even though two storage elements irreducible to a single equivalent are in the circuit. Replacement of the ideal voltage source by a practical one with nonzero Thevenin equivalent resistance in this circuit, however, results in a second-order circuit, as shown in Exercise 7.1.2.

7.1.1. Find the equation satisfied by the mesh current i_2.

ANS: $d^2i_2/dt^2 + 7(di_2/dt) + 6i_2 = dv_g/dt$

7.1.2. Place a 1-Ω resistor in series with the ideal voltage source v_g in Fig. 7.3, and find the describing equations for v_1 and v_2.

ANS:
$$\frac{d^2v_1}{dt^2} + \frac{11}{5}\frac{dv_1}{dt} + \frac{4}{5}v_1 = \frac{2}{5}\frac{dv_g}{dt} + \frac{4}{5}v_g;$$

$$\frac{d^2v_2}{dt^2} + \frac{11}{5}\frac{dv_2}{dt} + \frac{4}{5}v_2 = \frac{4}{5}\frac{dv_2}{dt} + \frac{4}{5}v_g$$

7.1.3. Find v_1 and v_2 in Fig. 7.3 for $t > 0$ if v_g is a +6-V ideal dc source, $v_1(0-) = 1$ V, and $v_2(0-) = 4$ V.

ANS: $6 - 5e^{-t}$ V; $6 - 2e^{-2t}$ V

7.2 SECOND-ORDER EQUATIONS

In Chapter 6 we considered first-order circuits and saw that their describing equations were first-order differential equations of the general form

$$\frac{dx}{dt} + ax = f(t) \tag{7.5}$$

In Section 7.1 we defined second-order circuits as those with describing equations that were second-order differential equations, given generally by

$$\frac{d^2x}{dt^2} + a_1\frac{dx}{dt} + a_0x = f(t) \tag{7.6}$$

In (7.5) and (7.6) the a's are real constants. x may be either a voltage or a current, and $f(t)$, the forcing function, is a known time function determined by the independent

sources. As an example, for the circuit of Fig. 7.1, the describing equation is (7.4). Comparing this equation with (7.6), we see that $a_1 = 10$, $a_0 = 16$, $f(t) = 2v_g$, and $x = i_2$.

From Chapter 6 we know that the total response satisfying (7.5) is

$$x(t) = x_n(t) + x_f(t)$$

where $x_n(t)$ is the natural response, obtained by setting $f(t) = 0$, and $x_f(t)$ is the forced response, which satisfies the forced differential equation (7.5).

Let us see if this same procedure can be applied to the second-order equation (7.6). By a solution to (7.6), we shall mean any function x that satisfies (7.6) identically. That is, when x is substituted into (7.6), the left member becomes identically equal to $f(t)$ for all t within a specified solution region, usually $t > t_0$ for some initial time t_0.

If x_n is the natural response, it must satisfy the unforced (or *homogeneous*) differential equation

$$\frac{d^2x_n}{dt^2} + a_1\frac{dx_n}{dt} + a_0x_n = 0$$

And if x_f is the forced solution, it satisfies the original forced equation

$$\frac{d^2x_f}{dt_2} + a_1\frac{dx_f}{dt} + a_0x_f = f(t)$$

Adding (7.7) and (7.8) and rearranging the terms, we may write

$$\frac{d^2}{dt^2}(x_n + x_f) + a_1\frac{d}{dt}(x_n + x_f) + a_0(x_n + x_f) = f(t)$$

Comparing (7.6) and (7.9), we see that the sum of x_n and x_f is indeed a solution, as it was in the first-order case. That is, x satisfying (7.6) is made up of two components, a natural response x_n satisfying the homogeneous equation (7.7) and a forced response x_f satisfying the original equation (7.8) or (7.6). As we shall see, the natural response will contain two arbitrary constants and, as in the first-order case, we will choose a trial form forced solution so that it will end up having no arbitrary constants. We consider methods of finding the natural and forced responses in the next two sections.

If the independent sources are such that $f(t) = 0$ in (7.6), the forced response is zero and the solution of the differential equation is simply the natural response. A reader who has had a course in differential equations will note that the natural response and the forced response may also be called, respectively, the *complementary solution* and the *particular solution*. The natural or complementary solution contains the arbitrary constants needed to match initial conditions, as will be discussed in Section 7.5, and the particular solution, as its name implies, contains no free constants.

7.2.1. Show that $x_1 = K_1e^{-2t}$ and $x_2 = K_2e^{-3t}$ are each solutions of

$$\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 0$$

regardless of the values of the constants K_1 and K_2.

7.2.2. Show that $x_1 + x_2 = K_1e^{-2t} + K_2e^{-3t}$ also satisifies the equation of Exercise 7.2.1, regardless of the values of K_1 and K_2.

7.2.3. Does the result of (7.7) to (7.9) that $x_n + x_f$ is a solution as long as x_n is a natural response and x_f a forced response hold for third-order circuits? For higher order?

Yes; yes

7.2.4. For the *nonlinear* second-order differential equation

$$\frac{d^2x}{dt^2} = [1 + f(t)]x$$

show that $x_n = K_1e^{+t} + K_2e^{-t}$ is the natural response and $x_f = x_n = e^{+2t} - e^{-2t}$ is a forced response to the forcing term $f(t) = 3$; then see if $x_n + x_f$ satisfies the equation. Does the result of (7.7) to (7.9) that $x_n + x_f$ is a solution as long as x_n is a natural response and x_f a forced response hold for *nonlinear* circuits?

No; no

7.3 NATURAL RESPONSE

The natural response x_n, a comonent of the total solution $x = x_n + x_f$, must satisfy the unforced equation, which we repeat from (7.7):

$$\frac{d^2x}{dt^2} + a_1\frac{dx}{dt} + a_0x = 0 \qquad (7.10)$$

The trial form for the natural solution

$$x_n(t) = Ke^{st} \qquad (7.11)$$

worked well previously for the first-order case. Let us see if it will help with the second-order case. Any natural solution must satisfy (7.10) so substituting x_n from (7.11) gives

$$Ks^2e^{st} + Ksa_1e^{st} + Ka_0e^{st} = 0$$

or

$$Ke^{st}(s^2 + a_1s + a_0) = 0 \qquad (7.12)$$

If the trial form works, this equation must be true for each t or else (7.10) would fail to hold. Equation (7.12) will hold if either $K = 0$ or the factor in parentheses is zero (e^{st} cannot be zero for any time t). If $K = 0$ we have the null solution, $x_n(t) = 0$ for each t. This corresponds to the special case of no stored energy. More generally, we will not have $x_n = 0$ at the initial time, so we cannot have $K = 0$. This leaves only the possibility that the other factor in (7.12) is zero, or

$$s^2 + a_1s + a_0s = 0 \qquad (7.13)$$

This is the characteristic equation introduced in Chapter 6 for first-order circuits. Note that it can easily be derived from (7.10) simply by replacing derivatives by corresponding powers of s. That is, the second derivative of x is replaced by s^2, the first derivative by s, and the zeroth derivative of x, x itself, is replaced by $s^0 = 1$.

Since (7.13) is a quadratic equation, we have not one solution, as in the first-order case, but in general two solutions, say s_1 and s_2, given by the quadratic formula as

$$s_{1,2} = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0}}{2}$$

Unlike the first-order case, we have two natural solutions of the form Ke^{st}, which we denote by

$$x_{n1} = K_1e^{s_1t}$$

$$x_{n2} = K_2e^{s_2t}$$

Either of these two solutions (7.15) will satisfy the unforced equation for any value of K_1 or K_2, because substituting either one into (7.10) reduces it to (7.12). And we know that since s_1 and s_2 each satisfy the characteristic equation (7.13), (7.12) will indeed be true with either $s = s_1$ or $s = s_2$.

As a matter of fact, because (7.10) is a linear equation, *any linear combination* of the two distinct natural solutions (7.14) is also a solution. That is,

$$x_n = x_{n1} + x_{n2} = K_1e^{s_1t} + K_2e^{s_2t}$$

is a solution of (7.10) for any pair of values K_1 and K_2 as long as s_1 and s_2 satisfy the characteristic equation. To verify this, we need only substitute this expression for x into (7.10). This results in

$$\frac{d^2}{dt^2}(x_{n1} + x_{n2}) + a_1\frac{d}{dt}(x_{n1} + x_{n2}) + a_0(x_{n1} + x_{n2})$$

$$= \left(\frac{d^2x_{n1}}{dt^2} + a_1\frac{dx_{n1}}{dt} + a_0x_{n1}\right) + \left(\frac{d^2x_{n2}}{dt^2} + a_1\frac{dx_{n2}}{dt} + a_0x_{n2}\right)$$

$$= 0 + 0 = 0$$

Since (7.16) includes the individual solutions (7.15a) and (7.15b) as special cases, it is called the *general solution of the homogeneous equation* if s_1 and s_2 are distinct (i.e., not equal) roots of the characteristic equation (7.12). Note that if we put $K_2 = 0$ in (7.16) we have x_{n1}, and $K_1 = 0$ results in x_{n2}.

Example 7.2

As an example, the homogeneous equation corresponding to (7.4) in Example 7.1 is repeated here,

$$\frac{d^2i_2}{dt^2} + 10\frac{di_2}{dt} + 16i_2 = 0$$

and the characteristic equation is

$$s^2 + 10s + 16 = 0$$

The roots are $s = -2$ and $s = -8$, so the general solution of the homogeneous equation is given by

$$i_2 = K_1e^{s_1t} + K_2e^{s_2t} = K_1e^{-2t} + K_2e^{-8t}$$

The reader may verify by direct substitution that (7.18) satisfies (7.17) regardless of the value of the unspecified constants K_1 and K_2.

The numbers s_1 and s_2 are called the *characteristic exponents* of the circuit. In Example 7.2 they are the negative reciprocals of the time constants to be considered in Chapter 8. There are two time constants in (7.18), compared to one in the first-order case.

For second-order systems the characteristic equation is a quadratic equation. Unlike first-order systems, whose characteristic equation always has one real solution, its solutions s_1 and s_2 may be real or complex numbers. The nature of the roots is determined by the discriminant in the characteristic equation, which may be positive (corresponding to real distinct roots), negative (complex roots), or zero (real repeated roots). To simplify discussion of these cases, we recast the characteristic equation (7.13) as

$$s^2 + 2\zeta\omega_0 s + \omega_0^2 = 0$$

ω_0 is called the *undamped natural frequency* and ζ the *damping ratio*. The justification of these names will soon be apparent. Comparing (7.13) to (7.19), $\omega_0^2 = a_0$ and $2\zeta\omega_0 = a_1$. Substituting these into (7.14), s_1 and s_2 may be expressed in terms of the undamped natural frequency ω_0 and damping ratio ζ.

A circuit with characteristic equation $s^2 + a_1 s + a_0 = 0$ has:

Undamped natural frequency: $\omega_0 = +\sqrt{a_0}$

Damping ratio: $\zeta = \dfrac{a_1}{2\sqrt{a_0}}$

Characteristic exponents: $s_{1,2} = \left[-\zeta \pm \sqrt{(\zeta^2 - 1)}\right]\omega_0$

For instance, a circuit with characteristic equation $s^2 + 2s + 16$ has undamped natural frequency $\omega_0 = +\sqrt{16} = 4$ rad/s and damping ratio $\zeta = 2/(2)(4) = \frac{1}{4}$. The characteristic exponents for this circuit are

$$s_{1,2} = \left(-\tfrac{1}{4} \pm \sqrt{-\tfrac{15}{16}}\right)(4) = -1 \pm j\sqrt{15}$$

Here $j = \sqrt{-1}$ is the notation we use for the complex number of unit length directed along the imaginary axis in the complex plane (see Appendix B for a review of complex numbers).

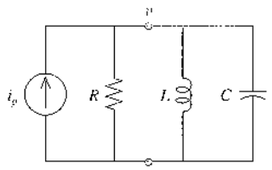
Example 7.3

The circuit of Fig. 7.4 is called a *parallel RLC circuit*. The nodal equation for this circuit is

$$\frac{v}{R} + \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + i_L(t_0) + C\frac{dv}{dt} = i_g$$

Differentiating and dividing both sides by C, the describing equation for v is

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}v = \frac{1}{C}\frac{di_g}{dt}$$



Parallel *RLC* circuit.

The characteristic equation is then

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$

By (7.20), the undamped natural frequency of this circuit is

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

and the damping ratio of this parallel *RLC* circuit is

$$\zeta_p = \frac{1}{2R}\sqrt{\frac{L}{C}}$$

If the damping ratio $\zeta < 1$, the discriminant $\zeta^2 - 1 < 0$, and by (7.20), the characteristic exponents are complex. If $\zeta > 1$, by (7.20) there will be two real characteristic exponents. Because it forms the boundary between these distinct behaviors, $\zeta = 1$ is called *critical damping*.

Setting $\zeta = 1$ and solving for R, we may define the *critical parallel resistance* R_{cp}.

$$R_{cp} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

For $R > R_{cp}$ the damping ratio $\zeta < 1$, which is described below as the underdamped case, and for $R < R_{cp}$ we have $\zeta > 1$, the overdamped case.

The three possible cases, that the characteristic exponents are real and distinct ($\zeta > 1$), real and repeated ($\zeta = 1$), or complex conjugates ($\zeta < 1$), lead to sharply different behavior of the natural response in a second-order circuit. We now consider these three cases, denoting the real part of the general characteristic exponent s as σ and the imaginary part ω so that

$$s = \sigma + j\omega$$

Overdamped Case ($\zeta > 1$)

With $\zeta > 1$ the characteristic exponents s_1 and s_2 in (7.20) are real and distinct, since the discriminant $\zeta^2 - 1 > 0$. In this case each s has only a real part,

$$s_1 = \sigma_1, \qquad s_2 = \sigma_2$$

and the natural response is given by a sum of two real exponentials,

$$x_n(t) = K_1 e^{\sigma_1 t} + K_2 e^{\sigma_2 t}$$

Examples include the circuit of Fig. 7.1 analyzed in Example 7.1 and the parallel *RLC* circuit discussed in Example 7.3 when R is less than the critical parallel resistance R_{cp}.

The two real characteristic exponents σ_1 and σ_2 may be positive, negative, or zero. For circuits consisting solely of *passive* elements driven by *independent* sources, however, it is not possible that either characteristic exponent be positive, for a positive σ_1 or σ_2 implies that $x_n(t)$ must increase in magnitude, eventually exceeding all bounds as t gets larger. This cannot happen, for in a passive circuit with independent sources killed (we are interested in the natural response), the total energy stored in the circuit can never increase beyond the initial stored energy.

Damping is the gradual loss of initial stored energy, that is, decay of the natural response toward zero. Passive circuits with distinct real roots have natural responses with two terms each exponentially decaying to zero. Thus $\zeta > 1$ is referred to as the *overdamped* case. In the overdamped case there is sufficient damping so that each term in the natural response steadily loses amplitude. In more lightly damped cases, the damping may be accompanied by oscillations, as we see next.

If $\zeta < 1$, by (7.20) the characteristic exponents are complex:

$$s_1 = \sigma + j\omega, \qquad s_2 = \sigma - j\omega, \qquad \omega \neq 0$$

They must be complex *conjugates* (real parts equal, imaginary parts equal but opposite in sign). This follows immediately from application of the quadratic formula to the characteristic equation (7.14). Then

$$x_n(t) = e^{\sigma t}(K_1 e^{+j\omega t} + K_2 e^{-j\omega t})$$

We need to use some properties of complex quantities to clarify this form and recognize its structure. For a real circuit, one with purely real element laws and sources, x_n will be real as well. This requires K_1 and K_2 to be complex conjugates, since otherwise the imaginary part of x_n in (7.22) would not be zero. Since the sum of a complex number and its conjugate is twice its real part,

$$x_n = \left(K_1 e^{-j\omega t} + K_1^* e^{-j\omega t}\right) e^{\sigma t} = 2\,\mathrm{Re}\left(K_1 e^{+j\omega t}\right) e^{\sigma t}$$

where $K_2 = K_1^*$ is the complex conjugate of K_1. The real part of a product of complex numbers is the product of their real parts minus the product of imaginary parts. Using the Euler identity,

$$e^{j\omega t} = \cos\omega t + j\sin\omega t$$

we may expand the real part of the product in (7.23) as

$$\mathrm{Re}\left(K_1 e^{+j\omega t}\right) = \tfrac{1}{2}B_1\cos\omega t - \tfrac{1}{2}B_2\sin\omega t$$

where for convenience we have defined

$$\mathrm{Re}(K_1) = \tfrac{1}{2}B_1, \qquad \mathrm{Im}(K_1) = \tfrac{1}{2}B_2$$

or

$$B_1 = 2\,\mathrm{Re}(K_1), \qquad B_2 = 2\,\mathrm{Im}(K_1)$$

Equation (7.22) may then be rewritten in terms of real quantities as

$$x_n(t) = B_1 e^{\sigma t}\cos\omega t + B_2 e^{\sigma t}\sin\omega t$$

While the overdamped case consists of two steadily decaying terms, in the present case (7.25) the decay is modulated by (multiplied by) sinusoidal oscillations of frequency ω rad/s. Since the damping is oscillatory rather than steady, this is called the *underdamped* case, which occurs when the damping ratio is less than its critical value of unity.

Note that the amplitude of these oscillations decays as $e^{\sigma t}$, so the damping rate varies inversely with σ. The extreme case is $\sigma = 0$, leading to no decay at all. This is *undamped* case, which occurs when the damping ratio $\zeta = 0$. The undamped natural response (7.25) is a constant-amplitude sinusoid with frequency, by (7.20), $\omega = \omega_0$. This explains why ω_0 is called the undamped natural frequency.

The last type we may have consists of characteristic exponents that are real and equal, a case that arises when the damping ratio ζ is unity in (7.20). This is the dividing line between the overdamped and underdamped cases and is referred to as the *critically damped* case

$$s_1 = s_2 = \sigma$$

In this case the general form for the natural solution given in (7.16) does not contain two free constants, since both terms may be collapsed into one. If we are to be able to meet initial conditions imposed by two storage elements, we must further generalize the trial form for the natural solution. Consider the trial form

$$x_n = h(t)e^{st}$$

Thus far we have used $h(t) = 1$. To see if another $h(t)$ may also work, we substitute (7.25) into the unforced differential equation (7.10). Using the product rule to compute derivatives, after gathering like terms, we obtain

$$e^{st}\left[h(s^2 + a_1 s + a_0) + \frac{dh}{dt}(2s + a_1) + \frac{d^2h}{dt^2}\right] = 0$$

Evaluating this equation for the present repeated root case $s = \sigma$, the first term in brackets is zero since σ satisfies the characteristic equation, and the second term is zero since

$$s + a_1 s + a_0 = (s - \sigma)^2 = s^2 - 2\sigma s + \sigma^2$$

shows that $a_1 = -2\sigma$, so $2s + a_1 = 2\sigma - 2\sigma = 0$. Thus (7.26) is satisfied if and only if

$$\frac{d^2h}{dt^2} = 0$$

This equation in $h(t)$ has solutions

$$h(t) = K_1 + K_2 t$$

Our amended trial form for this case is, from (7.26) and the above,

$$x_n(t) = K_1 e^{\sigma t} + K_2 t e^{\sigma t}$$

While the first term is the familiar exponential trial form, the second differs by an additional factor of t and is referred to as a *t-multiplied form.* The natural solution in the critically damped case consists of a sum of an exponential term and its associated t-multiplied form.

Example 7.4

Consider the series *RLC* circuit of Fig. 7.2. The describing equation was found to be

$$\frac{d^2 i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{1}{LC} i = \frac{dv_g}{dt}$$

The characteristic equation is

$$s^2 + \frac{R}{L} s + \frac{1}{LC} = 0$$

By (7.20), the undamped natural frequency of this circuit is

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

and the damping ratio for the series *RLC* case is

$$\zeta_s = \frac{R}{2\sqrt{C/L}}$$

The *critical series resistance* R_{cs} is found by solving for R with $\zeta = 1$:

$$R_{cs} = 2\sqrt{\frac{L}{C}}$$

As R increases the damping ratio ζ_s increases, and for values of R greater than R_{cs} the circuit is overdamped (two real characteristic exponents). The opposite relation holds for the parallel *RLC* circuit discussed in Example 7.3, in which the damping ratio ζ_p decreases with increased parallel resistance, and R greater than R_{cp} leads to underdamped behavior.

To illustrate the series *RLC* case, set $L = 1$ H and $C = \frac{1}{4}$ F in Fig. 7.2 and consider how the natural solution depends on R. The unforced equation in this case is

$$\frac{d^2 i}{dt^2} + R\frac{di}{dt} + 4i = 0$$

with characteristic equation

$$s^2 + Rs + 4 = 0$$

By (7.20), $\omega_0 = 2$ rad/s and $\zeta = R/4$. The critically damped case is $R = 4\ \Omega$. In this case $s = \sigma = -2$ is the repeated characteristic exponent, and

$$i_n = K_1 e^{-2t} + K_2 t e^{-2t} \qquad \text{(critically damped)}$$

The overdamped case requires that $\zeta = R/4 > 1$. Selecting $R = 5\ \Omega$ for illustration, the roots are $s_1 = -4$ and $s_2 = -1$, and

$$i_n = K_1 e^{-t} + K_2 e^{-4t} \qquad \text{(overdamped)}$$

The underdamped case $\zeta < 1$ requires that $R < 4$. For instance, with $R = 2$, we have $s_1 = \sigma + j\omega = -1 + j\sqrt{3}$ and $s_2 = \sigma - j\omega = -1 - j\sqrt{3}$. The natural response in this case is

$$i_n = K_1 e^{-t} \cos\sqrt{3}t + K_2 e^{-t} \sin\sqrt{3}t \qquad \text{(underdamped)}$$

For the undamped case $\zeta = 0$ we must have no series resistance $R = 0$. Then $s_1 = \sigma + j\omega = +j2$ and $s_2 = \sigma - j\omega = -j2$, and by (7.25),

$$i_n = B_1 \cos 2t + B_2 \sin 2t \qquad \text{(undamped)}$$

Graphs for these four distinct responses are shown in Fig. 7.5.

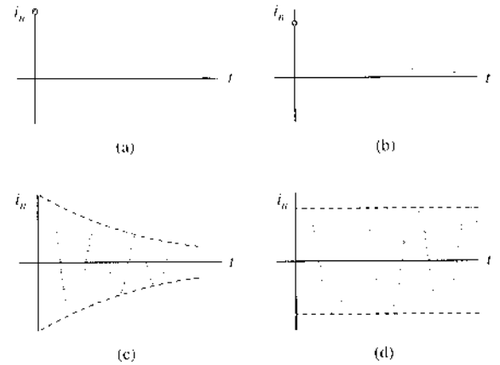


Fig. 7.5 Natural responses for second-order circuit: (a) overdamped; (b) critically damped; (c) underdamped; (d) undamped.

Example 7.5

In the parallel *RLC* circuit of Fig. 7.4 and Example 7.3, let $R = \frac{5}{3}\ \Omega$, $L = 2$ H, and $C = \frac{1}{10}$ F with no source, $i_g = 0$. If $v(0-) = 4$ V and there is no current through the inductor at time $t = 0-$, find $v(t)$ for $t > 0$.

With no independent source, there will only be a natural response. The describing equation, from Example 7.3, is

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}v = \frac{1}{C}\frac{di_g}{dt}$$

or

$$\frac{d^2v}{dt^2} + 6\frac{dv}{dt} + 5v = 0$$

The characteristic equation is

$$s^2 + 6s + 5 = 0$$

or $s_1 = -1$ and $s_2 = -5$. This is an overdamped natural response,

$$v = K_1e^{-t} + K_2e^{-5t}$$

The capacitive voltage is continuous, so $v(0+) = v(0-) = 4$ V. By KCL,

$$\frac{3}{5}v + i_L + \frac{1}{10}\frac{dv}{dt} = 0$$

where i_L is defined with its reference arrow pointing downward in Fig. 7.4. Evaluating this equation at $t = 0+$ yields

$$\frac{3}{5}(4) + 0 + \frac{1}{10}\left.\frac{dv}{dt}\right|_{0+} = 0$$

where we have used continuity of inductive current to evaluate $i_L(0+)$. Then at $t = 0+$, $v = 4$ V and $dv/dt = -24$ V/s. Applying these conditions to (7.34), we have

$$v(0+) = 4 = K_1 + K_2$$

$$\left.\frac{dv}{dt}\right|_{0+} = -24 = -K_1 - 5K_2$$

Adding gives

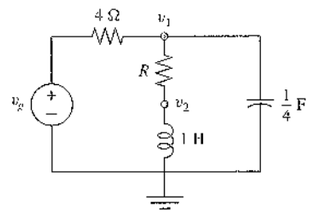
$$-20 = -4K_2$$

or $K_2 = 5$. Then back-substituting, $K_1 = -1$ and the solution is

$$v = 5e^{-5t} - e^{-t}\ \text{V}$$

7.3.1. What unforced second-order differential equation yields characteristic exponents $s_1 = -10$ and $s_2 = -3$? What are ω_0 and ζ? Repeat for $s_1 = -2 + j5$ and $s_2 = -2 - j5$.

Ans. $d^2x/dt^2 + 13(dx/dt) + 30x = 0$; $\omega_0 = 5.48$ rad/s, $\zeta = 1.19$; $d^2x/dt^2 + 4(dx/dt) + 29x = 0$; $\omega_0 = 5.39$ rad/s; $\zeta = 0.37$



7.3.2. In the series *RLC* circuit of Fig. 7.2, let $v_g = 0$, $R = \frac{20}{9}\ \Omega$, $L = \frac{1}{9}$ H, and $C = \frac{1}{4}$ F. Find the natural solution i_n.

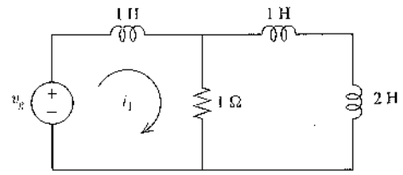
Ans. $i_n = K_1e^{-2t} + K_2e^{-18t}$

7.3.3. Write nodal equations for the circuit shown. Solve the v_1 nodal equation for v_2, and use this to eliminate v_2 from the differentiated v_2 nodal equation, yielding the describing equation for v_1. For what value of $R > 0$ is the circuit critically damped?

Ans. $d^2v_1/dt^2 + (R+1)(dv_1/dt) + (R+4)v_1 = dv_g/dt - Rv_g$; $R = 5\ \Omega$

7.3.4. Find the natural response i_{1n} for the circuit shown.

Ans. $i_{1n} = K_1e^{-(2+\sqrt{2})t} + K_2e^{-(2-\sqrt{2})t}$



7.4 FORCED RESPONSE

The forced response x_f of the general second-order circuit must satisfy the original, or forced, differential equation (7.8), repeated here:

$$\frac{d^2x_f}{dt^2} + a_1\frac{dx_f}{dt} + a_0x_f = f(t)$$

There are a number of methods for finding x_f, but for our purposes we use the procedure of guessing a trial form solution that has worked well for us thus far. Examining (7.35), we note that for this equation to be true, a linear combination of x_f and its derivatives must equal $f(t)$. Consider a trial solution in which x_f is guessed to be a linear combination of $f(t)$ and its derivatives. Since a linear combination of linear combinations is still a linear combination, this trial form might well work. In other words, the differentiations of the linear combination of $f(t)$ and its derivatives required on the left side of (7.35) may result in the cancellations required to leave only $f(t)$ itself, which is what is needed to satisfy the equation. The adequacy of any trial form will, in the final analysis, be judged by its ability to satisfy the differential equation.

Example 7.6

As an example, let us consider the dc source $v_g = 16$ V in the circuit of Fig. 7.1. Then, by (7.4),

$$\frac{d^2i_2}{dt^2} + 10\frac{di_2}{dt} + 16i_2 = 32$$

The characteristic equation is

$$s^2 + 10s + 16 = (s + 2)(s + 8) = 0$$

with natural response

$$i_{2n} = K_1 e^{-2t} + K_2 e^{-8t}$$

Since the forcing term $f(t)$ is a constant $f(t) = 32$, its first, second, and in fact all its derivatives, are zero. Then the most general linear combination of $f(t)$ and all its derivatives is simply a constant:

$$i_{2f} = A$$

Substituting this trial forced solution into (7.36) yields

$$0 + 0 + 16A = 32$$

The trial form works if we set $A = 2$. Then the total solution is

$$i_2 = i_{2n} + i_{2f} = K_1 e^{-2t} + K_2 e^{-8t} + 2$$

Knowledge of the initial energy stored in the inductors (or the initial inductive current) can be used to evaluate K_1 and K_2, as discussed in Section 7.5.

In the case of constant forcing functions, as in this example, we may obtain the forced solution directly from the circuit diagram. In dc steady state all currents and voltages will be constant, including the unknown in the describing equation. Thus the constant forced solution must be identical to the dc steady-state value. Recall that this may be found by replacing inductors by short circuits and capacitors by open circuits. This may easily be verified for the circuit of Fig. 7.1 used in Example 7.6 by replacing the inductors by short circuits, so by KVL around the outer loop

$$i_{2f} = \frac{v_g}{8} = \frac{16}{8} = 2 \text{ A}$$

Example 7.7

As another example, suppose that in Fig. 7.1 we have $v_g = 20\cos 4t$ V. Then, by (7.4),

$$\frac{d^2 i_2}{dt^2} + 10\frac{di_2}{dt} + 16 i_2 = 40\cos 4t$$

The natural response is given by (7.37), as in Example 7.6. To find the forced response, we need the general linear combination of $f(t)$ and all its derivatives. In the case under consideration, $f(t) = 40\cos 4t$ and all its derivatives are of the form $C\cos 4t$ or $D\sin 4t$. Summing all these terms together, we arrive at a trial forced solution of the form

$$i_{2f} = A\cos 4t + B\sin 4t$$

That is, every possible linear combination of $40\cos 4t$ and all its derivatives must indeed be of this form.

We now determine whether this trial form works and, if so, what the values of the constants A and B must be. Substituting (7.39) into the forced differential equation (7.38), we have

$$40B\cos 4t - 40A\sin 4t = 40\cos 4t$$

Since this must be true for all t, the coefficients of like terms must be the same on both sides of the equation. In the case of the $\cos 4t$ terms we have

$$40B = 40$$

and for the $\sin 4t$ terms we have

$$-40A = 0$$

Thus $A = 0$ and $B = 1$, so that

$$i_{2f} = 0\cos 4t + 1\sin 4t = \sin 4t$$

Superposing the natural and forced solutions, the general solution of (7.38) is given by

$$i_2 = i_{2n} + i_{2f} = K_1 e^{-2t} + K_2 e^{-8t} + \sin 4t$$

This may readily be verified by direct substitution.

Some of the more common forcing functions $f(t)$ are listed in the first column of Table 7.1. The general form of the corresponding forced response is given in the second column. This is the trial forced solution. In each case the trial form is the general linear combination of $f(t)$ and its derivatives. If the forcing term is a sum of entries in this table, superposition may be used to find the component of the forced response due to each separately. The forced response is then a sum of these components.

Table 7.1 Trial Forced Solutions for Common Forcing Functions

Forcing Term $f(t)$	Trial Forced Solution $x_f(t)$
k	A
t	$At + B$
t^2	$At^2 + Bt + C$
t^n	$At^n + Bt^{n-1} + \cdots + Ft + G$
$e^{\sigma t}$	$Ae^{\sigma t}$
$e^{\sigma t}$	$Ae^{\sigma t}$
$\sin\omega t$, $\cos\omega t$	$A\sin\omega t + B\cos\omega t$
$e^{\sigma t}\sin\omega t$, $e^{\sigma t}\cos\omega t$	$e^{\sigma t}(A\sin\omega t + B\cos\omega t)$
$te^{\sigma t}\sin\omega t$, $te^{\sigma t}\cos\omega t$	$te^{\sigma t}(A\sin\omega t + B\cos\omega t) + e^{\sigma t}(C\sin\omega t + D\cos\omega t)$

Example 7.8

Consider a circuit whose describing equation is

$$\frac{d^2i}{dt^2} + 3\frac{di}{dt} + 2i = 4 + 60e^{-7t}$$

Let i_{f1} and i_{f2} be the two components of the forced response satisfying

$$\frac{d^2i_{f1}}{dt^2} + 3\frac{di_{f1}}{dt} + 2i_{f1} = 4$$

$$\frac{d^2i_{f2}}{dt^2} + 3\frac{di_{f2}}{dt} + 2i_{f2} = 60e^{-7t}$$

The sum $i_f = i_{f1} + i_{f2}$ clearly satisfies the forced differential equation, as can be seen by adding the last two equations. Using the trial forms from Table 7.1, for i_{f1} we have a constant $i_{f1} = A$, from which

$$2A = 4$$

or $i_{f1} = 2$. For i_{f2} we have the trial form $i_{f2} = Ae^{-7t}$, or

$$Ae^{-7t}[+49 - 3(7) + 2] = 60e^{-7t}$$

or $A = 2$. The forced solution is then

$$i_f = i_{f1} + i_{f2} = 2 + 2e^{-7t}$$

which can be verified by direct substitution into the forced differential equation.

In the circuits and describing equations studied so far, we have used trial forms for the natural and forced solutions that have been chosen completely independently of one another. The form for the natural solution depended only on the left side of the describing equation (through the coefficients of the characteristic equation), while the form for the forced solution depended only on the right side [$f(t)$ itself]. There is one condition under which the natural and forced trial solution forms cannot, however, be determined independently.

Consider the following equation, for which *the trial forced solution happens to match a term in the natural solution.*

$$\frac{d^2v}{dt^2} + 3\frac{dv}{dt} + 2v = 6e^{-t} \qquad (7.43)$$

The characteristic equation is

$$s^2 + 3s + 2 = (s + 1)(s + 2) = 0$$

and natural solution

$$v_n = K_1e^{-t} + K_2e^{-2t} \qquad (7.44)$$

From Table 7.1, for $f(t) = 6e^{-t}$ the trial forced solution [consisting of the general linear combination of $f(t)$ and all its derivatives] is

$$v_f = Ae^{-t} \qquad (7.45)$$

But by (7.41) we know that the function Ae^{-t} is in fact a special case of the *natural* solution ($K_1 = A$, $K_2 = 0$) and as such must satisfy the *unforced* version of (7.40), not the forced equation. The usual trial forced solution does not work in this circumstance. We must somehow modify (7.42) if we are to have a forced solution.

A similar problem arose with repeated roots of the characteristic equation, and the solution there proved to be to t-multiplication of the trial form. Guided by that experience, we will try the t-multiplied version of (7.42) as a new trial form,

$$v_f = Ate^{-t}$$

Computing the derivatives gives

$$\frac{dv_f}{dt} = Ae^{-t} - Ate^{-t}$$

$$\frac{d^2v_f}{dt^2} = -Ae^{-t} + Ate^{-t} - Ae^{-t}$$

Substituting these expressions for v_f and its derivatives into the forced equation (7.40), the t-multiplied forms cancel, leaving

$$Ae^{-t}(-2 + 3) = 6e^{-t}$$

The new trial form thus works if we set $A = 6$, or

$$v_f = 6te^{-t}$$

The existence of an A for which the new, t-multiplied trial forced solution works (satisfies the forced equation) is sufficient to justify our guess. Equation (7.43) is the forced solution, and the total solution is

$$v = v_n + v_f = K_1e^{-t} + K_2e^{-2t} + 6te^{-t}$$

Generalizing, *when the trial forced solution matches a term in the natural solution, it is necessary to t-multiply the trial forced solution.* If there is still a match, t-multiply again, and so on, until there is no match.

Finally, if the trial forced solution matches a *t-multiplied form* in the natural solution, it may be necessary to t-multiply the trial solution further, as demonstrated in Example 7.9 and in Exercise 7.7.3.

Example 7.9

$$\frac{d^2v}{dt^2} = f(t)$$

The characteristic equation is $s^2 = 0$, or $s_1 = s_2 = 0$. This is a repeated root case, and by (7.28) the natural solution is

$$v_n = K_1e^{st} + K_2te^{st} = K_1 + K_2t$$

First consider the case where the forcing term is $f(t) = 2$. The trial forced solution is $v_f = A$. This matches one of the terms (K_1) of the natural solution, so we t-multiply v_f, yielding $v_f = At$. This still matches a term in the natural solution (K_2t), so, recognizing that At would still satisfy the natural, not the forced, differential equation,

we once again t-multiply, arriving at

$$v_f = At^3$$

This does not match a term in v_n. Substituting into the forced differential equation gives

$$2A = 2$$

or $A = 1$. The total solution is

$$v = K_1 + K_2 t + t^2$$

Differentiating twice, it is clear that v satisfies the original forced differential equation for any values of the free constants K_1 and K_2.

Next try the forcing term $f(t) = t$. The trial forced solution from Table 7.1 is $At + B$. Since this matches terms in the natural solution, we t-multiply to get the new trial form, $v_f = At^2$ (only the highest-order term in the t-multiplied form need be retained since the lower ones will satisfy the unforced, not the forced, differential equation). This trial form no longer matches a term in the natural solution, but upon substitution into the differential equation,

$$\frac{d^2 v_f}{dt^2} = \frac{d^2}{dt^2}(At^2) = 2A$$

which does not satisfy the forced equation for any A. The problem is that the original trial solution matched a t-multiplied term of the natural solution. Thus one final t-multiplication is needed, yielding the new trial form $v_f = At^3$. Upon substitution once more,

$$\frac{d^2}{dt^2}(At^3) = 6At$$

which satisfies the forced differential equation in this case [$f(t) = t$] with $A = \frac{1}{6}$. The forced solution is then $v_f = \frac{1}{6}t^3$ and the total solution is

$$v = K_1 + K_2 t + \tfrac{1}{6}t^3$$

Once again we may easily check by differentiating twice, noting that the result does indeed satisfy the original differential equation for any K_1 and K_2. Determining these values so that the solution not only satisfies the describing equation, but also agrees with the initial stored energy in the storage elements, will be taken up in the next section.

7.4.1. Find the forced response if

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 3x = f(t)$$

where $f(t)$ is given by (a) 6, (b) $2e^{-3t} + 6e^{-4t}$, and (c) $4e^{-t} + 2e^{-3t}$.

Answer: (a) 2; (b) $2e^{-4t} - te^{-3t}$; (c) $t(2e^{-t} - e^{-3t})$

7.4.2. Find the forced response if

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = f(t)$$

where $f(t)$ is given by (a) $6e^{-2t}$ and (b) $6te^{-2t}$. [*Suggestion:* In (b), try $x_f = At^3e^{-2t}$.]

Answer: (a) $3t^2e^{-2t}$; (b) t^3e^{-2t}

7.4.3. Find the total response if

$$\frac{d^2x}{dt^2} + 9x = 18\sin 3t$$

and $x(0) = dx(0)/dt = 0$.

Answer: $\sin 3t - 3t\cos 3t$

7.4.4. Find the forced response for i_2 in the circuit of Fig. 7.1 if the source function is $v_g = 16t^2 + \frac{11}{2}$.

Answer: $2t^2 - \frac{5}{2}t + 2$

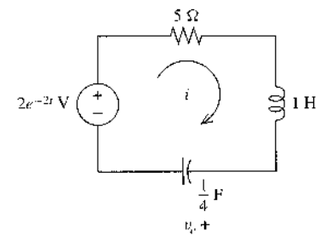
7.5 TOTAL RESPONSE

The total response of a circuit is the sum of its natural and forced response, and since the natural response contains undetermined constants, so must the total response. In the first-order case studied in Chapter 6, there was a single undetermined constant in the natural response, and its value was specified by the need to have the total solution agree with the initial energy in the storage element (i.e., initial current in the inductor or initial voltage across the capacitor).

The same principle applies to second-order circuits, with two storage elements each supplying an initial value needed to define the two undetermined constants contained in the total response. These two initial inductive currents and/or capacitive voltages may be used to determine the two initial conditions required to solve the second-order describing equation.

Consider, for instance, the series *RLC* circuit of Fig. 7.6. Suppose that we are given the initial voltage $v_c(0) = -6$ V and inductive current $i(0) = 1$ A. By KVL,

$$5i + 4\int_0^t i(\tau)\,d\tau + v_c(0) + \frac{di}{dt} = 2e^{-2t}$$



Forced series *RLC* circuit.

Differentiating, the describing equation for $i(t)$ is

$$\frac{d^2i}{dt^2} + 5\frac{di}{dt} + 4i = -4e^{-2t}$$

We will find the total solution by summing natural and forced solutions. The characteristic equation is

$$s^2 + 5s + 4 = (s+1)(s+4) = 0$$

with natural response for this overdamped (distinct real roots) series RLC circuit

$$i_n = K_1 e^{-t} + K_2 e^{-4t}$$

From Table 7.1, the trial forced solution is $i_f = Ae^{-2t}$, and substituting into (7.44) yields

$$(4 - 10 + 4)Ae^{-2t} = -4e^{-2t}$$

or $A = 2$. The total solution is then

$$i = i_n + i_f = K_1 e^{-t} + K_2 e^{-4t} + 2e^{-2t}$$

Evaluating the total solution (7.45) at $t = 0$ gives

$$i(0) = K_1 + K_2 + 2$$

Differentiating the total solution and evaluating at $t = 0$, the initial value of the derivative of the unknown must be

$$\left.\frac{di}{dt}\right|_0 = -K_1 - 4K_2 - 4$$

These two equations may be used to determine K_1 and K_2 if we can relate the initial conditions $i(0)$ and $di/dt|_0$ to the given initial circuit variables, the inductive current and capacitive voltage. The former is easy, since $i(0) = 1$ A is just the inductive current. To relate $di/dt|_0$ to the given values, we write KVL around the loop and evaluate at $t = 0$:

$$5i(0) + \left.\frac{di}{dt}\right|_0 + v_c(0) = 2$$

or

$$\left.\frac{di}{dt}\right|_0 = 2 + 6 - 5(1) = 3$$

Using these values, (7.46) may be rewritten as

$$K_1 + K_2 = -1$$

$$-K_1 - 4K_2 = 7$$

Adding (7.47a) and (7.47b) yields

$$-3K_2 = 6$$

or $K_2 = -2$. Substituting back into (7.47a), we have $K_1 = 1$. The total solution (7.45) is now fully specified as

$$i = e^{-t} - 2e^{-4t} + 2e^{-2t}\,\text{A}$$

The initial capacitive voltages and/or inductive currents to determine the required initial conditions may be given, as in the previous illustration, or it may be necessary to determine them from the circuit diagram. As introduced in Chapter 6, initial conditions are often set by the action of switches in the circuit or of sources that suddenly step on or off. We next present two examples in which the initial values are computed rather than given, the first in a circuit containing a switch and the second containing a step source.

Example 7.10

We wish to determine the capacitive voltages v_1 and v_2 for $t > 0$ in the circuit of Fig. 7.7, given that it is in dc steady state at $t = 0-$, just before the switch acts. We begin by determining the capacitive voltages at $t = 0-$. For $t < 0$ the switch is closed, and in dc steady state, by KVL around the outer loop,

$$v_1(0-) - v_2(0-) = 12(1) = 12\ \text{V}$$

But since the switch has shorted out node 2 for $t < 0$, $v_2(0-) = 0$ V, and hence $v_1(0-) = 12$ V. By continuity of capacitive voltages, $v_1(0+) = v_1(0-) = 12$ V and $v_2(0+) = v_2(0-) = 0$ V. These values permit us to compute the initial conditions at $t = 0+$ needed to solve the describing equation for $t > 0$.

At $t = 0$ the switch opens, and the two node equations for the circuit as it is configured in the desired solution interval $t > 0$ are

$$\frac{dv_1}{dt} + (v_1 - v_2) = 12$$

$$2\frac{dv_2}{dt} + 3v_2 - v_1 = 0$$

To combine these into a single describing equation, we solve (7.48a) for v_2,

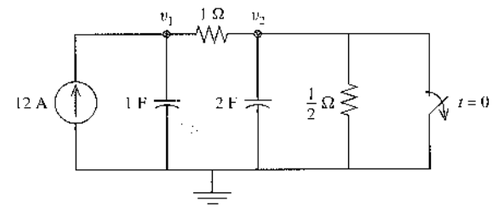
$$v_2 = \frac{dv_1}{dt} + v_1 - 12$$

and substitute this expression for v_2 into (7.48b), yielding

$$2\frac{d}{dt}\left(\frac{dv_1}{dt} + v_1 - 12\right) + 3\left(\frac{dv_1}{dt} + v_1 - 12\right) - v_1 = 0$$

or, after division by 2,

$$\frac{d^2v_1}{dt^2} + \frac{5}{2}\frac{dv_1}{dt} + v_1 = 18$$



Circuit for Example 7.10.

This is the describing equation for v_1. Once the circuit is broken by solving this for v_1, v_2 will follow immediately from (7.49).

The characteristic equation for (7.50) is

$$s^2 + \tfrac{5}{2}s + 1 = (s+2)\left(s+\tfrac{1}{2}\right) = 0$$

and the natural solution is

$$v_{1n} = K_1 e^{-2t} + K_2 e^{-t/2}$$

The trial forced solution is $v_{1f} = A$, which, after substitution into (7.50), gives $A = 18$. The total solution is then the sum

$$v_1 = K_1 e^{-2t} + K_2 e^{-t/2} + 18$$

Evaluating the unknown v_1 and its derivative at $t = 0+$,

$$v_1(0+) = K_1 + K_2 + 18$$

$$\left.\frac{dv_1}{dt}\right|_{0+} = -2K_1 - \frac{1}{2}K_2$$

To help determine these initial conditions, note that the element law for a capacitor may be written

$$\frac{dv}{dt} = \frac{i}{C}$$

Then since v_1 is the voltage across the 1-F capacitor,

$$\left.\frac{dv_1}{dt}\right|_{0+} = \frac{i_1(0+)}{1}$$

By KCL at node 1,

$$i_1(0+) + (v_1(0+) - v_2(0+)) = 12$$

or

$$i_1(0+) = 12 - 12 = 0$$

and by (7.53) $dv_1/dt|_{0+} = 0$. We are now ready to solve (7.52) for K_1 and K_2:

$$K_1 + K_2 = -6$$

$$-2K_1 - \frac{1}{2}K_2 = 0$$

Doubling the second equation and adding gives

$$-3K_1 = -6$$

or $K_1 = 2$, and from the first equation, $K_2 = -8$. The total solution is

$$v_1 = 2e^{-2t} - 8e^{-t/2} + 6 \text{ V}$$

for the unknown node voltage v_1. The other node voltage v_2 is given in (7.49) as

$$v_2 = \frac{d}{dt}(2e^{-2t} - 8e^{-t/2} + 6) + (2e^{-2t} - 8e^{-t/2} + 6) - 12$$
$$= -2e^{-2t} - 4e^{-t/2} - 6 \text{ V}$$

which completes the analysis.

A second-order describing equation will, in general, require initial conditions consisting of the values of the unknown and its first derivative. The steps required to determine the value of the first derivative at time $t = 0+$ are not always obvious. In Example 7.9 we used the element law for a capacitor in the form

$$\left.\frac{dv_c}{dt}\right|_{0+} = \left.\frac{1}{C}i_c\right|_{0+}$$

to determine the initial derivative of the unknown v_1. Similarly, for an inductor, the element law in the form

$$\left.\frac{di_L}{dt}\right|_{0+} = \left.\frac{1}{L}v_L\right|_{0+}$$

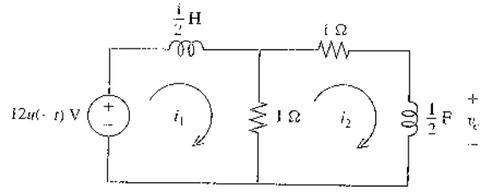
is often useful in determining the initial condition $di_L/dt|_{0+}$. To use these equations, it is necessary to know the circuit currents and voltages at $t = 0+$ so that their right-hand sides may be evaluated. In problems with switches or unit steps, we usually know only the capacitive voltages and inductive currents at time $t = 0+$, these by continuity from the moment $t = 0-$ (just before the switches act or the steps have their jump time). To determine the other circuit currents and voltages at time $t = 0+$, we may draw a circuit diagram for the time instant $t = 0+$ in which inductors are shown as current sources (their currents at $t = 0+$ are known) and capacitors as voltage sources. This $t = 0+$ circuit diagram will contain only resistors and sources, and it will be straightforward to determine all remaining currents and voltages at $t = 0+$, including those needed to evaluate (7.54). This procedure for determining initial conditions is illustrated in the next example.

Example 7.11

The circuit of Fig. 7.8 is in dc steady state at time $t = 0-$, and we require $i_1(t)$ for $t > 0$. Noting that the source is switched on for $t < 0$, $v_c(0-) = 12$ V and the inductive current is $i_1(0-) = 12$ A. For $t > 0$ the circuit is source-free, and the mesh equations are

$$\frac{1}{2}\frac{di_1}{dt} + (i_1 - i_2) = 0$$

$$i_2 + 2\int_{0+}^{t} i_2(\tau)\,d\tau + v_c(0+) + (i_2 - i_1) = 0$$



Circuit for Example 7.11.

Differentiating the second mesh equation, we obtain

$$2\frac{di_2}{dt} - \frac{di_1}{dt} + 2i_2 = 0$$

To get the describing equation for i_1, solve (7.55a) for i_2 and substitute into (7.56):

$$2\frac{d}{dt}\left(\frac{1}{2}\frac{di_1}{dt} + i_1\right) - \frac{di_1}{dt} + 2\left(\frac{1}{2}\frac{di_1}{dt} + i_1\right) = 0$$

or

$$\frac{d^2i_1}{dt^2} + 2\frac{di_1}{dt} + 2i_1 = 0$$

The characteristic equation is

$$s^2 + 2s + 2 = 0$$

with complex conjugate roots $s_1 = -1 + j$ and $s_2 = -1 - j$. This is an underdamped system, and by (7.25) the natural solution is

$$i_{1n} = B_1e^{-t}\cos t + B_2e^{-t}\sin t$$

which is also the total solution since the forcing term in the describing equation is zero. To evaluate the constants B_1 and B_2, we need the initial conditions $i_1(0+)$ and $di_1/dt|_{0+}$. While $i_1(0+) = 12$ A is known, we will compute the initial derivative with the aid of the circuit diagram for $t = 0+$ shown in Fig. 7.9. Note that the inductor is shown as a current source of value $i_1(0-) = i_1(0+) = 12$ A, and the capacitor is shown as a voltage source of value $v_c(0-) = v_c(0+) = 12$ V. This is a resistive circuit, and all currents and voltages can be found easily. In particular, to use (7.54b) we need the voltage $v_L(0+)$. The right mesh equation in Fig. 7.9 is

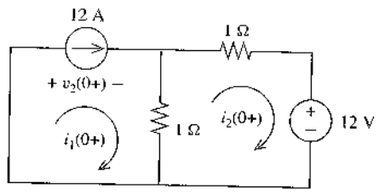
$$i_2(0+) + 12 + (i_2(0+) - 12) = 0$$

or $i_2(0+) = 0$. Then by KVL around the outer loop,

$$v_L(0+) + 1(0) + 12 = 0$$

or $v_L(0+) = -12$ V. The required initial condition is found from the above and (7.54b) to be

$$\left.\frac{di_1}{dt}\right|_{0+} = 2v_L(0+) = -24$$



Circuit for $t = 0+$.

With the initial value of i_1 and its derivative now known, we can determine the constants in the total solution. Evaluating (7.57) at the initial time yields

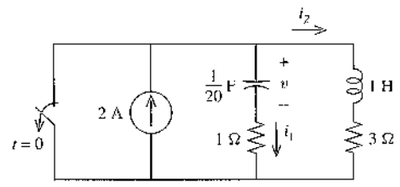
$$i_1(0+) = B_1 = 12 \text{ A}$$

Differentiating (7.57) and evaluating at the initial time, we have

$$\left.\frac{di_1}{dt}\right|_{0+} = -B_1 + B_2 = -24$$

Using $B_1 = 12$, this implies that $B_2 = -12$, and the solution (7.57) is

$$i_1 = 12e^{-t}(\cos t - \sin t) \text{ A}$$



7.5.1. Find $x(t)$ for $t > 0$, where

$$\frac{dx}{dt} + 2x + \int_0^t x\,dt = f(t)$$

$$x(0) = -1$$

and (a) $f(t) = 1$ and (b) $f(t) = t^2$.

(a) $(2t - 1)e^{-t}$; (b) $3(1 + t)e^{-t} + 2t - 4$

7.5.2. Find i_1, i_2, and v for $t = 0-$ and $t = 0+$. Do any of these values change discontinuously at $t = 0$?

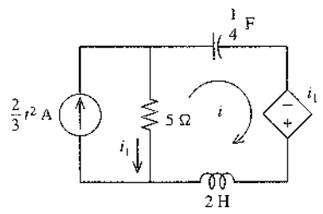
At $t = 0-$, $i_1 = i_2 = v = 0$. At $t = 0+$, $i_1 = 2$ A, $i_2 = 0$, $v = 0$. i_1 changed discontinuously (it is neither a capacitive voltage nor inductive current and thus may do so).

7.5.3. Find a_1, a_0, and $f(t)$ so that the total response $x(t)$, $t > 0$ for

$$\frac{d^2x}{dt^2} + a_1\frac{dx}{dt} + a_0x = f(t)$$

is $x = e^{-t} - e^{-2t} - 3\cos t + \sin t$.

3; 2; $10\sin t$



7.5.4. Find the total response i if the initial conditions are $i(0+) = 1$ and $di/dt|_{0+} = 0$. The controlled source has transresistance $r = 1$ Ω.

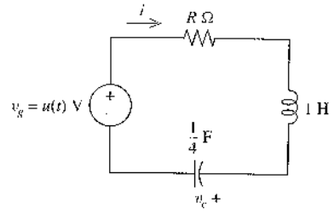
$6e^{-t} - 2e^{-2t} + 2t - 3$

7.6 UNIT STEP RESPONSE

The unit step response of second-order circuits exhibits much more variety than that of first-order circuits, shown in Chapter 6 to be characterized by a single exponential transient. The striking difference among the forms of the natural responses of overdamped, critically damped, and underdamped second-order circuits suggests that their

step responses will show equal variety, since the natural response is a component of the step response as well.

Example 7.12



Circuit for Example 7.12.

Consider the series *RLC* circuit of Fig. 7.10 driven by a source $v_g = u(t)$. We will determine the response $v_c(t)$ to the unit step source. For $t < 0$ there is no source to stimulate a response, and $i(0-) = v_c(0-) = 0$. For $t > 0$

$$1\frac{di}{dt} + Ri + 4\int_{0+}^{t} i(\tau)\,d\tau + v_c(0+) = 1$$

or, after differentiating,

$$\frac{d^2i}{dt^2} + R\frac{di}{dt} + 4i = 0$$

This is an unforced second-order describing equation for the series current i, which was taken up in Example 7.4 for several different values of R. The critical resistance for this series *RLC* circuit is, by (7.29), $R_{cr} = 2\sqrt{4/1} = 4\ \Omega$. For this value of R, by (7.30),

$$i = K_1 e^{-2t} + K_2 t e^{-2t} \qquad \text{(critically damped)}$$

The initial conditions $i(0+)$ and $di/dt|_{0+}$ are needed to determine the constants K_1 and K_2. By continuity of inductive current,

$$i(0+) = 0$$

Evaluating (7.58) at $t = 0+$, the integral term is zero since its upper and lower limits are equal, $v_c(0+) = v_c(0-) = 0$ by continuity of capacitive voltage, and

$$\left.\frac{di}{dt}\right|_{0+} = 1 - Ri(0+) = 1$$

Evaluation of (7.59) at $t = 0+$ with $i(0+) = 0$ yields $K_1 = 0$, and upon differentiation of (7.59), we have

$$\left.\frac{di}{dt}\right|_{0+} = 1 = K_2$$

so the mesh current in this critically damped case is

$$i = te^{-2t}\ \text{A}$$

Note that the mesh current is purely transient; there is no steady-state component. This may be predicted directly from the circuit diagram Fig. 7.10, recalling that the dc steady-state response and the step response coincide as t goes to infinity. In dc steady state the capacitor is an open circuit; thus no current can flow, and the steady-state mesh current in response to a unit step must be zero.

Here we are principally interested in the response v_c. By KVL,

$$Ri + \frac{di}{dt} + v_c = u(t)$$

or, for $t > 0$,

$$v_c = 1 - Ri - \frac{di}{dt}$$

For the critically damped case with $R = 4\ \Omega$,

$$v_c = 1 - 4(te^{-2t}) - \frac{d}{dt}(te^{-2t})$$

or

$$v_c(t) = 1 - e^{-2t} - 2te^{-2t}\ \text{V}$$

Note that the step response consists of a transient natural response of the form (7.59), plus a dc steady-state response of 1 V. In dc steady state, the capacitor is an open circuit and 1 V is the value of the unit step source voltage across the open circuit. This critically damped step response is shown in Fig. 7.11.

We next determine an overdamped unit step response v_c with $R = 5\ \Omega$. From (7.31),

$$i = K_1 e^{-t} + K_2 e^{-4t} \qquad \text{(overdamped)}$$

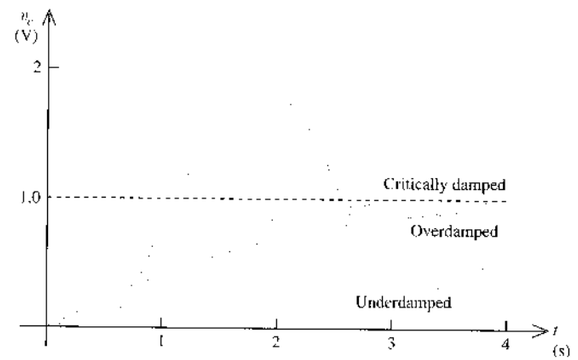
and using the initial conditions we computed in (7.60) yields

$$i(0+) = 0 = K_1 + K_2$$

$$\left.\frac{di}{dt}\right|_{0+} = 1 = -K_1 - 4K_2$$

Adding these last two equations yields $K_2 = -\frac{1}{3}$, and substituting into the first, we have $K_1 = \frac{1}{3}$. The mesh current i in this overdamped case is

$$i = \tfrac{1}{3}(e^{-t} - e^{-4t})\ \text{A}$$



Three step responses compared.

Using this in (7.63), we have

$$v_c = 1 - \frac{5}{3}(e^{-t} - e^{-4t}) - \frac{1}{3}\frac{d}{dt}(e^{-t} - e^{-4t})$$

$$= 1 - \tfrac{4}{3}e^{-t} + \tfrac{1}{3}e^{-4t} \text{ V}$$

Finally, we determine the unit step response for an underdamped case with $R = \frac{1}{4}\ \Omega$. The characteristic equation is

$$s^2 + \tfrac{1}{4}s + 1 = 0$$

with complex conjugate characteristic exponents $s_{1,2} = -\frac{1}{8} \pm j\sqrt{255/8}$, or

$$i = K_1 e^{-t/8} \cos\sqrt{\frac{255}{8}}\, t + K_2 e^{-t/8} \sin\sqrt{\frac{255}{8}}\, t \quad \text{(underdamped)}$$

Using the same initial conditions as in the previous cases gives

$$i(0+) = 0 = K_1$$

$$\left.\frac{di}{dt}\right|_{0+} = 1 = -\frac{1}{8}\, K_1 + \sqrt{\frac{255}{8}}\, K_2$$

which yields the mesh current

$$i = \frac{8}{\sqrt{255}}\, e^{-t/8} \sin\sqrt{\frac{255}{8}}\, t \text{ A}$$

and, by (7.63),

$$v_c = 1 - \frac{1}{4}\left(\frac{8}{\sqrt{255}} e^{-t/8} \sin\sqrt{\frac{255}{8}}\, t\right) - \frac{d}{dt}\left(\frac{8}{\sqrt{255}} e^{-t/8} \sin\sqrt{\frac{255}{8}}\, t\right)$$

$$= 1 - e^{-t/8}\left(\cos\sqrt{\frac{255}{8}}\, t + \frac{1}{\sqrt{255}} \sin\sqrt{\frac{255}{8}}\, t\right) \text{ A}$$

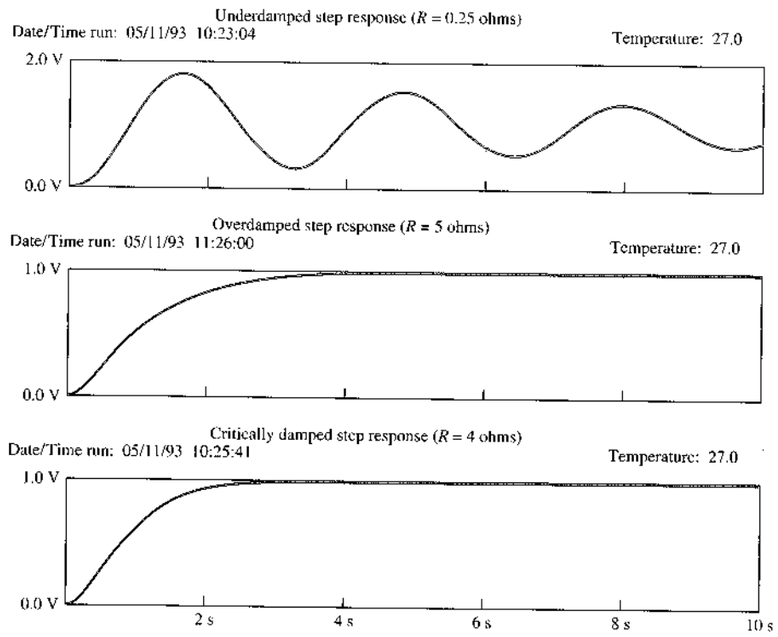
The overdamped and underdamped step responses are also shown in Fig. 7.11.

Comparing the three step responses, we observe that the critically damped circuit demonstrates the fastest convergence to the steady state of 1 V. The transient in the overdamped case contains two terms, one of which has a longer time constant ($\tau = 1$ s compared to $\frac{1}{2}$ s for the critically damped case) and thus decays to zero more slowly than the critically damped case. The transient in the underdamped response is oscillatory, with the amplitude of these oscillations decaying to zero with an even longer time constant, $\tau = 8$ s. Thus, although the underdamped response first reaches the value of 1 V much faster than the other responses, it "overshoots" this value, and its actual convergence to steady state is even slower than the overdamped case.

These results may be confirmed easily using SPICE. The input file for the circuit of Fig. 7.10 is

```
Series RLC step response
VG      1   0   DC   1
*RS = 0.25 OHMS (UNDERDAMPED), 4 OHMS
*(CRIT. DAMPED), 5 OHMS (OVERDAMPED).
RS      1   2     0.25
LS      2   3     1       IC=0
CS      3   0     0.25    IC=0
.TRAN   1  10   UIC
.PROBE
.END
```

The results of three runs with the three different values for R_s are shown in Fig. 7.12. These graphs were generated by the PSpice graphics postprocessor program PROBE. They closely match the calculated expressions (7.64) to (7.66).



PROBE output for three PSpice runs.

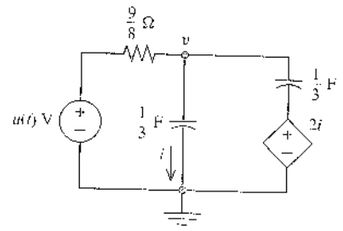
While the methods introduced in this chapter and Chapter 6 may be generalized for studying higher-order circuits (those with three or more storage elements and characteristic exponents), there are more efficient methods for breaking such circuits. These techniques will be introduced in the context of Laplace transform analysis in Chapter 12. The time-domain methods of Chapters 6 and 7, based on the direct solution of differential equations, are effective tools for analyzing first- and second-order circuits. They are less frequently the method of choice for analyzing higher-order circuits.

7.6.1. Find $v(0+)$, $dv/dt|_{0+}$, and the unit step response v of the parallel *RLC* circuit of Fig. 7.4 if $i_g = u(t)$, $R = 1\ \Omega$, $L = \frac{9}{14}$ H, $C = \frac{1}{9}$ F.

$$v(0+) = 0;\ dv/dt|_{0+} = 9;\ v = \tfrac{9}{5}(e^{-2t} - e^{-7t})$$

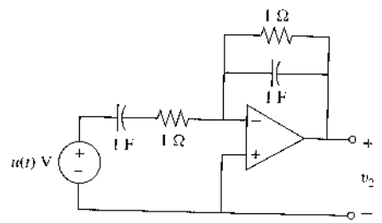
7.6.2. Find $v(0+)$, $dv/dt|_{0+}$, and the unit step response v of the circuit shown.

$$v(0+) = 0;\ dv/dt|_{0+} = 0;\ v = 1 - \tfrac{1}{5}(e^{4t} + 4e^{-t})$$



7.6.3. Find the describing equation for v_2 and the unit step response.

$$d^2v_2/dt^2 + 2(dv_2/dt) + v_2 = 0;\ v_2 = -te^{-t}$$



7.7 DESIGN OF SECOND-ORDER CIRCUITS

Second-order circuits are characterized by second-order differential equations and may have undamped sinusoids, damped sinusoids, or pairs of real exponentials as their natural responses. In contrast to the single real exponential natural response found in first-order circuits, second-order circuits are capable of much richer behavior. In particular, the ability to generate a sinusoidal response where none was input to the circuit may be used to generate tones, as is demonstrated in the first design example.

Design Example 7.1

Touchtone telephone dialing uses combinations of two tones to identify each pushbutton on the keypad. One is from the low group, consisting of 697 Hz, 770 Hz, 852 Hz, and 941 Hz, and the other is from the high group, 1209 Hz, 1336 Hz, and 1477 Hz. For instance, when key one is depressed, a combined 697- and 1209-Hz tone is sent. Our goal here is to design a circuit whose natural (unforced) response is a voltage that is the sum of a 697-Hz sinusoid and a 1209-Hz sinusoid.

We begin this design by noting as in Fig. 7.5 that pure sinusoids are the natural solutions to undamped second-order differential equations. The equation

$$\frac{d^2}{dt^2}x(t) + \omega_0^2 x(t) = 0$$

has natural solution as in (7.25) with $\sigma = 0$,

$$x_n(t) = B_1 \cos \omega_0 t + B_2 \sin \omega_0 t$$

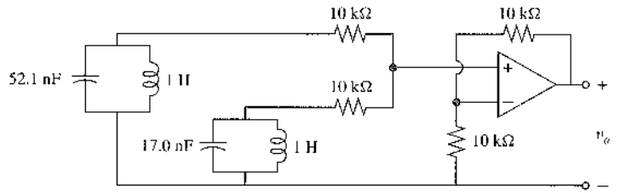
The parallel *LC* circuit in Fig. 7.4 with $\omega_0 = 1/\sqrt{LC}$ is one such circuit. Thus if we add together two such natural responses, one with

$$\omega_{01} = 2\pi(697) = 4379 \text{ rad/s} = \frac{1}{\sqrt{L_1 C_1}}$$

and the other

$$\omega_{02} = 2\pi(1209) = 7596 \text{ rad/s} = \frac{1}{\sqrt{L_2 C_2}}$$

we will produce the desired sum of tones. Setting $L_1 = L_2 = 1$ H for convenience, we get $C_1 = 52.1$ nF and $C_2 = 173$ nF. A noninverting summer that adds these tones together is shown in Fig. 7.13.



Touchtone pushbutton one-tone generator.

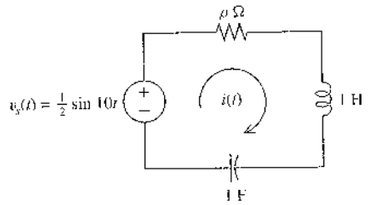
The reader should note that this is a purely *schematic* design, not intended as a practical physical one. A physical inductor will contain a significant amount of series resistance, as discussed in Chapter 5, and this will cause damping to attenuate the op amp inputs. Other circuit designs which are more practical as sinusoidal signal generators will be developed in Chapter 18.

Design Example 7.2

Design a circuit, called an *analog computer* circuit, which solves the differential equation

$$\frac{d^2}{dt^2}y(t) + \rho\frac{d}{dt}y(t) + y(t) = 5\cos 10t$$

with zero initial conditions. Use only passive elements and a single independent source. $y(t)$ should be a current somewhere in the circuit, and ρ a parameter of an element in the circuit. Show the SPICE plots of the solution for the interval [0, 2 s] for $\rho = 0.1$. Use the SIN independent source transient specification described in Appendix D.3.



Circuit to compute a differential equation.

Consider the series circuit of Fig. 7.14. In terms of general circuit parameters, the mesh equation is

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int i(t)\,dt = v_s(t)$$

Differentiating and dividing by L yields

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{1}{LC}i = \frac{1}{L}\frac{d}{dt}v_s(t)$$

This equation is of the desired form. Setting $L = C = 1$ and $R = \rho$ and identifying the mesh current $i(t)$ as the unknown $y(t)$ in the differential equation (7.67) that we are solving by the simple analog computer circuit of Fig. 7.14, we need only match forcing terms. Since the right-hand sides of (7.67) and (7.68) match for $L = 1$ with

$$v_s(t) = \tfrac{1}{2}\sin 10t$$

this is the desired voltage source in the circuit. The SPICE input file is

```
Design Example 7.2
*
VS      1     0    SIN(0 0.5  1.59)
* (0    0.5    1.59) = (xoff xampl freq)
R       1     2    .1
L       2     3    1   IC=0
C       3     0    1   IC=0
.TRAN       .05    2    UIC
.PLOT       TRAN   I(VS)
.END
```



```
**** 03/06/95 10:31:17 ********* Evaluation PSpice (January 1991) ************

 DESIGN EXAMPLE 7.2

 ****     TRANSIENT ANALYSIS                TEMPERATURE =   27.000 DEG C

*******************************************************************************

  TIME         I(VS)
(*)---------    -1.0000E-01  -5.0000E 02   0.0000E-00   5.0000E 02   1.0000E-01
  0.000E+00  4.475E-08 .                          *                          .
  5.000E-02 -6.243E-03 .                        *   .                        .
  1.000E-01 -2.282E-02 .                     *      .                        .
  1.500E-01 -4.523E-02 .                  *         .                        .
  2.000E-01 -6.797E-02 .               *            .                        .
  2.500E-01 -8.546E-02 .            *               .                        .
  3.000E-01 -9.361E-02 .         *                  .                        .
  3.500E-01  8.936E-02 .          *                 .                        .
  4.000E-01  7.349E-02 .               *            .                        .
  4.500E-01  5.030E-02 .                  *         .                        .
  5.000E-01  2.461E-02 .                     *      .                        .
  5.500E-01  2.415E-03 .                        *   .                        .
  6.000E-01 -1.082E-02 .                           * .                        .
  6.500E-01 -1.299E-02 .                           * .                        .
  7.000E-01 -3.983E-03 .                            *.                        .
  7.500E-01  1.430E-02 .                        *   .                        .
  8.000E-01  3.654E-02 .                     *      .                        .
  8.500E-01  5.672E-02 .                  *         .                        .
  9.000E-01  6.997E-02 .               *            .                        .
  9.500E-01  7.185E-02 .               *            .                        .
  1.000E+00  6.132E-02 .                  *         .                        .
  1.050E+00  4.128E-02 .                     *      .                        .
  1.100E+00 -1.570E-02 .                        *   .                        .
  1.150E+00  9.871E-03 .                           * .                        .
  1.200E+00  2.922E-02 .                              *                       .
  1.250E+00  3.871E-02 .                              *                       .
  1.300E+00  3.673E-02 .                                *                     .
  1.350E+00  2.347E-02 .                                *                     .
  1.400E+00  2.987E-03 .                                 *                    .
  1.450E+00 -1.881E-02 .                              *                       .
  1.500E+00 -3.664E-02 .                           *                          .
  1.550E+00 -4.511E-02 .                        *                             .
  1.600E+00 -4.140E-02 .                        *                             .
  1.650E+00 -2.678E-02 .                           *                          .
  1.700E+00 -4.017E-03 .                              *                       .
  1.750E+00  2.220E-02 .                                *                     .
  1.800E+00  4.545E-02 .                                   *                  .
  1.850E+00  6.089E-02 .                                     *                .
  1.900E+00  6.552E-02 .                                      *               .
  1.950E+00  5.780E-02 .                                    *                 .
  2.000E+00  4.019E-02 .                                 *                    .
```

FIGURE 7.15 SPICE plot for Design Example 7.2.

The output is shown in Fig. 7.15. Note that the forced response, which is a sinusoid at $\omega = 10$ rad/s (1.59 Hz), is superposed with the natural response, an underdamped sinusoid with much lower frequency.

Design Example 7.3

Design an *active* analog computer circuit that solves the equation

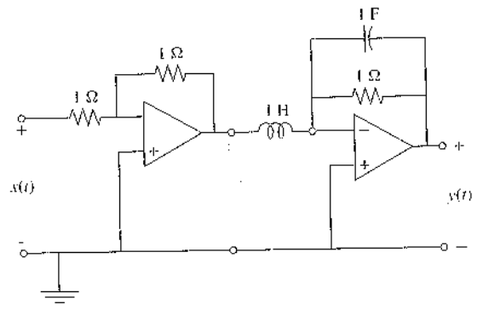
$$\frac{d^2}{dt^2}y(t) + \frac{d}{dt}y(t) = x(t)$$

where $x(t)$ is an arbitrary source voltage and $y(t)$ another voltage somewhere in the circuit.

As with Design Example 7.2, we may get a second-order equation by differentiating a mesh or node equation containing both a derivative and an integral. In the present case, however, the integral must involve the input, not the response variable. A simple passive circuit cannot have just this source dependency. The active circuit of Fig. 7.16 does, however. Using the virtual short and open principles and summing currents into the inverting input node to zero yields

$$C\frac{dy}{dt} + \frac{y}{R} + \frac{1}{L}\int [-x(t)]\,dt = 0$$

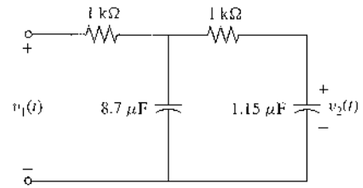
which for $R = L = C = 1$ yields the desired equation. A unity-gain inverting amplifier has been used to get the proper sign for the input signal $x(t)$.



Circuit to compute another differential equation.

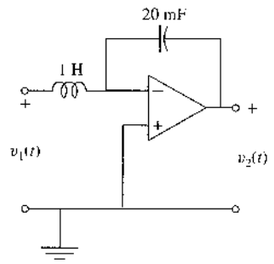
There are many other uses to which second-order circuits may be put. We did not consider filters, perhaps the most important use of all, as this topic is considered in detail in Chapter 14, and general methods for second-order filter design are discussed at length in Chapter 18. The design examples and exercises here, though, begin our exploration of second-order circuit design.

7.7.1. Design a circuit whose step response includes real exponentials with time constants $\tau = 10$ ms and $\tau = 1$ ms.



7.7.2. Using a single op amp, design a double-differentiator circuit, one that satisfies

$$\frac{d^2}{dt^2}v_2(t) = -50v_1(t)$$



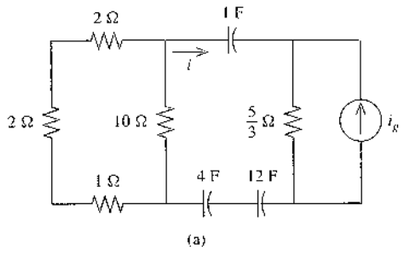
SUMMARY

Second-order circuits are those characterized by a single second-order differential equation. They can be identified by the presence of two equivalent storage elements (inductors and/or capacitors). Their behavior may be studied by writing and solving their describing equations in the time domain using the techniques introduced in Chapter 6 for first-order circuits. The natural behavior of second-order circuits is, however, much more diverse and interesting, ranging from pairs of real exponentials and t-multiplied exponentials to sinusoids both damped and undamped.

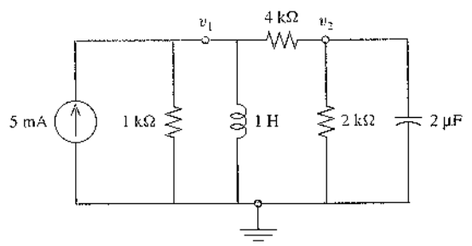
- The natural response of a second-order circuit is $K_1e^{p_1t} + K_2e^{p_2t}$, where p_1 and p_2 are the distinct roots of the characteristic equation, or if p_1 is a repeated root, the second term is replaced by the t-multiplied form $K_2te^{p_1t}$.

- The natural response can also be expressed in terms of the undamped natural frequency ω_o and damping factor ζ. ω_0 is the sinusoidal frequency in the corresponding undamped case ($\zeta = 0$), and ζ indicates the rate of exponential growth or decay.
- An overdamped system ($\zeta > 1$) has two real exponentials as its natural response, an underdamped system ($\zeta < 1$) has a damped sinusoid, and a critically damped system ($\zeta = 1$) has a real exponential and its t-multiplied form. An undamped system, with $\zeta = 0$, has a pure sinusoid as its natural response.
- The forced response is found by substituting a trial forced solution, which depends on the forcing function, into the differential equation and matching coefficients.
- The total response is found by adding the natural and forced responses, using the initial conditions to determine the coefficients in the natural response terms.
- The unit step response settles to its steady-state value most rapidly in the critically damped case. Overdamped circuits have extended periods of undershoot or overshoot, while underdamped circuits exhibit damped oscillations (ringing) around the steady-state value.

7.1. Write describing equations for the indicated variables. Are these second-order circuits?



7.2. Write a second-order differential equation satisfied by v_1.



7.3. Write a second-order differential equation satisfied by v_2 in the diagram for Problem 7.2.

7.4. Repeat Problem 7.3 if the capacitor is replaced by a $\frac{1}{4}$-H inductor.

7.5. Insert a 1-Ω resistor in series with v_g in Fig. 7.3 thereby making the source a practical rather than an ideal one. Show that in this case v_2 satisfies the second-order equation,

$$5\frac{d^2v_2}{dt^2} + 11\frac{dv_g}{dt} + 4v_2 = 4\frac{dv_2}{dt} + 4v_2$$

7.6. Find the describing equation for i.

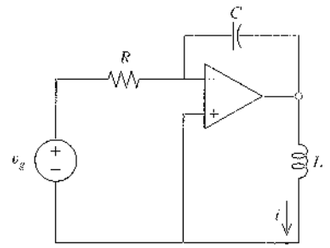


FIGURE P7.6

7.7. Find the describing equation for v_{out}. Is this a second-order circuit? A first-order circuit?

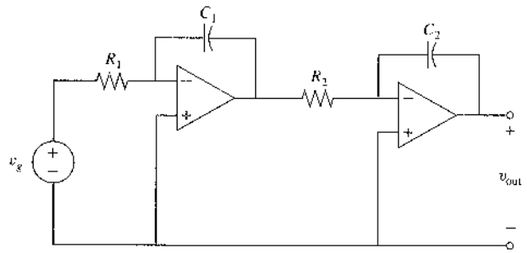


FIGURE P7.7

7.8. Consider the series RLC circuit shown in which the resistance is time varying. (a) Write the describing equation for i. (b) If i_n and i_f are natural and forced solutions, is their sum a solution?

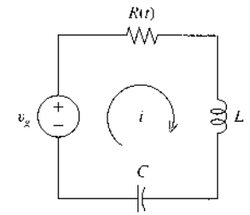


FIGURE P7.8

7.9. For a second-order differential equation

$$\frac{d^2x}{dt^2} + a_1\frac{dx}{dt} + a_0x = f(t)$$

let x_{n1} and x_{n2} be natural solutions and x_{f1}, x_{f2} be two distinct forced solutions. Show that *any* linear combination of x_{n1} and x_{n2} is still a natural solution. What *specific* linear combinations of x_{f1} and x_{f2} will be forced solutions?

7.10. Suppose the general second-order describing equation of Problem 7.9 is forced with an $f(t)$ that is a linear combination of m forcing functions

$$f(t) = b_1f_1(t) + b_2f_2(t) + \cdots + b_mf_m(t)$$

Show that the forced solution is the same linear combination of forced solutions to each $f_i(t)$ separately.

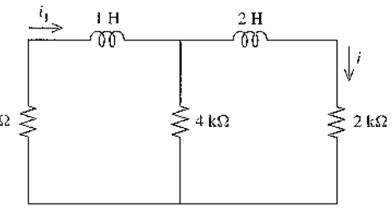
7.11.

$$\frac{d^2x}{dt^2} + a_1\frac{dx}{dt} + a_0x = f(t)$$

Find the natural response [$f(t) = 0$] if $a_1 = 14$, $a_0 = 49$, $x(0) = 0$, and $dx/dt|_0 = 2$.

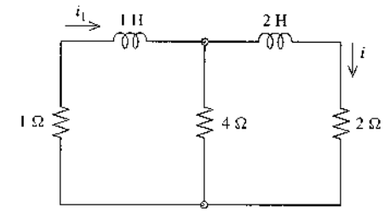
7.12. Write a second-order unforced differential equation with natural response $k_1e^{-t}\sin 2t + k_2e^{-t}\cos 2t$. Find k_1 and k_2 if the initial conditions are $x(0) = 0$ and $dx/dt|_0 = 4$.

7.13. Find i for $t > 0$ if $i_1(0) = 3$ mA and $i(0) = 1$ mA.

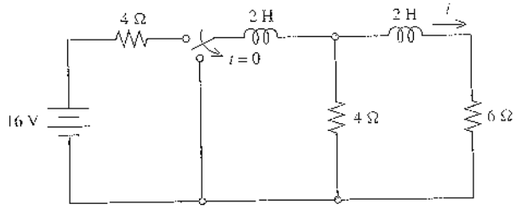


7.14. Find the natural response $i(t)$, $t > 0$ of the circuit of Problem 7.1(a) if $i(0) = 3$ A ($i_g = 0$).

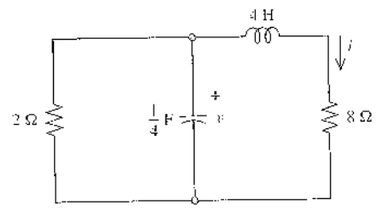
7.15. Find i for $t > 0$ if $i_1(0) = 9$ A and $i(0) = 3$ A.



7.16. Find i for $t > 0$ if the circuit is in steady state at $t = 0-$.

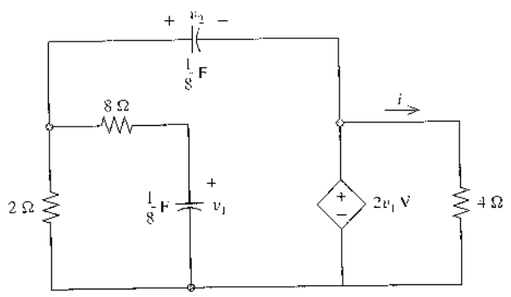


7.17. Find i for $t > 0$ if $i(0) = 4$ A and $v(0) = 8$ V.



7.18. Find v in Problem 7.17 if $i(0)$ is changed to 1 A.

7.19. Find i for $t > 0$ if $v_1(0) = v_2(0) = 12$ V.



7.20. Find the forced solution $x_f(t)$ if $f(t) = 16e^{-2t} + 4$.

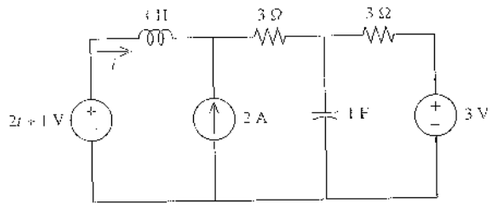
$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x = f(t)$$

7.21. Find the forced solution to the equation of Problem 7.20 if $f(t) = 4e^{-4t}$.

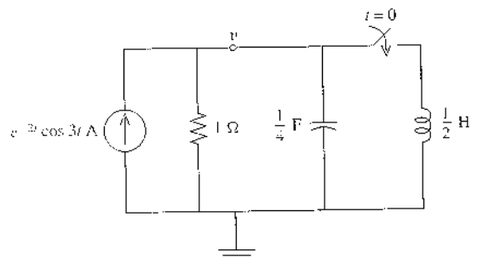
7.22. Find the forced solution to the equation of Problem 7.20 if $f(t) = 4te^{-4t} + 4e^{-t}$.

7.23. Sketch a circuit whose describing equation is $\frac{d^2v}{dt^2} + 2\frac{dv}{dt} + v = 2\sin 2t$.

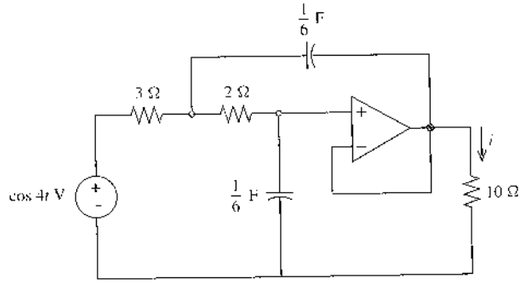
7.24. Find the forced solution $i_f(t)$.



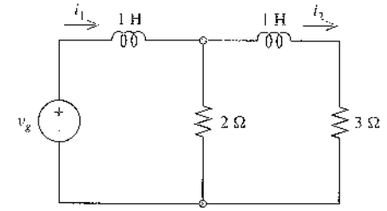
7.25. Find the forced solution $v_f(t)$.



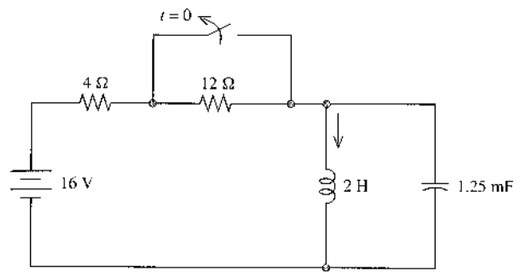
7.26. Find the forced solution $i_f(t)$.



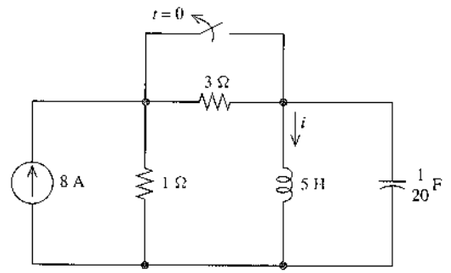
7.27. Find i_2 for $t > 0$ if $i_1(0) = 3$ A, $i_2(0) = -1$ A, and (a) $v_g = 15$ V, (b) $v_g = 10e^{-2t}$ V, and (c) $v_g = 5e^{-t}$ V.



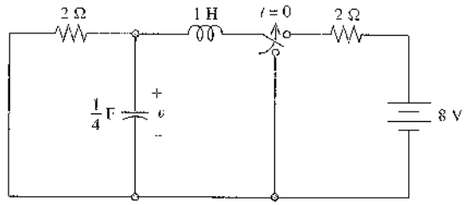
7.28. Find i for $t > 0$ if the circuit is in steady state at $t = 0-$.



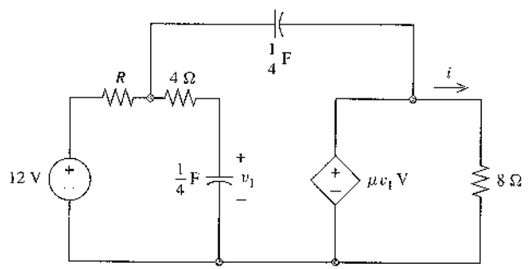
7.29. Find i for $t > 0$ if the circuit is in steady state when the switch is opened at $t = 0$.



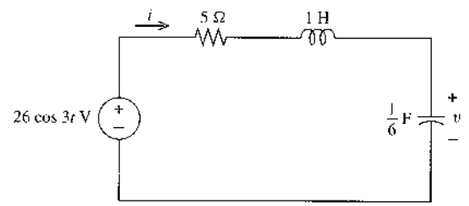
7.30. Find v for $t > 0$ if the circuit is in steady state at $t = 0-$.



7.31. Find i, $t > 0$, if there is no initial stored energy and (a) $R = 2\ \Omega$, $\mu = 2$; (b) $R = 2\ \Omega$, $\mu = 1$; and (c) $R = 1\ \Omega$, $\mu = 2$.

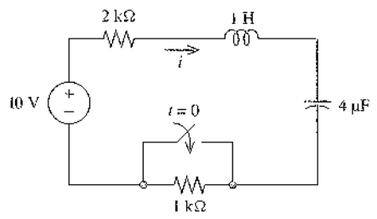


7.32. Find i for $t > 0$ if $i(0) = 2$ A and (a) $v(0) = 6$ V.



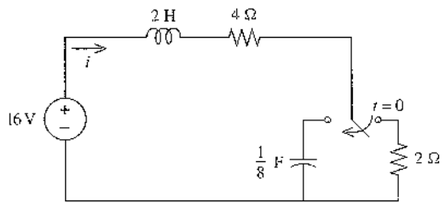
For Problems 7.33 to 7.40, assume dc steady state at $t = 0-$ and find the total solution for $t > 0$ for the indicated variable.

7.33.

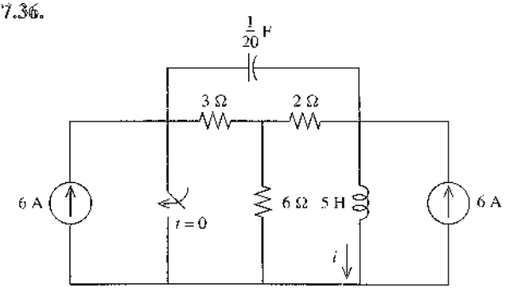


7.34. Solve Problem 7.33 if the switch closes at $t = 10$ s instead of $t = 0$ s.

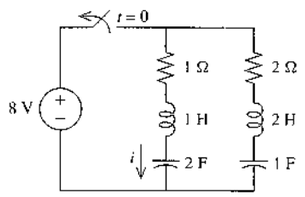
7.35.



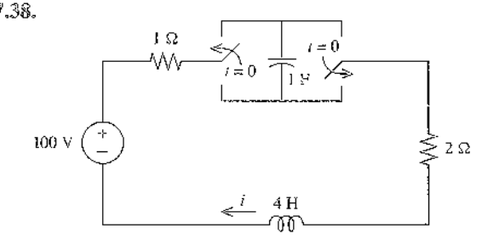
7.36.



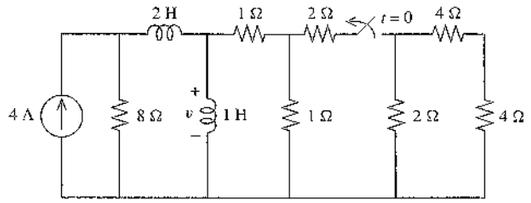
7.37.



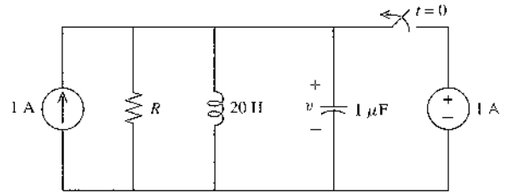
7.38.



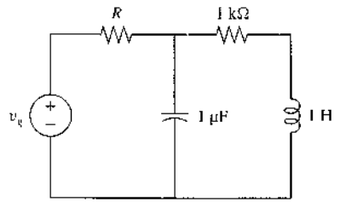
7.39.



7.40. Find the total solution $v(t)$ if R is set to the critical resistance (critical damping).

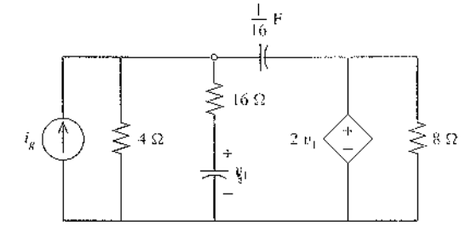


7.41. Find R so that the circuit is critically damped.

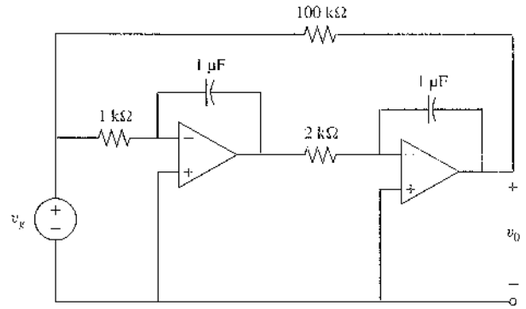


7.42. Find the unit step responses of the circuits of Problem 7.1. Check using SPICE.

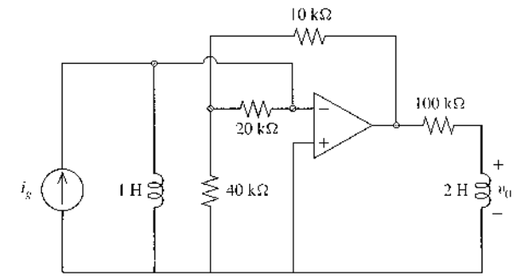
7.43. Find the unit step response. Check using SPICE.



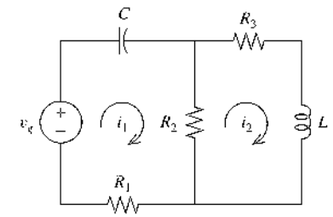
7.44. Find the unit step response. Check using SPICE.



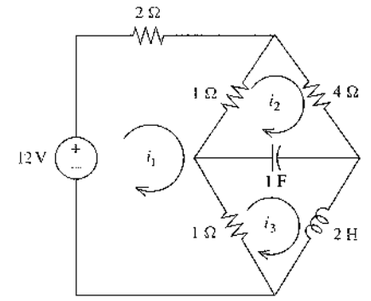
7.45. Find the unit step response. Check using SPICE.



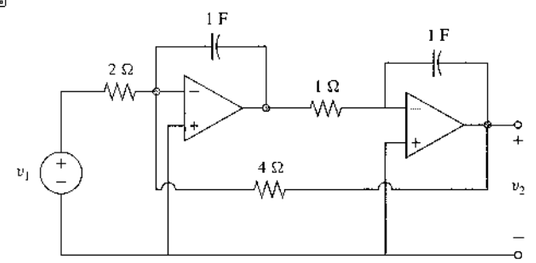
Write a second-order differential equation satisfied by i_2.



Find the mesh currents for $t > 0$ assuming all initial conditions are zero at $t = 0$.

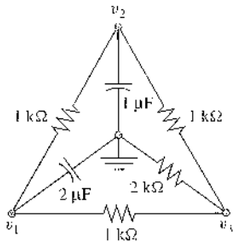


Find the forced response $v_2(t)$ if $v_1(t) = 4 \cos 2t$.

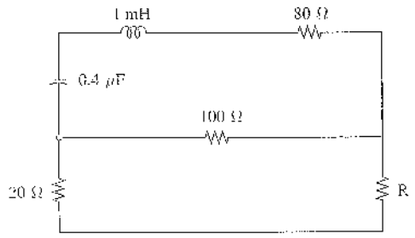


Write two different second-order differential equations that both have the same forced solution $x_{1f} = x_{2f} = 2t + 1$ but that have different forms for their natural solutions. Solve both equations assuming zero initial conditions.

Find the natural response $v_1(t)$, $t > 0$ (a) if $v_1(0) = v_2(0) = 0$; (b) if $v_1(0) = v_2(0) = 1$ V. [*Suggestion:* Solve node 1 equation for v_3, and use this to eliminate v_3 from the node 2 and 3 equations.]



Find a value for R such that the natural responses of this circuit all consist of damped sinusoids $Ke^{\sigma t}\cos(\omega t + \theta)$ with $\omega = 3000$ rad/s. Use SPICE to help search for the solution.



Design a circuit which solves the differential equation

$$\frac{d^2y}{dt^2} - 2\frac{dy}{dt} + 2y = 0$$

Specify what variable in your circuit $y(t)$ is, and how the initial conditions $y(0)$, $dy/dt|_{t=0}$ are determined.

An electric motor is modelled as a 22-Ω resistor in parallel with a 1/2-H inductor. Design a passive circuit to connect the motor to an ideal voltage source so that the resulting circuit is critically damped at its natural frequency of 60 Hz.

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Charles Proteus Steinmetz
1865–1923

I have found the equation that will enable us to transmit electricity through alternating current over thousands of miles. I have reduced it to a simple problem in algebra.

Charles Proteus Steinmetz

8

Sinusoidal Sources and Phasors

The use of complex numbers to solve ac circuit problems—the phasor method considered in this chapter—was first done by the German-Austrian mathematician and electrical engineer Charles Proteus Steinmetz in a paper presented in 1893. He is noted also for the laws of hysteresis and for his work in manufactured lightning.

Steinmetz was born in Breslau, Germany, the son of a government railway worker. He was deformed from birth and lost his mother when he was 1 year old, but this did not keep him from becoming a scientific genius. Just as his work on hysteresis later attracted the attention of the scientific community, his political activities while he was at the University of Breslau attracted the police. He was forced to flee the country just as he had finished the work for his doctorate, which he never received. He did electrical research in the United States, primarily with the General Electric Company. His paper on complex numbers revolutionized the analysis of ac circuits, although it was said at the time that no one but Steinmetz understood the method. In 1897 he also published the first book to reduce ac calculations to a science.

8.1 Properties of Sinusoids
8.2 RLC Circuit Example
8.3 Complex Sources
8.4 Phasors
8.5 Current–Voltage Laws for Phasors
8.6 Impedance and Admittance
8.7 Kirchhoff's Laws and Impedance Equivalents
8.8 Phasor Circuits
Summary
Problems

In Chapters 6 and 7 we analyzed circuits containing storage elements and have seen that the complete response is the sum of a natural and a forced response. The natural response is obtained after killing all independent sources and therefore is not a function of these sources, which are also known as *excitations* (unlike dependent sources, they excite responses without requiring coupling with other sources). The forced response, on the other hand, depends directly on the functional form of excitation applied to the circuit. In the case of a dc source, the forced response is a dc (constant) response, an exponential input evokes an exponential forced response, and so on.

Perhaps the single most important type of excitation is the sinusoid. Sinusoids are found everywhere in nature: for example, in the motion of a pendulum, the propagation of light or sound waves through space, and the vibration of strings or steel beams. As we have seen, any undamped second-order circuit produces a sinusoidal natural response, and any underdamped second-order circuit will have a decaying sinusoid as its natural response.

In electrical engineering and technology, sinusoidal time functions are found at the core of many, perhaps most, important applications. The carrier signals generated for communication systems are sinusoids, and the sinusoid is dominant in the electric power industry. Indeed, as we shall see later in the study of Fourier series, almost every useful signal in electrical engineering can be represented as a sum of sinusoidal components.

Because of their importance, circuits with sinusoidal excitation, or *ac circuits*, are considered in detail in this and several subsequent chapters. Since for all linear circuits the natural response is independent of the excitation and can be found by the methods of earlier chapters, we concentrate on finding the forced response to sinusoidal excitation. This response is important in itself since for stable circuits it is the ac steady-state response that remains after the time required for the transitory natural response has passed.

Being interested only in the forced response, we shall not limit ourselves, as we did in Chapters 6 and 7, to first- and second-order circuits. As we shall see, higher-order *RLC* circuits may be handled, insofar as the forced response is concerned, in much the same way as resistive circuits were handled in Chapter 2. The discovery of this technique

for analyzing ac circuits with no more difficulty than dc circuits, called phasor analysis, is one of the greatest intellectual achievements in the field of engineering analysis. Modern engineering, electrical and otherwise, would be unthinkable without the power and versatility of phasor analysis, the principal subject of this chapter.

8.1 PROPERTIES OF SINUSOIDS

We devote this section to a review of some of the properties of sinusoidal functions. Let us begin with the sine wave,

$$v(t) = V_m \sin \omega t$$

which is sketched in Fig. 8.1. The *amplitude* of the sinusoid is V_m, which is the maximum value that the function attains. The *radian frequency*, or *angular frequency*, is ω, measured in radians per second (rad/s).

The sinusoid is a periodic function, defined generally by the property that there is a smallest number T such that for all t,

$$v(t + T) = v(t)$$

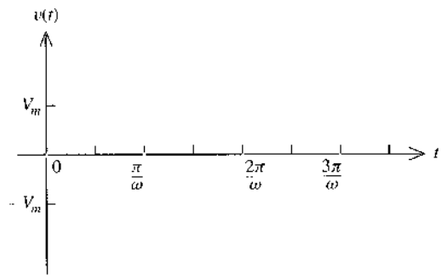
where T is the *period*. That is, the function goes through a complete cycle, or period, which is then repeated every T seconds. In the case of the sinusoid, the period is

$$T = \frac{2\pi}{\omega}$$

as may be seen from (8.1) and (8.2). Thus in 1 s the function goes through $1/T$ cycles, or periods. Its *frequency* f is then

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

cycles per second, or *hertz* (abbreviated Hz). The latter term, named for the German



Sinusoidal function.

physicist Heinrich R. Hertz (1857–1894), is now the standard unit for frequency. The relation between frequency and radian frequency is seen by (8.4) to be

$$\omega = 2\pi f$$

A more general sinusoidal expression is given by

$$v(t) = V_m \sin(\omega t + \phi)$$

where ϕ is the *phase angle*, or simply the *phase*. To be consistent, since ωt is in radians, ϕ should be expressed in radians. However, degrees are a very familiar measure for angle. Therefore, we may write

$$v = V_m \sin\left(2t + \frac{\pi}{4}\right)$$

or

$$v = V_m \sin(2t + 45^\circ)$$

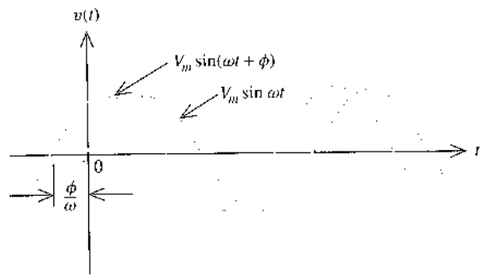
interchangeably, even though the latter expression contains a formal mathematical inconsistency. In the absence of the small circle indicating degrees, the unit for angle will be taken to be radians.

A sketch of (8.6) is shown in Fig. 8.2 by the solid line, along with a sketch of (8.1), shown dashed. The solid curve is simply the dashed curve displaced ϕ/ω seconds, or ϕ radians to the left. Therefore, points on the solid curve, such as its peaks, occur ϕ rad, or ϕ/ω s, earlier than corresponding points on the dashed curve. Accordingly, we shall say that $V_m \sin(\omega t + \phi)$ *leads* $V_m \sin \omega t$ by ϕ rad (or degrees). Note that a positive leading phase ($\phi > 0$) implies a left shift of the graph of the function. In general, the sinusoid

$$v_1 = V_{m1} \sin(\omega t + \alpha)$$

leads the sinusoid

$$v_2 = V_{m2} \sin(\omega t + \beta)$$



Two sinusoids with different phase.

by $\alpha - \beta$. An equivalent expression is that v_2 *lags* v_1 by $\alpha - \beta$. In Fig. 8.2, $V_m \sin \omega t$ lags $V_m \sin(\omega t + \phi)$ by ϕ radians. Positive lagging phase implies a right shift in the graph of the function.

As an example, consider

$$v_1 = 4 \sin(2t + 30^\circ)$$

$$v_2 = 6 \sin(2t - 12^\circ)$$

Then v_1 leads v_2 (or v_2 lags v_1) by $30 - (-12) = 42^\circ$. The graph of v_1 is left shifted by 42° relative to v_2.

Thus far we have considered sine functions rather than cosine functions in defining sinusoids. It does not matter which form we use since

$$\cos\left(\omega t - \frac{\pi}{2}\right) = \sin \omega t$$

$$\sin\left(\omega t + \frac{\pi}{2}\right) = \cos \omega t$$

The only difference between sines and cosines is thus the phase angle. For example, we may write (8.6) as

$$v(t) = V_m \cos\left(\omega t + \phi - \frac{\pi}{2}\right)$$

Equations (8.7) are trigonometric identities for quarter-period shifts. Half-period shifts change sign,

$$\sin(\omega t \pm \pi) = -\sin \omega t$$

$$\cos(\omega t \pm \pi) = -\cos \omega t$$

while full-period shifts have no effect:

$$\sin(\omega t \pm 2\pi) = \sin \omega t$$

$$\cos(\omega t \pm 2\pi) = \cos \omega t$$

Example 8.1

To determine how much one sinusoid leads or lags another of the same frequency, we must first express both as sine waves or as cosine waves with positive amplitudes. For example, let

$$v_1 = 4 \cos(2t + 30^\circ)$$

$$v_2 = -2 \sin(2t + 18^\circ)$$

Then, by (8.8a),

$$-\sin \omega t = \sin(\omega t + 180^\circ)$$

we have

$$\begin{aligned} v_2 &= 2 \sin(2t + 18^\circ + 180^\circ) \\ &= 2 \cos(2t + 18^\circ + 180^\circ - 90^\circ) \\ &= 2 \cos(2t + 108^\circ) \end{aligned}$$

Comparing this last expression with v_1, we see that v_1 leads v_2 by $30° - 108° = -78°$, which is the same as saying that v_1 lags v_2 by 78°.

The sum of a sine wave and a cosine wave of the same frequency is another sinusoid of that frequency. To show this, consider

$$A\cos\omega t + B\sin\omega t = \sqrt{A^2+B^2}\left(\frac{A}{\sqrt{A^2+B^2}}\cos\omega t + \frac{B}{\sqrt{A^2+B^2}}\sin\omega t\right)$$

which by Fig. 8.3 may be written

$$A\cos\omega t + B\sin\omega t = \sqrt{A^2+B^2}\,(\cos\omega t\cos\theta + \sin\omega t\sin\theta)$$

By a formula from trigonometry, this is

$$A\cos\omega t + B\sin\omega t = \sqrt{A^2+B^2}\cos(\omega t - \theta) \tag{8.9a}$$

where, by Fig. 8.3,

$$\theta = \tan^{-1}\frac{B}{A} \tag{8.9b}$$

A similar result may be established if the sine and cosine terms have phase angles other than zero, indicating that, in general, the sum of two sinusoids of a given frequency is another sinusoid of the same frequency.

The conversion of the sum of a sine and cosine may be reversed. By (8.9),

$$M\cos(\omega t - \theta) = A\cos\omega t + B\sin\omega t \tag{8.10}$$

where

$$A = M\cos\theta \tag{8.10a}$$

$$B = M\sin\theta \tag{8.10b}$$

The decomposition of a sinusoid into sine and cosine components is called the *quadrature representation* of the sinusoid.

Note that we must be clear on what is meant by (8.9b), since some mathematics books take this expression as the principal value of the arctangent and place θ in a specific quadrant. We mean that the terminal side of the angle θ is in the quadrant where the point (A, B) is located.

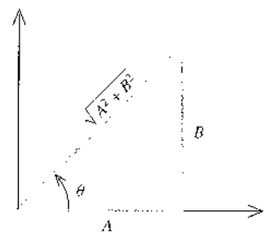


FIGURE 8.3 Triangle useful in adding two sinusoids.

Example 8.2

As an example, we have

$$-5\cos 3t + 12\sin 3t = \sqrt{5^2+12^2}\cos\left(3t - \tan^{-1}\frac{12}{-5}\right)$$

$$= 13\cos(3t - 112.6°)$$

since $\tan^{-1}(12/-5)$ is in the second quadrant, because $A = -5 < 0$ and $B = 12 > 0$.

8.1.1. Find the period of the following sinusoids:
(a) $4\cos(5t + 33°)$
(b) $\cos(2t + \pi/4) + 3\sin(2t - \pi/6)$
(c) $6\cos 2\pi t$

(a) $2\pi/5$; (b) π; (c) 1

8.1.2. Find the amplitude and phase of the following sinusoids:
(a) $3\cos 2t + 4\sin 2t$
(b) $(4\sqrt{3}-3)\cos(2t+30°) + (3\sqrt{3}-4)\cos(2t+60°)$ [*Suggestion:* In (b), expand both functions and use (8.9).]

(a) 5, −53.1°; (b) 5, 36.9°

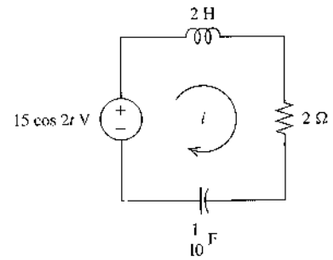
8.1.3. Find the frequency of the following sinusoids:
(a) $3\cos(6\pi t - 10°)$
(b) $4\sin 377t$

(a) 3; (b) 60 Hz

8.2 *RLC* CIRCUIT EXAMPLE

As an example of a circuit with sinusoidal excitation, let us find the forced mesh current i in the series RLC circuit shown in Fig. 8.4. By KVL,

$$2\frac{di}{dt} + 2i + 10\int_{0+}^{t} i(\tau)\,d\tau + v_c(0+) = 15\cos 2t$$



Series RLC circuit with sinusoidal excitation.

Differentiating and dividing by 2,

$$\frac{d^2i}{dt^2} + \frac{di}{dt} + 5i = -15\sin 2t$$

From Table 7.1, the trial forced solution is

$$i = A\cos 2t + B\sin 2t$$

Substituting this into (8.11) gives

$$(-4A\cos 2t - 4B\sin 2t) + (-2A\sin 2t + 2B\cos 2t) + 5(A\cos 2t + B\sin 2t) = -15\sin 2t$$

and matching coefficients of the cos and sin terms,

$$A + 2B = 0$$

$$-2A + B = -15$$

Doubling the first of these equations and adding to the second yields $5B = -15$ or $B = -3$, and substituting into the first, $A = +6$. The forced response is

$$i = 6\cos 2t - 3\sin 2t\,A$$

These two quadrature terms may be combined into one using the conversion formulas (8.10):

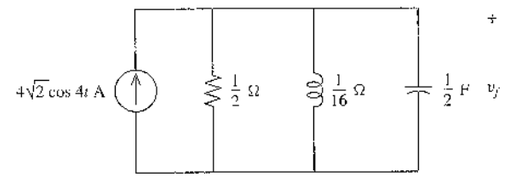
$$M = \sqrt{6^2 + 3^2} = 3\sqrt{5}$$

$$\theta = \tan^{-1}\left(-\frac{3}{6}\right) = -26.6°$$

or $$i = M\cos(\omega t - \theta) = 3\sqrt{5}\cos(2t + 26.6°)\text{ A}$$

The method used to arrive at our solution (8.14) is straightforward but, the reader may agree, somewhat laborious for such a simple single-loop circuit. In particular, the necessity of solving a set of simultaneous equations in the unknown quadrature coefficients A and B is noted. For high-order circuits it may be anticipated that use of this procedure would be even more cumbersome. In the remainder of this chapter we develop an approach offering considerable computational advantage, which bypasses the need to solve two simultaneous equations for the two quadrature coefficients A and B and which allows us to treat RLC circuits in the same way that we treated purely resistive circuits in earlier chapters. The method used above, based on the time-domain solution of differential equations, will always be available, but for complex problems it will seldom be preferred to the method of phasors to be introduced next.

8.2.1. Find the forced response v_f.
$2\cos(4t + 45°)$ V



8.2.2. Repeat Exercise 8.2.1 if the inductor is removed from the circuit.
$2\cos(4t - 45°)$ V

8.3 COMPLEX SOURCES

An alternative method for treating circuits with sinusoidal sources, which will be our focus of interest for this and the next several chapters, involves replacing the given sources by *complex sources*, those whose source functions have real and imaginary parts. Currents and voltages in circuits excited by complex sources will themselves be complex valued.

Since we rely heavily on complex numbers and their manipulation, readers unfamiliar with complex numbers or who seek to refresh their understanding should consult Appendix B. For convenience, we review several key definitions and properties of complex numbers before going on.

Each complex number is a point in the *complex plane*. The complex number A is written in *rectangular form* as

$$A = a + jb$$

where $j = \sqrt{-1}$ is the complex number of unit length along the imaginary axis in the complex plane. The real numbers a and b are the *real part*, denoted $a = \text{Re}\ A$, and the *imaginary part*, $b = \text{Im}\ A$, of the complex number A.

The same complex number A may be represented in *polar form* as

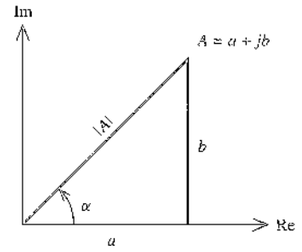
$$A = |A|/\underline{\alpha}$$

where $|A|$, the *magnitude* of A, and $/\alpha$, the *angle* of A, are given by

$$|A| = \sqrt{a^2 + b^2}$$

$$\alpha = \tan^{-1}\frac{b}{a}$$

The relation between rectangular and polar forms is shown in Fig. 8.5.



Rectangular and polar representations.

The important *Euler identity* is given by

$$e^{j\theta} = \cos\theta + j\sin\theta$$

or, using (8.18a) and (8.17), in its polar form,

$$e^{j\theta} = 1\underline{/\theta}$$

The latter form of the Euler identity may be used to generate a useful alternative way to write complex numbers in polar form. Since

$$A = |A|\underline{/\alpha} = |A|(1\underline{/\alpha})$$

then by (8.18b) we have the *exponential polar form* for a complex number:

$$A = |A|e^{j\alpha}$$

Rectangular, polar, and exponential polar forms of the same complex number A will each prove convenient in different contexts, and it is essential to be able to move easily among them.

Example 8.3

Consider the complex number A given in polar form as $A = 4 + j3$. Then $|A| = \sqrt{4^2 + 3^2} = 5$ and $\alpha = \tan^{-1}\frac{3}{4} = 36.9°$. The polar form is thus

$$A = 5\underline{/36.9°}$$

and the exponential polar form is

$$A = 5e^{j36.9°}$$

The three forms each describe the same complex number A, the same point A in the complex plane.

The behavior of the *complex exponential function* $e^{j\omega t}$ is central to our studies. By the polar form of the Euler identity (8.18b),

$$e^{j\omega t} = 1\underline{/\omega t}$$

Examining this expression, the magnitude of the complex exponential is always unity, while its angle increases uniformly at the rate of ω radians per second. Thus *the complex exponential $e^{j\omega t}$ traces out unit circles in the complex plane, beginning on the positive real axis at time $t = 0$ and moving counterclockwise, completing one full circle (or period) every $T = 2\pi/\omega$ seconds.* The projection of this point onto the horizontal axis is its real part ($\cos\omega t$) and that onto the vertical axis is its imaginary part ($\sin\omega t$).

The general scaled and phase-shifted complex exponential $V_m e^{(j\omega t+\phi)}$ shown in Fig. 8.6(b) is similar, except that at $t = 0$ its initial phase is ϕ radians and it traces out circles of radius V_m. By horizontal projection,

$$\mathrm{Re}\left[V_m e^{j(\omega t+\phi)}\right] = V_m\cos(\omega t + \phi)$$

and by vertical projection,

$$\mathrm{Im}\left[V_m e^{j(\omega t+\phi)}\right] = V_m\sin(\omega t + \phi)$$

which can be verified by inspection of the rectangular form of the Euler identity.

We next turn to the subject of main interest, that is, the application of complex numbers to electric circuits. Consider the linear circuit C of Fig. 8.7, which has independent source v_g. We wish to determine the forced response to the excitation v_g. Assume that all other independent sources in C have been killed, along with all initial conditions. Now let $f_1(t)$ and $f_2(t)$ be any two real functions of time. Suppose that when we use the source $v_g = f_1(t)$ we measure the forced response $i_1(t)$, and when $v_g = f_2(t)$ is used, the forced response is $i_2(t)$, as shown in Fig. 8.7(a) and (b). These responses to the real inputs f_1 and f_2 are real. By the proportionality principle, scaling the excitation f_2 by the constant j will scale the response i_2 by the same factor. Then, by superposition, the response to the sum $f_1 + jf_2$ will be the corresponding sum of separate responses

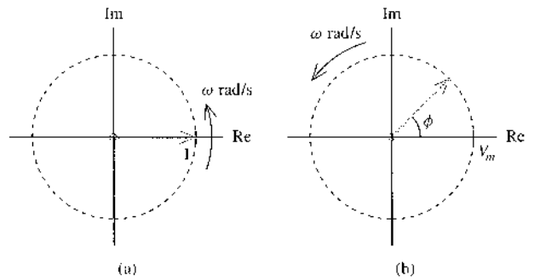
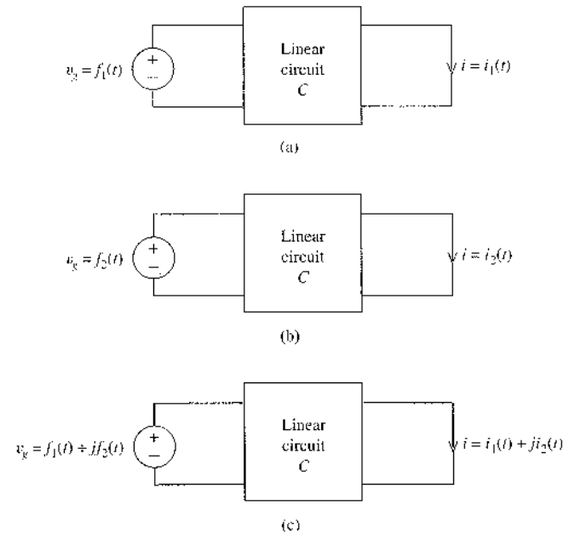


FIGURE 8.6 (a) Complex exponential $e^{j\omega t}$; (b) general complex exponential $V_m e^{j(\omega t+\phi)}$.



(a) Response to f_1; (b) response to f_2; (c) response to the linear combination $f_1 + jf_2$.

$i_1 + ji_2$, as shown in Fig. 8.7(c). Here we have used the fact that proportionality and superposition work equally well for complex quantities as for real ones.

Figure 8.7 reveals an important property of linear systems. $v_g = f_1 + jf_2$ is an arbitrary complex source, and $i = i_1 + ji_2$ its forced response. Since i_1 is also the separate response to f_1, which is the real part of the excitation v_g, *the response to the real part of a complex source is the real part of the response.* Similarly, noting that i_2 is the response to f_2, the imaginary part of v_g, *the response to the imaginary part of a complex source is the imaginary part of the response.*

This association is particularly significant when the complex source takes the form of a complex exponential $v_g = V_m e^{j(\omega t+\phi)}$. By Table 7.1 we know that the forced response to this source will be a complex exponential of the same frequency ω, which we denote as $i = I_m e^{j(\omega t-\theta)}$. *Once the response to the complex exponential source $V_m e^{j(\omega t+\phi)}$ has been found, we may immediately write down the response to the sinusoidal sources $V_m \cos(\omega t+\phi)$ as its real part and the response to $V_m \sin(\omega t+\phi)$ as its imaginary part.*

Example 8.4

Let us revisit the RLC circuit that was the subject of Section 8.2. Replacing the sinusoidal source $15\cos 2t$ V by the complex source $15e^{j2t}$ V, (8.11a) is

$$2\frac{di}{dt} + 2i + 10\int_{0+}^{t} i(\tau)\,d\tau + v_c(0+) = 15e^{j2t}$$

Differentiating and dividing by 2 as in (8.11b) gives

$$\frac{d^2i}{dt^2} + \frac{di}{dt} + 5i = 15je^{j2t}$$

Example 8.5

Substituting the trial form $i = Ae^{j2t}$, we have

$$(-4 + j2 + 5)Ae^{j2t} = 15je^{j2t}$$

Solving for A yields

$$A = \frac{15j}{1+j2} = 3\sqrt{5}/\underline{26.6^\circ}$$

The response i to this complex source is Ae^{j2t}, or

$$i = (3\sqrt{5}/\underline{26.6^\circ})e^{j2t} = 3\sqrt{5}e^{j(2t+26.6^\circ)}\ \text{A}$$

where the exponential polar form (8.19) has been used in the last expression.

Since the real part of $15e^{j2t}$ is $15\cos 2t$, the response to the source $v_g = 15\cos 2t$ V must be

$$\text{Re}[3\sqrt{5}e^{j(2t+26.6^\circ)}] = 3\sqrt{5}\cos(2t + 26.6^\circ)\ \text{A}$$

which agrees with our previous calculation. Moreover, since the imaginary part of $15e^{j2t}$ is $15\sin 2t$, had we instead applied the sinusoidal source $v_g = 15\sin 2t$, the response would have been

$$\text{Im}[3\sqrt{5}e^{j(2t+26.6^\circ)}] = 3\sqrt{5}\sin(2t + 26.6^\circ)\ \text{A}$$

As another example, let us find the forced response i_f of

$$\frac{d^2i}{dt^2} + 2\frac{di}{dt} + 8i = 12\sqrt{2}\cos(2t + 15^\circ)$$

First we replace the real excitation by the complex excitation,

$$v_1 = 12\sqrt{2}e^{j(2t+15^\circ)}$$

where for convenience the phase is written in degrees. The complex response i_1 satisfies

$$\frac{d^2i_1}{dt^2} + 2\frac{di_1}{dt} + 8i_1 = 12\sqrt{2}\ e^{j(2t+15^\circ)}$$

and it must have the general form

$$i_1 = Ae^{j2t}$$

Therefore, we must have

$$(-4 + j4 + 8)Ae^{j2t} = 12\sqrt{2}e^{j2t}e^{j15^\circ}$$

or
$$A = \frac{12\sqrt{2}\ e^{j15^\circ}}{4+j4} = \frac{12\sqrt{2}\ /\underline{15^\circ}}{4\sqrt{2}\ /\underline{45^\circ}} = 3\ /\underline{-30^\circ}$$

which gives

$$i_1 = (3/\underline{-30^\circ})e^{j2t}$$

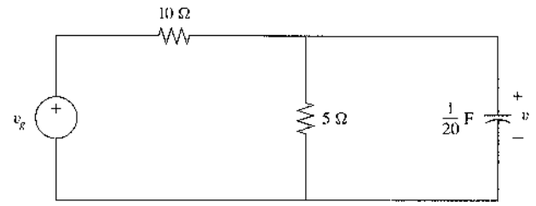
The response to the original excitation is the real part of the response to this complex excitation:

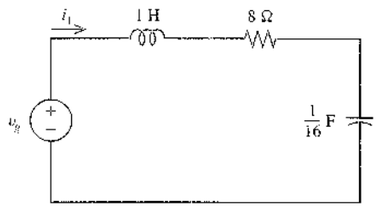
$$i_f = \text{Re}\ (i_1) = 3\cos(2t - 30^\circ)$$

In summary, given a circuit excited by a sinusoid, the forced response may be found by replacing the sinusoid by a complex exponential whose real (or imaginary) part is the given sinusoid. The describing equation will be easier to solve, since the trial forced solution will only have one undermined constant, rather than two, requiring the solution of simultaneous equations for the coefficients. Once solved, we may easily relate the complex excitation response to our desired sinusoidal response by taking the real part of the response if the original sinusoidal source is the real part of the complex exponential source, or the imaginary part if it is the imaginary part of the complex source.

8.3.1. (a) From the time-domain equations, find the forced response v if $v_g = 10(e^{j8t})$ V. (b) Using the result in part (a), find the forced response v if $v_g = 10\cos 8t$ V.

(a) $2e^{j(8t-53.1^\circ)}$ V; (b) $2\cos(8t - 53.1^\circ)$ V





8.3.2. Find the forced response v in Exercise 8.3.1 if $v_g = 10\sin 8t$ V. [*Suggestion:* $\sin 8t = \mathrm{Im}(e^{j8t})$.]

$2\sin(8t - 53.1^\circ)$ V

8.3.3. Using the method of complex excitation, find the forced response i if $v_g = 20\cos 2t$ V.

$2\cos(2t + 36.9^\circ)$ A

8.3.4. Repeat Exercise 8.3.3 if $v_g = 16\cos 4t$ V.

$2\cos 4t$ A

8.4 PHASORS

The results obtained in the preceding section may be put in much more compact form by the use of quantities called *phasors*, which we introduce in this section. The phasor method of analyzing circuits is credited generally to Charles Proteus Steinmetz (1865–1923), a famous electrical engineer with the General Electric Company in the early part of the twentieth century.

We are concerned with the forced response of a circuit to sinusoidal excitation at frequency ω. Each sinusoidal source may be expressed as a cosine

$$v_g(t) = V_m \cos(\omega t + \phi)$$

Suppose that we replace each such source by a complex exponential source given by

$$v_{g1}(t) = V_m e^{j(\omega t + \phi)}$$

Comparing these two expressions, we see that *the complex source has the same frequency ω, and that the original source is the real part of the complex source we have chosen to replace it.*

Consider the forced response in the new circuit. Since it is excited by complex exponential sources of frequency ω, all currents and voltages will also be complex exponentials of frequency ω. This follows from the form of the trial forced solution $Ae^{j\omega t}$. Then each current will be of the form

$$i_1(t) = \mathbb{I}e^{j\omega t} \tag{8.20a}$$

and each voltage of the form

$$v_1(t) = \mathbb{V}e^{j\omega t} \tag{8.20b}$$

where $\mathbb{I}$ and $\mathbb{V}$ are complex numbers. We define $\mathbb{I}$ and $\mathbb{V}$ as *phasors*, that is, *the complex numbers that multiply $e^{j\omega t}$ in the expressions for currents and voltages.* To distinguish them from other quantities, phasors are printed in boldface. The units for phasors are the same as the currents and voltages they are associated with; thus in (8.20) $\mathbb{I}$ inherits the units of i_1, usually amperes, and $\mathbb{V}$ has the same units as the voltage v_1, usually volts.

Phasors are thus defined in terms of the response to complex excitations, but the importance of phasors lies in their direct link to sinusoidal responses. In the sinusoidal circuit with which we began this discussion, all forced responses are sinusoids at frequency ω. Let one such response be $v(t)$. After substituting complex exponential sources, the same response variable will be $v_1(t) = \mathbb{V}e^{j\omega t}$, where $\mathbb{V} = |\mathbb{V}|\underline{/\theta}$ is its voltage phasor. But we recall from the last section that the response to the real part of a source is just the real part of the response. Since the sinusoidal source $V_m\cos(\omega t + \phi)$ is the real part of the complex source

$$V_m \cos(\omega t + \phi) = \mathrm{Re}[V_m e^{j(\omega t + \phi)}] \tag{8.21}$$

it follows that the sinusoidal response $v(t)$ is just the real part of the complex response $v_1(t)$, or

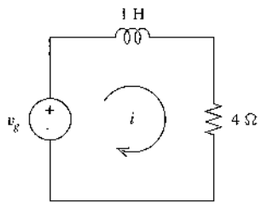
$$v(t) = \mathrm{Re}(\mathbb{V}e^{j\omega t})$$

Using the polar exponential form for $\mathbb{V}$ and taking the real part yields

$$v(t) = \mathrm{Re}(|\mathbb{V}|e^{j\theta}e^{j\omega t}) = |\mathbb{V}|\cos(\omega t + \theta) \tag{8.22}$$

Note the direct relationship between the the sinusoid $v(t)$ and its phasor $\mathbb{V}$. By (8.22), $|\mathbb{V}|$ is the amplitude of $v(t)$ and θ its phase angle. *The amplitude of the sinusoid is the magnitude of its phasor, and the phase angle of the sinusoid is the angle of its phasor.* Thus we can immediately write down the sinusoidal current or voltage once its phasor has been computed.

Example 8.6



Circuit for Example 8.6.

Suppose in Fig. 8.8 that $v_g = 6\cos 2t$ V. Since $v_g = \text{Re}(6e^{j2t})$, we use $v_{g1} = 6e^{j2t}$ V as our complex exponential source. The describing equation is

$$\frac{di}{dt} + 4i = 6e^{j2t}$$

The trial forced solution is $i_1 = Ae^{j2t}$, and applying this to the equation above gives

$$(j2 + 4)Ae^{j2t} = 6e^{j2t}$$

or
$$A = \frac{6}{4 + j2} = 1.34\underline{/-26.6^\circ}$$

The forced solution is $i_1 = Ae^{j2t}$, or

$$i_1 = (1.34\underline{/-26.6^\circ})e^{j2t} \text{ A}$$

The phasor representation of i_1 is $\mathbb{I} = 1.34/-26.6^\circ$ and the sinusoidal response is

$$i = \text{Re}(i_1) = 1.34\cos(2t - 26.6^\circ) \text{ A}$$

Starting with the cosine form $V_m\cos(\omega t + \phi)$ for each source, we conclude that each response is also a cosine (8.22) of the same frequency, whose amplitude and phase are just the magnitude and angle of the associated phasor. To preserve this association, if the sinusoidal source is given in the sine form $V_m\sin(\omega t + \phi)$, we first convert to the cosine form using (8.7b), repeated here as

$$\cos(\omega t + \phi - 90^\circ) = \sin(\omega t + \phi)$$

Table 8.1 shows the relation between these two sources and their phasors.

Complex Sources and Source Phasors for Cosine and Sine Sources

Source	Complex Source	Source Phasor
$A\cos(\omega t + \phi)$	$Ae^{j(\omega t + \phi)}$	$A\underline{/\phi}$
$A\sin(\omega t + \phi)$	$Ae^{j(\omega t + \phi - 90^\circ)}$	$A\underline{/\phi - 90^\circ}$

Example 8.7

To find the forced response $i(t)$ in Fig. 8.9(a), we replace the source $36\cos(2t + 30^\circ)$ V by the complex source $36e^{j(2t+30^\circ)}$ V and the source $2\sin(2t - 15^\circ)$ A, following Table 8.1, by $2e^{j(2t-105^\circ)}$ A. Analyzing Fig. 8.9(b), the single mesh equation is

$$4i_1 + 3\frac{d}{dt}\left[i_1 - 2e^{j(2t-105^\circ)}\right] = 36e^{j(2t+30^\circ)}$$

or
$$\frac{di_1}{dt} + \tfrac{4}{3}i_1 = 12e^{j(2t+30^\circ)} - j4e^{j(2t-105^\circ)}$$

Combining terms on the right-hand side gives

$$(12e^{j30^\circ} - j4e^{-j105^\circ})e^{j2t} = (12\underline{/30^\circ} - 4\underline{/-15^\circ})e^{j2t} = (9.6\underline{/47.1^\circ})e^{j2t}$$

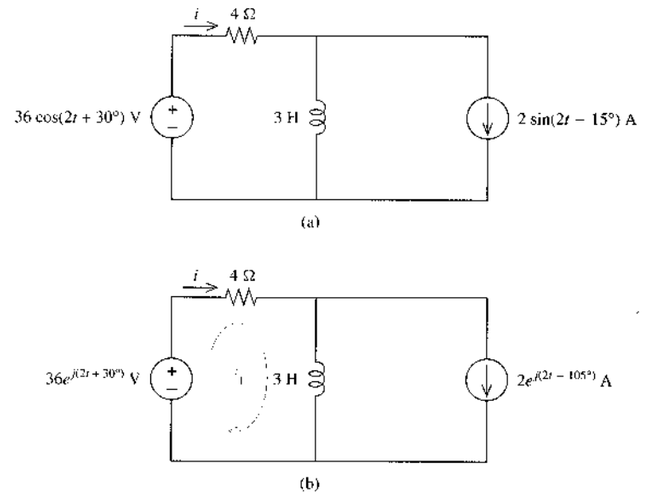
Using the trial form $i_1 = Ae^{j2t}$, we obtain

$$(j2A + \tfrac{4}{3}A)e^{j2t} = (9.6\underline{/47.1^\circ})e^{j2t}$$

and solving for A yields

$$A = \frac{9.6\underline{/47.1^\circ}}{4/3 + j2} = 4.0\underline{/-9.2^\circ}$$

Thus we have $i_1 = (4.0\underline{/-9.2^\circ})e^{j2t}$ A. The phasor associated with this current is $4.0\underline{/-9.2^\circ}$ A, and it follows that the corresponding sinusoidal current $i(t)$ has amplitude 4.0 A and phase angle -9.2°,



(a) Original circuit; (b) after source replacement.

or $$i(t) = 4.0\cos(2t - 9.2^\circ)\ \text{A}$$

One point of considerable practical significance is illustrated by Example 8.7. Our work with phasors will save us considerable time and effort compared to using real sinusoids, but only if we are equipped with the right tools. To use the phasor method, we need to be able to perform calculations such as (8.23),

$$\frac{9.6\underline{/47.1^\circ}}{4/3 + j2} = 4.0\underline{/-9.2^\circ}$$

rapidly and efficiently. This requires adequate computational support for the arithmetic operations required and repeated so frequently in the course of phasor circuit calculations: addition, subtraction, multiplication, and division of complex numbers, and also rectangular-to-polar and polar-to-rectangular conversion. *It is highly desirable to have access to an electronic calculator that supports complex data types, four-function complex arithmetic, and single-keystroke rectangular-to-polar and polar-to-rectangular conversions.* Although any calculator having sine, cosine, arctangent, and square root keys will permit us to do all required calculations, those explicitly set up to support complex arithmetic have a decided edge in convenience. Many fewer keystrokes will be required to complete a given problem, and keystrokes translate into time and diversion of mental focus from the concepts in the circuit problem at hand. Each major calculator manufacturer offers a line of suitable machines, beginning at relatively inexpensive prices.

Finally, in our work with phasors we rely on the principle that the response to the real part of a complex source (i.e., the response to a cosine source) can be computed as the real part of the response (i.e., the cosine part of a complex exponential). It is also possible to use another result from Section 8.3, that the response to the imaginary part of a complex source is the imaginary part of the response, to develop a second type of phasor analysis based on sines and imaginary parts rather than cosines and real parts. Since either type of phasor analysis is sufficient, the alternative form of phasor analysis is not considered in this book, except in one of the problems at the end of the chapter. We stick with phasors based on cosines and real parts, as in (8.22) and in the examples.

EXERCISES

8.4.1. Find the phasor representation of (a) $6\cos(2t + 45^\circ)$, (b) $4\cos 2t + 3\sin 2t$, and (c) $-6\sin(5t - 65^\circ)$.

Answer (a) $6\underline{/45^\circ}$; (b) $5\underline{/-36.9^\circ}$; (c) $6\underline{/25^\circ}$

8.4.2. Find the time-domain function represented by the phasors (a) $10\underline{/-17^\circ}$, (b) $6 + j8$, and (c) $-j6$. In all cases, $\omega = 3$.

Answer (a) $10\cos(3t - 17^\circ)$; (b) $10\cos(3t + 53.1^\circ)$; (c) $6\cos(3t - 90^\circ)$

8.5 CURRENT–VOLTAGE LAWS FOR PHASORS

In this section we show that relationships between phasor voltage and phasor current for resistors, inductors, and capacitors are very similar to Ohm's law for resistors. In fact, the phasor voltage is proportional to the phasor current, as in Ohm's law. Consider a circuit in which all currents and voltages are of the form $Ae^{j\omega t}$. This will be the case when sinusoidal sources have been replaced by complex exponentials and we are interested in the forced response only. For the resistor of Fig. 8.10,

$$v = \mathbb{V}e^{j\omega t}$$
$$i = \mathbb{I}e^{j\omega t}$$

where $\mathbb{V}$ and $\mathbb{I}$ are phasors. By Ohm's law applied to (8.24),

$$\mathbb{V}e^{j\omega t} = R\mathbb{I}e^{j\omega t}$$

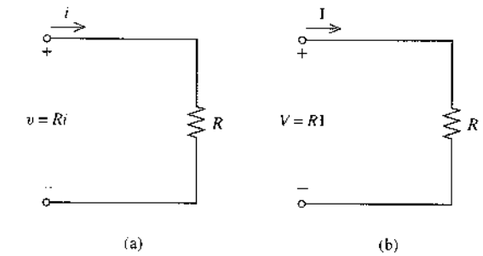
or, canceling the $e^{j\omega t}$ factors,

$$\boxed{\mathbb{V} = R\mathbb{I}}$$

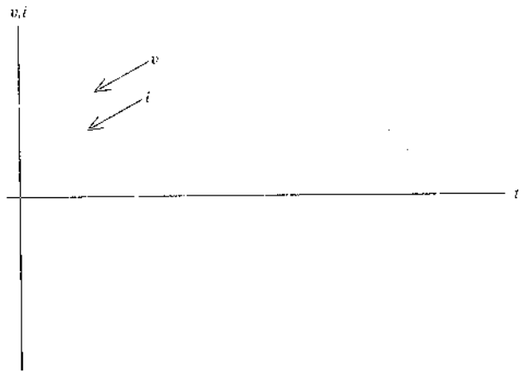
Thus the phasor or frequency-domain relation for the resistor is exactly like the time-domain relation. The voltage–current relations for the resistor are illustrated in Fig. 8.11. With $\mathbb{V} = V_m e^{j\phi_v}$ and $\mathbb{I} = I_m e^{j\phi_I}$, (8.26) becomes

$$V_m e^{j\phi_v} = (RI_m)e^{j\phi_I}$$

that is, the magnitude of the voltage phasor equals the magnitude of the current phasor scaled by R, and the angles are the same. Recalling that the magnitude of a phasor is the amplitude of its sinusoid, and its angle is the phase angle of the sinusoid, we have Fig. 8.11. Note that since $\phi_V = \phi_I$, the current and voltage are *in phase*.



Voltage–current relations for a resistor R in the (a) time and (b) frequency domains.



Voltage and current waveforms for a resistor.

Example 8.8

As an illustration, suppose that the voltage

$$v = 10\cos(100t + 30^\circ)\ \text{V}$$

is applied across a 5-Ω resistor, with the polarity indicated in Fig. 8.10(a). Then the phasor voltage is

$$\mathbb{V} = 10\underline{/30^\circ}\ \text{V}$$

and the phasor current is

$$\mathbb{I} = \frac{\mathbb{V}}{R} = \frac{10\underline{/30^\circ}}{5} = 2\underline{/30^\circ}\ \text{A}$$

Therefore, in the time domain we have

$$i = 2\cos(100t + 30^\circ)\ \text{A}$$

This is simply the result we would have obtained using Ohm's law.

In the case of the inductor, substituting the complex current and voltage into the time-domain relation,

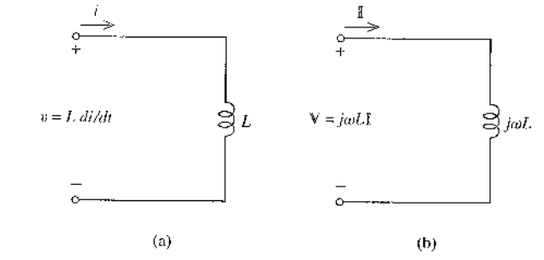
$$v = L\frac{di}{dt}$$

gives

$$V_m e^{j(\omega t+\phi_V)} = L\frac{d}{dt}\left[I_m e^{j(\omega t+\phi_I)}\right]$$
$$= j\omega L I_m e^{j(\omega t+\phi_I)}$$

Again, dividing out the factor $e^{j\omega t}$ and identifying the phasors, we obtain the phasor relation

$$\mathbf{V} = j\omega L\mathbf{I}$$



Voltage–current relations for an inductor L in the (a) time and (b) frequency domains.

Thus the phasor voltage $\mathbb{V}$, as in Ohm's law, is proportional to the phasor current $\mathbb{I}$, with the proportionality factor $j\omega L$. The voltage–current relations for the inductor are shown in Fig. 8.12.

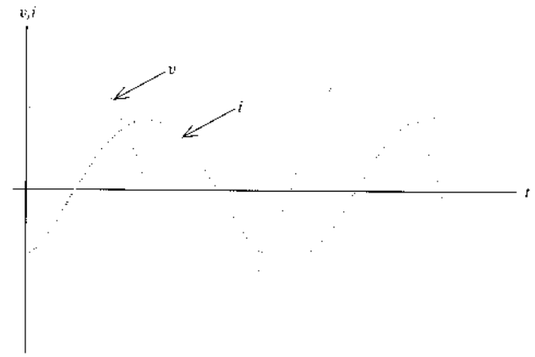
If the current in the inductor is given by $i = I_m\cos(\omega t + \phi_I)$, then by (8.27) the phasor voltage is

$$\mathbb{V} = (j\omega L)(I_m\underline{/\phi_I})$$
$$= \omega L I_m\underline{/\phi_I + 90^\circ}$$

since $j = 1\underline{/90^\circ}$. Therefore, in the time domain we have

$$v = \omega L I_m\cos(\omega t + \phi_I + 90^\circ)$$

We see that in the case of an inductor the current *lags* the voltage by 90°. Another expression that is used is that the current and voltage are 90° out of phase. This is shown graphically in Fig. 8.13.



Voltage and current waveforms for an inductor.

Finally, let us consider the capacitor. Substituting the complex current and voltage into the time-domain relation,

$$i = C\frac{dv}{dt}$$

gives the complex relation

$$I_m e^{j(\omega t+\phi_I)} = C\frac{d}{dt}\left[V_m e^{j(\omega t+\phi_V)}\right]$$
$$= j\omega C V_m e^{j(\omega t+\phi_V)}$$

Again dividing by $e^{j\omega t}$ and identifying the phasors, we obtain the phasor relation

$$\mathbb{I} = j\omega C\mathbb{V} \tag{8.28}$$

or

$$\mathbb{V} = \frac{\mathbb{I}}{j\omega C}$$

Thus the phasor voltage $\mathbb{V}$ is proportional to the phasor current $\mathbb{I}$, with the proportionality factor given by $1/j\omega C$. The voltage–current relations for a capacitor in the time and frequency domains are shown in Fig. 8.14.

In the general case, if the capacitor voltage is given by $v = V_m\cos(\omega t + \phi_V)$, then by (8.28) the phasor current is

$$\mathbb{I} = (j\omega C)(V_m/\phi_V)$$
$$= \omega C V_m/\phi_V + 90°$$

Therefore, in the time domain we have

$$i = \omega C V_m\cos(\omega t + \phi_V + 90°)$$

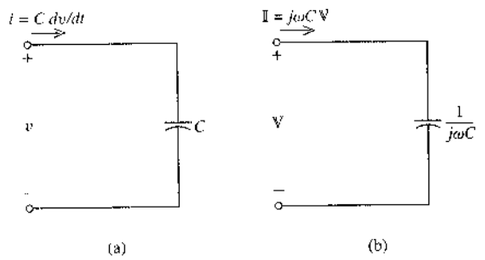
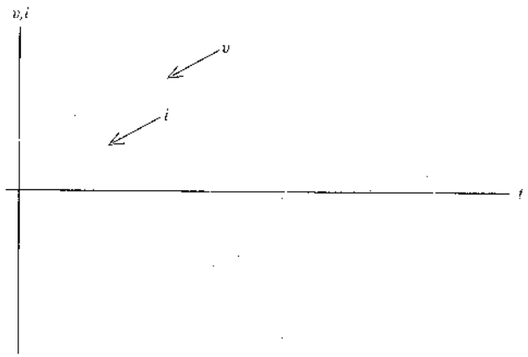


FIGURE 8.14 Voltage–current relations for a capacitor in the (a) time and (b) frequency domains.



Voltage and current waveforms for a capacitor.

which indicates that in the case of a capacitor the current and voltage are out of phase, with the current *leading* the voltage by 90°. This is shown graphically in Fig. 8.15.

Example 8.9

As an example, if $v = 10\cos(100t + 30°)$ is applied across a 1-μF capacitor, then by (8.28) the phasor current is

$$\mathbb{I} = j(100)(10^{-6})(10/30°)\text{ A}$$
$$= 1/120°\text{ mA}$$

The time-domain current is then

$$i = \cos(100t + 120°)\text{ mA}$$

and therefore the current leads the voltage by 90°, as it must in all capacitors.

8.5.1. Using phasors, find the ac steady-state current i if $v = 12\cos(1000t + 30°)$ V in (a) Fig. 8.10(a) for $R = 4$ kΩ, (b) Fig. 8.12(a) for $L = 15$ mH, and (c) Fig. 8.14(a) for $C = \frac{1}{2}\,\mu$F.

Answer: (a) $3\cos(1000t + 30°)$ mA; (b) $0.8\cos(1000t - 60°)$ A; (c) $6\cos(1000t + 120°)$ mA

8.5.2. In Exercise 8.5.1, find i in each case at $t = 1$ ms.

Answer: (a) 0.142 mA; (b) 0.799 A; (c) −5.993 mA

8.5.3. For what L in Fig. 8.12 will a sinusoidal voltage of amplitude 20 V produce a 4-mA amplitude current at $\omega = 500$ rad/s?

Answer: 10 H

8.6 IMPEDANCE AND ADMITTANCE

Let us now consider a general circuit with two accessible terminals, as shown in Fig. 8.16. If the time-domain voltage and current at the terminals are given by

$$v = V_m \cos(\omega t + \phi_V)$$

$$i = I_m \cos(\omega t + \phi_I)$$

the phasor quantities at the terminals are

$$\mathbf{V} = V_m \angle \phi_V$$

$$\mathbf{I} = I_m \angle \phi_I$$

We define the ratio of the phasor voltage to the phasor current as the *impedance* of the circuit, which we denote by Z. That is,

$$\mathbf{V} = Z\mathbf{I}$$

and, by (8.31),

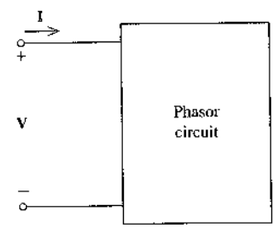
$$Z = |Z| \angle \phi_Z = \frac{V_m}{I_m} \angle \phi_V - \phi_I$$

where $|Z|$ is the magnitude and ϕ_Z the angle of Z. Evidently,

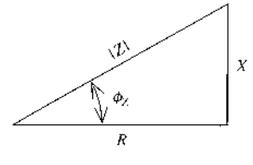
$$|Z| = \frac{V_m}{I_m}, \qquad \phi_Z = \phi_V - \phi_I$$

The magnitude of the impedance is the ratio of magnitudes of voltage and current phasors; the angle is the difference of the voltage and current phasor angles.

Impedance, as is seen from (8.32), plays the role, in a general circuit, played by resistance in resistive circuits. Indeed, (8.32) looks very much like Ohm's law; also like resistance, impedance is measured in ohms, being a ratio of volts to amperes.



General phasor circuit.



Graphical representation of impedance.

It is important to stress that impedance is a complex number, being the ratio of two complex numbers, but it is *not* a phasor. That is, it has no corresponding sinusoidal time-domain function as current and voltage phasors do. Impedance is a complex constant that scales one phasor to produce another.

The impedance Z is written in polar form in (8.33); in rectangular form it is generally denoted by

$$Z = R + jX$$

where $R = \text{Re } Z$ is the *resistive component*, or simply *resistance*, and $X = \text{Im } Z$ is the *reactive component*, or *reactance*. In general, $Z = Z(j\omega)$ is a complex function of $j\omega$, but $R = R(\omega)$ and $X = X(\omega)$ are real functions of ω. Like Z, both R and X are measured in ohms. Evidently, comparing (8.33) and (8.34), we may write

$$|Z| = \sqrt{R^2 + X^2}$$

$$\phi_Z = \tan^{-1} \frac{X}{R}$$

$$R = |Z| \cos \phi_Z$$

$$X = |Z| \sin \phi_Z$$

These relations are shown graphically in Fig. 8.17.

Example 8.10

As an example, suppose in Fig. 8.16 that $\mathbf{V} = 10\angle 56.9°$ V and $\mathbf{I} = 2\angle 20°$ A. Then we have

$$Z = \frac{10\angle 56.9°}{2\angle 20°} = 5\angle 36.9° \ \Omega$$

In rectangular form this is

$$Z = 5(\cos 36.9° + j \sin 36.9°)$$

$$= 4 + j3 \ \Omega$$

The impedances of resistors, inductors, and capacitors are readily found from their V–I relations of (8.26), (8.27), and (8.29). Distinguishing their impedances with subscripts R, L, and C, respectively, we have, from these equations and (8.32),

$$Z_R = R$$

$$Z_L = j\omega L = \omega L \angle 90°$$

$$Z_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C} = \frac{1}{\omega C} \angle -90°$$

In the case of a resistor, the impedance is purely resistive, its reactance being zero. Impedances of inductors and capacitors are purely reactive, having zero resistive components. The *inductive reactance* is denoted by

$$X_L = \omega L \tag{8.35}$$

so that

$$Z_L = jX_L$$

and the *capacitive reactance* is denoted by

$$X_C = -\frac{1}{\omega C} \tag{8.37}$$

and thus

$$Z_C = jX_C$$

Since ω, L, and C are positive, we see that *inductive reactance is positive and capacitive reactance is negative*. In the general case of (8.34), we may have $X = 0$, in which case the circuit appears to be resistive; $X > 0$, in which case its reactance appears to be inductive; and $X < 0$, in which case its reactance appears to be capacitive. These cases are possible when resistance, inductance, and capacitance are all present in the circuit, as we shall see. As an example, the circuit with impedance given by $Z = 4 + j3$, which we have just considered, has reactance $X = 3$, which is of the inductive type. In all cases of passive circuits, the resistance R is nonnegative.

The reciprocal of impedance, denoted by

$$Y = \frac{1}{Z} \tag{8.38}$$

is called *admittance* and is analogous to conductance (the reciprocal of resistance) in resistive circuits. Evidently, since Z is a complex number, then so is Y. The standard representation is

$$Y = G + jB \tag{8.39}$$

The quantities $G = \mathrm{Re}(Y)$ and $B = \mathrm{Im}(Y)$ are called *conductance* and *susceptance*,

respectively, and are related to the impedance components by

$$Y = G + jB = \frac{1}{Z} = \frac{1}{R + jX}$$

The units of Y, G, and B are all siemens, since in general Y is the ratio of a current to a voltage phasor.

To obtain the relation between components of Y and Z, we may rationalize the last member of the previous equation, which results in

$$G + jB = \frac{1}{R + jX}\frac{R - jX}{R - jX} = \frac{R - jX}{R^2 + X^2}$$

Equating real and imaginary parts results in

$$G = \frac{R}{R^2 + X^2}$$

$$B = -\frac{X}{R^2 + X^2}$$

Therefore, we note that R and G are *not* reciprocals except in the purely resistive case ($X = 0$). Similarly, X and B are never reciprocals, but in the purely reactive case ($R = 0$) they are negative reciprocals.

Example 8.11

As an example, if we have

$$Z = 4 + j3$$

then

$$Y = \frac{1}{4 + j3} = \frac{4 - j3}{4^2 + 3^2} = \frac{4}{25} - j\frac{3}{25}$$

Therefore, $G = \frac{4}{25}$ and $B = -\frac{3}{25}$.

Further examples are

$$Y_R = G$$

$$Y_L = \frac{1}{j\omega L}$$

$$Y_C = j\omega C$$

which are the admittances of a resistor, with $R = 1/G$, an inductor, and a capacitor.

Exercises

8.6.1. Find the impedance seen at the terminals of a series RL subcircuit in both rectangular and polar form.

Answer: $R + j\omega L$; $\sqrt{R^2 + \omega^2 L^2} \angle \tan^{-1} \omega L/R$

8.6.2. Find the admittance seen at the terminals of a series RL subcircuit in both rectangular and polar form.

$$\frac{R}{R^2+\omega^2L^2} - j\frac{\omega L}{R^2+\omega^2L^2};$$

$$\frac{1}{\sqrt{R^2+\omega^2L^2}}\angle -\tan^{-1}(\omega L/R)$$

8.6.3. Find the conductance and susceptance if Z is (a) $3 + j4$, (b) $0.4 + j0.3$, and (c) $(\sqrt{2}/2)\angle 45°$.

(a) 0.12, −0.16; (b) 1.6, −1.2; (c) 1, −1

8.7 KIRCHHOFF'S LAWS AND IMPEDANCE EQUIVALENTS

If a complex excitation, say $V_m e^{j(\omega t+\theta)}$, is applied to a circuit, then complex voltages, such as $V_1 e^{j(\omega t+\theta_1)}$, $V_2 e^{j(\omega t+\theta_2)}$, and so on, appear across the elements in the circuit. KVL applied around a typical loop results in an equation such as

$$V_1 e^{j(\omega t+\theta_1)} + V_2 e^{j(\omega t+\theta_2)} + \cdots + V_N e^{j(\omega t+\theta_N)} = 0$$

Dividing out the common factor $e^{j\omega t}$, we have

$$\mathbb{V}_1 + \mathbb{V}_2 + \cdots + \mathbb{V}_N = 0$$

where

$$\mathbb{V}_n = V_n\angle\theta_n, \qquad n = 1, 2, \ldots, N$$

are the phasor voltages around the loop. Thus *KVL holds for phasors.* A similar development will also establish KCL. At any node with N connected branches,

$$\mathbb{I}_1 + \mathbb{I}_2 + \cdots + \mathbb{I}_N = 0$$

where

$$\mathbb{I}_n = I_n\angle\theta_n, \qquad n = 1, 2, \ldots, N$$

Thus *KCL holds for phasors.*

In circuits having sinusoidal excitations with a common frequency ω, if we are interested only in the forced, or ac steady-state, response, we may find the phasor voltages or currents of every element and use Kirchhoff's laws to complete the analysis. The analysis is therefore identical to the resistive circuit analysis of Chapters 2, 4, and 5, with impedances replacing resistances and phasors replacing time-domain currents and voltages. Once we have found the phasors, we can convert immediately to the time-domain sinusoidal answers in the usual fashion.

Example 8.12

As an example, consider the circuit of Fig. 8.18, which consists of N impedances connected in series. By KCL for phasors, the single phasor current $\mathbb{I}$ flows in each element. Therefore, the voltages shown across each element are

$$\mathbb{V}_1 = Z_1\mathbb{I}$$

$$\mathbb{V}_2 = Z_2\mathbb{I}$$

$$\vdots$$

$$\mathbb{V}_N = Z_N\mathbb{I}$$

and by KVL around the circuit,

$$\mathbb{V} = \mathbb{V}_1 + \mathbb{V}_2 + \cdots + \mathbb{V}_N$$
$$= (Z_1 + Z_2 + \cdots + Z_N)\mathbb{I}$$

Since we must also have, from Fig. 8.18,

$$\mathbb{V} = Z_{eq}\mathbb{I}$$

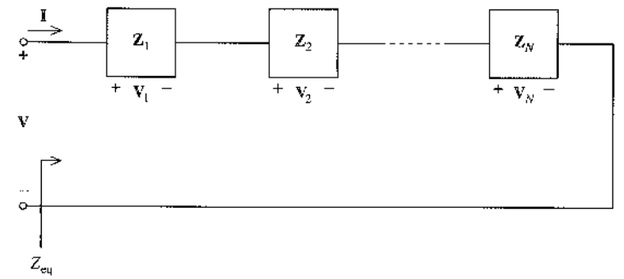
where Z_{eq} is the *equivalent* impedance seen at the terminals, it follows that

$$Z_{eq} = Z_1 + Z_2 + \cdots + Z_N$$

as in the case of series resistors.

Similarly, as was the case for parallel conductances in Chapter 2, the equivalent admittance Y_{eq} of N parallel admittances is

$$Y_{eq} = Y_1 + Y_2 + \cdots + Y_N$$



Impedances connected in series.

In the case of two parallel elements ($N = 2$), we have

$$Z_{eq} = \frac{1}{Y_{eq}} = \frac{1}{Y_1 + Y_2} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

In like manner, voltage and current division rules hold for phasor circuits, with impedances and frequency-domain quantities, in exactly the same way that they held for resistive circuits, with resistances and time-domain quantities. The reader is asked to establish these rules in Exercise 8.7.2.

Example 8.13

For example, let us return to the RLC circuit considered in Section 8.2. The circuit and phasor circuit are shown in Fig. 8.19(a) and (b). By KVL in the phasor circuit,

$$Z_C\mathbb{I} + Z_L\mathbb{I} + R\mathbb{I} = 15\underline{/0}$$

or

$$(2 - j)\mathbb{I} = 15\underline{/0}$$

from which the phasor current is

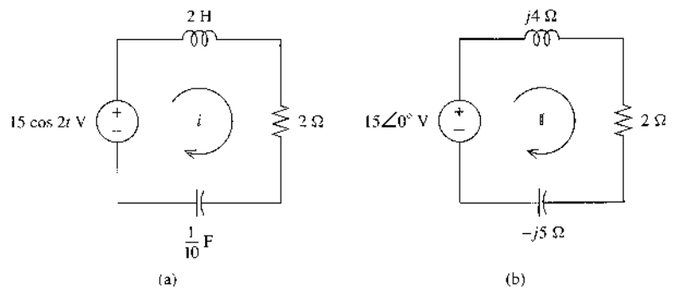
$$\mathbb{I} = \frac{15}{2 - j} = 3\sqrt{5}\underline{/26.6^\circ} \text{ A}$$

Therefore, in the time domain we have the same result as before, although now computed with considerably less effort:

$$i = 3\sqrt{5}\cos(2t + \underline{/26.6^\circ}) \text{ A}$$

An alternative method of solution is to observe that the impedance Z seen at the source terminals is the impedance of the inductor, $Z_L = j\omega L = j4$, the resistor $Z_R = R = 2$, and the capacitor, $Z_C = 1/j\omega C = -j5$, all connected in series. Therefore,

$$Z = j4 + 2 + (-j5) = 2 - j\ \Omega$$



(a) Time-domain circuit; (b) phasor circuit.

and

$$\mathbb{I} = \frac{\mathbb{V}}{Z} = \frac{15\underline{/0}}{2 - j} = 3\sqrt{5}\underline{/26.6^\circ} \text{ A}$$

as obtained earlier.

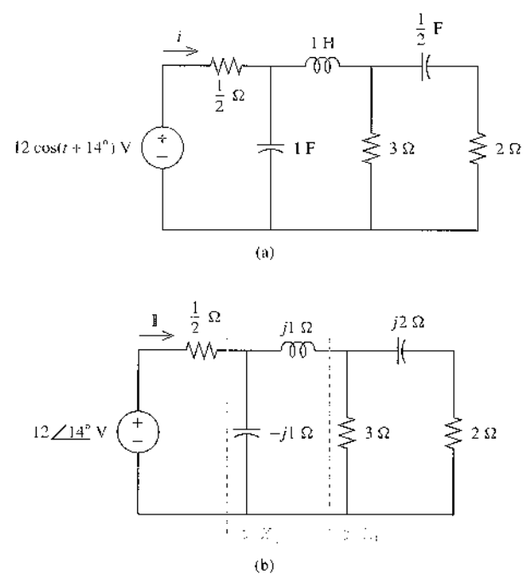
This ability to treat inductors, capacitors, and resistors alike as elements of the generic *impedance* without distinction among them when writing circuit equations is one of the main strengths of the phasor method. RLC circuits may be simplified using series–parallel equivalents, current–voltage dividers, and Thevenin–Norton transformations just as pure resistive circuits were earlier. Series impedances add, currents through parallel impedances divide in proportion to their admittances, and so on, regardless of the specific identities (R, L, or C) of the impedances involved.

Example 8.14

We wish to find $i(t)$ in steady state in Fig. 8.20. Using series–parallel impedance equivalents, we have

$$Z_1 = \frac{3(2 - j2)}{3 + (2 - j2)} = 1.45 - j0.621\ \Omega$$

$$Z_2 = \frac{(-j1)(Z_1 + j1)}{-j1 + (Z_1 + j1)} = 0.583 - j0.75\ \Omega$$



(a) Time-domain circuit; (b) phasor circuit.

so the equivalent impedance is

$$Z_{eq} = \tfrac{1}{2} + Z_2 = 1.083 - j0.75\ \Omega$$

and

$$\mathbb{I} = \frac{12/14^\circ}{Z_{eq}} = \frac{12/14^\circ}{1.083 - j0.75} = 9.11/\underline{48.7^\circ}\ \text{A}$$

Recalling that $\omega = 1$ rad/s yields

$$i(t) = 9.11\cos(t + 48.7^\circ)\ \text{A}$$

This is the desired ac steady-state current.

8.7.1. Derive (8.41).

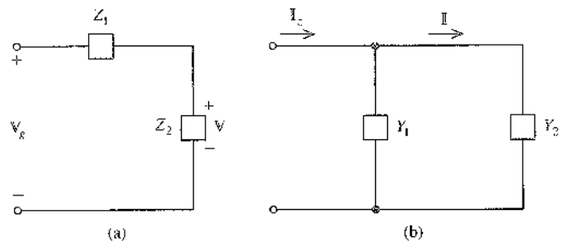
8.7.2. Show for circuit (a) that the voltage-division rule,

$$\mathbb{V} = \frac{Z_2}{Z_1 + Z_2}\mathbb{V}_g$$

and for circuit (b) that the current-division rule,

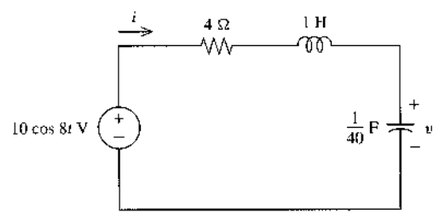
$$\mathbb{I} = \frac{Y_2}{Y_1 + Y_2}\mathbb{I}_g = \frac{Z_1}{Z_1 + Z_2}\mathbb{I}_g$$

are valid, where $Z_1 = 1/Y_1$ and $Z_2 = 1/Y_2$.



8.7.3. Find the steady-state current i using phasors.

Answer: $2\cos(8t - 36.9^\circ)$ A



8.7.4. Find the steady-state voltage v in Exercise 8.7.3 using phasors and voltage division.

Answer: $10\cos(8t - 126.9^\circ)$ V

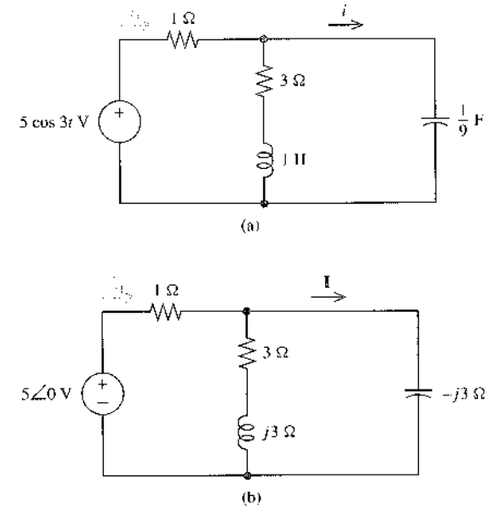
8.8 PHASOR CIRCUITS

As the discussion in Section 8.7 suggests, we may omit the steps of finding the describing equation in the time domain, replacing the excitations and responses by their complex forcing functions and then dividing the equation through by $e^{j\omega t}$ to obtain the phasor equation. We may simply start with the *phasor circuit*, which we define as *the time-domain circuit with the voltages and currents replaced by their phasors and the elements identified by their impedances*. The describing equation obtained from this circuit is then the *phasor equation*. Solving this equation yields the phasor of the answer, which then may be converted to the time-domain answer.

The procedure from starting with the phasor circuit to obtaining the phasor answer is identical to that used earlier in resistive circuits. The only difference is that impedances replace resistances.

Example 8.15

As an example, let us find the steady-state current i in Fig. 8.21(a). The phasor circuit, shown in Fig. 8.21(b), is obtained by replacing the voltage source and the currents by their phasors and labeling the



RLC time-domain and phasor circuits.

elements with their impedances. In the phasor circuit the impedance seen from the source terminals is

$$Z = 1 + \frac{(3 + j3)(-j3)}{3 + j3 - j3}$$

$$= 4 - j3\ \Omega$$

Therefore, we have

$$\mathbb{I}_1 = \frac{5\underline{/0}}{4 - j3} = \frac{5\underline{/0}}{5\underline{/-36.9^\circ}} = 1\underline{/36.9^\circ}$$

and, by current division,

$$\mathbb{I} = \frac{3 + j3}{3 + j3 - j3}\mathbb{I}_1 = \sqrt{2}\underline{/81.9^\circ}\ \text{A}$$

In the time domain, the answer is

$$i = \sqrt{2}\cos(3t + 81.9^\circ)\ \text{A}$$

In the case of a dependent source, such as a source kv_x volts controlled by a voltage v_x, it will appear in the phasor circuit as a source $k\mathbb{V}_x$, where $\mathbb{V}_x$ is the phasor representation of v_x, because $v_x = V_m\cos(\omega t + \phi)$ in the time domain will become $V_m e^{j(\omega t+\phi)}$ when a complex excitation is applied. Then dividing $e^{j\omega t}$ out of the equations leaves v_x represented by its phasor $V_m e^{j\phi}$. In the same way, $kv_x = kV_m\cos(\omega t + \phi)$ is represented by its phasor $kV_m e^{j\theta}$, which is k times the phasor of v_x.

Example 8.16

As an example of a circuit containing a dependent source, let us consider Fig. 8.22(a), in which it is required to find the steady-state value of i. The corresponding phasor circuit is shown in Fig. 8.22(b). Since phasor circuits are analyzed exactly like resistive circuits, we may apply KCL at node a in Fig. 8.22(b), resulting in

$$\mathbb{I} + \frac{\mathbb{V}_1 - \frac{1}{2}\mathbb{V}_1}{-j2} = 3\underline{/0} \qquad (8.43)$$

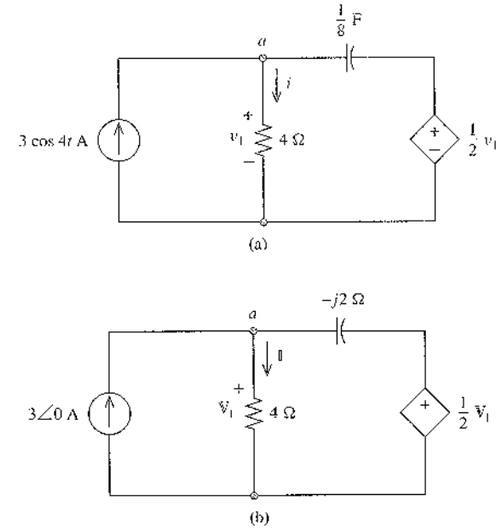
By Ohm's law we have $\mathbb{V}_1 = 4\mathbb{I}$, which substituted into (8.43) yields

$$-j2\mathbb{I} + \tfrac{1}{2}(4\mathbb{I}) = -j6$$

or
$$\mathbb{I} = \frac{-j6}{2 - j2} = \frac{6\underline{/-90^\circ}}{2\sqrt{2}\underline{/-45^\circ}} = \frac{3}{\sqrt{2}}\underline{/-45^\circ}\ \text{A}$$

Therefore, we have

$$i = \frac{3}{\sqrt{2}}\cos(4t - 45^\circ)\ \text{A}$$



(a) Circuit containing a dependent source; (b) corresponding phasor circuit.

In the case of an op amp, the phasor circuit is the same as the time-domain circuit. That is, an ideal op amp in the time-domain circuit appears as an ideal op amp in the phasor circuit, because the time-domain equations

$$i = 0, \qquad v = 0$$

which characterize the current into and the voltage across the input terminals retain the identical form,

$$\mathbb{I} = 0, \qquad \mathbb{V} = 0$$

in the phasor equations.

Phasor analysis is indeed a tool of great power and versatility. This may already be apparent from our work in this chapter and will be further underscored by our reliance on phasor analysis to explore important issues such as ac steady-state power in the next several chapters. Like any tool, however, its range of applicability is limited, and it is useful to remind ourselves of these limits.

The first caution is that phasors are useful for finding the *forced* response only. If the natural response is desired, another tool must be used. Also, if the total response is required, phasor analysis is useful in determining the forced component only. The second caution concerns the use of phasors to determine steady-state response. For many circuits it is the case that natural responses all decay to zero as $t \to \infty$. In such circuits the forced response and the steady-state response are synonymous, and phasors (designed for determining the forced response) may be used to find the steady-state response. Such circuits are called *stable*. For the remaining or *unstable* circuits, the natural response does not in general decay to zero, and thus the steady-state response cannot be identified with the

forced response. *Phasor analysis may be used to find the steady-state response only with stable circuits.* Examples of stable circuits include all circuits that are made up exclusively of passive elements and independent sources and that happen to have no undamped natural responses (no uncoupled LC subcircuits lacking resistance). Examples of unstable circuits are those with undamped natural response, and circuits possessing a natural response that grows, rather than decays, with time. Growing natural responses are possible only in linear circuits containing controlled sources. Stability is discussed in Chapter 14.

In general, unstable circuits do not possess steady-state responses at all (their natural responses either grow or oscillate periodically, so the circuit never goes to steady state no matter how long we wait). Thus it is meaningless to try to define a steady-state response for an unstable circuit, and for such circuits phasors may be used to compute the forced, but not the steady-state, response.

Example 8.17

Let us determine the describing equation for i in Fig. 8.23. The controlled source has transconductance $g = \frac{3}{2}$ S. By KVL around the left loop,

$$i + \int_{0+}^{t} \left(i - \frac{3v}{2}\right) d\tau + v_c(0+) + \frac{d}{dt}\left(i - \frac{3v}{2}\right) = +4\cos 3t$$

Substituting $v = i$ and differentiating yields

$$\frac{d^2i}{dt^2} - 2\frac{di}{dt} + i = -24\sin 3t$$

The natural response is found from the characteristic equation

$$s^2 - 2s + 1 = (s-1)^2 = 0$$

with repeated root at $s = +1$, so the natural response is of the form

$$i_n = K_1 e^t + K_2 t e^t$$

Note that except for the special case $K_1 = K_2 = 0$, all natural responses will grow rather than decay as $t \to \infty$, so the circuit cannot be expected to possess a steady state. This is an example of an unstable circuit. We may still use phasor analysis to determine the forced response, but since the circuit is unstable we will not equate the forced with the (nonexistent) steady-state response. To determine the forced response, consider the phasor circuit of Fig. 8.23(b). By KVL,

$$\mathbf{I} + \frac{j8}{3}\left(\mathbf{I} - \frac{3}{2}\mathbf{V}\right) = 4\underline{/0}\ \text{V}$$

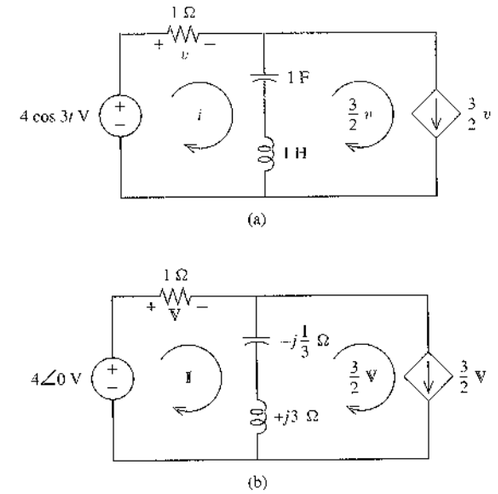
where we have summed the series impedances. Substituting $\mathbf{I} = \mathbf{V}$ and solving for $\mathbf{I}$ gives

$$\mathbf{I} = \frac{4\underline{/0}}{1 - j4/3} = 2.4\underline{/53.1^\circ}\ \text{A}$$

Thus the time-domain forced current is

$$i_f(t) = 2.4\cos(3t + 53.1^\circ)\ \text{A}$$

For almost all initial conditions the total current $i = i_n + i_f$ will become larger and larger with time, never attaining steady state.

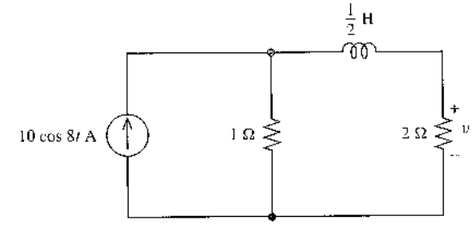


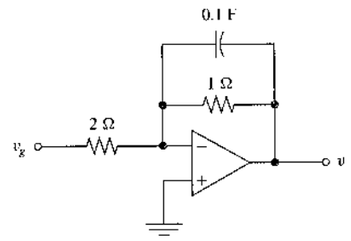
(a) Unstable circuit; (b) phasor circuit for computing forced (but not steady-state) response.

8.8.1. Solve Exercise 8.2.1 by means of the phasor circuit.

8.8.2. Find the steady-state voltage v using the phasor circuit.

Answer: $4\cos(8t - 53.1^\circ)$ V





8.8.3. Find the steady-state voltage v in Exercise 8.3.1 using the phasor circuit.

Answer: $2\cos(8t - 53.1^\circ)$

8.8.4. Find the steady-state voltage v using the phasor circuit given that $v_g = 4\cos 10t$ V.

Answer: $\sqrt{2}\cos(10t + 135^\circ)$ V

SUMMARY

The method of phasors permits efficient determination of the forced response in sinusoidal circuits while bypassing the difficulties of solving coupled differential equations. For stable circuits, the forced response is also the ac steady-state response. Ac steady-state reveals the long-term behavior of circuits excited by sinusoids, and as the excitation frequency varies, defines the circuit's frequency response.

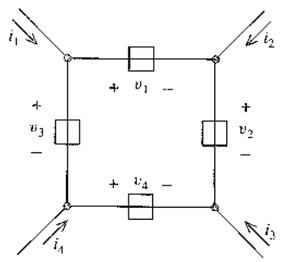
- If a sinusoidal source is replaced by a complex exponential source whose real part is identical to original sinusoidal, the real part of the response will be the response to the original sinusoidal source.
- A phasor is the (generally complex) number multiplying $e^{j\omega t}$ in the forced response of a circuit excited by complex exponentials.
- The amplitude of a sinusoid is the magnitude of the phasor, and the phase angle of a sinusoid is the angle of its phasor.
- The impedance Z of an RLC element is the ratio of its voltage phasor $\mathbb{V}$ to its current phasor $\mathbb{I}$. $Z_R = R$, $Z_L = j\omega L$, and $Z_C = 1/j\omega C$.
- Admittance Y is the inverse of impedance, $Y = 1/Z$.
- A phasor circuit is identical to the original circuit, except that RLC elements are labeled by their impedances, and each sinusoidal source is replaced by the phasor of its corresponding complex exponential source.
- The phasor circuit may be analyzed using KVL, KCL, and $\mathbb{V} = Z\mathbb{I}$ exactly as if it were a dc circuit that happened to have constant but complex resistances and sources. There are no time-varying quantities in the phasor domain.
- The sinusoidal response is just the real part of the phasor response after multiplication by $e^{j\omega t}$.
- A stable circuit is one whose unforced responses decay to zero for all values of its initial conditions.

8.1. Consider two currents $i_1 = 4\cos(2t + 17°)$ and $i_2 = -3\sin(2t + 0.71)$. Find their amplitudes, frequencies, phase angles in both degrees and radians, and periods. Sketch both on the same plot, determining which leads and by how many degrees.

8.2. Find the quadrature representations of the two currents of Problem 8.1.

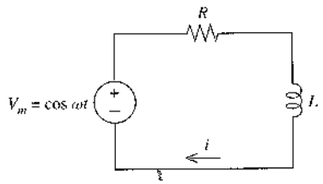
8.3. Using (8.8), find simple expressions for $\sin(\omega t + n\pi)$ and $\cos(\omega t + n\pi)$ where n is an integer. Show how your result follows from (8.8).

8.4. Suppose all currents and voltages have the same period. Find v_1 if $v_2 = 4\cos(3t - 70°)$, v_3 has the same amplitude as v_2 but leads v_2 by 28°, and v_4 has cosine quadrature component with amplitude 2 and sine component 0.

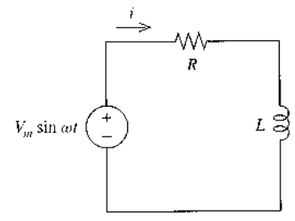


8.5. In the figure for Problem 8.4, is it true that the cosine quadrature components of the four currents sum to zero? The sine components? Justify.

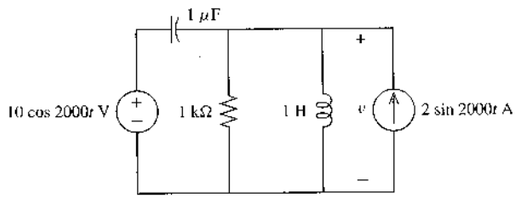
8.6. Find the forced response i if $L = 4$ mH, $R = 6$ kΩ, $V_m = 5$ V, and $\omega = 2 \times 10^6$ rad/s.



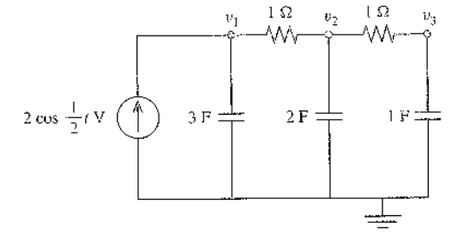
8.7. Write a differential equation for $i(t)$. Find the forced solution.



8.8. Write a differential equation for $v(t)$. Find the forced solution.



8.9. Write a single third-order differential equation for v_1 (*Hint*: Solve node equation 1 for v_2 and substitute into the other two node equations. Solve equation 3 for v_3.) Find the forced solution.



8.10. Sketch, in the complex plane, the locations of the complex exponential $(3/45°)e^{j4\pi t}$ at $t = 0$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$ and $\frac{1}{2}$ s.

8.11. If the complex exponential $Ae^{j\omega t}$ has value $2/80°$ at $t = 0$ and first reaches $2/-100°$ at $t = 2s$, what is its period? Find A and ω.

8.12. Express each complex number in three ways: rectangular form, polar form, and exponential polar form.

(a) $12/19°$
(b) $-40 + j18$
(c) $\frac{1}{4}/-180°$
(d) $4e^{j63°}$
(e) $16/90°$
(f) $14 + j2$

8.13. Evaluate the following expressions. Give your answer in polar form.

(a) $\frac{(4+j2)(2e^{-j18°})}{(6-j)^3}$
(b) $\frac{3/20°}{(2+j)+(3/120°)}$
(c) $(3 - j6)(3 - j6)^* + 4/18°$
(d) $[(2 - j2) + (7e^{j45°})][6/-25°]$

8.14. For each of the sinusoidal source functions specified, determine the complex exponential source that has given source as its *real* part.

(a) $4\cos 2t$
(b) $16\cos(100t - 26°)$
(c) $-3\sin(7t + 144°)$
(d) $2\cos(6t + 10°) + 3\sin(6t - 26°)$

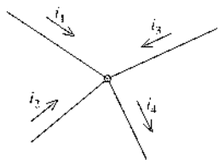
8.15. If the forced response to an excitation $v_g = f_1(t) + jf_2(t)$ in some circuit is $i(t) = i_1(t) + ji_2(t)$, where f_1, f_2, i_1, and i_2 are all real functions, what would the response to $v_{g_1} = -f_1(t)$ be? To $v_{g_2} = 3f_2(t) + j[f_1(t) - f_2(t)]$?

8.16. Using complex excitations and superposition, show that if the sinusoidal source $v_g = A\cos(\omega t + \phi_1)$ is delayed by θ radians, the response $i(t) = \beta\cos(\omega t + \phi_2)$ to v_g is also delayed by θ radians.

8.17. $$\frac{dx}{dt} + 2x = 4\cos 7t$$

Find the forced solution by replacing the sinusoidal forcing function by a complex exponential. Check by using the trial forced solution $A\cos 7t + B\sin 7t$.

8.18. Find i_4, using only the properties of sinusoids, if (a) $i_1 = 6\cos 3t$ A, $i_2 = 4\cos(3t - 30°)$ A, and $i_3 = -4\sqrt{3}\cos(3t + 60°)$ A, (b) $i_1 = 5\cos(3t + 30°)$ A, $i_2 = 5\sin 3t$ A, and $i_3 = 5\cos(3t + 150°)$ A, and (c) $i_1 = 25\cos(3t - 53.1°)$ A, $i_2 = 2\sin 3t$ A, and $i_3 = 13\cos(3t - 22.6°)$ A. (*Hint:* $\cos 22.6° = \frac{12}{13}$.)

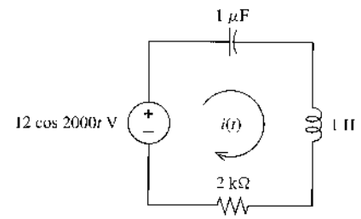


8.19. If the phasor $\mathbb{V}$ associated with a certain voltage $v(t)$ in some circuit is $\mathbb{V} = V_m\angle\theta_V$, what is the phasor if all independent sources are doubled? If all independent sources are lagged by 45°?

8.20. $$\frac{d^2x}{dt^2} + \frac{dx}{dt} - 6x = 4\sin t - 2\cos(t - 27.1°)$$

Find the forced solutions by replacing the sinusoidal forcing function by a complex exponential. Check by using the trial forced solution $A\cos t + B\sin t$.

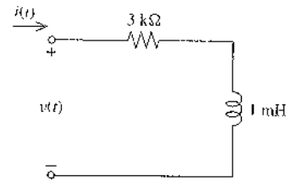
8.21. Write the describing equation as a second-order differential equation. Rewrite in terms of the phasor $\mathbb{I}$. Solve for $\mathbf{I}$, and then for the forced response $i(t)$.



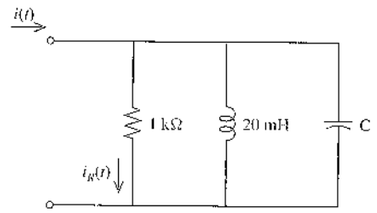
8.22. If $v(t) = 16\cos(4000t - 27°) + 3\sin(4000t - 114°)$ find the phasor associated with $v(t)$.

8.23. If $\omega = 20$ rad/s, find the time-domain functions represented by the phasors (a) $-5 + j5$, (b) $-4 - j3$, (c) $5 - j12$, (d) 10, and (e) $-j5$.

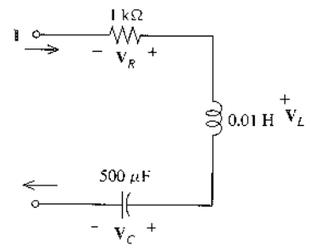
8.24. Find ω so that $i(t)$ lags $v(t)$ by 45°.



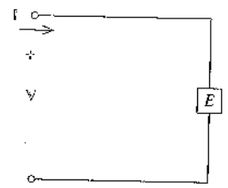
8.25. For what C will $i(t)$ lead $i_R(t)$ by 20° if the circuit is in steady state at frequency 2 kHz?



8.26. For phasor current $\mathbb{I} = 1\angle 0$ mA and $\omega = 1000$ rad/s, sketch $\mathbb{V}_R$, $\mathbb{V}_L$ and $\mathbb{V}_C$ in the complex plane.



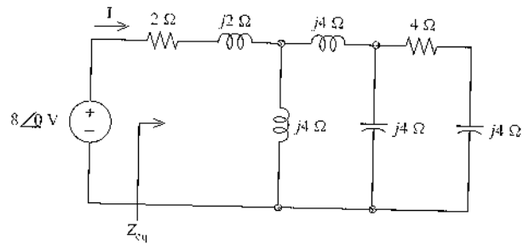
8.27. If $\omega = 1$ k rad, $\mathbb{V} = 20\angle 16°$ V and $\mathbb{I} = 6\angle -74°$ A, what single element is in E (type and value)?



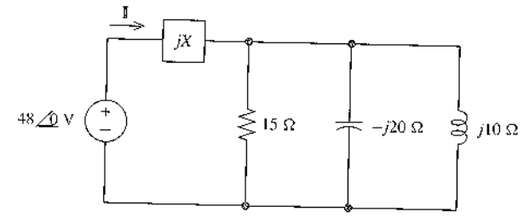
8.28. At what frequency ω are the magnitudes of the impedances of a 1-H inductor and 1-F capacitor equal?

8.29. Show that if $\text{Re}\ \mathbb{Z} = R$ is positive, then $\text{Re}\ \mathbb{Y} = \text{Re}[1/\mathbb{Z}] = G$ is also positive.

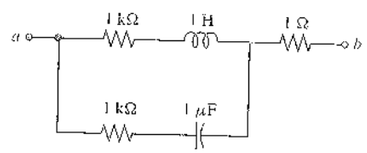
8.30. For the phasor circuit shown, find Z_{eq} and use the result to find the phasor current $\mathbb{I}$. If $\omega = 7$ rad/s, find the forced response i corresponding to $\mathbb{I}$.



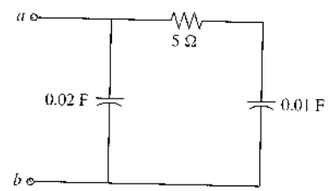
8.31. Find the reactance X so that the impedance seen by the source is real. For this case, find the steady-state current $i(t)$ corresponding to $\mathbb{I}$ if $\omega = 10$ rad/s.



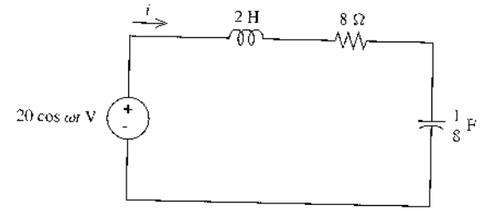
8.32. Find the impedance Z between a and b as a function of ω.



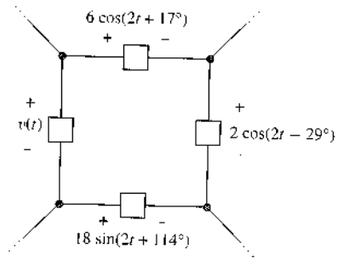
8.33. At what frequency ω is the susceptance B at terminals ab equal to $B = 2$ S?



8.34. Find the steady-state value of i if (a) $\omega = 1$ rad/s and (b) $\omega = 2$ rad/s. Note that in the latter case the impedance seen at the terminals of the source is purely resistive.

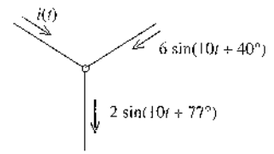


8.35. Convert all voltages to phasors. Use the fact that KVL holds for phasors to find $\mathbb{V}$, then $v(t)$.

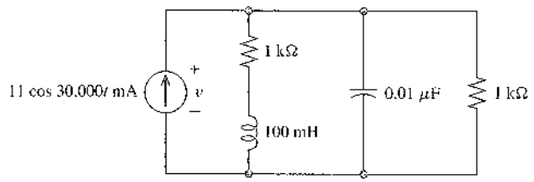


8.36. Repeat Problem 8.35 without using phasors.

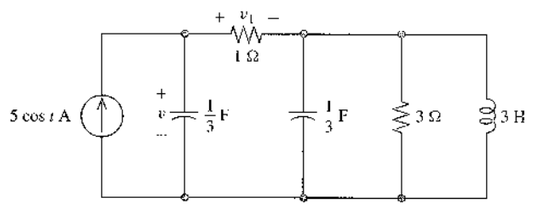
8.37. Convert all currents to phasors. Use the fact that KCL holds for phasors to find $\mathbb{I}$, then $i(t)$.



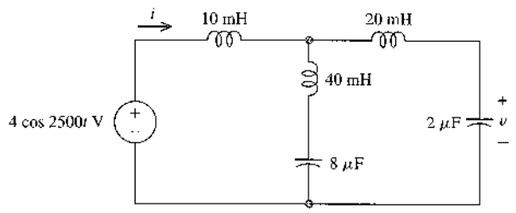
8.38. Find the steady-state value of v.



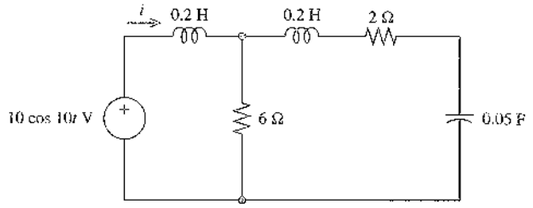
8.39. Find the steady-state values of v and v_1.



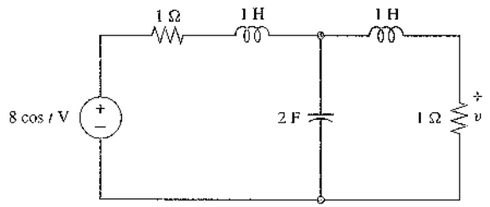
8.40. Find the steady-state values of i and v.



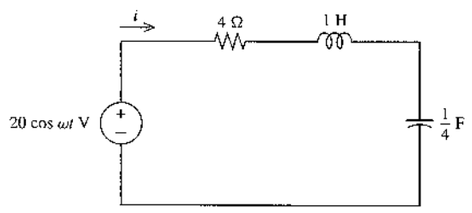
8.41. Find the steady-state current i.



8.42. Find the steady-state voltage v.

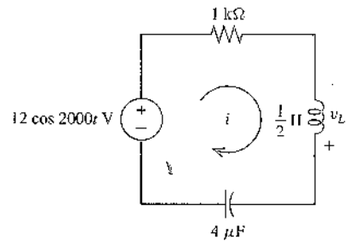


8.43. Find the steady-state value of i when (a) $\omega = 1$ rad/s, (b) $\omega = 2$ rad/s, and (c) $\omega = 4$ rad/s. [Note that (b) is the resonant case.]



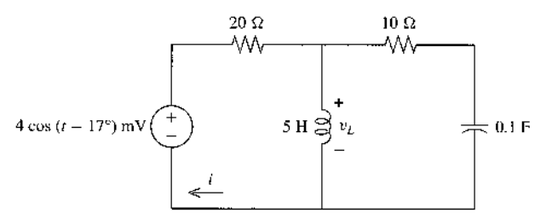
For Problems 8.44 through 8.48, sketch the phasor circuit and find the ac steady-state values of all indicated variables.

8.44.

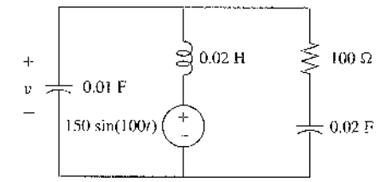


8.45.

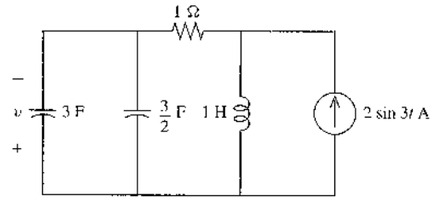
8.46.



8.47.



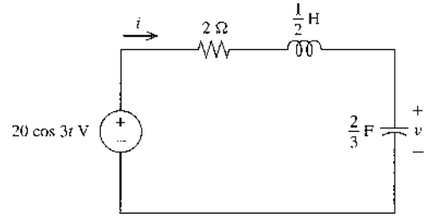
8.48.



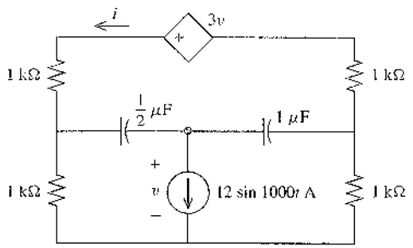
8.49. (a) Given a source $A\sin(\omega t + \phi)$ in an ac steady-state problem, suppose we use $A\angle\phi$ as the corresponding phasor source (rather than $A\angle\phi - 90°$ as prescribed in Table 8.1). It is no longer true that the ac steady-state response will be the real part of the complex response. How must this statement be modified, and why?

(b) Use this idea, sometimes called the "sine phasor" approach, to solve Problem 8.48 without using the 90° phase shift in the formula for the source phasor.

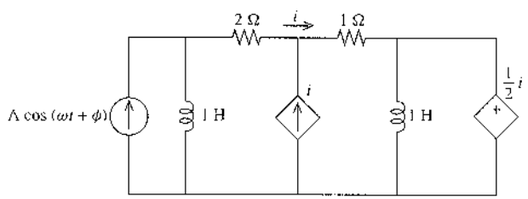
8.50. Find the *complete* response i if $i(0) = 2$ A and $v(0) = 6$ V. (*Suggestion:* Use phasors to get i_f and the differential equation to get i_n.)



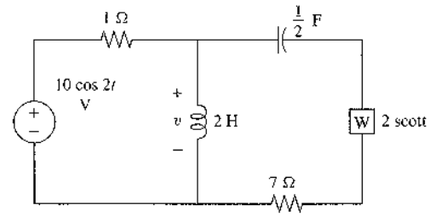
Find i.



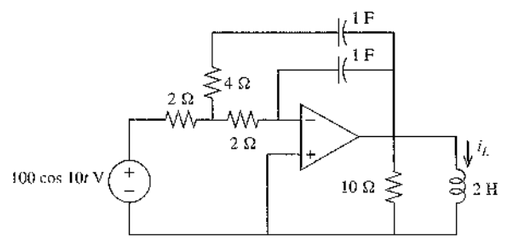
Find the ac steady-state value of $i(t)$ in terms of A and ϕ.



Suppose there is an element we call a *whizzer* which satisfies the element law $v = w\frac{d^2i}{dt^2}$. The constant w is called the whizzistance of the whizzer (units: scotts). What is the impedance of a w-scott whizzer? Derive laws for the equivalent whizzistance of whizzers in series and in parallel. The box is a 2-scott whizzer. Find the ac steady-state value of $v(t)$.



Find the ac steady-state value of $i_L(t)$.



9

AC Steady-State Analysis



Thomas Alva Edison
1847–1931

Genius is one per cent inspiration and ninety-nine per cent perspiration.

Thomas A. Edison

The greatest American inventor and perhaps the greatest inventor in history was Thomas Alva Edison, who changed the lives of people everywhere with such inventions as the electric light and the phonograph. He patented over 1100 inventions of his own and improved many other persons' inventions, such as the telephone, the typewriter, the electric generator, and the motion picture. Perhaps most important of all, he was one of the first to organize research, at one time employing some 3000 helpers.

Edison was born in Milan, Ohio, the youngest of seven children. He had only three months of formal education because his mother took him out of school and taught him herself. He asked too many questions to get along with the schoolmaster. He was exempt from military service because of deafness, and during the Civil War he roamed from city to city as a telegraph operator. During this time he patented improvements on the stock ticker and sold the patents for the then astounding price of $40,000. In 1876 he moved to Menlo Park, New Jersey, and from there his steady stream of inventions made him world famous. The electric light was his greatest invention, but to supply it to the world he also designed the first electric power station. His discovery of the Edison effect, the movement of electrons in the vacuum of his light bulb, also marked the beginning of the age of electronics.

9.1 *Circuit Simplifications*
9.2 *Nodal Analysis*
9.3 *Mesh Analysis*
9.4 *Sources with Different Frequencies*
9.5 *Phasor Diagrams*
Summary
Problems

In Chapter 8 we saw that the steady-state response of circuits excited by sinusoidal sources (ac circuits) could be efficiently computed by converting the time-domain circuit to its corresponding phasor circuit. The circuits studied were relatively simple and were analyzed directly from Kirchhoff's laws and use of the basic notion of impedance. In the present chapter we move beyond these relatively simple circuits. We shall see, in fact, that all the tools of circuit analysis introduced thus far apply equally well to phasor circuits as to their time-domain counterparts. First we consider circuit simplification methods: series–parallel source and impedance equivalents, Thevenin–Norton equivalents, and current–voltage division may all be applied directly to phasor circuits. In the subsequent two sections the general methods of nodal and mesh analysis are shown to carry over to phasor circuits without change. In Section 9.4 the use of superposition to study circuits excited by multiple sources of different frequencies is developed. Finally, phasor diagrams are introduced as effective graphical aids to understanding phase relationships in ac steady state, and the use of SPICE in ac steady state is discussed.

9.1 CIRCUIT SIMPLIFICATIONS

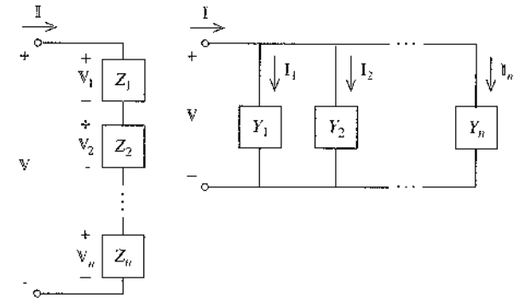
In Chapter 2 several useful methods for simplifying resistive circuits were introduced. These include series–parallel equivalents for resistors and for sources, current–voltage division, and Thevenin–Norton equivalents. Each of these methods was derived from the same foundation: Ohm's law describing the I–V behavior of individual elements and Kirchhoff's current and voltage laws governing their interconnection.

In ac steady state we may replace the original circuit by its phasor circuit counterpart, in which the sinusoidal sources are represented by their corresponding phasor sources and *RLC* elements by their impedances Z. In the phasor circuit the same

foundational laws are obeyed. Kirchhoff's laws apply equally well to phasors as to time-domain circuit variables, and the essence of Ohm's law, that current and voltage are linearly related through a simple constant of proportionality, holds equally well for impedances

$$\mathbb{V} = Z\mathbb{I}$$

where $\mathbb{V}$ and $\mathbb{I}$ are phasors. Since the laws upon which they are founded are equally valid for phasor circuits, we should anticipate that the same circuit simplifications derived for resistive circuits will hold for phasor circuits. Thus the full power of these methods may be brought to bear in ac steady-state circuit analysis, the only change being substitution of impedance Z for resistance R and the subsequent use of complex rather than real arithmetic.



(a) (b)

(a) Voltage division in phasor circuit; (b) current division.

We have already seen an important case of this principle in Chapter 8. Impedances in series were seen to be equivalent to a single impedance equal to the sum of the impedances, or more simply, series impedances add. Similarly, parallel admittances, in correspondence with parallel conductances in purely resistive circuits, also add. These equivalents may be used to derive the phasor form of current and voltage division. Consider the phasor circuit shown in Fig. 9.1(a). Since the n impedances are in series, $\mathbb{V} = Z\mathbb{I}$, where $Z = \mathbb{Z}_1 + \mathbb{Z}_2 + \cdots + \mathbb{Z}_n$. Then for each impedance, its voltage $\mathbb{V}_i = Z_i\mathbb{I}$, or

$$\mathbb{V}_i = \frac{Z_i}{Z_1 + Z_2 + \cdots + Z_n}\,\mathbb{V}$$

Comparing the voltage drops across two impedances gives Z_i and Z_j gives

$$\frac{\mathbb{V}_i}{\mathbb{V}_j} = \frac{Z_i}{Z_j}$$

The voltage across series impedances divides in direct proportion to their impedances.
Similarly, consider Fig. 9.1(b), in which the elements are labeled by their admittances Y_i. Since parallel admittances add, from $\mathbb{I} = Y\mathbb{V}$ with $Y = Y_1 + Y_2 + \cdots + Y_n$, we have $\mathbb{I}_i = Y_i\mathbb{V}$, or

$$\mathbb{I}_i = \frac{Y_i}{Y_1 + Y_2 + \cdots + Y_n}\mathbb{I}$$

and comparing two of these parallel currents,

$$\frac{\mathbb{I}_i}{\mathbb{I}_j} = \frac{Y_i}{Y_j}$$

The current through parallel impedances divides in direct proportion to their admittances.

Example 9.1

We seek $\mathbb{V}_1$ and $\mathbb{I}_2$ in the phasor circuit of Fig. 9.2. The impedances for each element are shown in the circuit diagram. To get $\mathbb{V}_1$, we use voltage division. The admittance of the parallel elements is

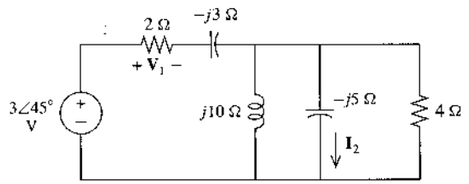
$$Y = -j\tfrac{1}{10} + j\tfrac{1}{5} + \tfrac{1}{4} = \tfrac{1}{4} + j\tfrac{1}{10}\ \text{S}$$

Thus the equivalent impedance of these elements is

$$Z = \frac{1}{Y} = 3.45 - j1.38\ \Omega$$

Then, by (9.2),

$$\begin{aligned}\mathbb{V}_1 &= \frac{2}{2 - j3 + (3.45 - j1.38)}(3\underline{/45^\circ}) \\ &= (0.286\underline{/38.8^\circ})(3\underline{/45^\circ}) \\ &= 0.858\underline{/83.8^\circ}\ \text{V}\end{aligned}$$



Circuit for Example 9.1.

We next use current division to get $\mathbb{I}_2$. The current into the parallel combination is $\mathbb{I} = \mathbb{V}_1/2$ or $\mathbb{I} = 0.429\underline{/83.8^\circ}$ A. Then, by (9.4),

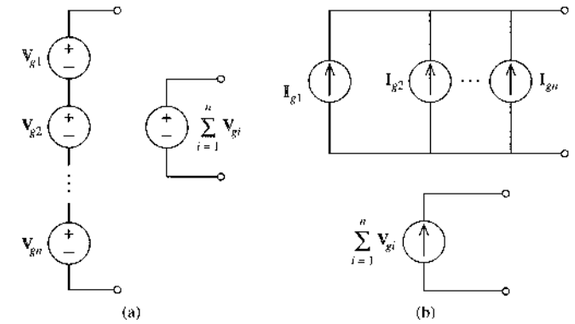
$$\begin{aligned}\mathbb{I}_2 &= \frac{j\frac{1}{5}}{\frac{1}{4} + j\frac{1}{10}}\mathbb{I} \\ &= (0.743\underline{/68.2^\circ})(0.429\underline{/83.8^\circ}) \\ &= 0.319\underline{/152^\circ}\ \text{A}\end{aligned}$$

Series and parallel equivalents may be found for sources as well as impedances in phasor circuits. Application of KVL in Fig. 9.3(a) shows that voltage sources in series are equivalent to a single source whose source function is the sum of the individual source functions, or *series voltage sources add.* Using KCL in Fig. 9.3(b), we have the corresponding result: *parallel current sources add.*

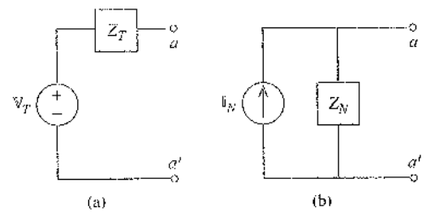
Thevenin and Norton equivalents in phasor circuits are found exactly as described in Chapter 2 for resistive circuits, with only the substitution of impedance Z for resistance R and subsequent use of complex arithmetic. Following the development of Section 2.6 with only this change, we have, from (2.20), that the Thevenin and Norton forms shown in Fig. 9.4 are equivalent if the relations

$$\text{(a)}\quad Z_T = Z_N$$
$$\text{(b)}\quad \mathbb{V}_T = Z_N\mathbb{I}_N$$

hold between the circuits. To find the Thevenin or Norton equivalent of any two-terminal subcircuit of a phasor circuit, we follow the prescription of (2.24), again only replacing R by Z. The two-terminal circuit A with open-circuit phasor voltage $\mathbb{V}_{oc}$ and short-circuit phasor current $\mathbb{I}_{sc}$ shown in Fig. 9.5 is equivalent to the Thevenin and Norton forms



(a) Series voltage sources and equivalent; (b) parallel current sources and equivalent.



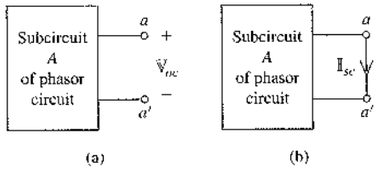
(a) Thevenin form; (b) Norton form.

shown in Fig. 9.4 if

$$\text{(a) } \mathbb{V}_T = \mathbb{V}_{oc}$$
$$\text{(b) } \mathbb{I}_N = \mathbb{I}_{sc}$$
$$\text{(c) } Z_T = Z_N = \frac{\mathbb{V}_{oc}}{\mathbb{I}_{sc}}$$

Z_T and Z_N can also be found as the impedance looking into terminals a–a' with all independent sources in the phasor circuit A killed.

Thevenin and Norton equivalents are used in phasor circuits to reduce complicated multielements circuits to simple two-element circuits, just as they were used in the resistive circuits studied previously. These equivalent circuits are even more generally useful in phasor circuits, since the restriction that the passive elements all be of the same type does not apply. Any mixture of *RLC* elements, that is, impedances, will do.

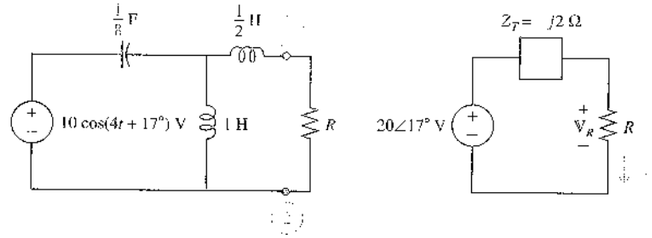


$\mathbb{V}_{oc}$ and $\mathbb{I}_{sc}$ for computing Thevenin and Norton equivalents of the circuit A.

Example 9.2

We wish to determine the value of R in Fig. 9.6(a) that will cause a sinusoidal current of amplitude 1 A to flow through this resistor. Our strategy will be to reduce this to the simple one-loop circuit shown in Fig. 9.6(b) by finding the Thevenin equivalent of everything except the resistor. The open-circuit voltage $\mathbb{V}_{oc} = \mathbb{V}_{12}$ with R removed is found by voltage division to be

$$\mathbb{V}_{oc} = \frac{j4(10/\underline{17^\circ})}{j4 - j2} = 20/\underline{17^\circ} \text{ V}$$



(a) Circuit for Example 9.2; (b) after Thevenin equivalent of circuit to left of terminals 1–2.

where $j4\ \Omega$ is the inductive impedance, $-j2\ \Omega$ is the capacitive, and the phasor source is $20/\underline{17^\circ}$ V. $\mathbb{V}_{oc}$ is the Thevenin source shown in Fig. 9.6(b). The short-circuit current is found by shorting terminals 1 and 2 together. The two inductors in this case are in parallel, and are equivalent to an inductance of $\frac{1}{3}$ H, with impedance

$$Z_L = j\omega L = j\tfrac{4}{3}\ \Omega$$

The common voltage across these inductors is, by voltage division,

$$\mathbb{V}_L = \frac{j4/3}{(j\frac{4}{3} - j2)(10/\underline{17^\circ})} = 20/\underline{-163^\circ} \text{ V}$$

The current through the $\frac{1}{2}$-H inductor is the short-circuit current

$$\mathbb{I}_{sc} = \frac{\mathbb{V}_L}{j(4)(\frac{1}{2})} = \frac{20/\underline{-163^\circ}}{(2/\underline{90^\circ})} = 10/\underline{107^\circ} \text{ A}$$

Thus Z_T in Fig. 9.6(b) is found to be

$$Z_T = \frac{\mathbb{V}_{oc}}{\mathbb{I}_{sc}} = \frac{20/\underline{17^\circ}}{10/\underline{107^\circ}} = 2/\underline{-90^\circ}\ \Omega$$

Having reduced the problem to one with a simple circuit, we turn to Fig. 9.6(b). The current through the resistor is

$$\mathbb{I}_R = \frac{20/\underline{17^\circ}}{R - j2}$$

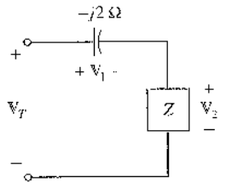
The magnitude of the phasor $\mathbb{I}_R$ is given by the ratio of magnitudes of its numerator and denominator,

$$|\mathbb{I}_R| = \frac{20}{\sqrt{R^2 + 4}}$$

If $|\mathbb{I}_R| = 1$ A, the amplitude of the corresponding sinusoidal current i_R will be 1 A as well. Thus we require that

$$\frac{20}{\sqrt{R^2 + 4}} = 1$$

or $R = 6\sqrt{11}\ \Omega$.

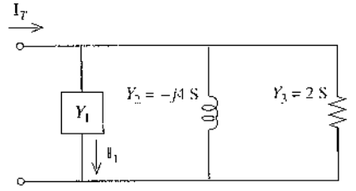


9.1.1. (a) Find $\mathbf{V}_2$ for $Z = 1 + j2\ \Omega$ and $\mathbf{V}_T = 1\underline{/0}$ V. (b) If $\mathbf{V}_1 = 3 + j$ V, for what Z will $\mathbf{V}_2 = 7 - j3$ V?

(a) $2.24\underline{/63.4^\circ}$ V; (b) $4.82\underline{/-132^\circ}$ V

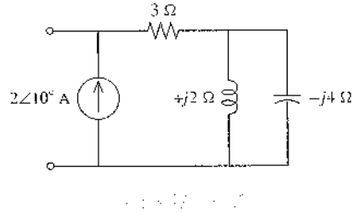
9.1.2. (a) Find $\mathbf{I}_1$ for $Y_1 = 2 - j$ S and $\mathbf{I}_T = 1\underline{/0}$ A. (b) If $\mathbf{I}_3 = 3\underline{/18^\circ}$ A, for what Y_1 will $\mathbf{I}_1 = 4 - j$ A?

(a) $0.35\underline{/24.8^\circ}$ A; (b) $2.75\underline{/-32.0^\circ}$ S



9.1.3. Find the Norton equivalent of the circuit shown.

$Z_N = 3 + j4\ \Omega$; $\mathbf{I}_N = 2\underline{/10^\circ}$ A



9.1.4. Find the Thevenin equivalent of the circuit of Exercise 9.1.3.

$Z_T = 3 + j4\ \Omega$; $\mathbf{V}_T = 10\underline{/63.1^\circ}$ V

9.2 NODAL ANALYSIS

As we have seen, the voltage–current relation (9.1)

$$\mathbf{V} = Z\mathbf{I}$$

for passive elements is identical in form to Ohm's law, and KVL and KCL hold in phasor circuits exactly as they did in resistive circuits. Therefore, the only difference in analyzing phasor circuits and resistive circuits is that the excitations and responses are complex quantities in the former case and real quantities in the latter case. Thus we may analyze phasor circuits in exactly the same manner in which we analyzed resistive circuits. Specifically, nodal and mesh analysis methods apply. We shall illustrate nodal analysis in this section and mesh analysis in the following section.

Example 9.3

To illustrate the nodal method, let us find the ac steady-state voltages v_1 and v_2 of Fig. 9.7. First we obtain the phasor circuit by replacing the element values by their impedances for $\omega = 2$ rad/s and the sources and node voltages by their phasors. This results in the circuit of Fig. 9.8(a). Since we are interested in finding $\mathbf{V}_1$ and $\mathbf{V}_2$, the node voltage phasors, we may replace the two sets of parallel impedances by their equivalent impedances, resulting in the simpler equivalent circuit of Fig. 9.8(b).

The nodal equations, from Fig. 9.8(b), are

$$2(\mathbf{V}_1 - 5\underline{/0}) + \frac{\mathbf{V}_1}{j1} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{-j1} = 0$$

$$\frac{\mathbf{V}_2 - \mathbf{V}_1}{-j1} + \frac{\mathbf{V}_2}{(1 + j2)/5} = 5\underline{/0}$$

which in simplified form are

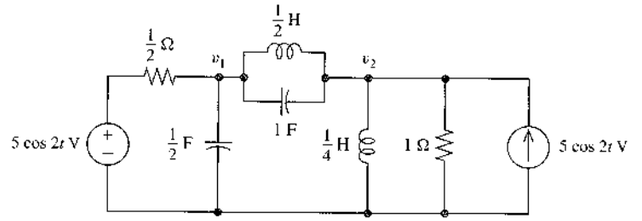
$$(2 + j2)\mathbf{V}_1 - j1\mathbf{V}_2 = 10$$

$$-j1\mathbf{V}_1 + (1 - j1)\mathbf{V}_2 = 5$$

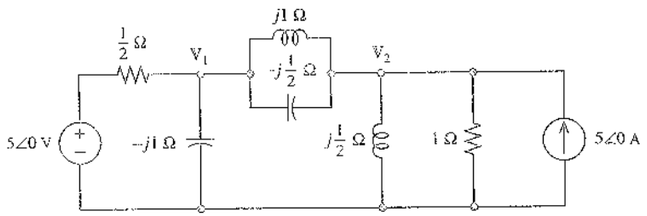
Solving these equations by Cramer's rule, we have

$$\mathbf{V}_1 = \frac{\begin{vmatrix} 10 & -j1 \\ 5 & 1 - j1 \end{vmatrix}}{\begin{vmatrix} 2 + j2 & -j1 \\ -j1 & 1 - j1 \end{vmatrix}} = \frac{10 - j5}{5} = 2 - j1 \text{ V}$$

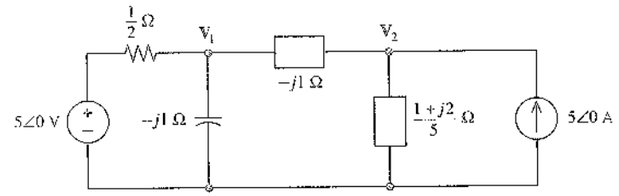
$$\mathbf{V}_2 = \frac{\begin{vmatrix} 2 + j2 & 10 \\ -j1 & 5 \end{vmatrix}}{5} = \frac{10 + j20}{5} = 2 + j4 \text{ V}$$



Circuit to be analyzed by the phasor method.



(a)



(b)

Two versions of the phasor circuit corresponding to Fig. 9.7.

In polar form these quantities are

$$\mathbb{V}_1 = \sqrt{5}/\underline{-26.6^\circ} \text{ V}$$

$$\mathbb{V}_2 = 2\sqrt{5}/\underline{63.4^\circ} \text{ V}$$

Therefore, the time-domain solutions are

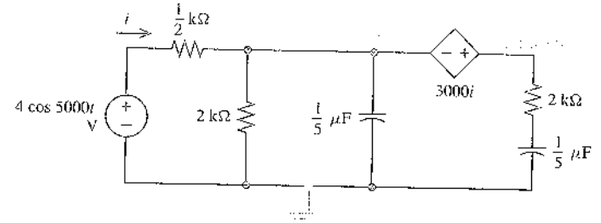
$$v_1 = \sqrt{5}\cos(2t - 26.6^\circ) \text{ V}$$

$$v_2 = 2\sqrt{5}\cos(2t + 63.4^\circ) \text{ V}$$

Example 9.3 illustrates an important feature of phasor analysis. By reducing element laws for storage elements that involve integrals and derivatives to the simpler $\mathbb{V} = Z\mathbb{I}$, we replace *calculus* by *algebra*. The subsequent nodal or mesh equations may always be formulated as a single vector-matrix equation and solved by the general methods of linear algebra rather than the more complicated methods involving differential equations. Of course, for simple circuits we need not rewrite as vector-matrix equations if we choose, but this general method will always be available for all phasor analysis problems.

Example 9.4

As an example involving a dependent source, let us consider Fig. 9.9, in which it is required to find the forced response i. Taking the ground node as shown, we have the two unknown node voltages v and $v + 3000i$, as indicated. The phasor circuit in its simplest form is shown in Fig. 9.10, from which we may observe that only one nodal equation is needed. Writing KCL at the supernode, shown



Circuit containing a dependent source.

dashed, we have

$$\frac{\mathbb{V} - 4}{\frac{1}{2}(10^3)} + \frac{\mathbb{V}}{\frac{2}{5}(1 - j2)(10^3)} + \frac{\mathbb{V} + 3000\mathbb{I}}{(2 - j1)(10^3)} = 0$$

Note that the gain of the dependent source is 3000 V/A. Also, from the phasor circuit we have

$$\mathbb{I} = \frac{4 - \mathbb{V}}{\frac{1}{2}(10^3)}$$

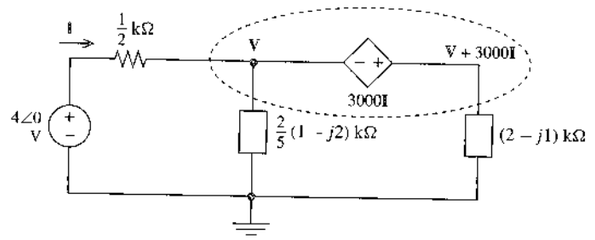
Eliminating $\mathbb{V}$ between these two equations and solving for $\mathbb{I}$, we have

$$\mathbb{I} = 24 \times 10^{-3}/\underline{53.1^\circ} \text{ A}$$

$$= 24/\underline{53.1^\circ} \text{ mA}$$

Therefore, in the time domain, we have

$$i = 24\cos(5000t + 53.1^\circ) \text{ mA}$$



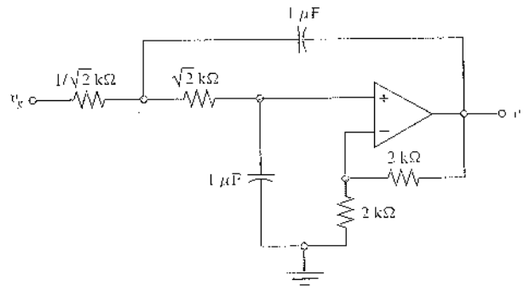
Phasor circuit of Fig. 9.9.

Example 9.5

As a final example illustrating nodal analysis, let us find the forced response v in Fig. 9.11 if

$$v_g = V_m \cos \omega t \text{ V}$$

We note first that the op amp and the two 2-kΩ resistors constitute a noninverting amplifier with gain $1 + \frac{2000}{2000} = 2$ (see Section 3.5).



Circuit containing an op amp.

Therefore, $v = 2v_2$, or $v_2 = v/2$, as indicated by the phasor $\mathbb{V}/2$ in the phasor circuit of Fig. 9.12.

Writing nodal equations at the nodes labeled $\mathbb{V}_1$ and $\mathbb{V}/2$, we have

$$\frac{\mathbb{V}_1 - V_m\underline{/0}}{(1/\sqrt{2})(10^3)} + \frac{\mathbb{V}_1 - (\mathbb{V}/2)}{\sqrt{2}(10^3)} + \frac{\mathbb{V}_1 - \mathbb{V}}{-j10^6/\omega} = 0$$

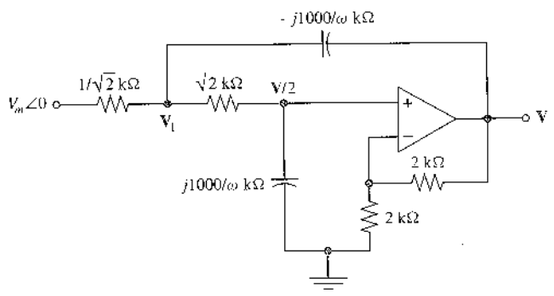
$$\frac{(\mathbb{V}/2) - \mathbb{V}_1}{\sqrt{2}(10^3)} + \frac{\mathbb{V}/2}{-j10^6/\omega} = 0$$

Eliminating $\mathbb{V}_1$, by solving the last for $\mathbb{V}_1$ and substituting into the previous equation, and solving for $\mathbb{V}$ results in

$$\mathbb{V} = \frac{2V_m}{[1 - (\omega^2/10^6)] + j(\sqrt{2}\omega/10^3)}$$

In polar form this is

$$\mathbb{V} = \frac{2V_m\underline{/\theta}}{\sqrt{1 + (\omega/1000)^4}}$$



Phasor circuit of Fig. 9.11.

where

$$\theta = -\tan^{-1}\frac{\sqrt{2}\omega/1000}{1 - (\omega/1000)^2}$$

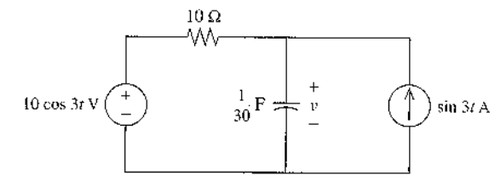
In the time domain we have

$$v = \frac{2V_m}{\sqrt{1 + (\omega/1000)^4}}\cos(\omega t + \theta)$$

We might note in this example that for low frequencies, say $0 < \omega < 1000$, the amplitude of the output voltage v is relatively large, and for higher frequencies, its amplitude is relatively small. Thus the circuit of Fig. 9.11 *filters* out higher frequencies and allows lower frequencies to "pass." Such a circuit is called a *filter* and is considered in more detail in Chapter 14.

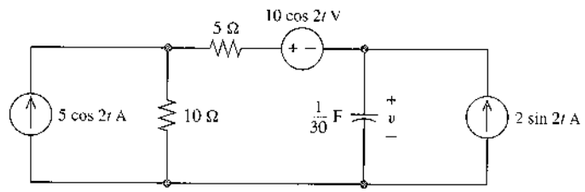
9.2.1. Find the forced response v using nodal analysis.

Answer: $10\sin 3t$ V



9.2.2. Find the steady-state value of v using nodal analysis.

Answer: $25\sqrt{2}\cos(2t - 81.9°)$ V

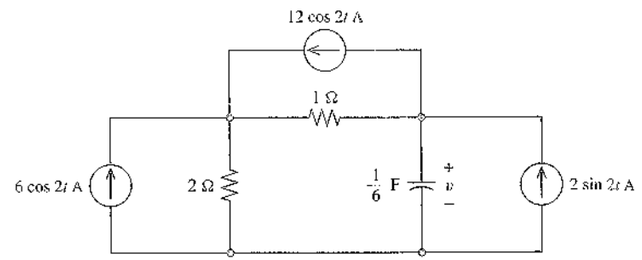


9.2.3. Find the amplitude of v in Example 9.5 if $V_m = 10$ V for (a) $\omega = 0$, (b) $\omega = 1000$ rad/s, (c) $\omega = 10{,}000$ rad/s, and (d) $\omega = 100{,}000$ rad/s.

Answer: (a) 20; (b) 14.14; (c) 0.2; (d) 0.002 V

9.2.4. Find the steady-state value of v using nodal analysis.

Answer: $3\sqrt{2}\cos(2t - 135°)$ V



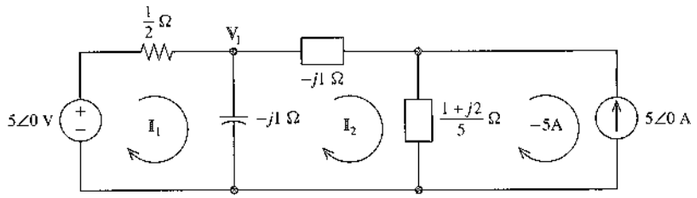
9.3 MESH ANALYSIS

As suggested in Section 9.2, the general methods of circuit analysis apply to phasor circuits as if they were resistive circuits, with resistance R replaced by the more general impedance Z. In this section we illustrate the application of mesh analysis to phasor circuits. Once again we note that the generalized form of Ohm's law using impedance $\mathbb{V} = Z\mathbb{I}$ permits replacement of calculus (differentiation and integration) by the simpler operations of algebra (multiplication and division).

Example 9.6

To illustrate mesh analysis of an ac steady-state circuit, let us find v_1 in Fig. 9.7, which was obtained, using nodal analysis, in Section 9.2. We shall use the phasor circuit of Fig. 9.8(b), which is redrawn in Fig. 9.13, with mesh currents $\mathbb{I}_1$ and $\mathbb{I}_2$, as indicated. Once the circuit is broken, the phasor voltage $\mathbb{V}_1$ may be obtained as

$$\mathbb{V}_1 = 5 - \frac{\mathbb{I}_1}{2} \qquad (9.\;\;)$$



Circuit of Fig. 9.8 redrawn for mesh analysis.

The two mesh equations are

$$\frac{1}{2}\mathbb{I}_1 - j1(\mathbb{I}_1 - \mathbb{I}_2) = 5$$

$$-j1(\mathbb{I}_2 - \mathbb{I}_1) - j1\mathbb{I}_2 + \left(\frac{1 + j2}{5}\right)(\mathbb{I}_2 + 5) = 0$$

Solving these equations for $\mathbb{I}_1$, we have

$$\mathbb{I}_1 = 6 + j2 \text{ A}$$

which substituted into (9.10) yields

$$\mathbb{V}_1 = 2 - j1 \text{ V}$$

This is the same result that was obtained in Section 9.2 and may be used to obtain the time-domain voltage v_1.

The same shortcut procedures for writing nodal and mesh equations discussed in Sections 4.3 and 4.5 for resistive circuits apply to phasor circuits. For example, in Fig. 9.13, if $\mathbb{I}_3 = -5$ is the mesh current in the right mesh in the clockwise direction, the two mesh equations are written by inspection as

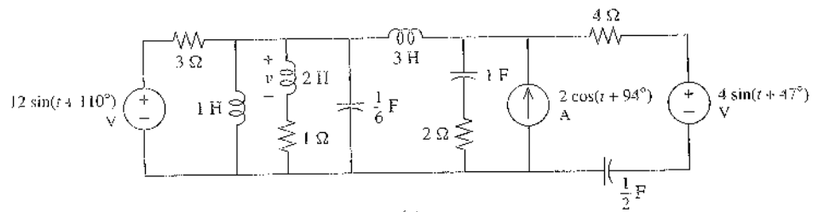
$$\left(\frac{1}{2} - j1\right)\mathbb{I}_1 - (-j1)\mathbb{I}_2 = 5$$

$$-(-j1)\mathbb{I}_1 + \left(-j1 - j1 + \frac{1 + j2}{5}\right)\mathbb{I}_2 - \left(\frac{1 + j2}{5}\right)\mathbb{I}_3 = 0$$

These are equivalent to (9.11) and are formed exactly as in the resistive circuit case. That is, in the first equation the coefficient of the first variable is the sum of the impedances around the first mesh. The other coefficients are the negatives of the impedances common to the first mesh and the meshes whose numbers correspond to the currents. The right member is the sum of the voltage sources in the mesh with polarities consistent with the direction of the mesh current. Replacing "first" by "second" applies to the next equation, and so on. The dual development, as described in Section 4.3, holds for nodal equations.

Example 9.7

This example illustrates that series–parallel impedance conversions, together with Thevenin–Norton transformations, may be of great benefit in simplifying a circuit before writing the general analysis equations (nodal or mesh). We shall apply mesh analysis to the circuit of Fig. 9.14(a), in which the desired response is the voltage v across the 2-H inductor. Since this circuit contains six meshes and one current source, simultaneous solution of five equations in five unknowns would be required if the mesh equations were written directly. By series–parallel conversions, the circuit may be redrawn



(a) Circuit for Example 9.7; (b) corresponding phasor circuit.

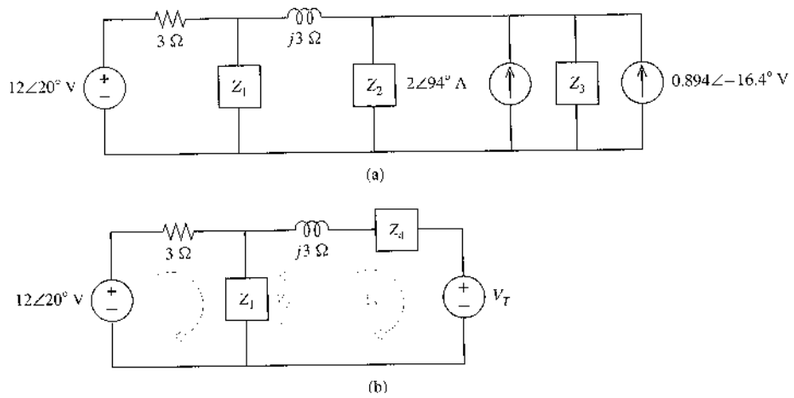
as in Fig. 9.15(a), where

$$Z_1 = j1 \| (1 + j2) \| (-j6) = 0.128 + j0.79\ \Omega$$

$$Z_2 = 2 - j1\ \Omega$$

$$Z_3 = 4 - j2\ \Omega$$

We have also converted the series combination of Z_3 and the $4/\underline{-43^\circ}$ V voltage source in Fig. 9.14(b) to the Norton form shown



(a) Simplified phasor circuit; (b) further simplified.

in Fig. 9.15(a), where

$$\mathbb{I}_N = \frac{4/\underline{-43^\circ}}{Z_3} = 0.894/\underline{-16.4^\circ}\ \text{A}$$

As a final simplification, we combine the parallel impedances Z_2 and Z_3 and the parallel current sources, and convert the result back to the Thevenin form, leading to the circuit shown in Fig. 9.15(b).

$$Z_4 = Z_2 \| Z_3 = \frac{(2 - j1)(4 - j2)}{6 - j3} = \frac{4}{3} - j\frac{2}{3}\ \Omega$$

$$\mathbb{V}_T = Z_4(2/\underline{94^\circ} + 0.894/\underline{-16.4^\circ}) = 2.81/\underline{41^\circ}\ \text{V}$$

The mesh equations, in vector-matrix form, are

$$\begin{bmatrix} 3.13 + j0.79 & -0.128 - j0.79 \\ -0.128 - j0.79 & 1.45 + j3.12 \end{bmatrix} \begin{bmatrix} \mathbb{I}_1 \\ \mathbb{I}_2 \end{bmatrix} = \begin{bmatrix} 12/\underline{20^\circ} \\ 2.81/\underline{-139^\circ} \end{bmatrix}$$

The determinant of the matrix is

$$\Delta = (3.13 + j0.79)(1.45 + j3.12) - (-0.128 - j0.79)^2 = 11.1/\underline{75.8^\circ}$$

Thus

$$\begin{bmatrix} \mathbb{I}_1 \\ \mathbb{I}_2 \end{bmatrix} = \frac{1}{11.1/\underline{75.8^\circ}} \begin{bmatrix} 1.45 + j3.12 & 0.128 + j0.79 \\ 0.128 - j0.79 & 3.13 + j0.79 \end{bmatrix} \begin{bmatrix} 12/\underline{20^\circ} \\ 2.81/\underline{-139^\circ} \end{bmatrix}$$

$$= \begin{bmatrix} 3.58/\underline{7.2^\circ} \\ 0.657/\underline{88.3^\circ} \end{bmatrix}$$

Having broken the circuit by finding the mesh currents, the desired unknowns are next found in terms of the mesh currents. The voltage across Z_1 in Fig. 9.15(b) is

$$\mathbb{V}_{Z1} = Z_1(\mathbb{I}_1 - \mathbb{I}_2) = (0.128 + j0.79)(3.58/\underline{7.2^\circ} - 0.657/\underline{88.3^\circ})$$

$$= 2.83/\underline{77.4^\circ}\ \text{V}$$

From Fig. 9.14 we see that this voltage is across an impedance of $j2\ \Omega$ (the desired voltage, $\mathbb{V}$) in series with another impedance of $1\ \Omega$. By voltage division,

$$\mathbb{V} = \frac{j2}{1 + j2} 2.83/\underline{77.4^\circ} = 2.53/\underline{104^\circ}\ \text{V}$$

which is the desired unknown phasor. The final time-domain result is

$$v(t) = 2.53\cos(t + 104^\circ)\ \text{V}$$

While some effort was required to arrive at this result, it should be remembered that the original circuit had a total of 10 passive elements and 3 independent sources distributed in 6 meshes. Had we tried to solve the phasor circuit directly with no simplifications,

or worse had we not used phasors at all, much more computational effort would have been required.

Example 9.8

As a final example, let us consider the circuit of Fig. 9.16(a), where the response is the steady-state value of v_1. The controlled source has transconductance $g = 2$ S. The phasor circuit is shown in Fig. 9.16(b), with the mesh currents as indicated.

Applying KVL around the supermesh labeled I, we have

$$-\mathbb{V}_1 - j1(-j1 + \mathbb{I}) + (1 + j2)(\mathbb{I} + 2\mathbb{V}_1) = 0$$

Also, from the figure we see that

$$\mathbb{V}_1 = j1(4 - \mathbb{I})$$

Eliminating $\mathbb{I}$ from these equations and solving for $\mathbb{V}_1$, we have

$$\mathbb{V}_1 = \frac{-4 + j3}{5} = 1\underline{/143.1^\circ} \text{ V}$$

Therefore, in the time domain, the voltage is

$$v_1 = \cos(2t + 143.1^\circ) \text{ V}$$

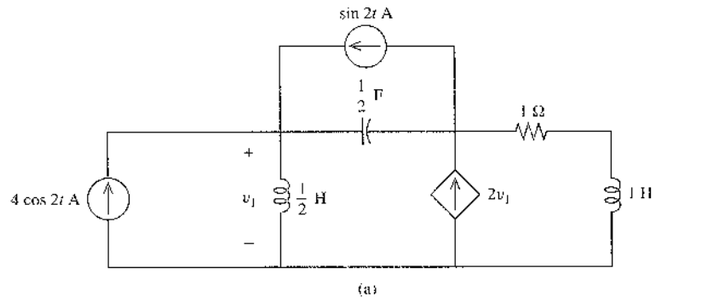
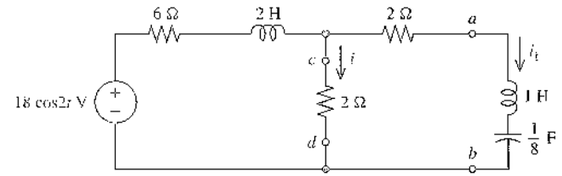


FIGURE 9.16 (a) Time-domain circuit; (b) phasor counterpart.

9.3.1. Find the forced response i in Fig. 9.9 using mesh analysis.

9.3.2. Solve Exercise 9.2.4 using mesh analysis.

9.3.3. Find the steady-state current i using mesh analysis.
Answer: $\sqrt{2}\cos(2t - 45^\circ)$ A



EXERCISE 9.3.3

9.4 SOURCES WITH DIFFERENT FREQUENCIES

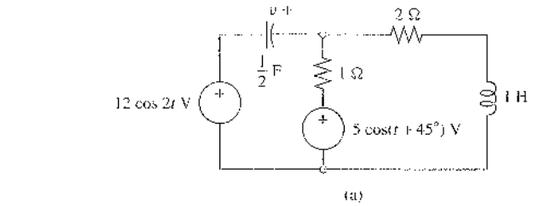
An ac circuit is, we recall, a circuit whose independent sources are all sinusoids. If all sources in an ac circuit are of the same frequency ω, the corresponding phasor circuit may be used to determine the forced response in the manner described in Sections 9.1 to 9.3. It may be computed by a single unified nodal or mesh analysis or by use of superposition (summing the component responses to each individual source or group of sources with all other independent sources killed).

Superposition is a general principle that may always be used to find the response of a linear circuit containing more than one source. Even if an ac circuit contains sources with different frequencies, superposition may still be used to find the forced response. For purposes of superposition, the sources are grouped so that each component problem contains only sources of a single frequency. Then, for each resulting component problem, a phasor circuit may be used to determine the phasor response, which is then converted to a sinusoidal response and added with the other component responses as the principle of superposition requires.

Example 9.9

To find v in Fig. 9.17(a), we will use superposition. The component phasor circuits are shown in Fig. 9.17(b) and (c). Note that the impedances in the two component problems are different for all elements but the resistors, since for the other (storage) elements impedance is frequency dependent. By voltage division,

$$\mathbb{V}_1 = -\left(\frac{-j1}{-j1 + Z_a}\right)(12/0) \qquad (9.4.2)$$



(a) Ac circuit; (b) phasor circuit for the $\omega = 2$ component; (c) phasor circuit for the $\omega = 1$ component.

where Z_a is the equivalent of the rightmost three impedances in Fig. 9.17(b):

$$Z_a = \frac{(1)(2 + j2)}{3 + j2} = 0.769 + j0.154\ \Omega$$

Using this in (9.12), $\mathbb{V}_1 = 10.5/\underline{138^\circ}$ V. Thus

$$v_1 = 10.5\cos(2t + 138^\circ)\ \text{V}$$

Turning to Fig. 9.17(c), again by voltage division,

$$\mathbb{V}_2 = \frac{Z_b}{Z_b + 1}(5/\underline{45^\circ})$$

where Z_b is the parallel equivalent of the impedances $-j2$ and $2 + j1$, or

$$Z_b = \frac{(-j2)(2 + j1)}{2 - j1} = 1.6 - j1.2\ \Omega$$

which, upon substitution, yields

$$\mathbb{V}_2 = \frac{1.6 - j1.2}{2.6 - j1.2}(5/\underline{45^\circ}) = 3.5/32.9^\circ\ \text{V}$$

The corresponding sinusoidal component is

$$v_2 = 3.5\cos(t + 32.9^\circ)\ \text{V}$$

and by superposition, the forced response v is the sum of its components:

$$v = v_1 + v_2 = 10.5\cos(2t + 138^\circ) + 3.5\cos(t + 32.9^\circ)\ \text{V}$$

Note in Example 9.9 that $\mathbb{V}_1$ and $\mathbb{V}_2$ were converted to sinusoids before being added together. *Phasors corresponding to different frequencies cannot be superposed; only their corresponding sinusoids can be superposed.* Recall that a given phasor corresponds to amplitude and phase information on a sinusoid of a specific frequency ω. If the frequencies of two phasors differ, it makes no sense to add them together. The magnitude of their phasor sum does not correspond to any sinusoidal amplitude, nor does the angle of their sum to any sinusoidal phase angle. When frequencies differ, the principle of superposition applies to the summing of time-domain components, not phasors. Within a component problem corresponding to a single frequency, however, phasors may also be superposed. This is illustrated in the next example.

Example 9.10

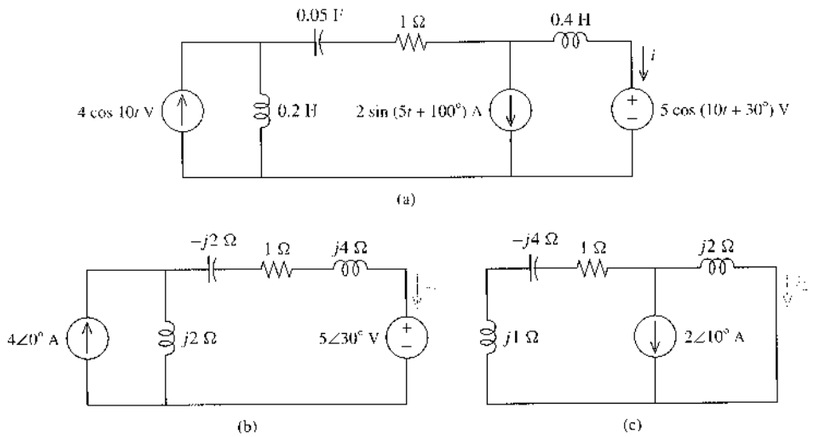
We seek the current i through the voltage source in the ac circuit shown in Fig. 9.18(a). We will use superposition, grouping the two sources at $\omega = 10$ rad/s together and calling this component of the response i_1; the remaining component due to the source at $\omega = 5$ rad/s is i_2. Figure 9.18(b) shows the phasor circuit for computing i_1 and Fig. 9.18(c) that for i_2. In each case the other sources have been killed, and the value of ω corresponding to the active sources used to compute the impedances. Using superposition, each component problem may be set up and solved independently.

The right mesh equation in Fig. 9.18(b) is

$$\mathbb{I}_1(-j2 + 1 + j4) + j2(\mathbb{I}_1 - 4/\underline{0}) = 5/\underline{30^\circ}\ \text{A}$$

or

$$\mathbb{I}_1 = \frac{5/\underline{30^\circ} + j8}{1 + j4} = 2.76/\underline{-8.4^\circ}\ \text{A}$$



(a) Ac circuit; (b) phasor circuit for the $\omega = 10$ component; (c) phasor circuit for the $\omega = 5$ component.

The corresponding ac component is

$$i_1 = 2.76\cos(10t - 8.4^\circ)\ \text{A}$$

Turning to the other component, use of current division in Fig. 9.18(c) yields

$$\mathbb{I}_2 = \frac{1 - j3}{1 - j1}(-2\underline{/10^\circ}) = 4.47\underline{/163^\circ}\ \text{A}$$

The second ac component is then

$$i_2 = 4.47\cos(5t + 163^\circ)\ \text{A}$$

The overall forced response, or ac steady-state response, is the sum of these components:

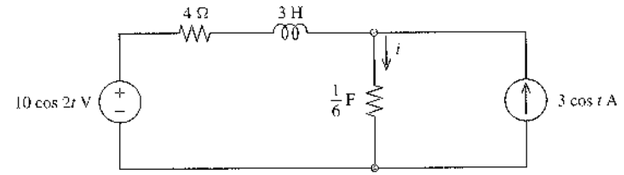
$$i = i_1 + i_2 = 2.76\cos(10t - 8.4^\circ) + 4.47\cos(5t + 163^\circ)\ \text{A}$$

In Example 9.10 we computed one component for each source frequency and summed these components. Note that since there are two sources in the phasor circuit Fig. 9.18(b) corresponding to the frequency $\omega = 10$ rad/s, we may choose to solve this component problem by once again invoking the principle of superposition. We could compute the subcomponents of $\mathbb{I}_1$ due to each of the two sources with the other killed and then sum these two phasors to get $\mathbb{I}_1$. Superposition may be freely applied within a single-phasor circuit, that is, one in which there is a single frequency ω.

In this section we have seen that superposition may be used to decompose ac steady-state problems involving independent sources at more than one frequency into component problems each containing sources at the same frequency. The component problems may then be solved with the help of phasors. In other instances thus far where superposition was discussed, it was also possible to choose to bypass superposition, solving in a single unified manner with all sources in place. This same choice is also available for ac circuits with multiple frequencies; but if we are to use phasors, we cannot choose to solve the problem all at once. The difficulty is that a single unified phasor circuit involving sources of different frequencies cannot be defined.

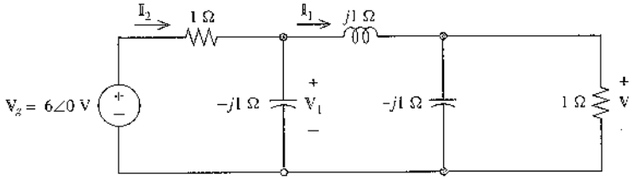
Phasors are defined as those complex numbers $\mathbb{I}$ and $\mathbb{V}$ used to specify currents and voltages when they are of the specific form $i = \mathbb{I}e^{j\omega t}$ and $v = \mathbb{V}e^{j\omega t}$. In circuits with distinct frequencies ω_1 and ω_2, the currents and voltages will not be of the required form. They will in fact be the sums of complex exponentials at each of the different frequencies. Thus, in ac circuits containing sources with different frequencies, we cannot hope to define a unified overall phasor associated with a given current or voltage. If we were to try to do so by coverting the original circuit to a phasor circuit containing sources at both ω_1 and ω_2, the dilemma would be apparent when we tried to assign values to the impedances. What would we use as the value of ω in computing impedance values for those elements whose impedance depends on ω? Clearly, neither ω_1 nor ω_2 by itself would do. A single phasor circuit with sources at different frequencies is meaningless. *For sources at different frequencies, superposition is not just a choice; it must be used to determine the ac steady-state response.*

9.4.1. Find the steady-state current i.
ANSWER: $2\cos(2t - 36.9^\circ) + 3\cos(t + 73.8^\circ)$ A



9.4.2. For the phasor circuit corresponding to Exercise 9.3.3, replace the part to the left of terminals a–b by its Thevenin equivalent and find the steady-state current i_1.
ANSWER: $\mathbb{V}_{oc} = \frac{9}{5}(2 - j1)$ V, $Z_{th} = \frac{1}{5}(18 + j1)\ \Omega$, $i_1 = \cos 2t$ A

9.4.3. Find $\mathbb{V}_1$, $\mathbb{I}_1$, and $\mathbb{I}_2$. Use the ladder method, assuming $\mathbb{V} = 1\underline{/0}$.
ANSWER: 3 V; $3 - j3$ A; 3 A



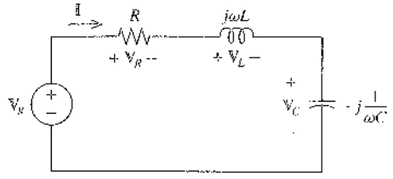
9.5 PHASOR DIAGRAMS

Since phasors are complex numbers, they may be represented by vectors in a plane, where operations such as addition of phasors may be carried out geometrically. Such a sketch is called a *phasor diagram* and may be helpful in analyzing ac steady-state circuits.

Example 9.11

To illustrate, let us consider the phasor circuit of Fig. 9.19, for which we shall draw all the voltages and currents on a phasor diagram. To begin, let us observe that the current $\mathbb{I}$ is common to all elements and take it as our *reference* phasor, denoting it by

$$\mathbb{I} = |\mathbb{I}|\underline{/0}$$



RLC series phasor circuit.

We have taken the angle of $\mathbb{I}$ arbitrarily to be zero, since we want $\mathbb{I}$ to be our reference. We may always adjust this assumed value to the true value by the proportionality principle, which permits the addition of a constant phase shift θ to every current and voltage (scaling of every phasor by $1\underline{/\theta}$).

The voltage phasors of the circuit are

$$\mathbb{V}_R = R\mathbb{I} = R|\mathbb{I}|$$

$$\mathbb{V}_L = j\omega L\mathbb{I} = \omega L|\mathbb{I}|\underline{/90^\circ}$$

$$\mathbb{V}_C = -j\frac{1}{\omega C}\mathbb{I} = \frac{1}{\omega C}|\mathbb{I}|\underline{/-90^\circ}$$

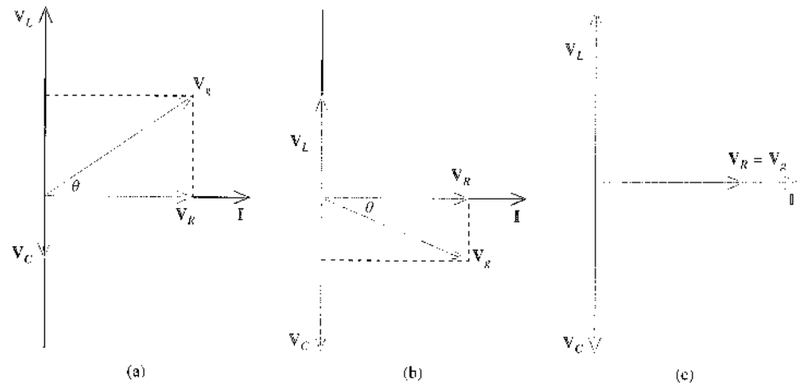
$$\mathbb{V}_g = \mathbb{V}_R + \mathbb{V}_L + \mathbb{V}_C$$

These are shown in the phasor diagram of Fig. 9.20(a), where it is assumed that $|\mathbb{V}_L| > |\mathbb{V}_C|$. The cases $|\mathbb{V}_L| < |\mathbb{V}_C|$ and $|\mathbb{V}_L| = |\mathbb{V}_C|$ are shown in Fig. 9.20(b) and (c), respectively. In all cases the lengths representing the units of current and voltage are not necessarily the same, so for clarity $\mathbb{I}$ is shown longer than $\mathbb{V}_R$.

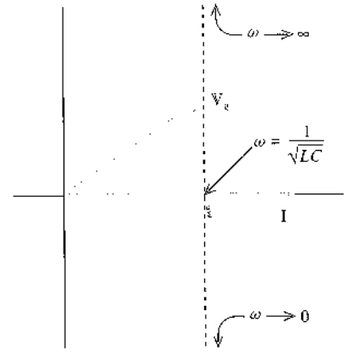
Recall that the phasor diagram is a snapshot at time $t = 0$ of circular motion in the complex plane. All vectors rotate together in the counterclockwise direction; thus a more counterclockwise phasor leads a less counterclockwise one in this motion.

In case (a) the net reactance is inductive, and the current lags the source voltage by the angle θ that can be measured. In (b) the circuit has a net capacitive reactance, and the current leads the voltage. Finally, in (c) the current and voltage are in phase, since the inductive and capacitive reactance components exactly cancel each other. These conclusions follow also from the equation

$$\mathbb{I} = \frac{\mathbb{V}_g}{Z} = \frac{\mathbb{V}_g}{R + j[\omega L - (1/\omega C)]}$$



Phasor diagrams for Fig. 9.19.



Locus of the voltage phasor for a fixed current response.

Case (c) is characterized by

$$\omega L - \frac{1}{\omega C} = 0$$

or

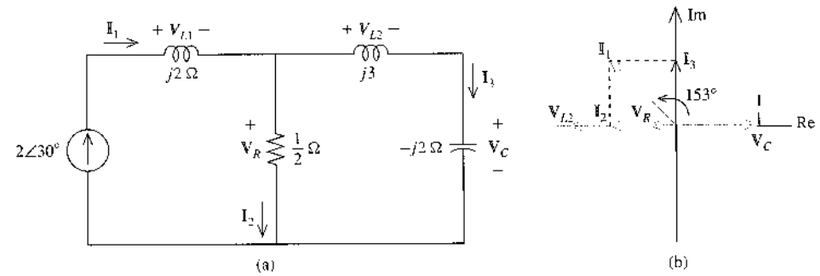
$$\omega = \frac{1}{\sqrt{LC}}$$

If the current in Fig. 9.19 is fixed, the real component of the voltage $\mathbb{V}_g$ is fixed, since it is $R|\mathbb{I}|$. In this case the *locus* of the phasor $\mathbb{V}_g$ (its possible location on the phasor diagram) is the dashed line of Fig. 9.21. The voltage phasor varies up and down this line as ω varies between zero and infinity. The minimum amplitude of the voltage occurs when $\omega = 1/\sqrt{LC}$, as seen from the figure. For any other frequency, a larger voltage is required to produce the same current.

Example 9.12

We wish to determine $\mathbb{V}_C$ in Fig. 9.22 and to visualize the phase relationships among all the currents and voltages. We will use $\mathbb{V}_C$ as our reference phasor, and since its magnitude is also unknown, we will temporarily set $\mathbb{V}_C = 1\underline{/0}$ as shown in Fig. 9.22(b). Since the current through a capacitor leads its voltage by 90°, $\mathbb{I}_3$ must lead $\mathbb{V}_C$ by 90° as shown in the phasor diagram. Using the reference value for $\mathbb{V}_C$,

$$\mathbb{I}_3 = \frac{\mathbb{V}_C}{-j2} = \frac{-1}{j2} = \frac{1}{2}\underline{/90^\circ}$$



(a) Phasor circuit; (b) phasor diagram using $\mathbb{V}_C$ as the reference phasor.

$\mathbb{V}_{L2}$, the voltage across an inductor, leads its current $\mathbb{I}_3$ by 90°, so it falls on the negative real axis in the phasor diagram as shown.

$$\mathbb{V}_{L2} = (j3)\left(\frac{1}{2}\underline{/90^\circ}\right) = \frac{3}{2}\underline{/180^\circ}$$

$\mathbb{V}_R$ is, by KVL, the vector sum of $\mathbb{V}_C$ and $\mathbb{V}_{L2}$. Since they both lie on the real axis the sum must as well:

$$\mathbb{V}_R = 1\underline{/0} + \frac{3}{2}\underline{/180^\circ} = \frac{1}{2}\underline{/180^\circ}$$

Current and voltage in a resistor are in phase, so $\mathbb{I}_2$ has the same angle in the phasor diagram as $\mathbb{V}_R$:

$$\mathbb{I}_2 = 2\mathbb{V}_R = 1\underline{/180^\circ}$$

Then since $\mathbb{I}_1$ is the vector sum of $\mathbb{I}_2$ and $\mathbb{I}_3$ as shown in the phasor diagram,

$$\mathbb{I}_1 = 1\underline{/180^\circ} + \frac{1}{2}\underline{/90^\circ} = 1.12\underline{/153^\circ}$$

Thus we see that $\mathbb{V}_C = 1\underline{/0}$, together with all the other phasors we computed based on this reference phasor, would have been produced if the current source were of value $1.12\underline{/153^\circ}$. By the proportionality principle, scaling the source will scale all phasors by the same factor. Choosing the scale factor α to match the actual source value $2\underline{/30^\circ}$,

$$\alpha = \frac{2\underline{/30^\circ}}{1.12\underline{/153^\circ}} = 1.79\underline{/-123^\circ}$$

we see that all the phasors in the phasor diagram must be enlarged by a factor of 1.79 and rotated 123° clockwise to complete the solution. In particular,

$$\mathbb{V}_C = \alpha(1\underline{/0}) = 1.79\underline{/-123^\circ}\ \text{V}$$

Note that rotating all the phasors in the vector diagram the same amount does not change their phase relationships: $\mathbb{I}_2$ still leads $\mathbb{I}_3$ by 90° and is still 180° out of phase with $\mathbb{V}_C$, and so on. The arbitrarily selected angle of the reference phasor is adequate to determine all phase relationships without the need for corrections. Corrections are needed only to determine the phase angle of a response rather than phase shift between two response variables.

Example 9.13

As a final example illustrating the use of phasor diagrams, let us find the locus of $\mathbb{I}$ as R varies in Fig. 9.23. The current is given by

$$\mathbb{I} = \frac{V_m}{R + j\omega L} = \frac{V_m(R - j\omega L)}{R^2 + \omega^2 L^2}$$

Therefore, if

$$\mathbb{I} = x + jy \tag{9.15}$$

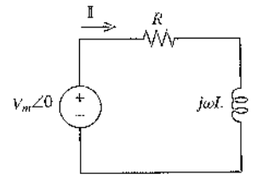


FIGURE 9.23 *RL* phasor circuit.

we have

$$x = \text{Re}\,(\mathbb{I}) = \frac{RV_m}{R^2 + \omega^2 L^2}$$

$$y = \text{Im}\,(\mathbb{I}) = \frac{-\omega L V_m}{R^2 + \omega^2 L^2}$$

The equation of the locus is the equation satisfied by x and y as R varies; thus we need to eliminate R between (9.16) and (9.17).

If we divide the first of these two equations by the second, we have

$$\frac{x}{y} = -\frac{R}{\omega L}$$

from which

$$R = -\frac{\omega L x}{y}$$

Substituting this value of R into (9.16), we have, after some simplification,

$$x^2 + y^2 = -\frac{V_m y}{\omega L}$$

This result may be rewritten as

$$x^2 + \left(y + \frac{V_m}{2\omega L}\right)^2 = \left(\frac{V_m}{2\omega L}\right)^2$$

which is the equation of a circle with center at $[0, -(V_m/2\omega L)]$ and radius $V_m/2\omega L$.

The circle (9.18) appears to be the locus, as R varies, of the phasor $\mathbb{I} = x + jy$. However, by (9.16), $x \geq 0$; thus the locus is actually the semicircle shown dashed on the phasor diagram of Fig. 9.24. The voltage $V_m\underline{/0}$, taken as reference, is also shown, along with the phasor $\mathbb{I}$. If $R = 0$, we have, from (9.16) and (9.17), $x = 0$ and $y = -V_m/\omega L$. If $R \to \infty$, then $x \to 0$ and $y \to 0$. Thus as R varies from 0 to ∞, the current phasor moves counterclockwise along the circle.

If $\mathbb{I}$ is as shown in Fig. 9.24, the current phasor may be resolved into two components, one having amplitude $I_m \cos\theta$ in phase with the voltage and one with amplitude $I_m \sin\theta$, which is 90° out of phase with the voltage. This construction is indicated by the dashed vertical line. As we shall see in Chapter 10, the in-phase component of the current is important in calculating the average power delivered by the source. Thus the phasor diagram gives us a method of seeing at a glance the maximum in-phase component of current. Evidently, this occurs at point a, which corresponds to $\theta = 45^\circ$. This is the case $x = -y$, or $R = \omega L$.

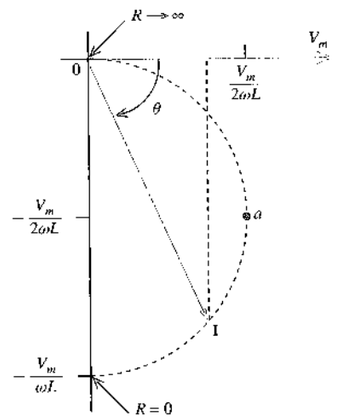


FIGURE 9.24 Locus of the phasor $\mathbb{I}$.

9.5.1. Eliminate ωL in (9.16) and (9.17) and show that as ωL varies, the locus of the phasor $\mathbb{I} = x + jy$ is a semicircle.

$$\left(x - \frac{V_m}{2R}\right)^2 + y^2 = \left(\frac{V_m}{2R}\right)^2, \qquad y \le 0$$

9.5.2. Find ωL in Exercise 9.5.1 so that Im $\mathbb{I}$ has its largest negative value. Also find $\mathbb{I}$ for this case.

R; $(V_m/\sqrt{2}\,R)/\underline{-45^\circ}$

The use of phasors substantially reduces the computational burden of ac steady-state circuit analysis compared to formulating and solving the describing equation as a high-order differential equation. Still, for *RLC* circuits with several nodes and loops, substantial computing is required. The complex arithmetic used to develop the solution becomes time consuming and tedious, even with the support of a calculator well suited to the purpose.

Fortunately, SPICE is equipped with an ac control statement, which, if included in the SPICE input file, invokes ac steady-state analysis. In this mode the SPICE program determines the solution to a phasor circuit that you have specified. The format for this control statement is

```
.AC    FVAR    NP    FLOW    FHIGH
```

In the ac analysis mode invoked by this statement, SPICE is set up to perform multiple ac steady-state analyses at user-specified sets of frequencies. We will have occasion to make full use of this capability when studying frequency response in subsequent chapters, but for present purposes, determination of the ac steady state at a single source frequency is sufficient. Since the control statement is set up to accommodate the more general purpose, however, we must be aware of the general format. `FVAR` must be replaced by one of the keywords `DEC`, `OCT`, or `LIN`. This keyword describes the manner in which the frequency variation of the set of frequencies to be used for repeated ac analysis is to be done: by decade, by octave, or linearly. `NP` is a number specifying the number of frequencies per decade, per octave, or in the `LIN` case, the total number of frequencies in the set. `FLOW` is the lowest frequency, and `FHIGH` the highest frequency to be analyzed, with units of hertz (Hz). For instance,

```
.AC    DEC    5    10    1000
```

specifies that an ac analysis is to be performed for each of five evenly spaced frequencies per decade, from 10 Hz to 1 kHz. The logarithmic units of decade and octave are not

needed for the present purpose of ac steady-state analysis; they will be taken up again in Chapter 14. The control statement

```
.AC    LIN    1    30K    30K
```

specifies that a single ac analysis be done at a frequency of 30 kHz. When `NP` is set to 1 as above, there is only one frequency in the analysis set. The resulting statement may seem like a roundabout way simply to specify an ac analysis at the single frequency of 30 kHz, but these are the current rules of SPICE. Perhaps in some future version improvements in this statement format will be made.

The phasors for all independent sources in the ac analysis must be specified on their element statements. The format is

```
VXXXX    N1    N2    AC    MAG    ANG
IXXXX    N1    N2    AC    MAG    ANG
```

`V` stands for independent voltage source, `XXXX` its name, and `N1` and `N2` the positive and negative nodes. Inclusion of the keyword `AC` indicates that this source is to be used in an ac analysis. `MAG` is the magnitude and `ANG` the angle of the phasor voltage source. The units for all angular quantities, such as `ANG`, throughout SPICE are degrees. The independent current source statement is similar to the voltage source, with the current source function reference arrow pointing out of node `N1`, through the source, and into node `N2`.

The output of a SPICE ac analysis is a set of phasor response variables. These may be printed as a table using the statement

```
.PRINT    AC    CVLIST
```

where `CVLIST` is a list of circuit variables. This list is formatted exactly as described in Chapter 4 for dc analysis, except that we specify the magnitude or phase of a variable by including `M` or `P` after the `V` (for voltage) or `I` (for current). For instance,

```
.PRINT    AC    VM(2)    VP(2)    IM(VDUMMY)
```

results in the printing of the magnitude and phase of the node voltage phasor $\mathbb{V}_2$ and the magnitude of the current phasor through the voltage source `VDUMMY`. If rectangular components of the output are preferred, substitution of `R` or `I` for `M` or `P` will result in the printing of the real or imaginary parts.

Recall that in some versions of SPICE, only currents through voltage sources may be output, and it is necessary to install a zero volt "look-in" or "dummy" voltage source in series with any other element whose current is required and that does not happen to have a voltage source already in series. If the rectangular rather than polar representation is desired, inclusion of `R` or `I` after the leading `V` (voltage) or `I` (current) will result in printing of the real part or the imaginary part, respectively. SPICE solves phasor circuits only; it is the responsibility of the user to convert the original sinusoidal sources to their phasor representations and then convert the response phasors back to sinusoids.

To illustrate the application of SPICE, consider the circuit of Fig. 9.7. Let us find the voltage of node 1 in polar form and the current of the 1-Ω resistor in rectangular form. A SPICE deck for calculating these values is

```
AC STEADY-STATE SOLUTION FOR CIRCUIT OF FIG. 9.7
*DATA STATEMENTS
V1 100 0 AC 5 0
R1 100 1 .5
C1 1 0 .5
C2 1 2 1
L1 1 2 .5
L2 2 0 .25
R2 2 0 1
I1 0 2 AC 5 0
*SOLUTION CONTROL STATEMENT FOR AC ANALYSIS [f = 2/(2*PI) Hz]
.AC LIN 1 .3183 .3183
*OUTPUT CONTROL STATEMENT FOR V(1) & I(R2)
.PRINT AC VM(1) VP(1) IR(R2) II(R2)
.END
```

The .PRINT statement is formatted for versions of SPICE that accept current references such as I(R2). If your version of SPICE only outputs currents through voltage sources I(VXXXXX), a dummy source should be inserted in series with R1. The solution printed in this case is

FREQ	VM(1)	VP(1)	IR(R2)	II(R2)
3.183E-01	2.263E+00	-2.657E+01	2.000E+00	4.000E+00

The corresponding sinusoids are $v = 2.26\cos(2t - 26.6°)$, and since $\mathbf{I}_R = 2 + j4 = 4.47\underline{/63.4°}$, $i_R = 4.47\cos(2t + 63.4°)$.

As a second example, consider finding the phasor current $\mathbf{I}$ in the circuit of Fig. 9.9, which contains a controlled source of transresistance $r = 3000\ \Omega$.

```
AC STEADY-STATE SOLUTION FOR FIG. 9.9.
*DATA STATEMENTS
V 1 0 AC 4 0
R1 1 2 0.5K
R2 2 0 2K
C1 2 0 0.2UF
H 3 2 V -3000
R3 3 4 2K
C2 4 0 0.2UF
```



```
*SOLUTION CONTROL STATEMENT FOR f = 5000/(2*3.1416)
.AC LIN 1 795.77 795.77
.PRINT AC IM(R1) IP(R1)
.END
```

The solution is

FREQ	IM(R1)	IP(R1)
7.958E+02	2.400E-02	5.313E+01

Thus the ac steady-state current is $0.024\cos(5000t + 53.1°)$.

In Chapter 4 the use of subcircuits in SPICE was introduced. The subcircuit definition, a set of statements enclosed by .SUBCKT and .ENDS control lines, may be included in the SPICE input file for the circuit containing the subcircuit, or it may be stored as a separate library file. The latter is particularly useful if the subcircuit is to be used in several different circuits. Inclusion of the control statement

```
.LIB    FILENM
```

will cause the contents of the text file FILENM to be linked to the main SPICE input file. FILENM must contain only subcircuits and, if we wish to link other library files to FILENM, .LIB control statements. For instance, an op amp model introduced as a subcircuit in Example 4.23 and repeated here is

```
.SUBCKT &    OPAMP   1   2   3
*NODE 1 is the + in, 2 the - in, and 3 the output
RIN    1    2   1MEG
E1     4    0   1   2   100K
R0     4    3   30
.ENDS
```

Since we will have occasion to use this model frequently, assume we have stored these five lines as a separate file named OPAMP.CKT. Example 9.16 illustrates how this library file can be used.

SPICE Example 9.16

Let us find the phasor output voltage of the op amp circuit of Fig. 9.11 if the input voltage is $v_g = 10\cos(1000t + 30°)$ V. A circuit file for the nodes of the op amp inverting input, op amp output, and input source, assigned as 3, 4, and 10, respectively, with nodes 1 and 2 as shown, is

```
AC STEADY-STATE SOLUTION OF FIG. 9.11
*DATA STATEMENTS USING OPAMP.CKT OF CHAPTER 4
.LIB OPAMP.CKT
VG 10 0 AC 10 30
R1 10 1 0.707K
R2 1 2 1.414K
C1 1 4 1UF
C2 2 0 1UF
R3 3 0 2K
R4 3 4 2K
XOPAMP 3 2 4 OPAMP
*SOLUTION CONTROL STATEMENT Ef = 1000/(2*3.1416) Hz]
.AC LIN 1 159.15 159.15
.PRINT AC VM(4) VP(4)
.END
```

This circuit file gives a solution

FREQ	VM(4)	VP(4)
1.592E+02	1.414E+01	-5.999E+01

For ac circuits containing sources at different frequencies, recall that a separate phasor analysis is required for the subset of sources at each distinct frequency, with all other sources killed. After conversion of the desired response phasors to sinusoids, the component results are superposed. Each phasor analysis requires, in general, a separate SPICE run with some editing of the SPICE input file in between. The phasor-to-sinusoid conversions and final addition must be done by hand. Based as it is on phasor analysis, SPICE has no facility to handle ac analysis of circuits with different frequencies more directly.

In Chapter 8 a fundamental caution in the use of phasors to compute ac steady state was pointed out. Phasors directly compute the forced response in ac circuits, which can be equated with ac steady-state response only if the circuit is stable. Stable circuits are those whose natural responses all decay to zero over time. If a circuit is not stable, its natural response will not all die away, and thus it will not in general possess a steady state regardless of how long we wait. If the circuit is not stable, we will get a numerical result using phasor analysis, but we should interpret this as the forced response only, not the ac steady state (which does not exist for unstable circuits).

Ac analysis in SPICE, which performs phasor analysis, inherits the same limitation. The result of an ac analysis in SPICE will always equate to the forced response, but this is also the ac steady-state response only if the circuit happens to be stable (which SPICE ac analysis does not tell us). Given the output of SPICE ac analysis, it is up to the user to verify independently that the circuit is stable before calling it the ac steady state. This requirement is illustrated in Example 9.17.

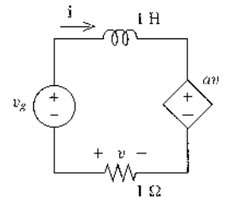


Figure 9.25 Circuit for Example 9.17.

Consider the single-loop circuit of Fig. 9.25, with describing equation for the mesh current i as follows:

$$\frac{di}{dt} + (1 - a)i = v_g$$

The characteristic equation is

$$s + (1 - a) = 0$$

leading to the natural solution

$$i_n = Ke^{-(1-a)t}$$

If the controlled source voltage gain $a < 1$, then i_n is a damped exponential, which goes to zero, and the system is stable. If $a > 1$, then i_n goes to (plus or minus) infinity, the circuit is unstable, and there is no steady state.

Let us perform ac analysis on this circuit using SPICE, with $v_g = \cos t$. Setting $a = 0.5$, our SPICE input file is

```
Circuit of Fig. 9.25
*
V       1       0       AC      1       0
L       1       2       1
H       3       3       V       0.5
R       3       0       1
.AC     LIN     1       .159    .159
.PRINT AC IM(V)    IP(V)
.END
```

which when run yields the output

FREQ	IM(V)	IP(V)
1.590E-01	8.951E-01	1.166E+02

The indicated forced response, $\mathbb{I} = 0.895/\underline{117^\circ}$ A or $i = 0.895\cos(t + 117^\circ)$ A, is the ac steady-state response of this stable circuit. Changing to the value $a = 2$ in the SPICE input file and rerunning, the resulting output is

FREQ	IM(V)	IP(V)
1.590E-01	7.075E-01	4.497E+01

Since this circuit is unstable, the forced response $\mathbb{I} = 0.707/\underline{45^\circ}$ A or $i = 0.707\cos(t + 45^\circ)$ A computed by SPICE can no longer also be identified as the ac steady-state response. There is no ac steady state in this unstable circuit.

The determination of whether a network is stable will be taken up more systematically in Chapter 14. For now, recall that *RLC* circuits with no dependent sources and

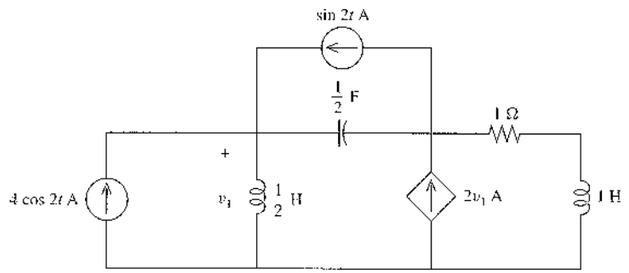
no undamped natural responses (some resistance in every loop) are all stable. Also, op amp circuits with negative feedback, such as the building block circuits of Chapter 3, are stable. For other circuits, we should interpret the results of any phasor analysis, including ac analysis by SPICE, with the proper caution. If an ac steady state exists, SPICE will find it, but we will not be warned if it does not.

9.6.1. Use SPICE to find the phasor representation of v in Exercise 9.2.4 for $\omega = 5$ rad/s.

Answer: $2.224\angle -158.2°$ V

9.6.2. Use SPICE to find the phasor current of the 1-H inductor.

Answer: $0.331\angle 47.8°$ A

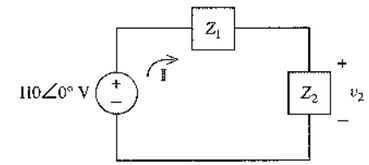


SUMMARY

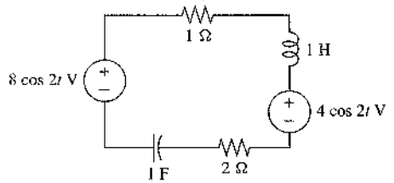
In this chapter phasor analysis is systematically applied to ac steady-state circuits. All of our familiar tools, nodal analysis, mesh analysis, current and voltage division, series and parallel equivalents, and so on, pass over to the phasor domain unchanged. Their use in the phasor domain is greatly simplified by the fact that all currents and voltages are constants, and all nonsource elements satisfy the same simple equation $\mathbf{V} = Z\mathbf{I}$.

- Series impedances add, parallel admittances add.
- Thevenin–Norton transformations and current or voltage division are unchanged in the phasor domain.
- Nodal and mesh analysis are unchanged in the phasor domain.
- For circuits with sinusoidal sources at two or more frequencies, superposition must be invoked to define a separate phasor circuit for each frequency. Each is solved independently. The overall ac steady-state response is the sum of the ac steady-state (sinusoidal) responses of all these phasor circuits.
- A phasor diagram is the plot of one or more phasors in the complex plane. The length of each phasor is the amplitude of its corresponding sinusoid, and the angle of a phasor is the phase angle of its corresponding sinusoid. All phasors rotate together in a counterclockwise direction at ω rad/s.
- SPICE may be used to analyze phasor circuits. The frequency, magnitude, and phase of each sinusoidal source is entered in its element statement, and a `.AC` control statement entered to invoke phasor analysis.

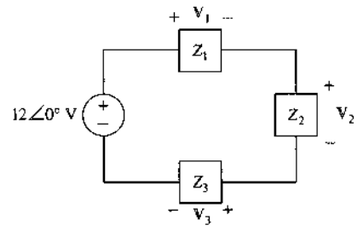
9.1. Find $\mathbf{V}_2$ using the voltage divider principle (9.2). Check by computing $\mathbf{V}_2 = Z_2\mathbf{I}$.



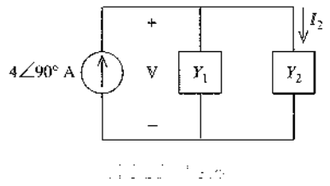
9.2. Find v_1 using voltage division.



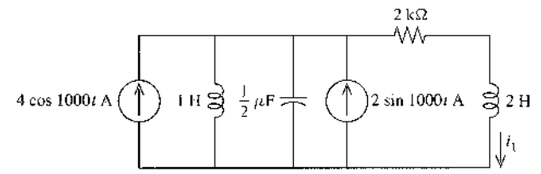
9.3. Specify three impedances Z_1, Z_2, Z_3 so that in this circuit $\mathbf{V}_1 = 6\angle 60°$, $\mathbf{V}_2 = 6\angle 0$, and $\mathbf{V}_3 = 6\angle -60°$ V.



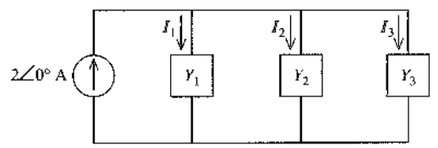
9.4. Find $\mathbf{I}_2$ using the current-divider principle (9.4). Check by computing $\mathbf{I}_2 = Y_2\mathbf{V}$.



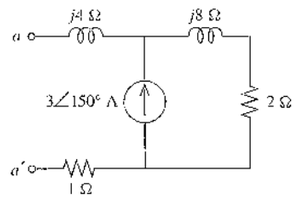
9.5. Find i_1 using current division.



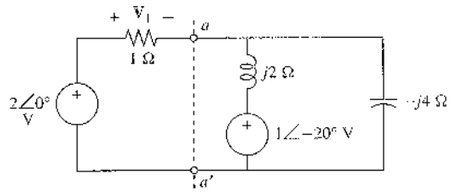
9.6. Specify three admittances Y_1, Y_2, Y_3 so that in this circuit $\mathbf{I}_1 = 1/0$ A, $\mathbf{I}_2 = 1\angle 60°$ A, and $\mathbf{I}_3 = 1\angle -60°$ A.



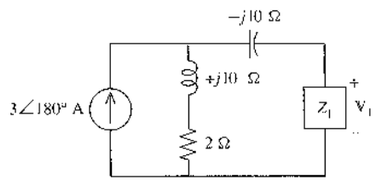
9.7. Find the Thevenin and the Norton equivalents of this subcircuit.



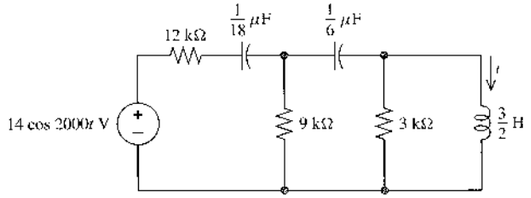
9.8. Find the Thevenin equivalent to the right of the dashed line; then use the voltage-divider principle to find $\mathbf{V}_1$.



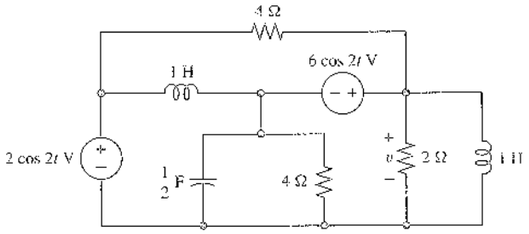
9.9. For what Z_1 is $\mathbf{V}_1 = 1/90^\circ$ V?



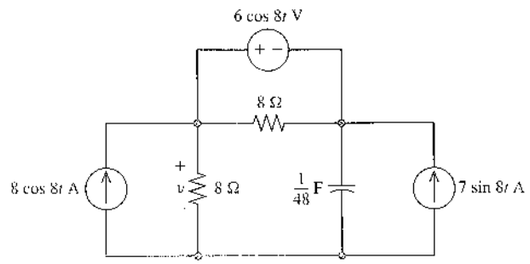
9.10. Find the steady-state current i using nodal analysis.



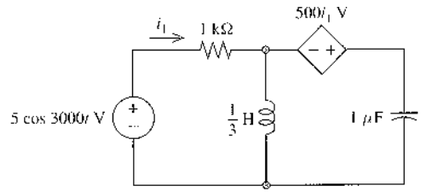
9.11. Find the steady-state voltage v using nodal analysis.



9.12. Find the steady-state voltage v using nodal analysis.

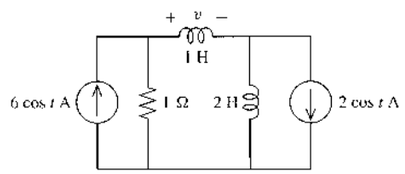


9.13. Find the steady-state current i_1 using nodal analysis.

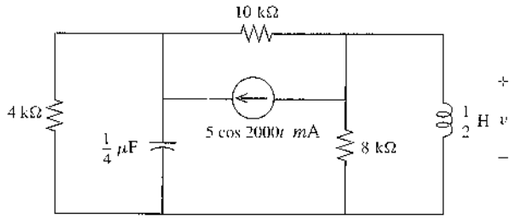


For Problems 9.14 through 9.18, solve for the indicated variables using nodal analysis.

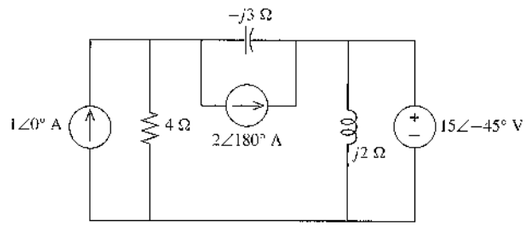
9.14.



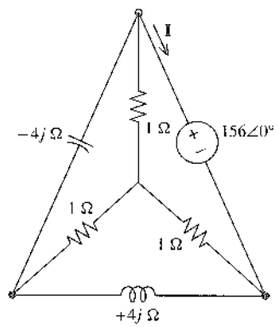
9.15.



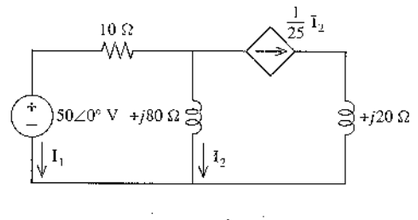
9.16.



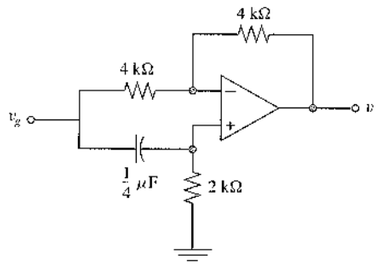
9.17.



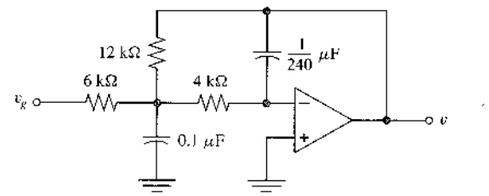
9.18.



9.19. Find the steady-state voltage v if $v_g = 2\cos 2000t$ V.



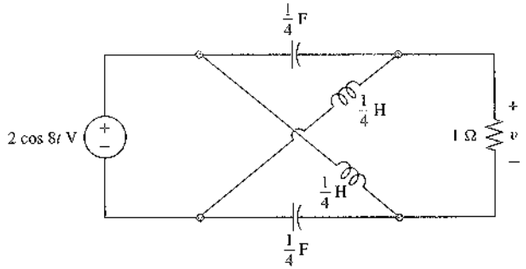
9.20. Find the steady-state voltage v if $v_g = 2\cos 5000t$ V.



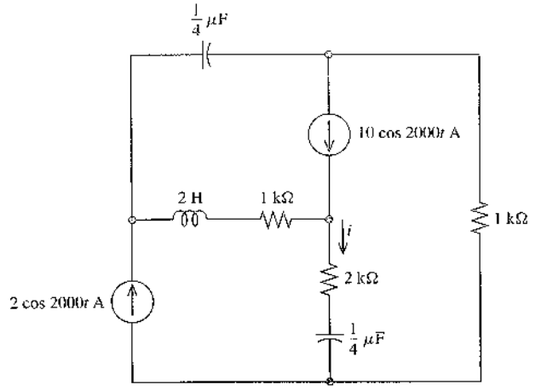
9.21. Solve Exercise 9.2.1 using mesh analysis.

9.22. Solve Exercise 9.2.2 using mesh analysis.

9.23. Find the steady-state voltage v.



9.24. Find the steady-state current i.



9.25. Show that if $\mathbf{Z}_1\,\mathbf{Z}_4 = \mathbf{Z}_2\,\mathbf{Z}_3$ in the "bridge" circuit shown, then $\mathbf{I} = \mathbf{V} = 0$ and therefore all the other currents and voltages remain unchanged for any value of $\mathbf{Z}_5$. Thus it may be replaced by an open circuit, a short circuit, etc. In this case, the circuit is said to be a *balanced bridge*.

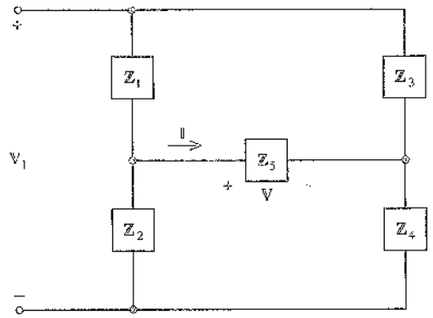


FIGURE P9.25

For Problems 9.26 through 9.30, solve for the indicated variables using mesh analysis.

9.26.

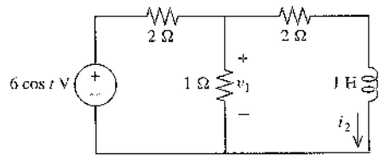


FIGURE P9.26

9.27.

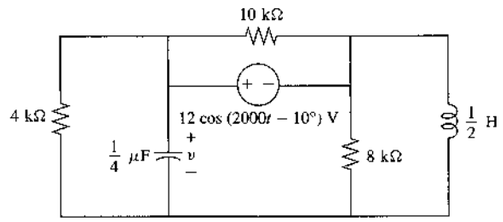


FIGURE P9.27

9.28.

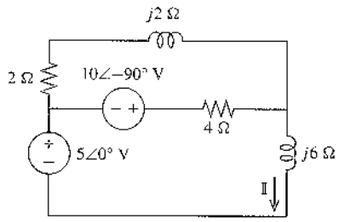


FIGURE P9.28

9.29. The controlled source has transconductance $g = (1/2)$ S.

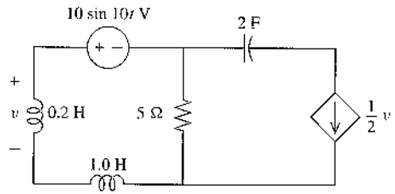


FIGURE P9.29

9.30.

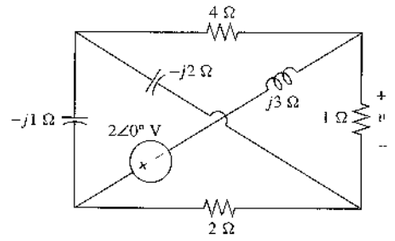
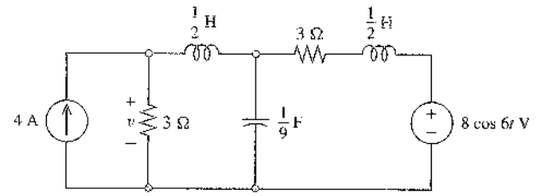


FIGURE P9.30

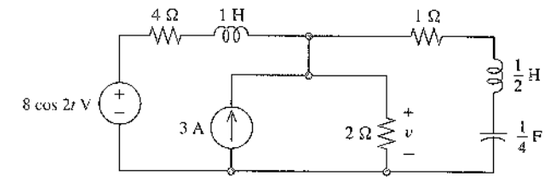
9.31. Solve Problem 9.18 using mesh analysis.

9.32. Mesh analysis is restricted to planar circuits. Sketch a circuit that cannot be analyzed by the mesh method and that contains the minimum number of nodes for a nonplanar circuit.

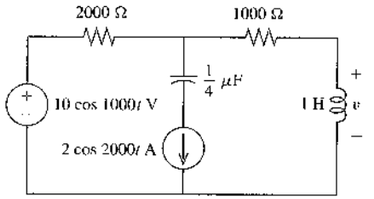
9.33. Find the steady-state voltage v.



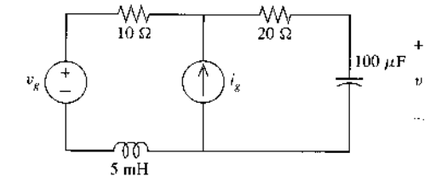
9.34. Find the steady-state value of v.



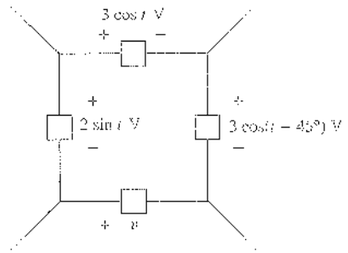
9.35. Find v.



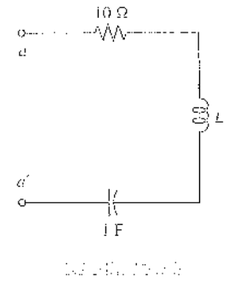
9.36. The sinusoidal sources v_g and i_g operate at $\omega = 1000$ and $\omega = 2000$ rad/s, respectively. If $v = 4\cos 1000t - 2\sin 2000t$, find v_g and i_g.



9.37. Sketch the three given voltages as a phasor diagram, and find $\mathbf{V}$, the phasor corresponding to v, from the diagram. Find v.



9.38. Sketch the impedances of the three elements in a phasor diagram. What must L be if the magnitude of the overall impedance is $|Z| = 14\ \Omega$? Assume $\omega = 10$ rad/s.



9.39. Sketch the locus of the total series impedance shown in Problem 9.38 in the complex plane as L varies from 0 to 1 H.

9.40. Find the locus of $\mathbf{I}$, the phasor associated with i, as ω varies from 0 to ∞. For which values of ω is $|\mathbf{I}|$ largest? Smallest?

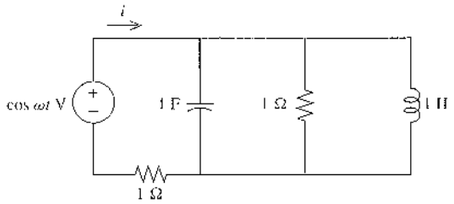


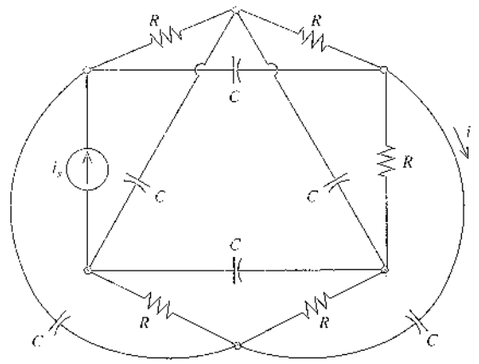
FIGURE P9.40

SPICE Problems

9.41. Use SPICE to solve Problem 9.15.

9.42. Use SPICE to solve Problem 9.27.

9.43. Find the phasor current $\mathbf{I}$ in this nonplanar circuit using SPICE. $R = 10\ \text{k}\Omega$, $C = 1\ \mu\text{F}$, and $i_s = 4\cos 377t$.



9.44. Find $v(t)$ for $\omega = 1, 2, 3, 4, 5$ rad/s. Use only one SPICE run, and use the ideal voltage-controlled voltage source op amp model of Fig. 3.7 with $A = 100{,}000$.

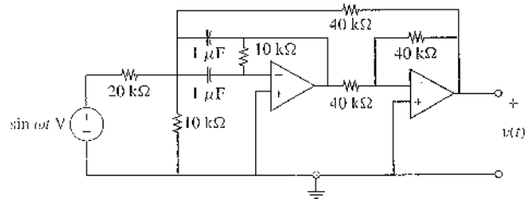


FIGURE P9.44

Verifying Solutions

9.45. Solve Problem 9.44 as specified, but use the improved op amp model of Fig. 3.9 with $A = 100{,}000$, $R_i = 1\ \text{M}\Omega$, and $R_0 = 30\ \Omega$. Enter the op amp model in the SPICE input file as a `.SUBCKT`.

9.46. For what L will $i(t) = 0$?

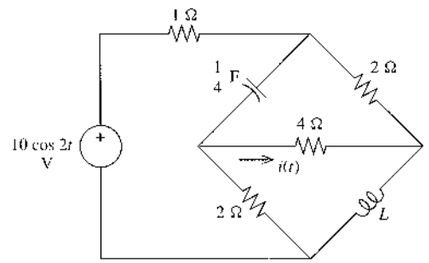
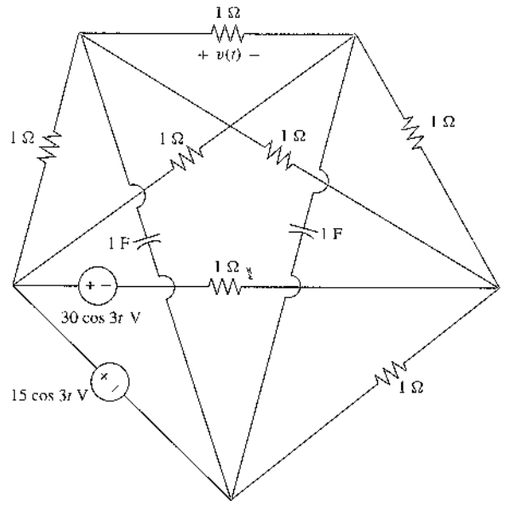
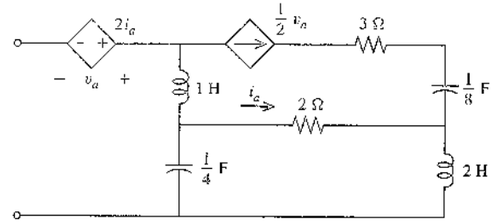


FIGURE P9.46

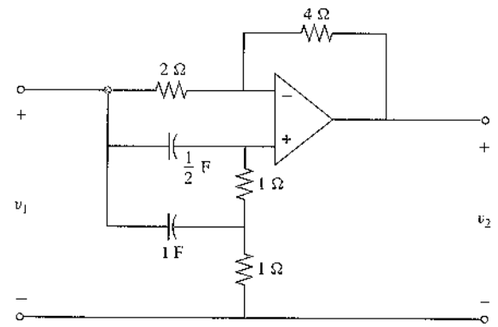
9.47. Find $v(t)$ in ac steady state.



9.48. Find the ac steady-state Thevenin equivalent, specifying $\mathbf{V}_T$ and Z_T.



9.49. Find v_2 in ac steady state if $v_1 = \cos \omega t$ V.



10

AC Steady-State Power



James Prescott Joule
1818–1889

The heating of a conductor depends upon its resistance and the square of the current passing through it.

James P. Joule

The man to whom we are indebted for the familiar expression i^2R for the power dissipated in a resistor is the English physicist James Prescott Joule, who published the result as Joule's law in 1841. He also shared in the famous discovery of the conservation of energy.

Joule was born in Salford, England, the second of five children of a wealthy brewer. He taught himself electricity and magnetism at home as a young boy and obtained his formal education at nearby Manchester University. His experiments on heat were conducted in his home laboratory, and to maintain the accuracy of his measurements he was forced to develop his own system of units. His chief claim to fame is that he did more than any other person to establish the idea that heat is a form of energy. Throughout most of his life Joule was an isolated amateur scientist, but toward the end of his years his work was recognized by honorary doctorates from Dublin and Oxford. In his honor the unit of energy was named the *joule*.

10.1 Average Power
10.2 RMS Values
10.3 Complex Power
10.4 Superposition and Power
10.5 Maximum Power Transfer
10.6 Conservation of Power
10.7 Reactive Power and Power Factor
Summary
Problems

In this chapter we consider power relationships for networks that are excited by periodic currents and voltages. We concern ourselves primarily with sinusoidal currents and voltages since nearly all electrical power is generated in this form. Instantaneous power, as we now well know, is the rate at which energy is absorbed by an element, and it varies as a function of time. The instantaneous power is an important quantity in engineering applications because its maximum value must be limited for all physical devices. For this reason, the maximum instantaneous power, or *peak power*, is a commonly used specification for characterizing electrical devices. In an electronic amplifier, for instance, if the specified peak power at the input is exceeded, the output signal will be distorted. Greatly exceeding this input rating may even damage the amplifier permanently.

A more important measure of power, particularly for periodic currents and voltages, is that of *average power*. The average power is equal to the average rate at which energy is absorbed by an element, and it is independent of time. This power, for example, is what is monitored by the electric company in determining monthly electricity bills. Average powers may range from a few picowatts, in applications such as satellite communications, to megawatts, in applications such as supplying the electrical needs of a large city.

Our discussion will begin with a study of average power. After introducing the convenient root-mean-square (rms) metric, complex power is defined and its uses are discussed. Superposition is related to power, and the notions of maximum power transfer and conservation of power explored. The chapter ends with power factor, an idea of great practical interest to companies supplying electrical power, and a comment on the use of SPICE for ac power calculations.

10.1 AVERAGE POWER

In linear networks that have inputs that are periodic functions of time, the steady-state currents and voltages produced are periodic, each having identical periods. Consider an instantaneous power

$$p = vi \tag{10.1}$$

where v and i are periodic of period T. That is, $v(t + T) = v(t)$, and $i(t + T) = i(t)$. In this case

$$\begin{aligned} p(t + T) &= v(t + T)i(t + T) \\ &= v(t)i(t) \\ &= p(t) \end{aligned} \tag{10.2}$$

Therefore, the instantaneous power is also periodic and p repeats itself every T seconds.

The period T_p of p (the *minimum* time in which p repeats itself) is not necessarily equal to T, however, but T must contain an integral number of periods T_p. In other words,

$$T = nT_p \tag{10.3}$$

where n is a positive integer.

Example 10.1

As an example, suppose that a resistor R carries a current $i = I_m \cos \omega t$ with period $T = 2\pi/\omega$. Then

$$\begin{aligned} p &= Ri^2 \\ &= RI_m^2 \cos^2 \omega t \\ &= \frac{RI_m^2}{2}(1 + \cos 2\omega t) \end{aligned}$$

Evidently, $T_p = \pi/\omega$, and therefore $T = 2T_p$. Thus, for this case, $n = 2$ in (10.3). This is illustrated by the graph of p and i shown in Fig. 10.1.

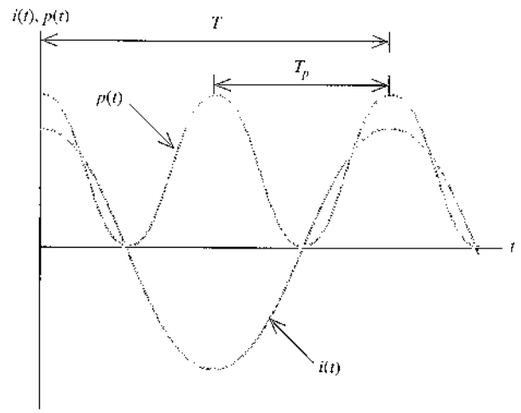


FIGURE 10.1 Instantaneous power and current waveforms (sinusoidal case).

If we now take $i = I_m(1 + \cos \omega t)$, then i is still periodic with period T.

$$p = Ri_m^2(1 + \cos \omega t)^2$$

In this case, $T_p = 2\pi/\omega$, and $n = 1$ in (10.3).

Mathematically, the average value of a periodic function is defined as the time integral of the function over a complete period, divided by the period. Therefore, the average power P for a periodic instantaneous power p is given by

$$P = \frac{1}{T_p} \int_{t_1}^{t_1+T_p} p \, dt \tag{10.4}$$

where t_1 is arbitrary (Fig. 10.2). If we integrate over an integral number of periods, say mT_p (where m is a positive integer), the total area is simply m times that of the integral in (10.4). Thus we may write

$$P = \frac{1}{mT_p} \int_{t_1}^{t_1+mT_p} p \, dt \tag{10.5}$$

If we select m such that $T = mT_p$ (the period of v or i), then

$$P = \frac{1}{T} \int_{t_1}^{t_1+T} p \, dt \tag{10.6}$$

Therefore, we may obtain the average power by integrating over the period of p, as in (10.4), or over the period of v or i.

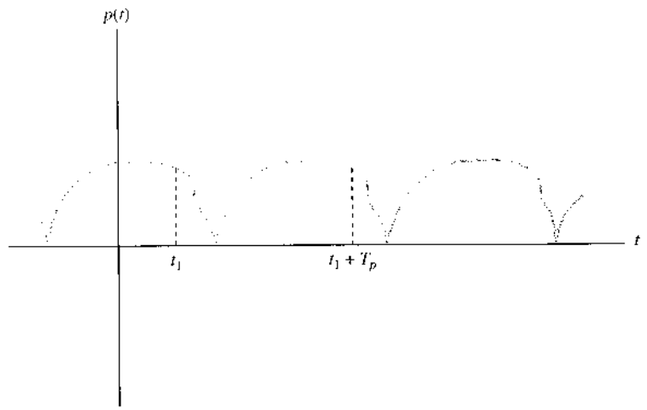


FIGURE 10.2 Periodic instantaneous power (nonsinusoidal case).

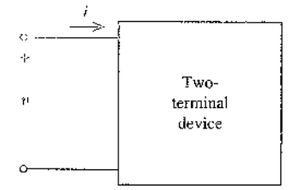


FIGURE 10.3 General two-terminal device.

Consider the general two-terminal device of Fig. 10.3, which is assumed to be in ac steady state. If

$$v = V_m \cos(\omega t + \phi_v) \tag{10.7}$$

we have

$$i = I_m \cos(\omega t + \phi_i) \tag{10.8}$$

for some I_m and ϕ_i, since in ac steady state all currents and voltages have the same frequency.

The average power delivered to the device, taking $t_1 = 0$ for convenience in (10.6), is

$$P = \frac{\omega V_m I_m}{2\pi} \int_0^{2\pi/\omega} \cos(\omega t + \phi_v) \cos(\omega t + \phi_i) \, dt$$

Using the trigonometric identity

$$\cos \alpha \cos \beta = \tfrac{1}{2} \cos(\alpha + \beta) + \tfrac{1}{2} \cos(\alpha - \beta) \tag{10.9}$$

this may be rewritten

$$P = \frac{\omega V_m I_m}{2\pi} \int_0^{2\pi/\omega} \left[\frac{1}{2} \cos(2\omega t + \phi_v + \phi_i) + \frac{1}{2} \cos(\phi_v - \phi_i) \right] dt \tag{10.9}$$

Since the integral runs over two periods of the first term and the average value of any sinusoid (with $\omega \neq 0$) is zero, the integral of the first term is zero. The second term is constant, and we have

$$P = \frac{V_m I_m}{2} \cos(\phi_v - \phi_i) \tag{10.10}$$

Thus the average power absorbed by a two-terminal device is determined by the amplitudes V_m and I_m and the angle θ by which the voltage v leads the current i.

In terms of the phasors of v and i,

$$\mathbb{V} = V_m \underline{/\phi_v} = |\mathbb{V}| \underline{/\phi_v}$$
$$\mathbb{I} = I_m \underline{/\phi_i} = |\mathbb{I}| \underline{/\phi_i}$$

and we have, from (10.10),

$$P = \tfrac{1}{2}|\mathbb{V}|\,|\mathbb{I}|\cos(\phi_v - \phi_i)$$

The average power absorbed by a two-terminal subcircuit is one-half the product of the magnitude of their current and voltage phasors times the cosine of the angle between them.

If the two-terminal device is a resistor R, then $\phi_v - \phi_i = 0$, and $V_m = RI_m$, so (10.10) becomes

$$P_R = \tfrac{1}{2}RI_m^2$$

It is worth noting at this point that if $i = I_{\text{dc}}$, a constant (dc) current, then $\omega = \phi_v = \phi_i = 0$, and $I_m = I_{\text{dc}}$ in (10.7b). The instantaneous power $vi = (RI_{\text{dc}})(I_{\text{dc}})$ is constant in this special case, so it equals its average, or

$$P_R = RI_{\text{dc}}^2$$

Example 10.2

Let us determine the average power absorbed by various parts of the ac steady-state circuit shown in Fig. 10.4(a), whose phasor circuit is drawn in Fig. 10.4(b). Combining impedances yields

$$\mathbb{Z} = 2 + (j2)(1 - j2) = 6 + j2\ \Omega$$

so

$$\mathbb{I} = \frac{4\underline{/0}}{6 + j2} = 0.632\underline{/-18.4^\circ}\ \text{A}$$

By current division,

$$\mathbb{I}_1 = \frac{j2}{1}\mathbb{I} = 1.26\underline{/71.6^\circ}\ \text{A}$$

so $\mathbb{V}_R = 1.26\underline{/71.6^\circ}$ V and

$$\mathbb{V}_C = (-j2)\mathbb{I}_1 = 2.52\underline{/-18.4^\circ}\ \text{V}$$

The phasor diagram for these currents and voltages is shown in Fig. 10.5.

We first determine the average power absorbed by the 1-Ω resistor. The magnitude of its voltage is $|V_R| = 1.26$ V, its current is $|I_1| = 1.26$ A, and the angle between these phasors is 0; they are

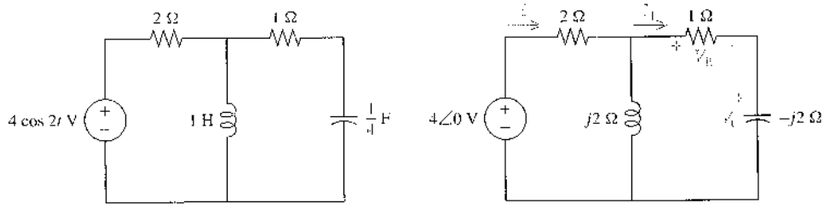


FIGURE 10.4 (a) Circuit for Example 10.2; (b) phasor circuit.

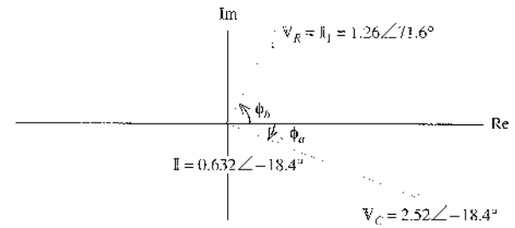


FIGURE 10.5 Phasor diagram for selected currents and voltages from the circuit of Fig. 10.4.

in phase. Hence by (10.11),

$$P_R = \tfrac{1}{2}|\mathbb{V}_R|\,|\mathbb{I}_1|\cos(\underline{/\mathbb{V}_R} - \underline{/\mathbb{I}_1}) = \tfrac{1}{2}(1.26)^2 = 0.80\ \text{W}$$

For the capacitor we have

$$P_C = \tfrac{1}{2}|\mathbb{V}_C|\,|\mathbb{I}_1|\cos\theta_b = 0$$

because the angle between voltage and current is $\theta_b = \underline{/\mathbb{V}_C} - \underline{/\mathbb{I}_1} = -90^\circ$. The net average power absorbed by all the impedances is determined by the voltage $4\underline{/0}$ across the impedance subcircuit and current I into it to be

$$P_Z = \tfrac{1}{2}(4)(0.632)\cos\theta_a = 1.26\cos(-18.4^\circ) = 1.20\ \text{W}$$

Finally, let us compute the power delivered by the source. Note from Fig. 10.3 that the terminal current and voltage used to compute power absorbed by a subcircuit using (10.11) must satisfy the passive sign convention. If they do not, (10.11) measures the *negative* of the power absorbed, that is, the power *delivered by* the subcircuit to the rest of the circuit. Considering the voltage source as a subcircuit, the terminal voltage $4\underline{/0}$ V and current $\mathbb{I}$ violate the passive sign convention since the current reference direction arrow points out of the positive end of the voltage reference direction. Thus the 1.20 W absorbed by the impedances above is also the power delivered by the source. Indeed, the total power delivered to all the elements absorbing power in a circuit is balanced by the power supplied, an intuitively reasonable conservation result to be discussed in more detail in Section 10.6.

Example 10.3

Let us find the power supplied by the two sources in Fig. 10.6. Combining the two parallel impedances gives

$$\mathbb{Z} = \frac{(1)(-j1)}{1 - j1} = \frac{1}{2} - j\frac{1}{2}\ \Omega$$

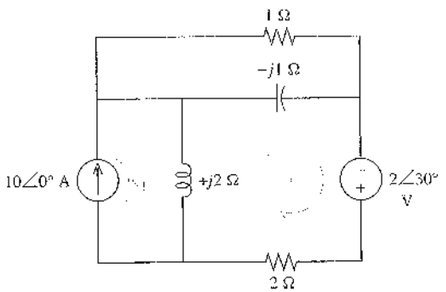


Figure 10.5 Circuit for Example 10.3.

The resulting circuit has two meshes and one current source; thus only one mesh equation is needed. Writing the mesh equation for $\mathbb{I}$, we have

$$\left(\tfrac{1}{2} - j\tfrac{1}{2} + 2 + j\right)\mathbb{I} - j2(10\underline{/0}) = 2\underline{/30^\circ}$$

or

$$\mathbb{I} = 7.23\underline{/54.3^\circ} \text{ A}$$

The voltage $\mathbb{V}_{S1}$ is identical to the voltage across the inductor:

$$\mathbb{V}_{S1} = j2(10\underline{/0} - \mathbb{I}) = 16.5\underline{/44.6^\circ} \text{ V}$$

The voltage $\mathbb{V}_{S1}$ and the current source function $10\underline{/0}$ together violate the passive sign convention relative to the current source. Thus the power delivered by this source, rather than absorbed by it, is given by (10.11):

$$P_{S1} = \tfrac{1}{2}|\mathbb{V}_{S1}||10\underline{/0}|\cos 44.6^\circ = \tfrac{1}{2}(16.5)(10)(0.712) = 58.7 \text{ W}$$

Since $\mathbb{I}$ and the voltage source function $2\underline{/30^\circ}$ also violate the passive sign convention, the power delivered by the voltage source is

$$P_{S2} = \tfrac{1}{2}|\mathbb{I}||2\underline{/30^\circ}|\cos(30^\circ - 54.3^\circ) = \tfrac{1}{2}(7.23)(2)(0.911) = 6.59 \text{ W}$$

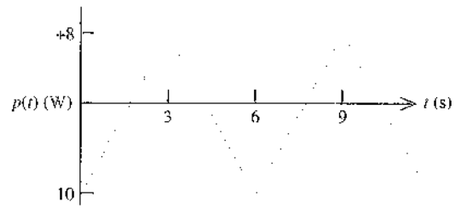
Both sources supply net power to the rest of the circuit. The power is in turn dissipated by the pair of resistors.

Since the current and voltage phasors are always in phase for resistors, by (10.11) resistors always absorb net power. For capacitors and inductors, the plus or minus 90° phase angle of their impedances drives their current and voltage 90° out of phase, so the storage elements neither absorb nor supply steady-state average power in any circuit. It is no surprise that none of the *RLC* elements supplies average power, since we know them to be passive elements. Because inductors and capacitors do not absorb net power either, they are referred to as *lossless* elements. Any element or subcircuit that absorbs average power is referred to as *lossy*. Clearly, any *LC* subcircuit (one containing only inductors and capacitors) is lossless, and any *RLC* subcircuit in which there is current flow into at least one resistor is lossy.

As we saw in Example 10.3, there is no such uniform rule concerning delivery or absorption of power that we may apply to sources. Sources can do either, depending on the circuit contexts in which they are placed. For instance, the same automobile battery will supply power to the starter motor subcircuit or absorb power from the alternator subcircuit depending on the switch positions in the rest of the circuit.

10.1.1. Determine the period T_p of the instantaneous power $p(t)$ and the average power P.

Answer: 6 S, −6 W



10.1.2. For a capacitor of C farads carrying a current $i = I_m \cos \omega t$, verify that the average power is zero from (10.6). Repeat this for an inductor of L henrys.

10.1.3. Find the average power delivered to a 10-Ω resistor carrying a current of:

(a) $i = 5|\sin 10t|$ mA

(b) $i = 10 \sin 10t$ mA, $0 \le t < \pi/10$ s
$= 0$, $\pi/10 \le t < \pi/5$ s; $T = \pi/5$ s

(c) $i = 5$ mA, $0 \le t < 10$ ms
$= -5$ mA, $10 \le t < 20$ ms; $T = 20$ ms

(d) $i = 2t$, $0 \le t < 2$ s; $T = 2$ s

Answer: (a) 125 μW; (b) 0.25 mW; (c) 0.25 mW; (d) $\frac{160}{3}$ W

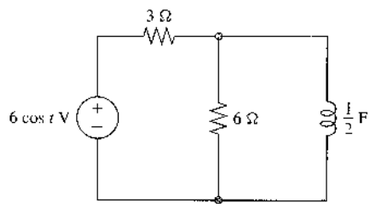
10.1.4. Find the average power absorbed by the capacitor, the two resistors, and the source.

Answer: 0; $\frac{10}{3}$, $\frac{2}{3}$; −4 W

10.1.5. If $f_1(t)$ is periodic of period T_1 and $f_2(t)$ is periodic of period T_2, show that $f_1(t) + f_2(t)$ is periodic of period T if relatively prime positive integers m and n exist such that

$$T = mT_1 = nT_2$$

Extend this result to the function $(1 + \cos \omega t)^2$, considered in this section, to find its period, $T = 2\pi/\omega$.



EXERCISE 10.1.6

10.2 RMS VALUES

We have seen in Section 10.1 that periodic currents and voltages deliver an average power to resistive loads. The amount of power that is delivered depends on the characteristics of the particular waveform. A method of comparing the power delivered by different

waveforms is therefore very useful. One such method is the use of *rms* or *effective* values for periodic currents or voltages.

The *rms* value of a periodic current (voltage) is a constant that is equal to the dc current (voltage) that would deliver the same average power to a resistance R. Thus, if I_{rms} is the *rms* value of i, we may write

$$P = RI_{\text{rms}}^2 = \frac{1}{T}\int_0^T Ri^2\,dt$$

from which the rms current is

$$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i^2\,dt} \tag{10.12a}$$

In a similar manner, the rms voltage is

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2\,dt} \tag{10.12b}$$

The term rms is an abbreviation for *root mean square*. Inspecting (10.12a), we see that we are indeed taking the square *root* of the average, or *mean*, value of the *square* of the current. From our definition, the rms value of a constant (dc) is simply the constant itself. The dc case is a special case ($\omega = 0$) of the sinusoidal current or voltage.

Suppose that we now consider a sinusoidal current $i = I_m \cos(\omega t + \phi)$. Then, from (10.12a) and (10.8), we find

$$I_{\text{rms}} = \sqrt{\frac{\omega I_m^2}{2\pi}\int_0^{2\pi/\omega} \cos^2(\omega t + \phi)\,dt} = \frac{I_m}{\sqrt{2}} \tag{10.13a}$$

Thus a sinusoidal current having an amplitude I_m delivers the same average power to a resistance R as does a dc current that is equal to $I_m/\sqrt{2}$. We also see that the rms current is independent of the frequency ω or the phase ϕ of the current i. Similarly, in the case of a sinusoidal voltage, we find that

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \tag{10.13b}$$

Substituting these values into (10.10), we have for any two-terminal subcircuit,

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\phi_v - \phi_i) \tag{10.14}$$

where P is the average power absorbed if the passive sign convention is satisfied and the power delivered by the subcircuit if it is not.

Rms values are often used in the fields of power generation and distribution. For instance, the nominal 115-V ac power which is commonly used for household appliances is an rms value. Thus the power supplied to most of our homes is provided by a 60-Hz voltage having a maximum value of $115\sqrt{2} \approx 163$ V. On the other hand, maximum values are more commonly used in electronics and communications.

Thus far we have defined the rms value of time-domain currents and voltages. Next let $\mathbb{V}$ and $\mathbb{I}$ be phasors, and define their associated *rms phasors* by

$$\mathbb{V}_{\text{rms}} = \frac{1}{\sqrt{2}}\mathbb{V} \tag{10.15a}$$

$$\mathbb{I}_{\text{rms}} = \frac{1}{\sqrt{2}}\mathbb{I} \tag{10.15b}$$

RMS phasors thus have the same angle as our usual phasors, but their magnitudes are reduced by the factor $1/\sqrt{2}$. Thus while the magnitude of a standard or amplitude phasor equals the amplitude of the associated sinusoid, the magnitude of an rms phasor equals the rms value of the associated sinusoid [by (10.12a) and (10.13a)]. An immediate use of the rms phasor may be seen by substituting (10.13) into (10.10), revealing that

$$P = |\mathbb{V}_{\text{rms}}|\,|\mathbb{I}_{\text{rms}}| \cos(\phi_v - \phi_i) \tag{10.16}$$

Average power may be computed as the product of rms phasor magnitudes times the cosine of the angle between current and voltage phasors without the factor of $\frac{1}{2}$ necessary with amplitude phasors as in (10.10).

Note that by dividing all phasors by $\sqrt{2}$, it is easy to show that rms phasors satisfy Kirchhoff's current and voltage laws, and also that

$$\mathbb{V}_{\text{rms}} = Z\mathbb{I}_{\text{rms}} \tag{10.17}$$

for any impedance. Thus, by designating all independent sources by their rms phasors and elements by their usual impedances, we have an *rms phasor circuit* in which rms currents and voltage phasors may be computed exactly like amplitude current and voltage phasors in amplitude phasor circuits.

Example 10.4

Let us find the power delivered by the 60-Hz ac generator to the load impedance in the circuit of Fig. 10.7(a). The rms source phasor is $(325/\sqrt{2})\underline{/0} = 230\underline{/0}$ V rms, and the rms phasor current is shown in Fig. 10.7(b). The equivalent load impedance is computed in the usual way:

$$Z_L = \frac{1}{j\omega C} \parallel 30 = \frac{(-j53)(30)}{30 - j53} = 22.7 - j12.9\ \Omega$$

In an rms phasor diagram, all phasors are rms, and the mesh current is

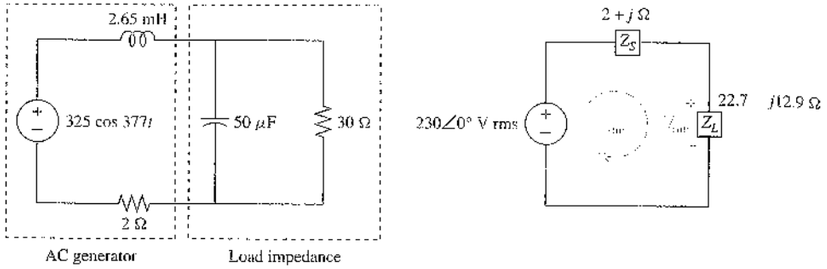
$$\mathbb{I}_{\text{rms}} = 230\underline{/0}/(24.7 - j11.9) = 8.39\underline{/25.7°}\ \text{A rms}$$

so

$$\mathbb{V}_{\text{rms}} = Z_L \mathbb{I}_{\text{rms}} = (22.7 - j12.9)\,(8.39\underline{/25.7°}) = 219\underline{/-3.91°}\ \text{V rms}$$

Then, by (10.16), the average power *absorbed* by the load impedance Z_L is

$$P = |\mathbb{V}_{\text{rms}}|\,|\mathbb{I}_{\text{rms}}| \cos(\phi_v - \phi_i) = (219)\,(8.39)\cos(-3.91° - 25.7°)$$



(a) Circuit for Example 10.4; (b) phasor circuit.

or $P = 1598$ W. This is also the average power delivered by the ac generator.

10.2.1. Find the rms value of a periodic current for which one period is defined by:

(a) $i = I, \quad 0 \le t < 2$ s
$\quad = -I, \quad 2 \le t < 4$ s
(b) $i = 2t, \quad 0 \le t < T$
(c) $i = I_m \sin \omega t, \quad 0 \le t \le \pi/\omega$
$\quad = 0, \quad \pi/\omega \le t \le 2\pi/\omega \ (T = 2\pi/\omega)$.

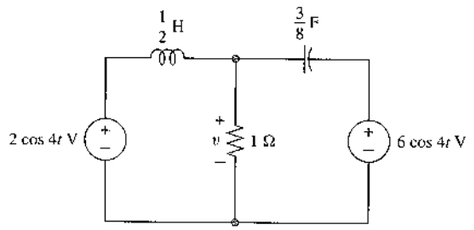
Answer: (a) I; (b) $2T/\sqrt{3}$; (c) $I_m/2$

10.2.2. Find the rms values of (a) $i = 10 \sin \omega t + 20 \cos(\omega t + 30°)$, (b) $i = 8 \sin \omega t + 6 \cos(2\omega t + 10°)$, and (c) $i = I(1 + \cos 377t)$.

Answer: (a) 12.25; (b) 7.07; (c) $I\sqrt{\frac{3}{2}}$

10.2.3. Find V_{rms}.

Answer: 4 V



EXERCISE 10.2.3

10.3 COMPLEX POWER

We have seen that the use of phasors reaps great benefits in the study of ac steady-state circuits. By using complex numbers to represent real time-domain currents and voltages, we have dispensed with the frequent invocation of those identities from trigonometry needed to combine sinusoids and, in addition, replaced the calculus operations of differentiation and integration by the far easier arithmetic operations of multiplication and division in the complex plane.

In our present efforts to extend ac steady-state analysis to include power calculations, the complications of trigonometry and calculus have reappeared, for instance in (10.8) and (10.9). To extend the full benefits of phasor analysis to the study of power in ac steady-state circuits, we will once again define a new complex quantity, which we will call complex power. Similar to the concise way in which phasors represent sinusoids, complex power will encapsulate two essential power-related quantities in one compact, easily manipulated representation.

Given a two-terminal subcircuit with rms voltage and current phasors $\mathbf{V}_{\text{rms}}$ and $\mathbf{I}_{\text{rms}}$, where

$$\mathbf{V}_{\text{rms}} = |\mathbf{V}_{\text{rms}}| \underline{/\phi_v}$$
$$\mathbf{I}_{\text{rms}} = |\mathbf{I}_{\text{rms}}| \underline{/\phi_i}$$

we define the *complex power* into the subcircuit as

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}^*_{\text{rms}} \tag{10.18}$$

where $\mathbf{I}^*_{\text{rms}}$ is the complex conjugate of the rms current phasor. Exploring this definition, the magnitude of the complex number $\mathbf{S}$ is given by

$$|\mathbf{S}| = |\mathbf{V}_{\text{rms}}||\mathbf{I}^*_{\text{rms}}| = |\mathbf{V}_{\text{rms}}||\mathbf{I}_{\text{rms}}| \tag{10.19a}$$

and its angle is

$$\underline{/\mathbf{S}} = \underline{/\mathbf{V}_{\text{rms}}} + \underline{/\mathbf{I}^*_{\text{rms}}} \tag{10.19b}$$

Since the angle of complex conjugate of any complex number is the negative of the angle of the number itself,

$$\underline{/\mathbf{S}} = \underline{/\mathbf{V}_{\text{rms}}} - \underline{/\mathbf{I}_{\text{rms}}} = \phi_v - \phi_i$$

Turning to the rectangular form for $\mathbf{S}$, its real part is $\text{Re}(\mathbf{S}) = |\mathbf{S}| \cos \underline{/\mathbf{S}}$, which, by (10.19), is

$$\text{Re}(\mathbf{S}) = |\mathbf{V}_{\text{rms}}||\mathbf{I}_{\text{rms}}| \cos(\phi_v - \phi_i) \tag{10.20}$$

Comparing the last with (10.6), we have an important connection with our earlier results. *The real part of the complex power is the average power.*

$$\text{Re}(\mathbf{S}) = P \tag{10.21}$$

Thus we have another way to compute the average power: compute the complex power $\mathbf{S}$; then take the real part to get P.

To avoid sign errors, the rms current and voltage phasors used to define the complex power $\mathbf{S}$ into a subcircuit will be assumed to satisfy the passive sign convention. In this

case the real part of $\mathbb{S}$ is interpreted as the power *absorbed* by the subcircuit. Thus if Re($\mathbb{S}$) is positive, the subcircuit is absorbing average power, and, if negative, it is delivering average power to the rest of the circuit. Occasionally, where the meaning is clear, we will use in (10.18) current and voltage phasors that together violate the passive sign convention, in which case the real part must be interpreted as the power delivered by the subcircuit to the rest of the circuit.

Example 10.5

In Example 10.3 we computed the average power for the two sources in the standard (non-rms) phasor circuit shown in Fig. 10.6. The rms phasor circuit corresponding to this problem is shown in Fig. 10.8. Note that the only difference is that the source phasors have been divided by a factor of $\sqrt{2}$. By proportionality, we may get the desired phasors in the rms circuit by scaling the results of Example 10.4 by the same factor, so from the Example 10.4 results we have

$$\mathbb{I}_{\text{rms}} = \frac{7.23}{\sqrt{2}}\underline{/54.3^\circ} = 5.11\underline{/54.3^\circ} \text{ A rms}$$

$$\mathbb{V}_{S1\text{ rms}} = \frac{16.5}{\sqrt{2}}\underline{/44.6^\circ} = 11.7\underline{/44.6^\circ} \text{ V rms}$$

The complex power into the voltage source is

$$\mathbb{S}_{S2} = (\sqrt{2}\underline{/30^\circ})(5.11\underline{/54.3^\circ})^* = 7.23\underline{/-24.3^\circ}$$

and the complex power into the current source is, after reversing the sign of the current in order to comply with the passive sign convention,

$$\begin{aligned}\mathbb{S}_{S1} &= \mathbb{V}_{S1\text{ rms}}(5\sqrt{2}\underline{/180^\circ})^* = (11.7\underline{/44.6^\circ})(7.07\underline{/180^\circ})^* \\ &= (11.7\underline{/44.6^\circ})(7.07\underline{/-180^\circ}) \\ &= 82.7\underline{/-135^\circ}\end{aligned}$$

Note that reversing the sign of a complex number adds or subtracts 180° to its angle in the complex plane, while taking complex conjugates reverses the sign of its angle. In the present case both these operations were performed.

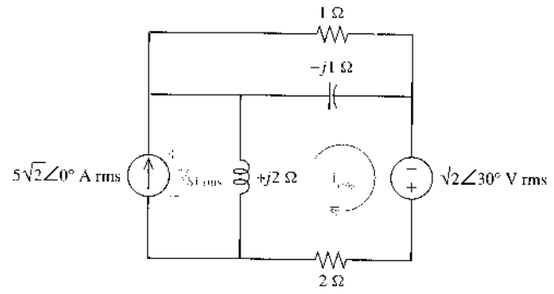


FIGURE 10.8 rms phasor circuit

The average power of the sources is then

$$P_{S2} = \text{Re}(\mathbb{S}_{S2}) = 7.23\cos(-24.3^\circ) = 6.59 \text{ W}$$

$$P_{S1} = \text{Re}(\mathbb{S}_{S1}) = 82.7\cos(-135^\circ) = -58.7 \text{ W}$$

Observing that the current and voltage used for P_{S2} violate the passive sign convention, the voltage source supplies 6.59 W of average power while the current source supplies 58.7 W to the rest of the circuit. This agrees with results of Example 10.4.

The definition of complex power permits useful expressions for the power absorbed by impedances to be easily derived. Given a single impedance or two-terminal subcircuit containing only impedances, using terminal rms phasors $\mathbb{V}_{\text{rms}}$ and $\mathbb{I}_{\text{rms}}$, which satisfy the passive sign convention,

$$\mathbb{V}_{\text{rms}} = Z\mathbb{I}_{\text{rms}} \tag{10.24}$$

and the power into the subcircuit is

$$\mathbb{S} = \mathbb{V}_{\text{rms}}\mathbb{I}^*_{\text{rms}} = Z\mathbb{I}_{\text{rms}}\mathbb{I}^*_{\text{rms}}$$

Recalling that the product of a complex number and its complex conjugate is the real number equal to its magnitude squared, we have

$$\mathbb{S} = Z|\mathbb{I}_{\text{rms}}|^2 \tag{10.25}$$

One implication of this useful formula is that the angle of $\mathbb{S}$ for an impedance is equal to angle of the impedance itself. Since Re($\mathbb{S}$) = P, another is that, taking real parts,

$$P = \text{Re}(Z)|\mathbb{I}_{\text{rms}}|^2 \tag{10.26}$$

or, substituting the generalized Ohm's law (10.22A),

$$P = \frac{\text{Re}(Z)}{|Z|^2}|V_{\text{rms}}|^2 \tag{10.27}$$

Note that if the impedance has an angle of plus or minus 90°, $P = 0$. This is the lossless case discussed earlier. If the real part of Z is positive, the impedance absorbs net power and is lossy. Combining these cases, impedances with angles from minus to plus 90° correspond to passive elements, while those with angles outside this range must be active elements. Any interconnection of passive elements is passive; thus we expect the angles of impedances constructed from passive elements to be 90° or less in magnitude.

In terms of the admittance, if the subcircuit satisfies

$$\mathbb{I}_{\text{rms}} = \mathbb{Y}\mathbb{V}_{\text{rms}}$$

then, by the definition of complex power,

$$\mathbb{S} = \mathbb{V}_{\text{rms}}\mathbb{I}^*_{\text{rms}} = \mathbb{Y}^*|\mathbb{V}_{\text{rms}}|^2 \tag{10.28}$$

and

$$P = \text{Re}(\mathbb{Y}^*)|\mathbb{V}_{\text{rms}}|^2 = \text{Re}(\mathbb{Y})|\mathbb{V}_{\text{rms}}|^2 \tag{10.29}$$

with (10.26a) following from the fact that taking complex conjugates does not change the real part of a complex number. Again using the generalized Ohm's law, (10.26a) may be expressed as

$$P = \frac{\text{Re}(Y)}{|Y|^2}|I_{\text{rms}}|^2$$

Example 10.6

To determine the power into the load in Fig. 10.9(a), we use the rms phasor circuit shown, where

$$Z_1 = \frac{(3)(-j2)}{3 - j2} = 0.923 - j1.38\ \Omega$$

$$Z_L = 2 + j1\ \Omega$$

By voltage division,

$$\mathbb{V}_{L\ \text{rms}} = Z_L \frac{50/\sqrt{2}}{Z_L + Z_1} = \frac{35.4(2 + j1)}{2.92 - j0.38}$$

$$= 26.9\underline{/34.1^\circ}\ \text{V rms}$$

Then by (10.26a) with

$$Y_L = \frac{1}{Z_L} = \frac{1}{2 + j1} = 0.4 - j0.2\ \text{S}$$

we have

$$P_L = \text{Re}(Y)|\mathbb{V}_{L\ \text{rms}}|^2 = 0.4(26.9^2) = 289\ \text{W}$$

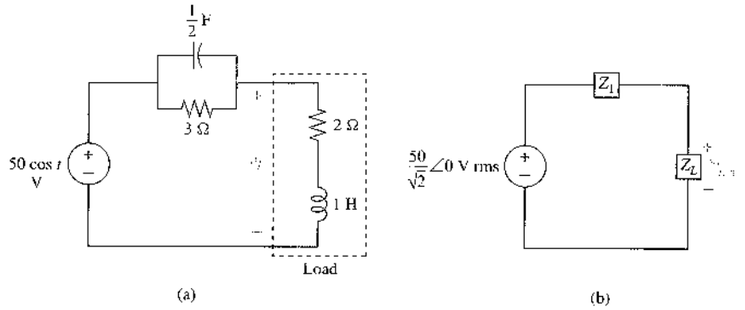


FIGURE 10.9 (a) Circuit for Example 10.6; (b) rms phasor circuit.

Example 10.7

We wish to find a value for the conductance G in Fig. 10.10 so that the source will deliver 1 W to the rest of the circuit. Since the power absorbed by the rest of the circuit must equal this value, by (10.26b), with Y the total admittance of all the *RLC* elements,

$$P = \frac{4\,\text{Re}(\mathbf{Y})}{|\mathbf{Y}|^2} = 1.0 \qquad (10.27)$$

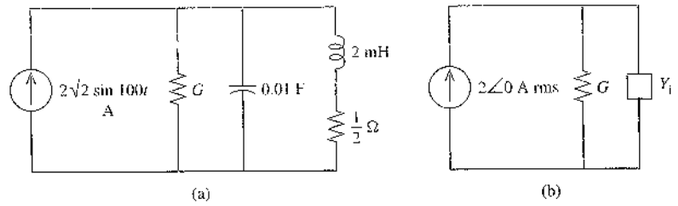


FIGURE 10.10 (a) Circuit for Example 10.7; (b) rms phasor circuit.

Combining the other admittances gives

$$Y_1 = j1 + \frac{(-j5)(2)}{2 - j5} = 1.72 + j0.31\ \text{S}$$

Since G and Y_1 are in parallel, the total admittance is

$$Y = G + Y_1 = (G + 1.72) + j0.31\ \text{S}$$

Substituting this into (10.27) yields

$$\frac{4(G + 1.72)}{(G + 1.72)^2 + 0.31^2} = 1$$

This yields a quadratic equation for G:

$$G^2 - 0.56G - 3.83 = 0$$

The roots of this quadratic are $G = 2.25$ and $G = -3.39$. Since $G > 0$ for passive conductors, the unique solution is $G = 2.25$ S, which is a $1/2.25 = 0.444$-Ω resistor.

Having identified the real part of the complex power $\mathbb{S}$ with the average power P, we next turn to the complex part, which we define as the *reactive power* Q. Thus

$$\mathbb{S} = \mathbb{V}_{\text{rms}}\mathbb{I}^*_{\text{rms}} = P + jQ \qquad (10.28)$$

where P is the average power in watts and Q the reactive power,

$$P = |\mathbb{V}_{\text{rms}}||\mathbb{I}_{\text{rms}}|\cos(\phi_v - \phi_i) \qquad (10.29a)$$

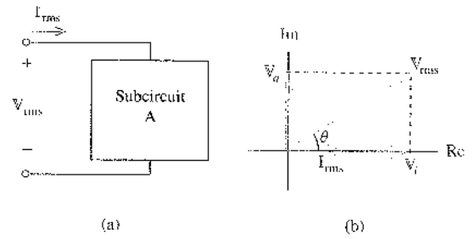
$$Q = |\mathbb{V}_{\text{rms}}||\mathbb{I}_{\text{rms}}|\sin(\phi_v - \phi_i) \qquad (10.29b)$$

The unit of reactive power is the *var*, or volt–ampere reactive.

To explore the meaning of reactive power, consider the phasor diagram shown as Fig. 10.11, in which the rms current and voltage phasors for the subcircuit A are shown. For convenience we take $\mathbb{I}_{\text{rms}}$ as the reference phasor and designate the angle between these phasors as $\theta = \phi_v - \phi_i$. Suppose that we resolve $\mathbb{V}_{\text{rms}}$ into two orthogonal components, one in phase with the current, designated $\mathbb{V}_i$, and the other 90° out of phase (or in *quadrature* with the current), $\mathbb{V}_q$. The magnitudes of these two components are

$$|\mathbb{V}_i| = |\mathbb{V}_{\text{rms}}|\cos\theta$$

$$|\mathbb{V}_q| = |\mathbb{V}_{\text{rms}}|\sin\theta$$



(a) (b)

FIGURE 10.11 (a) Subcircuit; (b) rms phasor diagram.

Now the complex power absorbed by the subcircuit is

$$\mathbb{S} = \mathbb{V}_{rms}\mathbb{I}^*_{rms} = \mathbb{V}_i\mathbb{I}^*_{rms} + \mathbb{V}_q\mathbb{I}^*_{rms}$$

The first term, or in-phase component, is purely real since the current and voltage are in phase, that is, have equal angles. The second term, or quadrature component, is purely imaginary. Thus the average power P is determined solely by the in-phase component and the reactive power Q solely by the quadrature component. Since we are interested here in the reactive power, we will focus on the quadrature component. In terms of the corresponding time functions,

$$i(t) = \sqrt{2}|\mathbb{I}_{rms}|\cos\omega t$$

$$v_q(t) = \sqrt{2}|\mathbb{V}_q|\cos(\omega t \pm 90^\circ)$$

$$= \sqrt{2}|\mathbb{V}_{rms}|\sin\theta\cos(\omega t \pm 90^\circ)$$

The phase angle is written as $\pm 90^\circ$ since the sign of this term depends on whether $\mathbb{V}_{rms}$ happens to lead $\mathbb{I}_{rms}$, the case shown in Fig. 10.11, or lag. The instantaneous power in the quadrature component is then the product

$$p_q(t) = v_q(t)i(t) = [2|\mathbb{V}_{rms}||\mathbb{I}_{rms}|\sin\theta]\cos\omega t\cos(\omega t \pm 90^\circ)$$

which may be simplified using the trigonometric identity introduced in Section 10.1 to be

$$p_q(t) = |\mathbb{V}_{rms}||\mathbb{I}_{rms}|\sin\theta\cos(2\omega t \pm 90^\circ)$$

From this expression we see that the instantaneous power in the quadrature component $p_q(t)$ is a sinusoid with amplitude equal to $|\mathbb{V}_{rms}||\mathbb{I}_{rms}||\sin\theta|$. By (10.29), this equals the magnitude of Q. Thus we arrive at a physical interpretation for the quantity Q: *the magnitude of the reactive power Q is the amplitude of the instantaneous power in the quadrature i–v component.* Since the instantaneous power in the quadrature component is a sinusoid centered at zero, it indicates energy being transferred periodically to the subcircuit and then back from the subcircuit to the rest of the network in equal amount. From earlier chapters we know this behavior to be characteristic of energy storage elements, neither supplying nor dissipating power on average, but rather exchanging it back and forth with the rest of the circuit. *Reactive power Q is thus a measure of the amount of periodic energy exchange taking place between a given subcircuit and the rest of the circuit.* By (10.29b), the sign of Q is positive for inductive reactances and negative for capacitive ones.

Example 10.8

Consider the series connection of a 1-Ω resistor, a $\frac{1}{2}$-F capacitor, a 1-H inductor, and an independent voltage source of $12\sqrt{2}\cos t$ V. The rms source phasor is

$$\mathbb{V}_{s\ rms} = 12/0$$

and the mesh current is

$$\mathbb{I}_{rms} = \frac{12/0}{1 - j2 + j1} = 8.5/\underline{45^\circ}$$

The complex power absorbed by each impedance is

$$\mathbb{S} = \mathbb{V}_{rms}\mathbb{I}^*_{rms} = \mathbb{Z}|\mathbb{I}_{rms}|^2$$

For the resistor, this is purely real with

$$\mathbb{S}_R = P_R + jQ_R = (1)(8.5^2) = 72 + j0$$

while for the capacitor

$$\mathbb{S}_C = P_C + jQ_C = (-j2)(8.5^2) = 0 - j144$$

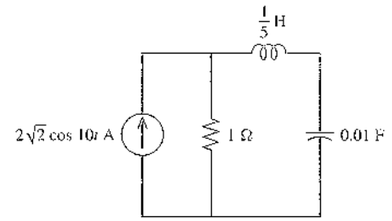
and for the inductor

$$\mathbb{S}_L = P_L + jQ_L = (j1)(8.5^2) = 0 + j72$$

Reversing the reference direction of $\mathbb{I}_{rms}$ to make it satisfy the passive sign convention together with $\mathbb{V}_{s\ rms}$ for calculation of the power absorbed by the source,

$$\mathbb{S}_V = P_V + jQ_V = (12/0)(8.5/\underline{-135^\circ})^* = -72 + j72$$

The reactive power of the resistor is zero, as anticipated since current and voltage in resistors are in phase. The storage elements are pure reactances and have 90° phase shift between their currents and voltages, and thus the complex power is purely reactive (zero average power). Note that the magnitude of the reactive power Q_C of the capacitor is twice that of the inductor Q_L. Since the impedance is larger by a factor of 2, the voltage across the capacitor is twice that across the inductor, and since both these voltages are in quadrature with the mesh current, the amplitude of the instantaneous power in the quadrature component is twice as large for the capacitor as for the inductor. The capacitor exchanges more energy per period, back and forth with the rest of the circuit, by a factor of 2. Finally, note that since the reactive power summed over the passive elements does not equal zero, the source must supply vars (as well as watts). That is, the source must develop an out-of-phase as well as an in-phase component. How the quadrature component may be adjusted to reduce the total current–voltage requirements at the source is discussed in Section 10.7.



10.3.1. The voltage across an impedance $\mathbb{Z} = 14\angle 37°\ \Omega$ has rms phasor $110\angle 30°$ V rms. Find the rms current phasor into the impedance and the complex power delivered to the impedance.

$\mathbb{I}_{\text{rms}} = 7.86\angle -7°$ A rms; $\mathbb{S} = 690.5 + j520.3$

10.3.2. Find the complex power delivered by each of the elements in this circuit.

$\mathbb{S}_{\text{source}} = -3.93 + j0.489$; $\mathbb{S}_R = 3.93 + j0$; $\mathbb{S}_L = j0.123$; $\mathbb{S}_C = -j0.612$

10.3.3. A parallel combination of 1-Ω resistor and L-H inductor is found to absorb equal watts and vars when $\omega = 10$ rad/s. What must L be?

100 mH

10.4 SUPERPOSITION AND POWER

In this section we consider power in networks containing two or more independent sources. Let i and v be the terminal current and voltage into a two-terminal subcircuit in a circuit containing two independent sources. The principle of superposition always applies to currents and voltages in any linear circuit, so

$$i = i_1 + i_2$$

$$v = v_1 + v_2$$

where the subscripts refer to the responses due to each source separately with the other killed. The instantaneous power absorbed by the subnetwork is, assuming i and v have reference directions that satisfy the passive sign convention,

$$p = iv = (i_1 + i_2)(v_1 + v_2)$$

or

$$p = i_1 v_1 + i_2 v_2 + (i_1 v_2 + i_2 v_1) \tag{10.32a}$$

$$p = p_1 + p_2 + (i_1 v_2 + v_1 i_2) \tag{10.32b}$$

where p_1 and p_2 are the instantaneous powers due to each source separately. If superposition were to apply to power, we would have p as the sum of p_1 and p_2, which from the last equation is not in general the case. *Superposition does not in general hold for instantaneous power.* That is, the instantaneous power into a subnetwork is not in general simply the sum of the instantaneous powers due to each excitation separately. Failure of superposition to work in general for power is a consequence of the nonlinear nature of power, its definition as the product of two circuit variables.

Next consider the same circuit in ac steady state. Suppose that both sources operate at the same frequency ω. If $\mathbb{I}_1$ and $\mathbb{I}_2$ are the rms current phasors and $\mathbb{V}_1$ and $\mathbb{V}_2$ the corresponding voltage phasors produced by the two sources, applying superposition to

current and voltage in the rms phasor circuit,

$$\mathbb{I} = \mathbb{I}_1 + \mathbb{I}_2$$

$$\mathbb{V} = \mathbb{V}_1 + \mathbb{V}_2$$

Thus the complex power into the subcircuit is

$$\mathbb{S} = \mathbb{V}\mathbb{I}^* = (\mathbb{V}_1 + \mathbb{V}_2)(\mathbb{I}_1 + \mathbb{I}_2)^*$$

or

$$\mathbb{S} = \mathbb{V}_1\mathbb{I}_1^* + \mathbb{V}_2\mathbb{I}_2^* + (\mathbb{V}_1\mathbb{I}_2^* + \mathbb{V}_2\mathbb{I}_1^*)$$

$$\mathbb{S} = \mathbb{S}_1 + \mathbb{S}_2 + (\mathbb{V}_1\mathbb{I}_2^* + \mathbb{V}_2\mathbb{I}_1^*)$$

where $\mathbb{S}_1$ and $\mathbb{S}_2$ and the separate complex powers due to each source. Since the term in parentheses in (10.32b) is not in general zero, *superposition does not hold in general for complex power.* Examining the real parts of both sides of this expression,

$$P = P_1 + P_2 + \text{Re}(\mathbb{V}_1\mathbb{I}_2^* + \mathbb{V}_2\mathbb{I}_1^*)$$

Since the last term in general is not zero, *superposition does not in general hold for average power P.* That is, the watts absorbed will not be the simple sum of those due to each source separately. Similarly, equating the imaginary parts of (10.32b) yields

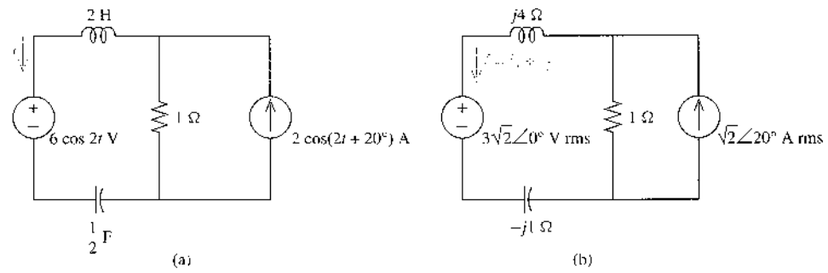
$$Q = Q_1 + Q_2 + \text{Im}(\mathbb{V}_1\mathbb{I}_2^* + \mathbb{V}_2\mathbb{I}_1^*)$$

Superposition does not in general hold for reactive power Q.

The failure of superposition shown above clearly extends to the case of more than two sources. Thus, in ac steady-state circuits containing multiple sources at the same frequency, power cannot be computed by superposing powers due to individual sources. This applies equally to all forms of power: instantaneous, average, reactive, or complex. Superposition may be used to find currents and voltages, but for sources at the same frequency, these components must be superposed *before* power is computed.

Example 10.9

To determine the net power supplied by the voltage source in Fig. 10.12(a), we note that since both sources in the circuit are at the same frequency $\omega = 2$ rad/s, we may determine the ac steady-state response using a single phasor circuit. The rms phasor circuit for $\omega = 2$ rad/s is shown in Fig. 10.12(b). Let $\mathbb{I}_1$ be the component of $\mathbb{I}$



(a) Circuit diagram; (b) rms phasor diagram.

due to the voltage source and $\mathbb{I}_2$ that due to the current source. Then

$$\mathbb{I}_1 = -\frac{3\sqrt{2}/0}{1+j3} = -0.424 + j1.27 \text{ A rms}$$

and by current division

$$\mathbb{I}_2 = \frac{\sqrt{2}/20^\circ}{1+j3} = 0.278 - j0.35 \text{ A rms}$$

Thus

$$\mathbb{I} = \mathbb{I}_1 + \mathbb{I}_2 = -0.146 + j0.92 \text{ A rms}$$

and the complex power absorbed by the voltage source is

$$\mathbb{S} = \mathbb{V}\mathbb{I}^* = (3\sqrt{2}/0)(-0.146 - j0.92) = -0.62 - j3.9$$

We conclude that the voltage source delivers 620 mW of average power and 3.9 vars of reactive power to the rest of the circuit.

The failure of superposition for ac steady-state power must be reconsidered when the sources are of different frequencies. Consider a subcircuit with terminal variables i and v in a circuit containing two sinusoidal sources at frequencies $\omega_1 \neq \omega_2$. By (10.30b), the instantaneous power is

$$p = p_1 + p_2 + (i_1 v_2 + v_1 i_2)$$

where p_1 and p_2 are the separate instantaneous powers due to each source with the other killed. Since the last term is a nonzero time function, superposition does not apply to instantaneous power in this case, any more than the previous case considered in which the frequencies were equal, $\omega_1 = \omega_2$.

When the frequencies are unequal, however, we cannot use (10.31) any longer since current or voltage phasors such as $\mathbb{V}_1$ and $\mathbb{V}_2$, which must be defined in different phasor circuits (corresponding to different frequencies ω_i) cannot be superposed. In this case, the average power is computed from (10.33) as

$$P = \frac{1}{T}\int_0^T p\,dt = \frac{1}{T}\int_0^T (p_1 + p_2 + i_1 v_2 + i_2 v_1)\,dt$$

or

$$P = P_1 + P_2 + \frac{1}{T}\int_0^T (i_1 v_2 + i_2 v_1)\,dt \tag{10.34}$$

Evaluating the first term in the integral, let

$$i_1 = I_{m1}\cos(\omega_1 t + \phi_{i1})$$
$$v_2 = V_{m2}\cos(\omega_2 t + \phi_{v2})$$

where we are interested in the case $\omega_1 \neq \omega_2$. Then

$$\frac{1}{T}\int_0^T (i_1 v_2)\,dt = \frac{I_{m1}V_{m2}}{T}\int_0^T \cos(\omega_1 t + \phi_{i1})\cos(\omega_2 t + \phi_{v2})\,dt \tag{10.35}$$

Using the identity (10.8), for $\omega_1 \neq \omega_2$, both resulting terms in the integrand are sinusoids with nonzero frequency, both have zero average values, and thus (10.35) evaluates to zero. By identical reasoning, the second term in the integral of (10.34) also evaluates to zero, and (10.35) simplifies to

$$P = P_1 + P_2$$

In other words, *superposition holds for average power P when the sources are of different frequencies.* Thus to compute the average ac steady-state power in a circuit excited by sources at two or more frequencies, we compute the average power for each frequency separately and then add average powers. By a similar argument, it can be shown that *superposition also holds for reactive power Q when the sources are of different frequencies.* Interestingly, when phasors can be superposed ($\omega_1 = \omega_2$), neither P nor Q can be, and when phasors cannot be superposed ($\omega_1 \neq \omega_2$), superposition works for both P and Q.

Finally, since both P and Q obey the superposition principle for distinct frequencies, so must $\mathbb{S}$, the complex power. Two examples illustrating the use of superposition of power in ac steady-state circuits excited by sources of different frequencies follow.

Example 10.10

We wish to determine the average power absorbed by the resistor in Fig. 10.13(a). This is the same circuit used in Example 10.9 [see Fig. 10.12(a)], except the frequency of one of the sources has been changed. Since the frequencies of the sources in the present case

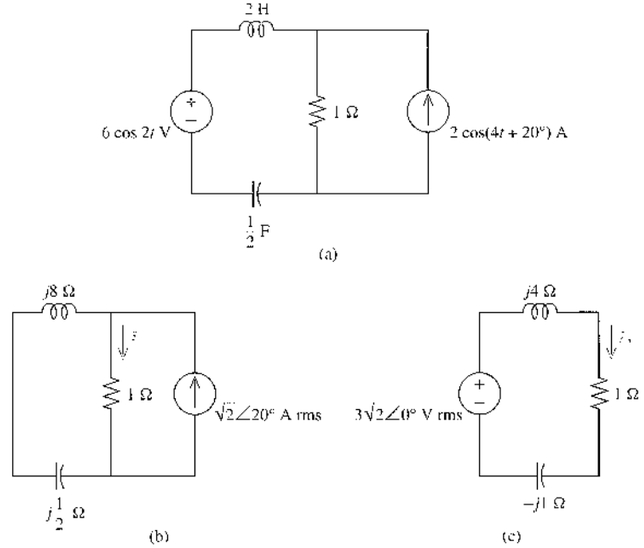


FIGURE 10.13 (a) Circuit diagram; (b) rms phasor diagram (ω = 4 rad/s); (c) rms phasor diagram ($\omega = 2$ rad/s).

differ, we must compute their individual responses from different phasor circuits. Then we may apply superposition to the average powers produced at each frequency.

By current division

$$\mathbb{I}_1 = (\sqrt{2}/\underline{20^\circ})\frac{j15/2}{1 + j15/2} = 1.40/\underline{27.6^\circ} \text{ A rms}$$

and the power into the resistor at $\omega = 4$ is

$$P_1 = \text{Re}(\mathbb{Z})|\mathbb{I}_1|^2 = R|\mathbb{I}_1|^2$$

$$= (1)(1.40)^2 = 1.96 \text{ W}$$

For $\omega = 2$ the mesh current is

$$\mathbb{I}_2 = \frac{3\sqrt{2}/0}{1 + j3} = 1.34/\underline{-71.6^\circ} \text{ A rms}$$

and the power into the resistor at $\omega = 2$ is

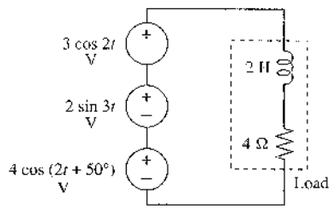
$$P_2 = R|\mathbb{I}_2|^2$$

$$= (1)(1.34)^2 = 1.80 \text{ W}$$

Thus the average power absorbed by the resistor is $1.96 + 1.80 = 3.76$ W.

Example 10.11

We wish to find the average and reactive power absorbed by the load in Fig. 10.14. To determine the power at $\omega = 2$, we kill the source at $\omega = 3$, and since the two remaining voltage sources are in series, the rms phasor current at $\omega = 2$ is



Circuit for Example 10.11.

$$\mathbb{I}_1 = \frac{1.5\sqrt{2}/0 + 2\sqrt{2}/\underline{50^\circ}}{4 + j4} = 0.795/-16.2^\circ \text{ A rms}$$

The complex power absorbed by the load at $\omega = 2$ is then

$$\mathbb{S}_1 = \mathbb{Z}|\mathbb{I}_1|^2 = 0.795^2(4 + j4) = 2.53 + j2.53$$

At $\omega = 3$, the rms source phasor is $\sqrt{2}/\underline{0}$ (selecting sine phasors for convenience) and impedance $\mathbb{Z} = 4 + j\frac{3}{2}$. Then

$$\mathbb{I}_2 = \frac{\sqrt{2}/0}{4 + j6} = 0.196/\underline{-56.3^\circ} \text{ A rms}$$

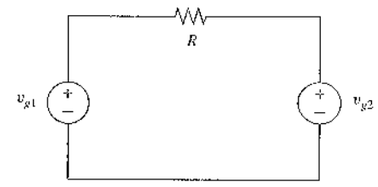
$$\mathbb{S}_2 = \mathbb{Z}|\mathbb{I}_2|^2 = 0.196^2(4 + j6) = 0.784 + j0.118$$

Since complex power at different frequencies superposes,

$$\mathbb{S} = \mathbb{S}_1 + \mathbb{S}_2 = 3.31 + j3.71$$

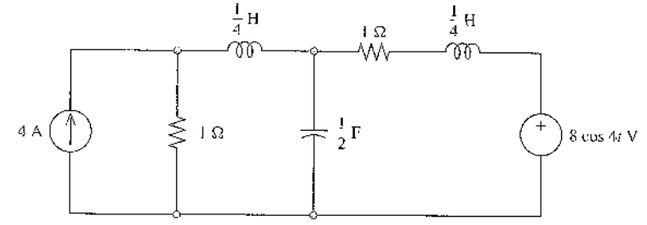
so the load absorbs 3.31 W of average power and 3.71 vars of reactive power.

In summary, for each frequency at which an ac steady-state circuit is excited, superposition may be applied at the level of currents and voltages, but not power. Then the average, reactive, or complex power may be found by superposing the power computed at each distinct frequency.

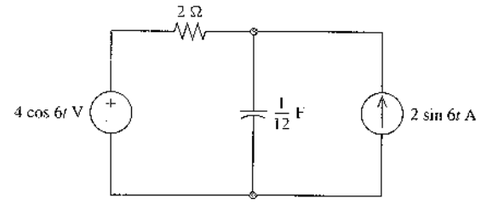


10.4.1. Find the average power delivered to the resistor if $R = 10\ \Omega$ and:
(a) $v_{g1} = 10\cos 100t$ and $v_{g2} = 20\cos(100t + 60^\circ)$ V
(b) $v_{g1} = 100\cos(t + 60^\circ)$ and $v_{g2} = 50\sin(2t - 30^\circ)$ V
(c) $v_{g1} = 50\cos(t + 30^\circ)$ and $v_{g2} = 100\sin(t + 30^\circ)$ V
(d) $v_{g1} = 20\cos(t + 25^\circ)$ and $v_{g2} = 30\sin(5t - 35^\circ)$ V
(a) 15 W; (b) 625 W; (c) 625 W; (d) 65 W

10.4.2. Find the average power absorbed by each resistor and each source.
8 W; 24 W; −8 W; −24 W

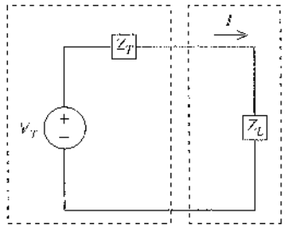


10.4.3. Find the average power absorbed by the resistor and each source.
8 W; −4 W; −4 W



10.5 MAXIMUM POWER TRANSFER

When designing a circuit, it is frequently desirable to arrange for the maximum possible power transfer to the load from the rest of the circuit. Consider the circuit of Fig. 10.15, consisting of a given two-terminal subcircuit connected to a load of impedance Z_L. We wish to specify Z_L so that the average power absorbed by this impedance is a maximum. Such problems arise, for instance, when specifying the impedance of an audio speaker system that will draw maximum power from the audio amplifier to



Load impedance connected to a subcircuit.

which it is attached. The process of specifying the load for maximum power from a given subcircuit is sometimes called impedance matching, for reasons that will soon be evident.

To simplify the analysis, we assume that the given two-terminal subcircuit is specified by its Thevenin equivalent, as shown in Fig. 10.15. The current into the load is given in terms of rms phasors by

$$\mathbb{I} = \frac{\mathbb{V}_T}{Z_L + Z_T}$$

and the average power absorbed by the load is, using (10.24b),

$$P_L = \mathrm{Re}(Z_L)|\mathbb{I}|^2$$

$$= \frac{\mathrm{Re}(Z_L)|\mathbb{V}_T|^2}{|Z_T + Z_L|^2}$$

Let $Z_L = R_L + jX_L$ and $Z_T = R_T + jX_T$. In terms of these variables, (10.36b) may be written

$$P_L = \frac{R_L}{(R_L + R_T)^2 + (X_L + X_T)^2}|V_T|^2$$

We wish to know the values of R_L and X_L that maximize P for fixed values of the remaining quantities. The maximizing choice for X_L is clearly $X_L = -X_T$, since this makes the denominator as small as possible. Setting $X_L = -X_T$ gives

$$\frac{\partial}{\partial R_L} P_L\Big|_{X_L = X_T} = \frac{\partial}{\partial R_L}\frac{R_L}{(R_L + R_T)^2}$$

the maximizing choice for R_L is found by setting this partial derivative to zero. It is easily checked that the partial above is zero for $R_L = R_T$ (and that the second partial derivative is negative, indicating a maximum). We conclude that the maximum power is transferred to a load impedance Z_L when the real parts of the source impedance $R_T = \mathrm{Re}(Z_T)$ and load impedance $R_L = \mathrm{Re}(Z_L)$ are equal, and their imaginary parts $X_T = \mathrm{Im}(Z_T)$ and $X_L = \mathrm{Im}(Z_L)$ are equal and opposite in sign. We may state both requirements together by concluding that *the maximum power is transferred to a load Z_L from a given source with Thevenin equivalent impedance Z_T if the load and Thevenin equivalent impedance are complex conjugates:*

$$Z_L = Z_T^*$$

Thus maximum power is transferred to a load if the magnitude of its impedance is matched to that of the source impedance Z_T and the angle of the load impedance matched to the negative of the angle of the source impedance. Load impedances whose magnitudes are large will sharply limit the current $\mathbb{I}$, resulting in less power delivery, while those with small impedance will have only a small fraction of the voltage $\mathbb{V}_T$ dropped across it, again reducing the power draw.

To determine the maximum power draw, using $Z_L = Z_T^*$ in (10.36b),

$$P_{\max} = \frac{\mathrm{Re}(Z_T)|\mathbb{V}_T|^2}{|2\ \mathrm{Re}(Z_T)|^2}$$

$$= \frac{|\mathbb{V}_T|^2}{4\ \mathrm{Re}(Z_T)}$$

or, in terms of $Y_T = 1/Z_T$,

$$P_{\max} = \frac{\mathrm{Re}(Y_T)|\mathbb{V}_T|^2}{4}$$

$$= \frac{|\mathbb{I}_T|^2}{4\ \mathrm{Re}(Y_T)}$$

It should be remembered that rms phasors are used in these expressions.

Example 10.12

Given the rms phasor circuit of Fig. 10.16(a), we first convert all but the load impedance to its Thevenin equivalent, as shown in Fig. 10.16(b). Z_T is the series equivalent of the impedances 3 and $j4$, or $3 + j4$. The open-circuit voltage with Z_L removed is

$$\mathbb{V}_T = (3)(4\underline{/0}) = 12\underline{/0}\ \text{V rms}$$

Thus, for maximum power transfer, we select $Z_L = Z_T^* = 3 - j4\ \Omega$. The maximum power draw is

$$P_{\max} = \frac{12^2}{(4)(3)} = 12\ \text{W}$$

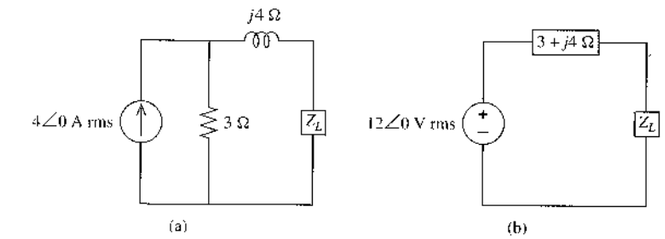
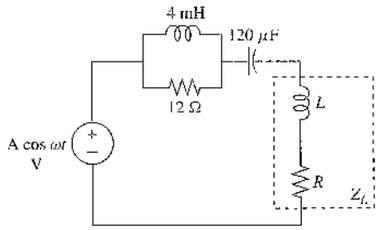


FIGURE 10.16 (a) Circuit for Example 10.12; (b) Thevenin equivalent.



10.5.1. Find values for load resistance R and and load inductance L so that the source with amplitude $A = 2\sqrt{2}$ V and frequency $\omega = 1000$ rad/s delivers maximum power to the load. Find the value of the power delivered to the load at these maximizing values for R and L.

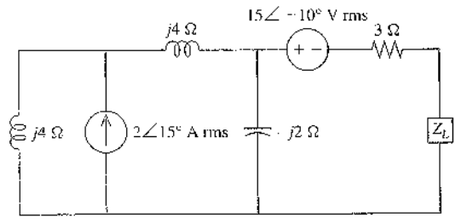
1.2 Ω; 4.73 mH; 833 mW

10.5.2. Repeat Exercise 10.5.1 for $A = 4\sqrt{2}$ V and $\omega = 1000$ rad/s. Note that the optimal values for the load impedance do not depend on the source strength A, but the maximum power delivered does.

1.2 Ω; 4.73 mH; 3.33 W

10.5.3. For the given rms phasor circuit, determine the load impedance Z_L that would absorb maximum power.

$4.0\angle 41.6°$ Ω



10.6 CONSERVATION OF POWER

Conservation laws play key roles throughout the physical sciences. In circuit theory, no principle is more fundamental than the conservation of electrical charge, an immediate consequence of which is Kirchhoff's current law. In this section we see how the general principles of conservation of power and energy apply to electrical circuits in ac steady state.

Consider an arbitrary $n+1$ node circuit with nodes numbered $0, 1, \ldots, n$. Let $\mathbb{V}_j$ be the node voltage rms phasor at node $j = 1, \ldots, n$ and node 0 be the reference node. The current flow from node j to node k will be designated $\mathbb{I}_{jk}$. By KCL at node j,

$$\sum_{k=0}^{n} \mathbb{I}_{jk} = 0$$

where $\mathbb{I}_{j0}$ is the current flow into the reference node and $\mathbb{I}_{jj} = 0$. Taking the complex conjugate of this KCL equation and multiplying by the node voltage $\mathbb{V}_j$ yields

$$\sum_{k=1}^{n} \mathbb{V}_j \mathbb{I}_{jk}^* + \mathbb{V}_j \mathbb{I}_{j0}^* = 0$$

where the term containing the reference node current is shown separately. There is one such equation for each node $j = 1, \ldots, n$. Adding all n equations together gives

$$\sum_{j=1}^{n} \left[\sum_{k=1}^{n} \mathbb{V}_j \mathbb{I}_{jk}^* + \mathbb{V}_j \mathbb{I}_{j0}^* \right] = 0$$

The current between every pair of nonreference nodes appears twice in the first double sum, as $\mathbb{I}_{jk}$ and its negative, $\mathbb{I}_{kj}$. Replacing $\mathbb{I}_{kj}$ by $-\mathbb{I}_{jk}$ for $k < j$, these two terms may be combined, producing the equation

$$\sum_{j=1}^{n} \left[\sum_{k=j+1}^{n} (\mathbb{V}_j - \mathbb{V}_k) \mathbb{I}_{jk}^* + \mathbb{V}_j \mathbb{I}_{j0}^* \right] = 0$$

Each term in this equation is recognized to be the complex power $\mathbb{S}_{jk}$ absorbed by the subcircuit connecting a particular node pair (j, k), and each node pair in the circuit appears exactly once. Thus this equation may be simplified to

$$\sum_{\text{node pairs } (j,k)} \mathbb{S}_{jk} = 0$$

Finally, we note that since we have already counted all nodes in the circuit, the subcircuit connecting nodes j and k can only consist of elements in parallel. Suppose that there are m such, for $m > 0$. These individual elements are shown in Fig. 10.17. The complex power into this parallel subcircuit is

$$\mathbb{S}_{jk} = (\mathbb{V}_j - \mathbb{V}_k) \mathbb{I}_{jk}^*$$

which, by KCL applied to the upper node, is equal to

$$\mathbb{S}_{jk} = \sum_{l=1}^{m} (\mathbb{V}_j - \mathbb{V}_k) \mathbb{I}_{jkl}^*$$

The terms inside the summation are the complex power absorbed by each of the m elements separately. Combining this with (10.38), we have the principle of *conservation of complex power: the sum of the complex powers absorbed by all the elements in a circuit equals zero.*

Conservation of complex power immediately implies conservation of average power P and reactive power Q, since these are the real and imaginary parts of $\mathbb{S}$. By putting

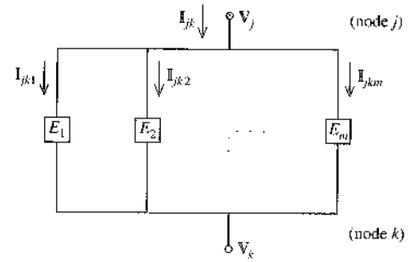
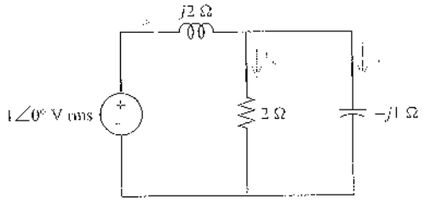


FIGURE 10.17 Elements connecting two nodes.

positive and negative terms on opposite sides of the equation, this conservation principle may be restated: *the net power absorbed by all elements in a circuit equals the net power delivered by all sources.* This principle, which holds equally for complex, average, and reactive power, is intuitively satisfying, since power absorbed must have come from somewhere and power delivered is delivered to somewhere. But it is important to have satisfied ourselves that such intuitively reasonable results follow from our basic definitions rather than simply to trust that they are probably true.

Example 10.13



Circuit for Example 10.13.

We wish to determine the "power budgets," that is, the power absorbed by or delivered by each element, in the rms phasor circuit shown in Fig. 10.18. The equivalent impedance is

$$\mathbb{Z} = j2 + \frac{(2)(-j1)}{2 - j1} = 0.4 + j1.2\ \Omega$$

and the current $\mathbb{I}$ is

$$\mathbb{I} = \frac{1/\underline{0}}{0.4 + j1.2} = 0.791/\underline{-71.6^\circ}\ \text{A rms}$$

At the source,

$$\mathbb{S}_V = \mathbb{V}\mathbb{I}^* = (1/\underline{0})(0.791/\underline{+71.6^\circ}) = 0.25 + j0.75$$

This complex power must be interpreted as the power delivered by the source, since $\mathbb{V}$ and $\mathbb{I}$ violate the passive sign convention. The power absorbed by the inductor is

$$\mathbb{S}_L = \mathbb{Z}_L|\mathbb{I}|^2 = (j2)(0.791)^2 = 0 + j1.25$$

We know that the complex power absorbed by a resistor is purely real and by a capacitor it is purely imaginary. By conservation of average power, the 0.25 W produced by the source must then be absorbed by the resistor, or

$$\mathbb{S}_R = 0.25 + j0$$

The source absorbs −0.750 var (produces +0.75 var) and the inductor absorbs an additional 1.25 var, so by conservation of reactive power the capacitor must produce 0.50 var or, in terms of power absorbed,

$$\mathbb{S}_C = 0 - j0.50$$

The last two results may be verified by direct calculations, since by current division

$$\mathbb{I}_R = \frac{(0.791/\underline{-71.6^\circ})(-j1)}{2 - j1} = 0.354/\underline{-135^\circ}\ \text{A rms}$$

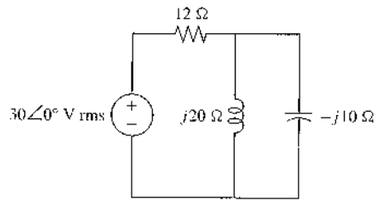
$$\mathbb{S}_R = Z_R|\mathbb{I}_R|^2 = (2)(0.354)^2 = 0.25 + j0$$

$$\mathbb{I}_C = \mathbb{I} - \mathbb{I}_R = 0.707/\underline{-45^\circ}\ \text{A rms}$$

$$\mathbb{S}_C = Z_C|\mathbb{I}_C|^2 = (-j1)(0.707)^2 = -j0.50$$

Finally, we note that since energy is the time integral of power and power is conserved, energy is conserved as well. The net energy produced by the sources in a circuit over a given period of time is absorbed by the other elements. Each source producing net energy may be studied further to determine the primary cause responsible for this energy production. In a storage battery, for instance, the primary cause is a chemical reaction converting electrochemical potential energy to electrical energy delivered by the battery to the rest of the electrical circuit. In a hydro generator it is the work done by the gravitational potential energy of falling water on a spinning shaft. Our study of electrical circuits, limited as it is to the electrical variables of current and voltage, will determine the amount of electrical energy produced by a source, but it is mute concerning its origins. All we can say from our circuit analysis is that if an element is a net producer of electrical energy, this energy must have been derived from some other, nonelectrical prime source.

EXERCISES



EXERCISE 10.6.1 Rms phasor circuit.

10.6.1. Determine the complex power absorbed by the resistor, inductor, and capacitor and show that their sum equals the complex power delivered by the source.

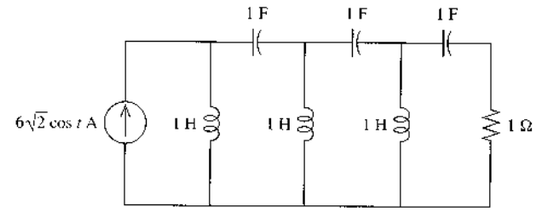
Answer: Power absorbed by each of the elements: $\mathbb{S}_R = 19.9$; $\mathbb{S}_L = +j33.1$; $\mathbb{S}_C = -j66.2$; $\mathbb{S}_V = -19.9 + j33.1$

10.6.2. A circuit consists of a 110-V rms source, a resistor, and an inductor. If the resistor absorbs 25 W and the inductor absorbs 40 var, what is the rms current produced by the source?

Answer: 429 mA rms

10.6.3. Use conservation of power to find the power delivered to the resistor by first finding the real part of the complex power supplied by the source.

Answer: 36 W



EXERCISE 10.6.3

10.7 REACTIVE POWER AND POWER FACTOR

The average power delivered to a load in the ac steady state, repeating (10.14), is

$$P = V_{\text{rms}} I_{\text{rms}} \cos\theta$$

where $\theta = \phi_v - \phi_i$. The power is thus equal to the product of the rms voltage, the rms current, and the cosine of the angle between the voltage and current phasors. In practice, the rms current and voltage are easily measured and their product, $V_{\text{rms}} I_{\text{rms}}$, is called the *apparent power*. The apparent power is usually referred to in terms of its units, volt-amperes (VA) or kilovolt-amperes (kVA), to avoid confusing it with the unit of average power, the watt. It is clear that the average power can never be greater than the apparent power.

The ratio of the average power to the apparent power is defined as the *power factor*. Thus if we denote the power factor by pf, then in the sinusoidal case

$$pf = \frac{P}{V_{\text{rms}} I_{\text{rms}}} = \cos\theta \tag{10.36}$$

which is dimensionless. The angle θ, in this case, is often referred to as the *pf angle*. We also recognize it as the angle of the impedance Z of the load.

In the case of purely resistive loads, the voltage and current are in phase. Therefore, $\theta = 0$, $pf = 1$, and the average and apparent powers are equal. A unity power factor ($pf = 1$) can also exist for loads that contain inductors and capacitors if the reactances of these elements are such that they cancel one another. Adjusting the reactance of loads so as to approximate this condition is very important in electrical power systems, as we shall see shortly.

In a purely reactive load, $\theta = \pm 90°$, $pf = 0$, and the average power is zero. In this case the equivalent load is an inductance ($\theta = 90°$) or a capacitance ($\theta = -90°$), and the current and voltage differ in phase by 90°.

A load for which $-90° < \theta < 0$ is equivalent to an RC combination, whereas one having $0 < \theta < 90°$ is an equivalent RL combination. Since $\cos\theta = \cos(-\theta)$, it is evident that the pf for an RC load having $\theta = -\theta_1$, where $0 < \theta_1 < 90°$, is equal to that of an RL load with $\theta = \theta_1$. To avoid this difficulty in identifying such loads, the pf is characterized as *leading* or *lagging* by the *phase of the current with respect to that of the voltage*. Therefore, an RC load has a leading pf and an RL load has a lagging pf. For example, the series connection of a 100-Ω resistor and a 0.1-H inductor at 60 Hz has $Z = 100 + j37.7 + 106.9/\underline{20.66°}$ Ω and a pf of $\cos 20.66° = 0.936$ lagging.

Example 10.14

In practice, the power factor of a load is very important. In industrial applications, for instance, loads may require thousands of watts to operate, and the power factor greatly affects the electric bill. Suppose, for example, that a mill consumes 100 kW from a 220-V rms line. At a pf of 0.85 lagging, we see that the rms current into the mill is

$$I_{\text{rms}} = \frac{P}{V_{\text{rms}} pf} = \frac{10^5}{(220)(0.85)} = 534.8 \text{ A}$$

which means that the apparent power supplied is

$$V_{\text{rms}} I_{\text{rms}} = (220)(534.8) \text{ VA} = 117.66 \text{ kVA}$$

Now suppose that the pf by some means is increased to 0.95 lagging. Then

$$I_{\text{rms}} = \frac{10^5}{(220)(0.95)} = 478.5 \text{ A}$$

and the apparent power is reduced to

$$V_{\text{rms}} I_{\text{rms}} = 105.3 \text{ kVA}$$

Comparing the latter case with the former, we see that I_{rms} was reduced by 56.3 A (10.5%). Therefore, the generating station must generate a larger current in the case of the lower pf. Since the transmission lines supplying the power have resistance, the generator must produce a larger average power to supply the 100 kW to the load. If the resistance is 0.1 Ω, for instance, then the power generated by the source must be

$$P_g = 10^5 + 0.1 I_{\text{rms}}^2$$

Therefore, we find

$$P_g = \begin{cases} 128.6 \text{ kW}, & pf = 0.85 \\ 122.9 \text{ kW}, & pf = 0.95 \end{cases}$$

which requires that the power station produce 5.7 kW (4.64%) more power to supply the lower pf load. It is for this reason that power companies encourage a pf exceeding, say, 0.9 and impose a penalty on large industrial users who do not comply.

Let us now consider a method of correcting the power factor of a load having an impedance Z or admittance $Y = 1/Z$. We may alter the power factor by connecting an element in parallel with the given load, as in Fig. 10.19. By placing the new element in parallel, the voltage delivered to the load will not be divided between the elements, as it would if the new element were placed in series.

Designate the new element by its admittance Y_1. The total admittance of the parallel combination is the sum of admittances

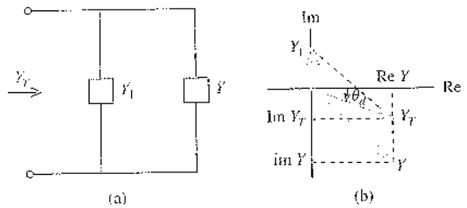
$$Y_T = Y + Y_1$$

We wish the new element to absorb no average power itself, so it must be a pure reactance, $\text{Re}(Y_1) = 0$ shown as a vertical vector in the phasor diagram. If our goal is to adjust the power factor pf to some desired value $pf = PF$, then, as shown in the diagram with $\cos\theta_d = PF$,

$$Y_1 = j\,[\text{Im}(Y) - \text{Im}(Y_T)]$$

where

$$\text{Im}(Y_T) = \text{Re}(Y)\tan\theta_d$$



(a) Circuit for power factor correction; (b) phasor diagram.

In other words, Y_1 must reduce the imaginary part of Y so that with fixed real part the angle of Y_T will have the desired power factor PF.

Example 10.15

As an example, let us change the power factor for the circuit of Example 10.14 (shown in Fig. 10.18) to 0.98 leading. The impedance of this circuit was found in Example 10.13 to be $Z = 0.4 + j1.2\ \Omega$, so its admittance is

$$Y = \frac{1}{Z} = 0.25 - j0.75 = 0.791\underline{/-71.6^\circ}\ \text{S}$$

Thus the circuit as given has a power factor of $\cos(71.6^\circ) = 0.32$, which is a lagging power factor (the angle of an admittance in this case -71.6°, is the angle by which the current leads the voltage). The desired Y_T must have an angle of $\theta_d = \cos^{-1} 0.98 = +11.5^\circ$ (Y_T of phase angle -11.5° would be 0.98 *lagging* rather than leading). The unique complex number that has real part 0.25 and angle $+11.5^\circ$ has as its imaginary part

$$\text{Im}(Y_T) = (0.25)(\tan 11.5^\circ) = 0.051\ \text{S}$$

Thus we must change the admittance from $Y = 0.25 - j0.75$ S to $Y_T = 0.25 + j0.051$ S. This requires that

$$Y_1 = Y_T - Y = j(0.051 + 0.75) = j0.801\ \text{S}$$

Thus, to correct the power factor to the desired value, we need to install an admittance of $j0.801$ S or, equivalently, an impedance of $1/j0.801 = -j1.25\ \Omega$, in parallel with the given impedance. For instance, if the working frequency for the circuit is $\omega = 1$ rad/s, then the power factor correction would be achieved by using a capacitor with impedance

$$\frac{1}{j(1)C} = -j1.25\ \Omega$$

or $C = 0.80$ F.

10.7.1. Find the apparent power for (a) a load that requires 20 A rms from a 115-V rms line and (b) a load consisting of a 100-Ω resistor in parallel with a 25-μF capacitor connected to a 120-V rms 60-Hz source.

(a) 2.3 kVA; (b) 197.9 VA

10.7.2. Find the power factor for (a) a load consisting of a series connection of a 10-Ω resistor and a 10-mH inductor operating at 60 Hz, (b) a capacitive load requiring 25 A rms and 5 kW at 230 V rms, and (c) a load that is a parallel connection of a 5-kW load with a power factor of 0.9 leading and a 10-kW load with a 0.95 lagging power factor.

(a) 0.936 lagging; (b) 0.87 leading; (c) 0.998 lagging

10.7.3. A load consisting of a series combination of a 100-Ω resistor and 1-H inductor is connected to a source delivering power at $\omega = 100$ rad/s. What capacitor C across this load will correct the power factor to (a) 0.9 leading; (b) 0.9 lagging?

(a) $C = 74.22\ \mu$F; (b) $C = 25.78\ \mu$F

When an ac analysis is requested of SPICE by inclusion of the .AC control statement, some additional information labeled "small-signal bias solution" accompanies the output. This includes a "total power dissipated" value. For instance, Example 9.17 used the source file

```
Circuit of Fig. 9.25
*
V     1     0   AC     1    0
L     1     2   1
H     2     3   V      0.5
R     3     0   1
.AC   LIN   1   .159   .159
.PRINT   AC   IM(V)   IP(V)
.END
```

The following data were found in the SPICE printout above the printout that we requested (the magnitude and phase angle of the current though the voltage source labeled V).

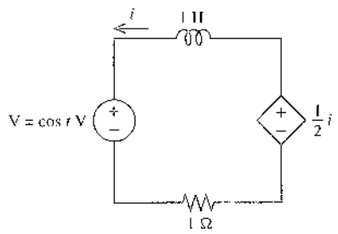
```
*** SMALL SIGNAL BIAS SOLUTION TEMP = 27.0 DEG C ***
NODE VOLTAGE          NODE VOLTAGE          NODE VOLTAGE
( 1) 0.0000          ( 2) 0.0000          ( 3) 0.0000

VOLTAGE SOURCE CURRENTS
NAME     CURRENT
V        0.000E+00

TOTAL POWER DISSIPATION 0.00E+00 WATTS
```

The power dissipation reported in this portion of the printout is *not* related to the ac power dissipated by the elements of the circuit. Whenever an ac analysis is requested, SPICE automatically first determines the dc operating point of the circuit, that is, the dc or constant values of all currents and voltages when the ac sources are killed. This operating point information is of great value in the case that nonlinear elements, such as transistors and diodes, are included in the circuit, and dc supplies are used to power them. This is the "small-signal bias solution" reported. The "total power dissipation" reported refers to steady-state dc power required to support this operating point. In a circuit with no dc sources all these values will be zero. Thus the power dissipation automatically reported by SPICE is of no help in determining ac steady-state power dissipations at nonzero frequencies.

Sadly, SPICE offers no specific control card or keyword for printing out ac steady-state power at any other frequency than $\omega = 0$ (dc) as described above. To determine the ac power absorbed by a circuit element using SPICE, we must request output of the current and voltage phasors; then use the formulas studied in this chapter to get complex, average, or reactive power (as required) from knowledge of the element current and voltage.



Circuit for Example 10.16.

We wish to determine the complex power supplied by the two sources in the circuit of Fig. 10.20. The SPICE input file for this circuit was listed at the beginning of this section. The output from this SPICE run is

```
FREQ         IM(V)        IP(V)
1.590E-01    8.951E-01    1.166E+02
```

The current phasor associated with the counterclockwise mesh current i in Fig. 10.20 is thus $\mathbb{I} = 0.895/116.6°$ A. Since this current phasor satisfies the passive sign convention relative to the independent source phasor voltage $\mathbb{V} = 1/0$, the complex power absorbed by the independent source is given by

$$\tfrac{1}{2}\mathbb{V}\mathbb{I}^* = \tfrac{1}{2}(1\underline{/0})(0.895\underline{/-116.6°}) = -0.2 - j0.4 \qquad (10.\ldots)$$

and the complex power absorbed by the dependent source is given by

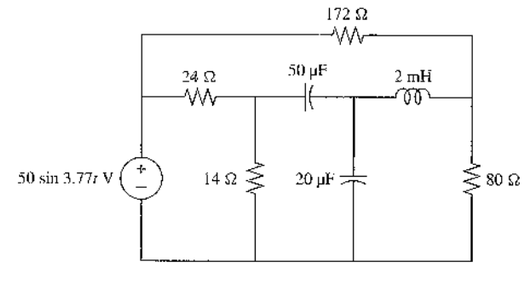
$$\tfrac{1}{2}\left(\tfrac{1}{2}\mathbb{I}\right)(-\mathbb{I})^* = -0.2 + j0 \qquad (10.\ldots)$$

Note that we used $-\mathbb{I}$, not $\mathbb{I}$, in the last calculation, since $-\mathbb{I}$ satisfies the passive sign convention for the dependent source (together with its voltage $\frac{1}{2}\mathbb{I}$). Also note that the phasors reported by SPICE are amplitude, not rms, phasors since we used amplitude phasors to describe the source. Thus we must include the factor $\frac{1}{2}$ in the power formulas (10.40) consistent with the use of non-rms standard or amplitude phasors. Examining our result, we see that the independent and dependent sources each deliver 200 mW to the rest of the circuit. By conservation of average power, the resistor must be absorbing this 400 mW. In addition, the independent source supplies 0.4 var to the inductor.

Finally, note that rms values may be used to describe the magnitude of sources in SPICE input files. In this case all calculated current and voltage phasor magnitudes will be in rms as well, and the factor of $\frac{1}{2}$ should be omitted from calculations of complex power such as (10.40a).

10.8.1. Find the power delivered to the 80-Ω resistor.

Answer: 13.0 mW



10.8.2. Check the results of Example 10.11 (Fig. 10.14) using SPICE.

Answer: 3.02 W, 2.74 var. Note that two SPICE runs are required, one at $\omega = 2$ rad/s and one at $\omega = 3$ rad/s (0.318 and 0.477 Hz).

SUMMARY

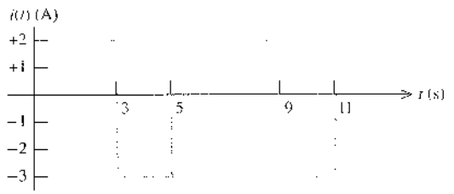
The power calculations are important not just to the electrical power generation and distribution industries, but also in the design of efficient circuits of all kinds and of miniaturized circuits which must dissipate heat over very small areas. Just as was the case with currents and voltages in Chapters 8 and 9, the calculation of power in ac steady-state circuits is greatly simplified by the use of phasors.

- For terminal variables $v(t)$ and $i(t)$ that satisfy the passive sign convention, instantaneous power $p(t) = v(t)i(t)$ greater than zero implies the circuit between the terminals is absorbing power at t, less than zero that it is supplying power to the rest of the circuit at t.
- Average power equals the product of current amplitude, voltage amplitude, and the cosine of the difference of their phase angles. If current and voltage satisfy the passive

sign convention, the average power greater than zero implies that power is on average dissipated, less than zero, and that it is being supplied to the rest of the circuit.

- The rms value of a sinusoid equals its amplitude divided by $\sqrt{2}$. An rms phasor is a phasor whose magnitude equals the rms value of the corresponding sinusoid.
- The real part of the complex power $S = \mathbf{V}_{\text{rms}}\mathbf{I}^*_{\text{rms}}$ is the average power (SI unit watt); the imaginary part is the reactive power (SI unit var).
- In a circuit containing sources at different frequencies, the total power for each component is the sum of the power delivery or dissipation at each frequency.
- Maximum power is delivered to a load impedance if it is the complex conjugate of the Thevenin equivalent impedance of the rest of the circuit.
- All forms of power are conserved: instantaneous, average, reactive, and complex.
- The power factor is the cosine of the phase angle between voltage and current. Unity power factor requires the least rms current to deliver a fixed amount of average power to a load at a given voltage level.

10.1. $i(t)$ flows through a 1-kΩ resistor. Find T, the period of $i(t)$, T_p, the period of the instantaneous power, and P, the average power.



10.2. Let $v(t) = 2\cos \pi t$ V for $3n \le t < 3n + 1$, $n = 0, \pm 1, \pm 2, \ldots$, and $v(t) = 0$ elsewhere. If $v(t)$ is across a 1-Ω resistor, sketch the instantaneous power $p(t)$ and find the average power P.

10.3. One cycle of a periodic current is given by

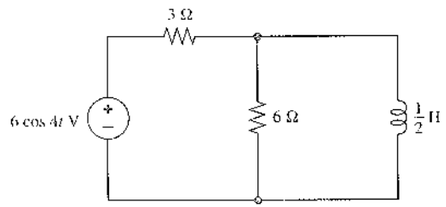
$$i = 10 \text{ A}, \quad 0 \le t < 1 \text{ ms}$$
$$= 0. \qquad 1 \le t < 4 \text{ ms}$$

If the current flows in a 20-Ω resistor, find the average power.

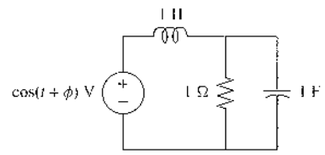
10.4. One period of a periodic current is as shown. If the current flows in a 20-Ω resistor, find the average power.

10.5. Find the average power dissipated in a resistor of resistance R if the current is $i = I_m(1 + \cos \omega t)$ A.

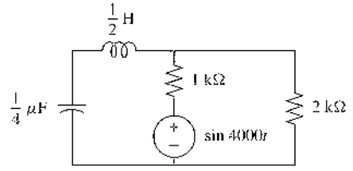
10.6. Find the average power absorbed by the resistors, the inductor, and the source.



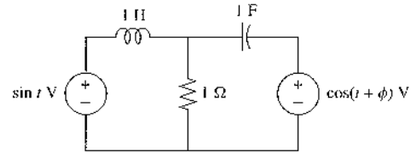
10.7. Find the average power delivered to the resistor when $\phi = 0$. How does the average power depend on the phase angle ϕ?



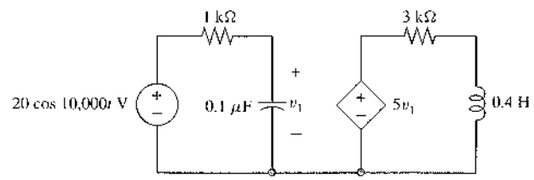
10.8. Find the average power delivered to each *RLC* element and the average power delivered by the source.



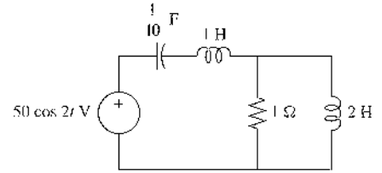
10.9. For what value of ϕ is the average power delivered to the 1-Ω resistor smallest?



10.10. Find the average power absorbed by the 3-kΩ resistor and the dependent source.



10.11. Sketch the rms phasor circuit and find the average power supplied by the source.

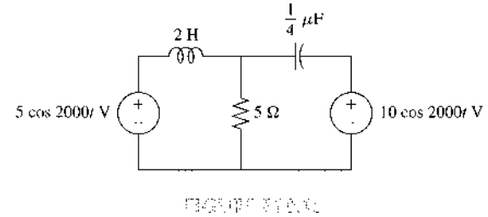


10.12. Find the rms value of the voltage
(a) $v = 5 + 12\sqrt{2}\cos t$ V
(b) $v = 12\cos(2t - 30°) + 4\cos 5t + \sqrt{2}\cos(8t + 60°)$ V
(c) $v = 4\cos t - 6\sqrt{2}\cos(2t + 45°) + \sqrt{10}\cos(5t - 10°)$ V

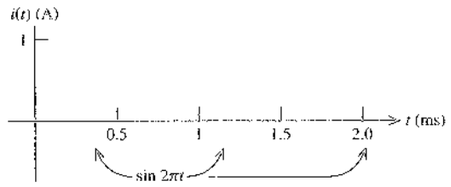
10.13. Find the rms value of a periodic current for which one period is given by
(a) $i = 4$ A, $\quad 0 < t < 1$ s
$\quad = 0, \quad 0 < t < 2$ s
(b) $i = 3t$ A, $\quad 0 < t < 2$ s
$\quad = -3(t - 4)$ A, $\quad 0 < t < 4$ s
(c) $i = I_m \sin \dfrac{2\pi t}{T}$ A, $\quad 0 < t < \dfrac{T}{2}$ s
$\quad = 0, \quad \dfrac{T}{2} < t < T$ s
(d) $i = I_m \sin \dfrac{\pi t}{T}$ A, $\quad 0 < t < T$ s

10.14. Find an expression for P, the average power absorbed by an impedance Z, in terms of $|V_{\text{rms}}|^2$, where V_{rms} is the rms voltage phasor across the impedance.

10.15. Sketch the rms phasor circuit and find the average power delivered to each element.



10.16. Determine the rms current; $i(t)$ is periodic with period 1 ms, and equals $\sin 2\pi t$ for $0 \le t < 0.5$, and zero for $0.5 \le t < 1$ ms.

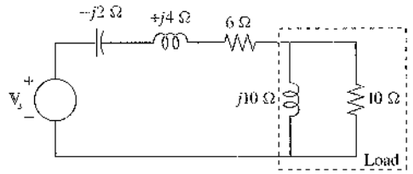


10.17. Show that the average power absorbed by an impedance Z may be expressed as $P = |\mathbb{I}_{rms}|^2 \text{ Re } Z$ where $\mathbb{I}_{rms}$ is the rms current phasor.

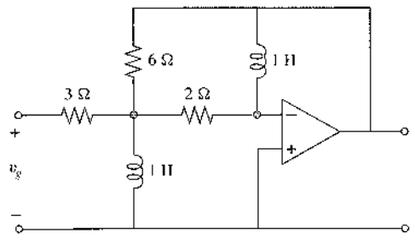
10.18. An impedance $Z = 2 - 3j\ \Omega$ carries a current $i(t) = 3\sin(10t - 14°)$ A. Find the complex power absorbed by Z.

10.19. A series combination of a resistor and capacitor absorbs 100 W and −42 vars at 60 Hz. If $R = 100\ \Omega$, what must C be?

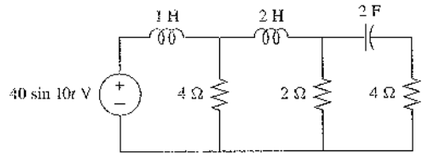
10.20. $\mathbb{V}_s$ is the rms source phasor. What must $|\mathbb{V}_s|$ be in order that the average power absorbed by the load be 25 W?



10.21. Find the complex power delivered to each inductor and resistor if $v_g = 0.10\sin(2t + 30°)$ V.

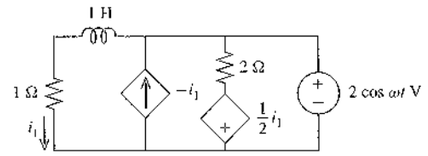


10.22. Determine the complex power supplied by the source. Use the ladder method to find the equivalent impedance.

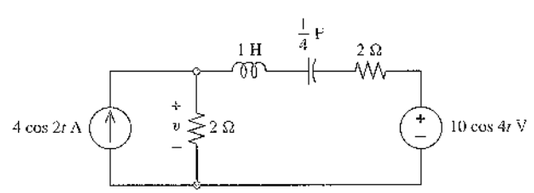


10.23. Solve Problem 10.8 by finding the complex power for each element and taking the real part.

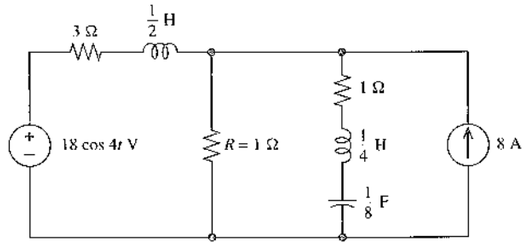
10.24. Find the reactive power absorbed by the inductor if (a) $\omega = 1$ rad/s; (b) $\omega = 0$.



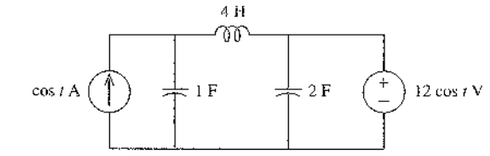
10.25. Find the rms value of the steady-state voltage v.



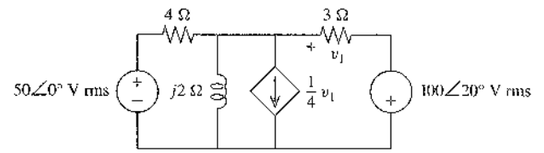
10.26. Find the rms value of the steady-state current in R.



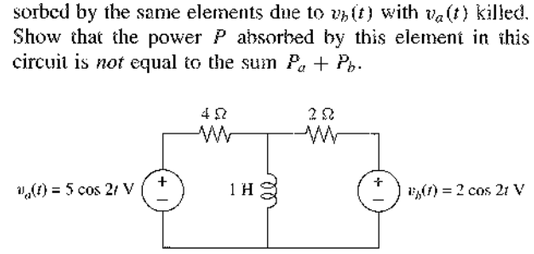
10.27. Find the complex power delivered by each source.



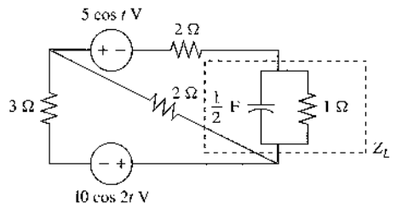
10.28. For the rms phasor circuit shown, find the complex power supplied by the controlled source.



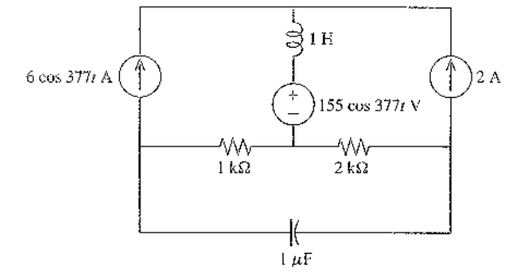
10.29. Find P_a, the average power absorbed by the 4-Ω resistor due to $v_a(t)$ with $v_b(t)$ killed, and P_b, the power absorbed by the same elements due to $v_b(t)$ with $v_a(t)$ killed. Show that the power P absorbed by this element in this circuit is *not* equal to the sum $P_a + P_b$.



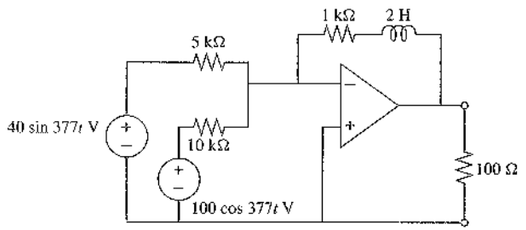
10.30. Compute the watts supplied to the load Z_L.



10.31. Determine the average power P delivered by each of the three sources.

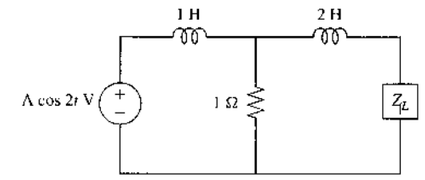


10.32. Find the power delivered to the 100-Ω load resistor.

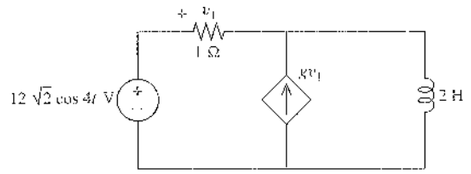


10.33. For a Thevenin equivalent circuit consisting of a voltage source $\mathbb{V}_g$ and an impedance $\mathbb{Z}_g = R_g + jX_g$, (a) show that the circuit delivers maximum average power to a load $\mathbb{Z}_L = R_L + jX_L$ when $R_L = R_g$ and $X_L = -X_g$, and (b) show that the maximum average power is delivered to a load R_L when $R_L = |\mathbb{Z}_g|$. This is the *maximum power transfer theorem* for ac circuits. (In both cases, $\mathbb{V}_g$ and $\mathbb{Z}_g$ are fixed, and the load is variable.)

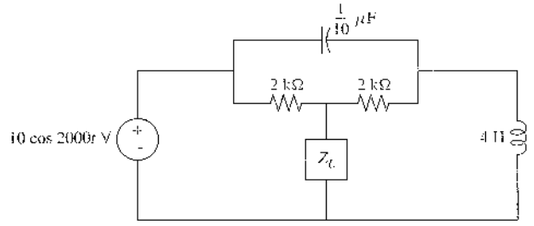
10.34. What must the amplitude A of the source be to supply 100 W to the purely resistive load $Z_L = 2\ \Omega$? What would it be reduced to if we could adjust Z_L to absorb 100 W using the lowest possible value source amplitude A?



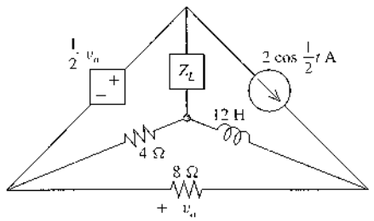
10.35. Find the value of transconductance g for which the real average power delivered by the independent source is maximized. Repeat for apparent power.



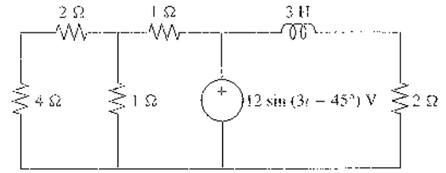
10.36. Find the load impedance Z_L that will absorb maximum real average power. Find the value of the maximum real average power into the load.



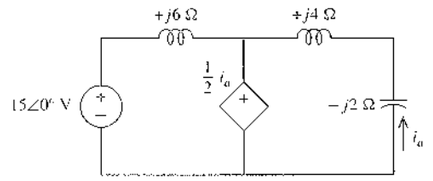
10.37. For maximum power transfer to the load Z_L, should Z_L be constructed as an RL circuit or an RC circuit?



10.38. Find the reactive power absorbed by the inductor. (*Hint*: Find the complex power supplied by the source.)



10.39. Verify conservation of complex power for this rms phasor circuit.

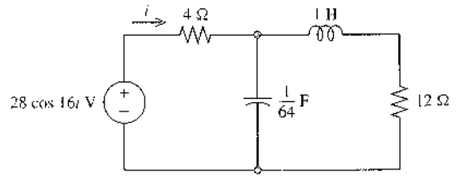


10.40. An ideal 220-V rms voltage source is put across a load impedance of $Z = 11 + j11\ \Omega$. How many vars does the source supply? If $\omega = 377$ rad/s, what capacitor C connected across the load will reduce the vars supplied by the source to zero?

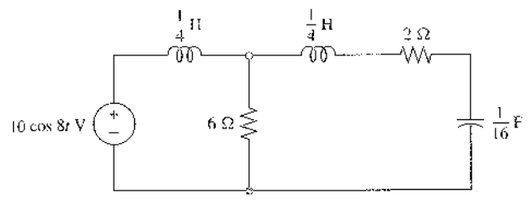
10.41. Show that for all $R > 0$, $C > 0$, and $\omega > 0$, the power factor of a series RC circuit and of a parallel RC circuit are leading.

10.42. Show that for all $R > 0$, $L > 0$, and $\omega > 0$, the power factor of a series RL circuit and of a parallel RL circuit are lagging.

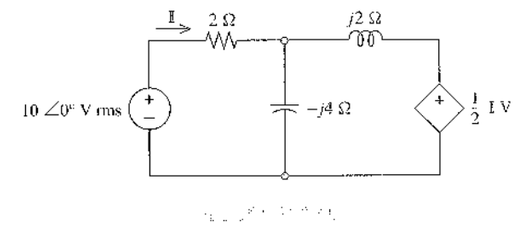
10.43. Find the rms value of the steady-state current i and the power factor seen from the source terminals. What element connected in parallel with the source will correct the power factor to 0.8 lagging?



10.44. Find the power factor seen from the terminals of the source and the reactance necessary to connect in parallel with the source to change the power factor to unity.



10.45. Find the power factor seen from the terminals of the independent source and the reactance which must be connected in parallel with the independent source to change the power factor to 0.8 lagging.



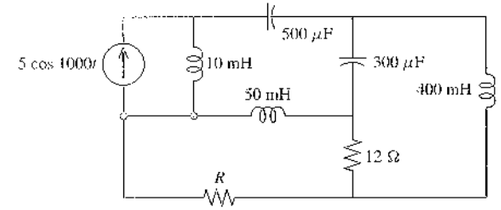
10.46. Three parallel passive loads, $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$, are receiving complex power values of $6 - j5$, $8 + j10$, and $2 + j7$ VA, respectively. If a voltage source of $50\underline{/0}$ V rms is connected across these loads, find the rms value of the current that flows from the source and the power factor seen by the source.

10.47. Two loads in parallel draw a total of 3 kW at a 0.9 lagging power factor from a 115-V rms 60-Hz line. One load is known to absorb 1000 W at a 0.8 lagging power factor. Find (a) the power factor of the second load and (b) the parallel reactive element necessary to correct the power factor to 0.95 lagging for the combined load.

SPICE Problems

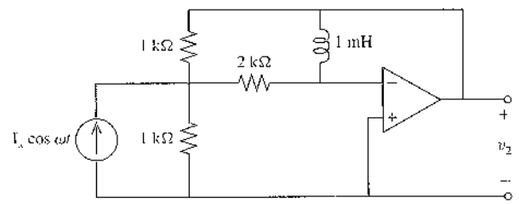
10.48. A load impedance $Z_L = 1 + j0.5$ is connected to a 10-mA rms ideal current source. What is the power factor of the load? How much could the voltage rating of the current source be reduced if a capacitor was put in series with the load, creating an overall power factor of 0.95 lagging? Use SPICE to determine the capacitance which corrects the pf to 0.95 lagging at 60 Hz.

10.49. Use SPICE to help determine the value of R for which the source supplies 1.00 mW to the rest of the circuit.

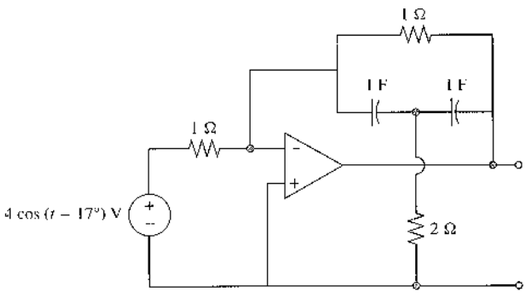


10.50. Repeat Problem 10.21 using SPICE. Use the ideal voltage amplifier op amp model of Fig. 3.7 with open-loop gain $A = 10^6$.

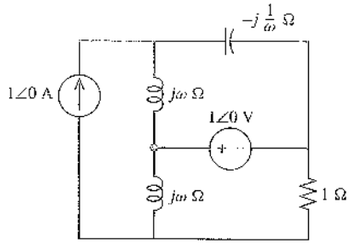
10.51. At what frequency ω is the rms value of v_2 maximized?

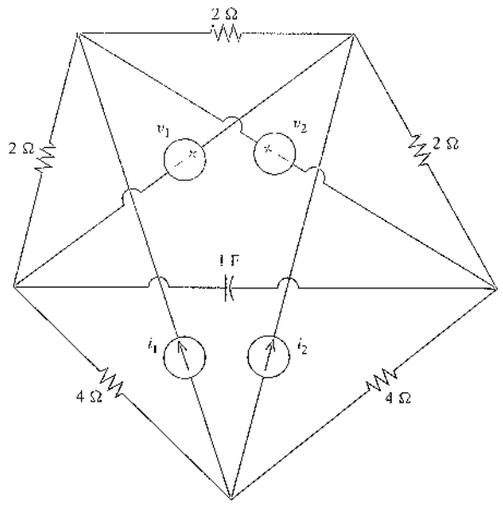


10.52. (a) Determine the complex power delivered by the independent source. (b) Does the answer to (a) depend upon the source phase angle $(-17°)$? Explain.



For what range ω is the current supplying more real average power than the voltage source?





Find the complex power delivered by each source. $i_1 = 2\cos 2t$ A, $i_2 = 6\sin 2t$ A, $v_1 = 12\sin(3t - 45°)$ V, $v_2 = 24\sin(3t - 45°)$ V.

11

Three-Phase Circuits



Nikola Tesla
1856–1943

He [Tesla] was the greatest inventor in the realm of electrical engineering.

W. H. Eccles

If Thomas A. Edison has a rival for the title of the world's greatest inventor, it is certainly the Croatian-American engineer, Nikola Tesla. When the tall, lanky Tesla arrived in the United States in 1884, the country was in the middle of the "battle of the currents" between Thomas A. Edison, promoting dc, and George Westinghouse, leading the ac forces. Tesla quickly settled the argument in favor of ac with his marvelous inventions, such as the polyphase ac power system, the induction motor, the Tesla coil, and fluorescent lights.

Tesla was born in Smiljan, Austria-Hungary (now Croatia), the son of a clergyman of the Greek Orthodox Church. As a boy Tesla had a talent for mathematics and an incredible memory, with the ability to recite by heart entire books and poems. He spent two years at the Polytechnic Institute of Graz, Austria, where he conceived the idea of the rotating magnetic field that was the later basis for his induction motor. At this point in Tesla's life his father died, and he decided to leave school, taking a job in Paris with the Continental Edison Company. Two years later he came to America, where he remained until his death. During his remarkable lifetime he held over 700 patents, settled the ac versus dc dispute, and was primarily responsible for the selection of 60 Hz as the standard ac frequency in the United States and throughout much of the world. After his death he was honored by the choice of *tesla* as the unit of magnetic flux density.

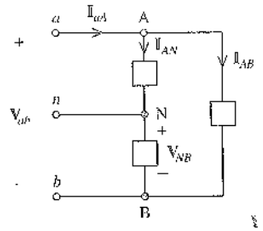
11.1 Single-Phase Three-Wire Systems
11.2 Three-Phase Wye–Wye Systems
11.3 Single-Phase versus Three-Phase Power Delivery
11.4 Delta Connection
11.5 Wye–Delta Transformations
Summary
Problems

As we have already noted, one very important use of ac steady-state analysis is its application to power systems, most of which are alternating current systems. One principal reason for this is that it is economically feasible to transmit power over long distances only if the voltages involved are very high, and it is easier to raise and lower voltages in ac systems than in dc systems. Alternating voltage can be stepped up for transmission and stepped down for distribution with transformers, as we shall see in Chapter 15. Transformers have no moving parts and are relatively simple to construct. In a dc system, on the other hand, the waveforms must first be converted to ac before transformers may be used to change voltage levels and then converted back to dc. This process requires expensive high-power switches and dissipates some of the energy it is converting.

Also, for reasons of economics and performance, almost all electric power is produced by *polyphase* sources (those generating voltages with more than one phase). In a single-phase circuit, the instantaneous power delivered to a load is pulsating, even if the current and voltage are in phase. A polyphase system, on the other hand, is somewhat like a multicylinder automobile engine in that the power delivered is much steadier. An economic advantage is that the weight of the conductors and associated components required in a polyphase system is appreciably less than that required in a single-phase system that delivers the same power. Virtually all the power produced in the world is polyphase power at 50 or 60 Hz. In the United States, 60 Hz is the standard frequency.

In this chapter we begin with single-phase three-wire systems, but we concentrate on three-phase circuits, which are by far the most common of the polyphase systems. In the latter case the sources are three-phase generators that produce a *balanced* set of voltages, by which we mean three sinusoidal voltages having the same amplitude and frequency but displaced in phase by 120°. Thus the three-phase source is equivalent to three interconnected single-phase sources, each generating a voltage with a different phase. If the three currents drawn from the sources also constitute a balanced set, the system is said to be a *balanced* three-phase system. A balanced system is required to reap the full benefits of polyphase power, and this is the case on which we concentrate our attention.

11.1 SINGLE-PHASE THREE-WIRE SYSTEMS



Double-subscript notation.

Before proceeding to the three-phase case, let us digress in this section to establish our notation and consider an example of a single-phase system that is in common household use. The circuits we have studied thus far are all single phase and are a good frame of reference for understanding the significance of multiphase.

In this chapter we shall find extremely useful the double-subscript notation introduced in Chapter 1 for voltages. In the case of phasors, the notation is $\mathbb{V}_{ab}$ for the voltage of point a with respect to point b. We shall also use a double-subscript notation for current, taking, for example, $\mathbb{I}_{aA}$ as the current flowing in the *direct* path from node a to node A. These quantities are illustrated in Fig. 11.1, where the direct path (no intermediate nodes) from A to B is distinguished from an indirect path from A to B through N.

Because of the simpler expressions for average power that result, we shall use rms values of voltage and current, and rms phasors, throughout this chapter. That is, if

$$\mathbb{V} = |\mathbb{V}|\underline{/0}\ \text{V rms}$$

$$\mathbb{I} = |\mathbb{I}|\underline{/-\theta}\ \text{A rms}$$

are the rms phasors associated with an element having impedance,

$$Z = |Z|/\theta\ \Omega$$

the average power delivered to the element is

$$P = |\mathbb{V}| \cdot |\mathbb{I}| \cos\theta$$

$$= |\mathbb{I}|^2 \operatorname{Re}(Z)\ \text{W}$$

In the time domain the voltage and current are

$$v = \sqrt{2}|\mathbb{V}| \cos\omega t\ \text{V}$$

$$i = \sqrt{2}|\mathbb{I}| \cos(\omega t - \theta)\ \text{A}$$

All phasors in this chapter will be understood to be rms phasors, whose magnitude corresponds to the rms value of the associated current or voltage, as introduced in Section 10.2.

Example 11.1

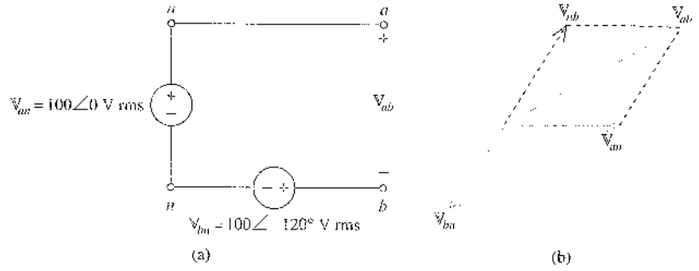
The use of double subscripts makes it easier to handle phasors both analytically and geometrically. For example, in Fig. 11.2(a), the voltage $\mathbb{V}_{ab}$ is

$$\mathbb{V}_{ab} = \mathbb{V}_{an} + \mathbb{V}_{nb}$$

This is evident without referring to a circuit since by KVL the voltage between two points a and b is the same regardless of the path, which in this case is the path a–n–b. Also, since $\mathbb{V}_{nb} = -\mathbb{V}_{bn}$, we have

$$\mathbb{V}_{ab} = \mathbb{V}_{an} - \mathbb{V}_{bn}$$

$$= 100 - 100\underline{/-120°}$$



(a) Phasor circuit; (b) corresponding phasor diagram.

which, after simplification, is

$$\mathbb{V}_{ab} = 100\sqrt{3}\underline{/30°} \text{ V rms}$$

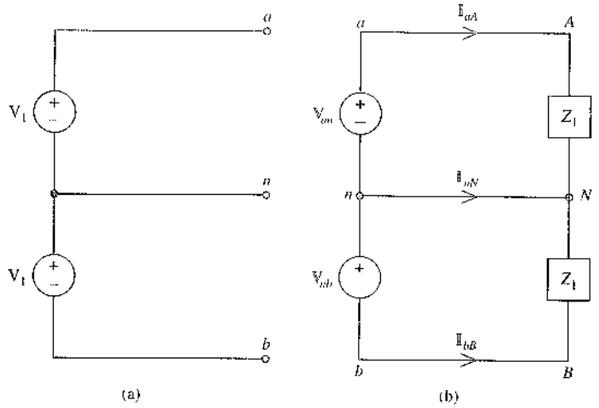
These steps are shown graphically in Fig. 11.2(b).

A single-phase three-wire source, as shown in Fig. 11.3, is one having three output terminals a, b, and a *neutral* terminal n that center-taps the voltage $\mathbb{V}_{ab}$:

$$\mathbb{V}_{an} = \mathbb{V}_{nb} = \mathbb{V}_1$$

This is a common arrangement in a North American residence supplied with both 115 V and 230 V rms, since if $|\mathbb{V}_{an}| = |\mathbb{V}_1| = 115$ V rms, then $|\mathbb{V}_{ab}| = |2\ \mathbb{V}_1| = 230$ V rms.

Let us now consider the source of Fig. 11.3(a) loaded with two identical loads, both having an impedance Z_1, as shown in Fig. 11.3(b). The currents in the lines aA



(a) Single-phase three-wire circuit; (b) single-phase three-wire circuit with two identical loads.

and bB are

$$\mathbb{I}_{aA} = \frac{\mathbb{V}_{an}}{Z_1} = \frac{\mathbb{V}_1}{Z_1}$$

$$\mathbb{I}_{bB} = \frac{\mathbb{V}_{bn}}{Z_1} = -\frac{\mathbb{V}_1}{Z_1} = -\mathbb{I}_{aA}$$

Therefore, the current in the neutral wire, nN, by KCL is

$$\mathbb{I}_{nN} = -(\mathbb{I}_{aA} + \mathbb{I}_{bB}) = 0$$

Now it is always the case that a wire carrying no current may be freely removed in any circuit without affecting any current or voltage, since its removal changes no KVL, KCL, or element law equation for that circuit. Thus in Fig. 11.3(b), the neutral could be removed without changing any current or voltage in the system. We use this procedure to simplify circuits throughout the chapter.

If the lines aA and bB are not perfect conductors but have equal impedances Z_2, then $\mathbb{I}_{nN}$ is still zero because we may simply add the series impedances Z_1 and Z_2 and have essentially the same situation as in Fig. 11.3(b). Indeed, in the more general case shown in Fig. 11.4, the neutral current $\mathbb{I}_{nN}$ is still zero. This may be seen by writing the two mesh equations

$$(Z_1 + Z_2 + Z_3)\mathbb{I}_{aA} + Z_3\mathbb{I}_{bB} - Z_1\mathbb{I}_3 = \mathbb{V}_1$$

$$Z_3\mathbb{I}_{aA} + (Z_1 + Z_2 + Z_3)\mathbb{I}_{bB} + Z_1\mathbb{I}_3 = -\mathbb{V}_1$$

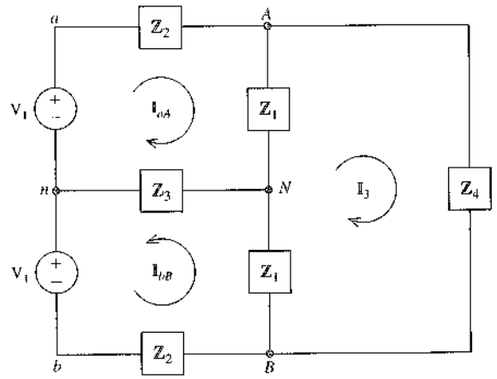
and adding the result, which yields

$$(Z_1 + Z_2 + Z_3)(\mathbb{I}_{aA} + \mathbb{I}_{bB}) + Z_3(\mathbb{I}_{aA} + \mathbb{I}_{bB}) = 0$$

or

$$\mathbb{I}_{aA} + \mathbb{I}_{bB} = 0 \tag{11.5}$$

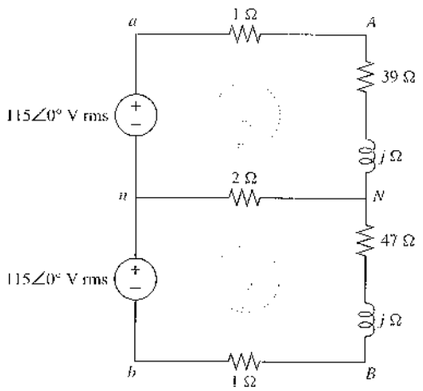
Since by KCL the left side of (11.5) equals $-\mathbb{I}_{nN}$, the neutral current is zero. This is a consequence of the symmetry of Fig. 11.4.



Symmetrical single-phase three-wire system.

If the symmetry of Fig. 11.4 is broken by having unequal loads at terminals A–N and N–B or unequal line impedances in lines aA and bB, there will be a neutral current.

Example 11.2



Unsymmetrical single-phase three-wire system.

For example, let us consider the circuit of Fig. 11.5. Note that the loads across terminals AN and NB are not equal, and the symmetry of Fig. 11.4 is broken. The mesh equations, in vector-matrix form, are

$$\begin{bmatrix} 42+j & -2 \\ -2 & 50+j \end{bmatrix} \begin{bmatrix} \mathbb{I}_1 \\ \mathbb{I}_2 \end{bmatrix} = \begin{bmatrix} 115\underline{/0} \\ 115\underline{/0} \end{bmatrix}$$

The determinant is

$$\Delta = (42+j)(50+j) - 4 = 2097\underline{/2.51^\circ}$$

and solving by matrix inversion yields

$$\begin{bmatrix} \mathbb{I}_1 \\ \mathbb{I}_2 \end{bmatrix} = \frac{115\underline{/0}}{\Delta} \begin{bmatrix} 50+j & +2 \\ +2 & 42+j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.85\underline{/-1.41^\circ} \\ 2.41\underline{/-1.21^\circ} \end{bmatrix}$$

so the neutral current $\mathbb{I}_{nN} = \mathbb{I}_2 - \mathbb{I}_1 = 0.44\underline{/177^\circ}$ A rms and is not zero.

11.1.1. Derive (11.5) by superposition applied to Fig. 11.4.

11.1.2. Find the complex power $\mathbb{S}_{AN}$ and $\mathbb{S}_{NB}$ delivered to the loads $Z_{AN} = 39 + j\ \Omega$, $Z_{NB} = 47 + j\ \Omega$, respectively, of Fig. 11.5.

Answer: $317 + j8.12$; $273 + j5.81$

11.1.3. Find the real average power P_{aA}, P_{bB}, and P_{nN} lost in the lines in Fig. 11.5.

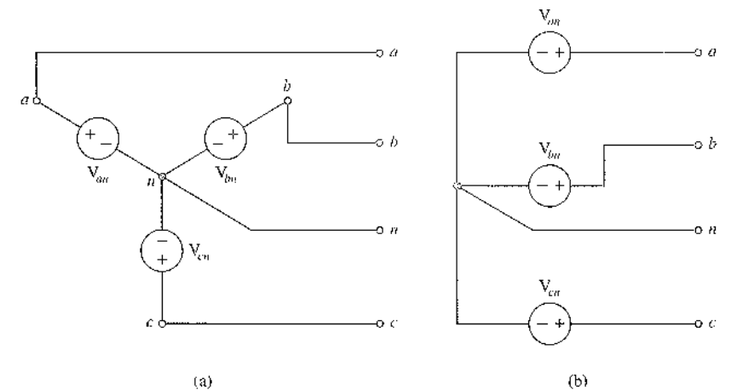
Answer: 8.12 W; 5.81 W; 0.39 W

11.1.4. Find the real average power P_{an} and P_{nb} delivered by the two sources in Fig. 11.5. Check the results in Exercises 11.1.2 and 11.1.3 for conservation of power.

Answer: 328 W; 277 W

11.2 THREE-PHASE WYE–WYE SYSTEMS

Let us consider the three-phase source of Fig. 11.6(a), which has *line* terminals a, b, and c and a *neutral* terminal n. In this case the source is said to be *wye (Y) connected*. An equivalent representation is that of Fig. 11.6(b), which is somewhat easier to draw.



Two representations of a wye-connected source.

The voltages $\mathbb{V}_{an}$, $\mathbb{V}_{bn}$, and $\mathbb{V}_{cn}$ between the line terminals and the neutral terminal are called *phase voltages* and in most cases we shall consider are given by

$$\begin{aligned} \mathbb{V}_{an} &= V_p\underline{/0} \\ \mathbb{V}_{bn} &= V_p\underline{/-120^\circ} \\ \mathbb{V}_{cn} &= V_p\underline{/120^\circ} \end{aligned}$$

or

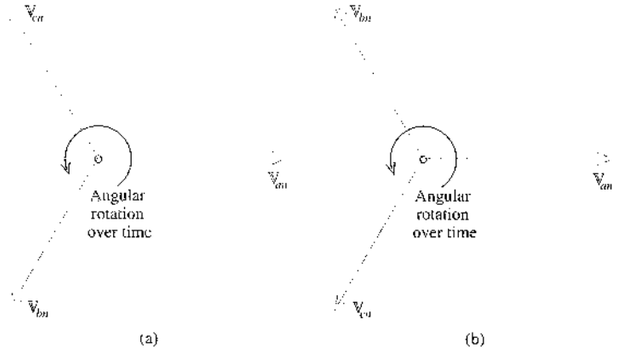
$$\begin{aligned} \mathbb{V}_{an} &= V_p\underline{/0} \\ \mathbb{V}_{bn} &= V_p\underline{/120^\circ} \\ \mathbb{V}_{cn} &= V_p\underline{/-120^\circ} \end{aligned}$$

In both cases, each phase voltage has the same rms magnitude V_p, and the phases are displaced 120°, with $\mathbb{V}_{an}$ arbitrarily selected as the reference phasor. Such a set of voltages is called a *balanced set* and is characterized by

$$\mathbb{V}_{an} + \mathbb{V}_{bn} + \mathbb{V}_{cn} = 0 \qquad (11.8)$$

as may be seen from (11.6) or (11.7).

The sequence of voltages in (11.6) is called the *positive*, or *abc, sequence*, while that of (11.7) is called the *negative*, or *acb, sequence*. Phasor diagrams of the two sequences are shown in Fig. 11.7, where we may see by inspection that (11.8) holds. Evidently, the only difference between positive and negative sequences is the arbitrary



(a) Positive and (b) negative phase sequences.

choice of the terminal labels, a, b, and c. Thus without loss in generality we consider only the positive sequence.

By Fig. 11.7(a), the voltages in the abc sequence may each be related to $\mathbb{V}_{an}$. The relationships, which will be useful later, are

$$\mathbb{V}_{bn} = \mathbb{V}_{an}/\underline{-120^\circ}$$
$$\mathbb{V}_{cn} = \mathbb{V}_{an}/\underline{120^\circ}$$

The *line-to-line* voltages, or simply *line* voltages, in Fig. 11.6 are $\mathbb{V}_{ab}$, $\mathbb{V}_{bc}$, and $\mathbb{V}_{ca}$, which may be found from the phase voltages. For example,

$$\begin{aligned}\mathbb{V}_{ab} &= \mathbb{V}_{an} + \mathbb{V}_{nb} \\ &= V_p/0 + V_p/\underline{+60^\circ} \\ &= (V_p) + (V_p)\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) \\ &= \sqrt{3}\ V_p\left(\frac{\sqrt{3}}{2} + j\frac{1}{2}\right) \\ &= \sqrt{3}\ V_p/\underline{30^\circ}\end{aligned}$$

In like manner,

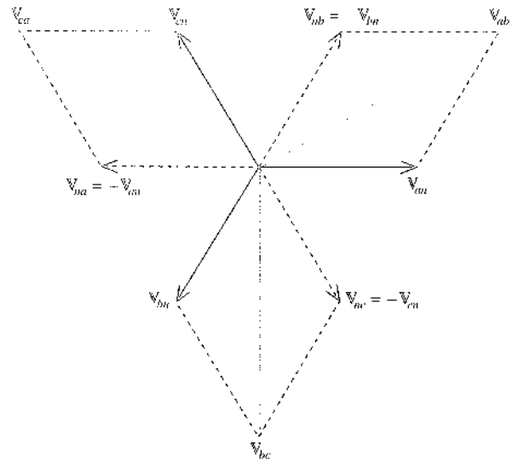
$$\mathbb{V}_{bc} = \sqrt{3}\,V_p/\underline{-90^\circ}$$
$$\mathbb{V}_{ca} = \sqrt{3}\,V_p/\underline{-210^\circ}$$

If we denote the magnitude of the line voltages by V_L, we have

$$V_L = \sqrt{3}\ V_p \tag{11.10a}$$

and thus

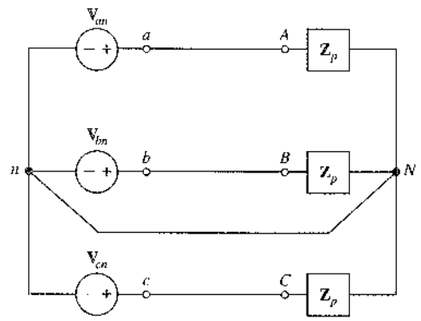
$$\mathbb{V}_{ab} = V_L/\underline{30^\circ}, \qquad \mathbb{V}_{bc} = V_L/\underline{-90^\circ}, \qquad \mathbb{V}_{ca} = V_L/\underline{-210^\circ} \tag{11.10b}$$



Phasor diagram showing phase and line voltages.

Examining Fig. 11.8, the line voltages are also a balanced set, whose magnitudes exceed the phase voltages by a factor of $\sqrt{3}$ and which are 30° out of phase with them (line voltage $\mathbb{V}_{ab}$ leads phase voltage $\mathbb{V}_{an}$ by 30°, $\mathbb{V}_{ca}$ leads $\mathbb{V}_{cn}$ by 30°, and $\mathbb{V}_{bc}$ leads $\mathbb{V}_{bn}$ by 30°). These results also may be obtained graphically from the phasor diagram shown in Fig. 11.8.

Let us now consider the system of Fig. 11.9, which is a *balanced* wye–wye three-phase four-wire system if the source voltages are given by (11.6). The term *wye–wye* applies since both the source and the load are wye-connected. The system is said to be balanced since the source voltages constitute a balanced set and the load is balanced (each *phase impedance* is equal, in this case, to the common value Z_p). The fourth wire



Balanced wye–wye system.

is the neutral line n–N, which since it carries no current in this symmetric circuit, may be omitted to form a three-phase three-wire system.

Applying KVL around loops containing the neutral wire, the line currents of Fig. 11.9 are evidently

$$\mathbb{I}_{aA} = \frac{\mathbb{V}_{an}}{Z_p}$$

$$\mathbb{I}_{bB} = \frac{\mathbb{V}_{bn}}{Z_p} = \frac{\mathbb{V}_{an}/-120^\circ}{Z_p} = \mathbb{I}_{aA}/-120^\circ$$

$$\mathbb{I}_{cC} = \frac{\mathbb{V}_{cn}}{Z_p} = \frac{\mathbb{V}_{an}/120^\circ}{Z_p} = \mathbb{I}_{aA}/120^\circ$$

The last two results are a consequence of (11.9) and show that the line currents also form a balanced set. Therefore, their sum is

$$-\mathbb{I}_{nN} = \mathbb{I}_{aA} + \mathbb{I}_{bB} + \mathbb{I}_{cC} = 0$$

confirming that the neutral carries no current in a balanced wye–wye four-wire system.

In the case of wye-connected loads, the currents in the lines aA, bB, and cC are also the *phase currents* (the currents carried by the phase impedances). If the magnitudes of the phase and line currents are I_p and I_L, respectively, then $I_L = I_p$, and (11.11) becomes

$$\mathbb{I}_{aA} = I_L/-\theta = I_p/-\theta$$

$$\mathbb{I}_{bB} = I_L/-\theta - 120^\circ = I_p/-\theta - 120^\circ$$

$$\mathbb{I}_{cC} = I_L/-\theta + 120^\circ = I_p/-\theta + 120^\circ$$

where θ is the angle of Z_p.

The average power P_p delivered to each phase of Fig. 11.9 is

$$P_p = V_p I_p \cos\theta$$

$$= I_p^2 \operatorname{Re}(Z_p)$$

and the total power delivered to the load is

$$P = 3P_p$$

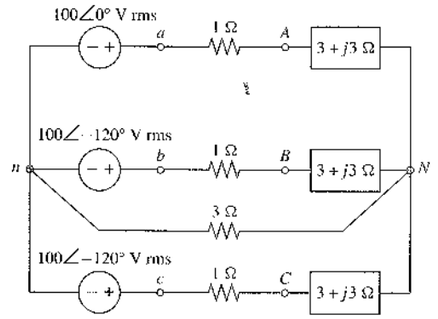
The angle θ of the phase impedance is thus the power factor angle of the three-phase load as well as that of a single phase.

Suppose now that instead of assuming that they are perfect conductors, a line impedance Z_L is used to model each of the lines aA, bB, and cC and that a neutral line impedance Z_N, not necessarily equal to Z_L, is inserted in line nN in series with the phase impedances. The two sets of impedances may be combined to form perfect conducting lines aA, bB, and cC with a load impedance $Z_p + Z_L$ in each phase.

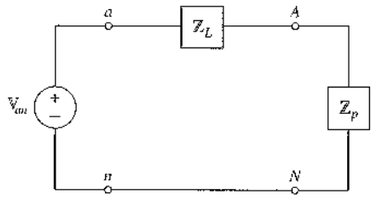
To determine the effect of Z_N, consider the full set of KCL, KVL, and element laws for this circuit. The currents and voltages solving these equations for the case $Z_N = 0$ include $I_{nN} = 0$, since the circuit in this case is balanced. But these same currents and voltages must solve the circuit equations for *any* Z_N, not just $Z_N = 0$. This follows from the fact that substituting the solutions for the case $Z_N = 0$ into the equations containing

arbitrary Z_N, no term in any equation is changed, since every term containing Z_n is multiplied by zero (i_{nN} or V_{nN}). By this argument we see that *in any circuit containing an impedance Z that draws no current, Z may be changed arbitrarily without affecting any current or voltage*. In particular, we may replace Z by a short circuit ($Z = 0$) or an open circuit ($Z = \infty$).

Example 11.3



Balanced system with line impedances.



Single phase for a per-phase analysis.

As an example, let us find the line currents in Fig. 11.10. We may combine the 1-Ω line impedance and $(3 + j3)$-Ω phase impedance to obtain

$$Z_p = 4 + j3 = 5/36.9^\circ \ \Omega$$

as the effective phase load. Since by the foregoing discussion there is no neutral current, we have

$$\mathbb{I}_{aA} = \frac{100/0}{5/36.9^\circ} = 20/-36.9^\circ \text{ A rms}$$

The currents form a balanced, positive sequence set, so we also have

$$\mathbb{I}_{bB} = 20/-156.9^\circ \text{ A rms}, \qquad \mathbb{I}_{cC} = 20/-276.9^\circ \text{ A rms}$$

This example was solved on a "per-phase" basis. Since the impedance in the neutral is immaterial in a balanced wye–wye system, we may imagine the neutral line to be a short circuit. We may do this if it contains an impedance or even if the neutral wire is not present (a three-wire system). We may then look at only one phase, say phase A, consisting of the source $\mathbb{V}_{an}$ in series with Z_L and Z_p, as shown in Fig. 11.11. (The line nN is replaced by a short circuit.) The line current $\mathbb{I}_{aA}$, the phase voltage $\mathbb{I}_{aA}Z_p$, and the voltage drop in the line $\mathbb{I}_{aA}Z_L$ may all be found from this single-phase analysis. The other voltages and currents in the system may be found similarly, or from the previous results, since the system is balanced.

Example 11.4

As another example, suppose that we have a balanced wye-connected source, having line voltage $V_L = 200$ V rms that is supplying a balanced wye-connected load with $P = 900$ W at a power factor of 0.9 lagging. Let us find the line current I_L and the phase impedance Z_p. Since the power supplied to the load is 900 W, the power supplied to each phase is $P_p = \frac{900}{3} = 300$ W, and from

$$P_p = V_p I_p \cos\theta$$

we have

$$300 = \frac{200}{\sqrt{3}} I_p (0.9)$$

Therefore, since for a wye-connected load the phase current is also the line current, we have

$$I_L = I_p = \frac{3\sqrt{3}}{2(0.9)} = 2.89 \text{ A rms}$$

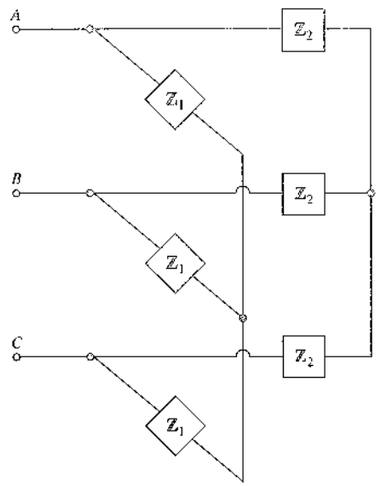
The magnitude of Z_p is given by

$$|Z_p| = \frac{V_p}{I_p} = \frac{200/\sqrt{3}}{3\sqrt{3}/(2)(0.9)} = 40\ \Omega$$

and since $\theta = \cos^{-1} 0.9 = 25.84°$ is the angle of Z_p, we have

$$Z_p = 40/25.84° \ \Omega$$

If the load is unbalanced but there is a neutral wire that is a perfect conductor, we may still use the per-phase method of solution for each phase. However, if this is not the case, this shortcut method does not apply.



11.2.1. $V_{ab} = 100/0$ V rms is a line voltage of a balanced wye-connected three-phase source. (a) If the phase sequence is *abc*, find the phase voltages. (b) Repeat for the *acb* phase sequence.

Answer: (a) $57.7/-30°$, $57.7/-150°$, $57.7/+90°$ V rms; (b) $57.7°/-30°$, $57.7°/+90°$, $57.7/-150°$ V rms

11.2.2. In Fig. 11.9 the source voltages are determined by Exercise 11.2.1(a) and the load in each phase is a series combination of a 30-Ω resistor, a 500-μF capacitor, and a 0.25-H inductor. The frequency is $\omega = 200$ rad/s. Find the line currents and the power delivered to the load.

Answer: $1.15/-83.1°$; $1.15/-203.1°$; $1.15/36.9°$ A rms; 120 W

11.2.3. Show that if a balanced three-phase three-wire system has two balanced three-phase loads connected in parallel, as shown, the load is equivalent to that of Fig. 11.9 with

$$Z_p = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

11.2.4. If, in Exercise 11.2.3, $Z_1 = 4 + j3\ \Omega$, $Z_2 = 4 - j3\ \Omega$, and the line voltage is $V_L = 200\sqrt{3}$ V rms, find the current I_L in each line.

Answer: 64 A rms

11.3 SINGLE-PHASE VERSUS THREE-PHASE POWER DELIVERY

It is interesting to compare the relative merits of the single-phase three-wire circuit of Section 11.1 and the three-phase circuit of Section 11.2 for delivering power to a load. Figure 11.12 shows the two circuits to be considered. Examining this figure, one

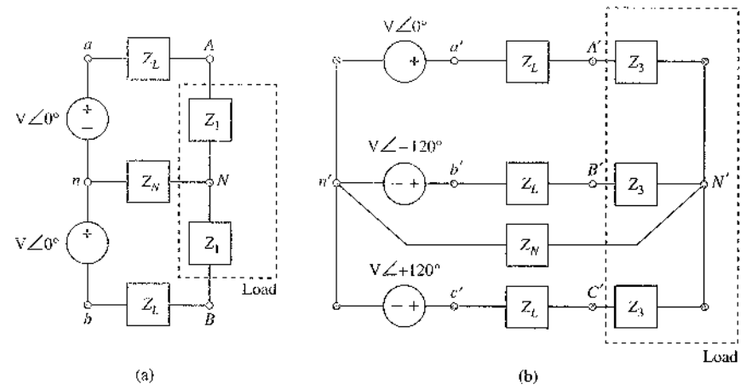


Figure 11.12 (a) Single- and (b) three-phase circuits delivering power to a load.

immediate observation is that the three-phase circuit has a higher element count: more wires, more phase sources, and more phase impedances. Simplicity favors the single-phase circuit. But we will look beyond element count in comparing the relative virtues of single- and three-phase systems.

Consider the line losses incurred by each system while producing the same amount of power at the same voltage level and power factor. In the single-phase case, $I_{nN} = 0$, and replacing Z_N by a short circuit in Fig. 11.12(a) gives

$$\mathbb{I}_{aN} = \frac{V/0}{Z_1 + Z_L}$$

$$P_{aN} = |\mathbb{V}_{aN}|\,|\mathbb{I}_{aN}| \cos\theta = \frac{V^2 \cos\theta}{|Z_1 + Z_L|}$$

where $\cos\theta$ is the power factor of the series line/load impedance. The total power delivered is

$$P_{aN} + P_{bN} = 2P_{aN} = \frac{2V^2 \cos\theta}{|Z_1 + Z_L|} \qquad (11.14)$$

In the three-phase case of Fig. 11.12(b) again, the neutral current $I_{n'N'} = 0$ and the per-phase calculation is as above. The total power delivered by the three-phase source is

$$P_{a'N'} + P_{b'N'} + P_{c'N'} = 3P_{a'N'} = \frac{3V^2 \cos\theta}{|Z_3 + Z_L|} \qquad (11.15)$$

The powers delivered by the single- and three-phase sources are equal if (11.14) and (11.15) are equal, or

$$\frac{|Z_3 + Z_L|}{|Z_1 + Z_L|} = \frac{3}{2}$$

For equal power delivery, the series phase impedance in the three-phase circuit, $Z_3 + Z_L$, is larger than the single-phase impedance by half. Since the phase voltages are the same,

the rms current in the single-phase lines must be larger by the same ratio:

$$\frac{|\mathbb{I}_{aN}|}{|\mathbb{I}_{a'N'}|} = \frac{V/|Z_1 + Z_L|}{V/|Z_3 + Z_L|} = \frac{3}{2}$$

The line losses are $|\mathbb{I}_{aN}|^2 \text{ Re}(Z_L)$ for each nonneutral line, or in the single-phase case the total line losses are

$$P_L = 2|\mathbb{I}_{aN}|^2 \text{ Re}(Z_L)$$

while in the three-phase circuit we have

$$P_{L'} = 3|\mathbb{I}_{a'N'}|^2 \text{ Re}(Z_L)$$

Substituting (11.17), this is

$$P_{L'} = 3 \cdot \left(\frac{2}{3}\right)^2 |\mathbb{I}_{aN}|^2 \text{ Re}(Z_L) = \frac{4}{3}|\mathbb{I}_{aN}|^2 \text{ Re}(Z_L)$$

Comparing (11.18) and (11.19), we conclude that $P_L = \frac{3}{2}P_{L'}$; in other words, with other factors equal (power produced, voltage level, power factor), *the single-phase circuit delivers power at the cost of half again as many watts dissipated in the lines. Three-phase power delivery is more efficient.*

Example 11.5

In Figure 11.12, let $V = 4$ kV rms, $Z_L = 2\ \Omega$, and the single-phase load impedance be $30 + j8\ \Omega$. Determine the phase impedance Z_3 for equal power from the single- and three-phase sources at equal power factors and the power delivered to the lines and loads. In the single-phase circuit, the rms line current $\mathbb{I}_{aN}$ is

$$\mathbb{I}_{aN} = \frac{4000/0}{32 + j8} = 121/\underline{-14.0^\circ}$$

The power delivered to the load is $2|\mathbb{I}_{aN}|^2 \text{ Re}(Z_1) = 2(121)^2(30) = 878$ kW and that dissipated by the lines $2|\mathbb{I}_{aN}|^2 \text{ Re}(Z_L) = 2(121)^2(2) = 58.6$ kW. In the three-phase case, the power dissipated by the lines at the same power factor is scaled by a factor of $\frac{2}{3}$; that is, $(\frac{2}{3})(58.6) = 39.0$ kW. Since powers delivered by the sources are equal, the power to the three-phase load must be increased by the line losses saved, or three-phase load power of $878 + \frac{1}{3}(58.6) = 897$ kW. The impedance Z_3 required must, by (11.16), satisfy

$$\frac{|Z_3 + 2|}{|32 + j8|} = \frac{3}{2}$$

or

$$|Z_3 + 2| = 49.5$$

For equal power factors, the angle of the impedance $Z_3 + 2$ must agree with that of $Z_1 + 2 = 32 + j8$, which is $\theta = 14.0^\circ$ by (11.20).

$$Z_3 + 2 = |Z_3 + 2|/\underline{\theta} = 49.5/\underline{14.0^\circ}$$

$$Z_3 = 49.5/\underline{14.0^\circ} - 2 = 46.0 + j12.0\ \Omega$$

Examination of the instantaneous power waveforms associated with these two circuits reveals another difference relevant to many practical applications. From Fig. 11.12(a), in the single-phase case

$$\mathbb{I}_{aN} = \mathbb{I}_{nB} = \frac{V/0}{Z_1 + Z_L}$$

and

$$i_{aN}(t) = i_{nB}(t) = \frac{\sqrt{2}V}{|Z_1 + Z_L|}\cos(\omega t - \theta)$$

where θ is, as before, the angle of the series impedance, $Z_L + Z_1$, and the factor of $\sqrt{2}$ is required since V is the magnitude of the rms phasor. The instantaneous power delivered to each of the two parts of the load is $p(t) = i_{aN}^2 \text{ Re}(Z_1)$ and is shown in Fig. 11.13(a). Note the instantaneous power delivered to the two load impedances is in phase, and the total power delivered to the load varies between its maximum and zero

$$p_{1T}(t) = 2\left(\frac{\sqrt{2}\ V}{|Z_1 + Z_L|}\right)^2 \cos^2(\omega t - \theta) \cdot \text{Re}(Z_1)$$

$$= \frac{2V^2 \text{ Re}(Z_1)}{|Z_1 + Z_L|^2}[1 + \cos 2(\omega t - \theta)]$$

periodically.

From Fig. 11.12(b), in the three-phase case,

$$\mathbb{I}_{a'N'} = \frac{V/0}{Z_3 + Z_L}$$

$$i_{aN}(t) = \frac{\sqrt{2}\ V}{|Z_3 + Z_L|}\cos(\omega t - \theta)$$

The three line currents form a balanced set, as shown in the phasor diagram in Fig. 11.14. These currents and hence the instantaneous powers to the three-phase impedances are 120° out of phase, as shown in Fig. 11.13(b). The total instantaneous power is the sum over

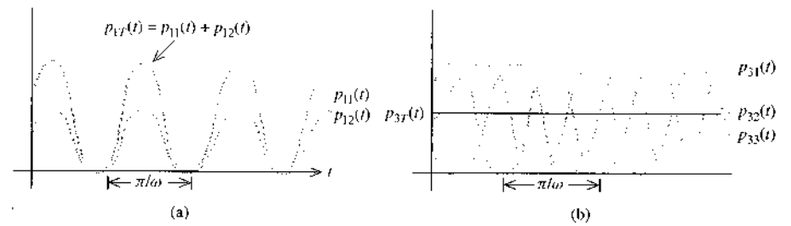
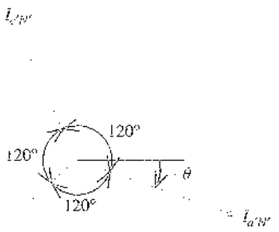


FIGURE 11.13 Instantaneous power delivery to single-phase three-wire and three-phase loads. (a) $p_{11}(t)$ and $p_{12}(t)$ are power to the single-phase load impedances and $p_{1T}(t)$ is total power. (b) $p_{31}(t)$, $p_{32}(t)$, and $p_{33}(t)$ are power to the three-phase impedances and $p_{3T}(t)$ is the total power.



Line currents associated with the three-phasor circuit of Fig. 11.12(b). θ is the angle of the phase impedance.

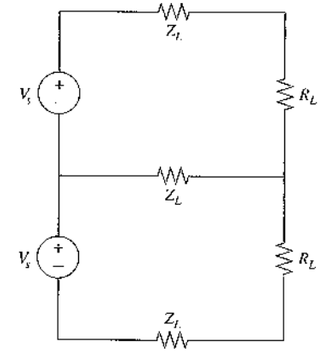
the three individual phases:

$$\begin{aligned} p_{3T}(t) &= \left(\frac{\sqrt{2}\,V}{|Z_3 + Z_L|}\right)^2 \\ &\quad \times \mathrm{Re}(Z_3)\left[\cos^2(\omega t + \theta) + \cos^2(\omega t + \theta + 120^\circ) + \cos^2(\omega t + \theta - 120^\circ)\right] \\ &= \frac{2V^2\,\mathrm{Re}(Z_3)}{|Z_3 + Z_L|^2} \\ &\quad \times \left[\tfrac{3}{2} + \cos 2(\omega t + \theta) + \cos 2(\omega t + \theta + 120^\circ) + \cos 2(\omega t + \theta - 120^\circ)\right] \end{aligned}$$

The cosine terms in (11.22) are evidently time functions representing a balanced set of phasors. Since a balanced set of phasors sums to zero, so must these three terms. The remaining term is constant, and thus $p_{3T}(t)$, the total instantaneous three-phase power delivered to the load, is constant, as shown in Fig. 11.13(b).

We conclude that single-phase power is delivered unevenly, varying periodically between zero and its maximum, while three-phase power is constant over time. Suppose that we were delivering power to an electric motor. Since the torque produced varies with the power delivered to it, a single-phase motor shakes as the power delivery fluctuates periodically, while a three-phase motor "pulls" as hard each instant of time as every other, with no induced vibrations. Or consider powering fluorescent lights, whose brightness varies with instantaneous power delivery. A pair of single-phase three-wire fluorescent fixtures would flicker periodically, while three three-phase fixtures mounted together would illuminate without flicker.

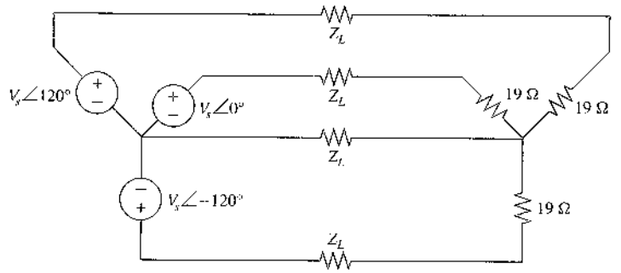
Thus not only does three-phase power delivery have an efficiency advantage over single phase, but a smooth power delivery advantage as well. It is no surprise that three-phase is favored the world over for delivery of bulk power. Only where small amounts of power and small fractional horsepower motors are sufficient is single-phase power the method of choice. This is the case for most residential wiring, for which the simplicity and low element weight of single-phase circuits are decisive. Dc is sometimes used to transport power over long distances, but multiphase ac is dominant in almost all power delivery systems.



EXERCISE 11.3.2

11.3.1. A three-phase source and load 100 km apart are connected by lines with impedance 0.01 Ω/km, as shown in the figure. What is the *efficiency* of this system, defined as power delivered to the load divided by total power produced by the source? At what distance is the efficiency 90%?

ANSWER: 0.95; 211 km



11.3.2. The figure shows a single-phase system of the same source voltage $|V_s|$ as in Exercise 11.3.1. For 100-km separation, for which R_L would the same power be delivered to the load as delivered by the three-phase system of Exercise 11.3.1 to its three-phase load? What is the efficiency of this system? (See Exercise 11.3.1.)

ANSWER: 11.95 Ω or 0.0837 Ω; 0.923 for $R_L = 11.95$ Ω; 0.0772 for $R_L = 0.0837$ Ω

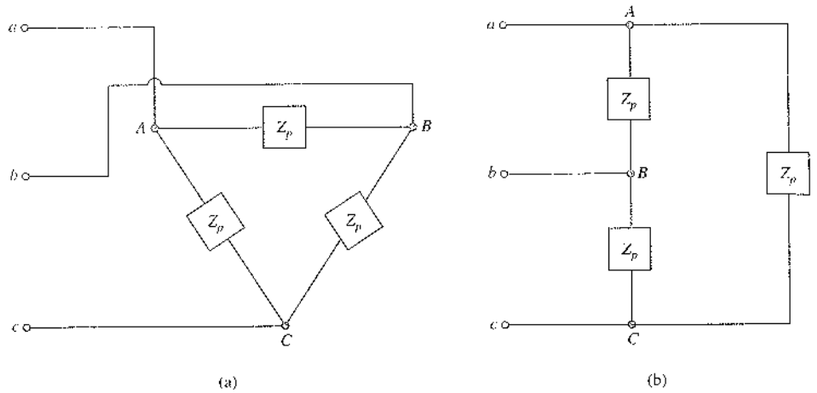
11.3.3. Consider a motor powered by a single-phase 50-Hz source. What is the period of the vibrational motion the motor will experience? Repeat for a balanced three-phase source and an unbalanced three-phase source both at 50 Hz.

ANSWER: 10 ms; no vibrational motion; 10 ms

11.4 DELTA CONNECTION

Another method of connecting a three-phase load to a line is the *delta*, or Δ, *connection*. A balanced Δ-connected load (with equal phase impedance Z_P) is shown in Fig. 11.15(a), in a way that resembles a Δ, and in an equivalent way in Fig. 11.15(b). If the source is Y- or Δ-connected, the system is a Y–Δ or Δ–Δ system.

An advantage of a Δ-connected load over a Y-connected load is that loads may be added or removed more readily on a single phase of a Δ, since the loads are connected directly across the lines. This may not be possible in the Y connection, since the neutral may not be accessible. Also, for a given power delivered to the load, the phase currents in a Δ are smaller than those in a Y. On the other hand, the Δ phase voltages are



Two diagrams of a delta-connected load.

higher than those of the Y connection. Sources are rarely Δ-connected, because if the voltages are not perfectly balanced, there will be a net source voltage, and consequently a circulating current, around the delta. This causes undesirable heating effects in the generating machinery. Also, the phase voltages are lower in the Y-connected generator, and thus less insulation is required. Obviously, systems with Δ-connected loads are three-wire systems, since there is no neutral connection.

From Fig. 11.16 we see that in the case of a Δ-connected load the line voltages are the same as the phase voltages. Therefore, if the line voltages are given by (11.10b), as before, the phase voltages are

$$\mathbb{V}_{AB} = V_L/\underline{30^\circ}, \qquad \mathbb{V}_{BC} = V_L/\underline{-90^\circ}, \qquad \mathbb{V}_{CA} = V_L/\underline{150^\circ}$$

where

$$V_L = V_p$$

If $Z_p = |Z_p|/\underline{\theta}$, the phase currents are

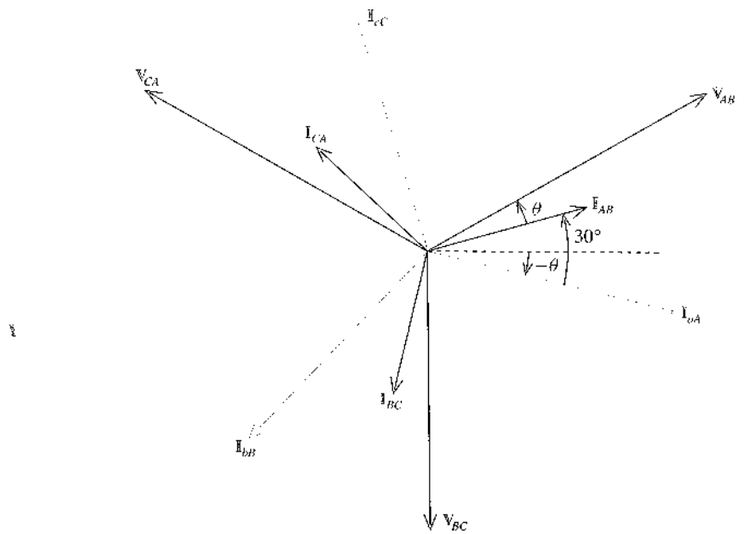
$$\mathbb{I}_{AB} = \frac{\mathbb{V}_{AB}}{Z_p} = I_p/\underline{30^\circ - \theta}$$

$$\mathbb{I}_{BC} = \frac{\mathbb{V}_{BC}}{Z_p} = I_p/\underline{-90^\circ - \theta}$$

$$\mathbb{I}_{CA} = \frac{\mathbb{V}_{CA}}{Z_p} = I_p/\underline{150^\circ - \theta}$$

where the rms phase current magnitude is

$$I_p = \frac{V_L}{|Z_p|}$$



Phasor diagram for a delta-connected load.

The current in line aA is

$$\mathbb{I}_{aA} = \mathbb{I}_{AB} - \mathbb{I}_{CA}$$

which after some simplification is

$$\mathbb{I}_{aA} = \sqrt{3}\, I_p/\underline{-\theta}$$

The other line currents, obtained similarly, are

$$\mathbb{I}_{bB} = \sqrt{3}\, I_p/\underline{-120^\circ - \theta}$$

$$\mathbb{I}_{cC} = \sqrt{3}\, I_p/\underline{-240^\circ - \theta}$$

Evidently, the relation between the line and phase current magnitudes in the Δ case is

$$I_L = \sqrt{3}\, I_p$$

and the line currents are thus

$$\mathbb{I}_{aA} = I_L/\underline{-\theta}, \qquad \mathbb{I}_{bB} = I_L/\underline{-120^\circ - \theta}, \qquad \mathbb{I}_{cC} = I_L/\underline{-240^\circ - \theta}$$

Thus the currents and voltages are balanced sets, as expected. The relations between line and phase currents for the Δ-connected load are summed up in the phasor diagram of Fig. 11.16.

Example 11.6

As an example of a three-phase circuit with a delta-connected load, let us find the line current I_L in Fig. 11.15 if the line voltage is 250 V rms and the load draws 1.5 kW at a lagging power factor of 0.8. For one phase, $P_p = \frac{1500}{3} = 500$ W, and thus

$$500 = 250 I_p (0.8)$$

or

$$I_p = 2.5 \text{ A rms}$$

Therefore, we have

$$I_L = \sqrt{3}\, I_p = 4.33 \text{ A rms}$$

Finally, in this section, let us derive a formula for the power delivered to a balanced three-phase load with a power factor angle θ. Whether the load is Y-connected or Δ-connected, we have

$$P = 3P_p = 3V_p I_p \cos\theta$$

In the Y-connected case, $V_p = V_L/\sqrt{3}$ and $I_p = I_L$, and in the Δ-connected case, $V_p = V_L$ and $I_p = I_L/\sqrt{3}$. In either case, then,

$$P = 3\frac{V_L I_L}{\sqrt{3}} \cos\theta$$

or

$$P = \sqrt{3}\, V_L I_L \cos\theta \qquad (11.29)$$

As a check on Example 11.6, (11.29) yields

$$1500 = \sqrt{3}(250) I_L (0.8)$$

or, as before,

$$I_L = 4.33 \text{ A rms}$$

11.4.1. Solve Exercise 11.2.2 if the source and load are unchanged except that the load is Δ-connected. [*Suggestion:* Note that in (11.23), (11.25), and (11.28), 30° must be subtracted from every angle.]

Answer: $2\sqrt{3}/-83.1°$; $2\sqrt{3}/156.9°$; $2\sqrt{3}/36.9°$ A rms, 360 W

11.4.2. A balanced Δ-connected load has $Z_p = 4 + j3\ \Omega$, and the line voltage is $V_L = 200$ V rms at the load terminals. Find the total power delivered to the load.

Answer: 19.2 kW

11.4.3. A balanced delta-connected load has a line voltage of $V_L = 100$ V rms at the load terminals and absorbs a total power of 4.8 kW. If the power factor of the load is 0.8 leading, find the phase impedance.

Answer: $4 - j3\ \Omega$

11.5 WYE–DELTA TRANSFORMATIONS

In many power systems applications it is important to be able to convert from a wye-connected load to an equivalent delta-connected load, and vice versa. For example, suppose that we have a Y-connected load in parallel with a Δ-connected load, as shown in Fig. 11.17, and wish to replace the combination by an equivalent three-phase load. If both loads were Δ-connected, this would be relatively easy, since corresponding phase impedances would be in parallel. Also, as we saw in Exercise 11.2.3, if both loads are Y-connected and balanced, the phase impedances may also be combined as parallel impedances.

To obtain Y-to-Δ or Δ-to-Y conversion formulas, let us consider the Y and Δ connections of Fig. 11.18. To effect a Y–Δ transformation, we need expressions for Y_{ab}, Y_{bc}, and Y_{ca} of the Δ in terms of Y_a, Y_b, and Y_c of the Y so that the Δ connection is equivalent to the Y connection at the terminals A, B, and C. That is, if the Y is replaced by the Δ, the same node voltages $\mathbb{V}_A$, $\mathbb{V}_B$, and $\mathbb{V}_C$ will appear if the same currents $\mathbb{I}_1$ and $\mathbb{I}_2$ are caused to flow. Conversely, a Δ–Y transformation is an expression of the Y parameters in terms of the Δ parameters.

Let us begin by writing nodal equations for both circuits. If node C is taken as reference, in the case of the Y network we have

$$Y_a \mathbb{V}_A - Y_a \mathbb{V}_D = \mathbb{I}_1$$

$$Y_b \mathbb{V}_B - Y_b \mathbb{V}_D = \mathbb{I}_2$$

$$-Y_a \mathbb{V}_A - Y_b \mathbb{V}_B + (Y_a + Y_b + Y_c)\mathbb{V}_D = 0$$

Solving for $\mathbb{V}_D$ in the third equation and substituting its value into the first two equations, we have, after simplification,

$$\frac{Y_a Y_b + Y_a Y_c}{Y_a + Y_b + Y_c}\mathbb{V}_A - \frac{Y_a Y_b}{Y_a + Y_b + Y_c}\mathbb{V}_B = \mathbb{I}_1$$

$$-\frac{Y_a Y_b}{Y_a + Y_b + Y_c}\mathbb{V}_A + \frac{Y_a Y_b + Y_b Y_c}{Y_a + Y_b + Y_c}\mathbb{V}_B = \mathbb{I}_2$$

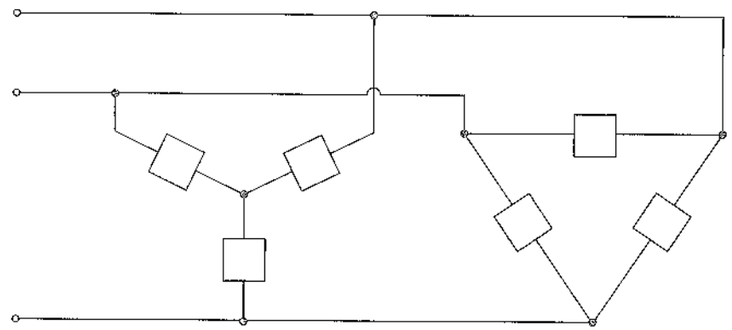
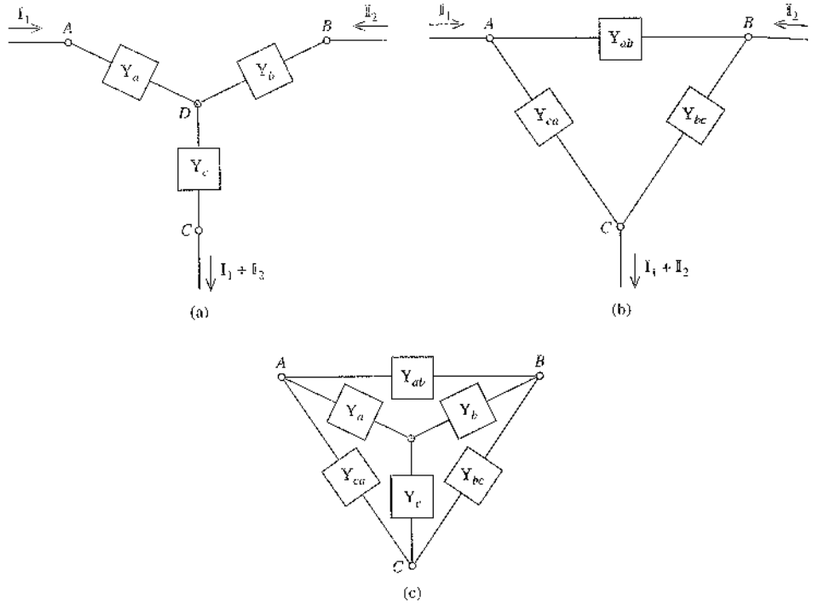


Figure 11.17 Wye- and delta-connected loads in parallel.



(a) Wye connection; (b) delta connection; (c) the two superimposed.

The nodal equations for the Δ circuit are

$$(Y_{ab} + Y_{ca})\mathbb{V}_A - Y_{ab}\mathbb{V}_B = \mathbb{I}_1$$
$$-Y_{ab}\mathbb{V}_A + (Y_{ab} + Y_{bc})\mathbb{V}_B = \mathbb{I}_2$$

Equating coefficients of like terms in these equations and (11.30), we have the Y–Δ transformation:

$$Y_{ab} = \frac{Y_a Y_b}{Y_a + Y_b + Y_c}$$
$$Y_{bc} = \frac{Y_b Y_c}{Y_a + Y_b + Y_c}$$
$$Y_{ca} = \frac{Y_c Y_a}{Y_a + Y_b + Y_c}$$

If we imagine the Y and Δ circuits superimposed on a single diagram as in Fig. 11.18(c), then Y_a and Y_b are *adjacent* to Y_{ab}, Y_b and Y_c are *adjacent* to Y_{bc}, and so on. Thus we may state (11.31) in words, as follows: *The admittance of an arm of the delta is equal to the product of the admittances of the adjacent arms of the wye divided by the sum of the wye admittances.*

To obtain the Δ–Y transformation, we may solve (11.31) for the Y admittances, a difficult task, or we may write two sets of loop equations for the Y and Δ circuits. In the latter case we shall have the dual of the procedure that led to (11.31). In either case, as the reader is asked to show in Problem 11.51, the Δ–Y transformation is

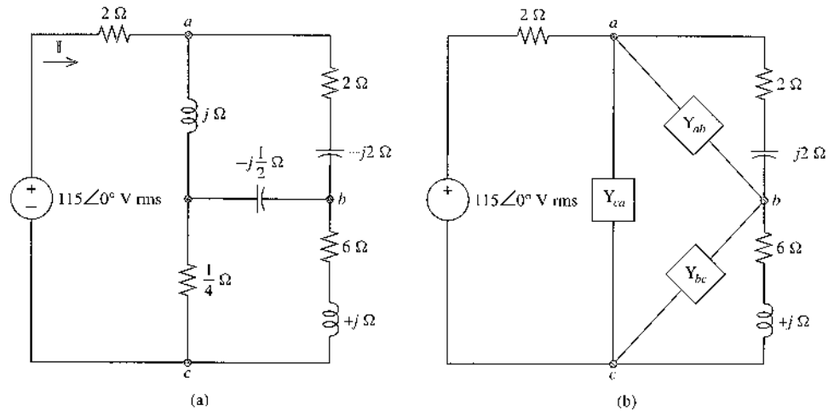
$$Z_a = \frac{Z_{ab}Z_{ca}}{Z_{ab} + Z_{bc} + Z_{ca}}$$
$$Z_b = \frac{Z_{bc}Z_{ab}}{Z_{ab} + Z_{bc} + Z_{ca}}$$
$$Z_c = \frac{Z_{ca}Z_{bc}}{Z_{ab} + Z_{bc} + Z_{ca}}$$

where the Z's are the reciprocals of the Y's of Fig. 11.18. The rule is as follows: *The impedance of an arm of the wye is equal to the product of the impedances of the adjacent arms of the delta divided by the sum of the delta impedances.* (By *adjacent* here, we mean "on each side of and terminating on the same node as." For example, in the superimposed drawing of the Y and Δ, Fig. 11.18(c), Z_a lies between Z_{ab} and Z_{ca} and all three have a common terminal A. Thus Z_{ab} and Z_{ca} are *adjacent* arms of Z_a.)

The Y–Δ conversion formulas (11.31) and (11.32) do not require balanced loads on even three-phase circuits. They are often of considerable use in simplifying general circuits containing impedances that are neither in series nor parallel, as illustrated in the next example.

Example 11.7

We seek the source phasor current $\mathbb{I}$. Note in Fig. 11.19 that using fundamental methods (nodal or loop analysis) three simultaneous equations would have to be solved. Comparing Fig. 11.18, we have



Using Y–Δ impedance conversion to simplify a circuit.

$Y_a = -j1$, $Y_b = j2$, and $Y_c = 4$ S. Then $Y_a + Y_b + Y_c = 4 + j$ and

$$Y_{ab} = \frac{(-j1)(j2)}{4+j} = \frac{2}{4+j}$$

$$Y_{bc} = \frac{j8}{4+j}$$

$$Y_{ca} = \frac{-4j}{4+j}$$

By Fig. 11.19(b) we see that after this Y–Δ transformation, Y_{ab} is in parallel with the admittance $1/(2 - j2)$, leading to a parallel equivalent admittance of

$$\frac{2}{4+j} + \frac{1}{2-j2} = \frac{8-3j}{10-6j}$$

Similarly, the equivalent admittance to Y_{bc} in parallel with the admittance $1/(6 + j)$ is

$$\frac{j8}{4+j} + \frac{1}{6+j} = \frac{-4+49j}{23+10j}$$

This yields the equivalent circuit with impedances shown in Fig. 11.20. Using series–parallel combinations, we have

$$Z_1 = \frac{10-6j}{8-3j} + \frac{23+10j}{-4+49j}$$

$$= 1.51 - j0.73$$

$$Z = 2 + \frac{[(-4-j)/4j]Z_1}{[(-4-j)/4j] + Z_1}$$

$$= 2 + \frac{(-0.25+j1)(1.51-j0.73)}{1.26+j0.27}$$

$$= 2.54 + j1.23$$

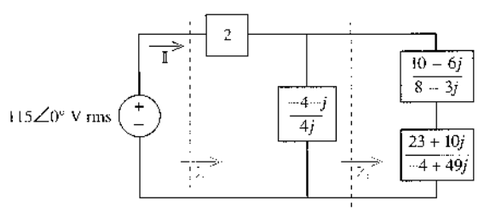


FIGURE 11.20 Equivalent of Fig. 11.19(b) with impedances shown.

Thus

$$I = \frac{115/\underline{0}}{Z} = \frac{115/\underline{0}}{2.54 + j1.23} = 40.7/\underline{-25.8^\circ} \text{ A rms}$$

The use of Y–Δ conversion in this unbalanced single-phase circuit permitted us to bypass writing and solving three complex equations in three unknowns.

If the load is balanced, the Y–Δ conversion rules simplify greatly. Setting $Y_a = Y_b = Y_c = Y_Y$ in (11.31), where Y_Y is the common phase admittance of the Y-connected balanced load, we see from (11.31) that $Y_{ab} = Y_{bc} = Y_{ca} = Y_\Delta$, the common admittance of the Δ-connected load, or

$$Y_\Delta = \frac{(Y_Y)^2}{3Y_Y} = \frac{1}{3}Y_Y$$

or, inverting, we have

$$Z_\Delta = 3Z_Y$$

The balanced Δ-connected phase impedance is three times the equivalent balanced Y-connected phase impedance.

Example 11.8

Find the magnitude of the balanced line voltages $\mathbb{V}_{AB}$, $\mathbb{V}_{BC}$, and $\mathbb{V}_{CA}$ (Fig. 11.21). This circuit is Y–Δ connected. We will convert to a Y–Y connected circuit so that simple per-phase calculations may be done. The load has Y equivalent impedances by (11.34):

$$Z_Y = \tfrac{1}{3}Z_\Delta = \tfrac{1}{3}(6 + j3) = 2 + j$$

Converting to this Y-connected load, the voltage across the equivalent phase a load impedance is, by voltage division,

$$\mathbb{V}_{AN} = \frac{2+j}{3+j}(12\underline{/0}) = 8.49/8.13^\circ$$

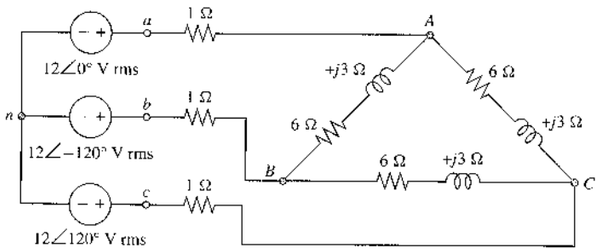


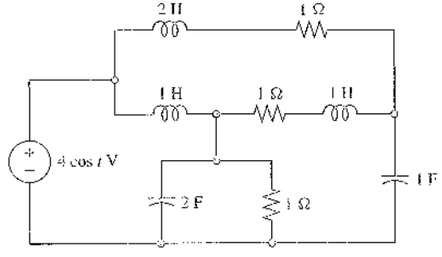
FIGURE 11.21 Circuit for Example 11.8.

The line voltages are, by (11.10a), $\sqrt{3}$ larger, so

$$|\mathbb{V}_{AB}| = \sqrt{3}\,|\mathbb{V}_{AN}| = \sqrt{3}\,(8.49) = 14.7 \text{ V rms}$$

which is the common magnitude of the balanced line voltages $\mathbb{V}_{AB}$, $\mathbb{V}_{BC}$, and $\mathbb{V}_{CA}$.

As demonstrated in this example, it is usually helpful to convert Y–Δ-connected systems to Y–Y so that currents, voltage, and power may be computed on the per-phase basis discussed in Section 11.2.



11.5.1. Find the input impedance seen by the source using a Y–Δ or Δ–Y transformation to simplify the circuit. From this result find the average power delivered by the source.

Ans. $(1 + j2)/5\ \Omega$; 8 W

11.5.2. A balanced three-phase source with $V_L = 100$ V rms is delivering power to a balanced Y-connected load with phase impedance $Z_1 = 8 + j6\ \Omega$ in parallel with a balanced Δ-connected load with phase impedance $Z_2 = 12 + j9\ \Omega$. Find the power delivered by the source.

Ans. 2.4 kW

11.5.3. Show that the Y–Δ transformation of (11.31) is equivalent to

$$Z_{ab} = \frac{Z_a Z_b + Z_b Z_c + Z_c Z_a}{Z_c}$$

$$Z_{bc} = \frac{Z_a Z_b + Z_b Z_c + Z_c Z_a}{Z_a}$$

$$Z_{ca} = \frac{Z_a Z_b + Z_b Z_c + Z_c Z_a}{Z_b}$$

or in words: The impedance of an arm of the delta is equal to the sum of the products of the impedances of the wye, taken two at a time, divided by the impedance of the *opposite* arm of the wye.

11.5.4. If the lines in Exercise 11.4.2 each have a resistance of 0.1 Ω, find the power lost in the lines.

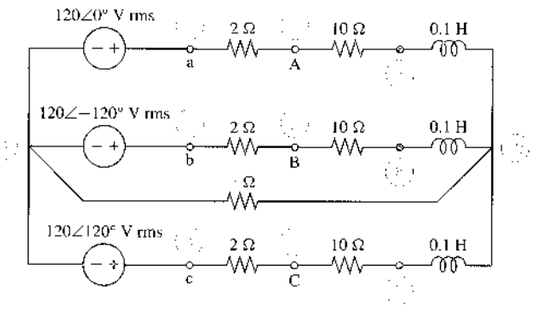
Ans. 1.44 kW

The analysis of three-phase networks presented previously has been restricted to balanced systems whose solutions can be expressed in terms of a single phase. SPICE is easily used for both balanced or unbalanced systems when applied to the entire network. SPICE makes no distinction between single- and three-phase circuits.

As a first example, consider finding the line voltages and phase currents at the load for the balanced Y–Y system of Fig. 11.22. The transmission line for interconnecting the generator and load has losses that are represented by 2-Ω resistors. A circuit file for this network is

```
3-PHASE Y-Y SYSTEM WITH TRANSMISSION LINE LOSSES
*DATA STATEMENTS VOLTAGES EXPRESSED IN RMS
VAN 1 0 AC 120 0
VBN 2 0 AC 120 -120
VCN 3 0 AC 120 120
RLOSSA 1 4 2
RLOSSB 2 5 2
RLOSSC 3 6 2
RLOSSN 10 0 2
RA 4 7 10
LA 7 10 0.1
RB 5 8 10
LB 8 10 0.1
RC 6 9 10
LC 9 10 0.1
.AC LIN 1 60 60
.PRINT AC VM(4,5) VP(4,5) IM(VAN) IP(VAN)
.PRINT AC VM(5,6) VP(5,6) IM(VBN) IP(VBN)
.PRINT AC VM(6,4) VP(6,4) IM(VCN) IP(VCN)
.PRINT AC IM(RLOSSN) IP(RLOSSN)
.END
```

The resulting output lists VM(4, 5) = VM(5, 6) = VM(6, 4) = 204.9 V rms, and VP(4, 5) = 32.8°, VP(5, 6) = −87.2°, VP(6, 4) = 152.8°, a balanced, positive sequence set as expected.



Balanced Y–Y system for SPICE analysis.

Also, IM(VAN) = IM(VBN) = IM(VCN) = 3.033 A rms; IP(VAN) = 107.7°, IP(VBN) = −12.3°, IP(VCN) = −132.3° forms a balanced set of phase currents. The magnitude of the neutral current is computed to be IM(RLOSSN) = 2.08×10^{-15} A rms, very close to its exact value of zero.

In this example we investigate the effects of unbalancing the circuit of Example 11.9. SPICE does not check for balanced conditions or perform "per-phase" calculations; hence it can deal with unbalanced circuits as easily as balanced ones.

Unbalanced conditions may arise from either an unbalanced source or load. If the source is unbalanced by reducing the phase a source magnitude by 10% to 108 V rms, replacing the VAN element line in the previous circuit file by

```
VAN 1 0 AC 108 0
```

the resulting currents and voltages are unbalanced as well. The output shows line voltages $\mathbb{V}_{AB} = 194.7/34.5°$, $\mathbb{V}_{BC} = 204.9/-87.2°$, and $\mathbb{V}_{CA} = 194.7/151.1$ [$\mathbb{V}_{AB}$ = V(4, 5), and so on; see Fig. 11.22]. Note that an imbalance of the phase source magnitudes leads to unbalanced phase as well as magnitude of the outputs. The line currents are also unbalanced in both magnitude and phase $\mathbb{I}_{Aa} = 2.74/107.4°$, $\mathbb{I}_{Bb} = 3.04/-12.1°$, $\mathbb{I}_{Cc} = 3.02/-132.3°$. The neutral current is no longer zero, with the output listing $\mathbb{I}_{Nn} = 0.29/115.5°$. This implies that if the neutral wire were removed, conditions would change. This is unlike the balanced case in which $\mathbb{I}_{Nn} = 0$, and this wire can be removed without altering any output. Removing the neutral wire requires only removing (or "commenting out" by adding an asterisk in the first column) the RLOSSN element line. With the neutral wire removed, rerunning SPICE, the currents have changed to $\mathbb{I}_{Aa} = 2.83/107.7°$, $\mathbb{I}_{Bb} = 2.99/-10.7°$, and $\mathbb{I}_{Cc} = 2.99/-134°$. Note that the degree of imbalance caused by this modest (10%) source imbalance is itself relatively small. Assuming balanced circuit conditions will not lead to large errors in the presence of minor source, line, or load imbalances.

11.6.1. Find the line voltage and phase current for the load of phase A of the system of Fig. 11.22 if the load of phase C is short-circuited (called a phase fault).

Answer: 204.9/32.8° V; 4.07/90.54° A

11.6.2. Repeat Exercise 11.6.1 if the 2-Ω neutral line between nodes 0 and 10 is removed.

Answer: 204.9/32.8° V; 5.33/82.41° A

SUMMARY

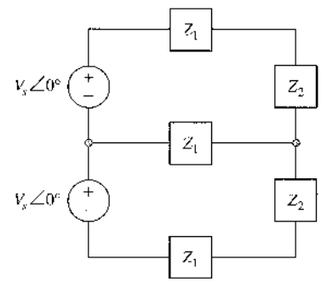
Three-phase power is by far the most common form of power distribution. Its advantages when compared to single-phase power delivery include smoother and more efficient power delivery. These benefits are maximized if the circuit is balanced; that is, corresponding circuit variables in the three individual phases are equal in magnitude and 120° out of phase. This chapter presents an introduction to balanced three-phase circuits.

- A Y-connected source is balanced if its phase voltages are equal in magnitude and 120° out of phase.
- The line voltages in a balanced wye-connected three-phase source are $\sqrt{3}$ times bigger than its phase voltages.
- Wye- or Δ-connected loads are said to be balanced when their three component impedances are identical.
- In a balanced Y–Y circuit, the neutral wire may be removed or inserted without affecting any current or voltage.
- In a balanced four-wire Y–Y circuit, calculations may all be done on a per-phase basis, that is, with no interaction between the circuit variables of different phases.
- The equivalent of a balanced Y-connected load with phase impedances Z_p is a Δ-connected load with impedances $Z_l = 3Z_p$.
- A Y–Δ system is best analyzed by first performing a Δ–Y load transformation, then using per-phase calculations.
- SPICE has no special routines for three-phase circuits. It does not recognize balanced circuits nor do per-phase calculations.

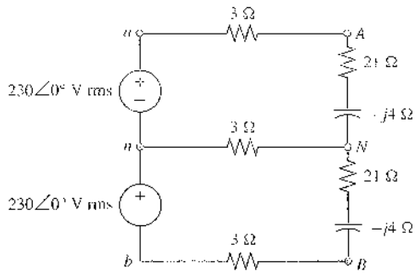
PROBLEMS

11.1. If in Fig. 11.3 $\mathbb{V}_{an} = \mathbb{V}_{nb} = 100/0$ V rms, the impedance between terminals A–N is $10/60°$ Ω, and that between terminals N–B is $10/-60°$ Ω, find the neutral current $\mathbb{I}_{nN}$.

11.2. For the symmetrical single-phase power delivery circuit shown, derive an expression for the loss fraction l, defined as the ratio of the real average power dissipated by the line impedances Z_l, to the total real average power produced by the sources, in terms of Z_l and Z_2.

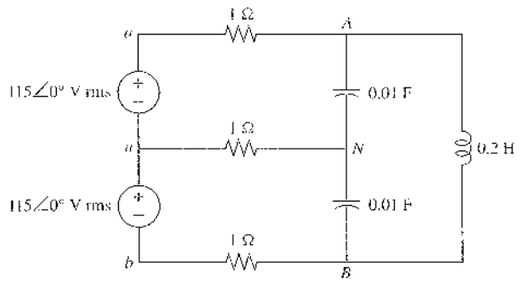


11.3. Find the currents $\mathbb{I}_{aA}$, $\mathbb{I}_{bB}$, and $\mathbb{I}_{nN}$ in this symmetric single-phase three-wire system. Sketch in a phasor diagram.



11.4. Find the real average power supplied by each of two voltage sources in the circuit of Problem 11.3.

11.5. Determine the currents $\mathbb{I}_{aA}$, $\mathbb{I}_{bB}$, and $\mathbb{I}_{nN}$, and sketch in a phasor diagram. The source frequency is 60 Hz.



11.6. Determine the Thevenin equivalent of the circuit of Problem 11.5 at the terminals AB when the inductor is removed. What value of inductance connected between A and B causes the complex power delivered by the sources to be purely real?

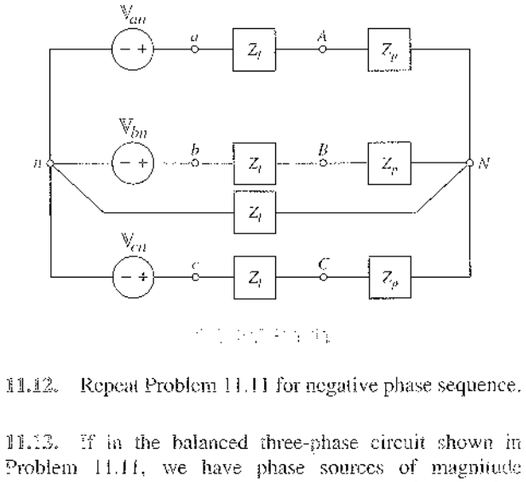
11.7. In Fig. 11.9 a balanced, positive-sequence source has $\mathbb{V}_{ab} = 120\underline{/0}$ V rms and $\mathbb{I}_{aA} = 10\underline{/-60^\circ}$ A rms. Find $\mathbb{Z}_p$ and the power delivered to the three-phase load.

11.8. A balanced three-phase Y-connected load draws 1.2 kW at a power factor of 0.6 leading. If the line voltages are a balanced 200-V rms set, find the line current I_L.

11.9. A balanced Y–Y system with $\mathbb{Z}_p = 3\sqrt{3}\underline{/30^\circ}$ Ω delivers 9.6 kW to the load. Find the line voltage V_L and the line current I_L.

11.10. In Fig. 11.9, the source is balanced, with positive phase sequence, and $\mathbb{V}_{an} = 100\underline{/0}$ V rms. Find $\mathbb{Z}_p$ if the source delivers 3.6 kW at a power factor of 0.6 leading.

11.11. For the balanced wye–wye system shown, with $\mathbb{V}_{an} = 115\underline{/0}$ V rms, $Z_l = 0$ Ω, $Z_p = 2 + j3$ Ω and positive phase sequence, compute the line voltages $\mathbb{V}_{ab}$, $\mathbb{V}_{bc}$, $\mathbb{V}_{ca}$ and line currents $\mathbb{I}_{aA}$, $\mathbb{I}_{bB}$, $\mathbb{I}_{cC}$, and sketch them in phasor diagrams.



11.12. Repeat Problem 11.11 for negative phase sequence.

11.13. If in the balanced three-phase circuit shown in Problem 11.11, we have phase sources of magnitude 4400 V rms, $Z_l = 0.01$ Ω/km, and $Z_p = 20 + j4$, find an equation relating line length in kilometers to rms line current, and sketch as a graph.

11.14. A balanced wye–wye system with 60-Hz phase sources of 1200 V rms and no line losses delivers 3.00 kW to a load consisting of a series combination of a 10-Ω resistor and L-henry inductor. Find L.

11.15. What is the magnitude of the source phase voltage needed to deliver 100 W at 60 Hz to a small three-phase motor whose Z_p is equivalent to a series combination of a 12-Ω resistor and 10-mH inductor?

11.16. A balanced wye–wye three-phase system with phase voltage of 115 V rms supplies power at a power factor of 0.85 lagging. Find the complex power supplied to the three-phase circuit if the line current is 2.3 A rms.

11.17. A balanced Y–Y three-wire, positive-sequence system has $\mathbb{V}_{an} = 200/0$ V rms and $\mathbb{Z}_p = 3 + j4$ Ω. The lines each have a resistance of 1 Ω. Find the line current I_L, the power delivered to the load, and the power dissipated in the lines.

11.18. A balanced Y-connected source, $\mathbb{V}_{an} = 200\underline{/0}$ V rms, positive sequence, is connected by four perfect conductors (having zero impedance) to an unbalanced Y-connected load. $\mathbb{Z}_{AN} = 10$ Ω, $\mathbb{Z}_{BN} = 10\underline{/-30^\circ}$ Ω, and $\mathbb{Z}_{CN} = 20\sqrt{2}\underline{/75^\circ}$ Ω. Find the neutral current and the power absorbed by the load.

11.19. In Fig. 11.12, let $V = 115$ V rms, $Z_L = Z_N = 0$ and $Z_3 = 20\underline{/+15^\circ}$. For what $Z_1 = |Z_1|\underline{/+15^\circ}$ will the line currents in the single- and three-phase circuits be equal? Compute the complex power delivered to the load for the single-phase current with this Z_1 and the complex power delivered to the three-phase load.

11.20. Consider Fig. 11.12 with V, Z_L, Z_N, and Z_3 as in Problem 11.19. For what $Z_1 = |Z_1|\underline{/15^\circ}$ will the real average power delivered to the single-phase load and three-phase loads be equal? Compute the line current magnitudes in both circuits if this Z_1 is used in the single-phase case.

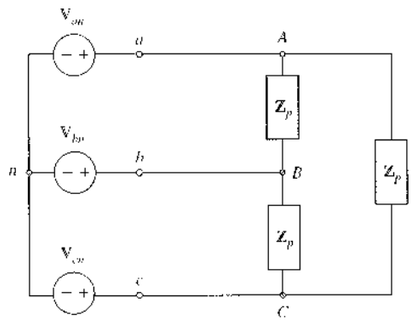
11.21. Why is the frequency of mechanical vibration of a single-phase motor twice that of its electrical supply?

11.22. Consider Fig. 11.12 with $V = 4.4$ kV rms, each source operating at a power factor of 1.0, and each circuit supplying 660 kW to its load. Sketch phasor diagrams for the line currents $\mathbb{I}_{aA}$, $\mathbb{I}_{bB}$, $\mathbb{I}_{a'A'}$, $\mathbb{I}_{b'B'}$, $\mathbb{I}_{c'C'}$ and the total instantaneous power absorbed by the single-phase and three-phase loads as a function of time.

11.23. In Fig. 11.15 the source is balanced with positive phase sequence and $\mathbb{V}_{an} = 100/0$ V rms. If the phase impedance is $3\sqrt{3}\underline{/30^\circ}$ Ω, find the line current and the power delivered to the load.

11.24. In Fig. 11.15 the positive sequence system has $\mathbb{V}_{an} = 200\underline{/0}$ V rms. Find $\mathbb{Z}_p$ if the source delivers 2.4 kW at a power factor of 0.8 lagging.

11.25. In the Y-Δ system shown, the source is positive sequence with $\mathbb{V}_{an} = 100\underline{/0}$ V rms and the phase impedance is $\mathbb{Z}_p = 3 - j4$ Ω. Find the line voltage V_L, the line current I_L, and the power delivered to the load.

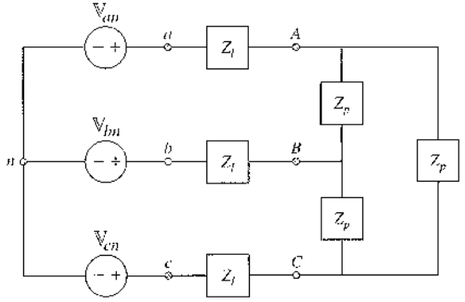


11.26. For the Y-Δ system of Problem 11.25, $\mathbb{Z}_p = 4 + j3$ Ω and the power delivered to the load is 19.2 kW. Find the line current I_L and the source phase voltages if $\mathbb{V}_{an}$ is the reference.

11.27. What advantages and disadvantages might six-phase power delivery systems have compared to three-phase?

11.28. For the delta-connected load shown in Fig. 11.15, find the line currents and real average power to the load if $Z_p = 3 - j$ and the line voltages are $\mathbb{V}_{ab} = 25\underline{/0}$, $\mathbb{V}_{bc} = 25\underline{/-120^\circ}$, $\mathbb{V}_{ca} = 25\underline{/+120^\circ}$.

11.29. For the balanced, positive phase sequence wye–delta circuit shown, assume $\mathbb{V}_{an} = 1200\underline{/0}$ V rms, $Z_l = 0.5$ Ω, and $Z_p = 8 + j2$ Ω. Find the line currents $\mathbb{I}_{aA}$, $\mathbb{I}_{bB}$, and $\mathbb{I}_{cC}$ and the complex power delivered to the load. Do not use wye–delta transformation.

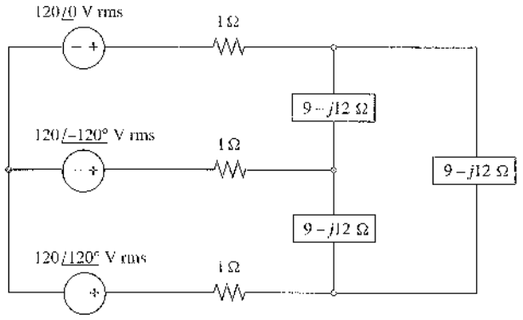


11.30. A balanced wye-connected source with line voltage 115 V rms supplies power to a balanced delta-connected load with phase loads Z_p each consisting of a parallel combination of a 75-Ω resistor, 2.5-mH inductor, and 10-μF capacitor. Find the complex power delivered to the load and the power factor at 60 Hz and at 400 Hz.

11.31. For the circuit of Problem 11.29, suppose $\mathbb{V}_{an} = 1200/0$, $Z_l = 0$ Ω, and the three-phase source delivers +2 kvars of reactive power to the load. Find Z_p and the real average power in watts delivered if the power factor is 0.92 lagging.

11.32. In the Y–Δ system of Problem 11.25 the source voltage $\mathbb{V}_{an} = 100\underline{/0}$ V rms and $\mathbb{Z}_p = 10/60°\ \Omega$. Find the line voltage, the line current, the load current magnitudes, and the power delivered to the load.

11.33. Find the line current I_L.



11.34. Find the power delivered to the load in Problem 11.33 if the magnitude of the source voltages is 200 V rms and $\mathbb{Z}_p = 9 - j9\ \Omega$.

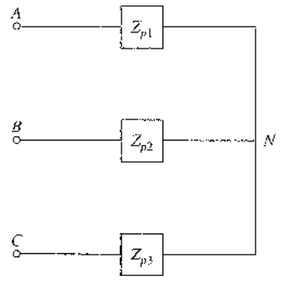
11.35. A balanced three-phase, positive-sequence source with $\mathbb{V}_{ab} = 200\underline{/0}$ V rms is supplying a Δ-connected load, $\mathbb{Z}_{AB} = 50\ \Omega$, $\mathbb{Z}_{BC} = 20 + j20\ \Omega$, and $\mathbb{Z}_{CA} = 30 - j40\ \Omega$. Find the line currents and the power absorbed by the load.

11.36. A balanced three-phase positive-sequence source with $\mathbb{V}_{ab} = 240\underline{/0}$ V rms is supplying a parallel combination of a Y-connected load and a Δ-connected load. If the Y and Δ loads are balanced with phase impedances of $8 - j8\ \Omega$ and $24 + j24\ \Omega$, respectively, find the line current I_L and the power supplied by the source, assuming perfectly conducting lines.

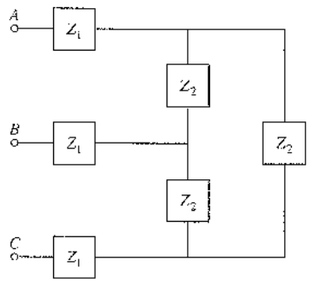
11.37. A balanced delta-connected load has $\mathrm{Re}(Z_p) = 14\ \Omega$. If a balanced wye-connected source delivers 1 kW at a power factor of 0.5 leading, find Z_p and the reactive power delivered to the load.

11.38. Draw an *RLC* circuit diagram for the wye-connected load that is equivalent at 60 Hz to a delta-connected load with each phase load consisting of a 300-Ω resistor in series with a 2-mH inductor. Repeat at 600 Hz.

11.39. Find the equivalent delta load if (a) the load is balanced $Z_p = 2 + j$; (b) $Z_{p1} = Z_{p2} = 2 + j$, $Z_{p3} = 2 - j$.



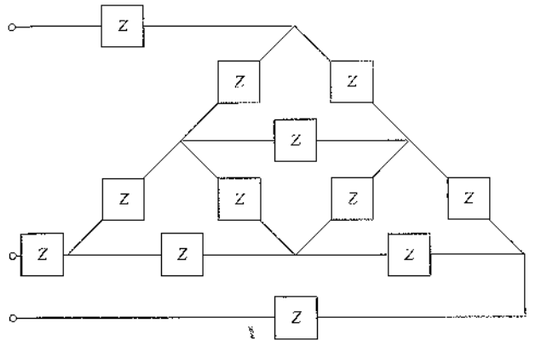
11.40. Find the equivalent delta- and wye-connected loads if $Z_1 = -j30\ \Omega$ and $Z_2 = 60 + j15\ \Omega$.



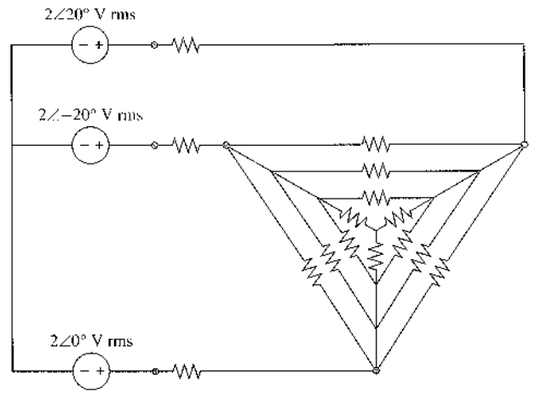
11.41. Solve Problem 11.5 by use of delta–wye load transformation.

11.42. A balanced three-phase wye-connected source with line currents 2 A rms is supplying 300 W to a balanced delta-connected load with phase impedance angle $\angle Z_p = +20°$. Find Z_p and the line voltage.

11.43. Each $Z = 1\ \Omega$. Find the delta-connected three-phase equivalent load. Check using SPICE.

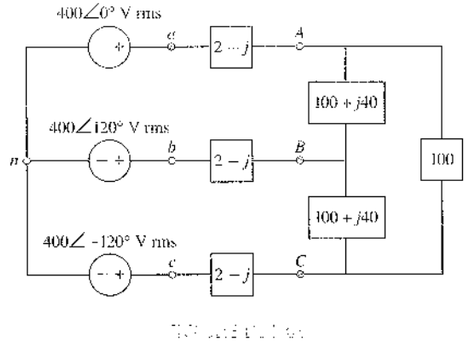


11.44. Find the total power delivered to the three-phase load by the unbalanced source. All resistors are 10 Ω. Check using SPICE.



11.45. Solve Problem 11.5 above using SPICE.

11.46. The circuit shown is operated at 60 Hz. Find the phase load currents I_{AB}, I_{BC}, and I_{CA} using SPICE.



11.47. For the circuit of Problem 11.46, suppose the lines are protected by circuit breakers set to trip when a line current I_{aA}, I_{bB}, or I_{cC} reaches 100 A. If the 100-Ω phase impedance between *A* and *C* were replaced by an *R*-Ω resistor, for what approximate value of *R* will a circuit breaker trip? Solve using SPICE. Guess at a trial value for *R*, reduce it if the limit is not exceeded and increase if it is. Find *R* to two significant digits.

11.48. Write three loop equations for this circuit using line current $\mathbb{I}_1$, $\mathbb{I}_2$, and $\mathbb{I}_3$. Include the neutral line in each loop. Rewrite for the case $\rho = 0$, calling the resulting balanced current variables $\mathbb{I}_{1b}$, $\mathbb{I}_{2b}$, and $\mathbb{I}_{3b}$. Now expressing the unbalanced currents $\mathbb{I}_1$, $\mathbb{I}_2$, and $\mathbb{I}_3$ in terms of a balanced and unbalanced component, $\mathbb{I}_1 = \mathbb{I}_{1b} + \mathbb{i}_1$, $\mathbb{I}_2 = \mathbb{I}_{2b} + \mathbb{i}_2$, $\mathbb{I}_3 = \mathbb{I}_{3b} + \mathbb{i}_3$, subtract the second (balanced) set of equations from the first. The resulting equations may be easily solved for the unbalanced components. Find $\mathbb{I}_1$, $\mathbb{I}_2$, and $\mathbb{I}_3$ for $V = 115$ V rms, $R = 1\ \Omega$, $Z = 1 + j\ \Omega$ and $\rho = -\frac{1}{2}\ \Omega$.

